APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R

3. DATE RECEIVED BY STATE	State Application Identifier

1. TYPE OF SUBMISSION

❍Pre-application ❍Application ●Changed/Corrected Application

2. DATE SUBMITTED 06/14/2024

Applicant Identifier

4. a. Federal Identifier AG066530

b. Agency Routing Identifier

c. Previous Grants.gov Tracking ID GRANT14185423

5. APPLICANT INFORMATION **UEI:** G88KLJR3KYT5

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street
City: Los Angeles
County/Parish:
State: CA: California
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Department:
Division:
Street1: 3720 S. Flower Street
Street2: Fl 3
City: Los Angeles
County/Parish: Los Angeles
State: CA: California
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 90089-0000
Phone Number: 2137401182
Fax Number:
Email: reymundd@usc.edu

6. EMPLOYER IDENTIFICATION NUMBER*(EIN) or (TIN)*: 1-951642394-A1

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍Women Owned ❍Socially and Economically Disadvantaged

8. TYPE OF APPLICATION:

❍New ❍Resubmission
●Renewal ❍Continuation ❍Revision

If Revision, mark appropriate box(es).

❍A. Increase Award ❍B. Decrease Award ❍C. Increase Duration
❍D. Decrease Duration ❍E. Other*(specify)*:

Is this application being submitted to other agencies? ❍ Yes ● No What other Agencies?

9. NAME OF FEDERAL AGENCY:
National Institutes of Health

10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER:
TITLE:

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Alzheimer's Disease Research Centers

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

13. CONGRESSIONAL DISTRICT OF THE APPLICANT:
CA-034

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Helena	Chang	Chui	

Position/Title: Chair and Professor of Neurology
Organization Name: University of Southern California
Department: KSOM Neurology
Division:
Street1: 1540 Alcazar Street
Street2:
City: Los Angeles
County/Parish: Los Angeles
State: CA: California
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 90033-0000
Phone Number: (323) 442-7686
Fax Number: (323) 442-7689
Email: helena.chui@med.usc.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested	$23,696,676.00
b. Total Non-Federal Funds	$0.00
c. Total Federal & Non-Federal Funds	$23,696,676.00
d. Estimated Program Income	$0.00

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?

a. YES ○ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:

DATE:

b. NO ● PROGRAM IS NOT COVERED BY E.O. 12372; OR

○ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

● I agree

The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.

18. SFLLL or other Explanatory Documentation. File Name: Mime Type:

19. Authorized Representative

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Organization Name: University of Southern California
Department: Office of Research Contracts a
Division: Office of Research
Street1: 3720 S. Flower Street
Street2: Fl 3
City: Los Angeles
County/Parish: Los Angeles
State: CA: California
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 90089-0000
Phone Number: 2137401182
Fax Number:
Email: reymundd@usc.edu

Signature of Authorized Representative

Rey Dominguez

Date Signed

06/14/2024

20. Pre-application File Name: Mime Type:

21. Cover Letter Attachment File Name: Mime Type:

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: Keck School of Medicine		Street2: 1540 Alcazar Street
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ● Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	Overall_Abstract1044447014.pdf	Mime Type: application/pdf
8. Project Narrative	Project_Narrative1044447058.pdf	Mime Type: application/pdf
9. Bibliography & References Cited	Overall_References1044447015.pdf	Mime Type: application/pdf
10. Facilities & Other Resources	Facilities_and_Other_Resources_FINAL1044446722.pdf	Mime Type: application/pdf
11. Equipment	Equipment1044446734.pdf	Mime Type: application/pdf
12. Other Attachments	PlanforEnhancingDiversePerspectives1044446733.pdf	Mime Type: application/pdf
12. Other Attachments	Tables_COMBINED_6_13_241044447095.pdf	Mime Type: application/pdf

Abstract

The USC- ADRC has three overarching goals: 1) **Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4,** focusing on vascular and metabolic factors that modify the effect of *APOE* ε4 on cognitive decline and AD ATNv biomarkers; **2) Promote local Alzheimer Disease and Related Disorders (ADRD) research at USC; 3) Further national collaborations.** The ADRC is led by PD/PIs (*Chui and Toga*) and Associate Director (*Yassine*). It has 6 required cores, the required Research Education Component (REC), and one optional imaging core. The *Administration Core* (*Chui, Toga, Yassine*) provides administrative and scientific oversight across the USC ADRC, fosters development projects and special initiatives (e.g., real-world data from ALZ-NET), and cultivates ADRC-affiliated studies. The *Clinical Core (Schneider and Joe)* performs standardized evaluations and diagnoses using the NACC Uniform Data Set (UDS) and enrolls to enrich *APOE ε4 in* our clinical cohort. The *Data Management and Statistical Core* (*Toga and Conti*) oversees the overall ADRC database (ADRC-DB), provides study- and core-specific databases, and curates our large imaging data sets as a local and national resource. The *Neuropathology Core* (*Hiniker and Hawes*) performs standardized neuropathological examinations, stores and distributes biological tissues; develops a digital pathology library; and elucidates vascular correlates of *APOE ε4* (e.g., cerebral amyloid angiopathy). The *Outreach, Recruitment, and Engagement Core* (*Aranda and Ringman*) works with the Clinical Core to recruit and retain the primary ADRC cohort focused on under-represented minority groups (especially Latinx). The 'Plan to Enhance Diverse Perspectives (PEDP)' uses GeneScreen, which combines a web-based outreach with home-based DNA kit. The *Biomarker Core* (*Rissman*) supplements NCRAD AD biomarkers with lipid and vascular biomarkers relevant to Aims 1 and 2. The *Imaging Core* (*Toga and Pappas*) provides high field (3T and 7T) MR imaging and ensures that MRI and amyloid/tau PET scans are compatible with national initiatives such as SCAN and CLARitI. The *Research Education Component* (*Yassine and Han*) mentors early-stage investigators, partnering with USC AD-RCMAR (*Zissimopoulos*) to train investigators from under-represented minority groups (PEDP-RCMAR). The USC ADRC is the hub for ADRD research at USC and contributes special expertise in vascular and metabolic factors that modulate the effect of *APOE ε4* on cognitive decline and AD.

Project Narrative - Overall

Three overarching goals of the USC Alzheimer Disease Research Center (ADRC) are 1) recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4, the major genetic risk factor for Alzheimer disease; 2) promote local research on Alzheimer Disease and Related Disorders; 3) further national collaborations**.** The USC ADRC is led by PD/PIs (*Chui and Toga*) and Associate Director (*Yassine*), has 7 cores, and a Research Education Component (REC). We have become a major hub for ADRD research in Los Angeles, and are known for expertise in vascular and metabolic treatments to lighten the effect of *APOE4* on cognitive decline and AD.

LITERATURE CITED

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FACILITIES AND OTHER RESOURCES

The facilities and other resources available to the research team include everything needed to undertake and complete the proposed ADRC research plan. Components that are essential for the success of this project include (1) the USC Memory and Aging Center in the Health Care Consultation Center II (HCC2), which is supported by the NIA-funded ADRC and a State-funded CADC (Clinical Core), (2) the Department of Population and Public Health Sciences (Biostatistical Component, Data Core), (3) Stevens Institute of Neuroimaging and Informatics (INI) which houses a Siemens 3T and 7T MRI scanners (Data and Imaging Core), (4) the Zilkha Neurogenetics Institute (ZNI) home to the Neuropathology Core, Center for Personalized Brain Health (CPBH), 5) the Alzheimer Therapeutic Research Institute (ATRI) in San Diego which houses the Biomarker Core, and (6) the Dworak-Peck School of Social Work and the Roybal Institute on Aging. At USC, there are a wealth of other research institutes and centers, as well as T32 training grants, focused on aging and ADRD.

ADRC Core	USC School	Department, Institute, or Center
Administrative	• Keck • School of Medicine • (KSOM)	• Department of Neurology • Stevens Institute of Neuroimaging and Informatics (INI) • Zilkha Neurogenetic Institute (ZNI)
Clinical	• KSOM	• Health Care Consultation Center • Center for Health Professions
Data Management and Statistical	• KSOM	• Department of Population and Public Health Sciences • Stevens Institute of Neuroimaging and Informatics
Imaging	• KSOM	• Stevens INI
Neuropathology	• KSOM	• Zilkha Neurogenetics Institute (ZNI)
Biomarker	• KSOM	• Center for Personalized Brain Health • Alzheimer Therapeutics Research Institute (ATRI)
Outreach Recruitment & Engagement	• Dworak-Peck School of Social Work	• USC Roybal Institute on Aging
Research Education Component	• KSOM • Dornsife Letters, Arts and Sciences	• USC-Buck Geroscience Training Program in the Biology of Aging • Training for the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease • Multidisciplinary Research Training in Gerontology • Minority Aging Health Economics Research Center (USC RCMAR)
Other ADRD Resources	• Keck SOM • Dornsife LAS • Davis Gerontology • Price School of Policy	• Alzheimer's Clinical Trials Consortium (ACTC) • Southern California Clinical and Translational Science Institute (SC-CTSI) • The Institute of Translational Genomics • The USC Michelson Center for Convergent Bioscience • Urban Air Pollution and Alzheimer's Disease: Risk, Heterogeneity, and Mechanism • Schaeffer Center for Health Policy and Economics (CHPE) • The Roybal Center for Health Policy • The Keck-Schaeffer Initiative for Population Health • Center for Economic and Social Research (CESR) • USC-Brookings Schaeffer Initiative for Health Policy
PEDP- DHS	• Rancho Los Amigos	• Rancho Research Institute

ADMINISTRATIVE CORE

Center for Health Professions (CHP)

The academic offices of Dr. Chui, Dr. Schneider, Dr. Ringman, Dr. Joe, and the ADRC administrator Michelle Aranda are situated in the Center for Health Professions (CHP). Dr. Chui has been assigned 4600 square feet of office space for the Department of Neurology. This includes 18 offices for neurology faculty and administrative staff, one conference room, and a dry lab. In addition, Dr. Schneider has 1400 square feet of office space assigned to his programs in the CHP building. This includes academic offices, a dry lab, a conference room, and additional office space for four staff. There are several computers, printers, scanners, and related peripherals that will be used for data analysis, administrative and publication needs. Face-to-face videoconferencing equipment is used to enable communications between the research teams. The intellectual environment is rich with other extramurally funded investigators who are doing complementary work to what is proposed here. The academic office for Dr. Toga is locate in Stevens Institute of Neuroimaging and Informatic (see below). Dr. Yassine's office is located in the Zilkha Neurogenetics Institute (see Center for Personalized Brain Health below).

CLINICAL CORE

Clinical Facilities: Health Care Consultation Center 2

The *USC Memory and Aging Center* comprises the USC *NIH Alzheimer's Disease Research Center* (USC ADRC, directed by Chui, Zlokovic, and Toga) and the state-funded *California Alzheimer's Disease Center* (USC CADC), directed by Dr. Schneider. These entities provide clinical care and continuity of research care for people with memory complaints and for noncognitive-impaired older people who have agreed to long-term follow-up and brain autopsy.



Figure 1. Health Consultation Center 2, ADRC Clinic

The *Health Consultation Center 2* (HCC 2) (Figure 1) on the Health Sciences Campus opened in 2005. It is a five-story, 238,000-square-foot, comprehensive ambulatory care facility. HCC2 is the main home for the clinical operations of the ADRC. The ADRC is located on the third floor, within the HCC2 department of neurology space, and includes over 15,000 square feet of dedicated clinical and clinical/research space. The overall space includes 26 examination rooms, family conference rooms, a patient education library, neuropsychological testing areas, a five-chair infusion room, a clean laboratory, a medical library and conference room, investigator and research office space, and medical record storage.

For the ADRC, there are three examination rooms, 3 study physician offices, two offices for research nurses, three rooms for neuropsychological assessment, four workrooms, one large clinical study coordinator office with shelves for research books, and double-locked medication storage for some studies, work cubicles for four additional research staff, two conference rooms, and a large wait area.

The CPBH Prevention Clinic is staffed by Dr. Yassine (MD), Jinseo Choi (Nurse Practitioner), and a registry data analyst. It will be additionally supported by a registry recruiter, a registry programmer, drug discovery specialist, an executive administrator, a genetic counselor, and an advanced practice provider. The clinic will focus on prevention care for lowering the risk of dementia and preserving brain health for E4 carriers by taking a personalized approach, tailoring a combination of pharmacological and behavior intervention to fit each patient's specific needs for preventing cognitive decline.

EHR integration

Digital cognitive assessments offer a host of workflow efficiency benefits — from straightforward administration and interpretation to real-time results and structured data capture. Integrating with all major EHR system takes the benefits even further, streamlining key cognitive testing processes, such as ordering assessments and reviewing results without leaving the EHR.

Linus Health Platform

The Linus enterprise platform is integrated into the Keck Medical Center Cerner EMR system. It provides next-generation validated assessments using a process-based approach – assessing both test outputs and a person's process of performing cognitive tasks and associated with changes in amyloid and tau biomarkers. The tablet digital platforms simplify test administration and review using a web portal to manage testing, an administrator iPad app for in-clinic digital assessments, and a remote assessments app. Main platform components include a portal as the primary tool for managing participants assigning tests, viewing results, and accessing general information regarding study activity.

The Linus Research Administrator App is accessed via iPad tablet and is used in an in-clinic setting to administer digital assessments to study participants. The remote assessments app is accessed via a web browser to allow participants to complete speech assessments and QNRs from anywhere using their own Internet-enabled devices. Participants will receive a web URL that is delivered to them via SMS or email with a link to the assessments.

Other important components of the Clinical Core include a second California Alzheimer Disease Center (CADC), directed by Dr. Segal-Gidan and located at Rancho Los Amigos National Rehabilitation Center, Downey, California; 2) Roybal Comprehensive Healthcare Center, Montebello, California and 4) the Huntington Medical Research Institute (HMRI) in Pasadena, California, where a longitudinal Aging Study is ongoing directed by Dr. Michael Harrington.

DATA MANAGEMENT AND STATISTICS CORE

Department of Population and Public Health Sciences

Dr. David Conti Co-leader of the Data and Biostatistics Core, is Professor in the Department of Population and Public Health Sciences. Ranked among the top academic departments in preventive medicine and public health sciences with the largest NIH-sponsored research of all US-based medical schools, the Department is home to worldwide authorities on chronic disease epidemiology (including cancer, genetic, and cardiovascular disease), biostatistics, environmental medicine, health behavior, and prevention, detection, and treatment methods. The Department faculty has conducted cutting-edge research work with impactful findings that have informed health guidelines, changed environmental health regulations, and improved clinical practice. The key conclusion of a recent comprehensive external peer review of the Department was the "intellectual excitement of faculty and students, a precious aura of collegiality, genuine interdisciplinary interest, and world-class academic scholarship that ranges over and integrates a multitude of interconnected scientific disciplines."

The Department houses several NIH-sponsored multidisciplinary institutes and research centers, including the Institute for Health Promotion and Disease Prevention Research (IPR), USC Institute on Inequalities in Global Health, USC Clinical and Translational Science Institute, Tobacco Center of Regulatory Science, Center for Genetic Epidemiology, Southern California Environmental Health Sciences Center (SCEHSC), and the USC Children's Environmental Health Center. Many of the Department faculty leading and engaged in these institutes and centers have served as critical catalysts to promote interdisciplinary research, education, and training across the Keck School departments and USC campuses. For instance, the NIEHS-funded SCEHSC developed an Environmental Neurology Initiative in 2009, which was led by Dr. JC Chen (co-director of Data Core, USC ADRC). This new initiative has led to a new research program in environmental neurosciences with continuing NIH support since 2011 (with the cumulative NIH funding over $25 million).

Space. The newest vanguard in the expansion of the USC Health Sciences Campus is a 120,000-square-foot facility—the Soto Street Building, which is home to the Department of Preventive Medicine. This three-story building houses offices, laboratories, classrooms, a student and faculty fitness center, and a café. The space also houses the PhD, Master's, and MPH programs, and other trainees.

Intellectual Environment for Education and Training. The Department offers four PhD programs (Biostatistics; Epidemiology; Health Behavior Research; Population, Health, and Place) to prepare leaders for the evolving biomedical research fields. The Department is also the home to two major NIH-funded T32 training

grants in Environmental Genomics (T32ES013678) and Cancer Control and Epidemiology (T32CA009492). In addition to Department seminars, many of the above-mentioned institutes and research centers sponsor regular events for education and training. For instance, the Division of Biostatistics runs a weekly seminar and regular workshops for statistical programming in R.

Stevens Hall

The USC Mark and Mary Stevens Neuroimaging and Informatics Institute (INI) is housed in a state-of-the-art facility designed specifically to accommodate our research. The overall academic environment is a remarkable, specially designed modern building of 35,000 gross sq ft. called Stevens Hall for Neuroimaging (SHN) (**Figure 1**) on the Health Sciences Campus. The facility houses a data center, a state-of-the-art theater, offices, and workspaces. The allocation of significant campus funding to create this Institute and its facilities creates the opportunity to establish broad and rich multidisciplinary expertise. The Keck School of Medicine has dedicated significant campus resources, capital infrastructure, and space to LONI and the USC Stevens Neuroimaging and Informatics Institute.



Figure 1. The newly renovated Stevens Hall for Neuroimaging (SHN)

SHN houses over 100 faculty, postdocs, students, and staff in a variety of seating configurations, from private offices to shared offices to open workspaces designed for collaboration. Great care was taken to design the people space to facilitate ongoing discussion and collaboration while providing a peaceful work environment for all.

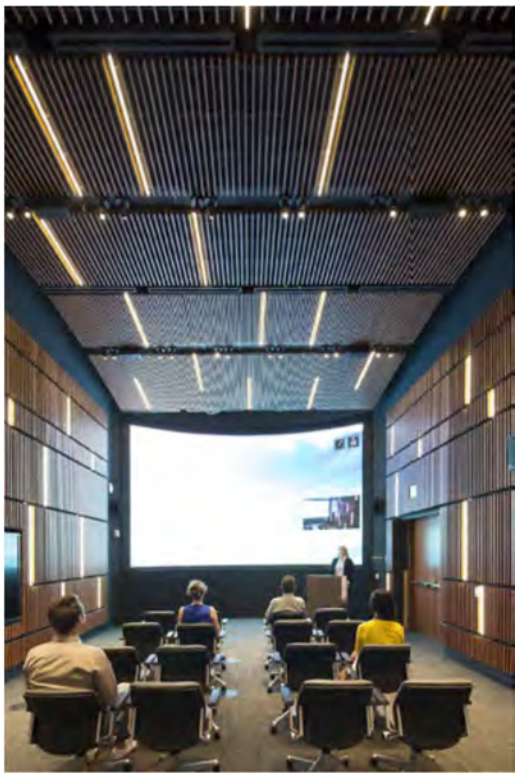


Figure 2. Data Immersive Visualization Environment (DIVE)

The Institute provides an extensive infrastructure designed and operated to facilitate modern informatics research and support for hundreds of projects, including several multi-site national and global efforts. We have redundancies built into all equipment and a completely secure facility to protect equipment and data. The resources described below provide networking, storage, and computational capabilities that will ensure a stable, secure, and robust environment. It is an unprecedented test bed to create and validate big data solutions. Because these resources have been designed, built, and continuously upgraded over the years by our systems administration team, we have the appropriate expertise and operating procedures in place to use these resources to their maximum benefit.

In addition to a data center, SHN houses a 50-high-definition ion theater – the Data Immersive Visualization Environment (DIVE) (**Figure 2**). The prominent feature of the DIVE is a large, curved display that can present highly detailed images, video, interactive graphics, and rich media generated by specialized research data. The DIVE display features a dominant image area, with consistent brightness across the entire display surface, high contrast, and 150° horizontal viewing angle. The display resolution target is 4k Ultra HD, 3840x2160 (8.3 megapixels), in a 16:9 aspect ratio. The DIVE is designed to facilitate research communication, dissemination, training, and high levels of interaction.

Adjoining the SHN on the north side is a brand-new MRI facility. The Center for Imaging Acquisition houses a Siemens Magnetom Prisma 3 Tesla MRI scanner and a Siemens Magnetom Terra 7T MRI scanner. The MAGNETOM Prisma 3T includes simultaneous 80mT/m @ 200T/m/s gradients, a new, high-end gradient system that delivers high gradient amplitudes and fast switching capabilities in a combination that is currently truly innovative. The Siemens Magnetom Terra 7T MRI system is an investigational device. Both MRI scanners are at the leading edge for neuroimaging.

The Laboratory of Neuro Imaging (LONI) (PI Arthur Toga) encompasses a diverse team of neuroscientists, engineers, and data scientists, all working toward a common goal: mapping the brain. LONI researchers study diseases, including neurodegenerative and neuropsychiatric disorders, and seek to understand changes in the brain across the lifespan. Originally focused on linking brain structure and function using sophisticated connectivity maps, LONI has expanded its work to include the application of 3D visualization, emerging mathematics, and large-scale computational architecture to biomedical datasets. LONI is a leader in the study of healthy neurological function, as well as brain development and aging, including disorders such as Alzheimer's disease, Parkinson's disease, schizophrenia, autism spectrum disorder, bipolar disorder, HIV, depression, and stroke.

Additional Office Spaces. Additional office spaces for faculty, staff, and students are available in Suite 200 of the Marina Towers in Marina Del Rey, CA. This building houses other high-tech research groups, such as the Information Sciences Institute, and serves as the home for the Imaging Genetics Center and ENIGMA, part of the USC Stevens Neuroimaging and Informatics Institute. Suite 200 has 13 faculty and shared offices, two director's offices, a conference room, four student workrooms, and a large kitchen.

The Enhancing Neuro Imaging Genetics through Meta Analysis (ENIGMA) Consortium (PI Paul Thompson) is a collaborative network of researchers working together on a range of large-scale studies that integrate data from 70 institutions worldwide. Organized into Working Groups that tackle questions in neuroscience, genetics, and medicine, ENIGMA studies have analyzed neuroimaging data from over 12,826 subjects. In addition, data from 12,171 individuals were provided by the CHARGE consortium for replication of findings in a total of 24,997 subjects. By meta-analyzing results from many sites, ENIGMA has detected factors that affect the brain that no individual site could detect on its own and that require larger numbers of subjects than any individual neuroimaging study has currently collected. ENIGMA's first project was a genome-wide association study identifying common variants in the genome associated with hippocampal volume or intracranial volume. Continuing work is exploring genetic associations with subcortical volumes (ENIGMA2) and white matter microstructure (ENIGMA-DTI). Working groups also focus on understanding how schizophrenia, bipolar illness, major depression, and attention deficit/hyperactivity disorder (ADHD) affect the brain.

Data Center

Data Center Infrastructure. The data center is approximately 3,000 square feet and was designed using cutting-edge high-density cooling solutions and high-density bladed compute solutions. A total of 48 racks are dedicated to research use, 10 of which are reserved for core services. The core services are on separate, dedicated, redundant power to ensure continuous operation.

The data center includes a PowerWare 9395 UPS system providing two 750kW/825kVA UPSs in an N+1 configuration for non-core racks and two 225kW/250kVA in a 2N configuration for core services racks. The UPS sends conditioned power to 300kVA Power Distribution Units (PDUs) located inside the data center. The PDUs feed 400A-rated Track Power Busways mounted above rows of racks, providing an "A" bus and a "B" bus for flexible overhead power distribution to the racks. There are VRLA batteries with 9 minutes of battery run time for the core services UPS and 6 minutes of battery run time for the non-core UPS (note that the generator requires less than 2 minutes of battery run time in order to fully take over the load in the event of an outage). A 750kW/938kVA diesel emergency generator located in a weatherproof, sound-attenuated enclosure adjacent to the building will provide at least 8 hours of operation before needing to be refueled.

There are 2 CRAC (Computer Room Air Conditioning) units in the Data Center that monitor and maintain the temperature, air distribution, and humidity in the data center. Each CRAC unit has 3 specially programmed motors to push out cold air into the room. There are rack cooling units between two racks located alternatively along each row of 24 racks, thus 13 on each row and 26 rack cooling units in total. Each of these rack cooling units has a set of 6 motorized fans that push cold air on either side into the racks. Both CRAC units and the 26 Rack cooling units have a constant supply of chilled water where the heat exchange occurs through convention. The chilled water supply is regulated by two Chillers and an additional emergency chiller located on our building roof.

Data Center Physical Site Security. The data center is secured by three levels of physical access to ensure

HIPAA compliance for data security. The facility is secured 24/7 with card and biometric access control devices. The interior of the data center contains glass break and motion detection sensors that report events directly to the campus Department of Public Safety (DPS). Only authorized personnel are allowed in, and guests are permitted only after checking in and only during business hours. The data center itself is additionally secured by a second layer of proximity card access. Only authorized staff are permitted to enter the data center facility. All racks are secured, allowing the functional presumption that any system might contain sensitive data, ensuring all systems receive the same level of protection, minimizing the physical attack surface for the entire data center.

Computational and Storage Resources. Rapid advancements in imaging and genetics technology have provided researchers with the ability to produce very high-resolution, time-varying, multidimensional data sets of the brain. The complexity of the new data, however, requires immense computing capabilities. As research paradigms and ways of attacking problems change, infrastructure must be easily mutable to meet researchers' needs. The Cisco Nexus and UCS data center platforms provide a high throughput and software-defined infrastructure platform necessary for keeping up with high rates of change.

The compute infrastructure within the data center boasts 4,096 cores and 38 terabytes of aggregate memory space. This highly available, redundant system is designed for demanding big data applications. Blades in the Cisco UCS environment are easy to replace. A failing blade sends an alert to Cisco, generating a replacement ticket automatically. Upon arrival, the new blade can go from the shipping box to fully provisioned and in production in as little as 5 minutes.

A separate graphical computing infrastructure is distributed among different commodity-based servers that primarily have graphical processing units to obtain high-resolution results from large and complex datasets retrieved through MRI scans of brain physiology. Each server has a set of graphical processing nodes and memory depending on its specifications. Each node has a different number of cores according to their version and specification, with a total number of 131,000 cores distributed among the GPU-based server farm.

The Cisco UCS blade solution described above allows the Institute to run the services of a much larger physical infrastructure in a much smaller footprint without sacrificing availability or flexibility. Each Cisco chassis hosts 8 server blades and has 160 gigabits of external bandwidth available per chassis. With Cisco Virtual Port Channel (VPC), bandwidth is dynamically allocated to blades as needed. Each of the 48 racks can hold up to UCS 6 chassis (48 blades) plus requisite networking equipment (4 fabric extenders). Thus, the new data center has adequate rack space to accommodate this project.

Institutions and scientists worldwide rely on the Institute's resources to conduct research. The USC Stevens Neuroimaging and Informatics Institute is architected using a fault-tolerant, high-availability systems design to ensure 24/7 functionality. The primary storage cluster consists of 52 Generation 6 Dell EMC Isilon nodes, with over 12 petabytes (PB) of high-performance storage. The secondary storage cluster is the Ceph storage system, which comprises several different commodity servers with a total storage of 9.5 petabytes (PB) and much more capacity to expand. The Ceph Storage Cluster is rapidly expanding and will soon be rendered as the primary storage cluster for our Institute. This storage solution combines all the modern-day technologies of high-density SSD and NVMe-based drives with the open-source-based system of Ceph that allows right-to-use without any overhead licensing fee and ensures a sustainable use of storage cost for resources equivalent to capacity. The Image & Data Archive has a dedicated cluster of 19 Isilon nodes with 2PB usable capacity and an 8-node Ceph Storage system with 1PB capacity, adding up to a total of 3PB for Image & Data Archive storage. A 3-node Cisco UCS C3260 storage cluster provides 1 petabyte of storage running an operating system and filesystem tailored specifically for large informatics workflows. Thus, the current total storage capacity of the data center is over 25 petabytes.

Data in these clusters moves exclusively over 10g links except node-to-node communication in the Isilon clusters, which is handled by 40GbE Ethernet (12PB cluster). Fault tolerance is as important as speed in the design of this data center. The Isilon storage clusters can each gracefully lose multiple nodes simultaneously without noticeably affecting throughput or introducing errors. The virtualization environment uses an 84 TB Unity 480F XT SAN appliance, with disks configured to provide ultra-high performance and high-performance tiers of storage. For development and research purposes, a 15TB EMC XtremIO all-flash SAN cluster and EMC VNX

15 terabyte SAN complements the storage environment, providing another avenue of redundant storage offered across differentiated networking to provide an additional layer of resilience for the data and virtualization infrastructure. All VM backend storage connectivity is iSCSI and ethernet-based, providing high performance while utilizing existing network infrastructure.

External services are load-balanced across four F5 BIG-IP load balancers. The F5 load balancers provide balancing services for websites and applications, as well as ICSA-certified firewall services. The core network is entirely Cisco Nexus hardware. Each of the Cisco Nexus 9504s supports 15 terabits per second of throughput. Immediately adjacent to this machine room is a user space with twelve individual stations separated by office partitions. These workspaces are manned by staff who constantly monitor the health of the data center as well as plan for future improvements. Each space also has a networked workstation for image processing, visualization, and statistical analysis.

Network Resources. Service continuity, deterministic performance, and security were fundamental objectives that governed the design of the network infrastructure. The Institute's intranet is architected using separate edge, core, and distribution layers, with redundant switches in the edge and core for high availability. While ground network connectivity is entirely Gigabit, server data connectivity is nearly all 10 Gigabit fiber and Twinax connected to an array of Cisco 9372 distribution switches, numerous Cisco Nexus 6628 switches, and Cisco Nexus 2248 fabric extenders. For Internet access, the Institute is connected to the vBNS of Internet2 via six fiber optic Gigabit lines using a variety of route paths to ensure that the facility's external connectivity will be maintained in the case of a single path failure.

The facility has two next-generation firewall appliances that provide network security and deep packet inspections. The USC Stevens Neuroimaging and Informatics Institute has also implemented virtual private network (VPN) services using SSL VPN and IPsec services to facilitate access to internal resources by authorized users. A VPN connection establishes an encrypted tunnel over the Internet between client and server, ensuring that communications over the web are secure.

The Institute has an extensive library of communications software for transmitting data and for recording transaction logs. The library includes software for monitoring network processes, automatically warning system operators of potential problems, restarting processes that have failed, or migrating network services to an available server. For instance, the laboratory has configured multiple web servers with Linux Virtual Server (LVS) software for high-availability web, application, and database service provisioning, as well as load balancing. A round-robin balancing algorithm is currently used such that if the processing load on one server is heavy, incoming requests, be it HTTP, JSP or MySQL, are forwarded to the next available server by the LVS software layer. Listeners on one virtual server monitor the status and responsiveness of the others. If a failure is detected, an available server is elected as master, and it assumes control and request forwarding for the entire LVS environment

To further enhance the security in the Image and Data Archive (IDA) ecosystem, IDA resources reside in a discrete network that isolates these resources from the overall INI network. This both simplifies access control management and increases its efficacy.

Offsite and Onsite Backup Resources. All critical system data and source code are backed up regularly to local nearline storage and cloned to LTO6 magnetic tape for offsite archival with Iron Mountain. Offsite backup data is kept current within one week to enable rapid redeployment of all services with a view of operations and data within a reasonable time frame. The EMC Isilon nodes retain snapshots of one month's worth of data to offer the most rapid but least disaster-resilient restoration. Snapshot retention allows rapid restoration of unintentionally overwritten or deleted data and does not require retrieval from archived tape. Nearline-stored backup data provides similarly rapid restoration of the prior year's onsite data. Offsite archived magnetic media, while the last resort, can be recalled the same day and offers retrieval of data from a date range as recent as the prior week to as old as the initial archival series of tapes over one year old. Onsite datasets grow rapidly constantly and require a flexible tape backup solution. Clones are pushed to tape by way of a pair of EMC backup accelerators to an expandable 34 LTO6 drive Quantum i6000 tape array. The array provides the backup parallelism the expansive data collection requires for offsite archival on such an aggressive schedule.

Virtualized Resources. Due to the rate that new servers need to be provisioned for scientific research, the Institute deploys a sophisticated high availability virtualized environment. This environment allows systems administrators to deploy new compute resources (virtual machines or VMs) in a matter of minutes rather than hours or days. Furthermore, once deployed, these virtualized resources can float uninhibited between all the physical servers within the cluster. This is advantageous because the virtualization cluster can intelligently balance virtual machines amongst all the physical servers. This permits resource failover if a virtual machine becomes I/O starved or a physical server becomes unavailable. The net benefit for the Institute is more software resources are being efficiently deployed on a smaller hardware footprint, which results in a savings in hardware purchases, rack space, and heat expulsion.

The software powering the virtualized environment is VMware's ESX 6. This is deployed on eight Cisco UCS B200 M3 servers and eight Cisco UCS B200 M4 servers. The B200 M3 servers each have sixteen 2.6/3.3 GHz CPU cores and 128GB of DDR3 RAM. The B200 M4 servers each have 64 2.3/3.6 GHz CPU cores and 512GB of DDR4 RAM. These sixteen servers reside within a number of Cisco UCS 5108 blade chassis with dual 8x 10 Gigabit mezzanine cards, providing a total of 160 Gigabits of available external bandwidth per chassis. Storage for the virtualization cluster is housed on the aforementioned Isilon clusters, XtremeIO, and VNX storage arrays. The primary bottleneck for the majority of virtualization solutions is disk I/O, and these storage arrays more than meet the demands of creating a highly available virtualized infrastructure whose capabilities and efficiency meet or greatly exceed those of a physical infrastructure. A single six-rack unit (6RU), eight-blade chassis can easily replicate the resources of a 600+ server physical infrastructure when paired with the appropriate storage solution such as the Isilon storage cluster.

Software

Workflow Processing. To facilitate the submission and execution of compute jobs in this compute environment, various batch-queuing systems, such as SGE, can be used to virtualize the resources above into a compute service. A grid layer sits atop the compute resources and submits jobs to available resources according to user-defined criteria such as CPU type, processor count, memory requirements, etc. The laboratory has successfully integrated the latest version of the LONI Pipeline with SGE using DRMAA and JGDI interface bindings. The bindings allow jobs to be submitted natively from the LONI Pipeline to the grid without the need for external scripts. Furthermore, the LONI Pipeline can directly control the grid with those interfaces, significantly increasing the operating environment's versatility and efficacy and improving the overall end-user experience.

IMAGING CORE

Center for Imaging Acquisition (CIA)

The Center for Imaging Acquisition is located in the Steven’s Institute for Neuroimaging and Informatics (INI) and houses a Siemens Magnetom Prisma 3 Tesla MRI scanner and a Siemens Magnetom Terra 7T MRI scanner. The MAGNETOM Prisma 3T includes simultaneous 80mT/m @ 200T/m/s gradients, a new, high-end gradient system that delivers high gradient amplitudes and fast switching capabilities in a combination that is currently truly innovative. The Siemens Magnetom Terra 7T MRI system is an investigational device. Both MRI scanners are at the leading edge for neuroimaging

The Magnetom Prisma 3T MRI is a high-performance 3T based on the Trio system. It allows for unparalled simultaneous 80mT/m @ 200T/m/s gradients. Its 60 cm bore diameter allows for magnet homogeneity at 40 cm DSV – 0.2 ppm. With a 64 – 128 channel receiver, it is really the ultimate MRI scanner for neuroimaging. High-performance gradients will allow for human connectomics. The Prisma offers as standard Diffusion Spectrum Imaging up to 514 diffusion directsions allowing characterization of crossing fibers. Higher resolution DTI with its reduced field of view – ZOOMit DTI for isotopic DTI. Higher SNR will also allow for reproducible 3D ASL perfusion and improved resting state fMRI data.



The new Magnetom Prisma is also FDA-approved for clinical imaging. This is important to allow data acquisition for patients with neurological diseases. This includes but is not limited to patients with Alzheimer's Disease, Parkinson's Disease, Multiple Sclerosis, Stroke, and Brain Tumor patients. The key part of this scanner, which is unique, is the powerful XR Gradients, which have a strength of 80 milliTesla per meter (mT/m) combined with a 200T-per-meter-per-second (T/m/s) slew rate. Furthermore, it employs advanced homogeneity and shimming functions for optimal MRI data acquisition. This will be critical for reproducibility in longitudinal studies of patients and subjects over time, such as those being studied by the USC ADRC and ADNI.

The Human Connectome Project is already working on methods for improving mapping the Connectome, such as Multi-shell HARDI reconstructed with a Fiber Orientation Density methodology. This will be enhanced by having the gradient performance of the 3T Prisma.

The USC Siemens Terra 7T MRI consists of a magnet system that generates the ultrahigh magnetic field. The electrical and mechanical shim are integrated into the gradient coil. The magnet comprises the superconducting magnet, including the system for cooling (interface for helium fill/refill, cold head), energizing (current probe) and monitoring the magnet during operation. In addition, it includes the cabling up to the point where the external lines are connected. The system consists of an actively shielded, highly homogeneous superconducting magnet (7.0 Tesla) housed in an 830 mm horizontal room temperature bore, low-loss helium cryostat, and zero helium boil-off technology. Field shimming is primarily accomplished using superconducting shim coils. Final shimming is performed with a small number of passive shims. The system is supplied with a helium-level monitor and an emergency quench heater control unit. A two-stage 4.2 K cryo refrigerator cooling system is employed, consisting of two independent cryocooler systems that eliminate the static cryogenic consumption.

Gradients & RF Receiver Technology. The gradient system essentially consists of a Gradient Power Amplifier (GPA) and a Gradient Coil (GC), both water-cooled to sustain a high-duty cycle. The Magnetom 7T uses the GPA model as is used on other Magnetom systems. The XR gradients are the same as Prisma and capable of 80mT/m @ 200T/m/s gradients. The gradient coil includes the full set of 5-second order electrical shim coils to adjust B0 homogeneity for patient and each measurement volume. The Siemens RF receiver technology has 64-channel receivers, and the USC 7T includes a Nova Medical 32-channel head coil. The system also has an 8–8-channel parallel RF transmit array (pTx). 3rd order shims provide four out of the seven possible 3rd order shims that are built into the SC72 gradients, along with 2nd 4-fold SPS cabinet, wiring for 3rd order shim, electric infrastructure expansion, SW interface to drive additional 3rd order shims.



Siemens Magnetom Terra 7T system. The Siemens 7T is an actively shielded whole body, highly homogeneous superconducting magnet. Construction of the 7T suite is complete. The 7T Terra system is designed for future clinical use with planned clinical clearances and can be switched to clinical tasks with Dual Mode.

The CIA is also equipped with a variety of patient monitoring and stimulus equipment. Patient physiologic monitoring is achieved with either a BIOPAC MP150 system or a Phillips Invivo Expression Patient Monitor (MR400). Stimulation equipment includes a Cambridge Research Systems BOLDscreen LCD display for fMRI and a Current Designs 4-button response box pad. Contrast is delivered via a Medrad Spectris Solaris EP Injector.

USC PET Imaging Science Center. PET imaging scans for the USC ADRC and overseen by the Imaging Core will take place at USC PET Imaging Science Center, in collaboration with Dr. Peter Conti, Center director. The USC PET Center, located directly adjacent to Keck Hospital of USC on the Health Sciences campus, uses a new Siemens Biograph PET/CT system. This center offers investigators centralized radiochemistry/chemistry support and technician staffing for PET studies. The cyclotron and radiochemistry laboratories contain all the equipment needed to support radiopharmaceutical development and production for molecular imaging studies. This center has helped make significant contributions to the understanding of the functional anatomy of motor control in normal health and neurodegenerative disorders. PET scans for this proposal will take place at the Health Consultation Center 2 Outpatient Imaging Center, located within walking distance of INI on the USC Health Sciences campus. The ADRC Imaging Core has already begun collecting PET data using amyloid and tau tracers across multiple ADRC-affiliated studies including), EEAJ-Jalisco (PI: Ringman), PPG (PI: Toga), SCAN and CLARiTI.

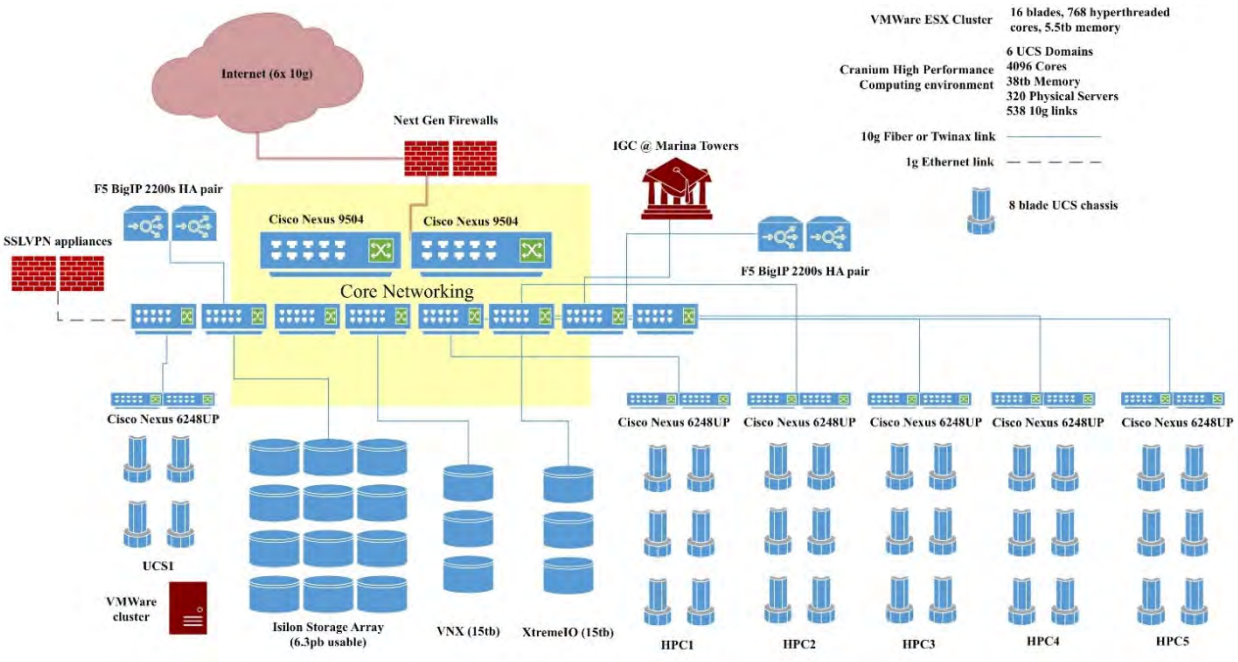


Figure 3. USC Stevens Neuroimaging and Informatics Institute network infrastructure and supercomputing environment.

NEUROPATHOLOGY CORE

USC Scientific Environment: The University of Southern California has outstanding resources in the fields of experimental neuropathology, cell biology, neurodegenerative disease research, and biochemistry. USC boasts an impressive faculty, with over 90 members named to the prestigious National Academies of Sciences, Engineering, and Invention and six Nobel laureates. Collaboration is highly encouraged both within and outside the campus, with successful partnerships with industry, government, and other academic institutions. The exceptional scientific environment at USC provides access to state-of-the-art facilities and cutting-edge technologies, allowing researchers to conduct innovative and groundbreaking research.

Zilkha Neurogenetics Institute

The Neuropathology Core, Center for Personalized Brain Health and the Los Angeles location of the Biomaker Core are housed in the Zilkha Neurogenetics Institute (ZNI).



Figure 1. The Zilkha Neurogenetic Institute on the USC Health Science Campus in Los Angeles, CA.

Zilkha Neurogenetics Institute: Upon Dr. Hiniker's arrival in January 2024, the ADRC Neuropathology (NP) Core moved to newly renovated laboratory space on the 5th floor of Zilkha Neurogenetics Institute (ZNI), a large state-of-the-art building located centrally within the USC Keck School of Medicine. ZNI is a global leader in neuroscience research that serves as a hub for neurodegenerative and psychiatric disease investigation. Research topics studied in ZNI's state-of-the-art laboratories include Alzheimer's, Parkinson's, Huntington's and other related diseases. Students and world-class researchers also study mechanisms, circuitry, and possible therapeutics for brain and neurological disorders.
This enriching environment houses innovative researchers engaged in studies complementary to the NP Core. This NP Core lab space is directly adjacent to Dr. Hiniker's research laboratory space, on the floor above Dr. Micheal Bienkowski's research lab where slide digitization will be performed (ZNI 401), and in the same building as Dr. Hussein Yassine's lab and the ADRC Biorepository Laboratory directed by Drs. Robert Rissman and Abhay Sagare (ZNI 210). This situation is ideal for direct oversight of the NP Core by Dr. Hiniker, digitization of the slide collection, and close collaboration between the NP and Biomarker cores.

ADRC NP Core Laboratory Space: The new ADRC NP Core occupies 5 adjacent rooms in ZNI that are next to Dr. Hiniker's research space and office. There is a freezer farm on the 5th floor as well as an additional freezer room in the ZNI basement. The NP Core space includes two designated -80 freezer farms (described below) for the ~700 frozen brain cases (ZNI 508 and ZNI B102, 202 and 194 sq. ft,); a fixed brain bank (described below) for the ~700 cases for which there are fixed hemibrains (ZNI 510, 173 sq ft); a room housing all glass slides and paraffin blocks for the ~1,100 cases for which there are neuropathology reports (described below) (ZNI 532, 58 sq. ft); a brain dissection laboratory (described below) for tissue procurement and request fulfillment (ZNI 514, 115 sq. ft), and a NP staff office space (ZNI 535, 122 sq. ft), described below. The NP Core has access to the biosafety cabinet in ZNI 210 for fresh or frozen tissue work. NP staff also utilize a breakroom shared with Dr. Hiniker's lab and one other lab.

Freezer Farms: Upon the arrival of Dr. Hiniker, the USC ADRC NP Core purchased 6 new PHCbi -86°C 208 V Upright Freezers (VIP ECO Series, Model # MDF-DU702VH-PA) that house the entire NP Core frozen tissue collection in a rigorously organized system and include an empty backup freezer for emergencies. 4 freezers are on of 5th floor of ZNI in a dedicated freezer farm (ZNI 508) with double air conditioning system that maintains air temperature at or below 70° C. We have a second freezer farm (ZNI B102, 194 sq. ft) with 2 PHCbi freezers and 2 Thermo Fisher ULT freezers (Model # RDE50086FA) purchased in 2019. All freezers are organized in a rack system with frozen tissue from the USC ADRC Neuropathology Core's inception in 1984. *We have frozen CNS tissue from >650 cases and lung, liver, lymph nodes, spleen, kidney, pancreas, adrenal, deltoid, intercostal and psoas muscle from >500 cases.* All freezers are hooked up to dedicated electrical lines, are on building

emergency power, and have IntelliSense Wi-Fi monitoring system to monitor the integrity and viability of stored samples around the clock. Dr. Hiniker and designated Core personnel are alerted by text and email in the event of a freezer or local power outage, a sustained temperature change, or any freezer error alert.

Fixed Brain Bank: The formalin fixed tissue, including ~700 fixed hemi and whole brains and 154 eyes, is being transferred to a designated fixed brain bank (ZNI 510). We have purchased and are in the process of installing a Spacesaver High-Density Mobile Mechanical Assist System (HDMS) for storage of the fixed CNS tissue. The HDMS system offers significantly greater storage capacity per square foot of floor space allowing us space to store ~1,200 cases in ZNI 510. Following the protocols Drs. Hiniker and Rissman used at UCSD, fixed and slabbed hemibrains will be packaged in formalin-resistant, heat-sealed plastic bags identified in multiple locations with the autopsy number, and filed in numeric order in bins also labelled with case numbers. Fixed eye tissue will be stored in a similar manner.

Slide and Paraffin Block Storage: All slides and paraffin blocks are housed in two new Tissue-Tek Lab Aid Ultra II storage cabinets, each of which accommodates up to 76,800 microscope slides or 28,160 blocks. Slides and blocks have been organized by case number in ascending order. Cabinets are kept locked and in a locked room. There is space for additional slides and blocks as the NP team completes immunohistochemistry of all 1,100 ADRC brain autopsies to fully restage older cases at NACC 11 standards. This effort is supported via institutional commitment of Keck School of Medicine Dean Meltzer to fund stains for a-synuclein, TDP-43 LATE staging, and Thal phasing on all cases.

Brain Dissection Laboratory: ZNI 514 (115 sq. ft) is a dedicated lab space for the ADRC NP Core with a bench with portable formalin fume hood, additional bench with sink, chemical and flammable storage, storage for ADRC reagents including dissection tools, and locked cabinet housing patient consents, which the ADRC diener checks prior to autopsy, which occurs in the Keck Hospital morgue, directly across the street (<500 ft) from ZNI. After the brain is removed at Keck Hospital, dissection and immediate harvest and flash freezing of the 8 most requested neuroanatomic regions (hippocampus, entorhinal cortex, frontal cortex, temporal lobe, cerebellar hemisphere, spinal cord and pituitary) occurs in the biosafety hood in ZNI 210. Teaching sessions for neurodegenerative gross pathology to neuropathology fellows and pathology residents also occurs in this space to supplement the ADRC brain cutting session which occurs at Keck Hospital. The NP Core brain dissection lab is also utilized for dissection of fixed special allocations for investigators and dissection of additional regions, allowing the team to upgrade older brain bank cases to NACC 11 staging standards.

Neuropathology Staff Office: ZNI 535 (122 sq. ft) is the neuropathology lab staff office space that has space for three ADRC staff with three desks, each with a Lenovo desktop computer. There is a dual-headed microscope for the scientific tech and Neuropathology fellow to read slides. There are two ADRC phones, a fax, scanner, and color printer. The laboratory and office spaces have networking and additional printing. There is shared access to conference rooms with audiovisual equipment. Dr. Hiniker has the adjacent office (Rm. 533, 122 sq. ft) with desk, filing space, personal microscope with dual head and camera, desktop and laptop computers, printer and scanner.

USC Center for Integrative Connectomics (CIC): The CIC, directed by Dr. Bienkowski and located in ZNI 401, is a translational center specializing in high-resolution 3D microscopy, spatial transcriptomics, and microscopy image informatics in rodent and human tissues. The CIC has established expertise in image databasing and informatics developing the USC Mouse Connectome Project and the lab is equipped with an Olympus VS120 Slide Scanner microscope connected to ADRC DMS Core infrastructure (INI servers) for USC ADRC digital pathology and machine learning analyses. In addition, the CIC is home to LifeCanvas SHIELD tissue clearing equipment for 3D imaging of thick mouse and human tissue samples and five other microscope imaging systems for advanced 3D microscopy (2 x Andor Dragonfly spinning disk confocals, LifeCanvas SmartSPIM lightsheet, Olympus FV-MPRS Multiphoton) and a Vizgen MERSCOPE for multiplexed spatial transcriptomics imaging.

EQUIPMENT

USC Neuropathology Laboratory:

Freezer Farm: (ZNI 508 and B102) 6 new PHCbi -86°C Upright Freezers (Model # MDF-DU702VH-PA) house the entire NP Core frozen tissue collection and on dedicated electrical lines, building emergency power, and

have IntelliSense Wi-Fi monitoring system with email and text alerts to the NP Core staff and leaders to continuously monitor the integrity and viability of stored samples. 2 Thermo Fisher ULT-80°C freezers (Model # RDE50086FA) house tissue for ADRC-related studies.

Fixed Tissue Repository: (ZNI 510) We are in the process of installing a Spacesaver High-Density Mobile Mechanical System (HDMS) for storage of fixed CNS tissue (space to store~1,200 hemibrains). All slides and paraffin blocks are housed in two new Tissue-Tek Lab Aid Ultra II storage cabinets, each of which accommodates up to 76,800 microscope slides or 28,160 blocks.

Brain Dissection Laboratory: (ZNI 514) Portable formalin fume hood, chemical and flammable storage cabinet, in-lab setup for cutting paraffin-embedded blocks into slides as well as performing manual immunohistochemical and histochemical staining. NP Core uses the biosafety hood in the Biofluid Biorepository (ZNI 210) for fresh and frozen tissue dissection.

Neuropathology Staff Office (ZNI 535): three staff desks, each with a Lenovo desktop computer. Dual-headed microscope for the scientific tech and Neuropathology fellow to read slides. There are two ADRC phones, a fax, a scanner, and a color printer. The laboratory and office spaces have networking and additional printing. There is shared access to conference rooms with audiovisual equipment. Dr. Hiniker's adjacent office (ZNI 533) has personal microscope with dual head and camera, desktop and laptop computers, printer, and scanner.

Bienkowski Laboratory: Five high-resolution, high-throughput microscope imaging platforms are available for use by the NP Core on the 4th floor of ZNI including:

1) Olympus VS120 Whole Slide Scanning microscope
2) Andor Dragonfly spinning-disk confocal microscope
3) LifeCanvas SmartSPIM lightsheet microscope
4) Olympus FVMPE-RS Multiphoton System
5) Vizgen MERSCOPE platform for spatial transcriptomics.

The Olympus VS120 Whole Slide Scanning microscope is ideal for high-throughput brightfield or epifluorescence imaging for image databasing. Digitization of tissue samples allows for rapid access and sharing of project data and is primarily used for the imaging of USC ADRC human neuropathology sections.

The Andor Dragonfly spinning disk confocal is an inverted microscope with 25X and 40X Nikon oil objectives, capable of high-throughput super-resolution imaging of up to 500µm thick SHIELD cleared tissue samples.

The SmartSPIM and FVMPE-RS multiphoton microscopes are ideal for 3D imaging of large tissue samples with three high-quality Olympus objectives: 1) a 10X multi-immersion objective (XLPLN10XSVMP), a 25X multi-immersion objective (XLPLN25XSVMP), and a 25Xwater immersion objective (XLPLN25XWMP2).

A new Vizgen MERSCOPE spatial transcriptomics platform. Using the Multiplexed Error Robust Fluorescent In Situ Hybridization (MERFISH) technique, the MERSCOPE is capable of imaging the subcellular expression of 1,000 genes within human tissue samples. MERSCOPE analysis software allows for clustering of single cell transcripts to demonstrate gene expression profiles that reveal cellular identity and transcriptional characteristics.

For microscopy and transcriptomics data processing, there are two dedicated Lambda Labs GPU Workstations (AMD Threadripper 3960x with 256GB RAM and 2x RTX A5000 24GB RAM graphics cards) installed with Aivia microscopy image analysis software (Leica).

Hiniker Laboratory: Additional equipment available as needed for the NP Core in the Hiniker lab (5th floor of ZNI adjacent to NP Core) includes a built-in chemical fume hood, two shared tissue culture rooms with four lab-owned CO_2 incubators, three designated tissue culture hoods, a Nikon Eclipse TS1 brightfield microscope, water bath, refrigerated centrifuge, Countess cell counter, and AMAXA nucleofector. Microscopes in the Hiniker lab include multiple light and fluorescent microscopes by Olympus, Zeiss, and Nikon, including a dual-head Nikon Eclipse E400. Additional molecular biology equipment includes four refrigerated microcentrifuges, a PCR thermal cycler (Applied Biosystems Proflex) and a Thermo QS3 qPCR machine. There are incubators for bacterial cell culture, as well as large and small format vertical and horizontal gel electrophoresis and transfer apparati, 3 eBlot wet transfer machines, iBlot transfer apparatus, a LI-COR Odyssey gel imaging system, Bio-Rad microplate reader to detect fluorescence, an Azure chemiluminescent imaging system, two 4°C fridges, and three -20°C freezers.

ZNI Shared Equipment: Cold rooms, autoclaves, a dark room, and an ice maker are located in shared spaces, and a shared Millipore water purification apparatus is available to the NP Core. Additional shared equipment on ZNI 5th floor include two Thermo Sorvall RC6+ floor centrifuges with rotors, a Beckman Optima XPR-100 ultracentrifuge with rotors, CFX-Opus RT-PCR machine, Molecular Devices Filtermax FS microplate reader, and a BioRad ChemiDoc XRS imaging system. There is are shaking and stationary incubators including a Thermo Shaker Max Q 435 HP, an Incumax IC250, a Gyromax 767, and a Thermo MaxQ 6000. There are multiple shared liquid N2 cryogenic storage tanks in the ZNI basement.

Relevant Core Facility Equipment:

USC Pathology Laboratory: Relevant equipment includes five Leica Bond-III IHC and ISH strainers, two Dako Artisan Link Pro special strainers, two cryostats, 12 microtomes, and multiple grossing stations, tissue processors, embedding stations, automated H&E stainers, automated cover slippers, and routine laboratory equipment.

The **Broad CIRM Center Optical Imaging Facility**: Relevant equipment includes four confocal microscopes, four widefield systems, a Zeiss AxioScan slide scanner, Optical Projection Tomography microscope, a Miltenyi Blaze lightsheet microscope and a Leica Thunder imaging system. The suite of confocal microscopes includes two Leica SP8-X systems; one outfitted with a pulsed laser for multiphoton excitation, the latest generation Leica Stellaris system with Fluorescent Lifetime Imaging (FLIM) and Stimulated Emission Depletion (STED) based super-resolution and a Zeiss LSM 800 upright confocal microscope. Image processing, analysis and visualization software includes Huygens (SVI) deconvolution, Imaris (Bitplane), Amira (Thermo Scientific) and VGStudio (Volume Graphics).

USC's **Mark and Mary Stevens Neuroimaging and Informatics Institute (INI)** has cutting-edge technology for database management, imaging, image analysis, and informatics. The computational and database resources available to the NP Core for their digital pathology library include a data center compliant with all HIPAA security and privacy standards. Computing infrastructure includes 3,328 cores and 26 terabytes of aggregate memory. with the main storage cluster composed of 33 EMC Isilon nodes with 2.8 usable petabytes of high performance storage. External F5 load balancers provide services for websites and applications on Cisco Nexus Hardware.

Children's Hospital Los Angeles:

Laboratory: Dr. Hawes, the Co-director of the NP Core, is also the Faculty Director of Immunohistochemistry at Children's Hospital which includes the Center for Pathology Research Services (CPRS). This laboratory has permanent dedicated research laboratory space in close proximity to Dr. Hawes' office and a full-time technician responsible for immunohistochemical (IHC) and special staining, Fusheng Yang, who has been in this position since 2016 and who is not only an expert in IHC and special stains but also in the processing and cutting human CNS tissue. The CPRS is fully capable to handling the required work for the USC ADRC NP Core and we have successfully been providing this service to this Core since 2016.

Dedicated CPRS Research Laboratory Equipment: Leica Bond RXm Fully Automated Academic Research Stainer; Thermo Scientific Gemini AS Automated H&E stainer*;* Tissue-Tek VIP tissue processor*;* Thermo Excelsior AS tissue processor*;* Thermo HM525 NX cryostat*;* 3 microtomes*;* Embedding stations x 2; Leica fluorescent Microscope DM4B*;* Olympus BX46 brightfield microscope; and Leica Aperio AT2 Scanner.

***Equipment Shared with Clinical Histology/IHC Laboratory*:** Leica BOND-MAX Fully Automated IHC and ISH Staining System; Leica BOND III Fully automated IHC and ISH Staining System; Ventana BenchMark ULTRA IHC/ISH System; Dako Artisan Link Pro Special Staining System; Ventana HE600 H&E stainer and coverslipper; Aperio GT 450 - Automated, High-Capacity Digital Pathology Slide Scanner.

University of Southern California Center for Integrative Connectomics:
Dr. Bienkowski is the director of the USC Center for Integrative Connectomics (CIC) within the USC Mark and Mary Stevens Neuroimaging and Informatics Institute (INI) and an active member of the Zilkha Neurogenetic Institute (ZNI). Dr. Bienkowski has assigned space consisting of office space in Stevens Hall for Neuroimaging (SHN) and over 1,000 sq ft of wet lab space in the Zilkha Neurogenetic Institute (ZNI), as described below. The wet lab space in ZNI houses a large histology laboratory, a surgical suite for stereotaxic surgeries, and microscopy rooms for five different high-resolution microscopy imaging systems.

Center for Personalized Brain Health (CPBH)

Dr. Yassine (MD) is a tenured Professor of Neurology, Internal Medicine, and Physiology and Neuroscience, and Director of the new Center for Personalized Brain Health (CPBH). He has dedicated laboratory spaces in Zilkha Neurogenetics Institute (ZNI 323) and Clinical Science Center (CSC 123A), offices in CSC (210) and ZNI (325), and a Prevention Clinic focusing on clinical studies and patient-facing intervention at the Keck Medical facility located at 9033 Wilshire Blvd in Beverly Hills, California.

Yassine Laboratory in ZNI is a 1760 sq. ft. research space equipped with sample processing, incubator, microscope, microplate reader, biosafety cabinets, space for 2 -80 freezers and 3 liquid N2 dewars; one room with light, temp and noise control for animal behavioral studies; and access to cores such as proteomics, lipidomics EP, and microscopy.

The Center for Personalized Brain Health Laboratory is located in the CSC building with additional equipment space, centrifuges, scintillation counters, a Millipore water purification system, an ice machine, and cold rooms. Dr. Yassine is acquiring an AF4 system (lipid analyzer) for measuring lipid fractions in plasma and cerebrospinal fluid. Upon completion of the Discovery and Translational Hub (DTH), the CPBH will be relocated to this seven-story, 202,000-square-foot building planned for a central location at the USC Health Sciences Campus. The DTH seeks to unite interdisciplinary research teams to fast-track discoveries and address the health concerns of adjacent communities. A major thrust of the work proposed for the hub would be creating innovation in precision medicine, which will tailor disease prevention and treatment to each individual's genetics, environment and lifestyle. Some floors within the designed structure will be dedicated to facilitating members of USC's neighboring communities to participate in clinical trials addressing their most pressing health issues; other floors will be dedicated to collaborative lab spaces would allowing biologists studying cells and tissues to work alongside computational scientists doing large-scale data analysis. These multimodal teams would join forces with health care providers at the adjacent Keck Hospital of USC to translate research findings to clinical contexts.

Computer. The CSC lab has numerous computers with laser jet printers, scanners, and associated peripherals that will be used for data analysis, administrative and publication needs.

CSC Procedure room. The CSC procedure room has 7 beds for performing research procedures such as lumbar punctures, phlebotomy, DEXA scans, obtaining vital signs, a sample processing unit, and two private participant interview rooms. It is staffed by research nurses and phlebotomists.

BIOMARKER CORE

The facilities and other resources at the Zilkha Neurogenetics Institute (ZNI) at the USC Health Science Campus, Los Angeles include everything needed to undertake and complete the USC centered portion of the proposed research project successfully.

Alzheimer Therapeutic Research Institute (ATRI), San Diego, California

Scientific Environment

The ATRI of USC was established in 2017 and is now one of the nation's leading research institutes for AD research. The graduate programs in neurosciences and clinical psychology together with the associated academic communities are regularly rated as among the best in the country.

The Alzheimer's Therapeutic Research Institute (ATRI) at the University of Southern California (USC) is composed of clinical trials and basic science translational research divisions for the study of AD and related disorders. The ATRI currently operates in approximately 95,000 square feet of leased space that houses the two divisions.

Institutional Support: ATRI Coordinating Center support services such as Contract and Grants management, Purchasing, and Shipping and Receiving are provided by the central USC departments. USC has provided resources for support of 4 full time Contracts Officers and 2 full time human resources staff to meet the contracting and personnel needs.

Software: ATRI research team has access to both open-source and licensed software, including statistical software such as R, and SAS. Each member of the research team has access to MS Office, Adobe Pro, and the usual suite of productivity software.

Computer: A 1GB Ethernet network through USC computing networks is currently available for all ATRI personnel. All faculty and staff are outfitted with a PC or Mac computer workstation linked up by a Gigabyte LAN network, and each office is provided a small footprint of black and white laser printer. In each suite, one common area is identified for a larger footprint of multi-function scanner printer copier.

Laboratory: The ATRI uses the services of a Central Laboratory to oversee routine safety hematology and biochemistry evaluations. Biological samples for biomarker studies collected from participants will be sent to the ATRI/ACTC Biospecimen Bank. The biorepository, which includes a bench laboratory, is housed in 45,500 sq. ft.of leased space located in the Sorrento Valley area of San Diego in close proximity to the ATRI Building. The laboratory and repository are directed by Dr. Rissman and this allows the ATRI to centralize and secure biospecimen storage for all trials the ATRI is involved in and will serve as a state of the art facility for other academic institutions and the pharmaceutical industry to store and share their residual trial specimens. Participating sites will be trained for preparation and shipment of samples to the Biospecimen Bank.

Rissman Biomarker Laboratory and Biorepository Resources

The Rissman laboratory is located at the Alzheimer's Therapeutic Research Institute (ATRI) located in the San Diego, CA campus of the University of Southern California (USC). Dr. Rissman's lab occupies approximately 8000 square feet of fully equipped lab space divided between 13 rooms furnished w/benches, certified fume hoods, safety cabinets, equipment, and supplies essential to molecular biology, biochemistry, tissue culture, neuropathology, histology, and immunohistochemistry. Dr. Rissman has two 100 sq.ft. rooms that are darkened for special microscopes, a shared 200 sq. room ft. room for freezers, and three 125 sq.ft. rooms including 2 cold rooms that are shared, one with an additional 100 sq.ft. room that holds a high-speed centrifuge. UC San Diego Information and Technology Services (ITS) provides state-of-the-art technology infrastructure and services, including telecom services, wi-fi, and high speed ethernet services to the Rissman laboratory and office spaces in the Medical Teaching Facility (MTF) on the School of Medicine campus at UC San Diego. Dr. Rissman occupies one 210 square foot office equipped with Dell Optiplex 7040 desktop computer, two twenty-four-inch high-resolution flat screen monitors, two HP color laser Jet Pro printers, one MacBook Pro laptop, and one 55" flat screen mirror- projection wall monitor. Rissman personnel occupy five 100 square foot offices equipped with all necessary equipment including printers, computers, scanners, fax machine, and telecom services, to perform administrative activities associated with active research projects conducted in the Rissman laboratory. Dr. Rissman is also the Biomarker Unit lead for the Alzheimer's Clinical Trials Consortium (ACTC) and oversees clinical trials sample processing and analysis. In addition to the ACTC lab, Dr. Rissman oversees the ACTC Biorepository which is a 7500 sq.ft. standalone facility in San Diego with -80 freezers and expertly trained staff for storing and managing human biological samples.

Rissman Lab Equipment

Cell, Molecular, and Neuroendocrine Lab Equipment:

Dr. Rissman's main basic research space at the USC ATRI is equipped with: BioRad Mini cells with Novex Power Ease 500 power sources (for SDS-Page and Western transfer); horizontal electrophoresis equipment for nucleic acid biochemistry; a C24 Incubator Shaker & Boekel incubator oven for DNA transformation in E. coli; Parr Cell Disruption Bomb, model 4634 (for cell fractionation); liquid nitrogen storage containers for cryo- preserving cell lines; a Zeiss Standard 25 ICS transmitted light microscope and an Olympus CK40 inverted microscope; one Eppendorf BioPhotometer spectrophotometer; a TECAN 96 PW Full Plate Washer and Sunrise Absorbance Reader for ELISAs. General lab equipment includes the following: one Beckman Coulter Optima LE-80K ultracentrifuge; two Sorvall refrigerated centrifuges; one Eppendorf 5804R refrigerated centrifuge and two Eppendorf 5415 C microcentrifuges; a Beckman Microfuge 12; a Heraeus Biofuge 15; Lauda E100, an Allegro Asperwash multi-tube IRMA bead washer; a Zymark Turbo VAP LV evaporator and a Savant Speed Vac Concentrator RH 200 lyophilizer; two computerized high-capacity gamma-counters; two Taylor- Wharton cryostorage refrigerators (20 XTA and 20 XTB-11M) and one Isotemp chromatography refrigerators; one REVCO Ultima -70 C freezer; Isotemp 202, and Precision 50 water baths; one C24 incubator shaker and single and multi-tube vortexers; Stoval and Roto-Shake Genie shakers and rotators; hot stir plates; incubator ovens; Rainin EDP3 electronic multichannel pipettes; automated pH meters; a Sartorius Ultra Micro balance for accurate weighing of microgram amounts of neuropeptides and molecular reagents; Fisher S-400, Mettler AE166, PE600 and AE50

weighing balances. Two newly renovated cold rooms are adjacent to Dr. Rissman's main laboratory, for performing biochemical analyses and running overnight immunoprobing and immunoprecipitation experiments, if required

Alzheimer's Biomarker and Human Biospecimen Equipment:

Dr. Rissman directs the Basic Science Translational Research Division at ATRI as well as the ACTC Biomarker Unit, which stores all clinical trials biofluids and human postmortem brains, respectively. These specimens are stored in -80 freezers that are overseen by Dr. Rissman's Cores staff. Dr. Rissman's laboratories are equipped with: RT and standard PCR machines, BioRad Bioplex Luminex machines, a MesoScale assay reader for running multiplex bioassays, Malvern Panalytical NanoSight NS300 Instrument for nanoparticle characterization, Quanterix Simoa HD-1 bioanalyzer, a Fujirebio Lumipulse G1200, a Roche ECOBAS analyzer, an AB SCIEX 5500 mass spectrometer coupled to a Shimatzu HPLC system, multiple -80 freezers for storage of samples and experienced technical personnel for handling processing samples and PK analyses. Dr. Rissman also has liquid nitrogen storage containers for cryo- preserving biospecimens; a TECAN 96 PW Full Plate Washer and Sunrise Absorbance Reader for ELISAs

General lab equipment

General lab equipment includes the following: one Beckman Coulter Optima LE-80K ultracentrifuge; two Sorvall refrigerated centrifuges; one Eppendorf 5804R refrigerated centrifuge and two Eppendorf 5415 C microcentrifuges; a Beckman Microfuge 12; a Heraeus Biofuge 15; Lauda E100, an Allegro Asperwash multi-tube IRMA bead washer; a Zymark Turbo VAP LV evaporator and a Savant Speed Vac Concentrator RH 200 lyophilizer; two computerized high-capacity gamma-counters; two Taylor-Wharton cryostorage refrigerators (20 XTA and 20 XTB-11M) and one Isotemp chromatography refrigerators; one REVCO Ultima -70 C freezer; Isotemp 202, and Precision 50 water baths; one C24 incubator shaker and single and multi-tube vortexers; Stoval and Roto-Shake Genie shakers and rotators; hot stir plates; incubator ovens; Rainin EDP3 electronic multichannel pipettes; automated pH meters; a Sartorius Ultra Micro balance for accurate weighing of microgram amounts of neuropeptides and molecular reagents; FisherS-400, Mettler AE166, PE600 and AE50 weighing balances. Two newly renovated cold rooms are adjacent to Dr. Rissman's main laboratory, for performing biochemical analyses and running overnight immunoprobing and immunoprecipitation experiments.

Microscopy Core Equipment:

Micron IV Retinal Imaging Microscope with Image Guided OCT2 System software and In-Sight Analysis Software; Kodak Imager EDAS 100; Leica Confocal Microscope; Leica Brighfield Microscope; Leica Ultra Microtome with Cryo cutting abilities; Leica Live Imaging Microscope DMI8

Alzheimer's Transgenic Animal Core, Behavioral, and TBI Equipment:

Dr. Rissman's lab has an operational transgenic mouse models core with 9 different mouse lines of neurodegeneration. 4 full-time staff employees work in the core and manage approximately 7000 mice. Dr. Rissman's mouse behavioral testing suite is equipped with: Water Maze, Barnes maze circular platform (Med-associates), T-maze and open field, zero maze, pole test, gait/stride analysis test, radial arm maze, home cage activity monitoring system (Kinder Scientific), rotorod (Med-associates), wire grip and balance beam, grip strength (San Diego Instruments), 8 SRLAB mouse startle chambers (San Diego Instruments), Noldus Ethovision XT tracking and image analyzer system on dedicated computer systems (Windows-based); Vet Equip Anesthetia Machine, AniMaze software plus camera, Open Field Activity System with a test station & additional rear Photobeams w/an Intel PII power pro system controller and data logger, SMART system, digital Stereotactic instrument; Dr. Rissman's mouse TBI suite is equipped with a controlled cortical impact CCI (open head injury) and simulated blast injury (closed head injury) apparatus.

OUTREACH RECRUITMENT AND ENGAGEMENT CORE

USC Suzanne Dworak-Peck School of Social Work (SSW)

The Leader of the ORE Core is Maria Aranda, MSW, PhD, MPA, Associate Professor, USC Suzanne Dworak-Peck School of Social Work. The SSW is strategically placed at the crux of critical societal problems in urban Los Angeles, a vastly diverse setting that reflects, in many ways, all modern complex urban environments. The

school's research activity is administered through the Hamovitch Center for Science in the Human Services (HRC), which was first endowed in 1988. HRC is home to several research centers. Dr. Aranda has a private office and access to classrooms, multi-media conference rooms, teleconferencing equipment, common spaces, and huddle rooms that can be used for scientific meetings and collaborative work.

HRC is recognized by USC as an organized research unit and maintains a strong infrastructure of staff experts in research administration, data analysis and database management, and writing and editorial services. Under the direction of the SSW Research Council, an elected faculty governance body, the HRC research administration staff, which features two nationally Certified Research Administrators (CRAs), provides investigators assistance with proposal development and submission, pre-award responses, and budget development, post-award project and budget management, and research compliance. PIs have access to Cayuse and TARA, USC's electronic research administration and information portals. These portals assist researchers and the research administration staff with various aspects of managing sponsored research projects, including pre-award and post-award activities. PIs have access to iStar, the USC online institutional review board (IRB) application system used to submit, track, and review all research studies involving human subjects. Access to other USC systems relevant to the research mission includes disClose for conflict of interest disclosure and management and Sophia for managing inventions, intellectual property, and patents.

HRC provides editorial support services through contracted consultants and online editorial companies for peer-reviewed manuscripts and proposals to assist faculty researchers in increasing the scientific and social impact of their research through such mechanisms as research altmetrics and op-eds in leading periodicals.

HRC employs a senior-level PhD biostatistician, a full-time PhD biostatistician, a part-time PhD bioengineering statistician, and a master's-level database manager to assist investigators and doctoral students in creating databases and datasets, developing design and analytic approaches, estimating sampling size requirements for specific research questions, planning, and performing interim and final analyses, and writing reports to summarize results for manuscript preparation. Additionally, HRC has access to senior biostatisticians and methodologists with expertise in specialized areas through its Southern California Statistical Support Network. HRC PIs with big datasets and complex computational models also have access to the USC High-Performance Computing Cluster's (HPCC) extensive computer facilities. HPCC features a diverse mix of computing and data resources. Two Linux clusters constitute the principal computing resource. In addition, HPCC has a central facility that provides more than 1.2 petabytes of combined disk storage and potential access to nearly a petabyte of tape storage. The HPCC computing resource consists of two shared-head nodes and various compute nodes. All PIs can obtain free accounts to supplement the HRC database management and data analysis resources to enable the successful performance of statistical analyses of longitudinal "big data." These statistical resources will be easily accessible to SSW researchers.

The HRC occupies more than 25,000 square feet of dedicated space in two locations for administration, investigators, and research project staff members and students; secure data storage facilities; three symposia spaces that include state-of-the-art video conferencing equipment; and an information technology (IT) network that contains 40 computer stations and two digital information and training laboratories offering academic and technological resources and services for students and the faculty. Offices are equipped with Dell Optiplex 990 desktop computers, which feature Microsoft Office Suite, SAS 9.2 for advanced statistical analysis, NVivo for qualitative data analysis, Bluejeans teleconferencing software, and a camera and microphone. A dedicated IT department maintains computers and local network services. The IT department provides daily backups of each computer. These backups and the data stored on these computers are secured by the university's large-data cyber security system with appropriate firewalls. The IT department provides technical assistance to investigators and is responsible for access and security on computers, laptops, and mobile devices.

USC Edward R. Roybal Institute on Aging (USC Roybal Institute)

Dr. Aranda is also the executive director of the USC Edward R. Roybal Institute on Aging, which is located in downtown Los Angeles within the USC Dworak-Peck School of Social Work. The USC Edward R. Roybal Institute on Aging is a university-wide Institute that is funded independently by the University of Southern California to advance research, policy, and practice to improve the lives of older adults, vulnerable adults, and their families from diverse communities locally, regionally, and globally. The USC Roybal Institute provides a range of support services for the development, implementation, and dissemination of research projects to USC Roybal-affiliated faculty. The goal is to improve administrative transactions to create an environment conducive to collaborative and interdisciplinary research while reducing the administrative burden. Core staff support the

success and scientific productivity of the faculty by (a) assisting with pre-award services such as preparing the non-scientific sections of the applications and assisting with IRB applications; (b) providing post-award services, including assisting with USC forms and business systems (purchasing, consultants & reimbursements), organizing seminars, conferences, workshops and meetings; and (c) assisting with copy editing of manuscripts, website management, dissemination of information including press releases, internal publications, and social media.

RESEARCH EDUCATION COMPONENT (REC)

Other USC T32 Training Programs

USC-Buck Geroscience Training Program in the Biology of Aging T32 (Keck School, PI Davies). This Geroscience training program in the Biology of Aging is a unique cooperative venture of the University of Southern California's Leonard Davis School of Gerontology (USC) and the Buck Institute for Research on Aging (Buck). In 2012, in a truly joint venture, we created the very first dedicated Biology of Aging Ph.D. program in the USA. Students take courses at both institutions (some in a fully interactive, online format) and undertake first-year lab rotations at both campuses before selecting a mentor and lab at USC or the Buck. Faculty at the USC Davis School have been training Ph.D. students in the Biology of Aging since 1964, but such students were previously admitted via faculty joint appointments in other schools/departments at USC: over 50 doctoral students have been trained in this way. Similarly, most Buck faculty had prior university appointments where they established extensive Ph.D. training records, and many have continued to train Ph.D.s from other universities after joining the Buck.

Training for the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease T32 (Keck School, PI Arthur Toga). Recognizing the critical need for a broad range of expertise in biomarker analysis at several levels of analysis, we propose a unique postdoctoral training program titled "Training for the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease" (AD). USC is home to leading experts who utilize biomarkers in their work investigating AD through a myriad of methods and analyses. Drawing from USC's eminent resources and expertise, this proposal will focus on preparing investigators for independent research careers in the multiscale and multimodal analysis of biomarkers in AD by integrating molecular and cellular methods, imaging tools and informatics, quantitative methods for clinical trial research, and large-scale population analyses. Our proposed training program draws preceptors and faculty from six schools, departments, institutes, and centers at USC, fulfilling these methodological areas of expertise: Mark and Mary Stevens Neuroimaging and Informatics Institute (INI) in the Keck School of Medicine; Department of Biomedical Engineering (BME) in the Viterbi School of Engineering; Zilkha Neurogenetic Institute (ZNI) in the Keck School of Medicine; Department of Psychology in the Dana and David Dornsife College of Letters, Arts and Sciences; Leonard D. Schaeffer Center for Health Policy and Economics, a unique collaboration between the USC Sol Price School of Public Policy and School of Pharmacy; and Leonard Davis School of Gerontology.

Multidisciplinary Research Training in Gerontology T32 (Davis School, PI Crimmins). This USC multidisciplinary research training program in Gerontology provides predoctoral and postdoctoral training within key disciplines in aging and familiarity with the theory and approaches of related disciplines. The intent is to inspire students to incorporate multidisciplinary models in their research. The training program involves predoctoral training in Biology, Gerontology, Psychology, and Preventive Medicine. Postdoctoral students can also be appointed through Economics, Sociology, and Policy Planning and Development. The focus of the program is on physical and mental health over the lifecycle and factors influencing health. The current program emphases reflect the research strength of the current faculty: cognitive decline and mental and emotional change associated with the aging process; physical and functional change across the lifespan; the role of social, contextual, environmental, and institutional factors in promoting or delaying physical and cognitive age changes; the biological pathways, including the role of genetic factors, promoting or delaying age changes. The training program is housed in a multidisciplinary independent School, the Davis School of Gerontology, devoted to the study of aging. The training is directed by a multidisciplinary faculty with a concentrated focus on aging research. The training environment is rich with resources and research opportunities.

Minority Aging Health Economics Research Center (USC RCMAR) (PI: Julie Zissimopoulos) in USC's Schaeffer Center for Health Policy and Economics works to increase the number, diversity, and academic success of researchers focusing on the health and economic well-being of minority elderly populations. The research examines the differences across racial and ethnic groups of elderly in a) healthcare decision-making, b) health

behaviors and outcomes, and c) financial behavior, including savings and work, and economic well-being.

USC Graduate and Post-Graduate Programs for Recruitment and Retention of Underrepresented Minority Students

Graduate Students

USC's Graduate Initiative for Diversity, Inclusion, and Access (DIA) Boot Camp is a writing-intensive 10-day workshop for incoming and first-year PhD students applying for Ford Predoctoral, GEM, Soros, NSF GRFP, and other major external and research fellowships. Participants will receive stipends and supplemental research awards based on specific performance goals.
Programs in Biomedical and Biological Sciences (PIBBS) recruits under-represented minority students, providing the entry portal to PhD programs at the Keck School of Medicine. Targeted fellowships for talented URM students have helped increase their representation, with URM students comprising 25% of the 2016-17 incoming class.
Suzanne Dworak-Peck School of Social Work provides Summer Fellowships to PhD and master's students to develop individual research projects and to attend and present at research conferences. The Research Council Pilot Study Program and Mini Research Fellowships match doctoral and masters-level graduate students with research faculty and provide the opportunity to lead pilot research studies.

Postdoctoral Students

The Provost's Postdoctoral Scholars Program for Faculty Diversity Program enhances the diversity of postdoctoral scholars and builds a pool of exceptional underrepresented minority candidates for the next generation of faculty. These scholars link the expertise of USC faculty and doctoral students with the knowledge and experience gained through their own research. Appointments are for up to two years ($50,000 per year salary plus fringe benefits, with a research and travel account of $5,000 per year).
USC-CONACYT Postdoctoral Scholars Program: In collaboration with Mexico's National Council of Science and Technology (CONACYT), this joint program provides postdoctoral scholars from Mexico with 1–2-year postdoctoral fellowships in fields related to science, technology, and innovation. The fellows receive a salary of up to $60,000 per year in addition to $8,000 USD towards a comprehensive benefits package.

USC RCMAR: The Minority Aging Health Economics Research Center (an NIA-funded Resource Center for Minority Aging Research [RCMAR]) provides mentorship to scholars with research focused on the health and economic well-being of minority elderly populations. The Center specifically serves as a postdoctoral student recruiting effort for individuals from under-represented minorities.

OTHER USC RESOURCES FOCUSED ON ADRD AND/OR AGING

The Alzheimer's Clinical Trials Consortium (ACTC) (NIA P01, PI Paul Aisen) supports the conduct of academic clinical trials across the continuum of Alzheimer's disease (AD). The ACTC leverages the depth and breadth of AD clinical research teams at USC, Harvard, and the Mayo Clinic and the considerable experience of investigators in AD clinical trials at the 35 ACTC member sites nationwide. The ACTC provides an optimized infrastructure to launch more efficiently, rapidly recruit, improve diversity, and complete NIH-supported clinical trials in AD and related dementias. It utilizes streamlined contracting processes and a centralized IRB with a specific focus on neurodegenerative diseases, and it incorporates state-of-the-art informatics and statistical analyses. The ACTC aims to improve the diversity of participants in AD clinical trials, incorporating learnings from its recent experience and utilizes a new Minority Outreach and Recruitment (MORE) Team to support both central and local partnerships with communities of color. The ACTC fosters continued innovation in AD trial design, providing expertise on novel cognitive, imaging, and biomarker outcomes to support future ACTC trial applications. The team evaluates promising exploratory measures embedded in its current trials, such as Tau PET imaging and computerized cognitive testing, and it works to incorporate these measures into a robust platform for future Proof of Concept (POC) trials to rapidly assess signals of efficacy. Units within the ACTC include: Administrative: Serves to coordinate the activities of the ACTC, interfacing with all of the other Units and Committees, the project teams and each of the sites. Essential components of the Administration Unit include Regulatory Affairs, Fiscal Management, Contracts, Human Resources, and Quality Assurance (QA); Biomarkers: The key objective of the Unit is to provide the infrastructure necessary for storage, pharmacokinetics screening, and optimization of tools for preventive studies; Biostatistics: The Biostatistics Unit provides the statistical leadership and oversight for all ACTC studies. The Unit aims to advance AD and related dementias (ADRD)

clinical trials research through the utilization and development of efficient clinical trial designs, a comprehensive quality assurance program, and state-of-the-art statistical methodologies and analytic approaches; Clinical Outcome Instruments: Establishing clinical meaningfulness on cognitive and functional measures used in AD trials is a critical FDA requirement. This entails advising applicants applying to utilize the ACTC infrastructure as to the best cognitive and functional measures that would be responsive to targeted change in clinical trials across the neurodegenerative disease continuum from preclinical, MCI and dementia trials; Inclusion, Diversity, and Education in Alzheimer's Disease Clinical Trials (IDEA-CT): The Inclusion, Diversity and Education in Alzheimer's disease - Clinical Trials (IDEA-CT) Committee is a standing ACTC unit, charged with developing goals, formulating a strategic plan, and serving as a source of oversight to support the ACTC's core values of inclusion, diversity and training in ADRD clinical trials; Informatics: Develops and maintains the ACTC Informatics Platform (ACTC-IP) to support the ACTC program and its scientific aims. The ACTC-IP serves as an extensible, secure, and scalable information, collaboration, and process management infrastructure to facilitate the effective execution of program- and study-related research activities by ACTC Units, investigators, study teams, and performance sites. The ACTC-IP provides facilities to collect, store, process, analyze, visualize, audit, and share scientific, operational, and administrative data using standardized data models, formats, and taxonomies; Medical Safety: Provides medical and regulatory supervision for all clinical trials coordinated by the ACTC. The ultimate goal of the ACTC Medical Safety unit is to develop better methods for clinical trials safety monitoring, including risk-based monitoring, semi-automated coding of adverse events and concomitant medications and real time support for clinical trial sites. The Medical Safety Unit is comprised of 4 physicians and 20 site monitors; MRI: Provides a comprehensive set of services that address all aspects of MRI needed for clinical trials. The ACTC MR Unit is modeled on the MR Core of ADNI, with a central core lab at Mayo under Dr. Jack, with Dr. Weiner's group providing image analysis services; Neuropathology: Captures the critical information available from autopsy examination of clinical trial subjects. The Neuropathology Unit works collaboratively with other neuropathologists committed to understanding neurodegenerative diseases in order to fulfill this mandate; PET: Provides a comprehensive set of services that address all aspects of PET needed for clinical trials supported by ACTC; Recruitment and Retention: The ACTC Recruitment Unit serves to integrate the strategic and scientific mission of efficient and representative recruitment for all ACTC trials with the necessary operational infrastructure to achieve study accrual.

Southern California Clinical and Translational Science Institute (SC-CTSI, Director and Principal Investigator Thomas Buchanan): Established in 2008 with support from the University of Southern California (USC), Children's Hospital Los Angeles (CHLA) and Clinical and Translational Science Awards (CTSA) from the National Institutes of Health (NIH), the SC CTSI is an organization devoted entirely to increasing the success of translational science. The SC CTSI is part of a national consortium of CTSA institutions that work together to improve the translational research process to get more treatments to more patients more quickly. In partnership with the Los Angeles County Department of Health Services and broad-based community health organizations throughout Los Angeles, the SC CTSI seeks to be a leader in engaging diverse communities and special populations in clinical and translational research to improve their health. It is led by a team of faculty and staff members who have deep experience in developing and supporting clinical and translational research in diverse populations. The SC-CTSI provides education, training, and online resources for workforce development, with a special focus on clinical research in diverse populations. Courses include:

- Research in Diverse Urban Populations is a certificate course on clinical and translational research in populations characterized by diversity in race, ethnicity, country of origin, and socioeconomic status.
- Study Population and Study Design in Clinical and Translational Research: Video lecture educates researchers on Study Populations, Target Populations, and Source populations, including challenges in diverse "mega-cities," such as ethnic and cultural differences, subpopulations, and factors that can bias findings and impair generalization of research findings to broader populations.
- Community-Engaged Research 101 covers health solutions among groups that have historically been underrepresented in medical studies. Clinical investigators learn how community-engaged research differs from traditional research and how it transforms the role of community stakeholders from research subjects to research participants, who help shape and drive research questions that address urgent health needs.
- Promotora Instructional Video: This Spanish-language video helps researchers develop community-engaged health interventions involving promotoras – community members trained to serve as health educators and service navigators. Lecturers review step-by-step research protocols and data collection tools that are integral to an evidence-based promotora program.

Core groups as part of the SC-CTSI include: Biostatistics, Epidemiology and Research Design: Through training and direct consultation, this group provides high quality, easy-to-access biostatistics service and guidance to investigator-initiated clinical studies and trials from concept to close-out; Central Administration: SC CTSI Administration provides leadership, direction and support to SC CTSI core groups in areas of strategic and financial planning, development and integration of programs and services, internal and external communications, program evaluation, human resources, contracts and grants, and space and facilities management; Clinical Research Informatics: This core group provides expertise in data integration, data sharing, and data security to help drive the integration of clinical care and clinical research. Clinical Research Informatics (CRI) fosters efficient, secure, and high-quality data access, management, and interoperability; Clinical Research Support: This core group promotes and supports efficient, safe, and high-quality investigator-initiated clinical trials. Teams at two locations – one on the USC Health Sciences Campus and the other at Children's Hospital Los Angeles – assist research teams with protocol development, activation, conduct and close-out; Communications: The Communications core group fulfills all internal and external communications needs for the SC CTSI, including but not limited to strategic communications with internal faculty and staff, university partners and community stakeholders; Community Engagement: The Community Engagement (CE) program bridges researchers and local communities, building sustainable partnerships to ensure that research initiatives meet real community needs; Evaluation & Improvement: This core group develops innovative approaches to data collection and continuous quality improvement to enhance SC CTSI functions and improve quality and efficiency of clinical research; Healthcare Delivery Science: The Healthcare Delivery Science (HDS) core group within the SC CTSI advances health system innovations through research awards, training, and collaborations with LA County Department of Health Services and other partners; Regulatory Knowledge & Support: RKS helps move USC and CHLA discoveries from research to commercialization. Because of the long timeline, heavy resource burden, and high regulatory hurdle of bringing new products to market, RKS works closely with investigators to identify critical steps in the go/no-go decision-making process; Research Development: The Research Development (RD) core serves as the research engine for the SC CTSI. RD provides support in various aspects of faculty and research development to help investigators conduct innovative research projects that test the feasibility of novel technologies, methodologies, and approaches to address challenges in clinical research and health problems of diverse communities; Workforce Development: This core group provides expertise in education, training, and career development with special focus on research in diverse populations.

The Institute of Translational Genomics (Keck School of Medicine, Director John Carpten) strives to make healthcare smarter based on a vision that future advances in personalized medicine will build from, benefit, and ultimately serve an incredibly diverse set of individuals. Medicine is undergoing one of its biggest transitions in recent history. For the first time, patients can be treated based not on what works best for an average person in the population but based on what will work best for each patient. Personalized medicine, also known as precision medicine, has been enabled by game-changing discoveries in genomics. Our deep understanding of DNA and how it varies from person to person is powering a new golden age of diagnostics, treatment, and predictive and preventive medicine. However, it is just the starting point in areas like cancer and rare diseases as high-throughput technology is utilized to integrate information from the molecular and systems level to ultimately improve patient outcomes. To realize the promise of precision medicine, it is important to understand the molecular and genetic diversity across all individuals. The Institute believes that precision medicine doesn't build or benefit from focusing on one population but instead requires studies involving diverse populations. The diversity of Los Angeles, in many ways, mirrors the future diversity of healthcare across America, highlighting its challenges and opportunities.

The USC Michelson Center for Convergent Bioscience (Director Steve Kay) brings together a diverse network of scientists and engineers from the USC Dornsife College of Letters, Arts, and Sciences, USC Viterbi School of Engineering, and Keck School of Medicine of USC to solve some of the greatest intractable problems of the 21st century – from cancer to neurological disease, to cardiovascular disease. The team of world-class scientists representing several disciplines – biological sciences, chemistry, computer science, engineering, medicine, nanomedicine, neuroscience, and physics – will be the catalyst that precipitates a sea change in research and development at USC and the world, establishing a model for innovation in precision medicine that quickly translates from the bench to the bedside in the form of new drug therapies, cures, high-tech diagnostics, and biomedical devices. With a generous $50 million gift from retired orthopedic spinal surgeon Gary K. Michelson and his wife, Alya Michelson, the USC Michelson Center for Convergent Bioscience is an opportunity for USC to transform and influence the course of scientific discovery and biomedicine for generations to come. The

Michelson Center for Convergent Bioscience (MCCB) brings together faculty from diverse fields to aggressively address new ways of merging the life and physical sciences. The dynamic nature of biological systems can now be described quantitatively, rather than the static views of the past. Convergence is moving us into a predictive era where mathematical models are combined with massive data collection and precision measurements to provide a deeper, scalable view of life. The Center is creating knowledge bases that greatly expand our ability to identify new entry points for treating disease. In parallel, engineering approaches can be used as a platform for unanticipated discoveries, providing a fresh perspective on disease states in addition to the design of therapeutic and diagnostic devices. Core facilities, centers, and institutes include:

Core Facilities – within Michelson Center

- Agilent Center of Excellence in Biomolecular Characterization
- Center for Peptide and Protein Engineering (CPPE)
- Core Center of Excellence in Nano Imaging (CNI)
- Dynamic Imaging Science Center (DISC)
- John O'Brien Nanofabrication Laboratory

Centers and Institutes – within Michelson Center

- The Bridge Institute
- Center for Discovery Informatics (CDI)
- Center for New Technologies in Drug Discovery & Development (CNT3D)
- Center for Peptide and Protein Engineering (CPPE)
- Convergent Science Institute in Cancer (CSI-Cancer)
- Translational Imagining Center (TIC)

The USC/UCLA Center on Biodemography and Population Health (NIA P30, PI Eileen Crimmins) is a multi-site center located at the Andrus Gerontology Center of the University of Southern California and the Program in Geriatric Medicine in the School of Medicine of the University of California at Los Angeles. It is one of 14 centers in the Demography and Economics of Aging Centers Program sponsored by the National Institute on Aging (NIA). The primary purpose of the CBPH is to provide a synergistic research environment for integrating and translating research findings from various disciplines to understand population health. The Center takes a multidisciplinary approach by linking demographers, biologists, economists, psychologists, epidemiologists, health policy specialists, medical researchers, and clinical geriatricians. This integration of biological, epidemiologic, and medical risk information, which characterizes Center research, is fundamental to understanding and projecting demographic trends and differences in population health.

Urban Air Pollution and Alzheimer's Disease: Risk, Heterogeneity, and Mechanism (NIA P01, PI Caleb Finch). This project's overarching goal is to further resolve the neurodegenerative role of traffic-related air pollutants (TRAP), a ubiquitous exposure in urban areas where most older Americans reside. The teams recently published epidemiologic data showed a strong association between elevated PM2.5 (particulate matter <2.5μm) and increased dementia risk in women >65 years, with a bias for ApoE4 homozygotes (JC Chen in Cacciottolo et al. 2017, PMID 28140404). In this report, experimental studies of female mice from the Finch-Sioutas Labs showed that exposure to nPM (a nano-sized subfraction of TRAP) was pro-amyloidogenic with an ApoE4 bias. Other changes include attrition of hippocampal CA1 neurites and myelin, which model selective damage in AD and cerebral ischemia. In a mouse stroke model, nPM exposure exacerbated cerebral ischemic damage (William Mack: Liu et al. 2016, PMID 27071057). Differences by sex and ApoE alleles suggest sources of heterogeneity in human responses to TRAP. The program project has four studies: epidemiological studies of two nationwide cohorts of women (Project 1, JC Chen: Women's Health Initiative Memory Studies; WHIMS) and men (Project 2, C Franz, and W Kremen: Vietnam Era Twin Study of Aging, VETSA) and two experimental studies of air pollution exposure (Project 3, Finch: mouse models of aging and AD; Project 4 Wm Mack, chronic cerebral hypoperfusion). The results of this program will advance understanding of TRAP contributions to AD risk and accelerated cognitive decline and provide a rationale for preventive intervention in the environmental neurotoxicology of AD.

Schaeffer Center for Health Policy and Economics (CHPE) (Director Dana Goldman). The Leonard D. Schaeffer Center for Health Policy & Economics, established in 2009 at the University of Southern California through a generous gift from Leonard and Pamela Schaeffer, reflects Mr. Schaeffer's lifelong commitment to solving healthcare issues and transforming the healthcare system. Our vision is to be the premier research and educational institution recognized for innovative, independent research that makes significant contributions to

policy and health improvement. Our mission is to measurably increase value in health through evidence-based policy solutions, research excellence, transformative education, and private and public-sector engagement. With an extraordinary breadth and depth of expertise, the Schaeffer Center has a vital impact on the positive transformation of healthcare. In 2016, the Schaeffer Center partnered with the Center for Health Policy at the Brookings Institution to establish the Leonard D. Schaeffer Initiative for Innovation in Health Policy. This unique partnership enhances the capacity of both organizations to develop evidence-based solutions to inform policymaking on some of the most pressing healthcare challenges facing the U.S. today — from the future of Medicare to reshaping the Affordable Care Act. The Schaeffer Center offers the human and technical capacity necessary to conduct breakthrough interdisciplinary research, exceptional policy analysis, and leading-edge training, with more than 30 distinguished scholars investigating a wide array of topics. Our work is augmented by a visiting scholar's program and partnerships with other universities, allowing outside researchers to benefit from our unparalleled infrastructure and data collection. The Schaeffer Center actively develops excellent research skills in new investigators who can become future innovators. The Center also supports the next generation of healthcare leaders in creating strong management, team building and communication skills.

The Roybal Center for Health Policy Simulation (Schaeffer Center, Director Dana Goldman) moves economic modeling findings into programs, practices, and policies. For over a decade, the Roybal Center for Health Policy Simulation has been developing economic models to answer salient policy questions surrounding the elderly in the United States. The Center is one of 10 established by the National Institute on Aging to move promising social and behavioral research findings out of the laboratory and into programs, practices, and policies that will improve older people's lives and society's capacity to adapt to societal aging. The Center's centerpiece effort is the Future Elderly Model (FEM), an economic-demographic microsimulation led by faculty at the University of Southern California, with collaborators from Columbia University, Harvard University, Stanford University, RAND Corporation, the University of Michigan, and the University of Pennsylvania. FEM is used to explore a variety of policy questions and has produced more than 50 peer-reviewed publications, including clusters of research highlighted in special issues of Health Affairs. FEM findings have been relied on by several government agencies, the White House, and Congress, as well as by private organizations interested in aging policy.

The Keck-Schaeffer Initiative for Population Health Policy (Keck School of Medicine and Schaeffer Center, Director Seth Seabury) combines empirical, multidisciplinary, and policy-driven research with clinical insight to develop high-value strategies to improve the health of vulnerable populations. It is a collaboration between the Keck School of Medicine of USC and the USC Schaeffer Center for Health Policy & Economics. Housed in the USC Roski Eye Institute and created with support from the USC Department of Emergency Medicine and the Schaeffer Center, KSI fosters multidisciplinary collaborations to generate new research and educational opportunities to study population health policy at the Keck School of Medicine.

Center for Economic and Social Research (CESR) (USC Dornsife, Director Arie Kapteyn) aims to improve social welfare by informing and influencing decision-making in the public and private sectors. CESR focuses on economic and health disparities, social inequality, and the needs of children, families and older Americans. Its research explores the impact of social services and public policy, shifting population trends and their implications, and how human behavior — attitudes, actions, and decision-making ability — affects well-being now and with age. The research includes collaborating with the Minority Aging Health Economics Research Center - Understanding Racial/Ethnic Inequalities in Wealth Trajectories in Middle and Late Life: Patterns and Explanations. CESR's impressive roster of researchers comes from many different fields of study, empowering us to look at complex topics from multiple angles. Centers and programs within CESR include: Behavioral Economics Studio; Behavioral and Health Genomics Center; Brain Health Observatory at USC; Center for Applied Research in Education; Center for Health and Labor Economics; Center for Self-Report Science; Center for the Study of Health Inequality; Institute for Food System Equity; Interplay of Genes and Environment Across Multiple Studies; Program for Children and Families; Program on Global Aging Health and Policy; Section on Clinical Research in Aging and Psychology; LABarometer.

Rancho Research Institute

Rancho Los Amigos National Rehabilitation Center (RLNRC)

Downey California

Geriatric Neurobehavior & Alzheimers Center
The Rancho GNAC offices are located within the newly renovated historic Harriman Building on the grounds of Rancho Los Amigos National Rehabilitation Center. The 1200 sq ft office includes a private office space where confidential interviews can be conducted. There are three exam rooms within the Harriman Building that are available by reservation for use for research studies.

Rancho Los Amigos National Rehabilitation Center (RLANRC) is a 159-bed intensive rehabilitation facility offering inpatient and outpatient services for persons with Spinal Injury, Traumatic Brain Injury, Stroke, Neurological, and Orthopedic conditions. RLANRC is part of the Los Angeles County Department of Health Services' system. RLANRC is uniquely capable of providing comprehensive rehabilitative care for severely disabled patients of all ages, offering the necessary total services within one setting that is unavailable elsewhere in the community. RLANRC's teams of highly qualified rehabilitation experts blend their extensive clinical expertise with the latest advanced technology: their creative spirit that has led to many important advances in rehabilitation. RLANRC's more than 20 Centers of Excellence provide specialized care to individuals with catastrophic injury and illness. RLANRC is currently finishing a $418 million renovation project with the new Don Knabe Wellness Center opened in October 2016 and new outpatient building opened in January 2018, providing state-of-the-art resources to our clinical staff. RLANRC's leadership is dedicated to providing clinical expertise, patient access, and a supportive research infrastructure to ensure project success.

RLANRC treats approximately 4,000 inpatients annually and sees over 71,000 outpatients. The majority of people served by RLANRC qualify for Medi-Cal (Medicaid) and come from a racial/ethnic minority group; RLANRC's SCI population is 56% Hispanic, 24% African American, 13% non-Hispanic Caucasian, and 7% Asian. On average, 3,000 outpatient visits occur per year in the outpatient SCI program. This program includes the Patient Centered Medical Home for SCI, which serves in- and outpatients with a wide range of spinal cord dysfunction. In addition, as the largest program of its kind in California, the Seating Center at RLANRC evaluates mobility needs through customized wheelchairs. Clinical professionals in the seating center are specially trained in wheelchair and equipment selection and actively collaborate with users and family members to promote successful mobility outcomes. Their services include evaluations of cushion (to increase comfort and reduce pressure injury), wheelchair (to assess suitability and make recommendations), and sitting posture (to recommend changes in equipment to improve posture). An average of 15 SCI patients are served per month.

The facilities and resources available at RLANRC, including testing laboratories, engineering spaces, restrooms, common areas and equipment are completely accessible and constructed to allow patients maximal independence of movement from one area to another, as part of the rehabilitation process. These resources are regularly used by a number of staff members that require accessibility accommodations (including full-time manually-propelled wheelchair (MWC) users). The areas designated for clinical testing have ample handicapped accessible parking as well as dedicated areas for van pick-ups and drop-offs. In addition, RLANRC has a designated Americans with Disabilities Act (ADA) compliant mass transit stop (bus) on-site. Personnel are available to assist individuals with hearing or visual impairments. All these resources will be made available to project staff and patients and caregivers, as needed**.**

In addition, RLANRC 's mission is to provide each patient with superior medical and rehabilitation services in a culturally sensitive environment. Many individuals who have received services at RLANRC throughout the years choose to live close to the facility as it offers them continuity of care, long after their initial hospitalization. Individuals with disabilities have become intricately entwined with the institution as consumers, peer mentors, volunteers and workers – enabling newly injured individuals to better transition to the community.

The Orthotic Department provides a full range of orthotic services to the various diagnostic areas served by RLANRC. Service is provided to both inpatients and outpatients. Staff orthotists are part of the multi-disciplinary team that works together to develop an assessment and treatment plan for clients. Practitioners consult with RLANRC therapists and physicians to determine the best design for each patient. The department's on-site fabrication facilities allow for quick response and follow-up to the needs of the medical center, as well as easy access for clinic patients to customized devices. Of particular relevance to this project is the *Orthotic and Prosthetic Technology Center*, which occupies 5,500 square feet and will be used to construct the proposed

manufacturing testbed for evaluating workplace technologies. The facility has a dedicated movement capture space (34 x 30 ft) and 2 patient care rooms as well as both traditional and modern orthotic manufacturing equipment (e.g., vacuum press, oven, glue benches, 3D scanners, CNC mills).

The Rehabilitation Engineering Program occupies 7,500 square feet of space divided evenly between offices/conference rooms, and workshops/engineering laboratories. Within these spaces are a computer lab, a machine shop, an electronics laboratory, a mechanical testing facility, and an orthotic and prosthetic fabrication and prototype shop. The facility is accessible and in compliance with ADA standards. Equipment, instruments, and machines are fully accessible and operated by three individuals with SCI. All of these are available to support the proposed activities.

The Pathokinesiology Laboratory at RLANRC is a 5,350 square foot facility with a fully accessible modern 1,782 square foot gait laboratory on the grounds of RLANRC. In addition to the 45 x 28 ft designated walkway and 50-meter outdoor track, the lab has designated areas available to obtain participant consent, modify wheelchairs, and provide activity education (over 250 square feet). Ample computers running custom codes for data acquisition and processing, video editing, and modeling are networked through a central data server. Multimedia equipment including several large screen LCD displays, video cameras, and DVD recorders are also available for the study.

Rancho Research Institute (RRI) of RLANRC will provide the primary administrative support services. The institute is a nonprofit corporation located on the hospital grounds for the purpose of administering research grants and contracts. RRI maintains the IRB and administers the Continuing Medical Education for RLANRC. Services provided to the project include bookkeeping, salary disbursement, maintenance of personnel records, determination of fringe benefits and salary schedules, and management of sub-contracts.

Equipment

See Facilities and Other Resources document.

PLAN TO ENHANCE DIVERSITY PERSPECTIVES (PEDP-DHS and PEDP-RCMAR)

PEDP-DHS: ORE Core Specific aims 1 and 2. The City of Los Angeles, with 3.9 million residents, is one of the most racially and culturally diverse communities in the world with 48.1% of its population being Latinx. We plan several strategies to increase diverse perspectives and participation in ADRC research for persons of Latinx ethnicity (see ORE core)

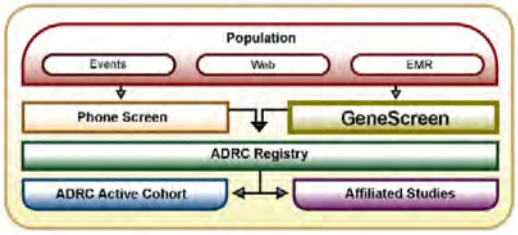


Figure 1. USC GeneScreen is a novel web-based program with a mail-in APOE genotyping test kit to identify individuals who are at increased risk for AD.

ORE CORE Specific Aim 1: Participant Recruitment & Registry Development. USC GeneScreen is a web-based platform to identify individuals who are interested in participating in AD research. It includes a mail-in DNA test kit for *APOE* genotyping. GeneScreen is staffed by fully bilingual Spanish-English speaking staff. The prevalence of *APOE ε4* is lower and recruitment more challenging in Latinx. Thus, we have been gratified to see a good Latinx response to GeneScreen since it was launched in December 2023.

ORE CORE Specific Aim 2*:* Enhance Recruitment from the LA County Department of Health Services (DHS) through Rancho Los Amigos National Rehabilitation Center (RLANRC). Our ADRC has a 40-year affiliation with the California Alzheimer's Disease Center based at RLANRC, an integral part of the DHS safety-net community health system that cares for a diverse, underserved, and largely Latinx population. The CADC at RLANRC (Director: Freddi Segal-Gidan PA, PhD) sees approximately 100 new and 300 follow-up patients with memory loss each year. As part of our PEDP-DHS, we will identify and invite RLANRC patients with MCI or mild dementia to participate in ADRC research studies (e.g., CLARiTI, clinical trials). We will identify appropriate candidates at the biweekly CADC-Rancho consensus conference and arrange for the ADRC Clinical Core team to go on site at RLANRC to meet with potential participants , obtain informed consent and complete the UDSv4. We hope that meeting patients in a familiar clinical setting will increase level of comfort about going to Keck for MRI and PET studies.

To increase referral of MCI patients to CADC-Rancho, the USC ADRC will partner with the newly established DHS Dementia Care Aware (DCA) program. Originally funded by MediCal but supported ongoing by DHS, the goal of DCA is to identify patients with cognitive impairment in the primary care setting and refer to specialty care as appropriate. DCA is directed by Heather Schickedanz, MD, Chair of Family Medicine at DHS Harbor-UCLA Medical Center. During the past 2 years, 4 DHS clinics have participated in DCA leading to the identification of 260 persons with cognitive impairment. The USC ADRC will partner with Dr. Schickedanz to train primary care physicians to identify and treat cognitive impairment and facilitate referrals to the RLANRC (see letter of support).

We will also partner with Amy Towfighi, MD, Director of Neurological Services across DHS, to explore ways to use the EHR to identify persons with cognitive impairment who can be approached for research participation (see letter of support). For her CTSI KL2 Award, Elizabeth Joe MD (one of Dr. Towfighi's mentees) was able to query the DHS EHR and identify 4000 persons with diagnoses of dementia.

In the upcoming funding period, the ORE Core will initiate an annual Spanish-language participant-appreciation event to complement the annual English-language Batey participant appreciation event already in place to educate and enhance retention. We have established a community Advisory Board (CAB) to provide feedback and recommendations to further both Aim 1 and 2 above.

PEDP-RCMAR: Research Education Component (REC) Specific Aim 1: Support URG scientists through a multidisciplinary training program integrated with USC RCMAR.

We will cast a wide net and target special opportunities to develop a pool of underrepresented minority group applicants who would benefit from participation in the combined ADRC REC/RCMAR program. When appropriate individuals are identified, we will reach out and cultivate their interest in joining REC/RCMAR. Scientists will be recruited from the main participating USC schools, departments and centers, our collaborators from Historically Black Colleges and Universities and Hispanic-serving institutions, as well as qualified applicants from other institutions. Many RCMAR scientists have successfully joined the faculty at USC after their RCMAR training experience.

Table A1: Federal Funded Grants Supported by Resources of the ADRC				
PI/PD	**Grant Title/Number**	**Source & Amount**	**Project Period**	**Role of ADC**
O'Bryan S, Toga A, Yaffe K	Health Disparities in AD and MCI among Mexican Americans **R01AG054073**	NIA $1,982328 (2021)	9/1/2017-5/31/2023	Imaging Core (LONI)
Zlokovic B	The role of pericytes in white matter disease **5R01NS100459**	NIA $722,149	9/30/2016-8/31/2023	
Bateman R, Schneider LS (site PI)	DIAN-TU: Next Generation Prevention Trial **5R01AG053267**	NIA $17,951,661	09/30/2017-05/31/2023	Clinical Core
Aisen P, Schneider LS (site PI)	Alzheimer's Clinical Trials Consortium **3U24AG057437**	NIA $28,125,414	12/02/2017-11/30/2028	Clinical Core
Yassine H	The delivery of essential fatty acids to the brain in Alzheimer disease **5R01 AG054434**	NIA $884,286	9/120/17-4/30/2023	Clinical Core
Han D	Finance, cognition, and default mode network in aging **R01AG058166**	NIA $2,204,479	5/1/2017-4/30/2026	REC
Han D	Racial difference in decision making among older adults **R01AG055430**	NIA $623,813	7/15/2017-10/31/2024	REC
Toga A Thompson P	Laboratory of Neuroimaging (LONI) **P41 EB015922-20**	NIA $1,217,435	9/30/2018-2/28/2023	Imaging Core
Finch CE Chen JC	Urban Air Pollution and Alzheimer's Disease: Risk, Heterogeneity, and Mechanisms **P01AG055367**	NIA $2,308,593	7/1/2018-6/30/2023	Data Core
Multi-PI: Marmarelis V Chui H, Billinger S, Zhang R	Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression **R01AG058162**	NIA $2,144,503	8/1/2018-7/31/2023	Clinical and Imaging Core
Multi-PI: Goldberg T Schneider L Harvey PD	Novel Cognitive and Functional Measure for Alzheimer's Disease Prevention Trials **1R01AG051346-01A1**	NIA 1,595,795	9/1/2018-5/31/2023	Clinical Core
Toga A.	Multiscale and Multimodal Analysis of Biomarkers in Alzheimer Disease (AD) **T32 AG058507**	NIA $138,129	5/1/2018-4/30/2023	ORE/REC
Goldman D, Zissimopoulos J	USC Alzheimer's Disease Resource Centers for Minority Aging Research (USC AD-RCMAR) **2P30AG043073**	NIA $574,864	9/1/2018-6/30/2023	ORE/REC
Mosqueda L	Uncovering the causes, contexts, and consequences of elder mistreatment in people with dementia **1R01AG060096**	NIA $400,157	8/15/2018-3/31/2023	Clinical Core
Aisen, P Sperling, R Johnson, K	Combination Anti-Amyloid Therapy for Preclinical Alzheimer's Disease **R01AG061848**	NIA $8,743,788	9/30/2018-5/31/2023	Admin Core Clinical Core

Apostolova, L Aisen, P (Site PI)	Early Onset Alzheimer's Disease Consortium – LEADS Study **U01AG057195**	NIA $1,444,845	9/30/2018- 5/31/2023	Clinical Core
Reiman E, Schneider L (site PI)	Alzheimer's Prevention Initiative ADAD Colombia Trial **5R01AG055444**	NIA via Banner $5,754,078	4/15/2018- 3/31/2024	Clinical Core
Bateman R, Pawluczyk S (site PI)	A Phase II/III Randomized, Double-Blind, Placebo-Controlled, Cognitive Endpoint, Multicenter Study of Potential Disease Modifying Therapies in Individuals at Risk for and with Dominantly Inherited Alzheimer's Disease ("Protocol") **5R01AG053267**	NIA $5,486,752	7/1/2018– 3/31/2024	Clinical Core
Finch CE	Age-sex-ApoE allele interactions in neuronal and white matter vulnerability to air pollution **1P01AG055367**	NIA $324,805	7/1/2018- 6/30/2023	Administrative Core
Ringman J	Motor, visual, and olfactory changes in genetic subtypes of Alzheimer disease **5R01 AG062007**	NIA $738,679	2/15/2019- 11/30/2023	Clinical Core
Schneider LS (PI)	(-)-Phenserine inhibition of neuronal death in Alzheimer's disease and developing brain-labeled plasma exosomes assays as biomarkers for a phenserine phase 1b ascending dose trial **R01AG062687**	NIA $3,846,717	04/1/2019- 02/29/2024	Clinical Core
Thompson P Ganesan V John J	India ENIGMA initiative for global aging & mental health **5R01 AG060610**	NIA $568,893	9/1/2019- 5/31/2024	Imaging Core
Zlokovic B	The role of pericytes in the aging brain and pathogenesis and treatment of neurodegeneration, and Alzheimer's A-beta and tau pathology **2RF1AG039452**	NIA $3,981,640	2/13/2019- 1/31/2024	
Nation D	Cerebrovascular resistance in cognitive aging and Alzheimer's disease risk **5R01AG060049-06**	NIA $494,996	8/1/202019- 4/30/2024	
Nation D	Endothelial progenitor cells and neurovascular injury in the aging brain **5R01AG064228-05**	NIA $633,127	9/1/2019- 5/31/2024	
Wang JJ	BBB Permeability Imaging in CADASIL **5R01NS114382**	NINDS $247,500–yr 2	2/1/2020- 1/31/2025	Imaging Core
Yassine H	ApoE, ABCA1 and endosomal dysregulation in AD **5R01AG067063**	NIA 751, 791	5/15/2020- 2/28/2025	
Zlokovic B	Activated protein C mechanisms of brain white matter protection and new therapies for brain white matter ischemic injury. **1R01NS117827**	NIA $147,637 NINDS 843,669	7/15/2020- 4/30/2025	Imaging Core
O'Bryant S Johnson L Rissman R Toga A	Health and Aging Brain among Latino Elders (HABLE-AT(N)) **1R01AG058533**	NIA $9,074,729 (2022)	8/1/2020- 4/30/2025	Imaging Core
Chui HC Zlokovic B Toga A	Alzheimer Disease Research Center- Imaging Supplement **3P30AG066530-01S1**	NIA $412,500	9/1/2020- 2/28/2025	Clinical Core Imaging Core

Ringman J	Phenotype and genotype of autosomal dominant Alzheimer disease in Jalisco Mexico **1R01AG069013**	NIA $672,193	9/15/2020-5/31/2025	Clinical Core
Schneider L	Testing early markers of cognitive decline and dementia derived from survey response behaviors **5R01AG068190**	NIA $814372	9/15/2020-5/3/2024	Clinical Core
Thompson P	FiberNET: Deep learning to evaluate brain tract integrity worldwide and in AD **1RF1 AG057892**	NIA $2,563,788	9/15/2020-8/31/24	Imaging Core
Thompson P Davatzikos C Huang H Saykin A	Ultrascale machine learning to empower discovery in Alzheimer disease biobanks **1U01 AG068057**	NIA $3,767,729	9/15/2020-8/31/2025	Data Core Imaging Core
DeCarli C, Fornage M (Site PI: Chui)	The clinical significance of incidental white matter lesions on MRI amongst diverse population with cognitive complaints **1U19NS120384**	NINDS $3,236,413	9/30/2020-8/31/2026	ORE Core Clinical Core Imaging Core
Wang LS Chui HC	Asian Cohort for Alzheimer Disease **R56AG069130**	NIA $243,350	9/30/2020-5/31/2022	Clinical Core
Yan LR Chui HC	Quantitative cerebral blood vessel imaging biomarkers for AD and VCID **1RF1AG072490-01**	NIA $2,408,375	4/1/2021-3/3/2024	Clinical Core Imaging Core
Chui H Bienkowski M Pachicano M	USCADRC Diversity Supplement Pachicano **3P30AG066530-02S1**	NIA $34,000	9/1/2021-2/28/2025	Imaging Core Pathology Core
Chui H Ringman J Guzman S	USC ADRC Diversity Supplement – Guzman **3P30AG066530-02S2**	NIA $57,173	11/1/2021-2/28/2025	Pathology Core
Wang JJ Kashani A Ringman J	Validation of Imaging and blood-based Small Vessel VCID Biomarkers in Multiethnic Populations **2UF1NS100614**	NINDS $2,623,751	9/30/2021-7//31/2023	Clinical Core Imaging Core
Bienkowski M	A Translational 3D map of hippocampal cell types to drive investigations of Alzheimer disease. **K01AG066847**	NIA $125,766	8/1/2021-7/31/2026	Development Project, Pathology Core Imaging Core
Zlokovic B	PICALM: Role in the pathogenesis and treatment of Alzheimer's vascular blood-brain barrier clearance. **R01AG076921**	NIA $2,410,664	4/1/2022-3/31/2025	
Toga A	Vascular contributions to dementia and genetic risk factors for Alzheimer disease **P01AG052350-06**	NIA 3,246490 Year 1	1/1/2022-12/3120/27	Clinical, Imaging, and Biomarker Cores
Yassine H Arvanitakis Z	Brain cPLA2 as a mechanism for neuroinflammation in AD/ADRD with and without APOE4 **1RF1AG076124**	NIA $2,417,667	5/15/2022-4/30/2025	
Yassine H Borzage M Braskie M	Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults **1RF1AG078362**	NIA $3,755,733	9/30/2022-8/31/2025	Clinical Core Imaging Core
O'Bryant S, Johnson,Rissman, Toga, Yaffe	The Health and Aging Brain Study -Health Disparities (HABS-HD) **1U19AG078109**	NIA $27,685,482	9/30/2022-8/31/2027	Imaging Core
Agadjanyan M Schneider L Sultzer D	Safety/tolerability/immunogenicity of Abeta DNA vaccine AV-1959D Phase I **R01AG074983**	NIA $2,680,709	2/15/2022-1/31/2027	Clinical Core

Kaplan H Finch CE Gatz M	Pathways linking infection, physical activity and ApoE and biological sex to dementia in 2 Native American Populations R01AG054442	NIA $3,199,665	9/1/2022 4/30/2027	Clinical Core
Southern S Ettenhoffer M Nation D	Rapid saliva test for noninvasive diagnostic screening of MCI and dementia 5R44**AG078044-02**	NIA $1,302,154	7/1/2022- 6/30/2024	
Karim R.	Circulating levels of persistent organic pollutants and subclinical atherosclerosis progress in postmenopausal women **R01ES033705**	NIEHS $701,248	9/20/2022- 7/31/2024	Development Project
Gold T Wang D	Reduced BBB Water Exchange as a Preclinical Biomarker of Small Vessel Disease **1RF1NS122028-01A1**	NINDS 2022 1,952,654	2/1/2022- 1/31/2025	
Zlokovic B	Alzheimer's Abeta, Tau, Apolipoproteins and Blood-brain barrier **5R01AG34467**	NIA 2023: $771,703	3/1/2023- 2/28/2025	
Wang D Jiang XJ Shi YG	**Imagi**ng Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID **1RF1AG084072**	NINDS 2023 2,487,668	9/1/2023- 8/31/2026	Imaging Core
Nation D Bondi M Mather M	Locus Coeruleus Imaging Markers in Preclinical Alzheimer disease, Cerebrovascular Disease and Cognitive Decline **1R01AG082073**	NIA 2023 $1623978	4/1/2023- 3/31/2024	
Wang LS Chui HC Jun G Park V	Asian Cohort for Alzheimer Disease **1U19AG079774**	NIA $8,325,999	7/1/2023- 6/30/2028	Clinical Core
Joe E	Barrers to early identification of dementia in a safety net healthcare system **1R03AG082997**	NIA 2023 $165,000	9/1/2023- 5/31/2025	Clinical Core ORE Core (PEDP-DHS)
Hiniker, A	Targeting proteostatic mechanisms to inhibit LRRK2-mediated neurodegeneration and neuropathology **1R01NS135607**	NIA 2023 $648,555	9/21/2023- 8/31/2028	Pathology Core
Aisen P, Petersen R Sperling R	Alzheimer's Clinical Trials Consortium **3U24AG057437**	NIA $28,125,414	12/02/17- 11/30/28	Clinical Core Biomarker Core

Table A2. Non-Federal (e.g. Foundation) Funding Supported by Resources of the USC ADRC				
Principal Investigator/ Project Director	**Grant Title/Number**	**Source & Amount**	**Project Period**	**Role of ADC**
Supported by Administration, Clinical, ORE and Imaging Cores				
Schneider, Lon	California Alzheimer's Disease Center (CADC) **15-10291**	State of California $281,800 /yr	07/01/23-06/30/28	ORE: Joint Education REC: Joint Training
Segal-Gidan, Freddi.	California Alzheimer's Disease Center (CADC) at Rancho Los Amigos National Rehabilitation Center **15-10295**	State of California $281,800 /yr	07/01/23-06/30/28	ORE: Joint Education REC: Joint Training
Hjelm, Brooke	Ocular Biofluid Genomics and Alzheimer's Disease (AD) Progression/Specificity	Dean's Pilot Fund USC KSOM $75,000	01/01/22-12/31/22	Clinical Core: Tear Biofluid samples
Langen, Ralph	Novel approaches to misfolded proteins in neurodegenerative disorders	Batey Family Foundation $100,000	2020	Clinical Core: Tear Biofluid samples
Yassine, Hussein	Ion Mobility Lipid Particle Analyzer	Batey Family Foundation $75,000	2020	Clinical & Biomarker Cores: CSF and plasma samples
Sheikh-Bahaei, Nasim Pappas, Ioannis	PET-Guided 7T MRS for early diagnosis of Alzheimer's disease.	Glazer Foundation $100,000	04/01/24-03/31/25	Clinical Core: Recruit 30 participants
Supported by Pathology Core				
Finch, Caleb (USC)	Is ***APOE4*** a transcription factor?	USC Departmental Grant ($20,000)	2020 to 2021	Frozen CNS = 62
James, Kristen (BioProtonics, Inc.)	Application of a novel MR-based diagnostic to measure cortical fine tissue texture changes	Self-Sponsored $150,000	2020 to 2022	Fixed CSF = 21
Lakkaraju, Aparna (UCSF)	*APOE4* and Retinal Pathology	Bright Focus Foundation EY030668 $500,000	2020 to 2025	Fixed Eyes = 28 Frozen CNS = 8

Table A3. Funding for Therapeutic Trials and Other Grants from Industry				
Principal Investigator/ Project Director	**Grant Title/Number**	**Source & Amount**	**Project Period**	**Role of ADC**
Site PI: Schneider, L.S. (GR1036881 – 53-5143-0369)	Anti-Amyloid Treatment in Asymptomatic Alzheimer's Disease (A4 Study) 2U19AG010483-22	ATRI subcontract funded by Eli Lilly $2,260,845	11/1/2015-12/15/2023	Enrolled 32
Site PI: Schneider, L.S. (GR1036882 – 53-5143-0438)	Anti-Amyloid Treatment in Asymptomatic Alzheimer's Disease (A4 Tau) 15-338730	ATRI subcontract funded by NIH $175,000	11/1/2015-11/30/2023	Enrolled 32
Site PI: Schneider, L.S. (GR1059059 – 53-5143-0438)	Anti-Amyloid Treatment in Asymptomatic Alzheimer's Disease (A4 OLE) 15-338730	ATRI subcontract funded by NIH $554,200	11/1/2015-12/31/2023	Enrolled 22
Site PI: Schneider, L.S. (GR1036873 – 53-5143-0195)	Longitudinal Evaluation of Amyloid Risk and Neurodegeneration - the LEARN study 15-396727	ATRI subcontract funded by Alzheimer's Association $51,455	11/01/2015-04/30/2024	Enrolled 8
Site PI: Schneider, L.S. (GR1054405 – 53-5143-0681)	Alzheimer's Disease Neuroimaging Initiative 3 (ADNI 3) WEI2015-01	ATRI NCIRE funded by NIH $248,605	8/1/2020 - 7/31/2023	Enrolled 49
Site PI: Schneider, L.S. (GR1056716 – 54-5134-0008)	A Placebo-Controlled, Double-Blind, Parallel-Treatment Arm, 216 Week Study to Evaluate Efficacy and Safety of Treatment with BAN2401 in Subjects with Preclinical Alzheimer's Disease and Elevated Amyloid (A45 Trial) and in Subjects with Early Preclinical Alzheimer's Disease and Intermediate Amyloid (AHEAD 3-45)	Eisai Inc. $143,280	1/1/2021 – 12/31/2025	Screened 141 Enrolled 21
Site PI: Schneider, L.S. (GR1057971)	A Placebo-Controlled, Double-Blind, Parallel-Treatment Arm, 216 Week Study to Evaluate Efficacy and Safety of Treatment with BAN2401 in Subjects with Preclinical Alzheimer's Disease and Elevated Amyloid (A45 Trial) and in Subjects with Early Preclinical Alzheimer's Disease and Intermediate Amyloid (AHEAD 3-45)	ATRI subcontract funded by NIH $148,200	5/1/2021 – 4/30/2025	Enrolled 4
Site PI: Schneider, L.S. (GR1051047 – 54-5143-5295)	Allopregnanolone as Regenerative Therapeutic for Alzheimer's disease: Phase 2 Clinical Trial	Univ of Arizona subcontract funded by Alzheimer's Association $147,988	7/11/2018-3/31/2021	Enrolled 11
Site PI: Schneider, L.S. (GR1001105 – 51-5143-2445)	A Phase 3 Multicenter, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of Aducanumab (BIIB037) in Subjects with Early Alzheimer's Disease	Biogen MA $422,154	10/14/2015 - 7/25/2022	Enrolled 5

Site PI: Schneider, L.S. (GR1055445 – 51-5143-8851)	Phase 3b open-label, multicenter, safety study of BIIB037 (aducanumab) in subjects with Alzheimer's disease who had previously participated in the Aducanumab studies 221AD103, 221AD301, 221AD302 and 221AD205	Biogen MA $154,501	10/13/2022– 10/12/2023	Enrolled 1
Site PI: Pawuczyk, S, Schneider, L.S. (GR1054846 – 54-5143-9285)	A Phase II/III randomized, double-blind, placebo-controlled, cognitive endpoint, multicenter study of potential disease-modifying therapies in individuals at risk for and with dominantly inherited Alzheimer's disease (DIAN TU) 1U01AG059798-01	The Washington University subcontract funded by NIH $372,326	7/1/2018 – 5/31/2025	Enrolled: 4
Site PI: Schneider, L.S. (GR1036980 – 53-5143-2687)	Phenserine inhibition of neuronal death in Alzheimer's disease and developing brain-labeled plasma exosomes assays as biomarkers for a phenserine phase 1b ascending dose trial (Exosome) 1R01AG062687-01	R01 funded by NIH $3,521,885	4/1/2019 – 2/28/2025 (NCE)	Enrolled 108
Site PI: Schneider, L.S. (GR1055914 – 54-5143-0418)	Development of novel measures of cognition and function for Alzheimer's disease prevention trials (NOMAD) 139862	Research Foundation for Mental Hygiene subcontract funded by NIH $307,166	9/1/2018 - 5/31/2025 (NCE)	Enrolled 63
Site PI: Schneider, L.S. (GR1036985 – 53-5143-2873)	Escitalopram for Agitation in Alzheimer's Disease (S-CitAD) 5R01AG052510-03	JHU subcontract funded by NIH $171,085	9/30/2016 – 5/31/2023	Enrolled 4
Site PI: Schneider, L.S. (GR1036889 – 53-5143-0648)	TRC-PAD Program: In-Clinic Trial-Ready Cohort (Infrastructure) 5R01AG053798-03	ATRI subcontract funded by NIH $60,000	5/1/19 - 5/30/24	Enrolled 17
Site PI: Schneider, L.S. (GR1055915 – 53-5143-0648)	TRC-PAD Program: In-Clinic Trial-Ready Cohort 5R01AG053798-03	ATRI subcontract funded by NIH $77,425	9/1/22- 5/30/24	Enrolled 17
Site PI: Schneider, L.S. (GR1001109 – 51-5143-4134)	A Phase 2 Randomized Double-Blind Placebo-Controlled Trial to Evaluate the Efficacy and Safety of BHV-4157 in Patients with Mild to Moderate Alzheimer's Disease (T2) BHV4157203	Biohaven $101,545	12/20/18 – 12/19/21	Enrolled 3
Site PI: Schneider, L.S. (GR1056355 – 51-5143-0868)	Donanemab in Early Symptomatic Alzheimer's Disease	Eli Lilly $767,383	4/21/21 – 4/30/25	Enrolled 6
Site PI: Schneider, L.S. ADNI 4	Alzheimer's Disease Neuroimaging Initiative 4 (ADNI 4)	ATRI NCIRE funded by NIH	11/1/23-	Enrolled: 13
Site PI: Schneider, L.S.	A Seamless Phase 2A-Phase 2B Randomized Double-Blind Placebo-Controlled Trial to Evaluate the Safety and Efficacy of Benfotiamine in Patients with Early Alzheimer's Ds (BENFO-TEAM) (Pro00071249)	NIH UCSD ADCS	6/7/2024- 2028	Enrolled: 0

Table A4: NIH Training Awards			
Principal Investigator	**Grant Title/Type of Award**	**Project Period**	**Funding Source/ Amount**
Crimmins, E	Multidisciplinary Research Training in Gerontology **T32AG000037-42**	9/1/77-4/30/29	NIA $605,709
Toga, A	Training in the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease (AD). **T32AG058507**	2018-2024	NIA $175,317
Zissimopoulos, J	USC Alzheimer's Disease Resource Centers for Minority Aging Research (USC AD-RCMAR) **2P30AG043073**	9/1/18 6/30/28	NIA $680,681
Zissimopoulos, J	Center for advancing sociodemographic & economic studies of Alzheimer and related dementias (CeASES-ADRD) **P30AG066589**	7/16/20 - 6/30/25	NIA $297,427
McEligot, A Braskie, M Toga, A	Neurocognitive Aging & Analytics Research Education (NAARE) **5R25AG069711**	6/1/2021-5/31/2026	NIA $362,604
Han, D	Mid-career development and mentoring on financial vulnerability and AD **K24AG081325**	3/21/2023 1/31/2028	NIA $199,977
Buchanan, T Kipke, M	Clinical Translational Science Institute (CTSI) **KL2-SC; 5UL1TR001855**	2016-3/31/2027	NCATS $9,009,695
Hiniker, A	Neurodegenerative Scholars (R13) program American Association of Neuropathologists (AANP)	2023-2028	

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Table A6. Underrepresented Group(URG)-Related Grants								
Grant Title and Number	**Type**	**PI**	**Award Period**	**Total Award Amount**	**URG PI**	**URG or Health Disparities Topic**	**Funding Agency**	**Target group(s)**
The Health & Aging Brain Study - Health Disparities (HABS-HD) Project **5U19AG078109**		O'Bryant S Johnson L Rissman R Toga A Yaffe K	2022-2024	$1,858,021		Mexican Americans Phenotypic differences	NIA	Mexican American
Health and Aging Brain among Latino Elders (HABLE-AT(N)) Study **5R01AG058533**		O'Bryant S Johnson L Rissman R Toga A	2020-2024	$9,074,729		Mexican Americans Blomarkers		Mexican Americans
Pathways linking infection, physical activity and ApoE and biological sex to dementia in 2 Native American populations **R01 AG05442**		Kaplan H	2022-2027	$3,199,665		Tsimane Bolivia APOE	NIA	Indigenous Bolivian population
Validation of imaging and blood-based small vessel VCID biomarkers in multi-ethnic populations **2UF1NS100614**		Wang D Kashani A Ringman R	2021 2023	$2,623,751		Latinos African Americans ASCVD	NIA	Mexican / Mexican American/ African American
Motor, visual, and olfactory changes in genetic subtypes of Alzheimer disease **5R01 AG062007**		Ringman	2019 2023	$738,679		Latinos Clinical and Motor-sensory	NIA	Mexican / Mexican American
ADRC Imaging Supplement **P30AG066530-01S1**		Chui Zlokovic Toga	2020-2025	$412,500		Latinos Imaging ATN	NIA	Latinos
Asian Cohort for Alzheimer's Disease (ACAD) Project **1U19AG079774-01**		Wang LS Chui HC Jun G Park V	2023-2028	$8,325,999		Asian GWAS	NIA	Asian Americans
Clinical significance of WMH (DVCID) **U19NS120384**		DeCarli C Chui H	2020-2026	$3,236,413		Latinx MRI Biomarkers	NINDS	Latinx
Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults **1RF1AG078362**		Yassine H Borzage M Braskie M	2022-2025	$3,755,733		Latinx T2DM	NIA	Latinx
Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID **1RF1AG084072**		Wang D Jiang XJ Shi YG	2023 2026	$2,487,668		Chinese Korean Clinical Core	NIA	Chinese Korean Vietnamese

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Helena Middle Name Chang Last Name*: Chui Suffix:

Position/Title*:	Chair and Professor of Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-7686 Fax Number: (323) 442-7689 E-Mail*: helena.chui@med.usc.edu

Credential, e.g., agency login: changchui

Project Role*: PD/PI Other Project Role Category:

Degree Type: MD Degree Year:

	File Name
Attach Biographical Sketch*:	Chui_NIH_biosketch_ADRC1044446715.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Arthur Middle Name W. Last Name*: Toga Suffix:

Position/Title*:	Provost Professor
Organization Name*:	University of Southern California
Department:	KSOM Neuroimaging and Informatics Institute
Division:	Keck School of Medicine of USC
Street1*:	SSB 102
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (323) 442-7246 Fax Number: (323) 442-2230 E-Mail*: toga@usc.edu

Credential, e.g., agency login: TOGA22

Project Role*: PD/PI Other Project Role Category:

Degree Type: PhD, MS, BS Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Toga_May_20241044446534.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: Chui, Helena Chang

eRA COMMONS USER NAME (credential, e.g., agency login): changchui

POSITION TITLE: McCarron Professor of Neurology, University of Southern California

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Johns Hopkins University - Baltimore, MD	B.A.	06/1974	Chemistry
Johns Hopkins University - Baltimore, MD	M.D.	06/1977	Medicine

A. Personal Statement

During the past 34 years, my research interests have focused on aging, Alzheimer disease, and vascular contributions to cognitive impairment in late life. For 20 years, I served as principal investigator of the Aging Brain Program Project, a multi-institutional longitudinal study to characterize interactions between vascular and Alzheimer disease (P01AG12435-20; completed), using clinical-imaging-pathological correlations. We reported that lipids (e.g., HDL and LDL-cholesterol, docosahexanoic acid) are related to beta-amyloid deposition in the brain, measured by PiB PET (Ref #1; Reed et al., 2014; Ref #2 Yassine et al., 2016; Ref #3: Lee et al., 2019; Ref #4: Martinez et al., 2022). I have continued to pursue my research interests with Yassine, Marmarelis, and Yan. I also have 20 years of administrative experience, serving as Chair of the Department of Neurology and mPI of the Alzheimer Disease Research Center (ADRC: P50-AG05142; P30 AG066530) at USC. I am honored to serve with Dr. Toga as mPI of the proposed competitive renewal the USC ADRC.

P30-AG066530-03 (MPI: (Chui, Toga) 09/01/2020-03/28/2025
Alzheimer Disease Research Center (ADRC):

P01-AG052350-06 (A Toga) 5/15/2022 – 7/31/2027
Alterations in the neurovascular unit in persons carrying the apolipoprotein E4 allele.

U19-AG079774 (MPI: Wang, Chui, Jun, Park) 7/1/2023 – 6/30/2028
Asian Cohort for Alzheimer's Disease (ACAD)

U19-NS120384-01 (MPI: DeCarli, Fornage) 9/30/2020 – 8/31/2026
Diverse VCID

I am particularly interested in non-occlusive pathways leading from vascular-metabolic risk factors to vascular brain injury or AD pathology.

RF1-AG072490-02 (MPI: LR Yan, H Chui) 4/1/2021- 3/31/2024
Quantitative Blood Vessel Biomarkers in AD and VCID:

R01-AG058162-04 (MPI: V Marmarelis, H Chui, S Billinger, R Zhang) 8/1/2018-5/31/2024
Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression.

RF1-AG078362 (PI: H Yassine) 9/1/2022-8/31/2025
Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults.
-
UFINS100614 (D Wang, A Kashani, J Ringman) 8/2023-7/2026
Validation of imaging and blood-based small vessel VCID Biomarkers in multi-ethnic population.

B. Positions, Scientific Appointments and Honors

Positions

2003 - present	Chair, Department of Neurology, USC Keck School of Medicine
1994 - present	Professor (tenured), Department of Neurology, University of Southern California
1990 - 2003	Chair, Department of Neurological Sciences, Rancho Los Amigos Medical Center
1988 - 1994	Associate Professor, Department of Neurology, University of Southern California
1981 - 1987	Assistant Professor, Department of Neurology, University of Southern California
1979 - 1981	Resident, Department of Neurology, University of Iowa College of Medicine
1978 - 1979	Fellow, Behavioral Neurology, Dept. of Neurology, University of Iowa College of Medicine
1977 - 1978	Intern, Department of Psychiatry, University of Iowa College of Medicine

Scientific Appointments

2023	Reviewer, NIA Intramural Program
2017	Reviewer, NIH Pioneer Awards (NPA)
2016 - 2017	Ad hoc reviewer, NIA Clinical Neuroscience & Neurodegeneration Study Section
2008 - 2010	Member, NIA Neurological, Aging, and Musculoskeletal Epidemiology (NAME) Study Section
2004 - present	Co-Director, Alzheimer Disease Research Center of California (ARCC), Rancho Los Amigos National Rehabilitation Center
2003 - present	Director, Alzheimer Disease Research Center (ADRC), University of Southern California
1998 - 2004	Director, Alzheimer Disease Research Center of California (ARCC), Rancho Los Amigos National Rehabilitation Center
1990 - 1991	Chair, Geriatrics and Gerontology Review Subcommittee, National Institute on Aging
1987 - 1991	Study Section Member, Geriatrics and Gerontology Review Subcommittee A, National Institute on Aging
1985 - 1998	Co-Director, Southern California Alzheimer Disease Diagnostic and Treatment Center (ADDTC), Rancho Los Amigos Medical Center
1984 - 1994	Director, Cell Quantimetrics Core, NIA ADRC Consortium of Los Angeles and Orange Counties
1984 - 2001	Co-Director, Clinical Core, National Institute on Aging (NIA) ADRC Consortium of Los Angeles and Orange Counties

Honors

2023-2024 Los Angeles Top Doctor; 2018-USC Excellence in Professionalism Award; 2011-OPICA Founders Award; 2008-AAPA Physician-Physician Assistant Partnership Award; 2007-California Science Center Muses Woman of the Year Award; 2003-present, Best Doctors; 2001-Connolly's Top Doctor in America; 1991-Azheimer Association Research Award; 1997-Raymond and Betty McCarron Endowed Chair of Neurology; 1988-Teacher Investigator Development Award NIH; 1980-Senior Resident of the Year Award; 1977-Janet M. Glasgow American Medical Women's Association Achievement Citation; 1977-Alpha Omega Alpha (Johns Hopkins School of Medicine); 1973-Phi Beta Kappa; 1970-National Merit Scholar.

C. Contribution to Science (From 261 peer-reviewed publications)

Epidemiological studies report vascular risk factors (VRF) to be associated with increased incident AD. However, before the advent of amyloid PET scans, we could not determine whether VRF accentuated the AD pathology or caused subclinical vascular brain injury. In the Aging Brain Program Project (1P01 AG12435), we found that **plasma levels of LDL cholesterol are positively correlated, while levels of HDL cholesterol and DHA are negatively correlated, with levels of brain amyloid (assessed by amyloid-PET scans).** Among the LDL subfractions, greater plasma levels of large LDL particles are associated with greater cerebral amyloidosis. I continue to pursue associations between lipids and cerebral amyloidosis through collaborations in the USC ADRC (P30-AG066530).

1. Reed BR, Villeneuve S, Mack W, DeCarli C, **Chui HC** & Jagust W. Low HDL and High LDL serum cholesterol are associated with cerebral amyloidosis. JAMA Neurology 2014, 71(2):195-200.

2. Yassine H, Feng Q, Azizkhanian I, Rawat V, Castor K, Fonteh A. Harrington MG, Zheng L, Reed BR, DeCarli C, Jagust WJ, **Chui HC**. The association of serum docosahexaenoic acid with cerebral amyloidosis. JAMA Neurol. 2016; 73: 1208-1216.
3. Lee S, Parekh T, King SM, Reed B, **Chui HC**, Krauss RM, Yassine HN. Low-Density Lipoprotein Particle Size Subfractions and Cerebral Amyloidosis. J Alzheimers Dis. 2019;68(3):983-90. Epub 2019/03/19. doi: 10.3233/JAD-181252. PubMed PMID: 30883362.
4. Martinez AE, Weissberger G, Kuklenyik Z, He X, Meuret C, Parekh T, Rees JC, Parks BA, Gardner MS, King SM, Collier TS, Harrington MG, Sweeney MD, Wang X, Zlokovic BV, Joe E, Nation DA, Schneider LS, **Chui HC**, Barr JR, Han SD, Krauss RM, Yassine HN. The small HDL particle hypothesis of Alzheimer's disease. Alzheimers Dement. 2023;19(2):391-404. Epub 2022/04/14. doi: 10.1002/alz.12649. PubMed PMID: 35416404.

Subcortical ischemic brain injury, measure by volumetric MRI (white matter hyperintensities, silent infarcts, and lacunar infarcts) are associated with impairment in executive function, moreso than memory impairment, but the effects are additive with and are dwarfed by Alzheimer disease. Subclinical vascular brain injury became widely recognized with the advent of brain MRI, but the cognitive impact of white matter hyperintensities and silent infarcts was not known until the development of volumetric MRI deployed in prospective longitudinal studies such as the IVD program project (AG12435; 1996-2007), which I led as principal investigator.

5. **Chui HC**, Zarow C, Mack WJ, Ellis WG, Zheng L, Jagust WJ, Mungas D, Reed BR, Kramer JH, DeCarli CC, Weiner MW, Vinters HV. Cognitive impact of subcortical vascular and Alzheimer disease pathology. Annals Neurol 2006; 60: 677-687.
6. Reed BR, Mungas DM, Kramer JH, Ellis W, Vinters HV, Zarow C, Jagust WJ & **Chui HC.** Profiles of neuropsychological impairment in autopsy-defined Alzheimer's disease and cerebrovascular disease. Brain 2007;130(Pt 3):731-9.
7. Ramirez-Gomez L, Zheng L, Reed B, Kramer J, Mungas D, Zarow C, Vinters H, Ringman JM, **Chui H**. Neuropsychological Profiles Differentiate Alzheimer Disease from Subcortical Ischemic Vascular Dementia in an Autopsy-Defined Cohort. Dement Geriatr Cogn Disord. 2017;44(1-2):1-11. PMID: 28595184.
8. Pahlavian SH, Wang X, Ma S ... **Chui H**… Wang DJ, Yan L. Cerebroarterial pulsatility and resistivity indices are associated with cognitive impairment and white matter hyperintensity in elderly subjects: A phase-contrast MRI study. J Cereb Blood Flow Metab. 2020:271678X20927101. Epub 2020/06/06. doi: 10.1177/0271678X20927101. PubMed PMID: 32501154.

Atherosclerosis is associated with cortical atrophy, microinfarcts, and cognitive impairment, but not with Alzheimer pathology. For the first part of the 20th century, atherosclerosis was believed to be the most common cause of senile dementia. Since the 1970's, AD pathology replaced atherosclerosis as the most prevalent cause, and vascular dementia was re-conceptualized as due to infarcts large and small. We have found that atherosclerosis contributes independently of infarcts to cortical atrophy and cognitive impairment, which motivates a search for non-occlusive mechanisms of vascular brain injury. These discoveries resulted from the IVD program project (P01 AG12435), a prospective longitudinal study, including autopsy, which I led as principal investigator.

9. Jagust WJ, Zheng L, Harvey DJ, Mack WJ, Vinters HV, Weiner MW, Ellis WG, Zarow C, Mungas D, Reed BR, Kramer JH, Schuff N, DeCarli C & **Chui HC.** Neuropathological basis of magnetic resonance images in aging and dementia. Ann Neurol., 2008; 63(1):72-80. PMCID: 2624571
10. Villeneuve S, Reed BR, Madison CM, Wirth M, Marchant NL, Kriger S, Mack WJ, Sanossian N, DeCarli C, **Chui HC** *et al*. Vascular risk and Abeta interact to reduce cortical thickness in AD vulnerable brain regions. Neurology 2014; 83:40-47. PMCID: PMC 4114172.
11. Zheng L, Vinters HV, Mack MJ, Zarow C, Ellis WG & **Chui HC.** Cerebral atherosclerosis is associated with cystic infarcts and microinfarcts but not Alzheimer pathologic changes. Stroke, 2013; 44:2835-2841. PMCID: 4049465.

12. Zheng L, Vinters HV, Mack WJ, Weiner M, **Chui HC**. Differential effects of ischemic vascular disease and Alzheimer disease on brain atrophy and cognition. J Cereb Blood Flow Metabolism 2015; 1-9.

Hippocampal sclerosis is a common neuropathological finding among the elderly, but is often clinically misdiagnosed as Alzheimer disease because of prominent amnesic memory disturbance. These findings were made as part of the IVD program project (P01 AG12435), which I led as principal investigator.

13. Zarow C, Sitzer TE, **Chui HC**. Understanding hippocampal sclerosis in the elderly: epidemiology, characterization, and diagnostic issues. Curr Neurol Neurosci Rep. 2008 Sep;8(5):363-70. Review.
14. Zarow C, Wang L, **Chui HC**, Weiner MW, Csernansky J. MRI shows more severe hippocampal atrophy and shape deformation in hippocampal sclerosis than in Alzheimer disease. International Journal of Alzheimer Disease 2011:483972. PMCID: 3087502.
15. Zarow C, Weiner MW, Ellis WG, **Chui HC**. Prevalence, laterality, and comorbidity of hippocampal sclerosis in an autopsy sample. Brain Behav 2012;2:435-442.
16. Nelson PT, Dickson DW, Trojanowski JQ, Jack CR, **Chui HC**,..Schneider JA. Limbic-predominant age-related TDP-43 encephalopathy (LATE): consensus working group report. Brain. 2019;142(6):1503-27. Epub 2019/05/01. Doi: 10.1093/brain/awz099.PMCID: PMC6536849.

Improvements in neuropsychological assessment have included: 1) development of the modified Mini-Mental State Exam (3MS), 2) norms for the Spanish version of the Mattis dementia rating scale, and 3) development of the Los Angeles Chinese version of the Montreal Cognitive Assessment (MOCA). The 3MS has been widely used in epidemiological studies (e.g., Cardiovascular Health Study, Canadian Health Study on Aging, Women's Health Initiative Memory Study).

17. Teng EL, **Chui HC**. The Modified Mini-Mental State (3MS) examination. J Clin Psychiatry. 1987;48(8):314-8. Epub 1987/08/01. PubMed PMID: 3611032.
18. Lyness SA, Hernandez I, **Chui HC**, Teng EL. Performance of Spanish speakers on the Mattis dementia rating scale (MDRS). Arch Clin Neuropsychol. 2006;21(8):827-36. Epub 2006/11/03. doi: S0887-6177(06)00151-X [pii] 10.1016/j.acn.2006.09.003. PubMed PMID: 17079111.
19. Zheng L, Teng EL, Varma R, Mack WJ, Mungas D, Lu PH, **Chui HC**. Chinese-language montreal cognitive assessment for cantonese or mandarin speakers: age, education, and gender effects. Int J Alzheimers Dis. 2012;2012:204623. Epub 2012/07/26. doi: 10.1155/2012/204623. PubMed PMID: 22830073; PMCID: 3399373
20. Gatz M, Mack WJ, **Chui HC**, Law EM, Barisano G, Sutherland ML, Sutherland JD, Eid Rodriguez D, Quispe Gutierrez R, Copajira Adrian J, Bani Cuata J, Borenstein AR, Walters EE, Irimia A, Rowan CJ, Wann LS, Allam AH, Thompson RC, Miyamoto MI, Michalik DE, Cummings DK, Seabright E, Garcia AR, Hooper PL, Kraft TS, Finch CE, Thomas GS, Stieglitz J, Trumble BC, Gurven MD, Kaplan H. Prevalence of dementia and mild cognitive impairment in indigenous Bolivian forager-horticulturalists. Alzheimers Dement. 2023;19(1):44-55. Epub 2022/03/10. doi: 10.1002/alz.12626. PubMed PMID: 35262289; PMCID: PMC9458772.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Toga, Arthur W.

eRA COMMONS USER NAME (credential, e.g., agency login): TOGA22

POSITION TITLE: Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of Massachusetts	B.S.	02/1974	Psychology
St. Louis University	M.S.	05/1976	Neuroscience
St. Louis University	Ph.D.	04/1978	Neuroscience

A. Personal Statement

I am a Provost Professor at the University of Southern California (USC), Director of the USC Mark and Mary Stevens Neuroimaging and Informatics Institute, Director of the Laboratory of Neuro Imaging (LONI) and the founding editor-in-chief of the journal, NeuroImage. I was also the past Director of Admissions of the Neuroscience Graduate Program at UCLA. I have a career-long record of funding and publication in the analysis, modeling and atlasing of images and other data obtained from many species including humans. I have developed unique mathematical techniques for global and local warping of images, atlasing and multidimensional modeling as well as visualization of composite brain image sets. I direct several national and international collaborative studies that involve multi-site acquisition of data, coordination of analysis, and dissemination of data and results. The Institute and my laboratory coordinate neuroimage analysis and related informatics activities with national and international collaborators and, as such, is a worldwide resource by virtue of these distributive activities. I have experience and expertise in local and wide area computer networks, data transmission, storage and archiving as well as data compression and scientific visualization. I have developed a comprehensive supercomputing facility for brain mapping incorporating the latest technology in the analysis of structure and function of the brain in health and disease. I have a close working relationship not only with the clinical neuroscience departments (neurology, neurosurgery, psychiatry, neuroradiology) but also with the basic science programs throughout campus (e.g., psychology, physiology, computer science, electrical engineering and biostatistics). I have excellent rapport with other key collaborating scientists and colleagues throughout the world. I look forward to working on this proposed project.

Ongoing and recently completed projects that I would like to highlight include:

P01AG052350
09/01/2021-08/31/2026
Toga/Zlokovic (MPIs)
Vascular contributions to dementia and genetic risk factors for Alzheimer's disease. Program project to studying imaging and molecular biomarkers of neurovascular dysfunction in individuals at genetic risk for Alzheimer's disease both familial and sporadic.

P41EB015922
Toga (PI)
09/30/98-02/28/23
Laboratory of Neuro Imaging Resource (LONIR)

The goal of this competitive renewal application will go beyond current atlases and maps of brain that assume a static morphology and prohibit the examination of time varying changes. We will continue to develop the framework and tools to rigorously evaluate dynamic changes in brain structure and function focusing particularly on processes such as development, aging and the progression of specific diseases.

P30AG066530
Chui/Toga/Zlokovic (MPIs)
09/01/20-02/28/25
Alzheimer Disease Research Center (ADRC)
An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

R24MH114796
Toga/Duncan/Pouratian (MPIs)
09/05/18-05/31/23
Data Archive for the Brain Initiative (DABI)
The purpose of this proposal for a Data Archive for the Brain Initiative (DABI) is to develop web-accessible data archives to capture, store, and curate data related to the BRAIN Initiative proposals that collect invasive human neurophysiological data and make them broadly available and accessible to the research community for furthering research.

U24AG074855
Hohman/Cuccaro/Toga (MPIs)
09/01/21-08/31/26
Prime: Vanderbilt University
Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium
LONI's primary objective is to provide a system for housing, searching, and interrogating harmonized AD data and linkage with access to NIAGADS.

B. Positions, Scientific Appointments, and Honors

Positions

2016 - present	Ghada Irani Chair in Neuroscience, University of Southern California
2013 - present	Provost Professor and Director, USC Stevens Neuroimaging and Informatics Institute
2013 - present	Director, USC Laboratory of Neuro Imaging
2010 - 2013	Vice Chair, Dept. of Neurology, UCLA
2009 - 2013	Distinguished Professor, Dept. of Neurology, UCLA
2009 - 2012	Associate Vice Provost of Informatics, UCLA
2009 - 2012	Associate Dean, David Geffen School of Medicine at UCLA
2008 - 2009	Assistant Dean, David Geffen School of Medicine at UCLA
1993 - 2013	Co-Director, Division of Brain Mapping, UCLA, Neuropsychiatric Institute
1993 - 2013	Professor, UCLA, Dept. of Neurology
1987 - 2013	Director, Laboratory of Neuro Imaging, UCLA
1987 - 2003	Assistant Chairman for Research Affairs, UCLA, Dept. of Neurology
1987 - 1993	Associate Professor, UCLA, Dept. of Neurology
1980 - 1987	Research Assist. Professor, Washington University School of Medicine, Dept. of Neurology
1979 - 1980	Research Instructor, Washington University School of Medicine, Dept. of Neurology
1978 - 1979	Postdoctoral Fellow, Washington University School of Medicine, Dept. of Neurology

Scientific Appointments

- Nature-Scientific Data, Editorial Board Member
- Founding Editor-in-Chief (1992 - 2005), NeuroImage, Elsevier
- InSight - Board Member
- The Cerebellum - Section Editor, Neuroimaging
- Neuroinformatics - Editorial Board Member

- Anatomy & Embryology - Editorial Board Member
- Current Medical Imaging Reviews - Editorial Board Member
- Biology Image Library - Advising Editor to the Neuroscience Section
- Biomedical Computation Review - Editorial Advisory Board Member
- Brain Structure and Function - Editorial Board Member
- Journal of Neural Regeneration Research (NRR) - Editorial Board Member

Honors

2023 Ranked 3rd as World Scientist, Medical and Health Sciences/Neuroscience AD Scientific Index, 2016, 2019, 2020, 2022 Clarivate Analytics Highly Cited Researcher, 2022 Recipient of Highly Cited Researcher in the Category of Crossfield, 2022 Top 200 Best Scientists in Research.com Neuroscience, 2016 Ghada Irani Chair in Neuroscience at the University of Southern California, 2014 and 2015 World's Most Influential Scientific Minds – Thomson Reuters; 2013 Provost Professor at the University of Southern California; 2009 University Professor University of California; 2007 Pioneer in Medicine Award; 1999 Smithsonian Award for Scientific Innovation; 1999 Axiem Award (Silver Finalist) for National Medical Video; 1998 The Communication Award for Scientific Video; 1997 Giovanni DiChiro Award for Outstanding Scientific Research; 1997 Cum Laude Paper (1st prize), SPIE Medical Imaging; 1996 Best theoretical paper finalist, American Informatics Association; 1990 First prize for posters, American Association for Neurological Surgeons; 1990 Outstanding Contribution Award, National Computer Graphics Association; 1988 D&L Straus Scholar, UCLA School of Medicine;

C. Contributions to Science

My career long focus has been in the development and application of imaging and image analysis approaches in the study of brain development, aging, neurological disease and psychiatric disorders. I have been at the forefront of research and development of population based atlases depicting multimodality data to discover patterns and trends over time, across cohorts. I am the author and editor of widely used books and an encyclopedia of brain mapping.

1. Connectomics: We developed a number of diffusion imaging based approaches to collect and analyze connectome data. Our Human Connectome Grant was based on this quantitative methodology and is used in the analyses of numerous studies. This work includes structural architecture of connectivity functional connectivity and tractography comparisons in populations.

    a. Cabeen RP, Immonen R, Harris NG, Gröhn O, Smith G, Manninen E, Garner R, Duncan D, Pitkänen A & **Toga AW** 2020 A Computational Diffusion MRI Framework for Biomarker Discovery in a Rodent Model of Post-Traumatic Epileptogenesis. IEEE 17th International Symposium on Biomedical Imaging (ISBI) pp. 1013-1017. IEEE. doi:10.1109/ISBI45749.2020.9098575
    b. Kim SY, Liu M, Hong SJ, **Toga AW**, Barkovich AJ, Xu D & Kim H 2020 Disruption and Compensation of Sulcation-based Covariance Networks in Neonatal Brain Growth after Perinatal Injury. Cereb Cortex. doi: 10.1093/cercor/bhaa181 PMID: 32656563 PMCID: PMC7609941
    c. Contreras JA, Fujisaki K, Ortega N, Barisano G, Sagare A, Pappas I, Chui H, Ringman JM, Joe EB, Zlokovic B, **Toga AW**, Pa J. Decreased functional connectivity is associated with increased levels of Cerebral Spinal Fluid soluble-PDGFRβ, a marker of blood brain barrier breakdown, in older adults. Res Sq [Preprint]. 2023 Mar 7:rs.3.rs-2644974. doi: 10.21203/rs.3.rs-2644974/v1. PMID: 36945439; PMCID: PMC10029080
    d. La Rocca M, Barisano G, Garner R, Ruf SF, Amoroso N, Monti M, Vespa P, Bellotti R, Erdoğmuş D, **Toga AW**, Duncan D; EpiBioS4Rx Study Group. Functional connectivity alterations in traumatic brain injury patients with late seizures. Neurobiol Dis. 2023 Apr;179:106053. doi: 10.1016/j.nbd.2023.106053. Epub 2023 Mar 4. PMID: 36871641.

2. Image Analyses: I have developed a number of analytic methods to characterize the imaging phenotype beyond simple volumetric measures. We have created the algorithm and executable software and applied these approaches to a number of different studies.

    a. Clark K, Lynch K, & **Toga AW** 2019 Hippocampal shape maturation in childhood and adolescence. Cerebral Cortex, 29(9):3651-3665. doi: 10.1093/cercor/bhy244. PMID: 30272143 PMCID: PMC6686754

b. Ning K, Zhao L, Matloff W, Sun F, **Toga AW** 2019 Association of relative brain age with tobacco smoking, alcohol consumption, and genetic variants. Sci Rep. 10(1):10. doi: 10.1101/469924. PMC: 32001736 PMCID: PMC6992742
c. Zhao L, Matloff W, Ning K, Kim H, Dinov I, & **Toga AW** 2019 Age-related differences in brain morphology and the modifiers in middle-aged and older adults. Cereb Cortex. PMID: 30535294 PMCID: PMC6931275
d. Lynch KM, Sepehrband F, **Toga AW,** Choupan J. Brain perivascular space imaging across the human lifespan. Neuroimage. 2023 Mar 11;271:120009. doi: 10.1016/j.neuroimage.2023.120009. Epub ahead of print. PMID: 36907282.

3. Informatics: My groups has pioneered the multisite trial informatics solutions need to efficiently perform trials and studies across different acquisition sites. I have created the dissemination platforms to enable tens of thousands of users to gain access to unprecedented amounts of high quality, comprehensively described data.

   a. Zhao L, Batta I, Matloff W, O'Driscoll C, Hobel S & **Toga AW** 2019 Neuroimaging PheWAS (phenome-wide association study): a free cloud-computing platform for big-data, brain-wide imaging association studies. Neuroinformatics. doi: 10.1007/s12021-020-09486-4. PMID: 32822005 PMCID: PMC7897334
   b. Marino S, Zhao Y, Zhou N, **Toga AW**, Zhao L, Jian Y, Yang Y, Chen Y, Wu Q, Wild J, Cummings B, Dinov ID 2020 Compressive Big Data Analytics: An Ensemble Meta-Algorithm for High-dimensional Multisource Datasets. PLOS One. doi: 10.1101/2020.01.20.912485. PMID: 32857775 PMCID: PMC7455041
   c. Lam J, Cabeen RP, Tanna R, Navarro L, Heck CN, Liu CY, Lee B, Russin JR, **Toga AW**, Lee DJ. Gray Matter Atrophy: The Impacts of Resective Surgery and Vagus Nerve Stimulation in Drug-Resistant Epilepsy. World Neurosurg. 2021 May;149:e535-e545. doi: 10.1016/j.wneu.2021.01.141. Epub 2021 Feb 4. PMID: 33549931; PMCID: PMC8121141.

4. Alzheimer's Disease: I have developed methods and applications combining data from different subjects, scanners and projects for population maps relating structure, function and other data in Alzheimer's disease. These have formed the basis of maps that show the progression of Alzheimer's disease and other time varying brain changes.

   a. Alban SL, Lynch KM, Ringman JM, **Toga AW**, Chui HC, Sepehrband F, Choupan J; Alzheimer's Disease Neuroimaging Initiative. The association between white matter hyperintensities and amyloid and tau deposition. Neuroimage Clin. 2023 Mar 20;38:103383. doi: 10.1016/j.nicl.2023.103383. Epub ahead of print. PMID: 36965457; PMCID: PMC10060905.
   b. Weiner MW, Veitch DP, Miller MJ, Aisen PS, Albala B, Beckett LA, Green RC, Harvey D, Jack CR Jr, Jagust W, Landau SM, Morris JC, Nosheny R, Okonkwo OC, Perrin RJ, Petersen RC, Rivera-Mindt M, Saykin AJ, Shaw LM, **Toga AW**, Tosun D, Trojanowski JQ; Alzheimer's Disease Neuroimaging Initiative. Increasing participant diversity in AD research: Plans for digital screening, blood testing, and a community-engaged approach in the Alzheimer's Disease Neuroimaging Initiative 4. Alzheimers Dement. 2023 Jan;19(1):307-317. doi: 10.1002/alz.12797. Epub 2022 Oct 9. PMID: 36209495; PMCID: PMC10042173.
   c. Morris JC, Weiner M, Xiong C, Beckett L, Coble D, Saito N, Aisen PS, Allegri R, Benzinger TLS, Berman SB, Cairns NJ, Carrillo MC, Chui HC, Chhatwal JP, Cruchaga C, Fagan AM, Farlow M, Fox NC, Ghetti B, Goate AM, Gordon BA, Graff-Radford N, Day GS, Hassenstab J, Ikeuchi T, Jack CR, Jagust WJ, Jucker M, Levin J, Massoumzadeh P, Masters CL, Martins R, McDade E, Mori H, Noble JM, Petersen RC, Ringman JM, Salloway S, Saykin AJ, Schofield PR, Shaw LM, **Toga AW**, Trojanowski JQ, Vöglein J, Weninger S, Bateman RJ, Buckles VD. Autosomal dominant and sporadic late onset Alzheimer's disease share a common in vivo pathophysiology. Brain. 2022 Oct 21;145(10):3594-3607. doi: 10.1093/brain/awac181. PMID: 35580594; PMCID: PMC9989348.
   d. O'Bryant SE, Zhang F, Petersen M, Hall JR, Johnson LA, Yaffe K, Braskie M, Vig R, **Toga AW**, Rissman RA; HABS-HD Study Team. Proteomic Profiles of Neurodegeneration Among Mexican

Americans and Non-Hispanic Whites in the HABS-HD Study. J Alzheimers Dis. 2022;86(3):1243-1254. doi: 10.3233/JAD-210543. PMID: 35180110; PMCID: PMC9376967

5. Imaging Genetics: I have worked to develop methods for associated snps to imaging phenotypes and created software to provide access to databases containing these data. We have applied this approach to a number of disease and healthy cohorts.

   a. Lee J, Petrosyan S, Khobragade P, Banerjee J, Chien S, Weerman B, Gross A, Hu P, Smith JA, Zhao W, Aksman L, Jain U, Shanthi GS, Kurup R, Raman A, Chakrabarti SS, Gambhir IS, Varghese M, John JP, Joshi H, Koul PA, Goswami D, Talukdar A, Mohanty RR, Yadati YSR, Padmaja M, Sankhe L, Rajguru C, Gupta M, Kumar G, Dhar M, Jovicich J, Ganna A, Ganguli M, Chatterjee P, Singhal S, Bansal R, Bajpai S, Desai G, Bhatankar S, Rao AR, Sivakumar PT, Muliyala KP, Sinha P, Loganathan S, Meijer E, Angrisani M, Kim JK, Dey S, Arokiasamy P, Bloom DE, **Toga AW**, Kardia SLR, Langa K, Crimmins EM, Dey AB. Deep phenotyping and genomic data from a nationally representative study on dementia in India. Sci Data. 2023 Jan 20;10(1):45. doi: 10.1038/s41597-023-01941-6.
   b. Satizabal C, Adams H, Hibar D, White C, Stein J, Scholz M, Sargurupremraj M, …, **Toga AW**, … & Ikram M 2017 Genetic architecture of subcortical brain structures in over 40,000 individuals worldwide. Nature, doi: 10.1101/173831.
   c. Chen Q, Boeve BF, Schwarz CG, Reid R, Tosakulwong N, Lesnick TG, Bovee J, …, **Toga AW**, Trojanowski J, Weintraub S, Wong B, Boxer AI, Rosen H, Wszolek Z, Kantarci K 2019 Tracking White Matter Degeneration in Asymptomatic and Symptomatic MAPT Mutation Carriers. Neurobiol Aging, 83:54-62. doi: 10.1016/j.neurobiolaging.2019.08.011. PMID: 31585367 PMCID: PMC6858933
   d. Grasby KL, Jahanshad N, Painter JN, Colodro-Conde L, Bralten J, Hibar DP, Lind PA, ..., **Toga AW**, ..., Thompson PM & Medland SE; Enhancing NeuroImaging Genetics through Meta-Analysis Consortium (ENIGMA)—Genetics working group 2020 The genetic architecture of the human cerebral cortex. Science, 367(6484):eaay6690. doi: 10.1126/science.aay6690. PMID: 32193296 PMCID: PMC7295264

Complete List of Published Work in MyBibliography:
http://www.ncbi.nlm.nih.gov/pubmed/?term=toga,%20arthur[Author]

PHS 398 Cover Page Supplement

1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

PHS 398 Cover Page Supplement

5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☒ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

Introduction	
1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	Overall_Aims1044447016.pdf
3. Research Strategy*	Overall_Research_Plan1044447017.pdf
4. Progress Report Publication List	Progress_Report_Pubs_List1044446729.pdf
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	MPI_Plan1044446730.pdf
8. Consortium/Contractual Arrangements	Consortium_Agreement_ALL1044446731.pdf
9. Letters of Support	Letters_of_Support_ALL1044446732.pdf
10. Resource Sharing Plan(s)	
11. Other Plan(s)	USC_ADRC_Resource_Sharing_Plan1044447018.pdf
12. Authentication of Key Biological and/or Chemical Resources	
13. Appendix	
12. Appendix	

SPECIFIC AIMS: USC ADRC OVERALL

Over the past funding cycles, the USC ADRC, now celebrating its 40th anniversary, has contributed significantly to numerous areas of AD research. Our ADRC has previously focused on vascular factors related to AD and in this renewal we expand that focus to characterize factors that modify the effect of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers, emphasizing vascular, lipid, inflammatory, genetic, and ethnic influences. In the next 5-years, the USC ADRC will focus on three overarching Aims:

1) **Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4,** focusing on factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers. Our goal is to build a multi-sourced, participant registry with ApoE genotype, SEDoH, and digital cognitive assessment.
2) **Promote local ADRD research at USC:**
   a) Integrate research programs by holding annual events, pilot fundings, and harmonization of protocols.
   b) Share data, tissues, biofluids, and a participant registry through a common USC ADRC database.
   c) Train the next generation of ADRD researchers and educate participants and the public.
   d) Enhance Diverse Perspectives
3) **Further national collaborations:** Advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, comprehensive imaging to national initiatives.

The ADRC is led by two multi-principal investigators (mPI: Chui and Toga) and an Associate Director (Yassine). Chui has devoted her scientific career to understanding interactions between vascular disorders and AD, through clinical-imaging-pathological correlations. Toga is a leading neuroscientist in imaging and informatics, and developed or manages numerous programs in AD, including the informatics core for ADNI, SCAN, HABS-HD, and GAAIN resources that collect and manage big data from global partners/studies. Yassine is a translational physician-scientist with deep expertise in the role of apolipoproteins in AD, lipid metabolism, and neuroinflammation. He has chaired the national ADC REC committee and directs the new USC Center for Personalized Brain Health (CPBH). The ADRC is comprised of the six required cores, optional Imaging core, and a REC Core.

A. *Administration Core* [mPIs: Helena Chui (contact) and Arthur Toga; Associate Director: Hussein Yassine] fosters our overarching scientific goals, coordinates administrative functions aligned with NIA-ADRC program, attracts early-stage and new investigators through development projects, promotes collaborative local and national research, and promotes training/education with REC and ORE Cores.
B. *Clinical Core* (Lon Schneider, Elizabeth Joe) recruits and retains participants from diverse communities enriched for *APOE* ε4, performs standardized evaluations/diagnoses using the NACC Uniform Data Set, characterizes participants by *APOE* genotype and ATN(v) biomarkers, collects biofluids, obtains informed consent for autopsy, and maintains a research participant registry with the ORE and Data cores.
C. *Data Management and Statistical Core* (Arthur Toga/Dan Nation and David Conti/Xinhui Wang) provides a centralized data management framework, ensures timely and accurate data entry and management, and provides biostatistical support.
D. *Neuropathology Core* (Annie Hiniker, Deborah Hawes) performs standardized neuropathological diagnoses, distributes tissues, develops a digital pathology library, and interrogates the effect of *APOEε4* on AD neuropathology and perivascular inflammation.
E. *Outreach, Recruitment, and Engagement Core* (Maria Aranda, John Ringman) focuses on participant recruitment enriched for *APOE* ε4, develops a research participant registry in concert with the Clinical and Data Cores, enhances recruitment from public safety-net hospitals, and engages in bi-directional education, information and resource sharing.
F. *Biomarker Core* (Robert Rissman) coordinates collection, processing, storage, and sharing of biofluids, facilitates biomarker research through an integrated informatics toolkit, accelerates local, national, and international biomarker discovery efforts.
G. *Imaging Core* (Arthur Toga/Ioannis Pappas): Provides imaging for the ATN(v) framework for ADRC projects, performs high quality control and MRI analytics, develops technical and scientific approaches for ultra-high field 7T MRI, assists in PET protocol development, quality control, and analytics.
H. *Research Education Component* (Hussein Yassine, Duke Han) supports ADRC scientists by monitoring the progress of ADRC scientists in their early-career milestones.

RESEARCH PLAN

Overall Significance

Four major characteristics position the USC ADRC as a leader in the ADRD research community:

(A) Neuroscience and AD are major strengths at USC. The USC president designates brain health as one of the five moonshots. The ADRC supports major ADRD centers and institutes that the university has established for AD. USC is home to the first School of Gerontology in the United States (1975) and is among the first five inaugural Alzheimer Disease Research Centers (ADRC) funded by the National Institute on Aging (1984) (PI: CE Finch). During the past 20 years, the University of Southern California has continued to invest in this area and build a robust research infrastructure to support the discovery and treatment of Alzheimer disease (AD) and related disorders (ADRD). Cornerstones include the Zilkha Neurogenetics Institute (ZNI-2003), Stevens Institute for Neuroimaging and Informatics (INI-2013), Alzheimer Therapeutics Research Institute (ATRI-2015), and most recently, the Center for Personalized Brain Health (CPBH-2023). Further, the Vice President for Research (Puri), Senior Vice President for Health Sciences (Shapiro) and the Dean of the Keck School of Medicine (Meltzer) provide generous institutional support to the USC ADRC (see letter of support Overall).

(B) We have five major cross-center initiatives. 1) The ADRC supports ATRI/ACTC to provide informatics, develop biomarkers, recruit participants from URG and train the next generation of clinical ADRD researchers. 2) The ADRC works closely with the INI to develop new MR imaging sequences as well as to evaluate these in participants who have undergone previous standardized imaging among other measures. 3) The ADRC-INI encourages training to cultivate experts in neuroimaging, advance brain imaging modalities, informatics, analytics, and AI approaches, in addition to supporting national initiatives (Laboratory of Neuroimaging-LONI). 4) The ADRC REC works with RCMAR to cultivate and support demography research, addressing disparities supporting URM scientists. 5) The newly founded CPBH bridges ADRD research with our clinical enterprise, focused on *APOE* ε4, and supports the mission of the ADRC by providing institutional support to develop a registry, recruit translational physician scientists, and promote memory clinic growth.

(C) We built an outstanding team. During the current funding cycle, we have recruited established investigators and have mentored early-stage investigators at USC. Specifically, over the past year, the ADRC has recruited to its core leadership: Anne Hiniker MD, PhD (NP Core, from UCSD), Robert Rissman PhD (Biomarker Core, ATRI, from UCSD), David Conti PhD (Data Core, Department of Population and Public Health Sciences), Daniel Nation PhD (Data Core, School of Gerontology, returned to USC from UCI), and Duke Han PhD (REC, School of Letters, Arts, and Sciences). Continuity and mentorship are provided by Lon Schneider (Clinical Core), Maria Aranda (ORE Core), John Ringman (ORE Core), Debra Hawes (NP Core). We have designed specific training programs to foster the development of our early-stage faculty which include Elizabeth Joe (Clinical Core), Xinhui Wang (Data Core), Michael Bienkowski (NP Core), and Ioannis Pappas (Imaging Core). As a team of multi-disciplinary investigators and the source of integrated, standardized, multi-dimensional data, the USC ADRC serves as a local catalyst and national collaborator in AD/ADRD research. Succession planning for the overall ADRC leadership is outlined in the Administrative Core.

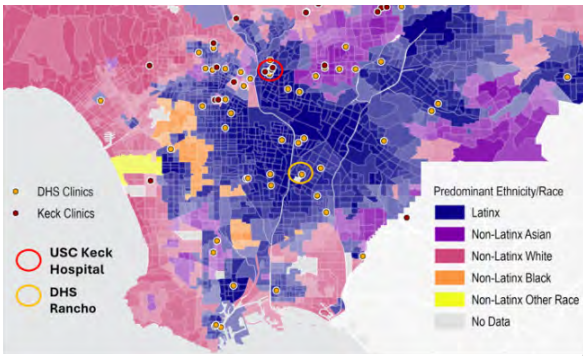


Figure 1: Predominant ethnicity/race in catchment areas surrounding USC Keck and DHS Rancho hospitals.

(D) Our population is diverse. USC is in the urban heart of Los Angeles, one of the most diverse cities in the world. Keck School of Medicine (KSOM) is part of Keck Medicine of USC (KMUSC), which is the University's medical enterprise and one of only two university-based medical systems in Los Angeles. KMUSC owns and operates Keck/Norris Hospitals of USC and two community hospitals (USC Verdugo Hills Hospital and USC Arcadia Hospital), as well as more than 40 outpatient locations. KSOM is the academic partner for Los Angeles General Hospital (LAG), the largest safety net hospital in Los Angeles. The ADRC has demonstrated success with engaging diverse communities in ADRD research. The USC-ADRC, Keck Hospital, Los Angeles

General Hospital (LAG) and Rancho Los Amigos National Rehabilitation Center (RLNRC) are located in catchment areas, where Latinos comprise the majority ethnic group.

The overarching aims of the USC ADRC are aligned and responsive to the National Alzheimer's Project Act (NAPA) inaugurated in 2011 and updated in December.

Table 1: USC ADRC Responsiveness to National Alzheimer Project Act (NAPA) Initiatives

NAPA Goal	Title of Initiative	ADRC Goal
1	Prevent and Effectively Treat ADRD	Aim 1: Characterize factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers. Aim 2: Support local research in ADRD.
2	Enhance Care Quality and Efficiency	Aim 1: Support ALZ-NET research data collection for patients being clinically treated with Lecanemab and other anti-amyloid antibodies in the real world, with a focus on early detection of ARIA. Aim 2: Leadership role in roll-out of digital cognitive assessment in the Keck Healthcare system.
3	Expand Support for People with ADRD	Aim 2: Integrate multi-disciplinary team in ADRC clinical core and Neurology Memory Disorders Clinic. Aim 2: Collaborate with Family Medicine in provision of neurological expertise to CMS GUIDE program. Aim 2: Educate the public through educational seminars and Brain Health4Me (BHrMe), a healthy lifestyles program.
4	Enhance Public Awareness and Engagement	Aim 2: Support inclusive research, through GeneScreen, with a focus on outreach to underserved groups in Los Angeles County, including those seen in the Department of Health Services. Share data with national networks such as ALZ-NET. Organize numerous educational events.
5	Improve Data to Track Progress	Aim 3: Participate in national database initiatives (NACC, SCAN, ADNI, NCRAD, NIAGADS, CLARiTI).
6	Accelerate Action to Promote Healthy Aging and Reduce Risk Factors For ADRD	Aim 1: Characterize factors that modify the effect of *APOE* ε4 on cognitive decline and on AD ATN(vi) biomarkers, with an emphasis on vascular, metabolic, and ethnic influences, in concert with the new Center for Personalized Brain Health (CPBH).

Progress Aim 1: Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4, focusing on factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers. Our goal is to build a multi-sourced, participant registry with *APOE* genotype, SDoH, and digital cognitive assessment.

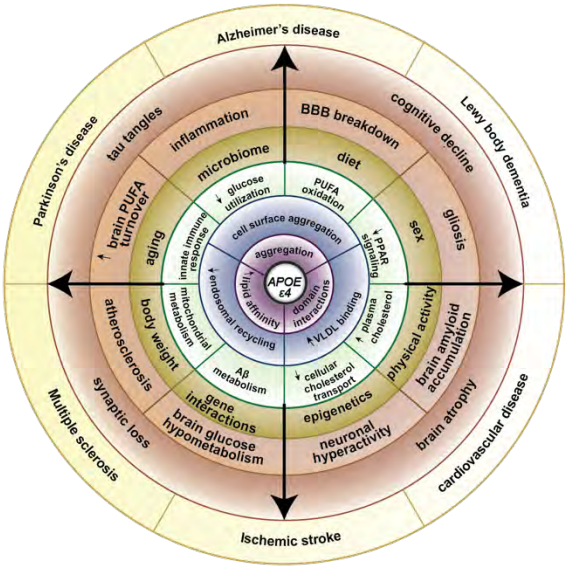


Figure 2: Pleiotropic Effects of *APOE* ε4 (1).

Background: The central scientific theme of the USC ADRC is to **characterize factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers.** Apolipoprotein E ε4 is the most common genetic risk factor for late-onset Alzheimer disease. The pathways leading to increased risk, however, are pleiotropic and still not entirely understood (1). Association with beta-amyloid plaques, tauopathy, and cerebral amyloid angiopathy are well-recognized, while alterations in lipid metabolism, neuroinflammation and vascular biology remain elusive.

APOE ε4 homozygosity represents a distinct genetic entity (2) which mimics that of a Mendelian disease with near full penetrance in AD neuropathological change (ADNC) by age 55 with a 50% likelihood of symptom onset by age 65, and a 60-80% lifetime risk for dementia diagnosis (red curve). Conversely, *APOE ε4* heterozygosity is more common and is associated with a range of ADNC and age at symptom onset severity (**Fig. 3**, two orange curves). Our aim is to better define factors (e.g, vascular, inflammatory, genetic and ethnic influences) that modify the effect of *APOE* ε4 on cognitive decline and ATN(v)i biomarkers.

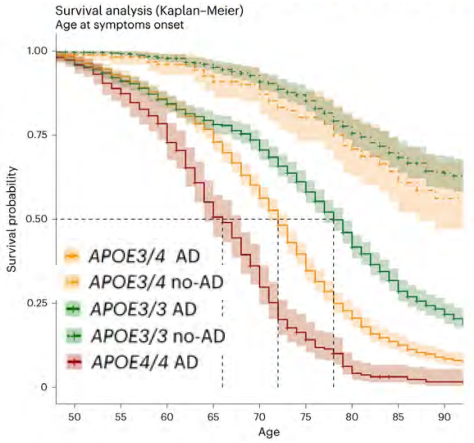


Figure 3: Survival analysis for age at symptom onset stratified by *APOE* genotype.

While *APOE* ε4 is the ancestral allele, its frequency differs across global populations. The *APOE* ε4 frequency is higher in Northern compared to Southern Europe, and is lower in Asia (3), Central America, and South America (4, 5). Of note, the risk of AD associated with *APOE* ε4/ε3 differs by ethnic ancestry, being higher in Asians and lower in Hispanic Whites, and Blacks (6). By virtue of its geographic catchment areas, the USC ADRC is well positioned to explore interactions between race/ethnicity and risk of AD associated with *APOE* ε4.

APOE ε4 is strongly associated with cerebral amyloid angiopathy (CAA), thereby contributing to VCID (vascular cognitive impairment and dementia) and amyloid-related imaging abnormalities (ARIA) (7). The *APOE* ε4 allele is strongly associated with Type II CAA where Aβ is deposited in capillaries as well as leptomeningeal arteries and cerebral arterioles (7, 8). Anti-amyloid immunotherapy is clearly associated with an increased risk of ARIA (9). In the CLARITY Phase III clinical trial, the incidence of ARIA-E doubled with one copy of and increased six-fold with two copies of *APOE* ε4 (8). The ability to diagnose CAA *in vivo*, which is a growing concern in the era of anti-amyloid antibody therapy, may be related to enlarged perivascular spaces (PVS) detectable on MRI (10, 11).

During the current funding cycle, ADRC publications have addressed various concerns relevant to the phenotypic expression of *APOE* ε4 (**Fig. 4**). We highlight four areas (vascular, neuro-inflammation, lipids, and air pollution) where the ADRC has advanced our understanding of the pleiotropic effects of *APOE* ε4 on brain health.

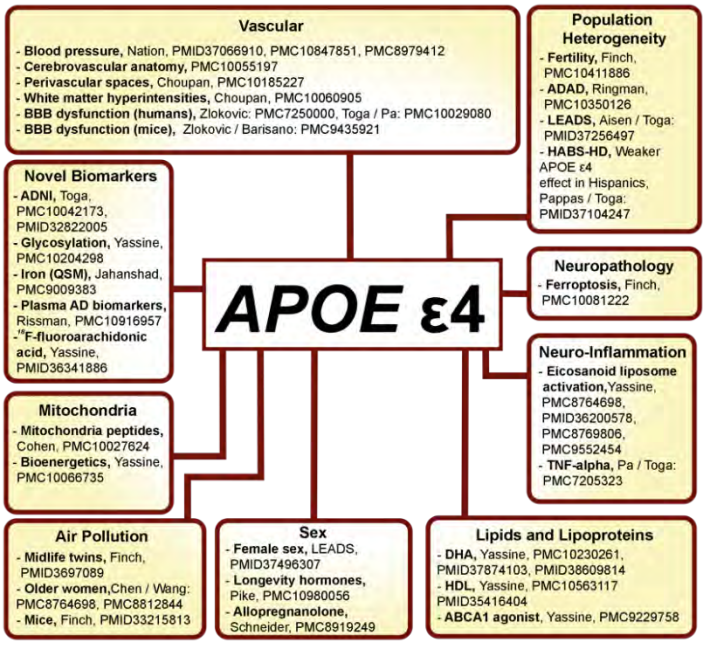


Figure 4: USC ADRC *APOE*-related publications from 2020 to 2024.

A) Vascular: 1) Increased blood pressure variability (BPV) is associated with *APOE* ε4. Sible and Nation (12) reported that non-demented carriers of *APOE* ε4 showed increased blood pressure variability (BPV), independent of mean BP. In turn, elevated BPV was associated with increased CSF phosphorylated tau, increased total tau, and decreased Aβ over time. This suggests that BPV correlates with alterations in AD biomarkers (13). *APOE* ε4 carriers with elevated BPV had the most accelerated tau accumulation at follow-up (14).

2) sPDGRFβ, a biomarker for pericyte-injury is associated with *APOE* ε4. The USC ADRC Biomarker Core focuses on soluble platelet derived growth factor receptor β (sPDGRFβ) which is a recognized biomarker for brain capillary pericytes, and is proposed as a vascular injury marker (v) for ATN(v). When injured, pericytes release sPDGFRβ, which is increased in CSF in humans with MCI (15, 16) and is not released from injured vascular smooth muscle cells or endothelia (17). High baseline levels of CSF sPDGFRβ predicted future cognitive decline in *APOE ε4* carriers, but not in non-carriers. This association remained after controlling for amyloid-β and tau and was correlated with increased activity of cyclophilin A-matrix metalloproteinase-9 (18).

3) Subcortical iron deposition detected by QSM imaging is associated with *APOE* ε4. Affiliated ADRC studies observed that disrupted iron homeostasis is associated with several neurodegenerative diseases, including Alzheimer's disease (AD), and may be partially modulated by genetic risk factors. We evaluated differences in subcortical quantitative susceptibility mapping (QSM), a type of MRI sensitive to cerebral iron deposition, between *APOE* genotypes in 27,535 participants from the UK Biobank (age: 45-82 years) (19). We found that *APOE* ε4 carriers had higher hippocampal and amygdala magnetic susceptibility, particularly among individuals aged 65 years or older. However, *APOE* ε2 (which protects against AD) carriers had a higher QSM only in the hippocampus, particularly in those under age 65.

B) Neuroinflammation: Neuroinflammatory lipids are associated with *APOE* ε4. Tissue from the USC ADRC (and in collaboration with Rush) enabled lipidomic analysis of post-mortem dorsolateral prefrontal cortex showing a significant interaction between the *APOE* ε4 allele and higher levels of both pro-inflammatory lipids and pro-

resolving eicosanoid lipids on measures of cognitive performance and on neuritic plaque burden. AD brain tissues displayed an activation of calcium-dependent phospholipase A2 (cPLA2), 5-lipoxygenase (5-LOX), and soluble epoxide hydrolase (sEH) enzymes (20, 21).

C) Lipids: 1) Small HDL particles may be protective against cerebral amyloidosis. The USC ADRC helped purchase a lipid particle analyzer (Ion Mobility) available to ADRC projects. Using lipid particle-size fractionation methods, Yassine et al. (22) demonstrated that CSF and plasma levels of small HDL particles are inter-correlated ($r=0.32$, $p < 0.0001$), which indicates that these smaller particles may cross through the BBB from plasma to the brain. Moreover, levels of smaller HDL particles are correlated with higher CSF Aβ42 levels and cognition. The small HDL hypothesis proposes that small HDL components can exchange into the brain, and the interaction between plasma-derived (ApoA-I) and brain-derived (apoE) apolipoproteins stabilize the function of the ATP-binding cassette 1 (ABCA1) to form apoE/apA-I small HDLs in the brain.

2) DHA supplementation for carriers of *APOE* ε4, a clinical trial. We tested the hypothesis that *APOE* ε4 is associated with reduced transport of DHA into the brain (23, 24). We conducted a Phase II, randomized clinical trial of DHA supplementation, stratified by *APOE* genotype. 365 cognitively unimpaired individuals between 55 and 80 were randomized to two grams of DHA per day or identically appearing placebo for two years. The primary outcome measure is the change in the ratio of CSF DHA to arachidonic acid after six months of supplementation. Secondary trial outcomes include change in functional and structural connectivity using resting state functional MRI at 24 months (n=365) (25).

D) Air pollution: Complex life-span interaction between air pollution and *APOE* genotype. ADRC-affiliated studies suggest a complex life-span interaction between geospatially located air pollution and *APOE* genotype. In the Vietnam Era Twin Study, higher exposure to fine particulate matter in air pollution (PM2.5) was related to poorer executive function in *APOE* ε4 carriers, but not in non-carriers (26). However, in older women, the adverse effects of PM2.5 exposures on dementia risk did not differ by *APOE* genotype (27). Mice exposed to nanosized particulate matter (nPM) displayed coordinated responses of Apoe-Apoc1-Tomm40 promoter group in the cerebral cortex (28). Further, humans and mice share 87% of potential binding sites for transcription factors in this promoter, which suggests that exposure to nPM may alter apolipoprotein expression in humans as well.

Active APOE Related Grants supported by the ADRC are presented in **Table 2**.

Table 2. ApoE Mechanism-Related Federal Funded Grants (Active in 2025+)

PI/PD	Grant Title/Number	Source & Amt (Total)	Project Period	Role of ADC
Yassine H	ApoE, ABCA1 and endosomal dysregulation in AD **5R01AG067063**	NIA $3,362,231	5/15/2020-2/28/2025	
Yassine H Arvanitakis Z	Brain cPLA2 as a mechanism for neuroinflammation in AD/ADRD with and without *APOE* ε4 **1RF1AG076124**	NIA $2,417,667	5/15/2022-4/30/2025	
Toga A	Vascular contributions to dementia and genetic risk factors (*APOE*) for Alzheimer disease **P01AG052350**	NIA 3,246490	1/2022-12/31/2027	Clinical, Imaging, and Biomarker Cores
Yassine H	Alzheimer's Abeta, tau, apolipoproteins and blood-brain barrier **5R01NS034467**	NIA $771,703/yr	3/1/2023 2/28/2025	
Kaplan H Finch CE Gatz M	Pathways linking infection, physical activity and *APOE* and biological sex to dementia in 2 Native American populations **R01AG054442**	NIA $3,199,665	9/1/2022 4/30/2027	Clinical Core

Progress Aim 2: Promote local ADRD research at USC. a) Integrate research programs by holding annual events, pilot fundings, and harmonization of protocols, b) Share data, tissues, biofluids, and a participant registry through a common USC ADRC database, c) Train the next generation of ADRD researchers and educate participants and the public, d) Enhance Diverse Perspectives

Promote local ADRD research at USC: AD and ADRD research are a major focus of the entire university research enterprise. It is encouraged and supported by USC leadership as a hallmark of the university's commitment to humanity's most important challenges. USC has been remarkably successful in building

infrastructure and expertise in ADRD (e.g, ZNI, INI, ATRI, CPBH, RCMAR, GERO (**Fig. 5**). Furthermore, USC currently has over 124 NIH funded grants focused on ADRD, of which 47 are supported by the ADRC (Overall Table 2_a1). **While the ADRC has been involved and supported these and many others, we are now poised to integrate them with a concerted effort to harmonize (when possible) data collection, shared protocols, data dictionaries and archival systems. This will result in a far more cohesive research community, yielding benefit to all projects.**

Integrate research programs. At the Keck School of Medicine (KSOM), the ADRC has created a nexus of cooperation with four Institutes/Centers: Zilkha Neurogenetics Institutes (ZNI), Steven's Institute of Neuroimaging and Informatics (INI), Alzheimer Therapeutics Research Institute (ATRI), Clinical Translational Science Institute (CTSI) and the Center for Personalized Brain Health (CPBH) (**Fig. 5**). The ADRC has an active role in integrating these programs via shared seminars, symposia and funding opportunities.

Major steps are already underway to link the ADRC to real world data from the Electronic Health Record at KSOM. Each year, KSOM faculty care for ~280,000 patients; one-third of which are URMs and 40% are over the age of 65. Initiatives linking data include: 1) ALZ-NET, 2) Digital Cognitive Assessment (Linus Health, Inc), and 3) Center for Medicare and Medicaid Services' prospective payment system for dementia [Guiding an Improved Dementia Experience (CMS GUIDE)]. The ADRC works with both **California Alzheimer Disease Centers** (**CADC**) [one at the USC Health Science Campus led by Schneider (CADC-HSC) and another at Rancho Los Amigos National Rehab Center (CADC-Rancho) led by Segal-Gidan] which refer potential participants, offer training sites, and co-sponsor educational events. We have established an auxiliary ADRC clinical research site at the **Roybal Comprehensive Medical Center**, located five miles from the USC Health Science Campus. Both Rancho and Roybal clinics are part of the **LA County Department of Health Services (DHS)** and serve predominantly Latinx patients.

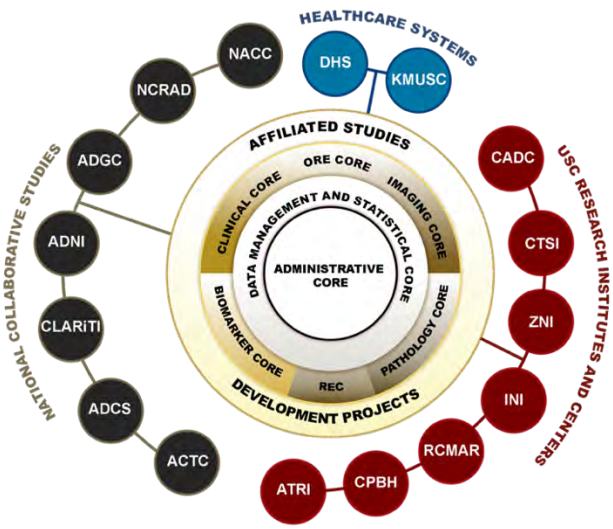


Figure 5: ADRC interactions with USC Research Institutes and Centers (Aim 2) and with National Collaborative Studies (Aim 3).

Share data, tissues, biofluids, and a participant registry: A) The Clinical Core provided well-characterized participants to affiliated studies and development projects. The NP Core distributed nearly 4,000 CNS tissues during the past four years (NP Core table 2). During the current funding period, we sent 282 plasma, 148 buffy coats, 154 PBMC, 153 DNA, 53 RNA to NCRAD, which were distributed to 12 investigators supported by 14 NIH Grants (NCRAD Sample Distribution Report, 2020 to 2024). The USC ADRC has supported and contributed to numerous other research projects and discoveries. Below we describe several of those, illustrating the breadth and depth of our AD/ADRD research.

B) USC School of Gerontology: Mitochondrial variants and Alzheimer disease. The Cohen lab at the USC School of Gerontology focuses on the role of mitochondrial DNA variants in aging and ADRD. Miller et al. recently reported that the mitochondrial SNP rs2853499 codes for a microprotein (SHMOOSE) associated with AD, neuroimaging, and transcriptomics (29). Levels of SHMOOSE, in cerebrospinal fluid (CSF) from USC ADRC subjects, were correlated with age, CSF tau, and brain white matter volume. UDS and CSF were provided from the USC ADRC Clinical Core; CSF tau and brain WMH were from the ADRC Biomarker and Imaging cores.

C) Cedar-Sinai Medical Center: Retinal pathology in AD. The NP Core of the USC ADRC provided postmortem ocular and brain tissue to the Koronyo-Hamaoui laboratory at Cedar Sinai Medical Center. Evaluation of postmortem retinas from a cohort of 56 human donors revealed an early and progressive decrease in vascular PDGRFβ in mild cognitive impairment (MCI) and AD compared to cognitively normal controls. A substantial pericyte loss together with significant Abeta deposition in retinal microvasculature and pericytes in AD were identified (30). Retinal PDGRFβ loss significantly associated with increased retinal vascular Aβ40 and

Aβ 42 burden. The identification of pericyte and PDGRFβ loss accompanying increased vascular amyloidosis in AD retina implies compromised blood-retinal barrier integrity and provides new targets for AD.

D) USC Zilkha Neurogenetics institute: Multi-institutional Program Project [Vascular and Genetic contributions to AD (P01 AG052350, Toga)]. A multi-institutional (USC ADRC, Washington University Knight ADRC, Banner Health Institute, and Mayo Clinic Arizona ADRC) Program funded by NIA and Alzheimer's Association (for PET imaging) tests the neurovascular hypothesis of AD in *APOE* ε4 carriers versus non-carriers. Carriers of the *APOE* ε4 allele recruited by the USC ADRC Clinical Core (16, 18) show evidence of breakdown of the BBB in the hippocampus and medial temporal lobe. Our findings suggest that breakdown of the BBB contributes to *APOE* ε4-associated cognitive decline independently of AD pathology and might be a therapeutic target among *APOE* ε4 carriers. CSF and plasma biomarkers are assayed by the ADRC Biomarker Core. UDS data, MRI images, and ATN(v) biomarker results are shared with the ADRC database.

Training the next generation of ADRD scientists (see Table in Administration Core). The ADRC REC supported 728 REC mentees, seven developmental projects and 22 REC scholars obtained NIH fundings (various levels), or hosted two symposia per year. Our mentees span multiple USC schools, backgrounds and career stages and include scholars from outside schools that attend our weekly seminar series. In the next funding cycle, we plan common development awards and career development workshops in close collaboration with Julie Zissimopoulos (AD-RCMAR).

Progress Aim 3: Further national collaborations. Advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, comprehensive imaging to national initiatives.

Advance ADRC research through contributing USC-generated data: Toga directs several national and international collaborative studies and informatics hubs that involve multi-site acquisition of data, coordination of analysis, and dissemination of data and results, including LONI, ADNI, the Human Connectome Project, and the Global Alzheimer's Association Interactive Network (GAAIN). These, and other programs are imaging and big data resources for innovation in brain mapping related to health and disease. We currently manage over 155 active projects and programs, serve 163,000 current users delivering over 420 million data downloads. Our data sharing policies are modelled on those promoted by ADNI whereby all digital data are distributed to anyone without embargo and without any impediment. Data are fully described with comprehensive provenance and presented via intuitive and efficient web-based portals. We link with and provide data to other national and global repositories and collaborate with many other institutions and projects.

A) USC Stevens Institute of Neuroimaging and Informatics: Multi-institutional Consortium [Vascular cognitive impairment and dementia (MarkVCID) (**UH2NS100614 & 4UH3NS100614: Wang, Ringman, Kashani)] and [Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (Asian VCID; 1RF1AG084072: PI Wang)].** The MarkVCID consortium; USC, Harvard, Johns Hopkins, Baylor, University of Kentucky, and Indiana University, identifies biomarkers related to the contribution of small vessel disease to cognitive impairment. **USC also focuses on developing novel retinal optical coherence tomography angiography (OCTA) measures and high field brain MRI methods,** including four novel MRI biomarkers: 1) 3D pCASL CBF, 2) high resolution black blood MRI of lenticulostriate arteries, 3) imaging water transport across the BBB, and 4) DTI along perivascular space (DTI-ALPS) (31). We focused on recruiting Latinx in the initial UH2 study, African Americans in UH3, and Asians in the recent RF1. The USC ADRC Clinical Core assists with subject recruitment; the PI has agreed to share UDS and imaging data with the ADRC database.

B) USC Stevens INI and USC ATRI: Health & Aging Brain among Latino Elders (**HABLE) (R01AG054073: PI Sid O'Bryant), Health & Aging Brain among Latino Elders-ATN (HABLE-ATN) (1R01AG058533: PI Sid O'Bryant), Health & Aging Brain Study-Health Disparities (HABS-HD) (1U19AG078109: PI Sid O'Bryant).** Together, Sid O'Bryant, Toga and Rissman share mPI leadership for HABLE, HABLE-ATN, and HABS-HD. The USC ADRC serves as an informatics hub and provides imaging/biomarker analysis. Numerous publications have elucidated cognitive phenotypes, cardiovascular risk factors, *APOE* ε4, WMH, and AD biomarkers (ATN) among Latino elders compared to non-Latino Whites (32-37).

C) USC School of Engineering - Multi-institutional Collaboration: Dynamic vasoreactivity (DVR) Study. Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression [R01AG058162: Marmarelis (USC), Chui (USC ADRC), Billinger (U Kansas ADRC), Zhang (UTSW)] incorporates principal dynamic modes to characterize hemodynamic responses to physiological changes in blood pressure that is impaired in early AD and MCI (38-40). Predictive dynamic modeling of cerebral hemodynamics expressed as a Composite Diagnostic Index differentiated the MCI/dementia due to AD from controls with high sensitivity and specificity (AAIC 2024). USC ADRC recruited 80 participants. UDS data, MRI, and amyloid PET results are shared in the ADRC database.

D) USC Stevens Institute of Neuroimaging and Informatics: Quantitative cerebral blood vessel imaging biomarkers for AD and VCID [1RF1AG072490; Yan LR (Northwestern University), Chui HC (USC ADRC)]. Goals: 1) develop and validate cerebral vascular biomarkers (VM) (e.g., arterial stiffness, pulsatility and morphology) in several arterial beds [leptomeningeal and lenticulostriate (LSA)], and 2) to correlate the severity and regional distribution of altered vascular biomarkers with changes in brain structure and function. Elevated carotid pulse-wave velocity (cPWV) was associated with cognitive impairment after controlling for age, gender, and years of education. All 40 participants were recruited by the ADRC Clinical Core.

Advance ADRC research through contributing USC generated expertise: In the last five years, we have produced over 400 peer-reviewed publications, participated in many meetings and review panels. We have served on the external advisory committees for other ADRC (OHSU-Chui; South Texas-Toga; NYU-Schneider, UAB-Yassine). Our expertise is evident in all traditional academic outlets. We have a large presence and present many papers at national meetings (ADNI, ATRI, ACTC, AD/PD, WDC, CTAD).

Innovation

(A) New recruitment and registry approaches to attract URM and *APOE* ε4 carriers. The USC ADRC created a unique and efficient recruitment portal called ***USC GeneScreen*** that includes a survey and saliva collection kit for *APOE* genotyping**.** The ADRC portal is building a population of *APOE* ε4 carriers across ethnicities representative of the LA area. It is available in both English (launched June 2023) and Spanish (launched October 2023). During the first six months, 482 surveys and 311 saliva kits were completed, yielding 18 homozygous, and 84 heterozygous *APOE* ε4 carriers. Among *APOE* ε4 carriers, 48% were Caucasian, 29% Latino, 7% African American, and 1% Asian. 29% of those carrying at least one copy of the *APOE* ε4 allele were enrolled into the ADRC. GeneScreen provides a highly effective method to sample participants by *APOE* genotype. The ORE Core of the USC ADRC is unique in that it is comprised of faculty and staff that are bicultural/bilingual Latinx and/or and sensitive to the relevant cultural issues.

(B) New Center for Personalized Brain Health (CPBH). The Center for Personalized Brain Health (CPBH) was launched in 2023 with a three-fold mission to: 1) conduct translational research to elucidate the pathophysiology of metabolic and inflammatory changes associated with *APOE* ε4 and develop novel strategies to mitigate their adverse effects, 2) to recruit and mentor early-stage physician-scientists, and 3) to support a clinic focused on personalized brain health for persons at an increased risk for AD. Recent research findings include insights related to associations between *APOE ε4* and pro-inflammatory effects of eicosanoids, small HDL particles, endosomal trafficking, cholesterol transport and cellular senescence.

(C) Administrative, Clinical and Imaging Cores: ALZ-NET. The USC ADRC is engaged in tracking real-world benefits and risk of anti-amyloid treatment within the ALZ-NET network. For patients who give consent, we share de-identified imaging in addition to clinical information with the ADRC and ALZ-NET databases. To date, we have initiated treatment with lecanemab in 38 patients and are working with the IRB and IT to put into effect appropriate informed consent and data transfer of serial MRI for ARIA surveillance.

(D) Pathology, Imaging, and Data Cores. (**1**) **Exceptionally Diverse Cohort.** The NP core includes 30% of autopsies from minority enrollees and more than 32% of the consented cohort are minorities compared to 9% of the NACC autopsies for which demographic information is available. 52% of our banked and genotyped cases have one or more *APOE* ε4 alleles versus 44% in the NACC dataset, and 29% of the UK Biobank autopsies. (**2**) **Additional tissues:** The Core harvests and banks additional CNS and non-CNS tissues not routinely collected by other sites (e.g. >150 fixed eyes, >500 frozen and ~ 1000 fixed spinal cord, and >500 cases from which frozen tissue is available on non-nervous system organs. (**3**) **Unique tissues:** We have a collection of brain and spinal

cord from 11 PSEN1 ADAD cases including six cases of the neuropathologically uncharacterized PSEN1 A431E mutation. (**4**) **Digitizing pathology:** Building upon previous work with animal models, we digitize pathology at a high throughput with a focus on *APOE ε4* /CAA. Whole slide images (WSIs) from each brain area will be collected from all deceased, consented ADRC participants to create an image database. We have an established workflow for hosting this type of data and will utilize a version of our iConnectome viewer (hosted by the USC INI) attached to the USC ADRC website and DB. (**5**) **Access:** With ongoing AD projects and the ADRC Data Core, this system will become a centralized nexus hosting both USC ADRC, and other project WSIs in addition to serving as an important resource to visualize neuropathology data from diverse patient populations.

(E) Biomarker Core: CSF and plasma assays. The ADRC will rely on NCRAD plasma Aβ (1-42, 1-40), pTau 181, pTau 217, Nfl and GFAP for its plasma measures of ATN. For our overarching scientific theme, the Biomarker will quantify HDL particle size (41), sPDGFRβ (marker of pericyte injury) (42), and placental growth factor (PlGF) (endothelial signaling in hypoxia) (43).

(F) Neuroimaging Core: Novel imaging. The USC ADRC has one of the few 7T Terra Siemens MRI machines available among all ADRCs. Novel imaging of the cerebrovascular system at 3T and 7T. The USC ADRC is focused on developing new imaging biomarkers that provide direct and specific information about the health and functional status of the cerebrovascular such as 1) BBB permeability using DCE-MRI(15), 2) water exchange across the BBB (44), 3) vascular compliance using ASL (45), 4) dynamic autonomic vascular reactivity, using end tidal CO_2 (38). Ultra-high field 7T MRI (Siemens Terra) allows resolution of smaller vessels and perivascular spaces (PVS). New sequences will provide better delineation of PVS and measurement of PWV and novel biomarkers related to glymphatic clearance and inflammatory effects of vascular amyloid.

(G) Real world data (RWD): The USC ADRC is working with KMUSC Information Technology to capture RWD from the EMR and PACS. Three projects are currently underway:

1. **Alzheimer's Network for Treatment and Diagnostics** (**ALZ-NET)**. The real-world research study ALZ-NET is a national effort, sponsored by the Alzheimer Association to study patients who are undergoing new FDA-approved clinical treatments for AD. At USC, we are currently treating 38 patients with lecanemab. For patients who give consent, clinical and imaging data will be de-identified and shared with ALZ-NET and with the USC ADRC Data/Imaging Cores. We will leverage our substantial clinical and imaging expertise to better delineate the role of *APOE* in CAA and risk for ARIA.

2. **Digital cognitive assessment**. In 2024, KMUSC signed a contract with Linus Health, Inc. for an AI-enhanced, quick digital cognitive assessment tool (FDA Class II Medical Device) built for primary care in consultation with memory specialists to enable earlier detection and intervention for MCI and to predict likelihood of AD pathology (see letter of support, Linus Health). Digital cognitive assessment data will be integrated into the KMUSC CERNER EMR, from whence it can be shared with the ADRC.

3. **CMS Guiding an Improved Dementia Experience (GUIDE).** GUIDE is an alternative payment model to test whether a comprehensive package of care coordination/management, caregiver support and education, and respite services can improve quality of life for people living with dementia and their caregivers. At CMS's invitation, KMUSC proposes to enroll to participate.

(H) Representativeness. Like many ADRCs, our ADRC clinical cohort was recruited largely as a convenience sample. Several ADRC-affiliated studies draw upon established, representative, community-based populations, [e.g., MarkVCID, the Los Angeles Latino Eye Study (LALES); ACAD, the Chinese Eye Study (CHES)]. We are developing a novel methodology to retrospectively characterize the Social and Environment Determinants of Health (SEDoH). Combined with GeneScreen and EMR RWD, this allows us to search by *APOE* genotype, SEDoH, digitized cognitive assessment, and EMR data.

(I) New partnership to enhance diverse training (REC/RCMAR). The REC core collaboration with the USC AD-RCMAR in terms of pilot funding and mentoring resources provides prospective REC mentees with a unique training opportunity that leverages multiple USC research and training entities. Given the ADRC and AD-RCMAR's focus on under-represented communities, this collaboration will strengthen efforts in supporting scholars from diverse backgrounds. The leveraging of other strong resources at USC for training and aging science also presents a unique aspect of the REC core at USC.

APPROACH

Approach Aim 1: Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4, focusing on factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers. Our goal is to build a multi-sourced, participant registry with *APOE* genotype, SEDoH, and digital cognitive assessment.

Building upon the considerable success of the past four years and more, and upon the solid foundation and commitment of the entire USC ADRC history we will continue our efforts to: 1) build unique cohorts that enable investigation of factors that influence the effect of *APOE* ε4, 2) further promote and support ADRD research at USC, and 3) contribute in unique and essential ways to the national landscape of AD research.

1.1 The USC-ADRC will recruit and follow a clinical cohort that reflects the ethnic diversity of our catchment area, enriched for *APOE* ε4 using our novel online USC GeneScreen. In addition to the variables collected in UDS4.0, we will gather additional lifestyle data and vascular, lipid, and metabolic biomarkers known to increase the risk of ADRD (**Fig. 6**). We will examine associations between putative risk factors (independent variables) on cognitive impairment (cross-sectional), decline (longitudinal follow up), and NCRAD ATN(v)i AD biomarkers (dependent variables), stratified by *APOE* genotype. The sample size of the ADRC clinical cohort is not powered for definitive conclusions, but may yield preliminary data for new grant applications with significantly larger sample size or novel gene by environment interactions. Going forward, we will geocode participants, and thereby obtain information about their SDoH.

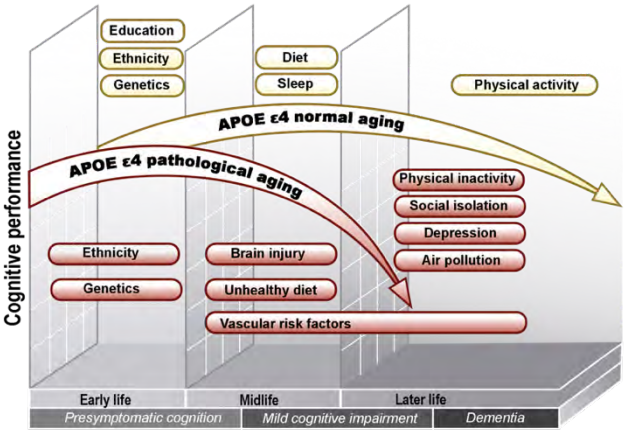


Figure 6: Candidate protective (gold) and risk factors (red) that may modify the effect of *APOE* ε4 on cognitive decline.

ADRC Core	Table 3: Aim 1 Factor modifying *APOE4* effects on cognitive decline and ATN Biomarkers
Admin	Cross Coordination across ADRC cores
Clinical	UDS 4.0 Blood, CSF collection NCRAD ATNvi biomarkers Fasting lipids, A1c, hsCRP
Data	NCRAD APOE genotype NCRAD AD biomarkers USC ADRC biomarkers
Path	Cerebral amyloid angiopathy Pathology ATNvi biomarkers
ORE	USC GeneScreen (*APOE*) Enhanced access to ADRC for pts n LA DHS
Biomarker	CSF ATN assays CSF and plasma Lipid particle Plasma ATN assays Plasma sPDGRFβ Plasma PlGF (Placental growth Factor)
Imaging	Blood pressure variability (BPV) Heart rate variability (HRV) BBB, CBF, WMH, PVS, PWV, CAA, ARIA, DVR

Each of the ADRC cores will contribute relevant variables to this initiative (**Table 3**), including UDS4.0, social determinants of health (SDoH), BP variability (BPV), a broad range of plasma biomarkers [NCRAD-ATN(vi) biomarkers], brain imaging [gray and white matter volumetrics, susceptibility weighted imaging (SWI) pulse wave velocity (PWV), water permeability (Kw)], perivascular spaces, and neuropathological ratings of cerebral amyloid angiopathy. We will select participants who represent a spectrum of *APOE* genotypes and SDoH for the USC ADRC Imaging database. We plan to acquire standardized MRI, amyloid PET and tau PET imaging, compliant with SCAN/CLARiTI for 120 participants imaged twice over two years (n=48 participants funded by the ADRC plus 72 participants funded by CLARiTI).

1.2 Recruitment based on diverse populations and electronic medical records. In the next five years we propose to **a)** expand the GeneScreen registry into the Latinx population, **b)** expand our recruitment efforts into both Keck and DHS, and **c)** refine tracking of recruitment efforts and outcomes (**Fig. 7**). We plan to establish a quality control and data harmonization pipeline to extract reliable and high-quality medical information from sources of RWD. The ADRC registry will be created and maintained on REDcap from consenting patients (Keck, DHS), community members, or past research participants. A subset will be invited to enroll in GeneScreen with the goal of enriching for *APOE ε4* carriers.

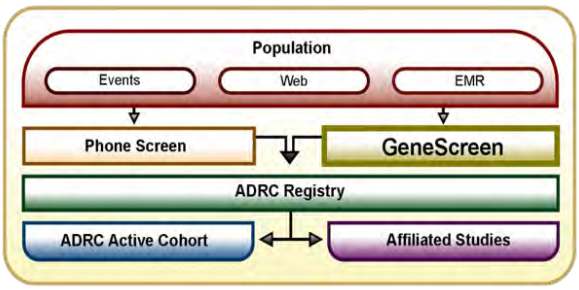


Figure 7. Expansion of the ADRC Registry.

Qualified investigators approved by the executive committee will result in participant communication informing them of a potential study opportunity and to obtain consent. Data will be shared with study investigators from consenting participants. Patients will be mailed recruitment letters, provided a website or phone number to join our registry, and may opt-out. We plan to recruit participants of low socioeconomic status based on the Social Vulnerability Index (SVI) and enhance representation of racial and ethnic minorities. For EMR data, our INI team will apply a novel weekly supervised learning approach called Gene-SGAN (46) to cluster the populations into several sub-populations, called AD subtypes. Our team has extensive research record applying advanced AI/ML to dissect disease heterogeneity in AD and aging, depression, and autism spectrum disorder (47)(48)(49).

In this cycle, Clinical core participants will use the Linus Health digital cognitive assessment platform that is being included into the Keck USC Medical Center's EMR (see letter from David Bates, CEO Linus Health). The platform integrates multimodal digital signals (i.e., fine motor control, drawing, voice, eye movements, gait, and balance) into validated AI and ML derived metrics that we will use for enhanced follow-up, screening for affiliated studies, and to combine with the UDSv4 and NACC data.

Approach Aim 2: Promote local ADRD research at USC. a) Integrate all USC programs that address ADRD. b) Share local tissue, biofluids, data, and participant registry through a common ADRC database. c) Train the next generation of ADRD researchers and educate the public. d) Enhance Diverse Perspectives

2.1) We will integrate USC programs through a weekly REC seminar, joint pilot fundings and annual events. The ADRC has created a USC-wide ADRD investigator registry that includes 728 faculty/trainees and mentees interested in ADRD (campus wide). The ADRC weekly REC seminars feature local research from several USC schools, guest speakers, and include case-conferences. Joint pilot fundings with RCMAR integrates diverse scholars with ADRC activities. Annually, the ADRC hosts the Finch Symposium, Zilkha Symposium, and Zach Hall Lecture. The ADRC participates in the annual USC ATRI/ACTC meeting (Aisen), RCMAR sponsored symposium "ADRD for Social Scientists" (Zissimopoulos), and the "Multidisciplinary Research in Gerontology Colloquium Series" (Cohen). The ADRC co-sponsors the annual So-Cal Alzheimer Center Symposium with UCI, USC, UCSD, UCLA and Cedars-Sinai. The frequency of seminars, topics, special issues, and ADRC support is listed under REC. We collect surveys after each event to assess representation, diversity and content.

2.2) Share local tissue, biofluids, data, and participant registry through a common ADRC database:

(a) Data. The DMSC will organize, annotate, and disseminate data in structured, accessible formats. We provide data collected anywhere within the ADRC umbrella so that those data can be used by others to answer new questions. We will store and share data to support reuse, and future unforeseen research endeavors. The number and range of AD and ADRC research projects and programs at USC is considerable. As these were investigator-initiated projects which then became linked with and dependent upon some of the resources and expertise of the ADRC, we have collectively and enthusiastically agreed to have all data from these efforts archived under a common data base. We will develop harmonization approaches to make data more comparable and to facilitate data aggregation of disparately collected data. We have experience doing this in ADSP-PHC and GAAIN projects. Common data elements will be utilized whenever possible or dictionaries to translate variables to equivalence.

(b) The **NP and biomarker core** will continue to provide fresh and frozen CNS and other tissue to a variety of investigators both inside and outside of the ADRC. NP Core will work with the DMSC to develop an accessible digital pathology library of our autopsy cohort that will later integrate with genetic and biomarker data from the Biomarker Core. Tissues and fluids will be catalogued in the **Biomarker and NP Cores,** but we will utilize application programming interfaces (APIs) to link Laboratory Information Management Systems (LIMS) to the database. This will allow tracking what is available, and what has been analyzed. We will develop this with a similar design to that used by ADNI, PPMI and other programs for which LONI serves as the informatics core.

(c) The **Imaging Core** will continue to work to establish protocol alignment with national programs and already have established **common MRI protocol** strategy where those studies who have unique pulse sequences as part of their protocols will still include those that facilitate data aggregation across studies. Common PET scan protocols are also being adopted to enable participation with national programs.

2.3) Train the next generation of ADRD scientists and educate the public. USC has six training grants directly relevant to the ADRC (Table 2a4**,** Admin Core). REC will provide training and mentoring of scientists in ADRD from all USC schools using a multidisciplinary empirical approach from basic science to clinical research. We will focus on developing and supporting scientists from diverse backgrounds in collaboration with the USC Alzheimer's Disease Resource Centers for Minority Aging Research (USC AD-RCMAR) (2P30AG043073; PI: Zissimopoulos) and national resources such as the NIA Bootcamp for REC trainees. REC will collaborate with the other cores and other USC training programs such as the RCMAR, the Clinical Psychology PhD program, and the Gerontology Training Program to develop individualized training. For educating the public, we hold the annual Batey Research Appreciation Day to celebrate and thank participants. The ADRC and CADC-Rancho sponsor a biannual "Many Faces of Dementia" conference for health care professionals. With USC Verdugo Hills Hospital, the ADRC organizes an annual Brain Health Day for the community**.**

2.4) Enhance diverse perspectives:

***PEDP-DHS*. Administrative, Clinical and ORE Cores.** Partner with the California Alzheimer Disease Center (CADC) and the LA County Department of Health Service (DHS) to enroll research participants from safety-net hospitals, focusing particularly on Latinx (PEDP-DHS) (see ORE Core)

***PEDP-RCMAR*. Administrative and REC core.** Partner with RCMAR to train investigators from under-represented groups (PEDP-RCMAR) and to fund development projects (see REC).

Approach Aim 3: Further national collaborations. Advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, comprehensive imaging to national initiatives.

3.1 Advance ADRC research through contributing USC-generated data. The USC ADRC will continue to contribute data to the NIA-ADC network, which includes over 35 ADC's and several centralized coordinating hubs: National Alzheimer Coordinating Center (NACC), National Centralized Repository for Alzheimer Disease (NCRAD), Alzheimer Disease Genetics Consortium (ADGC), Alzheimer Disease Neuroimaging Initiative (ADNI), Alzheimer Clinical Trials Consortium (ACTC), and Consortium for Clarity in ADRD Through Imaging (CLARiTI).

Most of these programs have strict protocol requirements and adherence is a prerequisite to upload to repositories (we are the informatics hub for several listed above). The data collected on our participants includes other data types and imaging that go beyond those required by these programs. Measures of lipid metabolism, vascular measures and other information collected at the USC ADRC will complement those uploaded to the national programs and will be available on the USC DB, linked via UDS number.

In addition, we play leadership roles in the following programs, ensuring our ability to further national collaborations: The Parkinson's Progression Markers Initiative (PPMI) to pinpoint biomarkers for Parkinson's disease progression, the Australian Imaging, Biomarkers and Lifestyle (AIBL), a longitudinal study investigating biomarkers and psychometric tools to observe AD progression. We are also part of the Longitudinal Early-onset AD Study (LEADS), an observational study to develop sensitive clinical and biomarker measures, and the ARTFL-LEFFTDS Longitudinal Frontotemporal Lobar Degeneration (ALLFTD) research study, which seeks to learn more about Frontotemporal Lobar Degeneration (FTLD) and understand the difference in brain changes due to FTLD compared to normal aging. The total number of projects where we are, in part or wholly, responsible for informatics is 156. **For these reasons, we are uniquely positioned to advance the missions of them all by capitalizing on the synergies between these and other studies and our ADRC**.

For those studies under the USC ADRC, requesters can apply for access, obtain permissions and have accounts automatically created with the philosophy that data is FAIR (50). Accounts enable logging, controlled access for reporting and for adherence to participant consent. The ADRC DB repository is a centralized and comprehensive informatics hub for well-described, searchable, and multimodal data, and provides authenticated researchers access to that data. Collectively, in other projects where we are responsible for informatics systems, access has been granted to over 160,000 active investigators from 169 countries supporting more than 420 million downloads of imaging, clinical, biomarker, and genetic data. We will create a web environment for ADRC projects, including the protocols, the overall design, data access request forms, Data Use Agreements (DUAs).

Derived data and return of results for linkage with raw data will be supported along with a formalized data provenance description and pointers to publications. Without adequate documentation and provenance including methods versions, parameters etc., the derived and results data are less than optimal. We will utilize similar approaches to those maintained as part of other national programs where we serve as informatics hub. One of the challenges with data aggregation is that participants can be enrolled in more than one study. This happens quite often. The USC ADRC is uniquely positioned to address this as we are the informatics hub of many of these projects.

Interoperability. Another challenge is interoperability, allowing data stores and systems to communicate with one another. We do this routinely and will continue connecting to other systems such as NCRAD, NIAGADS, ATRI (at USC), NACC and many others. Our systems synchronize data using a cadence that fits the needs of the project. We can create new APIs, as with the Biomarker Core LIMS, to function as a federated network.

Over the past years the USC ADRC Imaging Core has provided imaging protocols to the ADRC neuroimaging with emphasis on imaging of vascular pathology and the cerebrovascular system. Our efforts included the application of existing imaging protocols and the development of new imaging strategies to characterize and validate biomarkers of AD. In this ADRC phase, we will standardize the ADRC protocol to match the requirements of National Initiatives (SCAN, CLARitI) and support the development of new imaging strategies to characterize and validate biomarkers of AD with specific focus on ATN (vi) biomarkers and *APOE ε4*.

3.2 Advance ADRC research through contributing expertise. We will continue to support and innovate on **A)** imaging sequences focusing on vascular compliance and cerebrovascular risk. Such sequences include (1) ASL and ASL/blood-brain barrier imaging for measuring water-based blood brain barrier exchange and cerebral blood flow, (2) breath-hold induced CVR for measuring cerebrovascular reactivity, (3) T1-weighted structural imaging for perivascular spaces, and T2*-weighted images for measuring vessels and microhemorrhages, (4) 7T Ultra-high-resolution images, allowing delineation of structures susceptible to amyloid and tau pathology and smaller vessels (e.g., lenticulostriate) susceptible to vascular pathology and/or blood-brain barrier leakage, (5) ultra-high field spectroscopic sequences can also resolve neurotransmitter (glutamate, GABA) and inflammation (myo-inositol) information in regions with extensive amyloid and tau pathology. Each of these unique scans will often be on the same participants whose data are routinely included in other national programs (i.e., SCAN, CLARitI) where there are strict protocols. In this way we will enhance those national programs by extending the data collected with novel and important data types. **B**) ensuring proper imaging data provenance/dissemination, and **C**) providing analytics for all acquired images. MR-based and PET-based analytics will be obtained using automated pipelines. Derivative data will be used for A/T/N characterization of the ADRC cohort (A/T from amyloid/tau PET, N and v from MRI). This data will be archived in the ADRC DB repository for access and further analysis **D)** developing novel fluid biomarkers (HDL particles, eicosanoids, apoE glycosylation, exosomes, sPDGRFβ) among many others.

Overall impact. In the last five years, the USC ADRC had a significant impact on the ADRD community (**Admin Core Table 6**). USC currently has over 124 NIH funded grants focused on ADRD, of which 47 are supported by the ADRC. We have produced over 400 peer-reviewed publications, participated in many meetings and review panels. Our data core provided standardized, high-quality image processing and data on a national level. REC has created a USC-wide ADRD investigator registry that includes 728 faculty/trainees and mentees. The NP core updated >1,100 brain autopsies with 30% of autopsies from minority enrollees. The ORE core supported 255 events; reached 21,700 individuals (18,500 URG [60% Latinx]); and launched GeneScreen to enrich for *APOE* ε4. These efforts set us to accomplish our major goal of building a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4 with an emphasis on the factors that modify the effect of *APOE ε4* on cognitive decline and AD biomarkers. During the next cycle, we embrace new opportunities to integrate digital cognitive assessments and RWD obtained from EMR, move toward digitized pathology, utilize AI/ML for big data, among many other exciting initiatives. The impact of our ADRC is captured by how our data, tissues and information are disseminated, with the greater the number of downloads, the higher the publication count using these data are all accessible measures to assess impact. We have identified several opportunities to enhance and broaden our future impact through initiatives supported by cross center collaborations and more importantly through better integration of our teams and efforts.

Publications Reported for this Reporting Period

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In process at NIHMS	A call to address structural barriers to the representation of Hispanics/Latinos in clinical trials on Alzheimer's disease and related dementias: A micro-meso-macro perspective. Alzheimer's & dementia (New York, N. Y.). . NIHMSID: NIHMS1902891.
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MULTIPLE PI LEADERSHIP PLAN

The proposed Center represents a multi-disciplinary agenda including investigators from multiple departments and institutes across the USC Keck School of Medicine. The USC ADRC relies on a combination of expertise in Alzheimer's disease research, recruitment and retention of human subjects, data and biospecimen collection in humans, state-of-the art imaging and biomarker analysis, neuropathology, statistics and data analysis. A P30 center of this magnitude and with this number of investigators and participants requires leadership that embodies exceptional depth and breadth of knowledge, large-scale research leadership experience, and diverse resources. Such a project also requires a clear organizational system to ensure that each component meshes cleanly and effectively with the others. A rigorous system of leadership, organization, and oversight is absolutely necessary to ensure that the project does not become unwieldy. Utilizing a multiple PI leadership is, in our view, essential for the scientific and administrative goals of this project.

Each of the mPI, Associate Director, and leaders of the ADRC Cores is a highly experienced researcher, with outstanding support staff and resources, to enable them to accomplish the mission of this project effectively. Their proposed time and percentage effort are compatible with the responsibilities imposed by other obligations. Each of the PIs has available the time needed for commitment to this project. Together they form a strong leadership team, with established collaborative relationships between them and excellent rapport with the other key members of the research team.

Two mPI Leads – Chui -Toga

Helena Chui, MD is Professor and Chair of the Department of Neurology. Dr. Chui has worked in the field of dementia for over 40 years, specializing in clinical-imaging-pathological correlation of Alzheimer and cerebrovascular disease, and the relationship between vascular-metabolic risk factors and cognitive impairment. From 1995 to 2015, Dr. Chui was PI of the multi-institutional program project, "Aging Brain: Vasculature, Ischemia and Behavior" (P01-AG12435-19). She has served as the Principal Investigator of the ADRC (P50-AG05142) for 16 years, and as contact mPI for the ADRC (P30-AG066530) during the past 4 years. In this new cycle, Dr. Chui will continue in close collaboration with Dr. Toga (mPI) and with Dr. Yassine (Associate Director). She has worked closely with Toga, Yassine, as well as all Core leaders and affiliated investigators to establish the overarching scientific aims of the Center. Her expertise as a clinician and behavioral neurologist provides oversight to assure that the interface between the research participants is developed with integrity of all procedures and data collected, ethical treatment of all participants, and assurance that all data is made available to the scientific community. Dr. Chui will serve as contact mPI, organizing the ADRC Executive Committee meetings as well as the annual External Advisory Board meetings for reviewing the Center's progress. She is responsible for assuring that the Center's goals and objectives are met within the proposed timeframes and that additional objectives proposed by NIA and NACC are met as requested. Finally, she will be responsible with Dr. Toga for the fiscal and ethical integrity of the project operations, working with staff and University departments to assure that all requirements are being met in the operation of the Center. As a team, the combined expertise of Chui/Toga/Yassine as physician-scientist, neuro-computer-informatics scientist, and translational physician-scientist promise to guide the center into the next phase of growth and expansion of the USC ADRC research community.

Arthur Toga, PhD, is Provost Professor and Director of the Laboratory of Neuro Imaging (LONI), which developed the bioinformatics platform to identify biomarkers for the Alzheimer's Disease Neuroimaging Initiative (ADNI), as well as several other multicenter projects on Parkinson's disease, Huntington's disease, and other brain disorders. His formal training in neuroscience, computer science and informatics makes him not only well suited to lead both the Imaging and Data Cores, but also to serve as Co-PI/PD of this Center. Dr. Toga has a career-long funding and publication record as the director of several national and international collaborative studies that involve multi-site acquisition of data, coordination of analysis, and dissemination of data and results. He has also successfully recruited and mentored a vast number of young scientists, along with this leadership team, which is central to the aim of this Center. With this vast reach and well-earned respect as an experienced scientist, Dr. Toga will serve as an essential collaborator in leading the Center into the next five-year period with Drs. Chui. LONI has a very comprehensive administrative team with extensive experience managing very large, complex projects. LONI has been the lead on many multi-center U-type projects and has the necessary infrastructure and personal to insure a smooth and efficient operation from finances to science.

Governance and Organizational Structure

The PIs will be involved in all aspects of the project, scientific and administrative. Together, the PIs will have joint administrative responsibilities, including:

- Determining and coordinating research approaches & procedures and analyzing & interpreting research data
- Working closely with the project manager to ensure progress towards achieving all goals within 5 years
- Interacting with the External Advisory Board
- Meeting annual milestones for progress, and otherwise making satisfactory progress, toward accomplishing the P30 goals
- Overseeing the release of data and results according to the awardee's data sharing and release plan and consistent with NIH Resource Sharing Policy
- Releasing to the research community results, publications, data, models, and tools as soon as they have undergone basic quality control and quality assurance procedures in a time-delimited manner
- Providing information on progress of the Center to the program officer in a standard format on an annual basis, and as needed
- Planning for, facilitating, and participating in pre-arranged site visits by NIH staff members
- Participating in governance meetings and providing summary information of those meetings to the Program Officer
- Ensuring Core leads have access to administrative support, including financial and grants management, as provided by the Department of Neurology, Zilkha Neurogenetic Institute and Laboratory of Neuro Imaging

The two PIs (Drs. Chui [contact] and Toga), together, represent the forefront of research in every discipline needed to accomplish the immediate Specific Aims and long-term objectives of the ADRC. Both PIs share responsibility and authority for leading and directing the work within their components, as well as the overall center. The PIs are responsible for coordination among the NIA and Co-PIs and other investigators within the cores. The leadership of the ADRC and of the Cores can be changed according to the wishes of the MPIs, with the advice and consent of the Chair of the Internal Advisory Committee (Dr. Caleb Finch), Vice Dean for Research (Dr. Tom Buchanan) and the Dean of the Keck School of Medicine (Dr. Carolyn Meltzer).

In the unlikely event that conflict arises among the PIs, open discussions will occur between the two MPIs and should that fail to achieve resolution, the Chair of the Internal Advisory Board (Dr. Finch), Vice Dean for Research (Dr. Buchanan) and the Dean of the Keck School of Medicine (Dr. Meltzer) will also be brought in to facilitate mediation and resolution.

Associate Director - Yassine

Hussein Yassine, M.D. is a tenured Professor of Neurology, Medicine, and Physiology and Neuroscience, and Director of the new Center for Personalized Brain Health. He is an NIH-funded translational physician-scientist whose research combines imaging studies, clinical trials, observational cohorts and animal models with a focus on AD/APOE. During the past 10 years, Dr. Yassine has brought his deep knowledge related to apolipoproteins to bear on AD. Dr. Yassine is an inspirational leader and mentor, who currently leads the Research and Education (REC) core of USC ADRC. He chaired the national ADC REC steering committee (where he founded the mock study section boot camp). Dr. Yassine is the past co-chair of the Nutrition, Metabolism and Dementia Professional Interest Area (PIA) of the Alzheimer's Association and sits on the ISTAART Advisory council 2022-2024. He serves on NIH study sections and on NACC review panels. Dr. Yassine has immersed himself in all aspects of this competitive renewal. His contributions can be seen in our scientific theme, innovative recruitment approaches (e.g., GeneScreen, incorporation of the EHR as a source of real-world data), close interactions between cores, weekly REC seminars and annual symposia. During the next funding cycle, he will be considered as a successor.

Consortium/Contractual Arrangements – Children's Hospital of Los Angeles

The University of Southern California (USC) and Children's Hospital of Los Angeles (CHLA) will enter into a contractual agreement to carry out the tasks related to the proposed research. The majority of the research staff will be employed by USC and the MPI (Dr. Helena Chui and Dr. Arthur Toga) will be responsible for the overall scientific integrity of the ADRC. A formalized agreement whereby the specific level of effort from both consortium organizations and a categorical breakdown of costs such as personnel and other allowable expenses including F&A will be executed. Both organizations have experience in the coordination of multiple collaborators.

Dr. Debra Hawes is the consortium site PI responsible for ensuring proper conduct of the project at the consortium site of Children's Hospital Los Angeles.

The appropriate programmatic and administrative personnel of each organization involved in this grant application are aware of the NIH consortium agreement policy and are prepared to establish the necessary inter-organizational agreement(s) consistent with that policy. The Notice of Grant Award will further outline the consortium lead investigator responsible for ensuring proper conduct of the project or program at the consortium site. Further, we understand and acknowledge that NIH may request additional information before the award and may place a special condition(s) on the award.

Consortium/Contractual Arrangements – Rancho Research Institute

The University of Southern California (USC) and Rancho Research Institute (RRI) will enter into a contractual agreement to carry out the tasks related to the proposed research. The majority of the research staff will be employed by USC and the MPI (Dr. Helena Chui and Dr. Arthur Toga) will be responsible for the overall scientific integrity of the ADRC. A formalized agreement whereby the specific level of effort from both consortium organizations and a categorical breakdown of costs such as personnel and other allowable expenses including F&A will be executed. Both organizations have experience in the coordination of multiple collaborators.

Dr. Freddi Segal-Gidan is the consortium site PI responsible for ensuring proper conduct of the project at the consortium site of RRI. RRI will serve as a recruitment site for the ADRC.

The appropriate programmatic and administrative personnel of each organization involved in this grant application are aware of the NIH consortium agreement policy and are prepared to establish the necessary inter-organizational agreement(s) consistent with that policy. The Notice of Grant Award will further outline the consortium lead investigator responsible for ensuring proper conduct of the project or program at the consortium site. Further, we understand and acknowledge that NIH may request additional information before the award and may place a special condition(s) on the award.

LETTERS OF SUPPORT

Letter	Supporter	ADM	CLIN	DMS	PATH	ORE	BioM	IMG	REC
USC Senior Vice President Office of Research & Innovation USC Senior Vice President for Health Affairs USC Dean Keck School of Medicine	Ishwar Puri PhD Steven Shapiro MD Carolyn Meltzer MD	x	x	x	x	x	x	x	x
USC Associate Vice Dean for Strategic Alliances Co-Director Southern California Clinical Translational Science Institute (CTSI)	Michelle Kipke PhD	x	x	x		x			x
CEO Linus Health	David Bates PhD	x	x	x					x
PI, Health & Aging Brain Latino Elders (HABLE) Director, Institute for Translational Research University of North Texas	Sid O'Bryant PhD	x	x			x	x	x	x
Chief, Section of Neuropathology (Emeritus) University of California, Los Angeles (UCLA)	Harry Vinters, MD		x		x				
Director, Neurological Services Los Angeles County, Department of Health Services (LA-DHS)	Amytis Towfighi MD	x	x			x			
Director, Dementia Care Aware (DCA) Chair, Department of Family Medicine Harbor-UCLA Medical Center (LA-DHS)	Heather Schickedanz MD	x				x			
Chair, Community Advisory Board (CAB)	Nora Manzanilla	x	x			x			
Member, Community Advisory Board (CAB)	Steven Galindo	x	x			x			
Member, Community Advisory Board (CAB)	Kimberly Whyte	x	x			x			
Member, Community Advisory Board (CAB)	Ryo Fujita	x	x			x			

USC University of Southern California

RESEARCH & INNOVATION
Ishwar K. Puri, Ph.D.
Senior Vice President,
Research & Innovation
vpres@usc.edu

May 31, 2024

Cerise Elliott, Ph.D.
National Institute on Aging (NIA)
Telephone: 301-496-9350
Email: elliottce@mail.nih.gov

Nina B. Silverberg, Ph.D.
National Institute on Aging (NIA)
Telephone: 301-496-9350
Email: silverbergn@mail.nih.gov

Dear Drs. Elliot and Silverberg:

On behalf of the University of Southern California (USC), Ishwar Puri, Steve Shapiro, and Carolyn Meltzer, we are thrilled to express our institutional support for the renewal application for the USC Alzheimer's Disease Research Center. This application, a testament to USC's unwavering commitment to Alzheimer's disease research, is being submitted in response to the NIA solicitation, RFA-AG-21-019, Alzheimer's Disease Research Centers (ADRC).

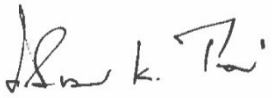
USC's ADRC, a pioneer in the field, is celebrating its 40th anniversary, a significant milestone that has contributed immensely to Alzheimer's Disease and Related Dementias (ADRD) research, interventions, and education. The proposed renewal, led by Drs. Helena Chui and Arthur Toga, will focus on characterizing factors that modify the effect of APOE e4 on cognitive decline and ATN(vi) biomarkers, with a special emphasis on vascular, lipid, inflammatory, genetic, and ethnic influences. Over the next five years, the USC ADRC will concentrate on three overarching aims: recruiting and following a clinical cohort that reflects the ethnic diversity of Los Angeles with the risk factor APOE e4, promoting local ADRD research at USC, and fostering further national collaborations such as the Clinical and Translational Science Awards (CTSA) Program, Alzheimer's Disease Neuroimaging Initiative (ADNI), or Alzheimer's Clinical Trials Consortium (ACTC). USC's ADRC will also continue to explore vascular and metabolic risk factors (VMRF), the neurovascular unit, brain structure and function, and the treatment and prevention of Alzheimer's disease (AD).

The USC ADRC is the cornerstone of USC's research efforts, and Dr. Chui's leadership since 2003 has been instrumental to its success. We are all thoroughly impressed by USC's significant ADRC milestones and synergistic collaborations. During the past 30 years, Dr. Chui's research interests have focused on aging, Alzheimer disease, and vascular contributions to cognitive impairment in late life. For the past 20 years, she served as principal investigator of the Aging Brain Program Project, a multi-institutional longitudinal study to characterize interactions between vascular and Alzheimer disease, using state-of-the-art clinical-imaging-pathological correlations. Her team is particularly interested in non-occlusive pathways leading from vascular risk factors to vascular brain injury or AD pathology. Recently, they showed that low HDL and high HDL-C are independently correlated with β-amyloid deposition in the brain and that HDL-C may have a protective effect against brain atrophy. During the past 10 years, Dr. Chui has served as Director of the Alzheimer Disease Research Center and Chair of the Department of Neurology at USC. Dr. Toga has a career-long record of funding and publication in the analysis, modeling and atlasing of images obtained from many species including humans. Dr. Toga directs several national and international collaborative studies that involve multisite acquisition of data, coordination of analysis, and dissemination of data and results. The USC Institute for Neuroimaging and Informatics (INI) and his laboratory coordinates neuroimage analysis and related informatics activities with national and international collaborators. It is a worldwide resource by virtue of these distributive activities. Dr. Toga has experience and expertise in local and wide area computer networks, data transmission, storage and archiving as well as data compression and scientific visualization.

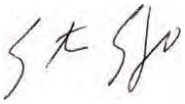
We have met with Drs. Helena Chui and Arthur Toga, and we have discussed several times how they will further improve the ADRC infrastructure, the new proposed ADRC aims, future activities for clinicians and scientists interested in ADRD, the recruitment of clinicians and scientists interested in ADRD, and the planned collaborations with other NIH-funded programs, such as the Clinical and Translational Science Awards (CTSA) Program, Alzheimer's Disease Neuroimaging Initiative (ADNI), or Alzheimer's Clinical Trials Consortium (ACTC). Drs. Helena Chui and Arthur Toga emphasize that these revisions, new incorporations, and planned collaborations between the various teams in different institutions lend themselves to a truly collegial and comprehensive program, which is precisely the outcome the USC ADRC always seeks to achieve. We, Ishwar Puri, Steve Shapiro, and Carolyn Meltzer, look forward to another five years of the ADRC's research productivity and to the contributions it will make to the research community.

To help the ADRC's mission move forward and provide tangible evidence of USC's support, the university is making a total of $2 million in institutional financial commitment to the ADRC. This commitment will foster innovation and empower young investigators. It is a testament to USC's steadfast dedication to excellence in neuroscience and fostering a nurturing environment for its investigators in ADRD. We sincerely appreciate the NIH's consideration of the exceptional proposal team, their efforts thus far, and the renewal of the ADRC, which is instrumental in our continued success in expanding USC ADRC's research focuses, interventions, and education aims.

Sincerely,



Ishwar K. Puri, PhD
Senior Vice President
Research & Innovation
Professor of Aerospace & Mechanical Engineering
University of Southern California



Steve Shapiro, MD
Senior Vice President for Health Affairs
University of Southern California



Carolyn Meltzer, MD
Dean, Keck School of Medicine
University of Southern California

USC University of Southern California

Michele D. Kipke, Ph.D.
Associate Vice President for Strategic Alliances
Office of the Senior Vice President of Health Affairs
University of Southern California
Professor of Pediatrics, Keck School of Medicine of USC

May 17, 2024

Dear Drs. Chui, Yassine, Aranda and Ringman,

As an Associate Vice President for Strategic Health Alliances and co-director of USC's Southern California Clinical and Translational Science Institute (SC CTSI), I am excited to partner with the USC Alzheimer's Disease Research Center's (ADRC). Our Office of Health Affairs is actively providing support and is poised to continue to support your Center's objectives under USC President Carol Folt's *Transforming Health Sciences* Moonshot initiative focused on interdisciplinary health sciences, healthy aging, and precision longevity. The USC ADRC and the newly founded *Center for Personalized Brain Health* are going to be critical to this initiative which will engage our health sciences schools (medicine, gerontology, pharmacy, dentistry, occupational sciences, biokinesiology and physical therapy, and social work) and our health system, Keck Medicine of USC.

I have facilitated interactions with our information technology office, led by Neil Bahroos, to boost the activities of your novel program GeneScreen. By having the ability to reach over 1 million active patients within Keck Medicine of USC's health system, there are real opportunities to engage our patient populations with dementia research. Your focus on subgroups such as APOE4 carriers aligns with our Precision Medicine Initiative.

We introduced you to USC's Spatial Sciences Institute which provided you with geospatial data that can be utilized in community-based and clinical dementia research to target high vulnerability areas and enhance diversity in research.

I am particularly excited about your *Outreach, Recruitment and Engagement Core*, which will be co-led by Dr. Maria Aranda, who is an expert in community-based research specifically addressing the health and social needs of marginalized groups in our local area. I have had the opportunity to work with Dr. Aranda in the USC Suzanne-Dworak Peck School of Social Work and look forward to co-sponsoring a community event on brain health and dissemination of ADRC findings to the public. Our SC CTSI has a very strong Community Engagement Core, with over 250 community partners on the Eastside, Central, and in South Los Angeles. We look forward to serving as a resource to support community events, such as the brain health event, and to assist with dissemination of ADRC findings to lay audiences in our local communities.

In summary, I am enthusiastic about your proposal and excited about our continued work together.

Best wishes,



Michele D. Kipke, PhD

May 29, 2024

Helena Chui MD
McCarron Professor and Chair
Department of Neurology
mPI, USC Alzheimer Disease Research Center
Keck School of Medicine
University of Southern California

Lon Schneider, MD, MS
Dela Martin Professor
Department of Psychiatry and Neurology
Lead, Clinical Core, USC ADRC
Keck School of Medicine
University of Southern California

Re: Digital Cognitive Assessment
Linus Health, Inc.

Dear Drs. Chui and Schneider:

Linus Health is pleased to partner with Keck Medicine of USC (KMUSC) and the USC Alzheimer Disease Research Center (ADRC) in bringing Digital Cognitive Assessment to real-world practice settings in primary and specialty care. Our AI-enhanced, quick digital cognitive assessment tool is FD- approved as a Class II Medical Device to improve early detection of cognitive impairment and differential diagnosis of underlying dementia etiology.

Evidence-based studies of Linus AD (3 min digital clock drawing test (DCT test) and Linus AD (6 min test ensemble) yield AUC = 0.89 compared to standard cognitive screening tests (e.g., MMSE, MoCA, RBANS, RAVLT) and AUC = 0,86 compared to β-amyloid PET imaging. Diagnostic accuracy is further improved with the addition of plasma AD biomarkers.

KMUSC is now licensed to use Linus AD Digital Cognitive Assessment (DCA) Tools. Keck is preparing to roll-out LinusAD (3 min) in Family Medicine and Neurology clinics. We are working with USC information technologists to integrate DCA data into the Keck Electronic Medical Record (EMR). For patients who have given consent, data can also be de-identified and exported to the ADRC Database, where it provides valuable resources for investigations using real world data.

Our team at Linus Health is delighted to assist you and your colleagues as you strive to expand the ADRC participant registry to Real-World Data (RWD) obtained from the healthcare EMR. Please don't hesitate to call on me or my team at any time.

Best wishes for the renewal of the USC ADRC.

Sincerely yours,

David J Bates

David Bates, PhD
CEO, Linus Health, Inc.

cc: David Peng, M.D. Chief Clinical Officer, Keck Medicine of USC

Sid E. O'Bryant, PhD
Dr. Joe and Peggy Schooler Endowed Chair
Executive Director, Institute for Translational Research
Department of Family Medicine
Texas College of Osteopathic Medicine
University of North Texas Health Science Center
Phone: (817) 735-2961
Fax: (817) 735-5071
sid.obryant@unthsc.edu

May 17, 2024

Helena Chui, MD
Arthur Toga, PhD
Lon Schneider, MD
Hussein Yassine, MD
Maria P. Aranda, PhD

RE: Consultant Role, University of Southern California, Alzheimer's Disease Research Center, P30 Renewal

Dear Drs. Chui, Toga, Schneider, Yassine, and Aranda,

It is with great enthusiasm I write this letter of support for the renewal of the Alzheimer's Disease Research Center (ADRC, P30) at the University of Southern California (USC), and agree to participate as a Consultant.

I am the Executive Director of the Institute for Translational Research and my lab is dedicated to performing clinical and biomarker research Alzheimer's disease and other neurodegenerative diseases, including Down syndrome, Lewy Body disease, Parkinson's disease, traumatic brain injury, and others. I focus on novel strategies for disease detection, screening into trials (therapeutic and prevention), and patient stratification for optimal treatment response.

As part of this work, my lab has a strong focus on the impact of ethnicity/diversity on cognitive loss during the aging process and runs the one-of-a-kind Health & Aging Brain among Latino Elders (HABLE) study, which is the most comprehensive study of Mexican-American brain aging to date. Our HABLE study currently collaborates with Arthur Toga, the USC's Mary and Mark Stevens Neuroimaging and Informatics Institute (INI) and the Laboratory of Neuro Imaging (LONI) to perform cognitive tests, blood work and brain scans on a diverse cohort of Mexican-American and non-Hispanic White people over a five-year period to monitor changes in health and behavior over time. INI/LONI are responsible for storing, processing, and analyzing 4,000 brain scans collected during the study.

In our current project, HABS-HD, we mounted the first large-scale, community-based project to study each of the AT(N) defined biomarkers in alignment with the NIA Health Disparities Research Framework using our network of community ties and relationships for recruitment. This approach, which emphasized outreach, engagement, and collaboration between the research team and key stakeholders in the community, led to the successful recruitment of over 3,000 participants (n=1,000 NHW, 1,000 MA, 1000 AA) aged 50 and older in the longitudinal study. Given our mutual interests in increasing awareness and engagement of diverse communities into research and to provide the outreach infrastructure to comprehensively understand the development and progression of AD and ADRDs across the life course, I believe I would be a key contributor to the ORE Core's objectives.

Sid E. O'Bryant, PhD
Dr. Joe and Peggy Schooler Endowed Chair
Executive Director, Institute for Translational Research
Department of Family Medicine
Texas College of Osteopathic Medicine
University of North Texas Health Science Center
Phone: (817) 735-2961
Fax: (817) 735-5071
sid.obryant@unthsc.edu

My involvement with your Center extends our existing relationships in several meaningful ways. First, our U19 HABS project is co-led by USC MPIs, O'Bryant, Toga, and Rissman. Second, Dr. Robert Rissman's lab at USC Alzheimer's Therapeutic Research Institute (ATRI) biobanks LONI specimens and conducts analyses. Third, Dr. Maria Aranda (ORE Core co-lead) and I serve on the UCSD/SDSU AD-Research Center on Minority Aging Research which targets Hispanic and immigrant populations. Last, I was one of the speakers for the REC 2024 Speaker Series led by Dr. Hussien Yassine.

As the goals of the USC ADRC overlap with those of my current projects, I am happy to lend this expertise, consulting with Drs. Aranda, Ringman, Rissman, Yassine, and other faculty and staff of the ADRC. Key topic areas to discuss are how might the Center optimize the recruitment and retention of Hispanics in their Center activities by creating a sustainable organizational infrastructure to enhance ORE Core aims, diagnosing cognitive impairment and dementia, elucidating the genetic architecture including the contribution of *APOE* to cognitive health, and the nature of biomarker changes in this growing population. I will also use my data and experience in the HABS project to advise on how the Center can best leverage their Clinical and Biomarker Cores to make insights into the nature of plasma and CSF biomarker changes related to cognition and AD/ADRD in their Hispanic population.

I wish you success in your renewal application.

Sincerely,



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Harry V. Vinters, MD,FRCPC,FCAP
Chief, Section of Neuropathology (Emeritus)
Distinguished Professor of Pathology and Laboratory Medicine & Neurology (Emeritus)
Formerly Daljit S. & Elaine Sarkaria Chair in Diagnostic Medicine 2005-2010
Editor Emeritus, "Brain Pathology" (2000-2006)

DEPARTMENT OF PATHOLOGY & LABORATORY MEDICINE
SECTION OF NEUROPATHOLOGY
DAVID GEFFEN SCHOOL OF MEDICINE AT UCLA
ROOM 18-170 NPI MC : 173216
10833 LE CONTE AVENUE
LOS ANGELES, CALIFORNIA 90095-1732
PHONE: (310) 825-6191 FAX: (310) 206-8290
HVINTERS@MEDNET.UCLA.EDU

May 14, 2024

Dr. Helena C. Chui,
Director, USC Alzheimer Disease Center,
Keck School of Medicine at USC,
Los Angeles, CA

RE: ***Renewal application for the USC ADRC P30 Grant***

Dear Helena,

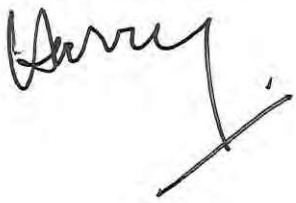
It is with great enthusiasm that I write this letter of support for the renewal application of your USC ADRC P30 Grant. As Distinguished Professor (Emeritus) of Pathology and Laboratory Medicine and Neurology at UCLA, it has been a pleasure collaborating with you and your colleagues at USC for decades now. As you remember, I was the Pathology Core leader of the multi-center study, the "***Aging Brain: Ischemia, Vasculature, and Behavior***" (P01-AG12435-15) on which you were the principal investigator at USC. Over the years we have collaborated in many other ways including, but not limited to, assisting your Pathology Core (then under the leadership of Carol Miller) with diagnostically challenging cases and holding joint rounds and case presentations between our institutions. Several highly cited publications resulted from our work together. We have ongoing collaborations with Drs. Ringman (formerly of UCLA) and Anne Hiniker, M.D., Ph.D, on autosomal dominant Alzheimer's disease, capitalizing on the availability of these rare cases having been collected for decades at both UCLA and USC. I am very encouraged by your successful recruitment of Dr. Hiniker to lead your Pathology Core and look forward to increasingly working with her.

As you know, the study of cerebral amyloid angiopathy (CAA) has been a research and clinical focus of mine and as such, I have become an internationally recognized leader in this field. In the new era of anti-amyloid antibody treatment, CAA is becoming an increasingly important topic in light of its association with Amyloid-Related Imaging Abnormalities (ARIA). There remain many questions to be answered regarding this association, the answers to which are likely to directly impact patient care. You and your Center's demonstrated expertise on vascular contributions to dementia, its new focus on *APOE*, its clinical trial and accumulating clinical experience with anti-amyloid therapies, and the existing infrastructure and new leadership of your Pathology Core place the USC ADRC in an excellent position to make important inroads in this critical area.

In forthcoming years, it will be my pleasure to continue working with your Center and Pathology Core as a consultant. In this role I look forward to advising Dr. Hiniker and other interested investigators at USC on how to optimally assess and quantify CAA and associated pathological changes. I also welcome the opportunity to continue providing input when requested on ways to approach difficult cases and for having joint interinstitutional Clinical Pathological Correlation (CPC) conferences as appropriate.

I wish you good luck in your application in these (scientifically !) exciting times! I'm not even sure you need 'luck' when you work with such a talented and energetic team of clinicians, scientists and investigators.

Sincerely,



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May 2, 2024

Helena Chui MD
McCarron Professor and Chair of Neurology
USC Keck School of Medicine
Director, USC Alzheimer Disease Research Center

RE: Collaboration with the USC ADRC

Dear Dr Chui,

It is with great enthusiasm that I write this letter of support for the application of the USC Alzheimer's Disease Research Center (ADRC) for continued funding. The USC ADRC has an extensive track record of ground- breaking research and its continued and growing focus on dementia among minoritized and socioeconomically disadvantaged populations parallels my own research and efforts in this area. As you know, I am the Director of Neurological Services for the Los Angeles County (LAC) Department of Health Services (DHS) and Chief of Neurology at the Los Angeles General Medical Center and have published extensively on gender, racial and ethnic, and socioeconomic health disparities in neurological disease, particularly with regards to cerebrovascular disease. I am also Director of the Healthcare Delivery Science core for the SC Clinical and Translational Science Institute.

In recent years, it has been a pleasure to serve as the primary mentor for Elizabeth Joe, MD, as a scholar through USC's Mentored Career Development in Clinical and Translational Science/KL-2 program. In addition to Liz's ongoing work with the ADRC's clinical core, she is deeply focused on disparities in dementia care within DHS from both a research and clinical perspectives. I am encouraged that she will expand her work to include the Geriatric Neurobehavior Clinic at Rancho Los Amigos Rehabilitation Center, a California Alzheimer's Disease Center (CADC), which was established in 1985 and provides a tertiary memory-clinic evaluation to people with dementia within DHS.

In my position, I am keenly aware of the ongoing disparities in dementia care in LAC and the barriers to timely diagnosis and treatment, particularly among Latinx persons, who comprise 49% of the population. Facilitating optimal diagnosis and treatment in this population is a priority for me and I recognize there are tremendous gaps in our knowledge

regarding the clinical and cognitive presentation of, genetics, and treatment responses in groups heretofore underrepresented in research. I am very excited by the newly established Dementia Care Aware Program, the goal of which is to increase identification of cognitive impairment in persons over 65 in primary care clinics in DHS.

In my role as Chief of Neurological Services for LAG, I look forward to an increased partnership with the USC ADRC and will support efforts to streamline the appropriate recruitment of DHS patients for potential research participation.

Sincerely,



Amytis Towfighi, MD
Professor of Neurology and Population and Public Health Sciences (Clinical Scholar)
James and Dorothy Williams Scholar
Keck School of Medicine, University of Southern California

Director of Neurological Services
Los Angeles County Department of Health Services
Associate Medical Director of Research
Chief, Neurology, Los Angeles General Medical Center

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Vice Chair, Clinical Affairs

Karen Olmos, MD MPH
Residency Program Director

Fathima Sarah Nazarkhan, MD
Bernadette Pendergraph, MD
Jyoti Puvvula, MD MPH
Associate Residency Program Directors

Claudia Alvarez, DO
Daniel B Castro, MD
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S. Diana Garcia, MD MPH
Gilberto Granados, MD MPH
Jesus Gutierrez, MD
James Huang, MD MPH
Amber Lara, DO
Robyn Mitchell, MD
Nahid Molaie, MD
Theresa Nevarez MD, MBA
Gloria Sanchez, MD
Faculty

Department of Family Medicine
1403 W Lomita Blvd., Suite 102
Harbor City, CA 90710
Tel: (310) 602-2550
Fax: (310) 326-7205

LETTER OF SUPPORT

RE: Collaboration with the USC ADRC

Dear Dr. Helena Chui,

It is with great enthusiasm that I write this letter supporting the application of the USC Alzheimer's Disease Research Center for continued funding. As Chair of the Department of Family Medicine at the Harbor-UCLA Medical Center and a Family Physician trained Geriatrician, I share many interests with you and the ADRC including optimizing the diagnosis and care for persons affected with cognitive impairment and dementia. Cognitive impairment and dementia is underdiagnosed in the Los Angeles County Department of Health Services safety net system. It has been a pleasure working with Drs. Freddi Segal-Gidan and Ellen Wong, colleagues with the USC-Rancho California Alzheimers Disease Center on the Dementia Care Aware (DCA) program. The goal of DCA, as you know, is to increase the identification of early cognitive changes, through routine screening in primary care. This will enable early evaluation and diagnosis, and, when necessary, referral to neurology specialty dementia clinics, such as the one at Rancho Los Amigos National Rehabilitation Hospital that is supported by the USC-Rancho California Alzheimers Disease Center. To date we have launched cognitive health screenings through DCA in four separate clinics, and 260 of 817 persons screened (32%) screened positive for cognitive impairment, meriting further assessment. And as of July 1, 2024 the LA County Dept of Health Services will be incorporating annual cognitive screening of all adults over age 65 as part of routine preventive care practice in all primary care clinics.

I appreciate the ADRC's goals of increasing our understanding of the clinical aspects and pathophysiology of dementia, and in particular, its focus on populations typically under-represented in research such as the Latinx population served by the LA County DHS clinics and hospitals. Latinx individuals tend to present relatively late in the course of their illness, if at all, and therefore they are a population well-suited to benefit from the institution of the DCA program. To improve health equity, I am supportive of the effort to facilitate referrals of persons screening positive for cognitive impairment for specialized clinical care and research opportunities at the USC-Rancho California Alzheimer's Disease Center directly from primary care clinics appropriate. These efforts should facilitate access to Spanish-language neuropsychological testing as well as state-of-the art diagnostic procedures such as blood biomarkers, amyloid- and tau-PET scans, and genetic testing as appropriate which would otherwise not be available to this patient population.

Please do not hesitate to contact me with questions. I look forward to our expanding work together!

Sincerely,



Heather Bennett Schickedanz, MD
Chair, Department of Family Medicine Harbor-UCLA
Associate Professor, UCLA David Geffen School of Medicine
Office: 310-602-2550 // Fax: 310-326-7205 // Pager: 310-501-0518
Email: HSchickedanz@dhs.lacounty.gov

May 8, 2024

Maria P. Aranda, PhD, Executive Director, USC Roybal Institute on Aging

Co-Lead, Outreach-Recruitment-Engagement Core

USC Alzheimer's Disease Research Cener

Department of Neurology, Keck School of Medicine of USC

Los Angeles, CA 90033

Dear Dr. Aranda,

I am honored to serve as Chair of the USC Roybal Institute on Aging Community Advisory Board which works closely with the USC Alzheimer's Disease Research Center. Our board installation last month at the bilingual USC Southern California Caregiver Celebration event in Los Angeles was inspiring and reflective of the important work that we need to do in the community on cognitive health, aging research, family caregiving, and research participation.

I will work closely with you, other advisory board members, and your colleagues on your various research projects to enhance the wellbeing of families affected by memory loss. **Thus, I support your research proposal to renew the USC Alzheimer's Disease Research Center for the next 5 years.** I truly believe that the ADRC is well-positioned to make a difference in people's lives by preventing or delaying cognitive decline and treating memory loss in our local communities. Focusing on people most at risk such as the Hispanic population is needed, and very much understudied even with all the advances we see in the news.

Your scientific team is on the cusp of making a difference, therefore, I am excited to serve in this advisory role to the ADRC. I understand that this role entails addressing how best to reach community people to let them know about the latest scientific advances in cognitive research, the center's great studies, and study participation opportunities. Unfortunately, Hispanics account for a very small percentage of clinical trials and participatory research. The

research proposal is vital because it helps identify factors that contribute to the development of effective strategies to address health disparities in the Hispanic population.

I fully support the research proposal because you and your team have a proven track record.

I provided around-the-clock care for my dear mother, Gladys, who was a patient of your neurology clinic, and diagnosed with dementia. I also am mother to my teenage daughter who assisted me in caregiving for my mother during her later years. I know first-hand what families experience in the face of dementia especially family caregivers who are also employed and need to balance caring and providing for my family. I juggled being the primary caregiver for my mother, managing caregivers/scheduling and a demanding job while raising my daughter. The strain on my time, financial and mental health impacted my life.

I consider the grant proposal a priority as it will provide resources that are desperately needed.

As Chair of the advisory board, I understand that I will meet with your ADRC outreach and scientific team under Dr. Aranda's leadership about 6 times per year (around 2 hours per meeting) over 5 years. I will receive remuneration at the rate of $100 per meeting.

In closing, I wish you the best in your grant submission. Let me know if there is anything I can do to help.

Best wishes,

Nora Manzanilla

Nora Manzanilla

206 S. Nicholson Ave.
Monterey Park, CA 91755
nora.manzanilla@lacity.org
(562) 397-0348 Mobile

May 10, 2024

Maria P. Aranda, PhD, Executive Director, USC Roybal Institute on Aging
Co-Lead, Outreach-Recruitment-Engagement Core
USC Alzheimer's Disease Research Center
Department of Neurology, Keck School of Medicine of USC
Los Angeles, CA 90033

Dear Dr. Aranda,

I am honored to serve as member of the USC Roybal Institute on Aging Community Advisory Board which works closely with the USC Alzheimer's Disease Research Center. Our board installation last month at the bilingual USC Southern California Caregiver Celebration event in Los Angeles was inspiring and reflective of the important work that we need to do in the community on cognitive health, aging research, family caregiving, and research participation.

I work closely with you, other advisory board members, and your colleagues on your various research projects to enhance the wellbeing of families affected by memory loss. **Thus, I support your research proposal to renew the USC Alzheimer's Disease Research Center for the next 5 years.** I truly believe that the ADRC is well-positioned to make a difference in people's lives by preventing or delaying cognitive decline, and treating memory loss in our local communities. Focusing on people most at risk such as the Hispanic population is needed, and very much understudied even with all the advances we see in the news.

I am a businessman located in the San Gabriel Valley, specifically, I am a real estate specialist with a unique sales credential called "Seniors Real Estate Specialist" that allows me to navigate the complexities of real estate transactions related to older adults with physical and cognitive vulnerabilities. My mission includes helping seniors and their families navigate the maze of financial, legal, and emotional issues that accompany the sale of their home to transition to congregate living arrangements. I prioritize senior's lived experience which carries a rich history of memories of the lives they have lived there. Unfortunately, some of these individuals face having to relocate in their later years because they experience memory loss and challenges living independently.

I would like to see more studies that help prevent or delay cognitive vulnerability in younger and middle-aged individuals who may be at risk for cognitive decline, for example, among people in my community with cardiovascular conditions, low levels of education, and monolingual Spanish-speakers. Your scientific team is on the cusp of making a difference, therefore, I am excited to serve in this advisory role to the ADRC. I understand that this role entails addressing how best to reach community people to let them know about the latest scientific advances in cognitive research, the center's great studies, and study participation opportunities. I add that it would be a great idea to

ascertain how men fare in recruitment, what is needed to get them to participate, and brain health strategies directed to men.

As member of the advisory board, I understand that I will meet with your ADRC outreach and scientific team under Dr. Aranda's leadership about 3 times per year (around 2 hours per meeting) over 5 years. I will receive remuneration at the rate of $100 per meeting.

In closing, I wish you the best in your grant submission. Let me know if there is anything I can do to help.

Best wishes,



Steven Galindo
225 S. Lake Ave. #100
Pasadena, CA 91101
steven.galindo@theagencyre.com
(626) 639-8548
License: California DRE #1002784

May 9, 2024

Maria P. Aranda, PhD, Executive Director, USC Roybal Institute on Aging
Co-Lead, Outreach-Recruitment-Engagement Core
USC Alzheimer's Disease Research Center
Department of Neurology, Keck School of Medicine of USC
Los Angeles, CA 90033

Dear Dr. Aranda,

Thank you for inviting me to serve as a USC Roybal Institute on Aging Community Advisory Board member. Our board installation last month at the bilingual USC Southern California Caregiver Celebration event in Los Angeles was inspiring and reflective of the important work that we need to do in the community on cognitive health, aging research, family caregiving, and research participation. I work closely with you, other advisory board members, and your colleagues on your various research projects to enhance the wellbeing of families affected by memory loss. Thus, I write to support your research proposal to renew the USC Alzheimer's Disease Research Center. This center is well-positioned to make a difference in people's lives by preventing or delaying cognitive decline and treating memory loss in our local communities. Focusing on people most at risk such as the Hispanic population is needed, and very much understudied even with all the advances we see in the news.

Your scientific team is on the cusp of making a difference, therefore, I am excited to serve in this advisory role to the ADRC. I understand that this role entails addressing how best to reach community people to let them know about the latest scientific advances in cognitive research, the center's great studies, and study participation opportunities. Moreover, I understand that I will meet with your outreach team under Dr. Aranda's leadership about 3 times per year (around 2 hours per meeting) and will receive remuneration at the rate of $100 per meeting.

May you be successful in your renewal, and I look forward to working with the ADRC.

Warm regards,

Kimberly Whyte

Kimberly Whyte
1425 W. 51 Street
Los Angeles, CA 90062
Whytekimberly6@gmail.com
(323) 847-3196 Mobile

May 9, 2024

Maria P. Aranda, PhD, Executive Director, USC Roybal Institute on Aging
Co-Lead, Outreach-Recruitment-Engagement Core
USC Alzheimer's Disease Research Center
Department of Neurology, Keck School of Medicine of USC
Los Angeles, CA 90033

Dear Dr. Aranda,

I am honored to serve as a member of the USC Roybal Institute on Aging Community Advisory Board which works closely with the USC Alzheimer's Disease Research Center. Our board installation last month at the bilingual USC Southern California Caregiver Celebration event in Los Angeles was inspiring and reflective of the important work that we need to do in the community on cognitive health, aging research, family caregiving, and research participation.

I work closely with you, other advisory board members, and your colleagues on your various research projects to enhance the wellbeing of families affected by memory loss. **Thus, I support your research proposal to renew the USC Alzheimer's Disease Research Center for the next 5 years.** I genuinely believe that the ADRC is well-positioned to make a difference in people's lives by preventing or delaying cognitive decline and treating memory loss in our local communities. Focusing on people most at risk such as the Hispanic population is needed, and very much understudied even with all the advances we see in the news.

I am a homegrown local artist from the East Los Angeles area who cares for his elderly Japanese mother who has been diagnosed with memory loss. Although my passion is art and music, nothing will come between me caring for my mother while she is experiencing this journey. I am extremely fortunate to have the support of my family and dear friends who have also experienced this journey. More importantly, I am surrounded by males, who have provided guidance, assistance and understanding.

My mission includes caregiving for my mother, informing younger generations of this heart wrenching disease through art and music. My voice will be able to assist families navigate the maze of caregiving and become educated about early testing for memory loss.

I would like to see more studies that help prevent or delay cognitive vulnerability in younger and middle-aged individuals who may be at risk for cognitive decline, for example, among people in my community with cardiovascular conditions, low levels of education, and monolingual Spanish-speakers.

Your scientific team is on the cusp of making a difference, therefore, I am excited to serve in this advisory role to the ADRC. I understand that this role entails addressing how best to reach community people to let them know about the latest scientific advances in cognitive research, the center's great studies, and study participation.

opportunities. I add that it would be a great idea to ascertain how men fare in recruitment, what is needed to get them to participate, and brain health strategies directed to men.

As a member of the advisory board, I understand that I will meet with your ADRC outreach and scientific team under Dr. Aranda's leadership about 3 times per year (around 2 hours per meeting) over 5 years. I will receive remuneration at the rate of $100 per meeting.

In closing, I wish you the best in your grant submission. Let me know if there is anything I can do to help.

Best wishes, Ryo Fujita

Ryo Fujita
424 South Lorena Street
Los Angeles, CA 90063
Ryo@dopetracks.com
(323) 683-6004 Mobile

RESOURCE SHARING PLAN

The ADRC overall and the Clinical, DMSC, Neuropathology, ORE, Biomarker, and Imaging Cores engage in broad sharing of clinical data, MRI and PET data, and biological samples, analytical methodology and disease models prior to publication.

We do this compliant to NIH guidelines and to maximize the availability and usability of the data and research resources generated by NIA's ADRC Program, and to comply with the NIH Genomic Data Sharing Policy and the NIA/NIH efforts aimed at increasing transparency, reproducibility, and translatability of research findings. We subscribe to the FAIR principles.

We follow the NIA best practice guidelines for processing and storage of biological specimens promulgated by the Biospecimen Task Force which can be found on the NACC web site.

Within the biorepository, we use a system of data access with defined levels of access privileges. We restrict access to research subjects' identities and medical, genetic, social, and personal histories to necessary biorepository staff members and those with IRB approved protocols who need such access as part of their duty or to persons permitted access by law. We monitor personnel compliance with access restrictions. We store human biospecimens only for research purposes according to approved protocols, not to serve individual biospecimen donor's needs or wishes.

Datasets

The Overview, Administration, and DMSC narratives describe these efforts: we will make all datasets used and generated by the individual Cores accessible and reusable by qualified individuals outside the ADRC. Most data is now available via web-based resources at LONI, NACC, and NCRAD. MRI and PET are anonymized and stored at LONI with links to underlying and anonymous clinical data including the UDS and other clinical data obtained by the Clinical Core. The NACC UDS clinical and neuropathology data and reports are shared via the NACC forms, uploaded to and shared via NACC and LONI/IDA. Supplemental ADRC data (e.g., VCS, PCD, Biomarkers) are entered through REDCap and shared through the IDA (DMSC)..

To fulfill the above data and resource sharing we utilize the following NIA-supported repositories: NACC, NCRAD, NIAGADS and the AMP-AD Knowledge Portal and as appropriate (SCAN, CLARiTI). Data can be made accessible via open or controlled access depending on the data type and data source and as determined by the informed consent documents for each study guided by the local IRB. A comprehensive listing of NIH data sharing repositories is available at NIH. Databases can be accessed by qualified researchers.

Biological Samples

All biological samples obtained or used to generate data with this award are available to qualified investigators through requests to the Neuropathology and Biomarkers Cores. Our goal is to ensure that investigators have timely, equitable, and appropriate access to human biospecimens and associated clinical data stored at NIH supported biorepositories without undue administrative burden.

The USC ADRC Executive committee has established clear, explicit guidelines for sample distribution (and associated clinical data sharing) consistent with ethical principles, prevailing laws, and consent form language. The guidelines address the following processes and procedures involved in tissue dissemination: the request process, review procedures, decision-making, distribution and future tracking. Guidelines will be applied to all new collections and, whenever possible, to existing collections. Access is guided by policies and procedures that address the following:

1. Scientific validity of the research proposal
2. Burden or demand on the tissue resource and staff
3. Investigator's agreement covering confidentiality, use, disposition, and security of biospecimens and associated data

4. Investigator's written agreement in a Material Transfer Agreement to comply with the NIH Research Tool Guidelines.
5. Investigator and institutional research qualifications
6. Ethical oversight where required by Federal regulations or local institutional requirements
7. The Executive Committee supports adequate grant funding for the biorepository. If applicable and where monetary charges are necessary, we charge only to recover costs as appropriate to retrieve and disseminate specimens (see Cost Recovery Guidelines).

In addition to the above, the following points are recommended for consideration while assessing access privileges:

8. Biospecimens and associated clinical data are to be appropriately matched with the specific scientific investigations for which they are intended.
9. Investigators requesting access need to be informed about site-specific information in order to correctly interpret their findings, e.g., how tissue was prepared, how the clinical diagnosis was derived, and how subjects were recruited (volunteers, clinic patients, etc.).
10. Additionally, if requested and available, for autopsy specimens agonal state information may be shared.

The local decision-making body (i.e., ADRC Executive Committee) takes local principles into account. Ethical considerations including conflicts of interest should come first among principles that guide the decision-making process. Should disputes or concerns arise over allocation decisions, investigators may request appeal and review by the ADRC Internal Advisory Committee (IAC).

PHS Assignment Request Form

Funding Opportunity Number: RFA-AG-24-001

Funding Opportunity Title: Alzheimer's Disease Research Centers (P30 Clinical Trial Not Allowed)

Awarding Component Assignment Request *(optional)*

If you have a suggestion for an awarding component (e.g., NIH Institute/Center) assignment, use the link below to identify the appropriate short abbreviation (e.g. "NCI" for National Cancer Institute") and enter it below in the boxes for "Suggested Awarding Components". All suggestions will be considered; however, not all assignment suggestions can be honored.

Awarding Components: https://grants.nih.gov/grants/phs_assignment_information.htm#AwardingComponents

Suggested Awarding Component: NIA

Study Section Assignment Request *(optional)*

If you have a suggestion for a study section assignment, use the link below to identify a study section(s). Enter the short abbreviation for that study section in the boxes for "Suggested Study Sections." Remove all hyphens, parentheses, and spaces. All suggestions will be considered; however, not all assignment suggestions can be honored.

For example, enter "CAMP" if you wish to suggest assignment to the NIH Cancer Molecular Pathobiology study section, or "ZRG1HDMR" if you wish to suggest assignment to the NIH Healthcare Delivery and Methodologies SBIR/STTR panel for informatics.

Study Sections: https://grants.nih.gov/grants/phs_assignment_information.htm#StudySection

Assign to Study Section:
Each entry is limited to 20 characters

PHS Assignment Request Form

Rationale for assignment suggestions *(optional)*

Entry is limited to 1000 characters

The contact for review of ADRC application is: Ramesh Vermuri, PhD (301) 496-9666, email: Ramesh.Vermuri@nih.gov

List individuals who should not review your application and why *(optional)*

Entry is limited to 1000 characters

Identify scientific areas of expertise needed to review your applications *(optional)*

Note: Do not provide names of individuals

Expertise:
Each entry is limited to 40 characters

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data? ●Yes ❍No

Provide an explanation of why the application does not involve human subjects research

Are Human Subjects Involved ●Yes ❍No

Is the Project Exempt from Federal regulations? ❍Yes ●No

Exemption Number ❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8

Other Requested Information

Human Subject Studies

Study#	Study Title	Clinical Trial?
1	USC Alzheimer's Disease Research Cohort	No

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Section 1 - Basic Information (Study 1)

1.1. * Study Title (each study title must be unique)

USC Alzheimer's Disease Research Cohort

1.2. * Is this study exempt from Federal Regulations? ❍Yes ●No

1.3. Exemption Number ❏1 ❏2 ❏3 ❏4 ❏5 ❏6 ❏7 ❏8

1.4. * Clinical Trial Questionnaire

1.4.a. Does the study involve human participants?	●Yes	❍No
1.4.b. Are the participants prospectively assigned to an intervention?	❍Yes	●No
1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?	❍Yes	●No
1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?	❍Yes	●No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics

2.1. Conditions or Focus of Study

- ❍ Cognitively impaired
- ❍ At-risk
- ❍ Mild Cognitive Impairment
- ❍ Dementia
- ❍ Alzheimer's Disease
- ❍ Apolipoprotein E

2.2. Eligibility Criteria

The eligibility criteria is outlined in a detailed summary that can be found in the Overall Protection of Human Subjects in Section 3.1.

2.3. Age Limits	Min Age: 18 Years Max Age: 120 Years
2.3.a Inclusion of Individuals Across the Lifespan	Inclusion_Across_the_Lifespan1044446723.pdf
2.4. Inclusion of Women and Minorities	Inclusion_Women_and_Minorities1044446724.pdf
2.5. Recruitment and Retention Plan	Recruitment_and_Retention_0614241044447278.pdf
2.6. Recruitment Status	Recruiting
2.7. Study Timeline	Study_Timeline_0614241044447285.pdf
2.8. Enrollment of First Participant	03/03/2025 Anticipated

Inclusion Enrollment Reports

Entry#	Enrollment Location Type	Enrollment Location
IER 1	Domestic	USC Health Science Campus, Rancho Los Amigos National Rehabilitation Center

Section 3 - Protection and Monitoring Plans

3.1. Protection of Human Subjects

Protection_of_Human_Subjects1044446728.pdf

3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site? ❍ Yes ● No ❍ N/A

If yes, describe the single IRB plan

3.3. Data and Safety Monitoring Plan

3.4. Will a Data and Safety Monitoring Board be appointed for this study? ❍ Yes ● No

3.5. Overall Structure of the Study Team

Overall_Structure_of_Study_Team1044446727.pdf

Section 4 - Protocol Synopsis

4.1. Study Design
4.1.a Detailed Description
4.1.b. Primary Purpose
4.1.c. Interventions

Type	Name	Description

4.1.d. Study Phase
Is this an NIH-defined Phase III Clinical Trial? ❍ Yes ❍ No

4.1.e. Intervention Model
4.1.f. Masking ❍ Yes ❍ No

❏ Participant ❏ Care Provider ❏ Investigator ❏ Outcomes Assessor

4.1.g. Allocation

4.2. Outcome Measures

Type	Name	Time Frame	Brief Description

4.3. Statistical Design and Power
4.4. Subject Participation Duration

4.5. Will the study use an FDA-regulated intervention? ❍ Yes ❍ No

4.5.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/Investigational Device Exemption (IDE) status
4.6 Is this an applicable clinical trial under FDAAA? ❍ Yes ❍ No

4.7. Dissemination Plan

Section 5 - Other Clinical Trial-related Attachments

5.1. Other Clinical Trial-related Attachments

Inclusion Enrollment Report 1

1. * Inclusion Enrollment Report Title: USC ADRC Cohort

2. * Using an Existing Dataset or Resource: ❍Yes ●No

3. * Enrollment Location Type: ●Domestic ❍Foreign

4. Enrollment Country(ies): USA: UNITED STATES

5. Enrollment Location(s): USC Health Science Campus, Rancho Los Amigos National Rehabilitation Center

6. Comments:

Planned

Racial Categories	Ethnic Categories				
	Not Hispanic or Latino		Hispanic or Latino		Total
	Female	Male	Female	Male	
American Indian/Alaska Native	0	0	0	1	1
Asian	11	9	1	0	21
Native Hawaiian or Other Pacific Islander	0	0	0	0	0
Black or African American	7	4	1	0	12
White	55	63	38	43	199
More than One Race	5	2	12	8	27
Total	78	78	52	52	260

Cumulative (Actual)

Racial Categories	Ethnic Categories									
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			Total
	Female	Male	Unknown/ Not Re-ported	Female	Male	Unknown/ Not Re-ported	Female	Male	Unknown/ Not Re-ported	
American Indian/Alaska Native	1	0	0	0	1	0	0	0	0	2
Asian	34	19	0	2	0	0	0	0	0	55
Native Hawaiian or Other Pacific Islander	0	1	0	0	0	0	0	0	0	1
Black or African American	23	8	0	1	0	0	0	0	0	32
White	216	175	0	61	41	0	0	0	0	493
More than One Race	16	6	0	26	11	0	0	0	0	59
Unknown or Not Reported	3	0	0	46	29	0	0	1	0	79
Total	293	209	0	136	82	0	0	1	0	721

Inclusion of Individuals Across the Lifespan

Children are excluded because they don't generally develop sporadic Alzheimer disease. We will include persons who are 18 or older who may be carriers of autosomal dominant AD or have Down's syndrome.

Inclusion of Women and Minorities

The USC ADRC included under-represented groups and historical minorities in numbers consistent with the population and Los Angeles County local communities from which participants are drawn. The ORE core and the clinical core proactively recruit Latinx from our community using a range of programs (see project narratives for the clinical core, the ORE core, and Section 3. Protection of Human Subjects). We plan for 35% or more of our clinical core participants will be of Latinx heritage. Currently, in the ADRC women represent 60% of our participants in the clinical core. We plan to continue to recruit women at that ratio to reflect their somewhat greater prevalence in the population with late-life cognitive impairment. Our inclusion and exclusion criteria for enrollment are clearly defined in the narratives for the overall, clinical core, ORE core, and Section 3. Protection of Human Subjects.

Recruitment and Retention Plan

See Clinical Core:

- Specific Aim 1: Participant recruitment and retention
- Approach C3.6: Participant retention and feedback

See ORE Core:

- Specific Aim 1: Engage in participant recruitment
- Specific Aim 2: Enhance recruitment from LA County DHS
- Approach C3.2.1 Outreach, Recruitment, and Retention Activities

Study Timeline

See Clinical Core:

- Specific Aim 1: Participant recruitment and retention
- Approach C.2.1: Recruitment and sample size
    - 52 new participants per year, enriched for *APOE ε*4 (Total n=260 over 5 years)
    - 24 new SCAN participants per year, enriched for *APOE ε*

See Neuropathology Core

- Specific aim 1: Rapid autopsies from diverse population
    - Tissue collection workflow (18-20 autopsies per year)

See ORE Core:

- Specific Aim 1: Engage in participant recruitment
    - 18 new Latinx participants/year, enriched for APOE ε4
- Specific Aim 2: Enhance recruitment from LA County DHS
    - 8 new Latinx SCAN participants/ year, enriched for APOE ε4

3.1 Protection of Human Subjects

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and NIH-defined phase 2, phase 3, and other clinical studies. The clinical intervention trials, however, are affiliated with, but not a funded part of the ADRC. Protection of human subjects is a primary concern to the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although specific details may differ among the individual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The USC ADRC is part of a network of approximately 30 Alzheimer Disease Centers (ADCs) funded by the National Institute on Aging. The ADCs contribute to a database, managed by the National Alzheimer Coordinating Center (NACC), and collaborate at national level: including the National Cell Repository for Alzheimer Disease (NCRAD), and the Alzheimer Disease Genetics Consortium (ADGC). Many ADRCs, including USC, collaborate with the Alzheimer's Disease Neuroimaging Initiative (ADNI) Alzheimer's Clinical Trials Consortium (ACTC), Alzheimer Disease Cooperative Study (ADCS), and the Dominantly Inherited Alzheimer's Network (DIAN). Participants in the USC ADRC are also approached about placing their de-identified clinical data, biospecimens, genetic and imaging data in national databases.

3.1.1. Risks to Human Subjects

a. Human Subjects Involvement, Characteristics, and Design

Participants for the ADRC may be from the ages of 50 and older, but for most studies will be from 60 to 90. As a product of the research strategies developed for the USC ADRC, a substantial proportion of participants are not cognitively impaired; another group has mild cognitive impairment; and a somewhat smaller group has mild dementia. Many participants in our cohorts will age into progressively more severe dementia. Because of their age as well as our recruiting emphasis on vascular, metabolic risk factors and *APOE* genotype, many will have chronic medical illnesses in addition to cognitive impairment. By design, some ADRC affiliated studies include individuals with disorders that place them at somewhat higher risk for stroke and cardiovascular illnesses such as hypertension and diabetes. Participants with serious unstable medical conditions, however, are excluded from these studies. Approximately 260 new participants to the ADRC and related studies will be enrolled over 5 years. Estimated numbers, recruitment methods for individual studies and inclusion and exclusion criteria are detailed in the Overall, Administrative, Clinical, ORE, Neuropathology, Biomarker, and Imaging Core narratives in addition to below.

<u>Description and justification of the proposed involvement of human subjects in the work outlined in the research strategy sections</u>

Participants are older adults with no cognitive impairment, age-related cognitive decline, mild cognitive impairment, or dementia who agree to participate in one or more of several studies undertaken, supervised, or affiliated with the USC ADRC. Participants are generally intended to be followed indefinitely and evaluated, tested for cognitive ability, approximately yearly, have MRIs, amyloid and tau PET scans, and blood specimens taken to assess for research biomarkers. Research lumbar punctures are offered but are optional.

The general purpose of the clinical research within the ADRC is to advance knowledge of at-risk, vascular, metabolic, and *APOE* genotype factors in the development of dementia and cognitive impairment. The purpose of the brain donation program is to enhance clinicopathological correlation, ante mortem diagnosis, and to obtain important biospecimens to support future research and to be shared across research centers. There are other observational studies, and experimental, clinical trials in which specific experimental therapeutic interventions will be made. For these studies participants with no cognitive impairment, mild cognitive impairment, and with dementia will be recruited and may be co-enrolled in the VCS and BRS if there is no contraindication.

<u>Describe and justify the sampling plan, as well as the recruitment and retention strategies and the criteria for inclusion or exclusion of any subpopulation</u>

As above, we recruit and enroll participants in the at-risk longitudinal cohort of the clinical core. We estimate we need to maintain a cohort of 500 to serve the needs of affiliated studies and to contribute effectively to the NACC UDS and NCRAD multicenter cohorts. We estimate that we need to recruit approximately 52 at-risk participants per year to increase and maintain the clinical core cohort at 500 (see clinical core research strategy for details). Our recruitment methods are described in the research strategy sections of the Overall, the ORE core, the clinical core, and below. Briefly, our recruitment strategies are multifaceted and involve community outreach, earned

media, the use of registries, outreach to physicians, public awareness, and education. For autopsy participants we also focus on our existing cohorts and the local community. We used actuarial methods to determine a minimum size for our brain donation (autopsy) subgroup. We estimate we need to enroll approximately 26 new participants per year, mainly from the clinical core cohort, to maintain or increase our autopsy rates (see Neuropathology and Clinical cores).

Our retention methods for ADRC clinical core participants and affiliated studies involve proactive contact by ADRC staff including study coordinators, the original intake coordinator, and nurses. We contact participants several times between their annual visits to assess their status, thank them, and encourage their continuing interest. The ORE core uses newsletters, updates, and participant appreciation days as well. Participants in affiliated clinical studies are often seen more frequently than general clinical core cohort participants which allows for more frequent interaction and encouragement. We can reimburse participants for travel and food in many of our studies under the IRB-approved protocols for those studies.

General eligibility criteria. Criteria for cognitively non-impaired: We generally use the "cognitively unimpaired" or "subjective memory complaint" criteria as established by the ADNI4 cohort study and now by the new UDSv4 criteria/guidelines (forms D.1, D.2). Criteria include nominally normal episodic memory or delayed recall; MMSE score ≥ 24 or MOCA ≥ 22, with no impairment of recall or orientation; no impairments on instrumental activities of daily living; no major depression, marked anxiety; adequate corrected vision, hearing, and communicative ability in order to allow for obtaining the outcomes assessments. Criteria for mild cognitive impairment: We generally follow Albert et al 2011 criteria, and further specify to the best of our judgment or with biomarker support whether MCI is due to Alzheimer disease. Criteria for dementia and Alzheimer disease generally follow 1984 and 2011 McKhann et al criteria. We use other published criteria and guidelines for other neurodegenerative and cognitive impairing conditions.

Eligibility criteria for brain donation (autopsy). (1) Informed consent for brain donation (autopsy) by participants with decisional capacity or by a legally authorized representative if not; (2) understanding and agreement with post-mortem procedures, (3) participation in the clinical core cohort, (4) evaluated and diagnosed with the UDS and initial data uploaded to NACC and the USC RedCap database.

General exclusion criteria for ADRC participants. Unwillingness to give informed consent; lack of assent by a participant without decisional capacity to give informed consent; unwillingness of the legally authorized representative or proxy decision-maker to provide surrogate consent; inability or unwillingness to undergo or contraindication to an MRI, functional impairment so severe that the participant cannot participate in required research activities; unstable medical conditions that impair participant's ability to participate in the studies.

Criteria for ORE core resource of Enhanced Recruitment from the LA County Department of Health Services at the Rancho Los Amigos National Rehabilitation Center (RLANRC). Our overall aim is to provide patients cared for in public-safety net hospitals (e.g., RLANRC) opportunities to participate in ADRC and affiliated research studies (e.g., NACC UDS, SCAN, CLARiTI, clinical trials). The sample pool includes participants seen at DHS hospitals and clinics. After informed consent, the ADRC DHS participant will be followed as usual and will provide additional data as noted below.

Procedures for obtaining informed consent for research on cognitively impaired human subjects who may not have the capacity to consent and to ensure future participation in research studies if or when the participant becomes unable to consent

Consent is demonstrated when an individual (either verbally or non-verbally) indicates what they are willing to do or allow a third party to do this for them. To give valid consent, participants need to be able to access, understand and process information relating to the decision they are making. Informed consent is therefore a fundamental requirement for participating in the USC ADRC. The mental capacity of an individual is at the center of the law relating to consent. For the ADRC studies, the physicians, nurse practitioner, or research nurses who are experienced in the field of dementia will determine whether a potential study participant has decisional capacity and is competent to consent to research prior to beginning any consent procedures. Informed consent will be presented and obtained in the language preferred by the potential participant and by appropriately trained and certified staff. Competency is assessed at each of the follow up visits. At the beginning of each of the follow up visits, the participant or responsible party (legally authorized representative) and/or study partner are provided a copy of the informed consent to review. They are given as much time as they need to answer any questions they might have about their continued participation. The research team during this time assesses whether the participant and study partner understand and are willing to continue participation in the study. When there is any

indication to suggest diminished capacity, we begin by discussing the research directly with the participant to the extent possible to gain as fair an idea as possible of what their understanding of the study and their inclination to participate. If a participant lacks capacity, we still ask for assent to study participation. If they refuse, then we will not continue the participant in the study. Additionally, we review the consent form with the participant's legally authorized proxy decision-maker or guardian, next of kin, or familial caregiver according to California State Law and require that that person consent to the study as well. If either party refuses, then we do not continue the participant in the study. Participants and LARs are given the option to revoke consent. As above, in cases where the participant no longer has the capacity to consent, consent is obtained from the legally authorized representative and assent is obtained from the participant per California State law AB2325 (California Health & Safety Code section 24178).

Procedures specifying how, by whom, and with whom brain donation (autopsy) will be discussed

Potential brain donors may be approached several times during their participation in the ADRC: in telephone screening by the intake coordinators and recruiters, by physicians, research nurse, nurse practitioner, and by study coordinators at initial visit, family conference, or follow-up visits. They are told about autopsy and what they will be asked to do. This includes: (1) completing an initial questionnaire asking about contact and health information, (2) participating as necessary in annual follow-ups or by telephone where memory testing may be done, (3) receiving time of death instructions (mortuary arrangements and funeral plans) to facilitate transfer of the body and to avoid delays. Participants are asked to give consent for USC to keep and to distribute blood and body fluid samples for future research and sharing information with other researchers.

b. Sources of Materials

Research material is obtained from living and deceased individuals in the form of specimens, tissue, records, or data as follows: The protocols of the ADRC require gathering a large amount of data specifically for research purposes. The major classes of data include medical diagnoses, findings on medical examinations, behavioral data on psychiatric disturbances, neuropsychological data on cognitive function, data on functional ability, demographic data, measurements of brain structure obtained with MRI, personal and family medical history, CSF, and genetic data. All this is gathered on individually identifiable participants and thus great care is taken to protect their privacy and the confidentiality of this information. The information is de-identified by the clinical core and data core before being sent to NACC, NCRAD, and other investigators. The data core narrative describes procedures for protecting participants' privacy and de-identifying participant and PHI data.

c. Potential Risks

In general, the risks to participating in the ADRC and affiliated studies are directly associated with the clinical data collection as the studies directly supported by the ADRC are observational and non-interventional, and there are minimal or no known risks regarding the observational components of the studies *per se.* There are risks involved in preparing for and traveling to the clinic for the studies. The risks involved in being assessed with noninvasive clinical and neuropsychological tests are minimal. Some participants may feel stressed or anxious, but this usually resolves after testing. There are also very low risks involved in venipuncture, including bruising and minor pain, anxiety or fatigue associated with neuropsychological testing, etc. The risks for MRI without contrast are rare and include risks of injury when patients with ferrous objects in their bodies undergo MRI. We do not plan to use MRIs with contrast for clinical core participants in this cycle.

Most clinical core participants will undergo a non-contrast MRI, a non-invasive procedure. Thousands of patients have undergone studies at USC on the equipment proposed for this study, using similar data acquisition strategies without any known harmful consequences. Magnetic field strengths, radio frequency power levels, gradient strength, and switching times are within the FDA approved range. The data acquisition sequences that will be used are provided by the manufacturer and are FDA approved. Some people may become claustrophobic in the confines of the magnets; those participants will be removed from the study. An MRI may cause possible anxiety for people due to the loud banging made by the machine and the confined space of the testing area.

Patients with pacemakers, aneurysm clips, artificial heart valves, ear implants, metal/foreign objects in their eyes, or a history of ferromagnetic material near vital structures will be excluded from the study because of the risk of pacemaker malfunction or movement of ferromagnetic material within the magnetic field of the imaging unit. Except for these patients there have been no adverse effects demonstrated following exposure to the magnetic field gradients and radiofrequency pulses used to generate the images. Some participants will undergo a 7T MRI in affiliated studies, under a separate protocol from the ADRC.

Many clinical core participants will undergo PET-CT scans, which include exposure to greater radiation than most diagnostic x-ray procedures. Radiation exposure can increase one's risk of developing cancer. The risk for cancer from the dose of radiation received in this study is known to be low. Injection site reactions could occur consisting of rash, irritation or pain. There are also minor risks with. venipuncture. The injection of the radioisotopes might include pain and bruising. It is possible that a pre-existing or unrelated medical abnormality is apparent on PET or MRI scans. If so, we will inform and help the participant get follow up care.

An important risk associated with the ADRC observational studies relates to loss of privacy or the potential exposure of personal medical information, particularly data related to risks of Alzheimer disease or dementia. The improper release of such information could have negative effects for participants. For example, loss of privacy, adversely affecting their ability to obtain health or life insurance and could carry negative effects for blood relatives as well. Participating in these studies generally does not require adding or withholding medications used to treat a participant's cognitive-impairing condition. The exceptions are for selected treatment studies with experimental medicines. This is addressed with the participant in separate IRB-approved and NIH-funded protocols and informed consent interviews and documents, as the clinical trials involve interventions with experimental drugs and the risks here are specific to the drug involved. The USC IRB reviews each protocol.

There are risks involving the optional lumbar puncture for clinical core participants. They include infection, bleeding, and headache due to CSF leakage. For prolonged post-LP headache or leakage an epidural blood patch may be placed.

The ADRC clinical core cohort is not a treatment study. There are no known alternative treatments and procedures for the conditions being studied. Hence, the risks and potential benefits of the alternative treatments and procedures to participants in the proposed research are not known.

3.1.2 Adequacy of Protection against Risks

a. *Recruitment and Informed Consent*

Written informed consent will be obtained for all participants in the ADRC. The experimental procedure and length will be explained to the participants who will be given consent forms and the time to read it. Participants will be allowed to ask questions and when their concerns have been satisfied will be asked to sign the form. In the absence of written consent, no research procedures will be performed. The major principle for approaching the participant is to ensure that they fully understand 1) the potential risks involved, 2) that participation in the research is purely voluntary; and 3) that refusal to participate will not, in any way, adversely affect their medical care. Potential benefits are also explained, but realistically, with care being taken not to induce false expectations. Individuals who are not actively involved in the research will witness the signing of the consent form. A copy of the signed consent form will be given to the participant.

Plans for recruitment and consent. Participants will be recruited in several ways, from existing longitudinal cohorts, or by new outreach (see Aim 1 (Gene Screen) and 2 Aim 2 (LA DHS facilities) of the ORE core and Aim 4 of the clinical core). In either case, potential participants may be approached initially by telephone, email, or in person, and the general nature of the research is discussed following guidelines approved by the USC IRB. Our protocols for obtaining consent include specific procedures for explicitly determining both the understanding of the research study and the willingness to participate. When there is an indication to suggest diminished capacity, we begin by discussing the research directly with the participants to the extent possible to gain as fair an idea as possible of what their understanding of the study and their inclination to participate. If the participant lacks decisional capacity, then we ask that they assent to study participation; if they refuse then they will not be enrolled.

A special problem is obtaining consent from people of limited decisional capacity or from those who are clearly incapable of consent. While this capacity to consent will not be at issue during recruitment of our cognitively normal or early MCI participants, it becomes one with early dementia patients and for those who experience cognitive decline during their illness such as might occur during follow-up evaluations of patients with MCI. This is discussed with the IRB for each new study we have done over the years. We and the USC IRB follow California State law AB2325 (California Health & Safety Code section 24178).

In addition, we review the consent form with the patient's legally authorized representative or guardian (if there is one), next of kin, or familial caregiver according to California State Law and require that this person consent to the study as well. If either refuse, then we do not enroll the person. We rely on the continuing informed consents from the participant or LAR decision-maker for continuing autopsy permission and for current and future

use of biological samples by qualified investigators including permission for sharing of cells, DNA, and genetic and phenotypic information, and storage in repositories. If the participant loses capacity to continue to provide informed consent, then we rely on the proxy decision-maker as described.

We follow the Alzheimer Disease Center Biospecimen Task Force guidelines for guidance on consent forms; use the NIH sharing policy and guidance on genetics research for guidance on language regarding genetics that we use in consent forms; and follow NIH policy on genome wide association studies.

b. Protections Against Risk

The following are our procedures for protecting against or minimizing potential risks, including risks to privacy of individuals and confidentiality of data, and an assessment of their likely effectiveness. This research does not involve vulnerable populations as defined in the DHHS regulations, Subparts B-D, i.e., pregnant women, human fetuses and neonates, prisoners, and children. It does involve cognitively impaired and older participants some of whom many have diminished decisional capacity and may be considered vulnerable.

Plans for ensuring necessary medical or professional intervention in the event of adverse effects to participants.

Research participants are overseen by physicians and are seen for assessment of adverse effects. All participants concerned about or reporting adverse events are evaluated and treated as necessary by the physician; adverse events are reported to the IRB, the NIH and FDA when required to ensure the safety of participants.

Plans to protect against and to ensure necessary medical intervention in the event adverse events occur for potential risks associated with ADRC participation including risks to confidentiality, risks of neuropsychological testing, of genotyping, MRI scans, and lumbar punctures are as follows:

Confidentiality. The ADRC takes the risk to subject confidentiality very seriously and has implemented a variety of measures to protect the confidentiality of data and the identities of our participants. In addition to several layers of safeguards for the electronic data (electronic firewalls, password protection of files, limiting access privileges, structural isolation of particularly sensitive components of the data base, use of pseudo ID numbers for sensitive data), normal prudence is observed with regard to the participant's physical records as well. Patient charts and records are either in locked or directly supervised areas during business hours, and always under lock and key during off hours. New staff members are specifically instructed in confidentiality issues and undergo required research, human subjects, and HIPPA training. A Federal certificate of confidentiality has been and will be applied for specific studies to add another set of protections against legal intrusions on subject privacy.

Neuropsychological testing. While many participants in the ADRC and affiliated studies are expected to have generally good cognitive abilities and, therefore, unlikely to fatigue during cognitive testing, we know that many will be anxious about the extent of their disabilities. This will be particularly true during follow-up evaluations where conversion to dementia or progression of dementia will be determined. All evaluations are held in confidence and shared with the patient and family at the initial evaluation or if diagnostic category is changed during follow-up. The neuropsychologists and psychometricians are sensitive to patient fatigue and anxiety and may end testing if a participant expresses discomfort. If so then, testing is often completed at a second visit.

Blood Sampling. The process of blood sampling is discussed with each participant. Blood samples are obtained in laboratory space by certified phlebotomists, technicians, or nurses and physicians, if needed.

APOE genotyping. Disclosure of *APOE* genotype has been not recommended by professional consensus groups for asymptomatic persons. In practice, *APOE* genotyping is available commercially and becoming more frequent for both symptomatic and asymptomatic persons in both medical and non-medical situations. We inform potential research participants that *APOE* genotype is part of the biospecimens that are collected but that an individual's genotype will not be disclosed to them unless they ask (e.g., GeneScreen). When *APOE* genotyping is not obtained and performed to CLIA standards we will not share these test outcomes with participants. However, for participants who know their genotype from elsewhere, other studies, or from commercial companies (e.g, 23&me) or who request to know their genotype we offer clinical (CLIA) genotyping from the USC clinical laboratory. Participants are counselled by ADRC physicians and a genetic counsellor, if requested, and provided with written educational materials like those used in the REVEAL, Generation studies, and by 23&me.

Other genetic studies. ADGC currently performs GWAS, whole genome wide sequencing, whole exome sequencing on DNA obtained from circulating white blood cells and sends DNA to NCRAD where this is also performed. We do not reveal these findings.

MRI imaging. All participants will answer detailed questions about potential ferro-magnetic implants or shrapnel that could result in injury during MRI. In addition, participants are directly observed during the imaging procedure and are in constant voice communication with the MRI technician. The clinical and research MRI scanners at USC are FDA approved for human use with respect to field and gradient switching times. All data acquisition sequences used will be FDA approved. Should a participant experience discomfort during a procedure, the study will be stopped. Participant confidentiality is maintained and no data which can be identified with an individual will be released to anyone not connected with this project without written consent of the subject. The exposure time and experimental factor will be recorded and filed with the consent forms for future reference. All procedures on humans will be subject to prior IRB approval.

PET/CT imaging. Participants are assessed for possible pregnancy. Those who are pregnant will not undergo PET/CT. If the participant is a woman who could become pregnant, then a pregnancy test will be performed before each PET/CT procedure. The ^{18}F radioactive tracers for amyloid-β and tau to be used in the ADRC are FDA approved. The tracer injected via a venous catheter. Images are obtained in the scanner about 90 to 100 minutes afterwards. The scan takes about 90 minutes. The exposure time and experimental factor will be recorded and filed with the consent forms for future reference. Participant confidentiality is maintained and no data which can be identified with an individual will be released to anyone not connected with this project without written consent of the participant. All procedures on humans will be subject to prior IRB approval.

Lumbar puncture. Risks of the lumbar puncture include infection, bleeding, CSF leakage, and headache, and will be explained to the participants and legally authorized representatives verbally and in the consent form before the procedure. Appropriate precautions will be taken to minimize these risks. Standard sterile protocol will be strictly followed. A small-bore atraumatic needle (22- or 24-gauge Sprotte brand) will be employed to help decrease the incidences of CSF leak headaches. After the procedure, participants will lie down in the procedure room for about an hour to decrease the risk of headache; will be given light food and drink; and will be instructed not to do any strenuous physical activity for 24 hours. This includes lifting, bending, doing housework, gardening, or doing exercise such as jogging or bicycling. They will be given instructions before they leave as to how to contact the ADRC staff if they should have any problems or concerns after the procedure. They will be contacted by the ADRC team on the day after the lumbar puncture to assess their status. Those who develop persistent positional headaches because of CSF leak will return for an evaluation and a possible blood patch.

Potential, rare risks of lumbar puncture include allergic reaction to lidocaine, infection, damage to nerves in your back, and bleeding that may affect the spinal cord or brain. The risk of these is very small. To minimize these risks, the lumbar puncture procedure will be performed by a physician specifically trained in the procedure.

Some lumbar punctures may be performed under fluoroscopy based on physician judgment or participant's preference. For those done under fluoroscopy there are additional risks similar to risks from other X-ray procedures. The radiation dose received varies depending on the individual procedure. On average, radiation received during a fluoroscopic procedure is about one-third of the annual ambient x-radiation dose that people receive from nature and man-made sources. Risks associated with exposure to very high levels of radiation include radiation-induced injuries to the skin and underlying tissues which occur shortly after the exposure, and radiation-induced cancers, which may occur sometime later in life. The probability that a person will experience these effects from a fluoroscopic procedure is very small. To minimize the radiation risk, fluoroscopy will be performed with the lowest acceptable exposure for the shortest time necessary.

Clinical and safety outcomes, clinical safety results will be recorded on ADRC case record forms and USC medical records. All data will be collected specifically for the ADRC and affiliated studies. After completion of the studies and confirmation of the accuracy of the research records, personal identifying information, and protected health information, will be separated from the research records, such that the research records will not be identifiable with any study participant. Paper records will be stored in a locked cabinet. Only study personnel with completed human subject protection training will have access to the obtained data.

3.1.3 Potential Benefits of the Proposed Research to Human Subjects and Others

The research procedures of ADRC studies do not generally directly benefit participants except insofar as some evaluations may inform a new clinical diagnosis and possibly improve the quality of participants' medical care. As described above the potential risks of participation are low to moderate and can be effectively mitigated. Participants are told that they are unlikely to accrue personal, tangible benefit from their research participation. Nonetheless, for many, participation is intrinsically rewarding because it is a way that they can contribute to

better understanding of illnesses that may have potentially affected them. Thus, there is a low degree of personal benefit, a substantial degree of general societal benefit to be weighed in relation to lesser, controllable risks.

3.1.4 Importance of the Knowledge to be Gained

Alzheimer's disease is an incurable, progressive, fatal illnesses. Vascular dementia may have a more variable, but also often progressive course. The knowledge to be produced by the studies associated with this grant is of significant value, particularly regarding the investigation of at-risk factors for neurodegeneration, dementia etiology and the potential for cerebrovascular disease to interact with Alzheimer pathology to produce cognitive impairment. The new information collected will help to better define diagnosis, prognosis, and potential treatments for Alzheimer disease, vascular dementia, and the mixture of both. The affiliated translational studies may help to better define potential treatments. The importance is that we may better understand how to prevent and treat aspects of cognitive decline with aging.

The risks to participants are reasonable in relation to the importance of the knowledge that reasonably may be expected to result. There are no known risks to appropriately selected individuals from MRI studies at the energy levels proposed in this study. The potential benefit of these studies is the advancement of scientific technique and knowledge. The success of these studies may provide additional insight useful for clinical prediction, diagnosis, and management of patients.

Overall Structure of the Study Team

See ADMIN Core

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Administrative Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: 1540 Alcazar Street		Street2:
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

Project/Performance Site Location 1

Organization Name: Rancho Research Institute

* Street1: 7601 E Imperial Hwy		Street2:
* City: Downey	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90242-3456
UEI: QAXKFGHBCMX3	* Project/Performance Site Congressional District: CA-040	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	Admin_Abstract_Final1044447019.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	Admin_References1044447022.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Administrative Core

The Administrative Core will provide the leadership, administrative mechanisms and oversight to enable the USC ADRC to achieve its overall aims and foster core interactions of the center. By providing a clear and transparent mechanism for communication, resource allocation, budget management and insuring all regulatory and participant safety processes are adhered to, the Administrative Core will further the overall missions as described in each of the components of this proposal. The Administrative Core has 4 specific aims: 1) **Guide the scientific direction of the ADRC** around its three overarching scientific goals to recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for carriers of *APOE* ε4**,** promote local ADRD research at USC, and further national collaborations, **2) Coordinate fiscal, operational, and regulatory functions across all cores** to ensure appropriate, optimal, and effective utilization of center resources. 3) **Promote participation across all Cores in collaborative national research initiatives**, including NACC, NCRAD, ADGC, ADNI, ACTC, DVCID, ACAD, SCAN, and CLARiTI, leveraging our metabolic, vascular disease, and neuroimaging expertise to clarify factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers, 4) **Attract early-stage and new investigators, and Support Training and Education** through mentorship, development projects (DP), seminars, and research symposia and provide public education on the importance of understanding and addressing genetic, metabolic and vascular risk profiles for brain health.

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RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Helena Middle Name Chang Last Name*: Chui Suffix:

Position/Title*: Chair and Professor of Neurology
Organization Name*: University of Southern California
Department: KSOM Neurology
Division:
Street1*: 1540 Alcazar Street
Street2:
City*: Los Angeles
County: Los Angeles
State*: CA: California
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 90033-0000

Phone Number*: (323) 442-7686 Fax Number: (323) 442-7689 E-Mail*: helena.chui@med.usc.edu

Credential, e.g., agency login: changchui

Project Role*: Other (Specify) Other Project Role Category: Admin Core Lead

Degree Type: MD Degree Year:

File Name

Attach Biographical Sketch*:

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Arthur Middle Name W. Last Name*: Toga Suffix:

Position/Title*: Provost Professor
Organization Name*: University of Southern California
Department: KSOM Neuroimaging and Informatics Institute
Division:
Street1*: SSB 102
Street2:
City*: Los Angeles
County: Los Angeles
State*: CA: California
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 90089-0000

Phone Number*: (323) 442-7246 Fax Number: (323) 442-2230 E-Mail*: toga@usc.edu

Credential, e.g., agency login: TOGA22

Project Role*: Other (Specify) Other Project Role Category: Admin Core Lead

Degree Type: PhD, MS, BS Degree Year:

File Name

Attach Biographical Sketch*:

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Hussein Middle Name Last Name*: Yassine Suffix:

Position/Title*:	Associate Professor of Medicine
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	2250 Alcazar St.
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-1909 Fax Number: (323) 442-2082 E-Mail*: hyassine@usc.edu

Credential, e.g., agency login: hyassine

Project Role*: Other (Specify) Other Project Role Category: Associate Director

Degree Type: MD Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Yassine_060220241044447124.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Caleb Middle Name E Last Name*: Finch Suffix:

Position/Title*:	Professor
Organization Name*:	University of Southern California
Department:	Leonard Davis School of Gerontology
Division:	
Street1*:	3715 McClintock Avenue
Street2:	
City*:	Los Angeles
County:	
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (213) 740-5156 Fax Number: E-Mail*: cefinch@usc.edu

Credential, e.g., agency login: CEFINCH

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD, BS Degree Year:

	File Name
Attach Biographical Sketch*:	Finch_NIH_Biosketch_Final1044447288.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Freddi Middle Name Last Name*: Segal-Gidan Suffix:

Position/Title*:	Associate Professor of Clinical Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	

Street1*:	OCH Rlm
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (562) 401-8130 Fax Number: (562) 803-6900 E-Mail*: segalgi@usc.edu

Credential, e.g., agency login: Segalgi

Project Role*: Other (Specify) Other Project Role Category: Site PI

Degree Type: PhD Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Segal_Gidan1044447024.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Yassine, Hussein

eRA COMMONS USER NAME (credential, e.g., agency login): hyassine

POSITION TITLE: Professor of Neurology, Medicine, Gerontology, Physiology & Neuroscience (with tenure)

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Beirut Arab University, Beirut, Lebanon	M.D.(w honors)	06/2003	Medicine
American University of Beirut, Beirut, Lebanon	Internship	06/2004	Pediatrics
Case Western University, Ohio	Residency	06/2007	Internal Medicine
Schwartz Center of Metabolism, Ohio	Research Scholar/ Postdoc	06/2007	Exercise Physiology, Aging
University of Arizona – Tucson	Research Fellow	06/2010	Endocrinology

A. Personal Statement

I am a tenured Professor of Neurology, Medicine, and Physiology and Neuroscience, and Director of the Center for Personalized Brain Health at USC. Our research combines imaging studies, clinical trials, observational cohorts and animal models with a focus on AD/APOE. During the past 10 years, our work has brought deep knowledge related to how apolipoproteins bear AD risk.

I would like to highlight my commitment to diversity and mentoring. Over the course of my 14 years as an independent investigator, I have been fully devoted to the training and championing of a diverse population of graduate and undergraduate students and post-doctoral fellows in my lab. I have mentored more than 50 scholars, and all have gone on to pursue further training in graduate, medical school, residency or fellowship. Of the 7 faculty, four are currently doing an academic postdoc, 1 is an Associate Professor on the teaching track and 2 at the Assistant Professor level. Mentoring for me is a privilege, and I feel lucky to not only have been mentored by amazing scientists myself, but to have trainees who trust me enough to mentor them. The successes of my trainees are what drive me as a mentor and scientist, and I believe this is how the science and people flourish. I chaired the national ADC REC steering committee in 2022 and helped lead the Mock Study section boot camp.

I am the past co-chair of the Nutrition, Metabolism and Dementia Professional Interest Area (PIA) of the Alzheimer's Association and sits on the ISTAART Advisory council 2022-2024 and serve on NIH study sections, Alz Association and NACC review panels. As an associate director of the USC ADRC, I have immersed himself in all aspects of this competitive renewal. My contributions can be seen in our scientific theme, innovative recruitment approaches (e.g., GeneScreen, Spatial-Geocoded maps, incorporation of the EHR as a source of real-world data), REC weekly seminars and annual symposiums.

Ongoing projects to highlight include:

NIA RF1AG078362 (mPI)
Yassine (contact PI)
09/30/22 - 10/01/27
Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults

NIA RF1AG076124
Yassine (PI)
05/15/22-05/31/27
Brain cPLA2 as a mechanism for neuroinflammation in AD/ADRD with and without APOE4
There is no overlap with the proposed investigation. The new application focuses on mitochondrial functions in synaptosomes in relation to cPLA2 activation profiles.

NIA R01AG067063
Yassine (PI)
05/15/20- 02/28/25
ApoE, ABCA1 and endosomal dysregulation in AD.

NIA R01AG054434
Yassine (PI)
09/01/17-04/30/24
The delivery of essential fatty acids to the Brain in Alzheimer's disease

ADDF GC-201711-2014197
Yassine (PI)
03/01/19-03/01/24
Effect of Supplementation with High Dose DHA on Cognitive Outcomes in APOE4 Carriers: “PreventE4 Trial”

NIA P30AG066530
Chui (PI), Yassine (REC Leader)
04/01/20-03/31/25
USC Alzheimer Disease Research Center (ADRC)

B. Positions, Scientific Appointments, and Honors

Position

2024 - Present	Professor of Neurology and Medicine, University of Southern California, Los Angeles, with tenure
2022 - 2024	Advisory Council, Alzheimer’s Association
2021 - present	Kenneth and Bette Volk Endowed Chair of Neurology
2020 - 2021	Chair, Steering committee, NIA, ADRC REC
2020 - present	USC ADRC REC leader
2019 - present.	Co-chair, Nutrition Metabolism and Dementia PIA, ISTAART, Alzheimer’s Association
2020 - 2024	Associate Professor of Neurology and Medicine, University of Southern California, Los Angeles, with tenure
2019 - 2020	Assistant Professor of Neurology, University of Southern California, Los Angeles
2012 - 2020	Assistant Professor of Medicine, University of Southern California, Los Angeles
2010 - 2012	Assistant Professor of Clinical Medicine, Section of Endocrinology, Diabetes and Hypertension, Department of Medicine, University of Arizona, Tucson
2009 - 2012	Co-appointment, Arizona Center on Aging
2008 - 2010	Fellow, Section of Endocrinology, Diabetes and Hypertension, University of Arizona
2007 - 2008	Postdoctoral Research Fellow, Division of Endocrinology
2007 - 2008	Chief Resident, Southern Arizona VA Healthcare System, Tucson, AZ
2004 - 2007	Resident, Department of Medicine, Case Western Reserve University, Cleveland, OH
2003 - 2004	Intern, Department of Pediatrics, American University of Beirut, Beirut, Lebanon

Honors

2009 Presidential Poster Award, Endocrine Society
2009 Department of Veterans Affairs VISN 18 Outstanding Research Award
2007 Resident Teacher of the Year Award, Case Western Reserve University, Cleveland, OH
2003 Dean's Honor List at graduation, Beirut Arab University, Beirut, Lebanon
2003 Dean's Distinguished Student Medicine Day Award, Beirut Arab University, Beirut, Lebanon

C. Contributions to Science

1. **Designing clinical trials for Alzheimer disease (AD) prevention using omega-3s:** May lab has a strong interest in brain lipid metabolism in APOE4. We demonstrated that the association of the omega-3 docosahexaenoic acid (DHA) with brain AD pathology is stage specific. In non-demented individuals, lower serum DHA levels are associated with brain amyloidosis. However, this association disappears in patients with mild AD. This suggested to us that interventions may have a role in prevention and not treatment. Dr. Yassine is the PI of PreventE4 trial (NCT03613844) to test high dose DHA in cognitively normal APOE4 carriers. We then examined the mechanisms for why persons with APOE4 AD had a greater ratio of arachidonic acid (AA) to DHA in the brain. This led to identification of cPLA2 activation as one of the major mechanisms/pathways of neuroinflammation in APOE4 AD.

    a. **Yassine HN,** Braskie MN, Mack WJ, Castor KJ, Fonteh AN, Schneider LS, Harrington MG, Chui, HC Literature-Based Association of Docosahexaenoic acid Supplementation with Alzheimer's Disease Stage in Apolipoprotein E ε4 carriers. JAMA Neurol. January 17, 2017. doi:10.1001/jamaneurol.2016.4899
    b. **Yassine HN.** Targeting prodromal Alzheimer's disease: too late for prevention? Lancet Neurol. 16(12):946-947, 2017. PMID: 29097165 PMCID: PMC5808883. doi.org/10.1016/S1474-4422(17)30372-1
    c. Arellanes, IC, ..**Yassine HN**. Brain delivery of supplemental docosahexaenoic acid (DHA): A randomized placebo-controlled clinical trial. EBioMedicine (Lancet). 2020;59 :102883. doi: 10.1016/j.ebiom.2020.102883. PubMed PMID:32690472 PubMed Central PMC7502665.
    d. Wang S, Li B, Solomon V, Fonteh A, Rapoport SI, Bennett DA, Arvanitakis Z, Chui HC, Sullivan PM, **Yassine HN**. Calcium-dependent cytosolic phospholipase A2 activation is implicated in neuroinflammation and oxidative stress associated with ApoE4. Mol Neurodegener. 2022 Jun 15;17(1):42. doi: 10.1186/s13024-022-00549-5. PMID: 35705959

2. **Lipid metabolism in AD**: Over the past 6 years, we identified that ApoE4 proteins propensity to aggregate in the lipid free form drive AD pathology using elegant experiments in human and mice brains. The significance of these findings is in identifying ApoE lipidation by inducing ABCA1 activity to reduce ApoE aggregation as a target to prevent AD pathology. More recently, we identified an association between aggregated apoE and activation of calcium dependent phospholipase A_2 as a mechanism for altered DHA and AA brain uptake.

    a. Martinez AE, Weissberger G, Kuklenyik Z, He X, Meuret C, Parekh T, …**Yassine HN**. The small HDL particle hypothesis of Alzheimer's disease. Alzheimer's & Dementia. 2022 Apr 13. doi: 10.1002/alz.12649. PMID: 35416404

    b. Rawat V*, Wang S*, Sima J, Bar R, Liraz O, Gundimeda U, Parekh T, Chan J†, Johansson JO, Tang C, Chui HC, Harrington MG, Michaelson DM, and **Yassine HN**. ApoE4 Alters ABCA1 Membrane Trafficking in Astrocytes. J Neurosci. 2019 Nov 27;39(48):9611-9622. doi: 10.1523/JNEUROSCI.1400-19.2019. PMCID: PMC6880458

    c. **Yassine, HN.,** Croteau, E., Rawat, V., Hibbeln, J. R., Rapoport, S. I., Cunnane, S. C., & Umhau, J. C. DHA brain uptake and APOE4 status: A PET study with [1-11C]-DHA. Alzheimer's research & therapy, 2017:*9*(1), 23. PMCID: PMC5364667

    d. **Yassine HN,** Feng Q, Chiang J, Petrosspour LM, Fonteh NA, Chui HC, Harrington MG, ABCA-1 mediated cholesterol efflux capacity to cerebrospinal fluid is reduced in patients with mild cognitive

impairment and Alzheimer's disease. Journal of the American Heart Association. 2016. 5(2): p. e002886. PMID: 26873692

3. **Effect of diet and exercise on metabolism:** My first project was during my research year at Case Western Reserve University in 2006 as a resident in internal medicine. I worked with Dr. John Kirwan to compare the effect of diet vs. exercise on insulin resistance and lipids in older adults. I led a randomized clinical trial that suggested that weight loss may not be a preferable approach for enhancing insulin sensitivity on older adults due to loss of muscle mass. In contrast, exercise enhanced insulin sensitivity and preserved muscle mass.
    a. **Yassine HN**, Marchetti CM, Krishnan RK, Vrobel TR, Gonzalez F & Kirwan JP 2009 Effects of exercise and caloric restriction on insulin resistance and cardiometabolic risk factors in older obese adults - a randomized clinical trial. J Gerontol A Biol Sci Med Sci., 64(1):90-5. PMCID: PMC2691195

4. **Structure and function of apolipoproteins:** After the completion of my clinical fellowship in 2009, I did a post-doctoral research fellowship in protein structure and function at the Lau Lab at University of Arizona. We identified novel biomarkers of diabetes complications. During this time, I became interested in functional and structural properties of lipoproteins, and identified that hypertriglyceridemia of diabetes enhances cholesterol efflux functions, and a mechanism that links apoC-III sialic acid with elevated plasma triglycerides. I was awarded the ATVB junior investigator award in 2014 for my work on apoC-III sialylations.

    a. **Yassine HN,** Belopolskaya A, Schall C, Stump CS, Lau SL & Reaven PD 2014 Enhanced cholesterol efflux to HDL through the ABCA1 transporter in hypertriglyceridemia of type 2 diabetes. Metabolism, 63(5):727-34. PMCID:PMC4014062.
    b. *Koska J, ***Yassine HN**, Trenchevska O, Sinari S, Schwenke D, Yen F, Billheimer D, Nelson R, Nedelkov D, Reaven PD, Disialylated Apolipoprotein C-III Proteoform is Associated with Improved Lipids in Prediabetes and Type 2 Diabetes. J Lipid Res. 2016. PMID: 26945091. *** First co-author**

Complete List of Published Work in MyBibliography:
http://www.ncbi.nlm.nih.gov/sites/myncbi/hussein.yassine.1/bibliography/45600448/public/?sort=date&direction=ascending.

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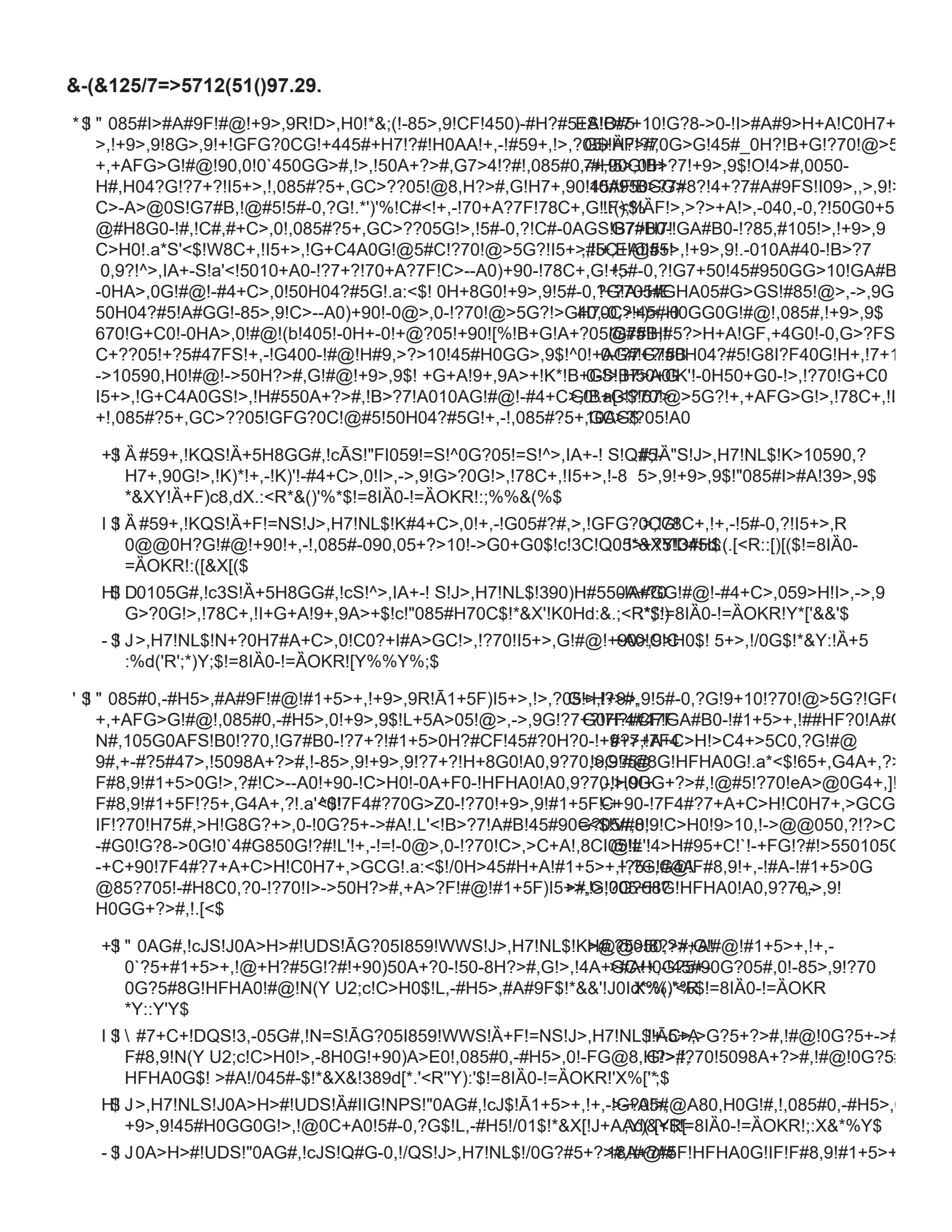
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BIOGRAPHICAL SKETCH

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NAME: Freddi Segal-Gidan

eRA COMMONS USER NAME (credential, e.g., agency login): Segalgi

POSITION TITLE: Associate Clinical Professor Neurology, Family Medicine & Gerontology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Reed College, Portland, OR	BA	05/1976	Psychology & biology
The Johns Hopkins University, Baltimore, MD	BS	05/1978	Health sciences
University of Southern California, Los Angeles, CA	PA Fellow	07/1987	Geriatric medicine
University of Southern California, Los Angeles, CA	PhD	05/1997	Gerontology-public policy
University of Southern California, Los Angeles, CA	Post Doc	12/2000	Neurobiology & Aging

A. Personal Statement

My professional career as clinician, researcher and educator has been devoted to improving the care of older adults, and for the past three decades focused primarily on those with dementia, along with other conditions that impact cognitive function I began as staff PA for the State of California Alzheimers Disease Diagnostic and Treatment Center (ADDTC) at Rancho Los Amigos, affiliated with USC, led the Education Core for the renamed Alzheimers Disease Research Center of California (ARCC), became the co-director of the Rancho/USC ARCC, and for the past fifteen years have been the director of the (renamed) USC-Rancho California Alzheimers Disease Center (CADC). I have been affiliated with the USC Alzheimers Diseae Reseach Center (ADRC) since my post-doctoral fellowship in Neurobiology and Aging with the USC-Rancho CADC serving as part of the Clinical Core for the USC ADRC. Most recently I served 3 yrs as the director of the USC ADRC's Outreach and Recruitment Core. I was a co-investigator for the LALES-cog project of the USC ADRC (2005-2009), participated in the clinical evaluation of research subjects for the NIH-NIA Aging Brain Program Project, and continue to perform clinical evaluations for patients involved in the USC-ADRC's longitudinal study. . I established the USC Alzheimers Disease Consortium which began the bi-annual Many Faces of Dementia conference, which for 20 years has provided ongoing education focused on the latest research and care practices in dementia for health professionals in the greater Los Angeles area. As the patient population served by the USC-Rancho CADC has evolved to include a majority of persons of Hispanic heritage my interest in the barriers faced by individuals with dementia and their families from these communities has increased. In 2019 the USC-Rancho CADC was awarded funds from the California Alzheimers Research Program to investigate the needs of caregivers of persons with an inherited form for Alzheimers disease of Mexican origin. That same year the Center received pilot funds to investigate dementia caregiver self care practices and the use of alternative therapies. My background as a geriatric trained clinician, my extensive experience working with patients and families with dementia and my long term association with both the USC-Rancho CADC and the USC ADRC provide me with the experience and leadership skills to be able to oversee an increasing number of interrelated research projects.

a. Kaskie B, Bobbit J, Herrera JW, Bhagianadh D, **Segal-Gidan F**, Brummel-Smith K. Cannabis Use among Persons with Dementia and Their Caregivers: Lighting up an Emerging Issue for Clinical Gerontologists, Clinical Gerontologist, 2021: 44 (1),42-52.
b. Cannabis Use and Older Adults: What Do We Know? (Symposium) Gerontology Society of America, annual meeting, Nov. 2019.

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2014-present	Associate Clinical Professor, Neurology Department, USC, Keck School of Medicine
2007-present	Director, Alzheimer's Disease Research Center of California (ARCC)/ California Alzheimer's Disease Center, Rancho Los Amigos National Rehabilitation Center/ USC, Downey, CA
2003-2007	Co-Director, Alzheimer Disease Research Center of California (ARCC), Rancho Los Amigos National Rehabilitation Center/ USC, Downey, CA
2000-2014	Assistant Clinical Professor, Neurology, Family Medicine & Gerontology USC, Keck School of Medicine
1998-2000	Director, Stroke Education and Research, Rancho Los Amigos National Rehabilitation Center, Downey, CA
1997-2003	Director, Inpatient Stroke Rehabilitation Services, Rancho Los Amigos National Rehabilitation Center, Downey, CA
1997	Adjunct Professor, School of Policy, Planning and Development, USC
1997-2001	Lecturer, Gerontology & Health Policy. L. Davis School of Gerontology, USC
1995	Lecturer, Health Policy, USC, School of Social Work. Los Angeles, CA
1991	Policy Intern: AARP-ANDRUS Leadership in Policy and Aging Programs, Kellogg Foundation Project: An In-Home Intervention Prog for Community-Dwelling Elders Over Age 75, LA, CA
1989-2000	Clinical Instructor, Neurology Department, USC, School of Medicine
1989-2000	Adjunct Lecturer & Clinical Instructor, Department of Family Medicine Physician Assistant Program, University of Southern California School of Medicine, Los Angeles, CA
1986-1989	Clinical Instructor, Department of Family Medicine, Physician Assistant Program, USC
1985-1989	Physician Assistant, Internal Medicine, California Primary Physicians, Los Angeles, CA
1984	Physician Assistant, Rehabilitation Medicine, Walker Institute, West Los Angeles, CA
1979-1984	Physician Assistant, Internal Medicine, Cranford L. Scott, M.D., Inglewood, CA
1978-1979	Physician Assistant, Wainaie Comprehensive Health Care Center, Wainaie, HI

Scientific Appointments

2020-present	Member, Provider Leadership Group, Brain Health Partnership, Us Against Alzheimers
2019-2021	Member, Health of the Profession Commission, American Academy of PAs
2008-present	Member, Medical Scientific Advisory Board, Alzheimers Los Angeles
1979- present	California PA License #10482

Honors

2023	Fulbright Specialist, HAN University, The Netherlands
2020	Founders Award, Many Faces of Dementia
2016	Fulbright Specialist Program, Roster Candidate
2008	Paragon Award PA-Physician Partnership, American Academy of Physician Assistants
1994	President's Annual Leadership Award, California Academy of Physician Assistants

C. Contributions to Science

1. An essential element toward improving care for older adults with dementia is the translation of research into clinical practice. Through written publications aimed at primary care providers that discuss the most current knowledge about dementia care and practice I have attempted to educate primary care providers to understand changes in practice to better meet the needs of patients with dementing illnesses and their families.

a) Sideman AB, Alagappan C, **Segal-Gidan F**, Alving L, Rosen HJ. A stakeholder-engaged process to design and implement a dementia diagnosis Toolkit in primary care settings. Alzheimers and Dementia. 2023.
b) Beavers-Kirby J and **Segal-Gidan F** (eds). Gerontology and Geriatrics for NPs and PAs; An Interprofessional Approach. Elsevier 2023.
c) **Segal-Gidan, F**. Cognitive Decline and Dementia: Primary Care Evaluation and When to Refer. Physician Assistant Clinics 3 (2018) 487-494.
d) **Segal-Gidan F**. Normal Pressure Hydrocephalus; A Case Presentation of Challenges in Recognition and Diagnosis. Clinicians Review 2015; 26(12):44-49.

2. Interprofessional team care (previously called multidisciplinary team care) has been recognized as an essential component for improving care and outcomes for older adults. Based on my experiences as part of an interprofessional dementia care team I was invited to serve as faculty of the HRSA funded Geriatric Workforce Enhancement Project (GWEP). This funding supported expansion of a geriatric interprofessional (IP) curriculum program involving health professional students that was developed under a prior HRSA funded primary care grant, as well as development of an additional IP student program and IP geriatric clinic.

a) Rodgers J**, Segal-Gidan F**, Reilly JM. Impact of an Interprofessional Health Student Education Program on Older Adult Participants. Gerontol Geriatr Med. 2022.
b) **Segal-Gidan F**, Resnik C, Reilly JM, Stevens GD. Enhancing Geriatric Workforce Competency: Impact of 3 Distinct Geriatric Interprofessional Education Programs. Health Interprofessional Practice and Education. 2022
c) Reilly JM, Halle A, Resnik C, Teo J, Williams B, Harris P, **Segal-Gidan F.** Qualitative analysis of an interprofessional in-home, community geriatric education program. Gerontology and Geriatric Medicine, March 2021.
d) Reilly JM, Stevens G, Halle A, Resnik C, Teoh J, Williams B, Joosten-Hagye D, **Segal-Gidan F.** Interprofessional, older adult, team-based home visits: A 6-year prospective analysis. Gerontol Geriatr Educ. 2020 May 3:1-11. doi: 10.1080/02701960.2020.1758081

3. Increasing involvement of individuals from underserved Hispanic communities has been a focus of my clinical career that I have tried to expand into the research arena. Funding through the USC ADRC supported expansion of the NIH-funded Los Angeles Latino Eye Study (LALES) research project that added a cognitive assessment component to the study of participants over age 55. In recent years my clinical practice has included a growing number of early onset dementia patients of Mexican heritage that carry a rare autosomal dominant genetic defect. In conjunction with researchers at the USC ADRC I am engaged in several research projects with this population to better understand the impact of EOAD on families and learn from their experiences to better meet the patient and caregiver needs.

a) Whithers M, **Segal-Gidan F,** Herrera J, Sanchez-Cortez K, Ringman J. My backpack is so heavy': Experiences of Latino Caregivers of Family with Early-Onset Alzheimer's. J Ameri Geriatri Soc, 2021 https://doi.org/10.1111/jgs.17091
b) Gatto NM, Varma R, Torres M, Wong TY, Johnson PL, Segal-Gidan F, Mack WJ. Retinal microvascular abnormalities and cognitive function in Latino adults in Los Angeles. Ophthalmic Epidemiol. 2012; 19(3):127-36. PMID: 22568425.
c) Segal-Gidan F, Varma R, Mack W. Factors influencing driving status in a Latino population, J Aging and Health, 22(3): 332-347, April 2010. PMID: 20142628.

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

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5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☒ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

Introduction	
1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	Admin_Aims1044447020.pdf
3. Research Strategy*	Admin_Research_Plan_0614241044447224.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
9. Letters of Support	
10. Resource Sharing Plan(s)	
11. Other Plan(s)	
12. Authentication of Key Biological and/or Chemical Resources	
13. Appendix	
12. Appendix	

SPECIFIC AIMS

The ADRC is led by two mPIs (Chui and Toga) and an Associate Director (Yassine), with the assistance of two administrators (MA Aranda and T Kolchins). The ADRC is comprised of the six required cores, an optional Imaging Core, and the required Research Education Component (REC).

The Administrative Core will provide the leadership, administrative mechanisms and oversight to enable our ADRC to achieve its overall aims and foster core interactions of the center. By providing a clear and transparent mechanism for communication, resource allocation, budget management and insuring all regulatory and participant safety processes are adhered to, the Administrative Core will advance the overall mission as described in each component of this proposal.

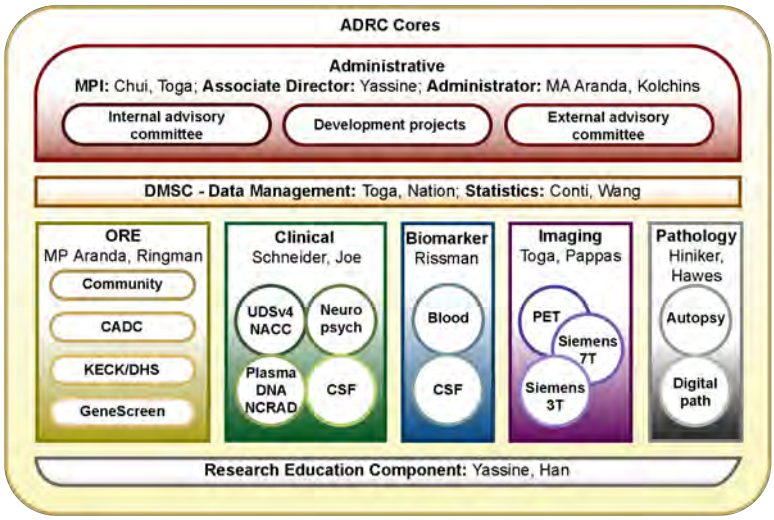


Figure 1. Organizational Structure of the ADRC.

The specific aims of the Administrative Core are as follows:

1. **Guide the scientific direction of the ADRC:** The Administrative Core will develop and foster three overarching scientific goals of the ADRC, working with our Internal Executive Committee (Executive Chair: Caleb Finch), Internal Advisory Committee (IAC), External Advisory Committee (EAC), and Community Advisory Board (CAB).

   A. Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4, focusing on factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers. Our goal is to build a multi-sourced, participant registry with *APOE* genotype, SDoH, and digital cognitive assessment.

   B. Promote local ADRD research at USC**:**
      1. Integrate research programs by holding annual events, pilot fundings, and harmonization of protocols.
      2. Share data, tissues, biofluids, and a participant registry through a common USC ADRC database.
      3. Train the next generation of ADRD researchers and educate participants and the public.
      4. Enhance Diverse Perspectives

   C. Further national collaborations**:** Advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, and comprehensive imaging to national initiatives.

2. **Coordinate fiscal, operational, and regulatory functions across all cores** to ensure appropriate, optimal, and effective utilization of center resources.

3. **Promote participation across all cores in collaborative national research initiatives** and resource sharing, including NACC, NCRAD, ADGC, ADNI, ACTC, DVCID, ACAD, SCAN, and CLARiTI playing a leadership role in research into *APOE* leveraging our metabolic, vascular disease, and neuroimaging expertise.

4. **Attract early-stage and new investigators, and support training and education** especially from under-represented groups to ADRD research through development projects (DP), working with the REC/RCMAR and all core leaders. REC and ORE Core, respectively, sponsor training workshops, research symposia for investigators, and provide public education on the importance of understanding and addressing genetic, metabolic and vascular risk profiles for brain health.

Research Strategy

Significance

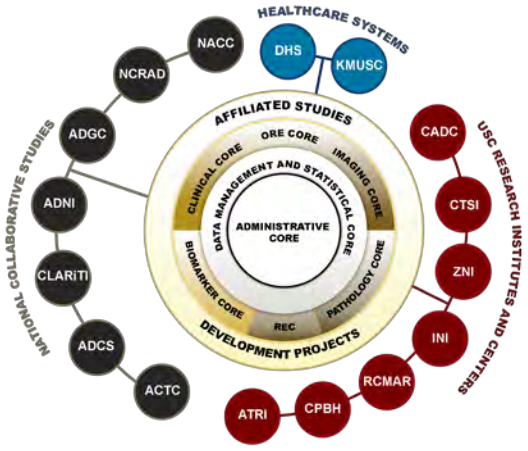


Figure 2: ADRC interactions with USC Research Institutes and Centers (Aim 2) and with National Collaborative Studies (Aim 3)

The Administrative Core provides the overall scientific direction and coordinates activities as well as resources across all ADRC Cores (**Fig. 2**). Second, the Admin Core serves as a catalyst and facilitator for seven university-wide research and training institutes and centers related to the ADRD (**Fig. 2**). Our ADRC goals include furthering the breadth of AD research across the university, beyond the respective entities described. Third, the Admin Core serves as the bi-directional interface with the NIA ADC program and the multitude of national ADRD initiatives (**Fig. 2**). Fourth, the ADRC connects with two clinical enterprises (DHS, KMUSC) which serve a diverse population residing in LA county.

Chui, Toga, and Yassine constitute the KSOM Research Council as: 1) directors of the ADRC (Chui, Toga, Yassine), 2) Chair of the Department of Neurology (Chui), 3) Director of the Stevens Institute on Neuroimaging and Informatics (Toga), and 4) Director, Center for Personalized Brain Health (Yassine). All three principals play leadership roles in the 2023 Dean's strategic affinity groups, which include neuroscience and neurodegenerative diseases. Of note, Chui's service as the Chair of Neurology advances the ADRC's mission in translational research to clinical care and the healthcare system, Keck Medicine of USC (KMUSC).

(A) Evolution of the Administrative Core. Our ADRC is celebrating 40 years of scientific progress. We have a long history of successful participation in this important national program. Throughout that history, we have successfully adapted to gradual changes in the field, program requirements, and opportunities at USC. The leadership of the ADRC has evolved as well. Twenty years ago, Dr. Finch stepped down as founding PI and transitioned that role to Dr. Chui. Five years ago, Drs. Toga and Zlokovic became mPIs along with Dr. Chui as our scientific focus on vascular contributions incorporated research on the blood-brain barrier, novel imaging methods, and a novel emphasis on informatics. Further, this year we are altering the leadership to accommodate a need for succession. The NIA has been clear that a realistic succession plan is important to maintain a healthy and vibrant leadership and to ensure stability. To that end, we have several younger investigators joining in leadership roles of each core and have appointed Dr. Yassine as an Associate Director of the ADRC. Finally, due to an administrative review of Dr. Zlokovic, we have been instructed by NIH and USC to remove Dr. Zlokovic from his PI role while that process is ongoing. We are confident that these personnel changes will enable our ADRC to continue to function successfully and productively for many years to come.

The ADRC is comprised of a diverse transdisciplinary team of investigators from four schools, six departments and three institutes, which range from basic neuroscience (Finch, Yassine, Pike), neuropathology (Hiniker, Rissman, Hawes), translational neuroscience (Yassine, Harrington, Nation, Cohen, Mather), neuroimaging (Toga, Wang, Shi, Braskie, Pappas), clinical trials (Schneider, Yassine), clinical neuroscience (Chui, Schneider, Ringman), neuropsychology (Han, Nation, D'Orazio), social science (MP Aranda), biostatistics (Conti, Wang), and genetic epidemiology (Conti, Shu).

All letters of support, collaboration and participation are located in Overall.

(B) Institutional Commitment. The institutional commitment to the ADRC is extensive and generous (**Table 1**). Beside stable, long-term support for physical space, the Vice President for Research (Ishwar Puri, PhD), the Dean of the Keck School of Medicine (Carolyn Meltzer, MD), the Senior Vice President for Health (Steven Shapiro, MD), and the CEO of Keck Medicine of USC (Rod Hanners, MHA) contribute significantly to supporting ADRC faculty, recruiting physician-scientists, providing research infrastructure, and facilitating interactions with the health care system (e.g., FDA-approved anti-amyloid therapy). Finally, as a clear testament to the importance and confidence in the ADRC, during the next funding cycle, the institutional leadership at USC has committed two million dollars to enable pursuit of new opportunities in the ADRC (see letter of support from Puri, Shapiro, and Meltzer).

Table 1. Institutional Support		
Organizational Unit	**Leadership**	**Support (*See Letter of support)**
Vice President for Research, Senior Vice President for Health Affairs, Dean, School of Medicine	Ishwar Puri, PhD Steven Shapiro, MD Carolyn Meltzer, MD	$2M over five years*
CEO, Linus Health	David Bates, PhD	Real-world Digital Cognitive Assessment*
Dean, Keck School of Medicine	Carolyn Meltzer MD	$100K/yr towards REC scholar (diversity) $6M Center for Personalized Brain Health (CPBH)
CEO, Keck Medicine of USC	Rod Hanners, MHA	Behavioral neurologist, nurse practitioner, memory clinic coordinator, nurse, social worker

Progress Aim 1: To guide the scientific direction of the ADRC. A) Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles enriched for *APOE* ε4 on cognitive decline and AD biomarkers, B) Promote local ADRD research at USC, and C) Further national collaborations.

The prime directive of the Administrative Core is to establish the scientific direction and operational coordination of all cores and components in the ADRC, as they interface with local and national initiatives in AD, along with other neurodegenerative and vascular disorders. Furthermore, our responsibility is to ensure that the scientific focus of the ADRC reflects new and important avenues for investigation. We propose the following Specific Aims:

(1.A) Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4. A major theme during the past funding cycle has focused on vascular contributions to cognitive impairment. Discoveries made in this area relate to ASL, WMH, PVS, glymphatics, and the neurovascular unit, resulting in 92 publications. We refined imaging protocols and analytics to measure the integrity of the blood brain barrier using dynamic contrast enhanced MR imaging (1), and sPDGFRβ (2). We also collected UDS data on 686 participants in 2020-2023 and contributed data to NACC, NCRAD, MarkVCID, DVCID, and SCAN. Our scientific theme is now focused on recruiting a cohort that reflects the diversity of Los Angeles and is enriched for *APOE ε4,* emphasizing vascular, metabolic, and inflammatory factors that mitigate the effect of *APOE* ε4 on cognitive decline and ATNvi. ADRC publications related to *APOE ε4* are summarized in the Overall (**Fig. 4**).

(1.B) Promote local ADRD research at USC. Through leadership roles at the university, monthly executive committee meetings, annual USC events, joint funding opportunities, promotion of weekly REC seminars and training grants, and the ADRC-database (ADRC-DB), the Admin Core serves as a catalyst to ADRD research at USC. Important developments/findings that have directly utilized ADRC resources are summarized in **Table 2**.

Our ADRC has had an enormous influence on the university's commitment to Alzheimer's disease research. As part of our research advancement efforts, the ATRI group, led by Paul Aisen, was recruited from UCSD in 2015. Numerous ATRI faculty are now integrated into the ADRC. In 2023, we recruited both Annie Hiniker MD, PhD from UCSD to lead the NP Core, and Dan Nation from UCI, who co-leads the DMSC among other roles. Alzheimer's research became part of our identity, with clear messaging, resources, recruitments, and investments. This is a testament to the influence of our ADRC. Alzheimer's disease research is now a significant focus across many schools (Keck Medicine, Davis Gerontology, Dornsife Letters, Arts, and Sciences, and Viterbi Engineering) at USC.

(1.C) Further national collaborations. The Administrative Core promotes and oversees ADRC engagement in multiple national initiatives. The Clinical and Path Cores contribute UDS data to NACC. The Clinical and Biomarker Cores send plasma to NCRAD, ADGC, and DVCID. In addition to serving as a participating site for ADNI, SCAN, CLARiTI, ACTC, and ADCS, the USC ADRC enforces a leadership role among several national collaborations. For example, Toga directs LONI which serves as the imaging repository for ADNI and SCAN. Rissman directs the Biomarker Cores for both the USC ADRC and ACTC. Contributions to scientific advancements which directly utilize resources of the USC ADRC are summarized in **Table 2** (AIM1 C).

During the current funding cycle, the ADRC contributed to over 398 publications. Several important contributions to research on AD, ADRD, and aging that directly leveraged ADRC core resources are summarized in **Table 2**.

TABLE 2: Research Findings that Directly Utililized ADRC Core Resources				
Findings	**Contribution of ADRC Core Resources**	**Overall Goal**		
		Aim 1A *APOE*	**Aim 1B Local**	**Aim 1C National**
APOE ε4 is associated with **breakdown of the BBB** in the hippocampus, which is apparent in cognitively unimpaired *APOE ε4* carriers and unrelated to CSF, PET amyloid-beta or tau biomarkers (1).	245 participants, predominantly recruited by the USC ADRC *Clinical Core*. DCE-MRI analyzed by the USC Imaging Core. CSF sPDGFRβ levels analyzed by the USC *Biomarker Core*.	x	x	x
CSF levels of **small HDL particles** are correlated with CSF Aβ42 and cognitive measures (3). *APOE ε4* associated with larger aggregated particles.	180 individuals aged ≥ 60 years from the USC ADRC *Clinical Core* and Huntington Memorial Research Institute (HMRI) Aging Program.	x		
CSF *APOE ε4* isoforms are the least **glycosylated** and percent CSF *APOE* glycosylation positively correlated with CSF Aβ42 levels (4).	106 matched CSF and plasma samples from the USC *Clinical Core*.	x		
Prevent ε4: Clinical trial of high dose DHA supplementation stratified by carriers of *APOE* ε4 (5).	365 individuals were recruited at USC. Secondary structural and fMRI outcomes analyzed by the ADRC *Imaging Core*.	x		
High baseline CSF levels of the BBB pericyte injury biomarker soluble **sPDGFRβ** predicted future cognitive decline in *APOE* ε4 carriers.	Participants are recruited by the USC ADRC *Clinical Core* team (68 enrolled during current ADRC funding cycle).	x	x	x
A mitochondrial DNA variation within the mitochondrial microprotein **SHMOOSE** increases the risk of AD by 30% and is associated with AD biomarkers, neuroimaging, and transcriptomics (6).	79 ADRC participants with CSF were evaluated by the *Clinical Core*. MRI, amyloid and tau PET were analyzed by the Imaging core.		x	
Pericyte loss (Koronyo-Hamaoui) together with significant Aβ deposition in retinal microvasculature and pericytes in AD. Retinal PDGFRβ loss significantly associated with increased retinal vascular Aβ40 and Aβ42 burden (7).	56 post-mortem eyes obtained from *Neuropathology Core* with UDS from *Clinical* Core and *DMSC Core*.		x	
(Koronyo-Hamaoui) Significant increases in Aβ42 forms and novel intraneuronal Aβ oligomers, which were closely associated with exacerbated **retinal macrogliosis, microgliosis, and tissue atrophy** (8).	Over 70 post-mortem eyes were obtained from the *Neuropathology Core* and UDS from *Clinical Core* and *DMSC*.		x	
MarkVCID (Wang). The ALPS index determined by DTI analysis along the perivascular space (DTI-ALPS) demonstrated favorable inter-scanner, inter-rater, and test-retest reliability (n=50), offering a potential biomarker for *in vivo* evaluation of glymphatic function (9).	Participants evaluated by the *Clinical Core* (UH2: n=63; UH3: n=164).		x	x
HABLE (O'Bryant). MRI measures of neurodegeneration from AT(N) differ among Mexican Americans (MA) vs. Non-Hispanic Whites (NWH) (10).	Imaging analyses (n=1305) performed by the USC ADRC *Imaging Core*.			x
HAB-HD (O'Bryant). Proteomic profiles were highly accurate for detecting neurodegeneration among MA and NHW, but N+ profile varied by diagnostic status and ethnicity (11).	Proteomic profiles (n=1,291) performed by the USC *Biomarker Core*.			x
DVR (Marmarelis). Quantitative analysis and a Composite Diagnostic Index of cerebral CO_2 vasoreactivity under resting spontaneous conditions revealed a significant vasoreactivity deficit in MCI/dementia due to AD (Marmarelis et al., AAIC 2023; 2024).	80 participants were recruited through the USC ADRC *Clinical Core* team.		x	x
ADVCID (Yan). Elevated **carotid pulse-wave velocity and pulsatility** of the lenticulostriate arteries were positively correlated with age and cognitive impairment [Tang,Yan et al., AAIC abstracts (#93187,#93252) 2024].	All participants (3T MRI n=40) and 7T MRI (n=25) were recruited by the ADRC *Clinical Core* team and scanned by the *Imaging Core*.		x	x

Progress Aim 2: Coordinate fiscal, operational, and regulatory functions across all cores to ensure appropriate, optimal, and effective utilization of center resources that are aligned with the NIA-ADRC Program and synergize with local ADRC-affiliated research.

(2) Progress in discovery rests in part by the availability of resources. Our responsibility in the Admin Core is to allocate resources and to ensure that they are apportioned appropriately. Efforts that do not meet milestones or activities that do not contribute are addressed, often through resource adjustment. For example, we have increased support and activities as part of our participant recruitment. Recovering from a slower recruitment period caused by the Covid-19 pandemic necessitated these changes (including the GeneScreen portal), and the results have been positive. Indeed, our current proposal reflects a continuation of these adjustments combined with a significant investment in digital pathology, Biomarker Core assays, and a comprehensive integrated informatics approach for both local needs and national program obligations.

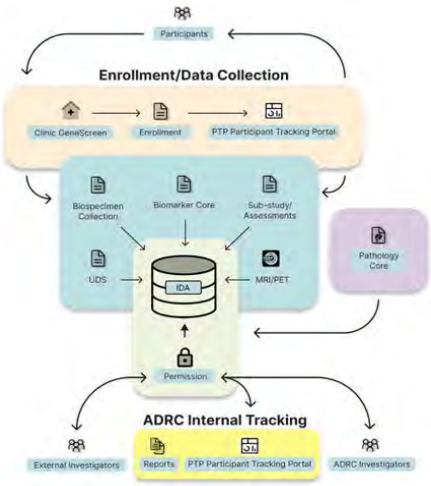


Figure 3. Flow chart of data and participant portal.

By integrating across multiple cores (**Fig. 3**), the ADRC supports deep phenotyping of the ADRC Clinical, Biomarker, Imaging, and NP Cores as well as supporting local and national investigators. These include the provision of high-value clinical, imaging, biomarker, and pathological resources. There is a symbiotic relationship between ADRC expertise, quality-control, resources and scientific questions and resources from independently funded studies which enables expansion of ADRC-trained coordinators, clinicians, statisticians, and imaging studies. Moreover, numerous PIs of affiliated-studies have granted permission to share clinical, imaging, and biomarker data with the ADRC-DB (IDA) database where they can be shared globally with other investigators (See DMSC). We augment the raw data collections with analytics into the ADRC-DB, which further encourages collaboration and avoids effort duplication. Through internal and external collaborations, the ADRC creates a whole that is greater than the sum of its parts. These dynamic interfaces promote the themes of the ADRC and support local, national, and international communities.

The ADRC is also a catalyst for other related programs. The Alzheimer's Association invests heavily in USC projects; philanthropy has been successful because of our ADRC supported research, and other private contributions have resulted because of the work we do as part of the ADRC.

Progress Aim 3: Promote participation across all Cores in collaborative national research initiatives and resource sharing, including NACC, NCRAD, ADGC, ADNI, ACTC, DVCID, ACAD, SCAN, and CLARiTI playing a leadership role in research into *APOE*, leveraging our metabolic, vascular disease, and neuroimaging expertise.

(3) During the past four years, we have contributed to NACC UDS (n=686 unique participants), NCRAD (282 plasma, 148 buffy coat, 154 PBMC, 153 DNA, 53 RNA, etc); DVCID (n=30), SCAN (n=22). Clinical Trials and the Clinical Core infrastructure, led by Lon Schneider, has supported USC ADRC participation in ADNI-3 (n=49), ADNI-4 (n=13), Aducanumab (Biogen, n=5), Solanezumab (A4, n = 32; LEARN (n=9), Lecanemab (AHEAD3-45; ATRI /Eisai, n= 21), Donanemab (Trailbazer, Lilly, n = 6), DIAN-TU (n=4), Exosome (n=108), NOMAD (n=63), and TRC-PAD (n=17) (see **Overall Table 2_a3**: **Therapeutic Trials**). Since its launch in May of 2023, GeneScreen has yielded 404 participants accrued by 1470 responses (see ORE core).

Progress Aim 4: Attract early-stage and new investigators and support training and education with the REC and ORE Core, respectively, sponsor training workshops, research symposia for investigators, and provide public education on the importance of understanding and addressing genetic, metabolic and vascular risk profiles for brain health.

(4) We attract new investigators in several ways. We created a REC listserv of our mentees and faculty that is utilized in our seminar announcements, conferences and workshops. We also provide mentoring opportunities and offer project funding. Further, USC received a five million dollar donation from the Epstein Family Foundation to support Breakthrough Awards for 5 years (2022-2027). Epstein Awards are $250K/year

each, which is five-fold larger than the ADRC Development projects. Chui served as a reviewer for the Epstein Awards in both 2022 and 2023. In 2022, four of 17 applications were funded. In 2023, four of 10 applications were funded. **Table 3** shows two projects that are supported by the ADRC. Importantly, these projects often result in the beginning of larger efforts with independent funding. This is monitored and used as one of the metrics of our ADRC success.

Table 3. Epstein Breakthrough Awards ($250,000/year; renewable for additional year)		
Year	**Faculty**	**Title and ADRC contribution**
2022-2024	Pinchas Cohen, MD Arthur Toga, PhD	"The microprotein SHMOOSE is a new therapeutic candidate for Alzheimer's disease." ADRC provides CSF samples and imaging data.
2022-2024	Mara Mather, PhD Jeiran Choupan, PhD	"Effects of Daily Resonance-Frequency Breathing on Brain Perivascular Space Volume" ADRC is supporting MRI analyses.

During the past four years, the ADRC designated $50K per year for two development projects to support an early-stage investigator (REC scholar or junior faculty member), or to a senior faculty member new to the field of AD. We received 20 applications and funded seven (**Table 4**).

Table 4. ADRC Development Projects ($50,000/year)		
Year	**Faculty Name and Project Title**	**Progress**
Year 1 DP 1.1	Michael Bienkowski, PhD "Informatics based approach to characterize individual heterogeneity and cell-type susceptibility in Alzheimer disease"	K01AG066847 award
Year 2	Nasim Sheikh-Bahaei, MD, PhD "PET-Guided 7T MRS for early diagnosis of Alzheimer's disease"	*funded by the Glazer Foundation through philanthropic*
Year 3 DP 3.1:	Roksana Karim, PhD "Estradiol Therapy, Plasma Biomarkers of Alzheimer's Disease & Cognitive Decline in Healthy Postmenopausal Women (ELITE Study)"	R01ES033705
Year 3 DP3.2: *REC Scholar*	Isaac Asante, PhD Title: "Effect of cPLA2 Inhibition on Inflammatory Mediators in *APOE ε4* Mice"	Publication: (12); USC SC-CTSI KL2 Award
Year 4 DP 4.1	Kai Chen, PhD "PET-MR Imaging of brain lipids and inflammation in *APOE ε4*"	Publication: (13) R21 submitted
Year 4 DP 4.2 REC Scholar	Yaqiong Chai, PhD "MRI-based analysis of glymphatic function and the association with Alzheimer disease progression."	Abstract: "Medical Imaging with Deep Learning (MIDL)" in Nashville, TN, 2023. Abstract: "World Congress of Neurology (WCN)" Montreal, Canada, 2023
Year 4 DP 4.3 *REC Scholar*	Amy Christensen, PhD "Beta-hydroxybutyrate supplementation as a prevention for AD neuropathology in *APOE* AD mice"	Fall NACC meeting October 19-20, 2023, poster was titled "Longevity treatments affect Alzheimer's neuropathology in aged male EFAD mice."

For year five, we awarded two Development Projects to 1) Qiulan Ma, PhD**:** "Novel mechanisms of synaptic failure in Alzheimer's disease pathogenesis," and 2) Tobias Ulmer, PhD**:** "The role and therapeutic potential of the minor CD33." We supported 21 REC scholars in this funding cycle who were promoted or successfully obtained NIH and local funding (**Table 5**).

Table 5: Role of the ADRC in bringing new investigators and stimulating funding			
Name	**Career Stage**	**Role of the ADRC in bring new investigators to the field**	**Funding**
Michael Bienkowski PhD	Early Stage Faculty	Development Award (Mentor: Toga)	K01AG066847
Lirong Yan PhD	Early Stage Faculty	Development Award (Mentor: Chui)	K25AG056594
Elizabeth Joe MD	Early Stage Faculty	Behavioral Neurology Fellowship (Mentor: Ringman/Chui)	CTSI KL2 Scholar REC Scholar R03AG082997
Isaac Asante PhD	Early Stage Faculty	Development Award (Mentor: Yassine)	CTSI KL2 Scholar REC Scholar
Maricarmen Pachicano	Doctoral Student	Diversity Supplement (Mentor: Michael Bienkowski)	Diversity Supplement to K01AG066847
Matt Borzage PhD	Early Stage Faculty	Promoted to Associate Professor (Mentor: Yassine)	RF1AG076124

Jiaming Liang PhD	Postdoctoral Scholar	Starting new position as Assistant Professor, University of Texas, Health Science Center at Houston School of Public Health (Mentor: Maria Aranda)	USC Provost Postdoctoral Scholar
Annie Nguyen, PhD	Early Stage Faculty	Promoted to Associate Professor at UCSD (Mentor: Duke Han)	K01AG064986
Kristan Leech, PhD	Early Stage Faculty	Assistant Professor in Physiology (Mentor: Duke Han)	K01AG073467
Aaron Lim, PhD	Postdoctoral Scholar	(Mentor: Duke Han)	T32AG000037
Joseph Saenz, PhD	Early Stage Faculty	Promoted to Assistant Professor at Arizona State University (Mentor: Duke Han)	K99AG058799
Gali Weissberger, PhD	Early Stage Faculty	Promoted to Assistant Professor at Bar Ilan University (Mentor: Duke Han)	T32AG000037
Kayla Tureson, MS	Doctoral Student	Clinical Core (Mentors: Ringman, Christopher Beam)	NIA Predoctoral Fellowship F31
Charisse Winston PhD	Early Stage Faculty	Promoted to Assistant Professor of Physiology and Neuroscience (Mentor: Robert Rissman)	K99MD015777
Kexin Yu PhD	Doctoral Student	Starting new position as Post doctoral fellow, Oregon Health & Science University (Mentor: Maria Aranda)	F99AG068492
Douglas Barthold, PhD	Assistant Professor	Assistant Professor, University of Washington (Mentor: Julie Zissimopoulos)	K01AG071843
Azizi Seixas, PhD	Assistant Professor	Associate Professor, University of Miami (Mentor: Julie Zissimopoulos)	R01AG075007
Sadaf Milani, PhD	Assistant Professor	Assistant Professor, University of Texas Medical Branch, (Mentor: Julie Zissimopoulos)	K01AG075254
Hoda Abdel Magid, PhD	Assistant Professor	Assistant Professor, USC, (Mentor: Julie Zissimopoulos)	R00HL161479
Douglas Barthold	Postdoctoral Scholar	Research Assistant Professor, University of Washington, School of Pharmacy, (Mentor: Julie Zissimopoulos)	K01AG071843
Ilene Hollin	Postdoctoral Scholar	Assistant Professor, College of Public Health at Temple University, (Mentor: Julie Zissimopoulos)	K01HL161330

Innovation

The Administrative Core plays a critical role in innovative projects that cross traditional organizational units (e.g., genetic screening in the ORE Core, private practice and public safety-net hospitals, and research using real-world, healthcare data. We describe several of these innovations below:

(A) Novel web-based USC GeneScreen to recruit a cohort enriched for carriers of *APOE ε4*. Mail-in buccal swab kit enables pre-screening for *APOE* genotype, resulting in efficient, targeted recruitment and enrollment. Outside funding (Krueger-Wyeth) is required to develop and will require ongoing ADRC funding, integration into ORE and Clinical Cores, and tracking by the Administrative Core.

(B) Early prediction of ARIA associated with anti-amyloid immunotherapy: Enabled by a collaboration between USC neurology and radiology private practices, ALZ-NET, with the ADRC Imaging core. With the FDA approval of Leqembi in June 2023, the ADRC Clinical Core collaborated with the KMUSC to launch the Leqembi Review Board, and to deliver patient-centered, coordinated clinical workflow. Currently, 38 patients are receiving Leqembi treatment. The Administrative Core is working with the IRB, radiology and hospital Information Technology (IT), such that given patient-consent, clinical and imaging data will be de-identified and transferred from the EMR to ALZ-NET, and the ADRC Imaging Core for the improvement of early ARIA detection.

(C) Increase DHS patient access to ADRC research through partnership between ORE and Clinical Cores with the Los Angeles County Department of Health Services (DHS) (***PEDP-DHS***). This builds on decades of collaboration between the ADRC, the California Alzheimer Disease Center (CADC) at Rancho Los Amigos National Rehabitation Center (CADC-Rancho), and DHS (see Overall for support letters: Towfighi, Schickedanz)**.**

(D) Incorporate real-world data as a future ADRC resource by partnering with KMUSC on digital cognitive assessment. The Administrative Core has been working on incorporating a digital cognitive assessment tool into EMR (see letter of support, David Bates, CEO Linus Health, Inc).

(E) Increase representativeness through directed recruitment strategies. The Administrative Core has been working with IRB and IT to use the EMR used by KMUSC and DHS as a basis for recruitment into the ADRC Clinical Core and participant registry. Social and Environmental Determinants of Health (SDoH) geocoding tools offered by the USC Spatial Science Institute can improve the representativeness of recruitment strategies by apriori-defined SEDoH characteristics, while GeneScreen improves efficiency to enrich recruitment by *APOE* genotype.

(F) New Center for Personalized Brain Health supporting recruitment of physician scientists, an annual symposium and pilot funding. The CPBH represents a crucial translational bridge between basic and clinical neuroscience that focuses on understanding and mitigating the adverse effects of *APOE ε4* on brain health.

Approach Aim 1: Guide the Scientific Direction of the ADRC. A) Recruit and follow a clinical cohort reflecting the ethnic diversity of Los Angeles enriched for *APOE* ε4 on cognitive decline and AD biomarkers, B) Promote local ADRD research at USC, and C) Further national collaborations.

Aim 1: Guiding the Scientific Direction of the ADRC. The ADRC Executive Committee (EXEC) composed of all core and project leaders meets monthly to assist the two mPIs and Associate Director in making the scientific and administrative decisions relating to the center. The Executive Committee is comprised of core leaders (Chui, Toga, Yassine, Schneider, Joe, Nation, Wang, Hiniker, Hawes, Aranda, Ringman, Rissman, Pappas, Han), and center administrators (MA Aranda, T Kolchins).

The agendas of these meetings are guided by specific metrics (**Table 6**) and include a: 1) review of the participant status to ensure recruitment milestones (numbers, inclusion criteria, ethnicity) are met, address bottlenecks as applicable to plan for remediation, 2) report for each core to identify new opportunities, or potential problems and discussion of solutions, 3) presentations regarding participation in national programs and ADRC obligations such a NACC, SCAN and Clariti throughout, 4) plans for seminars, workshops and other gatherings to promote local research, and 5) presentations from new investigators and updates on funded project.

Table 6: Impact and Metrics of ADRC Cores

Core	Functions	Leaders	Impact Current funding cycle	Metrics Next funding cycle
Admin	Provide the leadership, administrative mechanisms and oversight to enable our ADRC to achieve its overall aims and foster core interactions of the center.	Helena Chui MD Arthur Toga PhD Hussein Yassine, MD Michelle Aranda Tiffany Kolchins	Publications (n=398) NIH grants: R (n=39), U (n= 8) NIH APOE grants: R: n=4), P01:n=1) NIH training awards (n=7)	Special Projects APOE4 GeneScreen, ALZ-NET, PEDP-DHS, SCAN, CLARitI, Real-World Data Development Projects; Publications; Grants
Clinical	Supports a diverse clinical core cohort that is well-characterized, phenotyped, and willing to participate in research.	Lon Schneider MD Elizabeth Joe MD John Ringman MD	Clinical cohort of 436 diverse, well-diagnosed, biomarker enhanced participants, 40% male, 42% URG, 28% Latinx; 40% APOEε4 carriers; Diverse brain donation cohort of ~350	Enroll 52/ yr. diverse, at-risk & early AD participants, ~35% Latinx, ~40% APOEε4 carriers. Increase cohort to 500; enroll 26/ yr. brain donation
DMSC	Provide high quality services and resources to the ADRC	Arthur Toga PhD David Conti, PhD Dan Nation PhD	ADRD-Database for Cohort (n=7) and Affiliated (n=1) Studies; Statistical support to 8 NIH funded grants, 1 State grant, 5 pending	Usage ADRC database; Investigators Supported Biostatistical consultation Education and Training (T32)
Neuro-pathology	Provide gold-standard neuropathologic diagnosis for ADRC patients; banks and distributes nervous system tissue for ADRD research	Annie Hiniker MD, PhD Debra Hawes MD Michael Bienkowski PhD	Update >1,100 brain autopsies to NACC 11 standards; Distribute tissue to researchers including fixed/frozen brain, eyes, and spinal cord; Digitize autopsy slides to create free online ADRD library; Collect Latinx autopsy cases; Transition to online tissues requests	Process 20 brain tissues, eyes, and spinal cord per year; Utilize digital pathology to create online digital library and test effects of APOE ε4 on perivascular inflammation; Use of online tissue request system.
ORE	Build a liaison between the ADRC, our study participants, their caregivers/partners, and the larger community. Increase participation of URG in ADRC studies (PEDP-DHS)	Maria Aranda PhD John Ringman MD MS	Supported 255 events; Reached 21,700 individuals (18,500 URG [60% Latinx]); Launched GeneScreen (n= 400 genotyped: 31% APOE ε4)	Genotype 300 participants/y; Recruit 52/y into the active cohort per year. 40% APOE 4 carriers; Build an ethnically diverse cohort with focus on Latinx; Hold Spanish-language retention events
Biomarker	Standardize the collection and analysis of biofluid samples and brain images, along with the sharing of biomarker resources and derived data	Robert Rissman PhD Abhay Sagare, PhD	Analyzed 450+ CSF samples Distributed plasma (n=1091 participants) and CSF (n=382 participants) to NCRAD and other investigators. Transitioned to online tissue request system	Process samples for NCRAD, DVCID. Analyze vascular and lipid biomarkers, CSF biomarkers. Specimens shared, granted supports, Publications, junior investigators
Imaging	Provide standardized, high quality, SCAN-Compliant MR and PET imaging resources to the USC ADRC and AD research community	Arthur Toga PhD Ioannis Pappas PhD Nasim Sheikh-Bahaei MD, PhD	MR: 278 scans acquired MR: 117 scans processed Amyloid PET; 111 scans Amyloid PET: 33 analyzed Tau-PET: 60 scans Tau-PET: 31 analyzed	Acquire and process SCAN & CLARiTI 240 MRI scans 240 amyloid PET 240 Tau PET
REC	Direct the scientific careers of scientists into ADRD research. Partner with RCMAR to train scientists from URG. (PEDP-RCMAR)	Hussein Yassine MD Duke Han PhD Julie Zissomopoulos, PhD	Supported 728 REC mentees; 7 developmental projects 22 REC scholars obtain NIH fundings (various levels) 2 symposia per year	Weekly Seminar series 2 Development projects 2 Annual Symposia Recruit 2-5 new REC scholars per year

ADRC-Wide Strategic Initiatives: The center directors and the Executive Committee identified several cross-cutting initiatives across the ADRC.

(1.A) Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4. During the next five years, the USC ADRC cores will jointly develop a clinical cohort, ADRC database, and participant registry to support collaborative research to identify risk factors that modify the impact of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers. In addition to the variables collected in UDS4.0, we will gather additional lifestyle data and vascular, lipid, and metabolic biomarkers known to increase the risk of ADRD.

- Explore associations between putative risk factors and the effect of *APOE* ε4 on cognitive decline and ATNvi biomarkers, including developing an online *APOE* dementia risk calculator (combined ADRC Core Leaders).
- Recruit subjects for CLARiTI MRI, amyloid PET, and tau PET with approximately 50% APOE ε4 carriers (ORE, Clinical, Imaging, Data and Admin Cores).
- At the present time, the ADRC supports 4 R01 and 1 P01 NIH grants on *APOE ε4*.

(1.B) Promote local ADRD research at USC. The Cores will individually and collectively work to promote local research in ADRD and across the national ADC network. The ADRC interacts with and draws upon USC resources focused on ADRD and aging across a range of disciplines, which include basic neuroscience/neurobiology, neuroimaging, clinical trials, translational science and precision medicine, behavioral and social sciences, and health services research. USC *writ* large currently has over 124 NIH funded grants, 47 of which are supported by the ADRC [39 R-series, 8 U-series NIH awards (See **Overall: Table 2_a1: Federally-funded grants; Table 2_a2: Non-federal grants; Table 2.a3: Therapeutic Trials**)].

1. Integrate research programs by holding annual events (Table 8), development projects (See Approach Aim 4), and harmonization of protocols (See Clinical UDS and Imaging Core SCAN protocols)**.**

2. Share data, tissues, biofluids, and a participant registry through a common USC ADRC database.

(a) **Components of the ADRC registry:** In the next five years we propose to: (1) expand the GeneScreen into the Latinx population, (2) expand our recruitment efforts into both Keck and DHS, and refine tracking of recruitment efforts and outcomes, and (3) continue recruiting participants from local communities through both in person events and online campaigns. We envision the creation and growth of an ADRC mega-registry/database **(Fig. 4**) of prospective participants consenting to receive study notifications. Requests to access the registry will be reviewed by the Executive Committee in the Administrative Core.

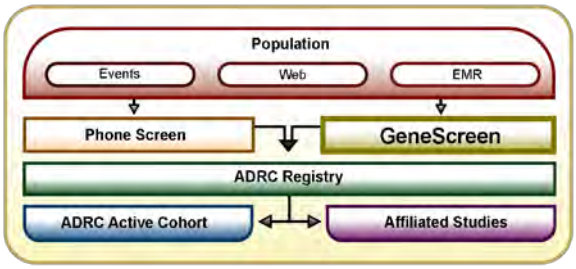


Figure 4. Components of the ADRC Registry.

(b) **Increase representativeness from real-world data (RWD):** We will establish a quality control and data harmonization pipeline to extract reliable and high-quality medical information from sources of RWD. The newly formed ADRC registry will be maintained on REDcap from consenting patients (Keck, DHS), community members, or past research participants. A subset will be invited to enroll in GeneScreen with the goal of enriching for *APOE ε4*. We plan to recruit participants of low socioeconomic status derived from the Social Vulnerability Index (SVI), inclusive of racial and ethnic minorities that are representative of Los Angeles.

(c) **Novel AI/ML approaches to EMR data:** Once Keck EMR data is cleaned, we will apply novel supervised learning approaches such as Gene-SGAN (14) to identify AD features or subtypes within our population. We have an extensive research record applying the advanced AI/ML to dissect disease heterogeneity in AD and aging, depression, and autism spectrum disorder (15-17).

(d) **Alzheimer's Network for Treatment and Diagnostics** (**ALZ-NET)**: The real-world research study ALZ-NET is a national effort, sponsored by the Alzheimer Association to study patients who are undergoing new FDA-approved clinical treatments for AD. At USC, we are currently treating 38 patients with Leqembi. For patients who give consent, clinical and imaging data will be de-identified and shared with ALZ-NET and with the USC

ADRC Data, and Imaging Cores. We will leverage our substantial clinical and imaging expertise to better delineate the role of *APOE* in CAA and risk for ARIA.

(e) ***PEDP-DHS*. Administrative, Clinical and ORE Cores:** Partner with the California Alzheimer Disease Center (CADC) and the LA County Department of Health Service (DHS) to enroll research participants from safety-net hospitals (PEDP-DHS) (for details see ORE Core, Specific Aim 2)

(f) **Administrative, Clinical, and Data Cores:** Build real-world registry and database through linkages with KMUSC EHR; integration of Digital Cognitive Assessment in Keck Medicine of USC (KMUSC) in partnership with Linus Health.

3. Train the next generation of ADRD researchers and educate participants and the public. The admin core will support REC and the ORE core with training programs, planning of events and funding opportunities.

4. Enhance Diverse Perspectives. PEDP-RCMAR. Administrative and REC Cores. Partner with RCMAR to train investigators from under-represented groups (PEDP-RCMAR) and to fund development projects (for details, see REC).

(1.C) Further national collaborations. We advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, and comprehensive imaging to national initiatives**.**

Approach Aim 2: Coordinate fiscal, operational, and regulatory functions across all cores to ensure appropriate, optimal, and effective utilization of center resources that are aligned with the NIA-ADRC Program and synergize with local ADRC-affiliated research.

Aim 2: The administrative core works collaboratively to coordinate the flow of scientific knowledge, participants, biological materials, data, and financial resources across the center. This collaboration enhances the multi-dimensional characterization of each resource and the ability to conduct translational research and clinical trials in the age of multi-modal biomarkers (e.g., CSF, plasma, imaging). Each ADRC core has aim(s) that support our unifying *APOE* scientific theme, local research, and national initiatives.

(a) **Oversight of research and grants administrative processes, including preparation of annual RPPR.** The Administrative Core (Chui, Toga, Yassine) oversees the research and grants administrative process of the ADRC. The leadership team meets monthly to review research progress, and monthly with the Executive Committee to review progress/challenges among ADRC Cores. At the ADRC meeting, we review updates from the ADC network, coordination and integration of across cores (see Aim 1), plan/review/select development project, preparations for external advisory board meeting, and submission of annual RPPR.

(b) **Coordination and integration of center components and activities.** The Imaging Data Archive (IDA) at Stevens INI houses the ADRC database for NACC UDS, NCRAD *APOE* genotype, NCRAD ATN, and USC ADRC Biomarker panel. We track recruitment participants using our KECK EMR I2B2 tool and a REDCap database. The DMSC has launched a REDCap internal tracking system for biofluids and imaging. The Administrative Core monitors and coordinates the workload for the Clinical Cohort between the Clinical, NP, Biomarker and Imaging Cores.

(c) **Assurance of compliance with human subjects, animal welfare, scientific integrity, data and sample sharing as appropriate, and financial policy requirements of NIH.** The Admin Core ensures that all key personnel have completed the educational certifications required by the USC Institutional Review Board, including HIPAA, Human Subjects, and Good Clinical Practices. The USC IRB uses online iStar to track research projects and compliance with human subjects and animal welfare. The Admin Core works with the Clinical Core to ensure that consent forms will allow broad sharing of biological samples.

(d) **Support and advice for the Center Directors (two MPIs; one Associate Director) in their oversight of the activities of the center.** The center directors seek support and advice from the following: 1) monthly executive committee (all core leaders), 2) quarterly Internal Advisory Committee (Meltzer, Finch, Buchanan), 3) annual External Advisory Committee (EAC), 3) semi-annual ADRC-DHS Community Advisory Board, and 4) the

Community Advisory Board (CAB). These committees advise the Directors in making scientific and administrative decisions related to the center.

(e) **Interaction with the scientific and lay communities to develop relevant goals for the center.** Collectively, the ADRC leadership is well connected to the relevant scientific and lay communities. Aranda is a member of the National Alzheimer Project Act (NAPA) Committee. Chui, Schneider, Hawes, Wang, and MA Aranda attend the bi-annual ADC directors meeting. Schneider informs the Executive Committee of priorities of the ACTC and ADNI. Chui serves on the Board of Directors of Alzheimer's Los Angeles. In addition, Chui (Chair), Schneider, Aranda, and Segal-Gidan serve as members of the Healthcare & Research Advisory Council of Alzheimer's Los Angeles. For the ADRC-DHS initiative (PEDP=DHS), we have assembled a five-member community advisory board (CAB) (see ORE core, **Approach Aim 3**).

(f) **Faculty recruitment, retention, tenure/promotion activities, including recognition of team science; and clear leadership succession planning.** The two mPIs report directly to Carolyn Meltzer MD, Dean, Keck School of Medicine (KSOM), and are empowered to recommend the hiring, retention, and promotion of faculty members: Chui as Chair of the Department of Neurology, Toga as Director of the Steven's Institute of Neuroimaging and Informatics. Tenure decisions require approval at three levels: Department/Institute, KSOM, and University. During the past four years, we have supported the promotion with tenure for Hussein Yassine, MD (granted April, 2024). Working closely with the Chair of Pathology, Dean Meltzer, Chui and Toga supported the key recruitment of Annie Hiniker PhD, and Robert Rissman MD from UC San Diego to lead the Neuropathology Core and the Biomarker Core, respectively. To address succession planning throughout the ADRC, we designated new leaders or co-leaders for each core and developed individual mentoring plans (including formal training courses, service on NIH study sections, experience on national ADC committees).

(g) **Coordination and organization of external and internal advisory committee meetings.** The External Advisory Board (EAC) is invited annually by the Admin Core to evaluate the programs of the ADRC, research progress, the effectiveness of communications within the ADRC, and interactions with NACC, NCRAD, and SCAN. The NIA Program Officer is invited to attend EAC meetings as a non-voting member. A copy of the advisory committee report is sent to the NIA with the annual Non-Competing Continuation. If successfully refunded, the center will include a list of newly appointed committee members with its Grant Progress Report.

During the last funding cycle, the EAC comprised William Jagust MD (Chair), Carl Cotman PhD (Neuroscience 2021, 2022), Donna Wilcock PhD (Neuroscience, 2023), Walter Kukull PhD (Data management and Biostatistics, 2021, 2022), Brittany Dugger PhD (Pathology), and Hector Gonzalez PhD (Outreach and Education). The Administrative Core selected the External Advisors and coordinated the EAC meetings which took place on 12/10/21, 11/29/22, and 11/7/23. Nina Silverberg PhD, Director, Alzheimer's Disease Research Centers Program was invited and attended all three of the EAC meetings.

In the current funding cycle, the Internal Advisory Committee (IAC) consists of Caleb Finch PhD (Founding ADRC director), Carolyn Meltzer MD (Dean, KSOM), and Tom Buchanan (Associate Dean for Research and Director, Southern California CTSI). Finch chairs the Internal Advisory Committee (IAC), is a standing member of the Executive Committee, and meets with us monthly. The other advisors are called upon as needed and annually in conjunction with the External Advisory Board.

Approach Aim 3: Promote participation across all Cores in collaborative national research initiatives and resource sharing, including NACC, NCRAD, ADGC, ADNI, ACTC, DVCID, ACAD, SCAN, and CLARiTI playing a leadership role in research into *APOE*, leveraging our metabolic, vascular disease, and neuroimaging expertise.

Aim 3: We will interact with other Centers, the Data Coordinating Center and other researchers to develop trans-ADRC and outside research projects. See **Table 7**.

Table 7. USC ADC Collaborations	
Cooperative Studies	**Activity**
National Alzheimer Coordinating Center (NACC)	Participating ADRC
National Centralized Repository for Alzheimer's Disease (NCRAD)	Send blood and DNA to NCRAD
Alzheimer Disease Genetics Consortium (ADGC)	Participating site via NCRAD for GWAS, WGS.
Alzheimer Disease Neuroimaging Initiative (ADNI)	Participating site; USC LONI serves as imaging archive
Alzheimer's Clinical Trial Consortium (ACTC)	USC is lead site (P Aisen); USC ADRC is participating site
Dominantly Inherited Alzheimer Network (DIAN-TU)	Participating site; Schneider sits on steering committee
Longitudinal Early Alzheimer Disease Study (LEADS)	USC LONI serves as imaging archive
Diverse VCID University of California, Davis (DVCID)	Participating site, focus on Latinx
Validation of imaging and blood-based small vessel VCID biomarkers in multiethnic populations (MarkVCID); 2UF1NS100614.	Participating site, focus on Latinx and African Americans
Asian Cohort for Alzheimer Disease (ACAD); U19AG079774.	Participating site for enrolling Chinese Asians
Collaborative Studies	**Multi-Institutions**
Vascular contributions to dementia and genetic risk factor for AD (VG-PPG); P01AG052350.	USC, Washington University, University of Arizona
Health and Aging among Latino Elders (HABLE); R01AG058533.	University North Texas Health Science Center; USC LONI serves as imaging archive
Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression (DVR); R01AG058162.	USC, Kansas University, University of Texas Southwestern
Pathways linking infection, physical activity and *APOE* and biological sex to dementia in 2 native American populations (Tsimane-Moseten Dementia Study); R01AG054442.	Chapman University, USC, University of Arizona
Locus ceruleus imaging markers in preclinical AD, cerebrovascular disease and cognitive decline; R01AG079774.	USC, University of California San Diego

(a) **Timely and routine transmissions of appropriate Center data sets to the NACC, NCRAD and SCAN.** The ADRC supports not only local, but also broad national and international data sharing. The Overview, Administration, and DMSC narratives describe these efforts. We will make all datasets used and generated by the individual cores accessible and reusable by qualified individuals outside of the ADRC. Most data are now available via web-based resources at LONI-IDA, NACC, and NCRAD. Data can be made accessible via open or controlled access depending on the data type and data source, as determined by the informed consent documents for each study guided by the local IRB.

Members of the Clinical and Pathology Cores will enter the Clinical and Neuropathology Uniform Data Set directly on the NACC website. NACC monthly reports are reviewed by the DMSC, Clinical, and NP Core leaders and signed-off by the center directors. The biostatistician and systems analyst in the DMSC monitor data accuracy and integrity. The Admin Core coordinates the review and sign-off process.

(b) **Coordination with NACC and NCRAD for ADGC DNA calls.** For potentially all ADRC participants, the Administrative Core works with the Clinical, and Neuropathology, Biomarker, and Data Cores to send and track distribution of biosamples (e.g., whole blood, DNA, and/or PBMC (peripheral blood mononuclear cells) to NCRAD for *APOE* genotyping, whole genome sequencing, and ATN biomarkers. The Clinical Core sends blood, NP Core send DNA, and the Biomarker Core sends PBMC to NCRAD.

Each core tracks the flow of biosamples through the DMSC (blood sample tracking system). At annual UDS follow-up visits, the Clinical Core determines the status of DNA sharing for each participant, so that blood can be obtained. The Administrative Core has also established a Material Transfer Agreement with NCRAD, necessary to retrieve 25 µg of DNA from NCRAD. The Clinical Core notifies the Administrative Core, which ensures reimbursement from NCRAD, replenishment of blood drawing kits, and mailing supplies. The Administrative Core organizes an annual meeting with NCRAD. The DMSC checks NACC on an ongoing basis for updates on genetic analyses, including *APOE* genotypes, GWAS, exome and whole genome sequencing.

(c) **Coordination with SCAN and CLARiTI for MRI and PET image sharing.** MRIs, amyloid- and tau-PET scans are anonymized and stored at the ADRC-DB with links to underlying and anonymous clinical data including the UDS and other clinical data obtained by the Clinical Core. The Imaging Core uploads compliant scans to SCAN and CLARiTI.

(d) **ADRC Clinical Core infrastructure supports participation in multi-center clinical trials** The Clinical Core supported USC ADRC participation in ADNI, AHEAD3-45; TRAILBLAZER, DIAN-TU, Exosome, NOMAD, TRC-PAD. Studies that are actively recruiting include ADNI-4, AHEAD 3-45, DIAN-TU, Exosome, NOMAD. (see Overall, **Table 3a**: Funding for Therapeutic Trial and Other Grants from Industry)

(e) **Coordination with NIA on media coverage of the latest research findings from the center.** The Administrative Core will work with representatives from NIA, ADEAR, as well as personnel in the USC public relations office to provide media coverage of the latest findings from the center.

Approach Aim 4: Attract early-stage and new investigators, and **support training and education.**

Aim 4: We attract early stage and new investigators especially from under-represented groups to ADRD research through development projects (DP), working with the REC and all Core Leaders. With the REC and ORE Core, respectively, sponsor training workshops, research symposia for investigators and provide public education on the importance of understanding and addressing genetic, metabolic and vascular risk profiles for brain health.Training grants serve as an excellent resource to attract new investigators. USC currently has seven active NIH-Training Awards relevant to ADRD: 1) Multi-disciplinary research training in gerontology (T32AG000037: Crimmins), 2) Training in the multi-scale and analysis of biomarkers in AD (T32AG058507: Toga), 3) AD resource center for minority aging research (AD-RCMAR) (2P30AG043073: Zissimopoulos), 4) Center for advancing sociodemographic and economic studies of ADRD (CeASES-ADRD) (P30AG066589: Zissimopoulos), 5) Neurocognitive aging and analytics research education (NAARE) (5R25AG069711: McEligot, Braskie, Toga), 6) Mid-career development and mentoring on financial vulnerability and AD (K24AG081325: Han), and 7) Southern California clinical translational science institute (SC-CTSI) (5UL1TR001855: Buchanan).

Table 8. Training Workshops, Research Symposia, Public Education

Target Audience	Name of Educational Event	Primary Organizer	Frequency
Research participants	Batey Research Participant Appreciation Day	Chui, Aranda	annual
Scientific community	Annual CBPH Symposium	Yassine	annual
	Finch Research Symposium	Yassine, Cohen, Chui, Davies	annual
	Southern California ADC Symposium	Grill, Yassine, Han	annual
REC scientists	Southern California ADRC Annual Symposium	Yassine, D'Orazio	annual
Health care providers	Many Faces of Dementia	Segal-Gidan	biannual
Public-at-large	Verdugo Hills Hospital	Danner, Chui	annual
	Saint Francis Hospital	Rose, Gunderson	annual
	Roybal Comprehensive Health Center	Yassine	bimonthly

(a) **Coordination and organization of development project (DP) advertisement, review, and submission of development projects to NIA for approval.** Each Fall, the ADRC will invite clinical and basic science investigators to submit letters of intent for a one-to-two year DP. The ADRC focuses on factors that modify the effects of *APOE* ε4 on cognitive decline and AD biomarkers, as well as local and inclusive research on ADRD. We seek new basic, clinical, and psychosocial approaches that inform pathogenesis, and advance prevention and treatment of early cognitive impairment in humans and animal models. We encourage the development of new approaches to intervention and translational research from pre-clinical to early phase trials.

The DP program is designed primarily for early-stage faculty investigators. Applicants must identify a USC ADRC collaborator/sponsor (member of the Executive committee) and schedule a consultation with the ADRC biostatistician. The ADRC sponsor informs the applicant of available resources and guides the research question and specific aims. Applicants may request up to **$100K/year (**direct costs) for one-to-two years. The ADRC will support approximately three-to-five new DP/year over five years (budget request = $150K/yr x 5 years).

(b) **A review panel to assist in selecting development projects (DP).** Potential development projects go through a four-step review process: 1) letter of intent with 500 words abstract, and NIH biosketch reviewed by the ADRC executive committee, including our biostatistician, 2) top three-to-five DP are invited to submit a three page semi-final application reviewed by the ADRC executive committee, 3) the top two-to-three proposals are sent out for external review, and 4) the revised final proposal is submitted together with the non-competitive renewal progress report for review and approval by the NIA. Each DP is reviewed by two external reviewers selected based on appropriate expertise. When appropriate we will ask members of our EAC to serve as reviewers.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	❍No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	❍No
Is the Project Exempt from Federal regulations?	❍Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447174.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT:

Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:

Clinical Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: 1540 Alcazar Street		Street2:
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	Clinical_Abstract1044447028.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	Clinical_References1044447027.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Clinical Core

The clinical core develops and maintains the clinical enterprise serving the ADRC's overarching themes, including factors that modify the effect of *APOE ε4* on dementia and ATN biomarkers, emphasizing vascular and metabolic risk. The core supports the ADRC's overall scientific goal, i.e., understanding mechanisms by which *APOE* genotype affects clinical phenotype and progression toward dementia.

The core provides well-characterized, longitudinally followed participants for affiliated research projects, USC researchers, and the broad scientific community by developing and maintaining a cohort at risk for dementia or its underlying pathology, characterizing participants by fluid and imaging biomarkers, and encouraging consent for brain donation. The core maintains an active registry of a diverse group of "study ready" potential participants who expressed interest in research participation.

The core contributes to national initiatives including NACC, NCRAD, ADGC, SCAN, and CLARiTI; to multi-site observational studies including ADNI, DIAN, MarkVCID, Diverse VCID, ACAD (Asian Cohort for AD); and to national and international trials including the USC ACTC, UCSD ADCS, A4 and AHEAD 3-45. The core catalyzes local research by supporting recruitment, providing oversight and training for investigator-initiated studies and trials (e.g., PPGII, DVR, NoMAD, GADSIL, DHA, phenserine, exosomes). To accomplish its mission, the core closely interacts with other ADRC cores: Major science findings supported by the core are discussed in the proposal.

Thus, the clinical core supports the USC ADRC's overarching goals A) defining factors that influence the effect of *APOE ε4* on cognitive decline and ATN biomarkers, B) supporting local ADRD research, and C) contributing to national research initiatives.

Literature Cited

1. Sible IJ, Nation DA, for the Alzheimer's Disease Neuroimaging I. Long-Term Blood Pressure Variability Across the Clinical and Biomarker Spectrum of Alzheimer's Disease. Journal of Alzheimer's Disease. 2020;77:1655-69. doi: 10.3233/JAD-200221.
2. Tureson KN, Beam CR, Medina LD, Segal-Gidan F, D'Orazio LM, Chui H, Torres M, Varma R, Ringman JM. Use of the Spanish English Neuropsychological Assessment Scale in older adult Latines and those at risk for autosomal dominant Alzheimer's disease. Journal of Clinical and Experimental Neuropsychology. 2023;45(6):553-69. doi: 10.1080/13803395.2023.2284971.
3. Jann K, Boudreau J, Albrecht D, Cen SY, Cabeen RP, Ringman JM, Wang DJJ. FMRI Complexity Correlates with Tau-PET and Cognitive Decline in Late-Onset and Autosomal Dominant Alzheimer's Disease. Journal of Alzheimer's Disease. 2023;95:437-51. doi: 10.3233/JAD-220851.
4. Charidimou A, Boulouis G, Frosch MP, Baron J-C, Pasi M, Albucher JF, Banerjee G, Barbato C, Bonneville F, Brandner S, Calviere L, Caparros F, Casolla B, Cordonnier C, Delisle M-B, Deramecourt V, Dichgans M, Gokcal E, Herms J, Hernandez-Guillamon M, Jäger HR, Jaunmuktane Z, Linn J, Martinez-Ramirez S, Martínez-Sáez E, Mawrin C, Montaner J, Moulin S, Olivot J-M, Piazza F, Puy L, Raposo N, Rodrigues MA, Roeber S, Romero JR, Samarasekera N, Schneider JA, Schreiber S, Schreiber F, Schwall C, Smith C, Szalardy L, Varlet P, Viguier A, Wardlaw JM, Warren A, Wollenweber FA, Zedde M, van Buchem MA, Gurol ME, Viswanathan A, Al-Shahi Salman R, Smith EE, Werring DJ, Greenberg SM. The Boston criteria version 2.0 for cerebral amyloid angiopathy: a multicentre, retrospective, MRI–neuropathology diagnostic accuracy study. The Lancet Neurology 2022;21(8):714-25. doi: 10.1016/S1474-4422(22)00208-3
5. Howard RS, Goldberg TE, Luo J, Munoz C, Schneider LS. Reliability of the NACC Telephone-administered Neuropsychological Battery (T-cog) and Montreal Cognitive Assessment for participants in the USC ADRC. Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring. 2023;15(1):e12406. doi: 10.1002/dad2.12406.

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Lon Middle Name Last Name*: Schneider Suffix:

Position/Title*:	Professor of Psychiatry
Organization Name*:	University of Southern California
Department:	KSOM Psychiatry
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-7600 Fax Number: (323) 442-7601 E-Mail*: lschneid@usc.edu

Credential, e.g., agency login: lschneid

Project Role*: Other (Specify) Other Project Role Category: Clinical Core Lead

Degree Type: MD, MS Degree Year:

	File Name
Attach Biographical Sketch*:	Schneider_Biosketch_ADRC_May_20241044446735.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Elizabeth Middle Name Last Name*: Joe Suffix:

Position/Title*:	Assistant Professor of Clinical Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764 -341 Fax Number: E-Mail*: elizabeth.joe@med.usc.edu

Credential, e.g., agency login: ejoe01

Project Role*: Other (Specify) Other Project Role Category: Clinical Co-Core Lead

Degree Type: MD Degree Year:

	File Name
Attach Biographical Sketch*:	JoeE_biosketch1044446685.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: John Middle Name M. Last Name*: Ringman Suffix:

Position/Title*:	Professor of Clinical Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-6845 Fax Number: E-Mail*: john.ringman@med.usc.edu

Credential, e.g., agency login: RINGMAN2

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: Degree Year:

	File Name
Attach Biographical Sketch*:	Bios-ketch_JRingman__5_29_241044446696.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Freddi Middle Name Last Name*: Segal-Gidan Suffix:

Position/Title*:	Associate Professor of Clinical Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	OCH Rlm
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (562) 401-8130 Fax Number: (562) 803-6900 E-Mail*: segalgi@usc.edu

Credential, e.g., agency login: SEGALGI

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year:

	File Name
Attach Biographical Sketch*:	
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Lina Middle Name M Last Name*: D'Orazio Suffix:

Position/Title*:	Assistant Professor of Clinical Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1520 SAN PABLO

Street2:	Bldg. #287
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (213) 740-4940 Fax Number: E-Mail*: lina.dorazio@med.usc.edu

Credential, e.g., agency login: LINADOR

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year:

	File Name
Attach Biographical Sketch*:	D_Orazio_Biosketch_0520241044446736.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Sonia Middle Name Last Name*: Pawluczyk Suffix:

Position/Title*:	Clinical Assistant Professor of Psychiatry
Organization Name*:	University of Southern California
Department:	KSOM Psychiatry
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-7600 Fax Number: (323) 442-7601 E-Mail*: pawluczy@usc.edu

Credential, e.g., agency login: PAWLUCZYK

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: MD Degree Year:

	File Name
Attach Biographical Sketch*:	Pawluczyk_NIH_biosketch_20241044447170.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Michael Middle Name Gordon Last Name*: Harrington Suffix:

Position/Title*:	Professor of Research Neurology
Organization Name*:	University of Southern California
Department:	
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	
State*:	CA: California
Province:	

Country*:	USA: UNITED STATES		
Zip / Postal Code*:	90033-0000		
Phone Number*: (213) 764-3341	Fax Number:	E-Mail*: mharring@usc.edu	
Credential, e.g., agency login: mikethescot			
Project Role*: Co-Investigator		Other Project Role Category:	
Degree Type:		Degree Year:	
		File Name	
Attach Biographical Sketch*:		Biosketch_MGH_5_30_241044446738.pdf	
Attach Current & Pending Support:			

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Schneider, Lon S.

eRA COMMONS USER NAME (credential, e.g., agency login): lschneid

POSITION TITLE: Professor of psychiatry, neurology, and gerontology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(If applicable)*	Completion Date	FIELD OF STUDY
Sarah Lawrence College, Bronxville, NY	AB	06/1974	Liberal Arts
Hahnemann Univ. Med. Col. (Drexel Univ.) Phila, PA	MD	06/1978	Medicine
LA County/ Univ. Southern California Medical Center	Residency	06/1082	Psychiatry
University of Southern California, Los Angeles, CA	Fellowship	06/1984	Geriatric psychiatry
American Board of Psychiatry & Neurology	Diplomate	12/1984	Psychiatry
ABPN, Added Qualifications, Geriatric Psychiatry	Diplomate	4/91,4/01, 12/22	Geriatric psychiatry
University of Southern California, Los Angeles, CA	MS	06/1995	Biometry, epidemiology

A. Personal Statement

Dr. Lon S. Schneider is the Della Martin Chair in Psychiatry and Neuroscience and professor of psychiatry, neurology, and gerontology at USC. He directs the USC State of California Alzheimer Disease Center, co-leads the clinical core of the USC NIA Alzheimer Disease Research Center. He holds a master's degree in biometry and epidemiology and has considerable expertise in clinical trials methods, instruments, and rating scales for dementia trials, metanalyses, and drug development programs. He is a co-director of the NIH UCSD ADCS; serves on the steering committees of the NIH ACTC, NIH ADNI, and the Banner Alzheimer Prevention Initiative; and is a project arm leader on the Washington Univ DIAN-TU directing an anti-tau antibody trial. He is internationally recognized in clinical drug development for Alzheimer's disease and cognitive disorders, neuropsychiatric, and behavioral disorders, has led numerous clinical trials for AD and MCI therapeutics, consults with numerous development programs, participated in the design and operations of several early and later phase development programs for AD starting with the NIA/Mt. Sinai/ Warner Lambert development of the first marketed drug for AD, tacrine. He has directed numerous multicenter trials in AD and major depression, including the paradigm shifting CATIE-AD effectiveness trial. He is responsible for the development of the ADCS-CGIC, the most common CGIC/CIBIC+ method for global impression of change in dementia trials. He and his colleagues examined evidence for secular changes in AD trials characteristics and outcomes and conducts trials simulations using large meta-databases to assess the effects of Alzheimer-related biomarkers and genotypes on the efficiency of targeted clinical trials to improve methods. He has been PI for an NIH-funded project using simulations to design AD trials and predict outcomes. He is a MPI of several NIH-funded early-stage phase 1a and 2b trials, including ER-β active phytoSERMs for cognitive symptoms in perimenopausal women; an allopregnanolone program (Brinton), a phase 1 MAD trial of phenserine as a neuroprotective, antiapoptotic agent, and MPI in a current first-in-human Phase 1 trial of AV-1959 DNA vaccine (Agadjanyan).

Dr. Schneider will continue as co-lead of the USC ADRC clinical core collaborating with co-leads of the other cores. This includes overseeing the professional medical staff, psychometricians, implementation and conduct of enrollment, retention, and diagnoses, overseeing clinical psychology research assistants and technicians. He reviews ADRC operations and new proposals regarding the use of core resources, serving on the ADRC executive committee, responding to NACC and NIA initiatives, requests for clinical data, and recruitment and retention of research participants.

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Ongoing and recently completed projects that I would like to highlight include:

R01 AG071217
Agadjanyan, Schneider, Sultzer (MPIs)
09/01/2021-08/31/2026
Safety/Tolerability/Immunogenicity of first-in-human Aβ DNA vaccine, AV-1959D Phase 1 trials in early-stage AD subjects: based on IND18953 cleared by FDA Phase 1 safety/tolerability/immunogenicity trials of DNA vaccine AV-1959D based on IND18953 cleared by the FDA. Completion of this study will allow us to initiate preventive trials in asymptomatic people at risk of MCI/AD in a future.

R01AG062687 NIA
Schneider (PI), Ballard (co-PI)
05/01/2019-2/29/2024, NCE
(-)-Phenserine inhibition of neuronal death in Alzheimer's disease and developing brain-labeled plasma exosomes assays as biomarkers for a phenserine phase 1b ascending dose trial.
This project addresses the urgent need to develop therapeutics to prevent, delay and treat Alzheimer's disease. It proposes the development of biological, plasma sourced, newly identified plasma exosomes containing proteins associated with programmed neuron death, synaptic arborization, and neuro-inflammation as companion biomarkers reflecting the state of brain activities in humans that are otherwise inaccessible to investigators for the development of (-)-phenserine tartrate as a treatment. We will conduct a Phase 1b clinical study of this promising neuroprotective therapeutic which pre-clinically acts by a novel mechanism to prevent neurons from dying in the presence of neuropathology that is otherwise fatal to neurons in Alzheimer's and other disorders.

R01AG051346 NIA
Goldberg, Schneider LS, Harvey P (MPIs)
09/01/2018-05/31/2024, 5/31/25 NCE
Novel Cognitive and Functional Measure for Alzheimer's Disease Prevention Trials
The goal of this project is to validate novel cognitive and everyday functional measures that have sharply attenuated practice effects and are not prone to ceiling effects for use in preclinical Alzheimer's disease (AD) trials in which participants are cognitively within normal limits.

R01 AG057684 NIA
Kennedy (PI), Schneider LS (co-PI)
09/15/2017-03/31/2023 (NCE)
In Silico Screening of Medications for Slowing Alzheimer's Progression
The project involves interrogating clinical trials datasets and outcomes and the Medicare database to assess for the potential effectiveness of current medications for improving Alzheimer's disease/ cognitive impairment

R01 AG054434 NIA
Yassine (PI)
09/01/2017-04/30/2024
The Delivery of Essential Fatty Acids to the Brain in Alzheimer's Disease
Conduct a double-blind placebo-controlled randomized trial of high dose DHA over 6 months in 160 cognitively healthy participants stratified by ApoE4 genotype to test the effect of ApoE4 genotype on CSF DHA levels in response to DHA supplementation.

P30 AG066530 NIA
Chui HC, Toga (MPI), Schneider (Director clinical core)
4/01/10 - 3/31/25
Alzheimer's Disease Research Center: Clinical Core
An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

03-75274 State of California
Schneider (PI)
7/01/12 - 6/30/26
California Department of Health Services

USC California Alzheimer Disease Center
This center is established to support the clinical care of and follow-up on patients with dementia and provide educational outreach to the community. Goals are to address the needs of our understudied and underserved communities and provide a model of statewide collaboration.

R01 AG055444 NIA
Reiman E, Tariot P (MPI)
4/15/18 - 3/31/23
Alzheimer's Prevention Initiative ADAD Colombia Trial
Major goals are to design a clinical trial protocol, analysis plans, and oversight of the overall study.

3U24AG057437 NIA
Aisen P (PI)
12/02/17 - 11/30/25
Alzheimer's Clinical Trials Consortium
The proposal aims to create an optimized infrastructure to optimally design, efficiently launch, rapidly recruit with improved diversity, and successfully complete NIH supported clinical trials in AD and related dementias.

U01 AG 010483 NIA and Lilly
Schneider (Site PI)
12/31/12 - 12/30/24
ADC-040-A4 Anti-Amyloid Treatment in Asymptomatic Alzheimer's Disease (A4 Study).
The A4 study is designed to test the hypothesis that solanezumab will slow decline on a cognitive composite measure designed to be sensitive to early decline in the preclinical stages of AD.

R01 AG052510 NIA
Lyketsos (PI)
10/01/16 – 9/30/24, 5/31/24 *S-citalopram for Agitation in Alzheimer's Disease (S-CitAD)*
A multicenter clinical trial investigating SSRIs, s-citalopram specifically for agitation in Alzheimer disease Role: Site PI

R01 AG053267 NIA
Bateman (PI)
10/01/16 – 9/30/26
DIAN-TU: Next Generation Prevention Trial
A multicenter clinical trial investigating solanezumab, gantenerumab, and a BACE1 inhibitor for dominantly inherited Alzheimer disease. Dr. Schneider serves as project arm leader overseeing the BACE1-I component. Role: Co-investigator, project arm leader

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2001 - present	Co-director (2001-2005), director (2006-present), clinical core, USC NIH Alzheimer Disease Research Center
1999 - 2011	PI, NIMH Clinical Antipsychotic Trial of Intervention Effectiveness Alzheimer dis, CATIE-AD
1999 - present	Director/ PI, USC AD Research and Clinical Center of California, and California AD Center
1997 - present	Professor of Gerontology (secondary) USC Leonard Davis School of Gerontology
1997 - present	Professor of Neurology (secondary), USC Keck School of Medicine
1997 - present	Professor of Psychiatry and the Behavioral Sciences, USC Keck School of Medicine
1997 - 2003	Vice President, Chief Financial Officer, USC Psychiatry and Psychology Associates, Inc.
1996 - 2007	Vice Chief of Psychiatry Services, USC University Hospital
1995 - 1997	Associate Professor of Gerontology (secondary), USC Leonard Davis School of Gerontology
1994 - 2000	Medical Director, Comprehensive Older Adult Service and Treatment, USC Univ. Hospital
1992 - 2001	Director, pharmacology program, Southern California, Alzheimer Disease Research Center, NIA
1990 - 1997	Associate Professor of Neurology (secondary),USC Keck School of Medicine
1990 - 1997	Associate Professor of Psychiatry and the Behavioral Sciences, USC Keck School of Med
1984 - 1990	Assistant Professor of Psychiatry and the Behavioral Sciences, USC School of Medicine

Other Experience

2014 - 2016	Editor-in-chief, *Alzheimer's & Dementia: Translational Research and Clinical Interventions* http://www.trci.alzdem.com/
2013 - present	Senior associate editor, *Alzheimer's & Dementia* http://www.alzheimersanddementia.com/
2011 - 2016	Section editor, BMC Psychiatry
1996 - present	Cochrane Dementia & Cognitive Impairment Group, editor, 1996 -
1994 - present	American Psychiatric Association, Committee for Practice Guidelines in Alzheimer's disease
1990 - 1991	Senior Scientific Advisor, National Institute of Mental Health, Division of Clinical Research

Honors

2020-	Della Martin Endowed Professor of Psychiatry and Neuroscience at USC
2015 -	Distinguished Life Fellow, America Psychiatric Association
2011 -	Fellow, American College of Neuropsychopharmacology
2007	Jay L. Foster Memorial Lecture. Univ of Pittsburgh Graduate School of Public Health
2006	Senator Mark Hatfield Award for Clinical Research, Alzheimer's Association
2006 -	Best Doctors in Southern California, *Los Angeles Magazine*
2006	21st Annual Research Awards, Los Angeles Alzheimer's Association
2004 - 2009	America's Top Doctors (Castle Connolly)
2003 - 2014	Distinguished Fellow, American Psychiatric Association
2002	American Association for Geriatric Psychiatry Senior Investigator Award
1996	11th Annual Research Awards, Los Angeles Alzheimer's Association
1993	1st Dorothy Kirsten French Memorial Lecture, French Foundation for Alzheimer Research
1992 - 2011	The Best Doctors in America and (2009) America's Top Doctors
1989	Tsai Memorial Lecture. Los Angeles/Brentwood VA Medical Center

C. Contributions to Science

1. Dr. Schneider has contributed to *improving methods in AD clinical trials* and cohort studies by leading in the design and operations of numerous early and later phase development AD programs; directed multicenter trials in AD and major depression, including the paradigm shifting CATIE-AD trial. He is responsible for the development of the ADCS-CGIC, the most common CGIC or CIBIC+ method.

a. **Schneider LS** 2008 Issues in design and conductance of clinical trials for cognitive-enhancing drugs. In: Animal and Translational Models of Behavioural Disorders, McArthur RA & Borsini F (ed) Elsevier, NY
b. **Schneider LS**, Clark C, Doody R, et al 2006 ADCS Prevention Instrument Project: ADCS-clinician's global impression of change scales (ADCS-CGIC), self-rated and study partner-rated versions. Alzheimer's Disease and Associated Disorders, 20(4) S3:124-38.
c. **Schneider LS** & Sano M 2009 Current Alzheimer disease clinical trials: methods and placebo outcomes. Alzheimer's & Dementia, 5(5):388-97. PMCID: PMC3321732
d. **Schneider LS** 2012 Recruitment methods for United States Alzheimer disease prevention trials. Journal of Nutrition, Health, and Aging, 16(4):331-5.

2. He and colleagues contributed evidence for secular changes in AD trials characteristics and outcomes and conducted work examining sources of bias using large meta-databases to assess the effects of Alzheimer-related biomarkers and genotypes on the efficiency of targeted clinical trials to improve methods.

a. **Schneider LS**, Kennedy RE, Cutter GR 2010 Requiring an amyloid-β1-42 biomarker for prodromal Alzheimer's disease or mild cognitive impairment does not lead to more efficient clinical trials. Alzheimer's & Dementia, 6:367-77. PMCID: PMC2947209
b. Kennedy RE, **Schneider LS,** Cutter GR 2012 Biomarker positive and negative subjects in the Alzheimer's Disease Neuroimaging Initiative cohort clinical characterization Current Alzh Research 9(10):1135-41
c. Kennedy RE, **Schneider LS,** Cutter GR 2013 Effect of APOE genotype status on targeted clinical trials outcomes and efficiency in dementia and mild cognitive impairment resulting from Alzheimer's Alzheimer's & Dementia:Journal of the Alzheimer's Association, 10(3):349-59. PMCID: PMC3900604
d. **Schneider LS**, Kennedy RE, Wang G, Cutter GR. 2015 Differences in Alzheimer disease clinical trial outcomes based on age of the participants. Neurology. Mar 17; 84(11):1121-7 PMCID: PMC437140520153

3. Dr. Schneider has contributed to the advancement of *treatment development* and regulatory science in several ways including development plans and guidelines.

a. Becker RE, Greig NH, Giacobini E, **Schneider LS**, Ferrucci L 2013 A new roadmap for drug development for Alzheimer's disease. Nature Reviews Drug Discovery, 13(2):156.

b. **Schneider LS**, Mangialasche F, Andreasen N, Feldman H, Giacobini E, Jones R, Mantua V, Mecocci P, Pani L, Winblad B, Kivipelto M 2014 Clinical trials and late-stage drug development for Alzheimer disease: appraisal from 1984 to 2014. Journal of Internal Medicine, 275(3):251-83. PMCID: PMC3956752

c. **Schneider LS** 2014 Rethinking FDA guidelines for early-stage Alzheimer's disease. Alzheimer's & Dementia: the Journal of the Alzheimer's Association, 10(2):247-50. PMCID: PMC3979530

d. **Schneider LS**, Goldberg TE. Composite cognitive and functional measures for early-stage Alzheimer's disease trials. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring.* 2020;12(1):e12017

4. He has made extensive contributions to the neuropsychopharmacology and therapeutics of Alzheimer disease and behavioral and psychological disorders.

a. **Schneider LS** 2005 Drug development, clinical trials, cultural heterogeneity in AD: the need for pro-active recruitment. Alzheimer's Disease and Associated Disorders, 19(4):279-83.

b. **Schneider LS**, Tariot PN, Dagerman KS, Davis SM, Hsiao JK, Ismail MS, Lebowitz BD, Lyketsos CG, Ryan JM, Stroup TS, Sultzer DL, Weintraub D, Lieberman JA, CATIE-AD Study Group 2006 Effectiveness of atypical antipsychotic drugs for patients with Alzheimer's disease. New Engl J of Med 355(15): 1525-38

c. Miller EA, Rosenheck RA & **Schneider LS** 2010 Assessing the relationship between health utilities, quality of life, and health care costs in Alzheimer's disease: CATIE-AD study. Cur Alzh Research, 7(4):348-57

d. **Schneider LS**, Frangakis C, Drye LT, Devanand DP, Marano CM, Mintzer J, Mulsant BH, Munro CA, Newell JA, Pawluczyk S, Pelton G, Pollock BG, Porsteinsson AP, Rabins PV, Rein L, Rosenberg PB, Shade D, Weintraub D, Yesavage J, Lyketsos CG. Heterogeneity of treatment response to citalopram for Alzheimer disease patients with aggression or agitation: the CitAD randomized clinical trial. American Journal of Psychiatry 2016; May 1;173(5):465-72. PMID: 26771737

5. Most recently, he has contributed leadership to numerous endeavors in Alzheimer pathology relating to COVID-19, signal detection, nutrition, and prevention.

a. **Schneider LS**, Qiu Y, Thomas RG, Evans C, Jacobs DM, Jin S, Kaye JA, LaCroix AZ, Messer K, Salmon DP, Sano M, Schafer K, Feldman HH. Impact of potential modifications to Alzheimer's disease clinical trials in response to disruption by COVID-19: a simulation study. Alzheimers Res Ther. 2021 Dec 20;13(1):201. doi: 10.1186/s13195-021-00938-w. PMID: 34930444; PMCID: PMC8686345.

b. Howard RS, Goldberg TE, Luo J, Munoz C, **Schneider LS**. Reliability of the NACC Telephone-administered Neuropsychological Battery (T-cog) and Montreal Cognitive Assessment for participants in the USC ADRC. Alzheimers Dement (Amst). 2023 Mar 8;15(1):e12406. doi: 10.1002/dad2.12406. PMID: 36911358; PMCID: PMC9994164.

c. Goldberg TE, Lee S, Devanand DP, **Schneider LS.** Comparison of relative change with effect size metrics in Alzheimer's disease clinical trials. J Neurol Neurosurg Psychiatry. 2023 Dec 14;95(1):2-7. doi: 10.1136/jnnp-2023-331941.PMID: 37979967.

d. Yassine HN, Samieri C, Livingston G, Glass K, Wagner M, Tangney C, Plassman BL, Ikram MA, Voigt RM, Gu Y, O'Bryant S, Minihane AM, Craft S, Fink HA, Judd S, Andrieu S, Bowman GL, Richard E, Albensi B, Meyers E, Khosravian S, Solis M, Carrillo M, Snyder H, Grodstein F, Scarmeas N, **Schneider LS.** Nutrition state of science and dementia prevention: recommendations of the Nutrition for Dementia Prevention Working Group. Lancet Healthy Longev. 2022 Jul;3(7):e501-e512. doi:10.1016/s2666-7568(22)00120-9. Epub 2022 Jul 4. PMID: 35821792; PMCID: PMC9273104.

List of Published Work:
https://pubmed.ncbi.nlm.nih.gov/?term=Lon+Schneider&sort=date (May 2024: N=537)
https://www.scopus.com/results/authorNamesList.uri?st1=Schneider&st2=Lon&origin=searchauthorlookup (May 2024, N=544)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: Joe, Elizabeth

eRA COMMONS USER NAME (credential, e.g., agency login): ejoe01

POSITION TITLE: Assistant Professor of Clinical Neurology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
University of California, Berkeley, Berkeley, CA	BA	05/2005	Earth and Planetary Science
Tufts University School of Medicine, Boston, MA	MD	05/2013	Medicine
Mount Auburn Hospital, Cambridge, MA	Resident	06/2014	Internal Medicine
University of Southern California, Los Angeles, CA	Resident	06/2017	Neurology
University of Southern California, Los Angeles, CA	Fellow	06/2018	Behavioral Neurology

A. Personal Statement

I have the knowledge, skills, and experience needed to successfully complete the proposed project for the NACC New Investigator Award. The overall goal of my research is to ensure that high quality healthcare is available for all people living with dementia, regardless of their background or where they access care. I am a behavioral neurologist and completed my neurology residency and fellowship at USC prior to transitioning to faculty in 2018. In my current position, I see patients at USC's Memory and Aging Center; Los Angeles General Medical Center, a large safety net hospital; and USC's Alzheimer's Disease Research Center (ADRC). I currently serve as the Project Director for the Vascular Cohort Study, part of the ADRC's primary longitudinal cohort, as well as several related observational studies. In this role I manage collection of clinical data from a cohort of more than 200 older adults ranging from cognitively normal to mild dementia, including overseeing regulatory and data quality issues as well as performing participant evaluations and lumbar punctures.

In addition to my role at the ADRC, I have a strong interest in health equity and patient-centered outcomes in dementia care. I am currently supported by a KL-2 award to study dementia epidemiology, referrals, and disparities within Los Angeles County's safety net health system using EHR-derived data, and was recently awarded a GEMSSTAR R03 to study patient factors contributing to delayed diagnosis of dementia in this health system. My work within DHS will facilitate recruitment of an increasingly diverse cohort as outlined in the Plan for Enhancing Diverse Perspectives.

Ongoing and recently completed projects that I would like to highlight include:

R03AG082997 Joe, Role: PI 9/1/2023-5/31/2025 Barriers to early identification of dementia in a safety net healthcare system

KL2TR001854 Patino-Sutton (PI) 6/1/2021-5/31/2024 Mentored Career Development in Clinical Translational Science (MCD-CTS/KL2) Scholar

UL1TR001855 Buchanan (PI), Role: Principal Investigator for pilot award 07/01/2020-06/30/2022 SC CTSI DHS-USC-UCLA Implementation Science Award: Group Medical Visits for Neurology Care

P30AG066530 Chui/Toga/Zlokovic (MPI), Role: co-investigator
04/01/2020-03/31/2025
Alzheimer Disease Research Center (ADRC)

R01AG058162 Marmarelis/Chui/Zhang/Billinger (MPI), Role: co-investigator 9/1/2018-5/31/2023, NCE 6/1/2023-5/31/2025
Dynamic vasomotor reactivity in Alzheimer’s disease

1P01G052350 B Zlokovic / A Toga (MPI), Role: co-investigator 9/30/2016 – 3/31/2027 Vascular contributions to dementia and genetic risk factors for Alzheimer disease.

1. Ringman JM, Joe E, Sheikh-Bahaei N, Miller C, Vinters HV, Guzman S, Chui HC. Cerebral Amyloid Angiopathy-related Inflammation Presenting With a Cystic Lesion in Young-onset Alzheimer Disease. Alzheimer Dis Assoc Disord. 2021 Jul-Sep 01;35(3):265-268. PubMed Central PMCID: PMC8289923.
2. Joe E, Medina LD, Ringman JM, O'Neill J. (1)H MRS spectroscopy in preclinical autosomal dominant Alzheimer disease. Brain Imaging Behav. 2019 Aug;13(4):925-932. PubMed Central PMCID: PMC6295351.
3. Harper DG, Joe EB, Jensen JE, Ravichandran C, Forester BP. Brain levels of high-energy phosphate metabolites and executive function in geriatric depression. Int J Geriatr Psychiatry. 2016 Nov;31(11):1241-1249. PubMed PMID: 26891040.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2019 - 2022	Residency Education Advisor, LAC+USC/USC Neurology Residency Program, Los Angeles, CA
2018 -	Assistant Professor of Clinical Neurology, University of Southern California, Los Angeles, CA
2016 - 2017	Education Chief Resident, LAC+USC/USC Neurology Residency Program, Los Angeles, CA
2007 - 2009	Environmental Scientist, Garcia and Associates, Oakland, CA
2006 - 2007	Research Assistant, T.C.Hoffmann & Associates, Oakland, CA

Honors

2004 - 2005	Diedrick Memorial Scholarship , Contra Costa Gems & Minerals Society
2001 - 2005	Chancellor's Scholar, U.C. Berkeley
2018	Young Investigator Award, Alzheimer's Greater Los Angeles
2018	Board Certification in Behavioral Neurology and Neuropsychiatry, Board Certification in Behavioral Neurology and Neuropsychiatry
2017	Young Investigator Award, Dominantly Inherited Alzheimer Disease Family Conference
2017	Resident Scholarship to the Annual Meeting, American Academy of Neurology
2017	Board Certification in Neurology, American Board of Psychiatry and Neurology
2015	Physicians & Surgeons License, Medical Board of California
2013	Ethel and Reuben Russman Prize in Geriatrics, Tufts University School of Medicine
2005	Outstanding Woman Student, Association for Women Geoscientists

C. Contribution to Science

1. Vascular and genetic contributions to dementia: As project director of our ADRC's primary longitudinal cohort, I oversee collection of clinical data and manage data quality for this and related studies as well as conduct research evaluations for our participants. This cohort is enriched in APOE4 carriers and people with vascular and metabolic risk factors and has yielded new insights on blood brain barrier dysfunction in Alzheimer disease pathogenesis.

    a. Contreras JA, Fujisaki K, Ortega N, Barisano G, Sagare A, Pappas I, Chui H, Ringman JM, Joe EB, Zlokovic B, Toga AW, Pa J. Decreased functional connectivity is associated with increased levels of Cerebral Spinal Fluid soluble-PDGFRβ, a marker of blood brain barrier breakdown, in older adults. Res Sq. 2023 Mar 7; PubMed Central PMCID: PMC10029080.
    b. Martinez AE, Weissberger G, Kuklenyik Z, He X, Meuret C, Parekh T, Rees JC, Parks BA, Gardner MS, King SM, Collier TS, Harrington MG, Sweeney MD, Wang X, Zlokovic BV, Joe E, Nation DA, Schneider LS, Chui HC, Barr JR, Han SD, Krauss RM, Yassine HN. The small HDL particle hypothesis of Alzheimer's disease. Alzheimers Dement. 2023 Feb;19(2):391-404. PubMed Central PMCID: PMC10563117.

c. Albrecht DS, Sagare A, Pachicano M, Sweeney MD, Toga A, Zlokovic B, Chui H, Joe E, Schneider L, Morris JC, Benzinger T, Pa J. Early neuroinflammation is associated with lower amyloid and tau levels in cognitively normal older adults. Brain Behav Immun. 2021 May;94:299-307. PubMed Central PMCID: PMC8793040.

d. Montagne A, Nation DA, Sagare AP, Barisano G, Sweeney MD, Chakhoyan A, Pachicano M, Joe E, Nelson AR, D'Orazio LM, Buennagel DP, Harrington MG, Benzinger TLS, Fagan AM, Ringman JM, Schneider LS, Morris JC, Reiman EM, Caselli RJ, Chui HC, Tcw J, Chen Y, Pa J, Conti PS, Law M, Toga AW, Zlokovic BV. APOE4 leads to blood-brain barrier dysfunction predicting cognitive decline. Nature. 2020 May;581(7806):71-76. PubMed Central PMCID: PMC7250000.

2. Barriers to early identification of cognitive impairment and dementia: We presented different clinical vignettes to a nationally representative volunteer survey panel and asked them which scenarios should be evaluated by a physician. We used randomization to demonstrate that older adults are less likely to think their own memory problems warrant workup compared to those of others, suggesting denial or a lack of insight influences decision to seek care. We used Medicare claims data to examine the relationship to show that beneficiaries who participate in an Annual Wellness Visit (AWV) have a higher incident diagnosis of dementia, suggesting that a structured cognitive assessment improves detection of cognitive impairment.

   a. Thunell JA, Jacobson M, Joe EB, Zissimopoulos JM. Medicare's Annual Wellness Visit and diagnoses of dementias and cognitive impairment. Alzheimers Dement (Amst). 2022;14(1):e12357. PubMed Central PMCID: PMC9473487.

   b. Jacobson M, Joe E, Zissimopoulos J. Barriers to seeking care for memory problems: A vignette study. Alzheimers Dement (N Y). 2022;8(1):e12238. PubMed Central PMCID: PMC8919244.

3. Patient-centered outcomes in aging and dementia: Even with emerging disease-modifying treatments becoming available, most clinical care for older adults with memory disorders is focused on symptom relief and improving quality of life. We evaluated the impact of how self- vs proxy-report modulates the association between depression and quality of life in older adults with dementia. We used magnetic resonance spectroscopy to identify early brain chemical changes in preclinical autosomal dominant Alzheimer's disease and in late life depression.

   a. Joe E, Segal-Gidan F, Cummings JL, Galasko D, Tomaszewski Farias S, Johnson DK, Ross L, Yaffe K, Wang X, Schneider LS, Chui H, Ringman JM. Association Between Self- and Proxy-Reported Depression and Quality of Life in Mild-Moderate Alzheimer's Disease. Am J Geriatr Psychiatry. 2024 Jan;32(1):58-67. PubMed Central PMCID: PMC10843510.

   b. Ringman JM, Joe E, Sheikh-Bahaei N, Miller C, Vinters HV, Guzman S, Chui HC. Cerebral Amyloid Angiopathy-related Inflammation Presenting With a Cystic Lesion in Young-onset Alzheimer Disease. Alzheimer Dis Assoc Disord. 2021 Jul-Sep 01;35(3):265-268. PubMed Central PMCID: PMC8289923.

   c. Joe E, Medina LD, Ringman JM, O'Neill J. (1)H MRS spectroscopy in preclinical autosomal dominant Alzheimer disease. Brain Imaging Behav. 2019 Aug;13(4):925-932. PubMed Central PMCID: PMC6295351.

   d. Harper DG, Joe EB, Jensen JE, Ravichandran C, Forester BP. Brain levels of high-energy phosphate metabolites and executive function in geriatric depression. Int J Geriatr Psychiatry. 2016 Nov;31(11):1241-1249. PubMed PMID: 26891040.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: **John Matthew Ringman**

eRA COMMONS USER NAME (credential, e.g., agency login): **RINGMAN2**

POSITION TITLE: **Professor of Clinical Neurology**

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of California, Berkeley	B.A.	05/1988	Neurobiology, Psychology
McGill University School of Medicine	M.D.	06/1993	Medicine
Baylor College of Medicine	Intern	06/1994	Internal Medicine
Baylor College of Medicine	Resident	06/1997	Neurology
UCLA Department of Neurology	Fellow	06/1999	Behavioral Neurology and Dementia
UCLA Department of Biomathematics	M.S.	02/2008	Masters of Science, Clinical Research

A. Personal Statement

From my undergraduate years studying psychology and neurobiology, I have been interested in behavioral neurology. Appreciating the overlap of disorders of cognition/behavior and those of movement, while a resident at Baylor College of Medicine I worked with Joseph Jankovic which ultimately led to the publication of a case series on persons with autism spectrum disorders and tics. I then came to UCLA in 1997 where I underwent subspecialty training in this area under the mentorship of Drs. Jeffrey Cummings and Mario Mendez. I then went to U.C. Irvine as a junior faculty member and my interest in dementia was focused when I began to encounter families we identified to have the A431E mutation in the *PSEN1* gene, a founder effect arising from Jalisco State in Mexico. Strengthened by collaborations with investigators in Mexico and under the guidance and additional mentorship from Claudia Kawas, M.D., I wrote and obtained a career development ("K") award from the NIA to perform functional MRI studies in presymptomatic persons carrying this and other autosomal dominant AD (ADAD) mutations. I returned to the UCLA Alzheimer's Disease Research Center in 2003 where I obtained a Masters degree in clinical research, improved my Spanish, gained experience in cross-lingual and cross-cultural research, and grew the observational ADAD study which included clinical, genetic, biochemical, and imaging assessments of persons at-risk for ADAD mutations. These efforts have directly resulted in the publication of at least 36 original research papers since 2004. In 2007 the Dominantly Inherited Alzheimer Network (DIAN, U01AG032438, PI: J Morris) was created that extended our study (and similar studies at other sites) into a multi-site collaboration. At UCLA we were one of the original DIAN sites and were the second highest enrolling site (the lead site, Washington University, being first). I joined the faculty at USC in May of 2015 and am continuing our studies with ADAD here via a U01 to perform the Human Connectome Protocol in persons of Mexican descent with or at-risk for specific ADAD mutations (the A431E *PSEN1* mutation and the V717I *APP* mutation) and late-onset AD (LOAD). This project has been extended through an R01 to become a longitudinal study, focusing on defining changes occurring in motor, visual, and olfactory systems and their neural bases in subtypes of AD. We are extending this genetic and phenotyping work with collaborators in Mexico through another R01 from Fogarty International. My continued work with this population of Mexican origin coupled with my clinical practice and work within ADRCs in Los Angeles, where 48% of the population is Latino, has inspired me to focus my interests on issues in the diagnosis and management of dementia in this population. As Co-Leader of the

Outreach and Education Core of the USC ADRC, I lend my interest and expertise in clinical and genetic aspects of dementia with a particular focus on the Latinx population and serve a critical role as a liaison between the ORE Core and the Clinical Core.

R01AG062007
Ringman (PI)
09/01/18-8/31/23 (on NCE to 11/24)
Motor, Visual, and Olfactory Changes in Genetic Subtypes of Alzheimer's Disease

R01AG069013
Ringman/Figuera (Co-PIs)
07/01/2020-6/30/25
Phenotype and Genotype of Autosomal Dominant Alzheimer's Disease in Jalisco, Mexico

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2019 - present	Chief, Division of Memory Disorders, Department of Neurology, USC
2016 - present	Helene and Lou Galen Endowed Professor
2015 - present	Professor of Clinical Neurology, Keck School of Medicine of USC
2015 - present	Director, Behavioral Neurology and Neuropsychiatry Fellowship Program, USC Department of Neurology
2013 - 2015	Clinical Professor, UCLA Department of Neurology
2010 - 2011	Interim Director, Mary S. Easton Center for Alzheimer's Disease Research at UCLA
2010 - 2014	Chairman, Medical & Scientific Advisory Council, Alzheimer's Association, Los Angeles
2009 - 2013	Associate Clinical Professor, UCLA Department of Neurology
2006 - present	Member, Medical & Scientific Advisory Council, Alzheimer's Association, Los Angeles
2006 - 2010	Chair, Publications Committee, National Alzheimer's Coordinating Center
2009 - 2015	Leader, Clinical Core, UCLA Alzheimer's Disease Research Center
2003 - 2008	Director, Kagan Alzheimer's Disease Treatment Development Program, U.C., Los Angeles, Department of Neurology
2003 - 2008	UCLA Site Principal Investigator, Alzheimer's Disease Cooperative Study
2003 - 2007	Assistant Professor, UCLA Department of Neurology
1999 - 2003	Assistant Clinical Professor (nontenured), U.C. Irvine Department of Neurology, Department Chair- Mark Fisher, M.D.
2000 - 2003	Consulting Neurologist (non-tenured), Long Beach Veterans Administration Medical Center, Acting Chair- Marc Ofstein, M.D, 4/02-08/03, Site Director, South West Parkinson's Disease Research, Education, and Clinical Center

Honors

2016	Fellow, American Academy of Neurology
2004	Turken Award for Research in Alzheimer 's disease
2003 - 2005	Shirley and Jack Goldberg Neuroscience Scholar
1998	First Prize, Gustaf Retzius Neuroanatomy Competition
1990	Francis McNaughton Book Prize for Neuroanatomy at McGill Medical School
1986	Phi Beta Kappa, U.C. Berkeley

C. Contributions to Science

1. My biggest contribution to the medical literature has been the characterization of clinical, cognitive, biochemical, and imaging changes occurring in the early- and presymptomatic stages of AD through the study of persons at-risk for inheriting fully-penetrant autosomal dominant mutations causing AD of early onset. To date, this research program has led directly to the publication of 30 original research articles. Among these papers is an early validation, in ADAD, of changes in Aβ42 and *tau* in the cerebrospinal (CSF) that had been previously demonstrated to occur in the presymptomatic phase of late-onset AD[1]. Additionally, in this paper, we showed evidence of increased oxidative stress in this phase of disease (the

a. **Ringman JM,** Sachs MC, Zhou Y, Monsell SE, Saver JL, Vinters HV. 2014 Clinical predictors of severe cerebral amyloid angiopathy and influence of APOE genotype in persons with pathologically verified Alzheimer disease. JAMA Neurol,71(7):878-883.
b. **Ringman JM**, Monsell S, Ng DW, et al. 2016 Neuropathology of Autosomal Dominant Alzheimer Disease in the National Alzheimer Coordinating Center Database. J Neuropathol Exp Neurol.

4. Since I was a neurobehavior fellow at UCLA and continuing when I led the Kagan AD Treatment Development Program there in 2003, I have obtained considerable experience and expertise in AD clinical trials. A site investigator for at least 14 such studies, this productivity was most evident in our publication of the investigator-initiated trial of curcumin for AD[12]. Though a negative biomarker trial, this study brought attention to the safety of curcumin and emphasized the importance of using readily bioavailable agents in AD. Prior to[13] and subsequently in collaboration with the DIAN Treatment Unit[14,15], we have been engaged in initiating trials to prevent ADAD in pre- and early-symptomatic mutation carriers.
   a. **Ringman JM**, Frautschy SA, Teng E, et al. 2012 Oral curcumin for Alzheimer's disease: tolerability and efficacy in a 24-week randomized, double blind, placebo-controlled study. Alzheimers Res Ther, 4(5):43.
   b. **Ringman JM**, Grill J, Rodriguez-Agudelo Y, Chavez M, Xiong C. 2009 Commentary on "a roadmap for the prevention of dementia II: Leon Thal Symposium 2008. Prevention trials in persons at risk for dominantly inherited Alzheimer's disease: opportunities and challenges. Alzheimers Dement, 5(2):166-171.
   c. Bateman RJ, Aisen PS, De Strooper B, et al. 2011 Autosomal-dominant Alzheimer's disease: a review and proposal for the prevention of Alzheimer's disease. Alzheimers Res Ther, 3(1):1.
   d. Mills SM, Mallmann J, Santacruz AM, et al. 2013 Preclinical trials in autosomal dominant AD: implementation of the DIAN-TU trial. *Rev Neurol (Paris)*,169(10):737-743.

5. Importantly, over the years working in Southern California and with the ADAD population, I have developed expertise in issues specific to the experience of dementia in Latinos. Through work at the ADRC at UCLA, we confirmed previous observations that Latinos have a younger onset of dementia that, in our dataset, appeared to be at least partly independent of their higher incidence of vascular risk factors[16]. This was contrary to my initial impression that such a difference might be accountable by cultural factors[17]. We also observed, in the NACC database, that CAA may be more common in Latinos but this observation awaits confirmation[10]. Most recently, we documented the utility of the Spanish-English Neuropsychological Assessment Scale (SENAS) in identifying the earliest cognitive changes occurring in ADAD as well as those of aging in a Latinx population[18] and we shared these results in a Conference focusing on dementia in Latin America in Mexico City in 2023 hosted by the Alzheimer's Association[19]. I bring this sensitivity to cultural and linguistic influences on dementia assessment and diagnosis in Latinos to the current project where I will perform a key role in the recruitment and assessment of subjects as well as the interpretation of cognitive assessments and other outcomes.
   a. Fitten LJ, Ortiz F, Fairbanks L, et al. 2014 Younger age of dementia diagnosis in a Hispanic population in southern California. Int J Geriatr Psychiatry, 29(6):586-593.
   b. Ringman JM, Flores DL. 2005 Earlier Alzheimer onset in Latino persons: ethnic difference vs selection bias. Arch Neurol,62(11):1786-1787.
   c. Tureson KN, Beam CR, Medina LD et al. 2023. Use of the Spanish English Neuropsychological Assessment Scale in older adult Latines and those at risk for autosomal dominant Alzheimer's disease. J Clin Exp Neuropsychol. 2023 Aug;45(6):553-569.
   d. Advancements in dementia research, diagnostics, and care in Latin America: Highlights from the 2023 Alzheimer's Association International Conference Satellite Symposium in Mexico City. Sosa AL, Brucki SM, Crivelli L, et al. Alzheimers Dement. 2024 May 27. PMID: 38801124

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: D'Orazio, Lina Mercedes

eRA COMMONS USER NAME (credential, e.g., agency login): LINADOR

POSITION TITLE: Assistant Professor of Clinical Neurology, University of Southern California

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Brown University	B.A.	05/2002	Psychology
University of Southern California	Ph.D.	12/2013	Clinical Psychology
VA Greater Los Angeles Healthcare System		8/2014	Postdoctoral Fellowship Health Psychology
Keck School of Medicine of USC		8/2016	Postdoctoral Fellowship Neuropsychology

A. Personal Statement

My role as co-investigator on this grant is to contribute to the study design involving neuropsychological assessment and cognitive data collection. As a bilingual (Spanish/English) and bicultural licensed clinical psychologist specializing in neuropsychology and cognitive assessment of Spanish speakers and Spanish/English bilinguals across the education spectrum, I oversee the selection and administration of cognitive testing in both English and Spanish, as well as oversight of ongoing scoring, quality control, and interpretation of these tests. In my role as the founding director of the Department of Neurology Cultural Neuropsychology Program at Keck Medicine of USC , I serve as a co-investigator and consultant for research involving cognitive assessment and/or Latinx populations across the medical school, including serving as a co-investigator for the neurology department's NACC Alzheimer's Disease Research Center and the ADC National Latino Task Force and contributor to the latest Spanish translation of the NACC Uniform Data Set 3 cognitive battery, as well as a member of the NIH Mark VCID (Biomarkers for Vascular Contributions to Cognitive Impairment and Dementia) Consortium. I have over 15 years of experience in neuropsychological research with English and Spanish speakers from the work listed above and from clinical and research positions I held prior to my USC Keck clinical faculty appointment, including cognitive and psychophysiological studies for Columbia University's Department of Psychiatry Behavioral Medicine Program, psycho-oncology research in the USC Department of Psychology and LAC Hospital, applied behavioral medicine research in the West Los Angeles VA Health Psychology Fellowship program, cultural psychiatry/global mental health research in the Department of Psychiatry and Human Behavior of the Warren Alpert Medical School of Brown University. Through these various clinical and research opportunities, I have cultivated my expertise in culturally informed neuropsychological and psychodiagnostic assessment, and in recruitment, management of multisite research protocols, data management and analysis, monitoring compliance with ethical and institutional standards and multidisciplinary collaborations. As a result of my comprehensive research experience, I possess the clinical, scientific, and administrative expertise needed for my role in designing the cognitive testing battery and overseeing its administration, scoring, interpretation, and quality control for the proposed study.

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2016 – present Director of Cross-Cultural Neuropsychology, Department of Neurology, USC Keck School of Medicine
2015 – present Assistant Professor of Clinical Neurology, USC Keck School of Medicine
2014 – 2016 Neuropsychology Postdoctoral Fellow/Instructor of Clinical Neurology, Department of Neurology, USC Keck School of Medicine
2013 – 2014 Health Psychology Postdoctoral Fellow, VA Greater Los Angeles Health Care System West LA Medical Center
2012 – 2013 Clinical Psychology Intern, VA Greater Los Angeles Health Care System West LA Medical Center
2011 – 2012 Teaching Assistant, Department of Psychology, USC
Courses: Developmental Psychology, Clinical Interviewing (Graduate Level), Assessment (Graduate Level)
2009 – 2012 Latino Mental Health Clinical Consultant, Psychological Services Center, Department of Psychology, USC
2008 – 2010 Research Fellow, Department of Obstetrics & Gynecology, USC Keck School of Medicine
2008 – 2009 Bilingual Neuropsychological Examiner, Alzheimer's Disease Research Center, Department of Neurology, USC Keck School of Medicine
2007 – 2009 Bilingual Geriatric Psychology Therapist and Neuropsychological Examiner, Psychological Services Center, Department of Psychology, USC
2007 – 2008 Teaching Assistant, Department of Psychology, USC
Courses: Minority Mental Health: Clinical Considerations, Assessment, and Treatment
2006 – 2008 Bilingual Family & Child Therapist and Neuropsychological Examiner, Psychological Services Center, Department of Psychology, USC
2006 – 2007 Teaching Assistant, Department of Psychology, USC
Courses: Adolescent Development, Introduction to Forensic Psychology, Developmental Psychology
2005 – 2006 Graduate Research Fellow, Department of Psychology, USC
2002 – 2005 Research Coordinator & Bilingual Neuropsychological Examiner, Behavioral Medicine Program, Department of Psychiatry, Columbia University
2001 – 2002 Research Assistant and Translator, Department of Psychiatry and Human Behavior, Warren Alpert Medical School of Brown University

Other Experience and Professional Memberships

2017 – present Neuropsychologist, NIH Mark VCID (Biomarkers for Vascular Contributions to Cognitive Impairment and Dementia) Consortium: Clinical Data, Physiological Data and Cognitive Assessments Subcommittee
2016 – present Neuropsychologist, Alzheimer's Disease Centers Latino Task Force
Alzheimer's Disease Centers, NIA NIH
2014 – present National Latino/a Psychological Association, Member
2014 – present Hispanic Neuropsychological Society, Member
2013 – present International Neuropsychological Society, Member
2009 – present Association for Contextual and Behavioral Science, Member
2008 – present Latinas Contra Cancer, Supporter
2006 – 2008 Keck School of Medicine Cervical Cancer Research Consortium
2005 – present American Psychological Association, Member
Division 35 Society for the Psychology of Women
Section 3 Concerns of Hispanic Women/Latinas
Division 38 Health Psychology
Division 40 Society for Clinical Neuropsychology
Division 45 Society for the Psychological Study of Ethnic Minority Issues
2004 – present Society of Behavioral Medicine, Member

Honors

2018 Yarrow Family Award, Department of Neurology

2012 American Psychological Association (APA) Minority Fellowship Program Tuition Support Award
2011 APA Minority Fellowship Program Dissertation Support Award
2011 APA Minority Fellowship Program 2011-2012 Fellow
2010 APA Minority Fellowship Program 2010-2011 Fellow
2009 USC Department of Psychology Summer Fellowship
2008 USC Keck School of Medicine Department of OB/GYN Research Fellow
2007 USC EDGE-SBE FIRST Summer Institute Fellowship

C. Contributions to Science *(Relevant to proposed study)*

Cross-cultural Neuropsychology & Neuropsychological Assessment of Spanish speakers

Hispanics are the largest minority group in the US and comprise approximately 7% of the population older than 65 years of age. They often encounter challenges related to healthcare access and quality of care. The prevalence of dementia among Hispanics is higher than that of non-Hispanic whites and they frequently present at a more advanced stage of illness. Cognitive evaluation should take into account not only socio-demographic information but also consider cultural factors to avoid misdiagnosis and guide management and support. Ideally, a provider who has appropriate knowledge and familiarity with Hispanic culture should conduct the neuropsychological assessment, and the tests used to measure cognitive functioning should have been developed or validated for Hispanics in the US.

1. Ramirez Gomez, L., Jain, F.A., & **D'Orazio, L.M**. (2017) Assessment of the Hispanic Cognitively Impaired Elderly Patient. In J. Ringman & R.W. Evans (Eds.), *Neurologic Clinics: Dementia* (pp.207-229). Philadelphia, PA: Elsevier Inc.
2. **D'Orazio, L.M.,** Medina, L.D., & Ringman, J.M. (2018). Performance on the Cognitive Abilities Screening Instrument (CASI) among Spanish and English Speakers at-risk for autosomal dominant Alzheimer's Disease. Poster presented at the American Academy of Neurology Annual Meeting. Los Angeles, CA.
3. Ashimatey, B.S., **D'Orazio, L.M.**, Ma, SJ, et al. (2020). Lower retinal capillary density in minimal cognitive impairment among older Latinx adults. *Alzheimer's & Dementia*; *12*:e12071. https://doi.org/10.1002/dad2.12071
4. Tureson, K. N., Beam, C. R., Medina, L. D., Segal-Gidan, F., **D'Orazio, L. M**., Chui, H., ... & Ringman, J. M. (2023). Use of the Spanish English Neuropsychological Assessment Scale in older adult Latines and those at risk for autosomal dominant Alzheimer's disease. *Journal of Clinical and Experimental Neuropsychology*. 1-17. https://doi.org/10.1080/13803395.2023.2284971

Vascular & Genetic Contributions to Alzheimer's Disease.

Age related diseases causing dementia are an increasing global, social and economic catastrophe that requires broad and aggressive research. Alzheimer's disease (AD) is the most common cause of cognitive impairment in older adults and affects over 5 million people in the US alone. Although vascular and genetic contributions to dementia and AD are increasingly recognized, the role of the cerebrovascular system in the pathogenesis of dementia and AD, and the underlying neurovascular mechanisms remain largely unknown and under-researched. The overall goals of this ongoing work are to advance current knowledge on the vascular and genetic contributions to dementia and AD by identifying the role neurovasculature plays in cognitive decline among individuals with the major genetic risk factor for late-onset AD (i.e., apolipoprotein E-ε4 (*APOE4*) gene) and early-onset autosomal dominant AD (ADAD), i.e., presenilin 1 (*PSEN1*) mutations).

1. Montagne, A., Nation, D. A., Sagare, A. P., Barisano, G., Sweeney, M. D., Chakhoyan, A., Pachicano, M., Joe, E., Nelson, A. R., **D'Orazio, L. M.**, Buennagel, D. P., Harrington, M. G., Benzinger, T., Fagan, A. M., Ringman, J. M., Schneider, L. S., Morris, J. C., Reiman, E. M., Caselli, R. J., Chui, H. C., … Zlokovic, B. V. (2020). APOE4 leads to blood-brain barrier dysfunction predicting cognitive decline. Nature, 581(7806), 71–76. https://doi.org/10.1038/s41586-020-2247-3
2. Ashimatey, B.S., **D'Orazio, L.M.**, Ma, SJ, et al. (2020). Lower retinal capillary density in minimal cognitive impairment among older Latinx adults. *Alzheimer's & Dementia*; *12*:e12071. https://doi.org/10.1002/dad2.12071
3. Nation, D.A., Sweeney, M.D., Montagne, A., Sagare, A.P., **D'Orazio, L.M.**, Pachicano, M., Sepehrband, F., Nelson, A.R., Buennagel, D.P., Harrington, M.G., Benzinger, T.L.S., Fagan, A.M., Ringman, J.M., Schneider, L.S., Morris, J.C., Chui, H.C., Law, M., Toga, A.W., Zlokovic, B.V. (2019) Blood-brain barrier breakdown is an early biomarker of human cognitive dysfunction. *Nature Medicine, 25(2),* 270-276.
4. Arellanes, I. C., Choe, N., Solomon, V., He, X., Kavin, B., Martinez, A. E., [including **D'Orazio, L.M.**]... & Yassine, H. N. (2021). Brain Delivery of Supplemental Docosahexaenoic Acid (DHA): A Randomized

Placebo-Controlled Clinical Trial (vol 59, 102883, 2020). EBIOMEDICINE, 66.

Long-term Psychosocial and Cognitive Effects of Cancer in English and Spanish Speaking Cancer Survivors. Understanding the emotional and psychosocial experiences of the millions of long-term cancer survivors is essential for providing them with optimal care. With respect to cognitive research, English and Spanish speaking breast cancer patients who had completed surgical treatment were evaluated before and after chemotherapy using a neuropsychological assessment battery and self-report measures of depression, anxiety, quality of sleep and fatigue. Findings indicated that **a third of postmenopausal patients demonstrated cognitive impairments prior to chemotherapy and that hormone replacement therapy appeared protective against cognitive difficulties and fatigue. Breast cancer patients with past or current depression prior to chemotherapy were more vulnerable to fatigue after treatment.**

1. McKinley P, Tager F, Hershman D, Neugut A, Habif U, & **D'Orazio LM** (2004). Fatigue, sleep and psychological status prior to adjuvant therapy for breast cancer. Poster presented at the meeting of the Society of Behavioral Medicine, Baltimore, MD.
2. McKinley P, Tager F, Hershman D, Neugut A, Habif U, & **D'Orazio LM** (2004). Pre-chemotherapy cognitive functioning in women with breast cancer. Poster presented at the meeting of the Society of Behavioral Medicine, Baltimore, MD.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: : Pawluczyk, Sonia

eRA COMMONS USER NAME (credential, e.g., agency login): PAWLUCZYK

POSITION TITLE: Assistant Professor of Clinical Psychiatry and the Behavioral Sciences

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
San Isidro National College, Buenos Aires, Argentina	B.S.	07/1961	
University of Buenos Aires Medical School, Buenos Aires, Argentina	M.D.	06/1973	Medicine
Los Angeles County + University of Southern California General Psychiatry Residency Program		06/1973	Psychiatry
University of Southern California, Institute for Psychiatry and the Law Fellowship Program	Fellow	06/1989	Forensic psychiatry
University of Southern California, Geriatric Psychiatry Fellowship Program	Fellow	06/1991	Geriatric psychiatry

A. Personal Statement

Dr. Sonia Pawluczyk is Assistant Professor of Clinical Psychiatry and the Behavioral Sciences at USC. She is medical director of the USC State of California Alzheimer Disease Center and a clinician with the USC NIA Alzheimer Disease Research Center. After receiving her MD diploma at the Medical School of the University of Buenos Aires, she completed the Curso Superior de Psiquiatria at Hospital Borda. She moved to Los Angeles and entered the residency training in Psychiatry at LAC+USC. She did a fellowship in Psychiatry and the Law and in Geriatric Psychiatry. She then joined the Department of Psychiatry and has been part of the Geriatric Studies Center since 1991. Dr. Pawluczyk practices at the Memory and Aging Center at USC and she is also a consultant at Rancho Los Amigos Hospital. Here expertise is in clinical research. She is a physician member of the ADRC Clinical Core and evaluates and manages research participants. She is a site PI for several clinical studies and trials. Dr. Pawluczyk is fluent in Spanish. For this grant, Dr. Pawluczyk will evaluate, assess patients, perform clinical research ratings, and ensure GCP compliance.

Ongoing and recently completed projects that I would like to highlight include:

NIH P50 AG05142 NIA Chui HC (PI), Schneider (PI clinical core) 4/01/10 - 3/31/25
Alzheimer's Disease Research Center: Clinical Core
An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.
Role: Co-Investigator

03-75274 State of California Schneider (PI) 7/01/23 - 6/30/26
State of California Alzheimer's Disease Center
A California CDPH funded center supports the clinical care of and follow-up on patients with dementia and provides educational outreach to the community. Goals are to address the needs of our understudied and underserved communities and provide a model of statewide collaboration.
Role: Co-Investigator

R01AG062687 NIA Schneider (PI), Ballard (co-PI) 05/01/2019-2/29/2024
(-)-Phenserine inhibition of neuronal death in Alzheimer's disease and developing brain-labeled plasma exosomes assays as biomarkers for a phenserine phase 1b ascending dose trial
This project addresses the urgent need to develop therapeutics to prevent, delay and treat Alzheimer's disease. It proposes the development of biological, plasma sourced, newly identified plasma exosomes containing proteins associated with programmed neuron death, synaptic arborization, and neuro-inflammation as companion biomarkers reflecting the state of brain activities in humans that are otherwise inaccessible to investigators for the development of (-)-phenserine tartrate as a treatment. We will conduct a Phase 1b clinical study of this promising neuroprotective therapeutic which pre-clinically acts by a novel mechanism to prevent neurons from dying in the presence of neuropathology that is otherwise fatal to neurons in Alzheimer's and other disorders.
Role: Co-Investigator

R01AG051346 NIA Goldberg, Schneider LS, Harvey P (MPIs) 09/01/2018-05/31/2023
Novel Cognitive and Functional Measure for Alzheimer's Disease Prevention Trials
The goal of this project is to validate novel cognitive and everyday functional measures that have sharply attenuated practice effects and are not prone to ceiling effects for use in preclinical Alzheimer's disease (AD) trials in which participants are cognitively within normal limits.
Role: Co-Investigator

U01 AG 010483 NIA and Lilly Schneider (Site PI) 12/31/12 - 12/30/23
ADC-040-A4 Anti-Amyloid Treatment in Asymptomatic Alzheimer's Disease (A4 Study)
The A4 study is designed to test the hypothesis that solanezumab will slow decline on a cognitive composite measure designed to be sensitive to early decline in the preclinical stages of AD.
Role: Co-Investigator

R01 AG052510 NIA Lyketsos (PI) 10/01/16 – 5/30/24
S-citalopram for Agitation in Alzheimer's Disease (S-CitAD)
A multicenter clinical trial investigating SSRIs, s-citalopram specifically for agitation in Alzheimer disease
Role: Co-Investigator

R01 AG053267 NIA Bateman (PI) 10/01/16 – 9/30/26
DIAN-TU: Next Generation Prevention Trial
A multicenter clinical trial investigating solanezumab, gantenerumab, and a BACE1 inhibitor for dominantly inherited Alzheimer disease.
Role: Site PI

B. Positions, Scientific Appointments, and Honors

1992-	Assistant Professor of Clinical Psychiatry and the Behavioral Sciences
1988-1992	Clinical Instructor of Psychiatry and the Behavioral Sciences
1989-	American Association for Geriatric Psychiatry
1986-	World Federation for Mental Health
1984-	American Psychiatric Association
1984-	Southern California Psychiatric Society

C. Contributions to Science

Dr. Pawluczyk's contributions to science have included participating in methods development in cohort studies, early phase 1 studies.

1. Schneider LS, Lyness SA, Pawluczyk S, Gleason RP, Sloane RB. Does blood pressure and age predict response to tacrine? A preliminary report. Psychopharmacology Bulletin 27:309-314,1991.

2. Pollock VE, Schneider LS, Zemansky MF, Gleason RP, Pawluczyk S. Topographic quantitative EEG amplitude in recovered alcoholics. Psychiatry Research 45:25-32, 1992.

3. Schneider LS, Olin J, Pawluczyk S. A double-blind crossover pilot study of l-deprenyl (selegiline) combination with cholinesterase inhibitor in Alzheimer's disease. American Journal of Psychiatry 150:321-323,1993

4. Know, ED, Pawluczyk, S. Pisa syndrome in a patient treated with quetiapine and risperidone. College of Psychiatric and Neurologic Pharmacists Annual Conference. Colorado Springs, CO. April 21-24, 2013

She has contributed extensively to assessments in clinical trials. Directing trials sites in AD and major depression.

1. Schneider LS, Frangakis C, Drye LT, Devanand DP, Marano CM, Mintzer J, Mulsant BH, Munro CA, Newell JA, Pawluczyk S, Pelton G, Pollock BG, Porsteinsson AP, Rabins PV, Rein L, Rosenberg PB, Shade D, Weintraub D, Yesavage J, Lyketsos CG. Heterogeneity of treatment response to citalopram for Alzheimer disease patients with aggression or agitation: the CitAD randomized clinical trial. American Journal of Psychiatry 2016; May 1;173(5):465-72. PMID: 26771737; PMCID: PMC6419726

2. Schneider LS, Hernandez G, Zhao L, Franke AA, Chen YL, Pawluczyk S, Mack WJ, Brinton RD. Safety and feasibility of estrogen receptor-β targeted phytoSERM formulation for menopausal symptoms: phase 1b/2a randomized clinical trial. Menopause. 2019 Aug;26(8):874-884. PMID: 30889096; PMCID: PMC6663614.

3. Hernandez GD, Solinsky CM, Mack WJ, Kono N, Rodgers KE, Wu CY, Mollo AR, Lopez CM, Pawluczyk S, Bauer G, Matthews D, Shi Y, Law M, Rogawski MA, Schneider LS, Brinton RD. Safety, tolerability, and pharmacokinetics of allopregnanolone as a regenerative therapeutic for Alzheimer's disease: A single and multiple ascending dose phase 1b/2a clinical trial. Alzheimers Dement (N Y). 2020 Dec 16;6(1):e12107. PMID: 33344752; PMCID: PMC7744018.

4. Lee E, Gatz M, Tseng C, Schneider LS, Pawluczyk S, Wu AH, Deapen D. Evaluation of Medicare Claims Data as a Tool to Identify Dementia. J Alzheimers Dis. 2019;67(2):769-778. doi: 10.3233/JAD-181005. PMID: 30689589; PMCID: PMC7164318.

Her substantial contributions to therapeutics have involved the development of Spanish instruments for multicenter trials for both eligibility criteria and outcomes.

1. Sano M, Mackell JA, Ponton M, Ferreira P, Wilson J, Pawluczyk S, et al. The Spanish Instrument Protocol: Design and implementation of a study to evaluate treatment efficacy instruments for Spanish-speaking patients with Alzheimer's Disease. Alzheimer's Disease and Associated Disorders 11 (2): S57-S64, 1997.

2. Olin JT, Pawluczyk S, Kaufman GT, Taussig 1M, Henderson VW, Schneider LS. A comparative analysis of Spanish- and English-speaking Alzheimer's disease patients: eligibility and interest in clinical drug trials. Journal of Clinical Geropsychology 3:183-190, 1997

3. Sano M, Egelko S, Jin S, Cummings J, Clark CM, Pawluczyk S, Thomas RJ, Schittini M, Thal LJ and the members of the Alzheimer's Disease Cooperative Studies: Efficacy Instruments for the Spanish Speaking Patients in Alzheimer's Disease Clinical Trials. Alzheimer's Disease and Associated Disorders 20(4):232-41, 2006.

4. Olin JT, Fox LS, Pawluczyk S, Taggart NA, Schneider LS. A pilot randomized trial of carbamazepine for behavioral symptoms in treatment-resistant outpatients with Alzheimer disease. Am J Geriatr Psychiatry. 2001 Fall;9(4):400-5. PMID: 11739066.

BIOGRAPHICAL SKETCH

NAME: Michael Gordon Harrington

eRA COMMONS USER NAME: mikethescot

POSITION TITLE: Professor of Research Neurology

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Glasgow University, Scotland, UK	MD	05/1976	Medicine
Royal College of Physicians and Surgeons, UK	MRCP	05/1979	Medicine
National Institutes of Mental Health, Bethesda, MD	Postdoctoral	05/1983	Biochemistry

A. Personal Statement

My primary research is to learn the mechanisms that underlie pathophysiology. Armed with better knowledge, my goal is to conceive corrective treatments. I am excited to continue working with the USC ADRC. This strong, multidisciplinary group enhances widespread collaboration with the national efforts to make progress towards diagnosis and eventual treatments.

1. **Alzheimer's disease and related dementias.** I classify the clinical status of study participants to identify pre-symptomatic Alzheimer's disease (AD) using the Uniform Data Set (UDS-3), in collaboration with Dr. Chui and the USC AD center. I have recruited individuals with classifications of AD, mild cognitive impairment (MCI), or cognitively healthy (CH), and collected their cerebrospinal fluid (CSF), blood, and urine. I further classified CH individuals into those with normal CSF amyloid and tau ratio (CH-NAT) or pathological amyloid and tau ratios (CH-PAT). We followed these CH participants longitudinally to assess if the CH-PAT stage predicts neurodegenerative disease: for 60 CH individuals ≥4 years after initial visit we find 40% of CH-PATs, but none of the CH-NATs, have declined cognitively. This confirms CH-PATs have AD pathology and are at increased risk of developing AD symptoms. CH-PAT individuals offer potential for non-invasive monitoring of older individuals at increased risk for AD.

   a) Fonteh AN, Cipolla M, Chiang AJ, Edminster SP, Arakaki X, **Harrington MG**. Front Physiol. 2020 Polyunsaturated Fatty Acid Composition of Cerebrospinal Fluid Fractions Shows Their Contribution to Cognitive Resilience of a Pre-symptomatic Alzheimer's Disease Cohort. Feb 14;11:83. doi: 10.3389/fphys.2020.00083. eCollection 2020.PMID: 32116789.

   b) Montagne A, Nation DA, Sagare AP, Barisano G, Sweeney MD, Chakhoyan A, Pachicano M, Joe E, Nelson AR, D'Orazio LM, Buennagel DP, **Harrington MG**, Benzinger TLS, Fagan AM, Ringman JM, Schneider LS, Morris JC, Reiman EM, Caselli RJ, Chui HC, Tcw J, Chen Y, Pa J, Conti PS, Law M, Toga AW, Zlokovic BV. Nature. 2020 APOE4 leads to blood-brain barrier dysfunction predicting cognitive decline. May;581(7806):71-76. doi: 10.1038/s41586-020-2247-3. Epub 2020 Apr 29. PMID: 32376954

   c) Castor KJ, Shenoi S, Edminster SP, Tran T, King KS, Chui H, Pogoda JM, Fonteh AN, **Harrington MG.** PLoS One. 2020 Urine dicarboxylic acids change in pre-symptomatic Alzheimer's disease and reflect loss of energy capacity and hippocampal volume. Apr

16;15(4):e0231765. doi: 10.1371/journal.pone.0231765. eCollection 2020. PMID: 32298384

d) Arakaki X, Lee R, King KS, Fonteh AN, **Harrington MG**. Alpha desynchronization during simple working memory unmasks pathological aging in cognitively healthy individuals. PLoS One. 2019;14(1):e0208517. doi: 10.1371/journal.pone.0208517. eCollection 2019. PubMed PMID: 30601822; PubMed Central PMCID: PMC6314588.

Ongoing and recently completed projects that I would like to highlight include:

U01NS100614 – Phase II
Wang/Kashani/Ringman (PD/PI), Role: co-investigator
08/01/2023-07/31/2026
"Validation of Imaging and Blood-based Small Vessel Biomarkers in Multi-ethnic Population"

R01AG084072
Wang/Jiang/Shi PD/PI, Role: co-investigator.
07/2023 – 06/2028
"Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID)"

1RF1AG078362
Yassine, PD/PI. Role: co-investigator
09/2022 – 08/2025
"Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults"

2P01AG052350-06
Zlokovic/Toga PD/PI. Role: co-investigator
09/2016-03/2027
"Vascular contributions to dementia and genetic risk factors for Alzheimer's disease"

R01NS072497
Harrington (PI)
07/01/19-06/30/24
"Dysfunction of sodium homeostasis in migraine."

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2018-on	Professor of Research Neurology, University of Southern California, Los Angeles, CA
2019-2021	Scientific Director, Neurosciences, Huntington Medical Research Institutes, Pasadena, CA
2016-2021	NINDS Headache CDE Working Group, Co-Chair of Biomarkers Committee.
2015-2019	Director Neurosciences, Huntington Medical Research Institutes, Pasadena, CA
2014-2021	Research Professor of Psychology, Fuller Theological Seminary, Pasadena, CA
2014-2018	Adjunct Research Professor of Neurology, University of Southern California, Los Angeles, CA
2015-2021	Visiting Associate, Zilkha Neurogenetic Institute, University of Southern California, Los Angeles
2014-2021	ZRG1 BDCN-W (03). Member of Study Section Meeting
2014	DoD: DIS-CDMRP Discovery-Chronic Migraine and Post-Traumatic Headaches, review panel
2014-2018	National High Magnet Field Laboratory User Advisory Committee 2011-2016, Tallahassee, FL
2006-2015	ZRG1 BDCN-A (90)S, ZRG1 ETTN-C (53) R, ZRG1-IFCN-B-03M, NSD-A, ad hoc reviewer.

2004-2021	Society for Neuroscience member
2002-on	International Headache Society member
2001-on	American Headache Society member
2001-2013	Editorial Board of Disease Markers
1999-2001	President, American Electrophoresis Society
1998-2021	Founder, Molecular Neurology Program, Huntington Medical Research Institutes, Pasadena
1998-2005	Director of Student Program, Huntington Medical Research Institutes, Pasadena, CA
1990-on	American Association for the Advancement of Science (AAAS), Member
1988-1997	Member of the Beckman Institute, Caltech, Pasadena, CA
1987-2001	American Electrophoresis Society
1983-1988	Visiting Associate and Unit Chief, NIMH, Bethesda, MD
1979-1983	Neurology Training (Registrar), Institute of Neurological Sciences, Glasgow, Scotland
1977-1979	Internal Medicine Training (SHO/Registrar), Glasgow Royal Infirmary, Scotland
1976-1978	Medicine and Surgery (JHO), Glasgow Royal Infirmary, Scotland

Honors

2022	Runner up Best Science paper 2020-2021 by American Headache Society
1999-2001	President, American Electrophoresis Society
1993	Elected Fellow of the Royal College of Physicians and Surgeons, Glasgow
1974	Merit Awards in Undergraduate Medicine and Surgery, Glasgow University

C. Contributions to Science

1. **My initial research** (1979) explored clinical and biochemical features of brain pathophysiology during my neurology training. I investigated blood biochemistry in patients with basal ganglia calcification and reported renal toxicity from normal doses of acyclovir. I saw patients with transient altered conscious levels who had bizarre presentations referred to as “funny (peculiar) turns”; to investigate, I prospectively tested glucose metabolism and, surprisingly, found insulinomas in 2 of 25 consecutive patients. I realized I wanted to pursue medical research for my career.
    a) **Harrington MG**, Macpherson P, McIntosh WB, Allam BF, Bone I. The significance of the incidental finding of basal ganglia calcification on computed tomography. *J Neurol Neurosurg Psychiatry*. 1981 Dec; 44(12):1168-70. PMCID: PMC491242.
    b) **Harrington MG**, Durward WF, Simpson JA. Renal impairment and acyclovir. *Lancet*. 1981 Dec 5; 2(8258):1281. PMID: 6118687.
    c) **Harrington MG**, McGeorge AP, Ballantyne JP, Beastall G. A prospective survey for insulinomas in a neurology department. *Lancet*. 1983 May 14; 1(8333):1094-5. PMID: 6133118.

2. **Biochemical brain pathophysiology.** With my mentor’s encouragement (Professor J.A. Simpson, Institute of Neurological Sciences in Glasgow), I started biochemical protein surveys (two-dimensional electrophoresis) of cerebrospinal fluid to find molecular measures of brain dysfunction. I was offered a position at NIH in Bethesda to continue this work, which provided >2,000 protein “spots” that could be quantified and assessed qualitatively. I found changes with gender, age, and every disorder studied. The most clinically useful result of this research was that in 1986 I discovered acidic, 30,000 molecular weight proteins that were diagnostic for Creutzfeldt-Jakob disease. I moved from NIH to Caltech to obtain amino acid sequences from each protein species. I sequenced the CJD biomarker, the 14-3-3 gamma protein, which became the first clinically useful biomarker discovered by the proteomics approach. This test is still a useful clinical tool to diagnosis prion disease.
    a) **Harrington MG**, Merril CR, Asher DM, Gajdusek DC. Abnormal proteins in the cerebrospinal fluid of patients with Creutzfeldt-Jakob disease. *N Engl J Med*. 1986 Jul

31; 315(5):279-83. PMID: 3523251.

b) Hsich G, Kenney K, Gibbs CJ, Lee KH, **Harrington MG**. The 14-3-3 brain protein in cerebrospinal fluid as a marker for transmissible spongiform encephalopathies. *N Engl J Med*. 1996 Sep 26; 335(13):924-30. PMID: 8782499.

c) Lee KH, **Harrington MG**. 14-3-3 and BSE. *Vet Rec*. 1997 Feb 22; 140(8):206-7. PMID:9061874.

d) **Harrington MG**, Fonteh AN, Biringer RG, Huhmer AF, Cowan RP. Prostaglandin D synthase isoforms from cerebrospinal fluid vary with brain physiology. Disease Markers 2006;22:73-81. PMID: 16410653.

3. **Migraine.** I have been investigating the basis for migraine pathophysiology for 25 years. We discovered a new mechanism for migraine based on altered sodium homeostasis and I continue to explore methods to correct the sodium disturbance in a rodent migraine model.

a) Gross NB, Abad N, Lichtstein D, Taron S, Aparicio L, Fonteh AN, Arakaki X, Cowan RP, Grant SC, **Harrington MG**. Endogenous Na+, K+-ATPase inhibitors and CSF [Na+] contribute to migraine formation. PLoS One. 2019;14(6):e0218041. doi: 10.1371/journal.pone.0218041. eCollection 2019. PubMed PMID: 31173612; PubMed Central PMCID: PMC6555523.

b) Meyer MM, Schmidt A, Benrath J, Konstandin S, Pilz LR, **Harrington MG**, Budjan J, Meyer M, Schad LR, Schoenberg SO, Haneder S. Cerebral sodium (23Na) magnetic resonance imaging in patients with migraine - a case-control study. Eur Radiol. 2019 Jul 1;. doi: 10.1007/s00330-019-06299-1. [Epub ahead of print] PubMed PMID: 31264011.

c) Cowan RP, Gross NB, Sweeney MD, Sagare AP, Montagne A, Arakaki X, Fonteh AN, Zlokovic BV, Pogoda JM, **Harrington MG.** Evidence that blood-CSF barrier transport, but not inflammatory biomarkers, change in migraine, while CSF sVCAM1 associates with migraine frequency and CSF fibrinogen. Headache. 2021 Mar;61(3):536-545. doi: 10.1111/head.14088. Epub 2021 Mar 16. PMID: 33724462

d) Ghaffari H, Grant SC, Petzold LR, **Harrington MG.** Regulation of CSF and Brain Tissue Sodium Levels by the Blood-CSF and Blood-Brain Barriers During Migraine. Front Comput Neurosci. 2020 Feb 4;14:4. doi: 10.3389/fncom.2020.00004. eCollection 2020.PMID: 32116618

4. **Brain-derived nanoparticles.** Between 2000 and 2006 I sequenced over 6,000 different proteins in CSF and found that one-quarter of them were integral membrane proteins. Using differential ultracentrifugation, I found many heterogeneous membranous structures between 30 and 300 nanometers in diameter. These nanoparticles include synaptic vesicles, large dense core vesicles, and many other forms, which we defined by transmission electron microscopy, immune EM, western blotting, and mass spectrometry, and some had functionally active enzymes, including cyclooxygenases a. This was the first report to demonstrate and define nanoparticles in CSF. We recently quantified that CSF has around 10^9 nanoparticles per mL (manuscript in preparation). This work has opened up studies to investigate the physiological roles of these previously unknown particles as well as how they inform about disease. Recent work defines lipids that differ in nanoparticles as Alzheimer's disease progresses from asymptomatic stages to dementia.

a) **Harrington MG**, Fonteh AN, Oborina E, Liao P, Cowan RP, McComb G, Chavez JN, Rush J, Biringer RG, Hühmer AF. The morphology and biochemistry of nanostructures provide evidence for synthesis and signaling functions in human cerebrospinal fluid. *Cerebrospinal Fluid Res*. 2009 Sep 7; 6:10. PMCID: PMC2746175.

<u>Complete list of published work in my bibliography</u>:
http://www.ncbi.nlm.nih.gov/sites/myncbi/michael.harrington.1/bibliography/44075935/public/?sort=date&direction=descending

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

PHS 398 Cover Page Supplement

5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

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1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	Clinical_Aims1044447025.pdf
3. Research Strategy*	Clinical_Research_Plan_0613241044447191.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
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10. Resource Sharing Plan(s)	
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SPECIFIC AIMS

The clinical core develops and maintains the clinical enterprise serving the USC ADRC's overarching themes, including factors that modify the effect of *APOE ε4* on dementia and ATN biomarkers, emphasizing vascular risk, lipids, genetics, and ethnicity. The core supports the ADRC's overall scientific goal for further understanding mechanisms by which *APOE* genotype affects clinical phenotype and progression toward dementia.

The core serves the ADRC by providing well-characterized, longitudinally followed participants for ADRC and affiliated research projects, USC researchers, and the wider scientific community. This includes developing and maintaining a cohort who are at risk for dementia or its underlying pathology due to *APOE ε4* genotype, characterizing them with fluid and imaging biomarkers, and encouraging them to consent for brain donation. The core maintains an active registry of a diverse group of "study-ready" potential participants who have expressed interest in research participation. We address informed consent, data sharing, biosamples, autopsy procedures, and appropriate disclosure of biomarkers and other results to our research participants as below.

The core contributes to important national initiatives including NACC, NCRAD, ADGC, SCAN, and CLARiTI; to multi-site studies including ADNI, DIAN, MarkVCID, DiverseVCID, ACAD (Asian Cohort for AD); and national and international trials including the USC ACTC, UCSD ADCS, A4, and AHEAD 3-45. The core catalyzes local research by supporting recruitment and providing oversight and training for investigator-initiated studies (PPGII, DVR, NoMAD, GADSIL) and trials (DHA, phenserine, exosomes).

The core closely interacts with other ADRC cores to accomplish its aims: the administrative core coordinates activities, manages resources, and ensures compliance; the data core integrates clinical and research data with other datasets and shares de-identified data; the neuropathology core supports brain donation, tissue-sharing, and returning results to families; Biomarker and Neuroimaging Cores to link clinical assessments to biomarkers, imaging, and an integrated ATN framework; ORE core for community outreach and recruitment; and REC core for training. Major science findings supported by the core are discussed in the overall research plan.

The Clinical Core supports the ADRC's overarching goals: 1) defining factors that influence the effect of *APOE ε4* on cognitive decline and ATN biomarkers, 2) supporting local ADRD research, and 3) contributing to national initiatives through the accomplishment of its Specific Aims:

Aim 1. **Participant recruitment and retention.** The Clinical Core will recruit and maintain a clinical cohort of approximately 500 diverse, community-dwelling older adults, enriched for *APOE ε4* carriers, for biomarker characterization, and post-mortem pathological study. The core works with the ORE core and GeneScreen to enrich the proportion of *APOE ε4* carriers; and with USC's California Alzheimer Disease Centers at the Health Science Campus (CADC-USC) and Rancho Los Amigos National Rehabilitation Center (CADC-RLANRC) to ensure diverse representation.

Aim 2. **Clinical assessment and phenotyping.** The core medical teams will evaluate participants' cognitive status and diagnoses using NACC UDSv4 clinical evaluation and neuropsychological assessment; assess social determinants of health and vascular risk; characterize participants phenotypically and with ATN(v) biomarkers; and share research data with NIA-supported programs (e.g., NACC, NCRAD, ADGC, CLARiTI).

Aim 3. Tissue and biomarker collection and biobanking. The core will collect blood and CSF (optional) biospecimens, amyloid PET, tau PET, and MRIs; consent and follow participants to autopsy; collaborate with the neuropathology, biomarker, imaging, and data cores to integrate clinical and biological data; facilitate the development of canonical and emerging molecular, genetic, and imaging biomarkers for prediction, diagnosis, and surrogate clinical progression, and share data with NIA-supported programs (as above).

Aim 4. **Develop/maintain the research participant registry.** Per this current RFA-AG-24-001, collaborating with the ORE and data cores, the Clinical Core will maintain a composite cohort of 'study-ready' individuals from diverse backgrounds who are managed to review their clinical, biomarker, and social determinants of health status for potential recruitment and participation in the ADRC, affiliated studies, and clinical trials.

Aim 5. **Training and education.** The core will collaborate with the administrative core, ORE core, REC and RCMAR to: 1) train next-generation researchers, 2) provide ongoing education and training for our current ADRC researchers and teams, and 3) educate the community by teaching study design, methods, GCP, clinical research quality, regulation, ethics, cultural sensitivity, diversity, and implicit bias in clinical research.

Impact. The ADRC's focus on APOEε4 carriage and cognitive impairment among the diverse Los Angeles Latinx community provides the unique opportunity to advance knowledge, understand the effects of genetic and molecular biomarkers on risk, metabolic and vascular pathology, and illness progression. The Clinical Core serves this goal and is responsive to NIH's Plan to Enhance Diversity Perspectives.

A. SIGNIFICANCE

The Clinical Core characterizes participants for affiliated local research and national collaborative studies by maintaining a cohort of 436 active participants to ensure its ability to support studies of *APOE ε4* and risk factors of cognitive decline, and to support local research on environmental, metabolic, and vascular risks. During the current grant period, the core supported studies that provided important insights into the interaction of the *APOE* ε4 allele with vascular and metabolic risk factors [See administrative core table 2].

Core participants are enrolled in affiliated studies of cerebral blood flow, direct contrast-enhanced (DCE)-MRI, pericyte injury, blood pressure variability, biomarkers, and therapeutics. These studies showed *APOE ε4* carriage is associated with: 1) impaired autonomic regulation of cerebral perfusion (DVR, R01AG058162, Marmarelis, Chui, Joe, et al), 2) breakdown of the blood-brain barrier as assessed by sPDFGRβ and K_{trans} measures of gadolinium across the capillary blood-brain barrier (VG PPG, P01AG052350, Toga et al), 3) increased BP variability, cognitive decline, and tauopathy,[1] and 4) cerebral edema and hemorrhage.

In addition to understanding the effect of *APOE,* core participants contribute to our understanding of dementia among Latinx elders. The SCAN (R01AG067418) initiative acquires standardized MRI sequences, amyloid, and tau PET scans across ADRCs. An administrative supplement to Keck USC has supported the acquisition of additional PET scans in Latinx participants in the Clinical Core. The core supports MarkVCID (UF1NS100614, Wang, Kashani, Ringman) providing representative Latinx participants to the study. The core supports the Genetic AD Subtypes in Latinos study (GADSIL, R01AG062007, Ringman et al), including a subset with autosomal dominant AD (ADAD); 87 are followed in the Clinical Core and are supported by a collaborative NIH Fogarty Grant with the University of Guadalajara (R01AG06901, Ringman). In the last year, results from GADSIL defined measures on the Spanish English Neuropsychological Assessment Scale (SENAS) that are impaired in early ADAD and with aging in a normal Latinx population, and how linguistic and education variables affected performance on the SENAS.[2] Imaging studies showed increased medial temporal lobe flortaucipir PET signal was associated with decreased resting state fMRI complexity, suggesting a new imaging biomarker for incipient AD.[3] The study of ADAD due to the A431E mutation in *PSEN1* ("Jalisco mutation") enabled the revelation of a modifier variant in *APP* which appears to cause spastic paraparesis at a very young age without associated amyloid or tau deposition on PET (2024 AAIC abstract). [See administrative core table 2].

The core facilitates sharing data and biospecimens with other investigators. Data and CSF from participants in the affiliated VG-PPG were shared with Cohen Lab and supported the discovery of a variant of mitochondrial DNA encoding a micropeptide, SHMOOSE, that increases the risk of AD, possibly through mitochondrial energetic pathways (Epstein Breakthrough Award, Miller, Cohen). The core facilitated studies of iPSCs (Ichida) phenotyping in the core WBCs immortalized at USC's Center for Regenerative Medicine and Stem Cell Research (CIRM). Cells from a person inheriting the A431E mutation in *PSEN1* were shared with Bart De Strooper laboratory, KU Leuven, where they were made into iPSCs and differentiated into neurons; treatment of these neurons with an antisense oligonucleotide (ASO) against the A431E *PSEN1* mutation decreasing Aβ 42/40 and increasing Aβ 38/42. Translational work targeting the A431E *PSEN1* mutation therapeutically is ongoing.

We plan to extend these advances in the next cycle by 1) increasing statistical power to detect genotype effects by enhancing the Clinical Core cohort for *APOE ε4* by recruiting through the USC GeneScreen program (launched in 2023), and 2) incorporating ATN biomarkers (i.e., NCRAD assayed markers), and newer markers of vascular and metabolic risk into our core clinical evaluation as described below.

A.1 Current Clinical Core cohort and participants. The active cohort includes 436 participants: age 73, 40% male, 56% white, 28% Latinx; 239 (55%) are cognitively unimpaired, CDR=0, with or without cognitive complaints; 133 (30%) have a UDSv3 diagnosis of MCI, CDR = 0.5; and 64 (15%) have a diagnosis of dementia, CDR ≥ 1 (Table 1). For etiological diagnosis 148 (74%) of the 200 impaired have AD, 36 (18%) cerebrovascular disease; and 10 (5%) DLB. The cohort is 40% *APOE ε4* carriers, including 6% homozygotes. We continue to increase *APOE ε4* carriers through the focused, web-based GeneScreen registry and ongoing recruitment by the Clinical and ORE Cores (**Aim 4**).

Longitudinal data, including clinical, cognitive, behavioral, functional, imaging, and biomarker characterization is collected. UDS data and plasma/WBCs for NCRAD assays are provided to NACC and NCRAD where it is combined with data from other ADRCs and made available for collaborative studies (**Aims 1 and 2**).

A.2 ATN biomarkers. Biomarkers acquired over the current cycle include: 1) plasma biomarkers from NCRAD (193 samples from the active NACC UDS cohort and 200 from minimal contact participants); 2) CSF biomarkers (92 participants, 121 samples); 3) amyloid and tau PET (153 and 75, respectively). We continue the collection of blood specimens during each baseline and follow-up evaluation from consented participants per protocol (**Aim 3**).

Table 1. Demographic and clinical characteristics of the clinical core

	Overall	Hispanic	White*	Black*	Asian*
N	436	120 (28%)	242 (56%)	26 (6%)	31 (7%)
Age, y (SD)	73.3 (11.9)	67.3 (13.6)	76.1 (10.4)	74.2 (8.8)	73.1 (11.6)
Educ, y med [Q1, Q3]	16 [14, 18]	14 [11, 16.5]	16 [16, 18]	16 [14, 17]	18 [15, 18]
Male, N	175 (40%)	48 (40%)	104 (43%)	7 (26%)	12 (3%)
MoCA, mean (SD)	23.7 (6.0)	21.6 (7.4)	24.4 (5.3)	26.4 (3.7)	23.2 (5.8)
CDR = 0, N	239 (55%)	63 (52%)	134 (56%)	17 (63%)	16 (48%)
CDR = 0.5, N	133 (30%)	31 (26%)	76 (32%)	8 (30%)	10 (30%)
CDR ≥ 1, N	64 (15%)	26 (22%)	32 (13%)	1 (4%)	5 (15%)
APOE ε4 carriage	40%	30%	42%	36%	35%
Follow up, y (SD)	5.2 (4.7)	3.7 (3.9)	6.2 (4.8)	4.7 (5.8)	5.1 (4.7)
*Non-Hispanic; 17 (4%) not classified who identified as other races or unknown					

Brain donation. Approximately 350 active core and minimal contact participants consented to brain donation (**Aim 1**). Characteristics of the participants are: 64% not cognitively impaired, 15% MCI, 21% dementia, mean age 79.5 (9.3 SD), 38% male, and 18% Latinx. There were 86 autopsies from 2019 to 2024. Autopsy reports and letters are provided to the family by Dr. Ringman; and NACC UDS neuropathology data is uploaded to NACC usually within 6 months (see neuropathology core). Consultations with family by ADRC physicians are offered with the report.

B. INNOVATION

The Clinical Core is innovative in: 1) measured enrichment of *APOE ε4* carriers to provide sufficient power to detect interactions between vascular and metabolic factors and the *APOE ε4* allele; 2) recruitment and retention of a diverse cohort that represents the Los Angeles metropolitan area, with emphasis on historically under-represented Latinx populations; 3) use of novel approaches to cerebrovascular, metabolic disease and lipids; 4) success at long-term follow-up; 5) the wide range of studies that are supported.

C. APPROACH

C.1 Aim 1: The core's primary mission is enrolling, following longitudinally, and coordinating the collection of clinical data for a diverse cohort of older adults at risk for, or with, early cognitive decline (**Aim 1**). As the only NIA ADRC in the Los Angeles area, the ADRC's catchment area covers Los Angeles County (Zip 90001-93591). Participants are primarily recruited from: 1) ORE core community events, 2) USC Memory and Aging Center, 3) the core's registries, including GeneScreen (**Aim 4**), and 4) recruitment efforts of affiliated studies.

C.1.1 ADRC's geographic catchment area. The Los Angeles County population is 9.72 M of whom 4.77 M are Latino, 1.47 M are Asian, 769,000 are Black, and 1.48 M (15%) are aged 65+ years (US Census 2023 estimate). The USC Keck Medical Center and Keck School of Medicine are in the Lincoln Heights/ Boyle Heights neighborhoods, three miles east of Downtown Los Angeles near the geographic center of LA County. Lincoln Heights/ Boyle Heights are socioeconomically disadvantaged, majority Latinx neighborhoods (~80%), and are a major focus of recruitment (Zips 90023, 31, 33, 63). The ADRC also draws from throughout the County and north to the Central Valley and south to North San Diego County. The figure depicts the neighborhood-level, predominant race/ethnicity of the area surrounding the ADRC.

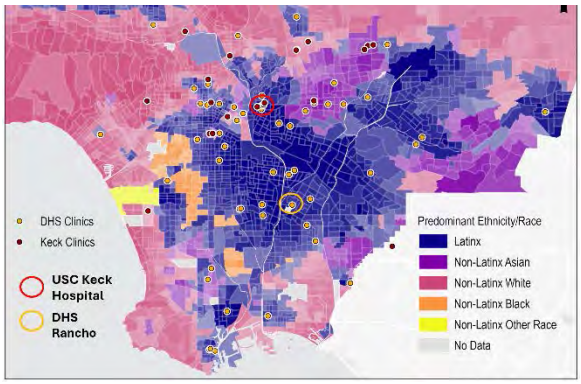


Figure 1. USC ADRC active cohort catchment area.

C.1.2 Clinical Core facilities. The Willametta Keck Day Clinic building houses the ADRC clinic, MRI suite, phlebotomy and preparation labs. The PET center is in the adjacent building. Neuropathology and biomarker cores are in Zilkha Neurogenetic Institute across the street. T3 and T7 MRIs are in the Stevens Neuroimaging and Informatics Institute 1000 feet up the street. Faculty offices are at CHP and ZNI. (see Facilities and Other Resources section).

C.1.3 Clinical Core team. The Senior/Key persons include Schneider, Joe, Ringman, Chui, Harrington, Pawluczyk, D'Orazio, McCleary. They are supported by a diverse, bilingual, and experienced team, including program managers, nurses, physician assistants, coordinators, assistants, and psychometrists (see Budget Justification).

C.1.4 Participant eligibility**.** Age ≥ 60 (for ADAD affiliated studies age 45+); English or Spanish speaking; willing to participate in annual UDSv4 clinical and neuropsychological evaluation, consent for genotyping, blood biomarkers, MRI, and PET imaging; willingness to share data, to consider referral to affiliated studies; having a study partner. Brain donation and lumbar puncture is optional. The cohort comprises participants from high-risk and historically underrepresented groups for research and others who disproportionately experience health burdens or low access to health care [see Human Subjects Clinical Trials Information section].

C.2 Aim 2: Clinical assessments and phenotyping. Participants will receive UDSv4 clinical evaluations, including neuropsychological, functional, social, and behavioral evaluations at yearly intervals (**Fig. 2**). They will be characterized by syndrome as meeting criteria for dementia, core criteria for MCI, or as not cognitively impaired by NIA-AA 2011 draft research criteria and per the new UDSv4 checklist (Form D1a). Participants with dementia or MCI are assessed for an etiologic diagnosis (Form D1b). Participants who do not meet the criteria for MCI or dementia, have a CDR-global= 0, and UDSv4 neuropsychological test results within age-adjusted normal ranges are considered as cognitively unimpaired. Diagnoses are supported by amyloid PET scans and/or plasma profiles consisting of a low Aβ42/Aβ40 ratio and/or high ptau181 or ptau217 level using NCRAD assays. We assess CAA risk by the Boston Criteria 2.0 for the clinical scans of persons who may receive lecanemab.[4] In mid-2024 we began MRIs, amyloid, and tau PET every other year for the active core cohort relying on funding from affiliated studies and in late 2024 will qualify and enroll core participants in CLARiTI as an added resource.

The UDSv4 introduces social determinants of health (SDOH), inclusive questions on gender, sexual orientation, and multi-racial identity, enabling an in-depth examination of diversity. It introduces forms for disease-modifying treatments, subjective cognitive decline, mild behavioral impairment, and a verbal learning test to assess cognition more broadly. Applying UDSv4 for current participants enhances understanding of our Clinical Core cohort.

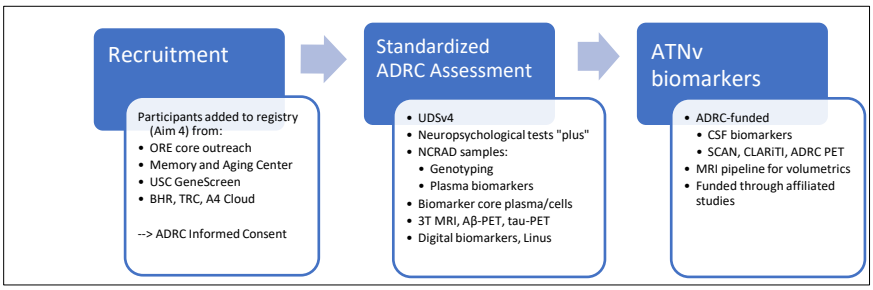


Figure 2. Workflow

C.2.1 Recruitment and sample size. Our goal is 52 participants per year, enriching for ≥ 21 (~40%) *APOE ε4* carriers (8 (~15%) homozygote), ≥ 18 (~35%) Latinx, and ~26 (~50%) early AD]. The planned sample size is based on a projected attrition of ~29 (~7%) per year, i.e., 143 participants over five years due to death, discontinuation, or shift to minimal contact status. Thus, the goal is based on ~29 to compensate for attrition and 23 to raise the sample to 500 active participants. We also target 26 (~50%) new cognitively unimpaired participants per year with most having a positive marker for amyloid plaque. Our goal for new volunteers for brain donations is 26 per year to maintain a pool of 350 (**Aim 1**).

Enrichment for recruitment of *APOE ε4* carriers is facilitated by USC GeneScreen (launched in late 2023) which uses targeted social media advertising to reach potential participants, who may consent to an at-home cheek swab collection kit for *APOE* genotyping. Those who return the kit are enrolled in the registry and are referred to the Clinical Core for consent and evaluation for eligibility (**Aim 4**). Social media advertising increases the targeting of traditionally underrepresented groups in research (**Aim 4**).

C.2.2 Procedures for informed consent. Participants or their legally authorized representatives must have a decisional capacity to provide informed consent and are formally assessed when that capacity is uncertain. Participants are asked to consent to provide biosamples for biomarker assays, banking, and for anonymously sharing data broadly. Study partners consent to provide demographic data, PHI, and perceptions of the participants. Participants are separately consented for MRI and PET. Willingness for brain donation consent is assessed annually with a goal for 26 consenting per year from the clinical core cohort and new participants (**Aim 1**, see Protection of Human Subjects section).

C.2.3 Brain donation. The Clinical and ORE cores collaborate to recruit and retain participants who agree to brain donation, including cognitively unimpaired, people with early AD, positive biomarkers, and older populations that are underrepresented for AD-related pathology (**Aim 1**). Clinicians interacting with core participants broach the topic at initial and follow-up visits. If participants are interested, then – after permission to contact is obtained – they are seen by a research nurse (Liberty Teodoro RN) who explains specifics and

initiates the separate consent process. We keep brain donation as optional to not inhibit broader and diverse participation in the ADRC. About 30% of core and minimal contact participants consented to brain donation.

Although core participants are prioritized for autopsy, those in affiliated studies including ADNI, DIAN-TU, MarkVCID, ACAD, USC ACTC, A4, AHEAD 3-45, clinical trials, ADAD, and persons with related CNS diagnoses who help enhance the ADRC's scientific mission, may be included even if they were not previously registered in the core. We collaborate with Washington University, St. Louis and MGH in sharing tissue from participants in ADNI, DIAN-TU, and ACTC trials. Tissue is shared with the broad research community as discussed by the neuropathology core (**Aim 3**).

C.2.4 COVID-19 pandemic. We dynamically adjusted to COVID-19 restrictions since 2020, used remote visits, and validated the T-Cog developed by NACC.[5] In 2020 and 2021, 451 of 651 total follow-ups were remote (**Fig. 3**). We continue to allow most core participants to choose remote evaluations, and endeavor to see participants who are concerned at least every other year. In 2023 we reduced remote follow-ups to 31 (12%), half of those in 2022.

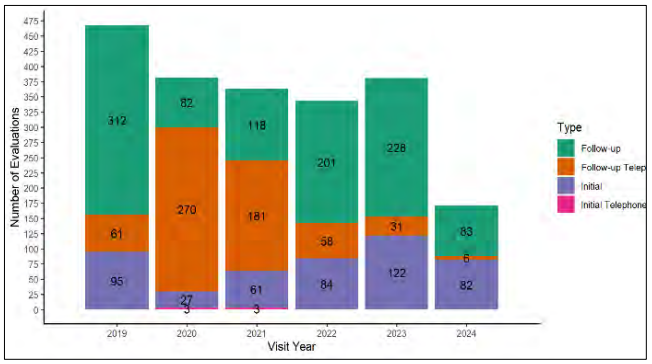


Figure 3. Initial and follow-up clinical core visits 2019- 2024

C.3 Aim 3: Biomarker, tissue collection, and biobanking. Procedures for fluid biomarkers. We collect blood samples at yearly or bi-yearly intervals depending on whether there is a change in symptoms or diagnosis. We coordinate with NCRAD in meeting sample-sharing requirements such as procedures, supplies, additional consents, MTAs, and budgetary issues. We follow ADRC guidance for best practices for collection, preparation, storage, and use of biospecimens; and follow uniform collection procedures outlined by the Standardization of Alzheimer's Blood Biomarkers (SABB) working group and the NACC procedures manual. 60mL of venous blood is collected in EDTA plasma and red top serum vacuum tubes. Samples are processed in the clinic lab or Biomarker Core lab, plasma aliquoted in 1.5 mL samples in polypropylene cryovial tubes, frozen, and stored within 2 hours. Split samples are shipped to NCRAD usually in batches based on clinic flow, including WBCs/DNA for the ADGC, with remaining samples stored at biomarker core. NCRAD assays are used to support ATN-informed diagnoses, phenotyping, quality control, validations, cross-validations, laboratory harmonizations, and novel assay development. White cells and iPSCs are stored at the biomarker core for future use. The NCRAD plasma sample is for Aβ40, Aβ42, pTau181, pTau217, NfL, GFAP (Quanterix Simoa HD-X) and Aβ42, Aβ40, p-Tau181 (Fujirebio Lumipulse) assays. Most participants receive additional biomarkers, including vascular markers, such as HgbA1c, lipids, sPDGFRb, placental growth factor (PlGF), and CRP performed at USC. Participants for in-depth biomarker characterization are selected to ensure enrichment for *APOE ε4* carriers and to support preclinical and early AD diagnoses.

Lumbar punctures are done usually in lateral decubitus using 22-gauge Sprotte needles to decrease post-LP adverse effects. 20 mL of CSF is collected in a polypropylene tube and delivered to the biomarker core for processing, aliquoting, and storage. Excluding 2020, we conducted an average of 30 lumbar punctures per year over the past grant cycle, approximately one-quarter of which were follow-up procedures.

C.3.1 Procedures for imaging. In the new grant cycle, MRI and PET imaging will be done at inception or at the next follow-up for participants who were not previously scanned within two years. Scans are prioritized by age, risk factors, *APOE* genotype, cognitive status, and autopsy consent. In the next cycle, we budgeted within the ADRC for 24 sets of scans per year (i.e., MRI, amyloid PET, and tau PET), and for 36 sets per year from CLARiTI for individual participants. Approximately 120 core participants received SCAN-comparable MRIs, amyloid PET, and tau PET through affiliated studies, including VG-PPG, DVR, ADNI, DiverseVCID, and clinical trials.

The Siemens Prisma 3T MRI at LONI is certified for ADNI4 and SCAN (see Facilities and Other Resources Statement). The protocol includes required sequences for CLARiTI, SCAN, and ADNI. The 60-minute scan includes: Accelerated Sagittal 3D T1 Sequence (SCAN compatible), Sagittal 3D FLAIR Sequence (SCAN compatible), Accelerated High-Resolution Hippocampus Scan, Axial resting-state MRI Sequence, Axial DTI AP Sequence, Axial 3D BBB/ASL, Sagittal TSE-VFA, and Axial T2* Star/GRE. Files are downloaded to the Imaging Core for quality assessment and storage, and data is shared with the USC ADRC database. To maximize the comparability of MRI scans obtained across studies we harmonized MRI protocols such that they include the common T1 and FLAIR Sequence of the SCAN protocol and an additional GRE sequence sensitive to microhemorrhages for all participants.

The Siemens Biograph PET/CT at the Keck PET center (qualified for ADNI4) is QC checked each day, calibrated, and normalized. We use florbetapir or florbetaben, and flortaucipir with target tracer doses of 370 MBq, 300 MBq, and 370 MBq respectively; and obtain 20-minute continuous (dynamic) scans at either 50-70 minutes (florbetapir) or 90-105 minutes (florbetaben and flortaucipir) post-injection.

C.3.2 Digital biomarkers. In this cycle, core participants will use the Linus Health digital cognitive assessment platform that is being included in the Keck USC Medical Center's EMR (see Facilities and letter from David Bates, CEO Linus Health). The platform integrates multimodal digital signals (i.e., fine motor control, drawing, voice, eye movements, gait, and balance) into validated AI and ML-derived metrics that we will use for enhanced follow-up, screening for affiliated studies, and to combine with the UDSv4 and NACC data. The platform relies on a tablet or smartphone to collect cognitive, behavioral, mood, and motor measures. In addition to assessments of core participants, in the future, we plan to approach minimal contact participants, Gene Screen respondents, and other registry participants for remote and in-person AI/ML-based assessment.

Participants are assessed at annual visits for hemodynamic risk factors (see Overview) including continuous beat-to-beat BP measurement using Biopacher repeated sit-stand maneuvers,[1] with data processed to yield derived measures including BP variability, arterial stiffening index, HR variability, baroreceptor sensitivity, and orthostatic hypotension.

C.3.3 Diagnostic approach. After participants complete scheduled procedures, the clinical teams review clinical assessment, neuropsychological testing, imaging, and biomarker results as available. Team conferences are held, particularly for newly enrolled participants to seek input from colleagues and consensus as needed. An innovative strength is the incorporation of additional (established) neuropsychological, fluid biomarkers, digital markers, and integrated neuroimaging. Research diagnoses and categories used are NIA-AA draft criteria 2011, NIA-AA research framework 2018, and UDSv4 (**Aim 2**).

C.3.4 Consensus and diagnostic conferences. Core meetings include consensus conferences to review and gain agreement for diagnosis and staging. Cases go to consensus when there is uncertainty. Clinicians present findings, and neuropsychological test results for further consideration from the team. Consensus diagnosis and ratings prevail and historically have been highly accurate with PPV > 90% with respect to autopsy findings. The core does its work through 1) weekly operations meetings (coordinators, nurses, and physicians); 2) weekly recruitment/ potential participants reviews (coordinators, administrators, nurses, physicians); 3) monthly reviews of active participants; bi-weekly meetings with data core for development, quality and process improvement, NACC and NCRAD data queries, and the internal REDCap database; 4) monthly conference calls with project coordinators on procedures, recruitment, and follow-up.

C.3.5 Return of results. The return of research results is essential for engaging participants and retention. Neuropsychological testing results are presented as basic one-page reports shared with participants. Tests done in CLIA labs, such as HgbA1c, are shared directly with participants. Research plasma molecular assay and DNA results conducted in a non-CLIA lab are not shared and participants are informed of this during the consenting process. *APOE* genotype is run under CLIA standards and shared with candidates for clinical treatment with amyloid-targeting antibodies and on request. Genotype status is disclosed to core participants in CLARiTI, ADNI, and most clinical trials, and to requestors using current clinical procedures and counselling detailed in those protocols. A genetic counselor will be available through funds allocated by the Dean to the Center for Personalized Brain Health. As research MRI scans undergo clinical review by a neuroradiologist relevant incidental findings are disclosed; otherwise, participants who request may review results with an ADRC physician. As amyloid PET scans are clinical procedures, relevant interpretations will be discussed proactively by an ADRC physician or a clinical interpretation if requested. Family members of deceased participants who donated their CNS receive a letter and an autopsy report within six months written by Dr. Ringman and follow-up if needed.

C.3.6 Participant retention and feedback. We use a range of strategies with the ORE core to engage participants as research partners. We have a participant-facing website that lists details on ongoing ADRC-affiliated studies. We host an annual participant appreciation day that generally has about 200 attendees and provides a venue to share updates from our USC ADRC studies and information about advances in the science of AD overall. Because participants in the autopsy cohort are older, may have trouble continuing annual visits and therefore may fall into minimal contact UDS status, we send birthday cards to strengthen the relationship with our center. We conduct biannual quality improvement surveys to ensure participants have positive experiences participating in research and that the informed consent process accurately shapes expectations about study procedures and results (**Aims 1 and 4**). The ORE core's Community Advisory Board includes clinical core participants who provide feedback and identify areas requiring improvement.

C.3.7 Collaborations and affiliated studies and trials. In addition to our participation in the NACC UDSv4, NCRAD, ADGC, SCAN, and CLARiTI, the core collaborates with many national efforts including observational studies (ADNI, DIAN-TU, MarkVCID, DiverseVCID, ACAD), and clinical trials (USC ACTC, UCSD ADCS, AHEAD 3-45). CLARiTI expands this aim through the collaboration of the clinical and ORE cores with the biomarker and imaging cores, allowing imaging and plasma biomarkers collection superimposed on the clinical core's UDSv4 standard data collection (**Aims 2 and 3**). For example, participating in CLARiTI improves our ability to phenotype participants in the ATN framework and to maximize the contributions of our participants in brain donation as this data is linked to neuropathology. The CLARiTI and SCAN collaborations will nearly triple our imaging capacity in addition to imaging collaborations from affiliated studies. In CLARiTI we will prioritize cognitively unimpaired and mildly impaired Latinx participants and oversample for *APOE ε4* carriage (goal: ≈ 50%), and older autopsy participants. The 2-year follow-up scans in CLARiTI will allow us to describe the temporal characteristics of AD and ischemic small vessel disease and the likelihood of co-pathology and proteinopathies indirectly in relation to *APOE* status in this underrepresented population.

C.3.8 Clinical Core resources used in affiliated projects. Discussed in the Specific Aims and Significance sections above, core resources including UDSv4 examination, neuropsychological test results, fluid biomarkers, PET and MRI scans, digital data and results, are shared with affiliated studies and projects. Registry results and potential participants in affiliated projects are shared with the projects within the limits of the registrants' consents (**Aim 4**, see Protection of Human Subjects section).

C.4 Aim 4. Develop and maintain a registry of 'study-ready' participants. We maintain an actively managed diverse cohort of potential participants for ADRC-affiliated studies and trials that includes demographic, clinical, and biomarker data. The voluntary registry provides participants with opportunities for a wide range of studies.

C.4.1 Registry structure**.** The core maintains a registry as parallel to but separate from the clinical core cohort. Registry participants are recruited from existing registries, USC GeneScreen, Trac-pad, Brain Health Registry (1R33AG062868 Weiner, PI) and the APT-Webstudy; as well as ongoing recruitment efforts through ORE core-led community outreach efforts and from affiliated studies. Core demographic and clinical variables for interested participants are entered into a HIPAA-compliant REDCap database that includes approximately 1500 individuals, mean age 67, 70% women, about 31% Hispanic, 41% white, 11% Black, 5% Asian, 0.7% American Indian or Alaska Native, 0.2% Hawaiian or Pacific Islander, 2.4% other, 7.4% unknown).

C.4.2 The minimal contact status cohort (UDSv3 M1.2 form) is maintained as a registry component (N=310, April 30, 2024). It includes participants who were entered in the UDS but asked for less contact due to travel, inconvenience, changed interest, circumstances, and the COVID-19 pandemic. Although they may decline annual in-person evaluations, they are willing to receive new information and consider new studies. They represent a well-characterized, approachable, and study-ready registry. Additionally, participants initially from the REDCap registry, direct referrals from other health centers and professionals may have entered the UDS directly but may shift to minimal contact status.

C.4.3 GeneScreen is a primary recruitment method for enhancing the participation of *APOE ε4* carriers and for tracking of those who may not be eligible or wish to enter the ADRC at the time but who agree to be followed via minimal contact approaches. GeneScreen enables genotyping, revising/updating clinical, SoDH, biomarkers, and diagnostic status in the registry. Launched in Spanish in October 2023 with the ORE core, enrollment of Latinx participants is enriched through outreach to patients at the CADC RLARC and CADC USC, focused enrollment into GADSIL, and targeted Facebook messages. The GeneScreen registry currently includes 434 of whom 40% are *APOE ε4* carriers. With the ORE core, we plan to enrich the registry by 300 with 50% within five years.

Registry participants may agree to evaluations remotely and referral to ADRC-affiliated studies. We use REDCap to track the number and reasons for those lost to follow-up and conduct longitudinal follow-up of registry participants. Participants in the registry are considered as "trial-ready cohorts" and are assessed remotely by telephone, videophone, web-based assessment, or another mobile assessment tool.

C.5 Aim 5. Training and education. The core trains current and next-generation researchers and educates the public community. It holds regular sessions with current staff to improve the skills of researchers, clinicians, and research associates/coordinators, teach research methods, documentation, GCP, improve/ensure data quality, QC, ethics, cultural sensitivity, and implicit bias training. Internal training is through regular team conferences. Core and ADRC leaders participate in weekly core participant diagnostic conferences, planning, referral conferences, weekly lecanemab eligibility conferences, and NACC-led online and virtual training programs.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	❍No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	❍No
Is the Project Exempt from Federal regulations?	❍Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447175.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Data Management and Statistical Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: SSB 102		Street2:
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90089-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	DMSC_Abstract1044447042.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	DMSC_References1044447092.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Data Management and Statistical Core

The overall objective of the Data Management and Statistical Core (DMSC) is to provide high quality services and resources to the Administrative, Clinical, Neuropathology, Biomarker, Imaging, and ORE Cores, the Research Education Component (REC), and ADRC faculty (including investigators involved in ADRC projects, development projects, and ADRC-affiliated investigators). The DMSC resources will provide an informatics and statistics environment to support cutting-edge dementia research both locally and nationally. There are three aims to this Core: 1) Data management. Focused on the provision of F.A.I.R. data management framework for ADRC-related research This effort will be built upon a unified ADRC archive that houses and integrates all ADRD research projects and programs at USC. 2) Program coordination with national efforts ensuring timely and accurate data entry to NCRAD, ADGC, SCAN, CLARiTI and GAAIN (Global Alzheimer's Association Interactive Network), among others. 3) Biostatistical support in the design, coordination and analysis of ADRC-related projects, including development projects supporting the REC, design of clinical studies and trials, integration and analysis of multimodal data, and support for data interpretation and manuscript preparation. 4) Education & training (in concert with the ORE Core and REC) via practical and didactic experiences on relevant study designs, statistical models, and informatics concepts and practices to trainees and investigators involved in dementia-related research. Finally, we will provide communications and documentation via the ADRC website.

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RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Arthur Middle Name W. Last Name*: Toga Suffix:

Position/Title*:	Provost Professor
Organization Name*:	University of Southern California
Department:	KSOM Mark and Mary Stevens Neuroimaging and Informatics Institute
Division:	
Street1*:	SSB 102
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (323) 442 7246 Fax Number: (323) 442 2230 E-Mail*: toga@usc.edu

Credential, e.g., agency login: TOGA22

Project Role*: Other (Specify) Other Project Role Category: DMS Core Lead

Degree Type: PhD, MS, BS Degree Year:

File Name

Attach Biographical Sketch*:

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: David Middle Name V. Last Name*: Conti Suffix:

Position/Title*:	Professor of Population & Public Health Scien
Organization Name*:	University of Southern California
Department:	KSOM Center for Genetic Epidemiology
Division:	
Street1*:	SSB 202 2001
Street2:	N. Soto Street
City*:	Los Angeles
County:	
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442#3140 Fax Number: E-Mail*: dconti@usc.edu

Credential, e.g., agency login: dconti

Project Role*: Other (Specify) Other Project Role Category: DMS Co-Core Lead

Degree Type: PhD, MS Degree Year:

File Name

Attach Biographical Sketch*: Biosketch_Conti_0614241044447245.pdf

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Daniel Middle Name Last Name*: Nation Suffix:

Position/Title*:	Professor of Gerontology and Medicine
Organization Name*:	University of Southern California
Department:	Gerontology
Division:	
Street1*:	University Park Campus
Street2:	GER 226A
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (213) 740#2219 | Fax Number: | E-Mail*: danation@usc.edu

Credential, e.g., agency login: DANIEL_NATION108

Project Role*: Other (Specify) | Other Project Role Category: DMS Co-Core Lead

Degree Type: PhD, MS | Degree Year:

	File Name
Attach Biographical Sketch*:	Nation_Biosketch1044446741.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. | First Name*: Xinhui | Middle Name | Last Name*: Wang | Suffix:

Position/Title*:	Assistant Professor Of Research Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764-3341 | Fax Number: | E-Mail*: xinhuiwa@usc.edu

Credential, e.g., agency login: XINHUI_WANG

Project Role*: Co-Investigator | Other Project Role Category:

Degree Type: PhD, MS | Degree Year:

	File Name
Attach Biographical Sketch*:	XWang_Biosketch_ADRC1044446742.pdf
Attach Current & Pending Support:	

1. Peng, C., Wang, J., Asante, I., Louie, S., Jin, R., Chatzi, L., Casey, G., Thomas, D.C., and **Conti, D.V.** (2019). A Latent Unknown Clustering Integrating Multi-Omics Data (LUCID) with Phenotypic Traits. Bioinformatics.
2. Jiang, L., Xu, S., Mancuso, N., Newcombe, P.J., and **Conti, D.V.** (2021). A Hierarchical Approach Using Marginal Summary Statistics for Multiple Intermediates in a Mendelian Randomization or Transcriptome Analysis. Am J Epidemiol.
3. **Conti, D.V.,** Darst, B.F., Moss, L.C., Saunders, E.J., Sheng, X., Chou, A., Schumacher, F.R., Olama, A.A.A., Benlloch, S., Dadaev, T., et al. (2021). Trans-ancestry genome-wide association meta-analysis of prostate cancer identifies new susceptibility loci and informs genetic risk prediction. Nat Genet 53, 65-75.
4. Darst, B.F., Shen, J., Madduri, R.K., Rodriguez, A.A., Xiao, Y., Sheng, X., Saunders, E.J., Dadaev, T., Brook, M.N., Hoffmann, T.J., et al. (2023). Evaluating approaches for constructing polygenic risk scores for prostate cancer in men of African and European ancestry. Am J Hum Genet 110, 1200-1206.

B. Positions and Honors

Positions, Employment and Honors

2019-present	Associate Director for Data Science Integration for the Norris Comprehensive Cancer Center at USC
2019-present	Kenneth T. Norris, Jr. Chair in Cancer Prevention
2014-present	Professor, Division of Biostatistics, Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, CA
2013 - 2014	Visiting Scientist, MRC Biostatistics Unit and Cancer Research UK, Cambridge Institute, University of Cambridge, Cambridge, UK
2013 - 2014	Visiting By-Fellow, Churchill College, University of Cambridge, Cambridge, UK
2009 - 2014	Associate Professor, Division of Biostatistics, Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, CA
2003 - 2009	Assistant Professor, Division of Biostatistics, Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, CA

C. Contribution to Science

1. Bayesian Hierarchical Modeling and Integrated Analysis: My current methodological research uses hierarchical model as the foundation to incorporate external and biological information. The latest of these approaches combines multiple types of omic data in a integrated analysis to estimate latent subgroups of individuals that are associated with the outcome. We have also investigated how this information can be used in a hierarchical manner to guide the search for interactions across multiple processors in a high performance-computing environment. Overall, these advances have facilitated integrated analyses beyond simple single SNP tests.

1. Jiang, L., Xu, S., Mancuso, N., Newcombe, P.J., and Conti, D.V. (2021). A Hierarchical Approach Using Marginal Summary Statistics for Multiple Intermediates in a Mendelian Randomization or Transcriptome Analysis. Am J Epidemiol.
2. Peng, C., Wang, J., Asante, I., Louie, S., Jin, R., Chatzi, L., Casey, G., Thomas, D.C., and Conti, D.V. (2019). A Latent Unknown Clustering Integrating Multi-Omics Data (LUCID) with Phenotypic Traits. Bioinformatics.
3. Baurley JW, Conti DV. A scalable, knowledge-based analysis framework for genetic association studies. *BMC Bioinformatics*. 2013;14:312. PMCID: PMC4015032
4. Liang WE, Thomas DC, Conti DV. Analysis and Optimal Design for Association Studies Using Next-Generation Sequencing with Case-Control Pools. *Genet Epidemiol*. 2012; PMCID: PMC4139478

2. Bayesian Stochastic Variable Selection and Incorporating Prior Information: While Bayesian hierarchical modeling allows for the construction of complex models from multiple, simple components, there is often uncertainty in what types of statistical models to include in the analysis and what variables should be included in those models. Prior information can be used to help guide the probabilistic selection of different models. For example, when analyzing multiple variants from genes in a specific pathway, prior information can indicate interactions that are more biologically relevant. To this end, we developed a Bayesian model averaging approach structured in a manner that attempts to take advantage of prior knowledge of the

metabolism of these classes of compounds and the various genes that regulate these pathways. For rare variant analysis, we have implemented a method that determines if regional rare variation in the aggregate is associated with disease and identifies which variants are most likely driving that association.

1. Newcombe PJ, Conti DV, Richardson S. JAM: A Scalable Bayesian Framework for Joint Analysis of Marginal SNP Effects. *Genet Epidemiol*. 2016;40:188–201. PMCID: PMC4817278
2. Quintana MA, Schumacher FR, Casey G, Bernstein JL, Li L, Conti DV. Incorporating prior biologic information for high-dimensional rare variant association studies. *Hum Hered*. 2012;74:184–195. PMCID: PMC4058572
3. Quintana MA, Conti DV. Integrative variable selection via Bayesian model uncertainty. *Stat Med*. 2013; PMCID: PMC4085544
4. Quintana, M.A., Berstein, J.L., Thomas, D.C., and Conti, D.V. (2011). Incorporating model uncertainty in detecting rare variants: the Bayesian risk index. Genet Epidemiol 35, 638-649.

3. Multi-SNP and Interaction Analysis: There has been a long-standing interest in genetic epidemiology to assess if SNPs in combination or in certain environments act synergistically to impact disease. We have developed approaches that use a Bayesian model selection framework and other promising, non-Bayesian techniques. These include methods that use principal components to account for the correlations between the SNPs, two-step approaches and Bayesian approaches that combine both case-only and case-control estimates.

1. Gauderman, W.J., Kim, A., Conti, D.V., Morrison, J., Thomas, D.C., Vora, H., and Lewinger, J.P. (2019). A Unified Model for the Analysis of Gene-Environment Interaction. Am J Epidemiol 188, 760-767.
2. Moss, L.C., Gauderman, W.J., Lewinger, J.P., and Conti, D.V. (2019). Using Bayes model averaging to leverage both gene main effects and G x E interactions to identify genomic regions in genome-wide association studies. Genet Epidemiol 43, 150-165.
3. Gauderman, W.J., Murcray, C., Gilliland, F., and Conti, D.V. (2007). Testing association between disease and multiple SNPs in a candidate gene. Genet Epidemiol 31, 383-395.
4. Conti, D.V., and Gauderman, W.J. (2004). SNPs, haplotypes, and model selection in a candidate gene region: the SIMPle analysis for multilocus data. Genet Epidemiol 27, 429-441.

4. Environmental Epidemiology: My statistical research relies on the thorough understanding of the underlying biology and the population being evaluated. This necessitates strong, collaborative interdisciplinary efforts for success and serves both methodological developments, as well as applied investigations. As an example, in my collaborative research in environmental epidemiology, my group has developed an approach for integrated analysis of multiple omic data. This approach has been applied to several applied investigations.

1. Goodrich, J.A., Wang, H., Walker, D.I., Lin, X., Hu, X., Alderete, T.L., Chen, Z., Valvi, D., Baumert, B.O., Rock, S., et al. (2024). Postprandial Metabolite Profiles and Risk of Prediabetes in Young People: A Longitudinal Multicohort Study. Diabetes Care 47, 151-159.
2. Goodrich, J.A., Walker, D.I., He, J., Lin, X., Baumert, B.O., Hu, X., Alderete, T.L., Chen, Z., Valvi, D., Fuentes, Z.C., et al. (2023). Metabolic Signatures of Youth Exposure to Mixtures of Per- and Polyfluoroalkyl Substances: A Multi-Cohort Study. Environ Health Perspect 131, 27005.
3. Stratakis, N., Conti, D.V., Jin, R., Margetaki, K., Valvi, D., Siskos, A.P., Maitre, L., Garcia, E., Varo, N., Zhao, Y., et al. (2020). Prenatal Exposure to Perfluoroalkyl Substances Associated with Increased Susceptibility to Liver Injury in Children. Hepatology.
4. Stratakis, N., Conti, D.V., Borras, E., Sabido, E., Roumeliotaki, T., Papadopoulou, E., Agier, L., Basagana, X., Bustamante, M., Casas, M., et al. (2020). Association of Fish Consumption and Mercury Exposure During Pregnancy With Metabolic Health and Inflammatory Biomarkers in Children. JAMA Netw Open 3, e201007.

5. Applied Genetic Analysis in Prostate Cancer: A large proportion of my research program is devoted to applied studies in genetic epidemiology, with a particular focus on multiethnic studies. Currently, the scientific emphasis is on large-scale consortiums performing meta-analyses and smaller scale studies with whole exome sequencing or whole genome sequencing. Multiple issues arise in the context of the applied analyses that motivate ongoing methodological developments. These include correct estimation of hidden population structure, methods for fine-mapping across multiple ethnic groups using summary statistics, and the construction and evaluation of polygenic risk scores across populations.

1. Wang, A., Shen, J., Rodriguez, A.A., Saunders, E.J., Chen, F., Janivara, R., Darst, B.F., Sheng, X., Xu, Y., Chou, A.J., et al. (2023). Characterizing prostate cancer risk through multi-ancestry genome-wide discovery of 187 novel risk variants. Nat Genet 55, 2065-2074.
2. Darst, B.F., Saunders, E., Dadaev, T., Sheng, X., Wan, P., Pooler, L., Xia, L.Y., Chanock, S., Berndt, S.I., Wang, Y., et al. (2023). Germline Sequencing Analysis to Inform Clinical Gene Panel Testing for Aggressive Prostate Cancer. JAMA Oncol 9, 1514-1524.
3. Conti, D.V., Darst, B.F., Moss, L.C., Saunders, E.J., Sheng, X., Chou, A., Schumacher, F.R., Olama, A.A.A., Benlloch, S., Dadaev, T., et al. (2021). Trans-ancestry genome-wide association meta-analysis of prostate cancer identifies new susceptibility loci and informs genetic risk prediction. Nat Genet 53, 65-75.
4. Conti, D.V., Wang, K., Sheng, X., Bensen, J.T., Hazelett, D.J., Cook, M.B., Ingles, S.A., Kittles, R.A., Strom, S.S., Rybicki, B.A., et al. (2017). Two Novel Susceptibility Loci for Prostate Cancer in Men of African Ancestry. J Natl Cancer Inst 109.

Complete List of Published Work in MyBibliography (over 300 publications):
https://www.ncbi.nlm.nih.gov/myncbi/david.conti.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Daniel Addison Nation

eRA COMMONS USER NAME (credential, e.g., agency login): DANIEL_NATION108

POSITION TITLE: Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of California Santa Cruz, CA	B.A.	2001	Psychology
University of Miami, Miami, FL	M.S.	2006	Psychology
University of Miami, Miami, FL	Ph.D.	2009	Clinical Psychology
University of California San Diego, CA	Internship	2009	Clinical Neuropsychology
University of California San Diego, CA	Postdoctoral	2009-11	Biological Psychiatry & Neuroscience
UCSD / VA Healthcare System	Postdoctoral	2011-13	Neuropsychology

A. Personal Statement

I am a clinical and translational scientist with a specific focus on vascular contributions to cognitive impairment and Alzheimer's disease (AD) dementia. The goal of this research is to improve our understanding the role of vascular factors and their treatment in cognitive decline and dementia. I have 20+ years of experience and success with clinical and observational studies involving serial neuroimaging, biofluid sampling, clinical/neuropsychological characterization, and longitudinal statistical analyses, including large, multi-site studies, with specific emphasis on neuroimaging and biofluid markers of cerebrovascular contributions to cognitive decline, including hemodynamic factors, cerebral perfusion, cerebrovascular reactivity and blood-brain barrier permeability. I am PI of 3 currently funded R01s and Co-Leader of a P01 program project, all focused on vascular factors related to cognitive decline. For the present renewal of the USC ADRC, I will serve as part of the leadership team and Data Management and Statistical Core (DMSC), and will provide expertise in the study of vascular risk factors and cerebrovascular dysfunction contributing to cognitive decline, including blood pressure variability and cerebrovascular reactivity, as well as statistical support for the DMSC. I would like to highlight the following studies relevant to this proposal.

Sible, I.J., **Nation, D.A.***, Alzheimer's Disease Neuroimaging Initiative (2022). Visit-to-visit blood pressure variability and longitudinal tau accumulation in older adults. *Hypertension.* 79(3):629-637.

Sible, I.J., **Nation, D.A.***, Alzheimer's Disease Neuroimaging Initiative (2022). Visit-to-visit blood pressure variability and CSF Alzheimer's disease biomarkers in older adults. *Neurology.* 98(24): e2446–e2453.

Kapoor, A., Jang, J., Dutt, S., Yew, B., Li, Y., Alitin, J.P., Gaubert, A., Ho, J.K., Blanken, A.E., Sible, I.J., Marshall, A., Shao, X Wang, D.J.J., **Nation, D.A.*** (2022). Older adults with enlarged perivascular spaces exhibit cerebrovascular reactivity deficits. *NeuroImage.* 264:119746.

Ho, J.K., Moriarty, F., Manly, J.J., Larson, E.B., Evans, D.A., Rajan, K.B., Hudak, E.M., Hassan, L., Liu, E., Sato, N., Hasebe, N., Laurin, D., Carmichael, P.H., **Nation, D.A.*** (2021). Blood-brain barrier crossing renin-angiotensin drugs and cognition in the elderly: a meta-analysis. *Hypertension*. 78(3):629-643.

B. Positions, Scientific Appointments, and Honors

2023 – present	Merle H. Bensinger Professor of Gerontology, Leonard Davis School of Gerontology, University of Southern California
2023 – present	Professor, Keck School of Medicine, University of Southern California
2019 – 2023	Associate Professor, Institute for Memory Impairments and Neurological Disorders, University of California Irvine
2019 – 2023	Associate Professor, Department of Psychological Science, University of California Irvine
2019 – 2023	Adjunct Professor, Department of Physiology & Neuroscience, Zilkha Neurogenetic Institute, University of Southern California, Keck School of Medicine
2019 – 2023	Adjunct Professor, Department of Psychology, University of Southern California, Dornsife College of Letters, Arts and Sciences
2019	Associate Professor, Neuroscience Graduate Program, University of Southern California
2019	Associate Professor, Department of Psychology, University of Southern California, Dornsife College of Letters, Arts and Sciences
2016 - 2019	Assistant Professor, Department of Physiology & Neuroscience, Zilkha Neurogenetic Institute, University of Southern California, Keck School of Medicine
2013 - 2019	Assistant Professor, Department of Psychology, University of Southern California, Dornsife College of Letters, Arts and Sciences
2011 - 2013	Postdoctoral Fellow, Neuropsychology, VA San Diego Healthcare System
2009 - 2011	T32-NIH Postdoctoral Fellow, Biological Psychiatry and Neuroscience, Department of Psychiatry, UCSD School of Medicine

Honors and Awards

2021 Standing Member, NIH Center for Scientific Review, CNN Study Section
2019 Primary Mentor, Canadian Institutes of Health Research Doctoral Foreign Study Award
2018 Primary Mentor, National Science Foundation Graduate Research Fellowship
2018 Outstanding Graduate Mentor Award, University of Southern California
2016 Keynote Speaker, Inaugural Wake Forest Alzheimer's Disease Center Conference
2013 Program Committee Member, International Neuropsychological Society Annual Meeting
2012 Licensed Psychologist, State of California
2011 Leon Thal Memorial Travel Award, Veterans Medical Research Foundation
2010 Program Committee Member, American Psychosomatic Society Annual Meeting
2009 Postdoctoral Research Training Award (T32), National Institute of Mental Health
2004-8 Outstanding Graduate Research Award

Other Reviewership: In addition to being a standing member of the CNN SRG since 2021, I have participated in several other NIH review panels (2020, RFA-NS-20-004: Molecular Mechanisms of Blood-Brain Barrier Function and Dysfunction in Alzheimer's Disease and Alzheimer's Related Dementias; 2020, The Blood-Brain Barrier, Neurovascular System and CNS Therapeutics, ZRG1 MDCN M91); 2019, Acute Brain Injury, ZRG1 BDCN-L(90)S; 2020, Clinical Neuroscience and Neurodegeneration [CNN]); 2019, Clinical Neurophysiology, Devices, Neuroprosthetics and Biosensors, ZRG1 ETTN-C(10)B; and ZRG1 ETTN-(19)B; 2021, Brain Injury and Neurovascular Pathologies, ZRG1 BDCN-Y(90)S; Cognitive and Neuropathological Signatures of Alzheimer's Disease, Brain Injury and Aging, ZRG1 BDCN-W(02)M. I conduct ad-hoc journal review for more than 20 journals, including *Nature Medicine*, *Brain*, *JAMA Neurology*, *Neurology*, *Diabetes Care*, *Stroke*, and others, and I am an editorial board member for *Journal of Alzheimer's Disease, Neuropsychology Review*, and *Neuropsychology*. I have also served as a grant reviewer for the Alzheimer's Association in 2015, 2019, 2020 and 2021, the California Department of Public Health in 2017, the Florida Department of Health in 2019, the Medical Research Council United Kingdom Research and Innovation in 2019, the Biotechnology and Biological Sciences Research Council in 2018, the Foundation Vaincre Alzheimer France in 2020, and the Deutsche Forschungsgemeinschaft (DFG), German Research Foundation in 2021.

C. Contributions to Science (Works cited below selected from over **115 publications**)

1. Arterial stiffness and blood pressure fluctuation in cognitive decline and Alzheimer's disease.
I led a series of studies linking markers of arterial stiffening and blood pressure fluctuation to cognitive decline and AD pathophysiologic change in cognitively normal older adults (Nation et al. 2013, *Neurology*; Nation et al., 2015, *JAMA Neurology*; Sible, Nation, 2022, *Neurology*; Sible, Nation, 2022, *Hypertension*).

1a. Sible, I.J., **Nation, D.A.***, Alzheimer's Disease Neuroimaging Initiative (2022). Visit-to-visit blood pressure variability and longitudinal tau accumulation in older adults. *Hypertension.* 79(3):629-637.

1b. Sible, I.J., **Nation, D.A.***, Alzheimer's Disease Neuroimaging Initiative (2022). Visit-to-visit blood pressure variability and CSF Alzheimer's disease biomarkers in older adults. *Neurology.* 98(24): e2446–e2453.

1c. **Nation, D.A.**, Edmonds, E.C., Bangen, K.J., Delano-Wood, L., Scanlon, B.K., Han, S.D., Edland, S.D., Salmon, D.P., Galasko, D.R., Bondi, M.W. (2015). Pulse pressure in relation to tau-mediated neurodegeneration, cerebral amyloidosis, and progression to dementia in the very-old. *JAMA Neurology*. 72(5):546-553. PMC4428938.

1d. **Nation, D.A.**, Edland, S.D., Bondi, M.W., Salmon DP, Delano-Wood L, Peskind ER, Quinn JF, Galasko DR (2013). Pulse pressure is associated with Alzheimer's biomarkers in cognitively normal older adults. *Neurology*, *81*, 2024-7. PMCID: PMC3854831.

2. Cerebrovascular reactivity and resistance in aging, cerebrovascular disease and Alzheimer's disease.
I have developed and studied new approaches to indexing cerebrovascular resistance and reactivity together with collaborators (Danny Wang, Co-I on this award) and have identified increased cerebrovascular resistance in early-stage Alzheimer's disease (Yew, Nation, 2017, *Brain*) and deficits in cerebrovascular reactivity in older adults with enlarged perivascular spaces (Kapoor et al., 2022, *NeuroImage*), hemodynamic risk (Sible et al., 2023, *American Journal of Hypertension*) and aging itself (Yew et al., 2022, *Neurobiology of Aging*).

2a. Yew, B., **Nation, D.A.*** (2017). Cerebrovascular resistance and cerebral amyloidosis: Effects on cognitive decline, cortical atrophy and progression to dementia. *Brain*. 140(7):1987-2001. PMC6059092

2b. Kapoor, A., Jang, J., Dutt, S., Yew, B., Li, Y., Alitin, J.P., Gaubert, A., Ho, J.K., Blanken, A.E., Sible, I.J., Marshall, A., Shao, X Wang, D.J.J., **Nation, D.A.*** (2022). Older adults with enlarged perivascular spaces exhibit cerebrovascular reactivity deficits. *NeuroImage.* 264:119746.

2c. Yew, B., Jang, J.Y., Dutt, S., Li, Y., Sible, I.J., Gaubert, A., Ho, J.K., Blanken, A.E., Marshall, A., Shao, X., Wang, D.J.J., **Nation, D.A.*** (2022). Cerebrovascular reactivity deficits in cognitively unimpaired older adults: vasodilatory versus vasoconstrictive responses. *Neurobiology of Aging.* 113:55-62.

2d. Sible, I.J., Jang, J., Dutt, S., Yew, B., Li, Y., Shao, X., Wang, D.J., **Nation, D.A.*** (2023). Older adults with higher blood pressure variability exhibit cerebrovascular reactivity deficits. *American Journal of Hypertension.* 36(1):63-68.

3. Treatment of vascular risk factors is linked to decreased risk for cognitive decline and dementia.
AT1-receptor blockers (ARBs) represent one class of antihypertensive medications that have been highlighted for their salutary effects on cognition. I have led a series of studies identifying links between treatments targeting central nervous system angiotensin pathways (Ho & Nation, 2018, *Neuroscience & Biobehavioral Reviews*) and diminished risk for memory decline (Ho et al., 2021, *Hypertension*), white matter lesion accumulation (Ho & Nation, 2017, *Alzheimer's Research & Therapy*). I also recently reported increased risk for AD dementia in older adults with uncontrolled diabetes, despite no increased risk in treated diabetics (McIntosh, Nation, 2019, *Diabetes Care*), underscoring the importance of vascular risk factor treatment in the prevention of dementia.

3a. Ho, J.K., Moriarty, F., Manly, J.J., Larson, E.B., Evans, D.A., Rajan, K.B., Hudak, E.M., Hassan, L., Liu, E., Sato, N., Hasebe, N., Laurin, D., Carmichael, P.H., **Nation, D.A.*** (2021). Blood-brain barrier crossing renin-angiotensin drugs and cognition in the elderly: a meta-analysis. *Hypertension*. 78(3):629-643.

3b. Ho, J., **Nation, D.A.*** (2017). Memory is preserved in older adults taking AT1-receptor blockers. *Alzheimer's Research & Therapy*. 9(1):33. PMC5405458

3c. Ho, J., **Nation, D.A.*** (2018). Cognitive benefits of Angiotensin IV and Angiotensin-(1-7): a systematic review of experimental studies. *Neuroscience & Biobehavioral Reviews*. 92:209-225.

3d. McIntosh, E., **Nation, D.A.***, for the Alzheimer's Disease Neuroimaging Initiative. (2019). Importance of treatment status in links between type 2 diabetes and Alzheimer's disease biomarkers. *Diabetes Care.* 42(5):972-979.

4. Angiogenesis and endothelial progenitor cell reserve in small vessel disease and cognitive decline.

I have identified endothelial progenitor cell and angiogenic signaling protein levels as an indicator of cerebral small vessel disease (Kapoor et al., 2021, *Frontiers in Aging Neuroscience*; Kapoor, Nation, 2022), and also a potential vascular compensatory factor related to cognitive resilience associated with better cognitive function and greater brain volume (Nation et al., 2018, *Journal of Alzheimer's Disease*).

4a. Kapoor, A. Gaubert, A., Marshall, A., Meier, I.B., Yew, B., Ho, J.K., Blanken, A.E., Dutt, S., Sible, I.J., Li, Y., Jang, J.Y., Brickman, A.M., Rodgers, K., **Nation, D.A.*** (2021). Increased levels of circulating angiogenic cells and signaling proteins in older adults with cerebral small vessel disease. *Frontiers in Aging Neuroscience.*

4b. Kapoor, A., **Nation, D.A.*** (2022). Elevated Platelet-derived growth factor-BB is associated with white matter hyperintensity burden in APOE4 carriers. *Cerebral Circulation - Cognition & Behavior*. 3:100131.

4c. **Nation, D.A.***, Tan, A., Dutt, S., McIntosh, E.C., Yew, B., Ho, J.K., Blanken, A.E., Jang, J.Y., Rodgers, K.E., Gaubert, A. (2018). Circulating progenitor cells correlate with memory, posterior cortical thickness and hippocampal perfusion. *Journal of Alzheimer's Disease*. 2018;61(1):91-101.

4d. Marshall, A.J., Gaubert, A., Kapoor, A., Tan, A., McIntosh, E., Jang, J.Y., Yew, B., Ho, J.K., Blanken, A.E., Dutt, S., Sible, I.J., Li, Y., Rodgers, K., **Nation, D.A.*** (2023). Blood-derived progenitor cells are depleted in older adults with cognitive impairment: a role for vascular resilience? *Journal of Alzheimer's Disease.*

5. The blood-brain barrier and genetic risk for cognitive decline, Alzheimer's disease and dementia.

I have played a leading role in a large program project aimed at identifying blood-brain barrier dysfunction as an independent and critical contributor to cognitive impairment (Nation et al., *Nature Medicine*, 2019), particularly among *APOE4* carriers (Montagne*, Nation* et al., *Nature*).

5a. **Nation, D.A.**, Sweeney, M., Montagne, A., Sagare, A.P., D'Orazio, L.M., Pachicano, M., Sepehrband, F., Nelson, A.R., Buennagel, D., Harrington, M.G., Benzinger, T.L.S., Fagan, A.M., Ringman, J.M., Schneider, L.S., Morris, J.C., Chui, H.C., Law, M., Toga, A.W., Zlokovic, B.V. (2019) Blood-brain barrier breakdown is an early biomarker of human cognitive dysfunction. *Nature Medicine*. 25(2):270-276.

5b. Montagne, A.*, **Nation, D.A.***, Sagare, A.P., Barisano, G., Sweeney, M.D., Chakhoyan, A., Pachicano, M., Joe, E., Nelson, A.R., D'Orazio, L.M., Buennagel, D.P., Harrington, M.G., Benzinger, T.L.S., Fagan, A.M., Ringman, J.M., Schneider, L.S., Morris, J.C., Reiman, E.M., Caselli, R.J., Chui, H.C., Chen, C., Pa, J., Conti, P.S., Law, M., Toga, A.W., Zlokovic, B.V. (2020). APOE4 leads to early blood-brain barrier dysfunction predicting human cognitive decline. *Nature*. 581(7806):71-76.

***Authors contributed equally.**

For full list of publications see link below:

https://www.ncbi.nlm.nih.gov/myncbi/daniel.nation.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Xinhui Wang

eRA COMMONS USER NAME (credential, e.g., agency login): XINHUI_WANG

POSITION TITLE: Assistant Professor of Research Neurology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of Science and Technology of China, Hefei, Anhui, P.R. China	B.S.	07/2000	Cellular and Molecular Biology
University of Southern California, LA, CA	M.S.	12/2006	Neuroscience
University of Southern California, LA, CA	M.S.	12/2007	Biostatistics
University of Southern California, LA, CA	Ph.D.	08/2015	Biostatistics

A. Personal Statement

As a formally trained researcher in Biostatistics, I have extensive experiences in conducting biostatistical and epidemiological studies that varied greatly in terms of their focus of research interests, scientific disciplines, study designs, as well as the statistical analysis approaches needed. I have been the principal biostatistician in charge of data management and data analysis for multiple projects. My expertise includes but not limited to designing databases and survey tools to gather data, coordinating complex databases from multiple sources for epidemiological studies and clinical trials, developing quality control protocols, and conducting statistical analysis. As an ADRC senior biostatistician in DMSC, I have provided statistical supports for multiple grant applications covering hypotheses formation, power calculation, data analysis plan preparation as well as statistical method consultation. I also provided statistical consultations for funded projects led by ADRC or ADRC-affiliated investigators and supervised junior biostatistician in team. Overall, my competences in data management and the ability to utilize and expand existing cutting-edge statistical methods in AD related research will strengthen the biostatistics supports for ongoing funded projects led by ADRC faculty and affiliates, contributing to the success of USC ADRC.

Ongoing projects that I would like to highlight include:

5P01AG055367
Chen, Finch (MPI), Role: Co-investigator
07/01/18-06/30/24
Traffic-related Air Pollutants and Alzheimer's Disease: Risk, Susceptibility and Mechanisms in Women

1RF1AG076124
Yassine, Role: Co-investigator
05/15/2022 – 04/30/2025
Brain cPLA2 as a mechanism for neuroinflammation in AD/ADRD with and without APOE4

P30 AG066530
Chui, Toga, Zlokovic (MPI), Role: Biostatistician

Citations:

a. Aranda MP, Liang J, **Wang X**, Schneider LS, Chui HC. The relationship of history of psychiatric and substance use disorders on risk of dementia among racial and ethnic groups in the United States. Front Psychiatry. 2023 Apr 24;14:1165262. doi: 10.3389/fpsyt.2023.1165262. PMID: 37168087; PMCID: PMC10165105.
b. **Wang X**, Younan D, Millstein J, Petkus AJ, Garcia E, Beavers DP, Espeland MA, Chui HC, Resnick SM, Gatz M, Kaufman JD, Wellenius GA, Whitsel EA, Manson JE, Rapp SR, Chen JC. Association of Improved Air Quality with Lower Dementia Risk in Older Women. *Proc Natl Acad Sci U S A.* 2022 Jan 11;119(2): e2107833119. doi: 10.1073/pnas.2107833119. PMID: 34983871.
c. **Wang X**, Younan D, Petkus AJ, Beavers DP, Espeland MA, Chui HC, Resnick SM, Gatz M, Kaufman JD, Wellenius GA, Whitsel EA, Manson JE, Chen JC. Ambient Air Pollution and Long-Term Trajectories of Episodic Memory Decline among Older Women in the WHIMS-ECHO Cohort. *Environ Health Perspect.* 2021 Sep;129(9):97009. doi: 10.1289/EHP7668. Epub 2021 Sep 13. PMID: 34516296; PMCID: PMC8437247.
d. Pahlavian SH, **Wang X**, Ma S, Zheng H, Casey M, D'Orazio LM, Shao X, Ringman JM, Chui H, Wang DJ, Yan L. Cerebroarterial pulsatility and resistivity indices are associated with cognitive impairment and white matter hyperintensity in elderly subjects: A phase-contrast MRI study. *J Cereb Blood Flow Metab*. 2020 Jun 5:271678X20927101. doi: 10.1177/0271678X20927101. Epub ahead of print. PMID: 32501154.

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2018 – present	Assistant Professor of Research Neurology, Department of Neurology, Keck School of Medicine, University of Southern California, Los Angeles, CA
2016 – 2018	Biostatistician III, Department of Research & Evaluation, Kaiser Permanente, Southern California, Pasadena, CA
2013 – 2016	Biostatistician II, Department of Research & Evaluation, Kaiser Permanente, Southern California, Pasadena, CA
2009 – 2013	Statistician I, Division of Environmental Health, Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, CA
2008 – 2009	Teaching Assistant, Division of Biostatistics, Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, CA
2007 – 2009	Research Assistant, Division of Biostatistics, Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, CA

Honors and Memberships

2022 – present	Member, Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment (ISTAART)
2015 – 2017	Member, American Association for the Advancement of Science
2008 – 2015	Member, American Statistical Association
2003	Provost’s Fellowship at University of Southern California

C. Contributions to Science

1. Air pollution effect on brain aging and cognitive function decline.

I have been working on multiple projects investigating the association between ambient air pollution and normal aging or neurodegenerative disease outcomes using multiple overlapping longitudinal cohorts of old women from Women’s Health Initiative Memory Study (WHIMS). In these studies, cognitive decline, all-cause

dementia, as well as brain MRI imaging data were the major outcomes. A variety of biostatistical methods have been implemented including generalized linear regression, linear mixed effect model, moderation and mediation effect assessment as well as Cox proportional hazard model for longitudinal data analysis. As a biostatistician faculty involved in these studies, I have provided supports for data management, data analysis as well as statistical methodology consultations for investigators at USC and collaborators at other universities.

a. **Wang X**, Younan D, Millstein J, Petkus AJ, Garcia E, Beavers DP, Espeland MA, Chui HC, Resnick SM, Gatz M, Kaufman JD, Wellenius GA, Whitsel EA, Manson JE, Rapp SR, Chen JC. Association of Improved Air Quality with Lower Dementia Risk in Older Women. *Proc Natl Acad Sci U S A.* 2022 Jan 11;119(2): e2107833119. doi: 10.1073/pnas.2107833119. PMID: 34983871.
b. **Wang X**, Younan D, Petkus AJ, Beavers DP, Espeland MA, Chui HC, Resnick SM, Gatz M, Kaufman JD, Wellenius GA, Whitsel EA, Manson JE, Chen JC. Ambient Air Pollution and Long-Term Trajectories of Episodic Memory Decline among Older Women in the WHIMS-ECHO Cohort. *Environ Health Perspect.* 2021 Sep;129(9):97009. doi: 10.1289/EHP7668. Epub 2021 Sep 13. PMID: 34516296; PMCID: PMC8437247.
c. Younan D, **Wang X**, Gruenewald T, Gatz M, Serre ML, Vizuete W, Braskie MN, Woods NF, Kahe K, Garcia L, Lurmann F, Manson JE, Chui HC, Wallace RB, Espeland MA, Chen JC. Racial/Ethnic Disparities in Alzheimer's Disease Risk: Role of Exposure to Ambient Fine Particles. *J Gerontol A Biol Sci Med Sci.* 2021 Aug 12:glab231. doi: 10.1093/gerona/glab231. Epub ahead of print. PMID: 34383042. PMCID: PMC9071399
d. Petkus AJ, **Wang X**, Beavers DP, Chui HC, Espeland MA, Gatz M, Gruenewald T, Kaufman JD, Manson JE, Resnick SM, Stewart JD, Wellenius GA, Whitsel EA, Widaman K, Younan D, Chen JC. Outdoor air pollution exposure and inter-relation of global cognitive performance and emotional distress in older women. *Environ Pollut.* 2021 Feb 15;271:116282. doi: 10.1016/j.envpol.2020.116282. Epub 2020 Dec 14. PMID: 33385889

2. Diabetes and associated health risk.

At Kaiser Permanente, I worked as a lead biostatistician in the group on multiple projects and trained junior biostatisticians in managing databases and conducting data analyses. The research work included studying the association between maternal Diabetes status and offspring disease risks using comprehensive Electronic Medical Record (EMR) data as well as performing clinical trials investigating Beta cell restoration through fat mitigation (BetaFat). Using Cox proportional hazard model controlling for other covariates and random effect due to sibling, we found that mothers with Type 1 Diabetes or had gestational diabetes diagnosed prior to 26 gestational weeks were associated with increased risk of Autism in offspring. In the finished BetaFat study, we compared metabolic measures at trial baseline between two arms among subjects with impaired glucose tolerance or recently diagnosed Type 2 diabetes. In these studies, I developed the expertise in handling complicated medical data from multiple sources and supervising data management in clinical trials.

a. **Wang X**, Martinez MP, Chow T, Xiang AH. BMI growth trajectory from ages 2 to 6 years and its association with maternal obesity, diabetes during pregnancy, gestational weight gain, and breastfeeding. *Pediatric Obesity* 2019; e12579. doi:10.1111/ijpo.12579
b. **Wang X**, Martinez MP, Chow T, Walthall JC, Guber KM, Xiang AH. Attention-Deficit Hyperactivity Disorder Risk: Interaction Between Parental Age and Maternal History of Attention-Deficit Hyperactivity Disorder. *J Dev Behav Pediatr*. 2019 Jun;40(5):321-329. doi: 10.1097/DBP.0000000000000669. PMID: 31206451.
c. Xiang AH, **Wang X**, Martinez MP, Page K, Buchanan TA, Feldman RK. Maternal Type 1 Diabetes and Risk of Autism in Offspring. *JAMA*. 2018 Jul 3;320(1):89-91. doi: 10.1001/jama.2018.7614.
d. Xiang AH, **Wang X**, Martinez MP, Walthall JC, Curry ES, Page K, Buchanan TA, Coleman KJ, Getahun D. Association of maternal diabetes with autism in offspring. *JAMA*. 2015 Apr; 313(14): 1425-34.

3. DNA methylation related work.

My doctoral training focused on methodology investigation of Illumina Infinium DNA methylation data, aiming at enriching signals from this high-throughput data for downstream analysis. Given the Beta-distribution characteristics of DNA methylation data, I developed a new probe filtering method, compared it with several commonly used methods using simulation, and evaluated them in real data sets. In another project, I utilized simulation designed on Beta-distributed data, systematically evaluated several batch effect

correction methods that were developed for Gaussian-distributed data. In addition, I had also been involved in several epidemiological studies investigating the association between DNA methylation and hazard exposure and/or adverse health outcomes. A variety of statistical methods targeting at analyzing high-throughput DNA methylation data were utilized including beta regression for beta-distributed outcomes, methods to control for multiple comparisons, feature selection and cluster analysis for high-dimensional data, etc.

a. **Wang X**, Laird PW, Hinoue T, Groshen S, Siegmund K. Non-specific filtering of Beta-distributed data. *BMC Bioinformatics*. 2014 Jun; 15:199.
b. Breton CV, Salam MT, **Wang X**, Byun HM, Siegmund KD, Gilliland FD. Particulate Matter, DNA Methylation in Nitric Oxide Synthase, and Childhood Respiratory Disease. *Environ Health Perspect*. 2012 Sep; 120(9):1320-6.
c. **Wang X**, Kang GH, Campan M, Weisenberger DJ, Long TI, Cozen W, Bernstein L, Wu AH, Siegmund KD, Shibata D, Laird PW. Epigenetic subgroups of esophageal and gastric adenocarcinoma with differential GATA5 DNA methylation associated with clinical and lifestyle factors. *PLoS One*. 2011; 6(10):e25985.
d. Breton CV, Byun HM, **Wang X**, Salam MT, Siegmund K, Gilliland FD. DNA methylation in the arginase-nitric oxide synthase pathway is associated with exhaled nitric oxide in children with asthma. *Am J Respir Crit Care Med*. 2011 Jul; 184(2):191-7.

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

PHS 398 Cover Page Supplement

5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

Introduction	
1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	DMSC_Aims1044447093.pdf
3. Research Strategy*	DMSC_Research_Plan_0614241044447225.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
9. Letters of Support	
10. Resource Sharing Plan(s)	
11. Other Plan(s)	
12. Authentication of Key Biological and/or Chemical Resources	
13. Appendix	
12. Appendix	

DATA MANAGEMENT AND STATISTICAL CORE: SPECIFIC AIMS

The overall objective of the Data Management and Statistical Core (DMSC) is to provide high quality services and resources to the Administrative, Clinical, Neuropathology, Biomarker, Imaging, and ORE Cores, the Research Education Component (REC), and ADRC faculty (including investigators involved in ADRC projects, development projects, and ADRC-affiliated investigators). The DMSC resources will provide an informatics and statistics environment to support cutting-edge dementia research both locally and nationally. Existing strengths of our Core include data management and biostatistical resources embedded within the USC Mark and Mary Stevens Neuroimaging and Informatics Institute (INI) and the USC Leonard Davis School of Gerontology and the Department of Population and Public Health Sciences.

The multimodal nature of data collected to investigate the etiology, pathogenesis, diagnosis, treatment, and prevention of AD requires robust informatics, data management and statistical approaches. Dr. Arthur W. Toga, as Core Leader, brings the unique data management and informatics resources of the INI. Leveraging several informatics tools, including the institute's state-of-the-art Image & Data Archive (IDA), the DMSC supports large-scale data management and dissemination of imaging, genetics, and other complex data for the Alzheimer Disease Neuroimaging Initiative (ADNI), 4 Repeat Tauopathy Neuroimaging Initiative (4RTNI), Alzheimer's Biomarkers Consortium — Down Syndrome (ABC-DS), ALLFTD, and other national and international initiatives. **Importantly, the INI provides the infrastructure for the DMSC and also supports the informatics needs of CLARiTI, SCAN and interfaces with ATRI, NACC, NIAGADS, NCRAD among others.** All activities of this Core will support the mission of the ADRC as well as the broader national AD research community. Drs. Dan Nation, Leon Aksman and David Conti bring extensive methodological expertise in designing longitudinal studies and conducting statistical analyses for multidimensional data relevant to dementia-related research. The statistics team has developed the Study Design and Biostatistical Consultation workflow (see Figure 4) which greatly strengthens the DMSC's Statistical Support function.

Aim 1. **Data management**. Provide a F.A.I.R. data management framework for ADRC-related research and supporting the Clinical, Neuropathology, Biomarker and Imaging Cores, with a centralized data system for:

a. A unified ADRC archive that houses and integrates **ALL ADRD** research projects and programs at USC.
b. Participant tracking dashboards to facilitate monitoring and scheduling assessments.
c. Standardized collection of local assessments, scales, and enrollment status.
d. Integration of local assessments, scales and measures data with Uniform Data Set (UDS) in a common database infrastructure, facilitating analyses across local and broader community levels.
e. De-identifying, storing, and tracking all data types - genetics, clinical, imaging, digital pathology and biosample data.
f. Controlled access to all UDS and Neuropathology data, genetic (*APOE*, GWAS, ES, WGS) data, images and sample availability including data system linkage to the biomarker repository and pathology core.
g. Comprehensive search across AD projects for cohort discovery and data examination.
h. Download and integrated processing with access control by project.

Aim 2. Program Coordination. Ensure timely and accurate data entry and management for the NACC-related UDS and Neuropathology databases as well as other data efforts such as NCRAD, ADGS, SCAN, CLARiTI, and GAAIN (Global Alzheimer's Association Interactive Network).

Aim 3. **Biostatistical Support**. Provide biostatistical consulting in the design, coordination, and analysis of ADRC-related projects, including development projects supporting the REC Core, design of clinical studies and trials, integration and analysis of multimodal data, and support for data interpretation and manuscript preparation.

a. Assist with study design, data collection, and statistical analyses of multimodal data.
b. Assist with image analysis and associations with other phenotype and genotype data.
c. Provide access to computational resources (see Resources & Environment) available in INI.

Aim 4. **Education & Training**. Provide education and training (in concert with the ORE Core and REC) via practical and didactic experiences on relevant study designs, statistical models, and informatics concepts and practices to trainees and investigators involved in dementia-related research. Provide communications and documentation via the ADRC website.

A. SIGNIFICANCE

In the next 5 years, the USC ADRC will focus on factors that modify the effect of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers, emphasizing vascular, lipid, inflammatory, genetic, and ethnic influences. The Data Management and Statistical Core will provide expertise and collaboration for all ADRC related aims and activities, as outlined in the aims above. Specifically:

A.1 The DMSC provides: (a) an overall and comprehensive data system for collecting and tracking participant enrollment and study data; (b) data management, data flow and reporting for UDS (including related interactions with NACC); (c) consultation service for study designs and statistical analyses; (d) computational resources to enable multimodal analyses including imaging and genetics; e) a central infrastructure for ADRC-related studies recruiting different subpopulations across diverse racial and ethnic ranges. These data are accessible via a secure and efficient web-based system to assist staff, scientists, and collaborators.

A.2 As a resource for other ongoing activities, additional biostatistical contributions are provided in design and analysis of ADRC clinical studies, clinical trials, and as a statistical resource for basic and translational scientists.

A.3 The DMSC contributes to national initiatives including NACC and ADGS. We are the informatics and data management nexus for several national and global projects in neurodegenerative disease. We serve as a community-centric repository for neuroimaging research studies for many programs, such as the International Consortium for Brain Mapping (2001 to present), the Alzheimer's Disease Neuroimaging Initiative (ADNI; 2004 to present), the Huntington's Disease Neuroimaging Initiative (2007 to 2012), Track TBI (2013 to 2019), the Parkinson's Progression Markers Initiative (PPMI; 2010 to present), and the Longitudinal Evaluation of Familial Frontotemporal Dementia (now ALLFTD; 2015 to present). We developed data harmonization tools and methods for cross-study data harmonization. Our database systems currently house data from more than 100,000 subjects and have provided over 400 million downloads to more than 160,000 investigators across the globe.

One of the more significant contributions of the DMSC is the aggregation and integration of data across independently funded projects focused on AD, which provides a valuable resource to compare and contrast data from different populations with variations in the data collected. These efforts will include a new initiative to obtain research quality MRI data on all clinical patients being treated with anti-amyloid therapy as part of monitoring for the presence of amyloid-related imaging abnormalities (ARIA). It should be noted and is described in the Imaging Core that the image acquisition protocol is harmonized to include a common set of sequences for all subjects across research projects so that data is comparable. **The centralized capabilities of the DMSC also provides access control and dissemination in an easy and efficient way, regardless of which project collected the data.** But we do so while ensuring that project specific data use requirements are met.

B. INNOVATION

An informatics, data and statistics core might, at first glance, seem mostly supportive and staid in its mission, without much to offer in terms of innovation. However, new models in data management, harmonization, mining, and dissemination have dramatically transformed how AD research is conducted. In fact, the AD research community has led the way in creating and demonstrating how revolutionary and beneficial informatics systems and policies can be in advancing the goals and pace of research. For example, ADNI created a system whereby all data are collected, comprehensively described, and made immediately available to any scientist interested in using the data—without embargo. That model was developed and implemented by the investigators of this core. We intend to continue to lead the field in innovation by continuing and extending this trend in data aggregation, integration, and dissemination. This will benefit the local members of our ADRC and scientific colleagues and will also enhance the efforts of other national and global data collection and networking projects. Over the past funding period, we have developed an integrated data system for collecting, managing, analyzing and sharing participant data and analysis results in response to the center needs, as well as the needs of collaborating sites and national initiatives. This data system allows us to merge the UDS and local data collection scales, measures, imaging and results and data collected by collaborators within a single data system, providing rapid and broad sharing of data and fostering collaboration within and between centers. We have developed a web portal, with

appropriate data protections, specifically designed to address the needs of data users, interested scientists, clinicians and others. Information about the projects, data, events, and other aspects of a successful ADRC are accessible and organized in a way that enables researchers and other collaborators to work efficiently. The portal also provides links to access other resources and biostatistical consultation. Biostatistical consultations support ADRC-affiliated investigators and projects by incorporating innovative statistical approaches in neuroimaging, bioinformatics, clinical trials, and longitudinal clinical and cognitive outcomes analyses.

Over the next 5 years, we will further expand this system to accommodate data collection, management and integration for new studies and additional imaging data types (e.g. digital pathology images, ophthalmic images). We will create a new participant tracking system that reads in the REDCap data streams currently used for such and attach dashboards and other graphical systems to better understand delays and chokepoints in the scheduling of assessments, providing support for recruiters and coordinators who must manage the complex needs of participants and resources. Access to dashboards and other participant tracking data will be coded and tightly controlled to internal ADRC personnel. We will also continue building our collection of electronic data collection forms, streamlining the data collection process, promoting data harmonization across studies, and ultimately making data available to qualified investigators. We created data mapping tools to relate our data with those of others in addition to preparing the UDS for transmission to NACC. Coupled with an open data sharing model, our DMSC will facilitate local, national, and global collaborations. Our group is a leader in similar efforts in AD research and other neurodegenerative diseases (Cousins et al., 2024, Veitch et al., 2024, Dage et al., 2023, Nutley et al., 2023, Zang et al/. 2023, Lynch et al., 2023, O'Bryant et al., 2022, Zhao et al., 2022, Duncan et al., 2019; Toga 2018; Zhou et al., 2018; Ning et al., 2018). For this reason, we are uniquely positioned to facilitate comparison of data across AD and PD, and FTD and Parkinson's, with data mining and analyses tools.

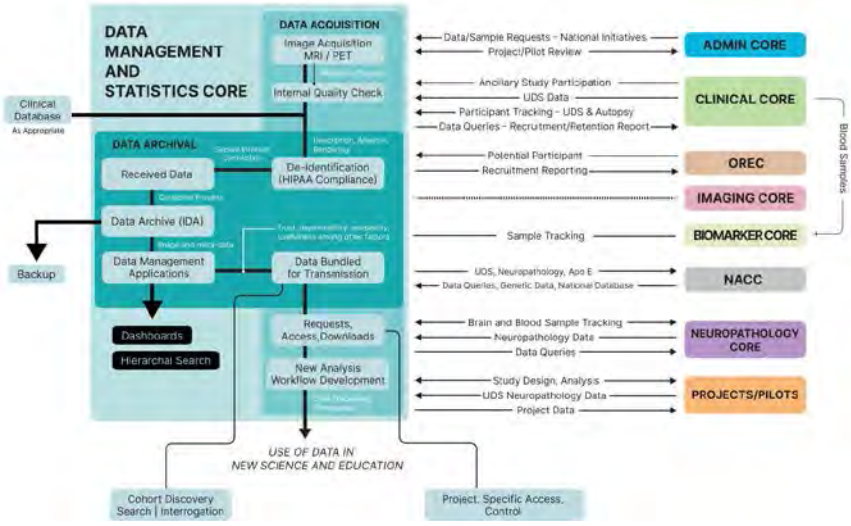


Figure 1. Data management and flow functions as described in Aim 1.

Effective communications are critical to the missions of the ADRC. The DMSC will also enhance the existing web site to foster enhanced collaboration with the research community. **Therefore, we position this core as central to not only data management and statistics, but to communications among ADRC Cores and projects, and external interested scientists (Fig. 1).**

C. APPROACH

C.1 Aim 1 Data Management. The DMSC now benefits from over thirty-five years of experience providing a secure, highly interactive environment for archiving, managing, and sharing neuroimaging and related data for large, multi-site studies serving research communities investigating Alzheimer's, Parkinson's, Huntington's and other diseases. The Image & Data Archive (IDA) is utilized by hundreds of thousands of investigators around the globe to safely de-identify, store, share and obtain medical imaging, genetics, cognitive assessments and related data (Crawford, Neu & Toga, 2015, Duncan et al., 2023; Neu et al., 2012; Toga et al., 2010 Liu et al., 2019; Duncan et al., 2018 Vietch et al., 2022; Toga, et al, 2018, 2023).

The ADRC acts as an umbrella organization for numerous projects and programs that conduct AD related research (see **Figure 2**). Each of these projects will have its data available using F.A.I.R. principles through the DMSC. Cohort discovery and interrogation of this data across projects will be possible. We will develop and deploy a comprehensive informatics and data management framework for ADRC-related research to include a

centralized data system for: **1)** A unified ADRC archive that houses **ALL AD** research projects and programs here at USC, **2)** participant tracking dashboards to facilitate monitoring and scheduling assessments, **3)** standardized collection of local assessments, scales, and enrollment status, **4)** integration of local assessments, scales, and measures data with the UDS in a common database infrastructure, facilitating analyses at local and broader community levels, **5)** de-identifying, storing and tracking all data types - genetics, clinical, imaging, digital pathology and biosample data, **6)** access to all UDS and Neuropathology data, genetic (APOE, GWAS, ES, WGS) data, images and sample availability including data system linkage to biomarker repository and pathology core, **7)** comprehensive search across AD projects for cohort discovery, data examination, **8)** download and integrated processing with access control by project.

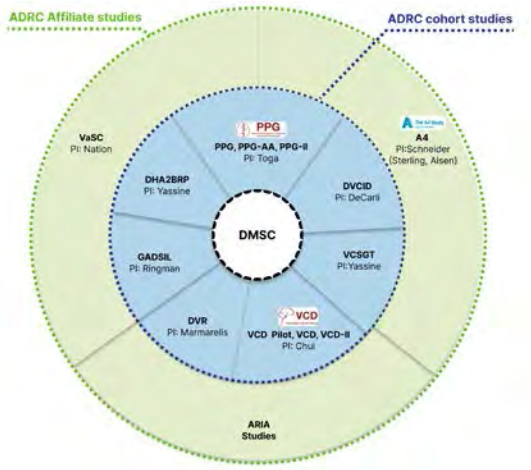


Table 1	
Study	**PI**
Genetic AD Subtypes in Latinos (GADSIL)	Ringman
Vascular Contributions to Dementia and Genetic Risk Factors (PPG)	Toga
Vascular Cohort Study (VCD) [Includes VCD Pilot, VCD, VCS-II]	Chui
Biomarkers of ABCA1 mediated functions in Alzheimer's (VCSGT)	Yassine
Diverse VCID (DVCID)	DeCarli
Model-based cerebrovascular markers (DVR)	Marmarelis
DHA Delivery to Brain Pilot Study (DHA2BRP)	Yassine
A4	Schneider (Sterling, Aisen)

Figure 2 and Table 1 depict some of the studies under the ADRC umbrella. Those with UDS are shown as Cohort studies where those that do not have UDS are Affiliate Studies.

C.1.1 Centralized tracking system. Over the past funding period, the DMSC has significantly enhanced the centralized participant tracking system, and we are linking systems for tracking brain, blood and other tissue samples of the Biomarker Core and Pathology Core (**Figure 3**). These tissue inventory systems include features to add/edit samples received, notate storage location, search for available tissues, and import and manage tissue requests and allocations. We will contribute, in an open way, with national databases and networks. This requires common data element mapping and translation depending on the dictionaries used by other databases. We will perform the mappings and share dictionaries and ontologies to ensure accurate data representations and provenance. As core leader of ADNI, PPMI and other global data sharing initiatives where we developed and implemented the data sharing methods and policies, we are strong advocates of open and unencumbered sharing of high quality well described data.

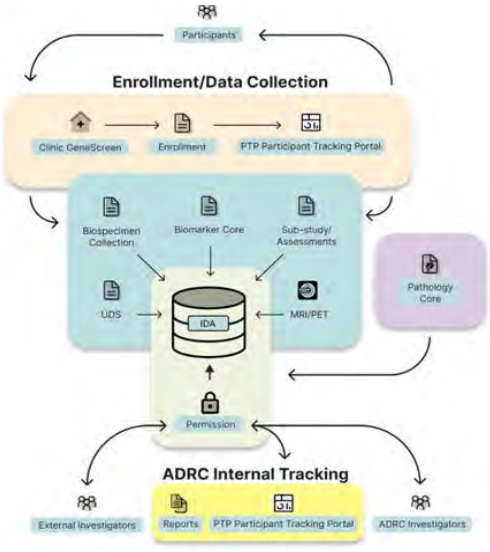


Figure 3. Flow chart of data and participant portal.

The data management infrastructure already established for ADRC projects will continue to be enhanced and utilized to meet the data collection, integration, secure archiving and sharing needs of the ADRC investigators and the research community. This infrastructure comprises:

C.1.2 LONI Image & Data Archive (IDA). The Laboratory of Neuroimaging (LONI) IDA was designed to ensure: 1) protection of patient privacy through integrated data de-identification components; 2) strict access controls to ensure data are only accessible to authorized individuals; 3) tracking of all data accesses to provide an audit trail so that project managers may understand by who and in what way data are being accessed;

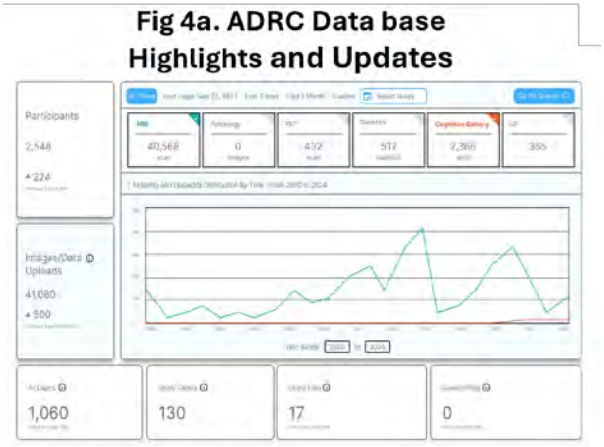
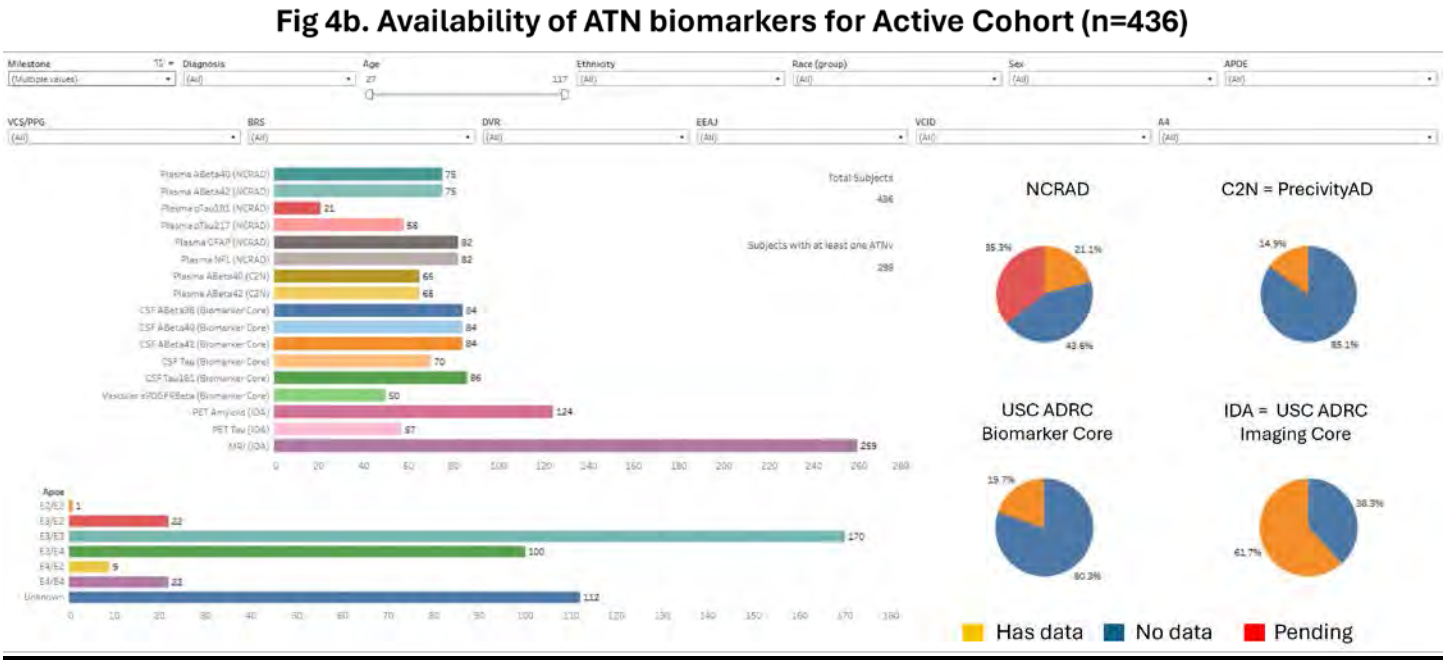


Figure 4a and 4b. Example dashboards can be selected from a menu of options enabling a variety of up-to-date data presentations.

4) ease of use through a platform-independent, user friendly interface, including dashboards (**Figure 4a and 4b**); 5) automated capture of image acquisition metadata; 6) image file format conversions via an integrated file format translation engine; 7) and in the case of image data, ease of inspecting image quality with the embedded image viewer. The IDA is easy to use, platform independent, and does not require installation of special software. The main components are: 1) simple and advanced query interfaces for searching the contents of the archive using demographic, image, and clinical metadata; 2) a collections interface for forming logical collections of data and downloading them (or passing them into the LONI Pipeline workflow environment); 3) an archiving interface for deidentifying and transmitting data to the archive; 4) project management interfaces to monitor study progress.



C.1.3 User Access Control. Only users specifically granted access to a project may access data or perform data-related activities. Data Use Agreements may be specific to the project, although we will work to harmonize those. IDA user access is multi-level and may be applied at a site-specific level or across all sites within a project. Users are allowed to query and download data, upload data, edit metadata, administer user access, and view project statistics depending upon their access levels. A user may have multiple access levels within the same project for different sites (i.e. the user can upload for site A but can download from both site A and site B). A graphical web interface for managing user access is integrated into the IDA, allowing project managers to administer user access for a single site or for multiple sites, depending upon the manager's permissions. All data accesses (uploads, downloads, edits, deletions) are logged.

C.1.4 Electronic Data Capture. The DMSC manages a REDCap server for collecting scales, measures, and results that supplement those required by the UDS. The DMSC collaborates with center investigators (e.g. VCS, VGPPG, DVR) to create, test and deploy custom data collection instruments and has developed software to automatically import data collected in REDCap into the LONI IDA, creating a comprehensive repository of study data and streamlining data integration and access. Once data collection instruments are developed, they can be reused across projects promoting data harmonization and aggregation efforts. REDCap is also used to collect

tracking information, providing the means to centralize enrollment, participation, and contacts across projects. Dashboards to facilitate the status of participants, what assessments have been or need to be scheduled and what data has been deposited into the archive will be developed.

C.2 Aim 2. Program Coordination.
Data entry and management for the NACC-related UDS and Neuropathology databases as well as other data efforts such as NCRAD, ADGS, SCAN, CLARiTI and GAAIN will be performed using the specific protocols of each. Given that the systems supporting several of these national programs were developed by LONI at USC, we have considerable knowledge and experience with them. The UDS and Neuropathology data are entered directly on the NACC website, using a routine centralized tracking system: This process includes: (1) a 2-step (Clinical Core and DMSC) visual check of all forms for completeness and accuracy, (2) data entry, (3) generation and tracking of data queries (missing data, logical errors, cross-database consistency, working with Clinical and Neuropathology Core staff), (4) generation of routine reports (e.g., subjects due and overdue for UDS follow-up, subjects targeted for blood draws), (5) work with Clinical and Neuropathology Core staff to respond to NACC-generated data queries, (6) export of UDS data from the NACC system, (7) import of UDS data into the central repository. All hard copies of data forms are stored by the respective core collecting the data (Clinical or Neuropathology). As specified in those cores, data are stored in locked file cabinets that are accessible only to essential core staff.

C.3 Aim 3. Biostatistical Support.
C.3.1 Overview. The DMSC provides consultation for study design and statistical data analyses as related to experimental, clinical, and community-based Alzheimer Disease Related Disorder (ADRD) research, including longitudinal cohort studies and clinical trials. In **Figure 5**, we show our current workflow to provide design and statistical support to our ADRC investigators in their competitive NIH applications, which include both pre-award and post-award services. While continuing the consultation services for ongoing ADRC projects, the DMSC has provided the design and statistical support to 8 NIH grants, (R01AG069013, R01AG058854, 1RF1AG068166, R25AG069711, P01AG052350, UF1 NS100614, 1RF1AG076124, R01ES033961) 1 California grant, and 5 others are pending, all of which leveraged the ADRC data. To support the development projects, the provision of Statistical Support will follow a similar workflow, with the interactions between the DMSC and investigative teams beginning at the formative stage of developmental projects.

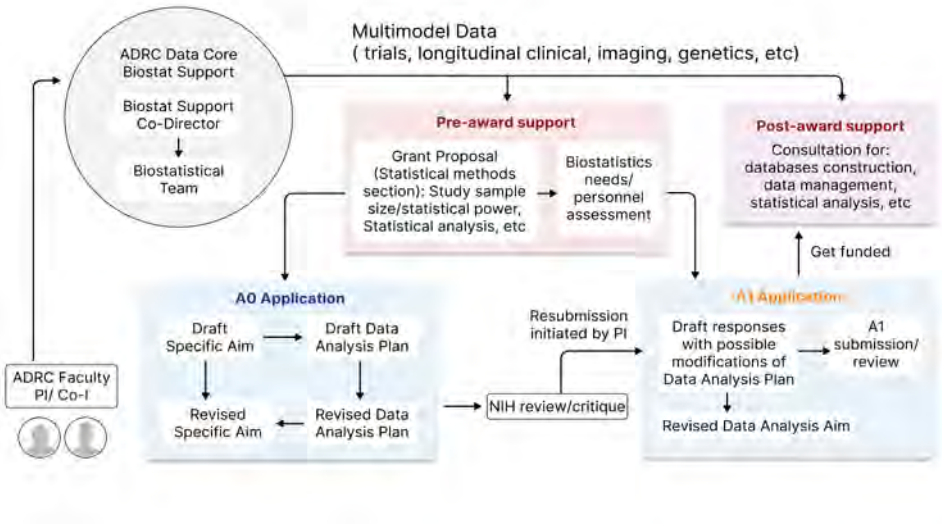


Figure 5. Workflow of AIM 3 for design and statistical support to ADRC investigators.

C.3.2 Operation and Flow. Our design and statistical (D&S) support includes both pre-award and post-award consultations (Figure 5). In the pre-award phase, once the request is triggered by the ADRC faculty, the D&S support team will assist with the study design and help the investigators formulate their research questions and draft the specific aims. These interactive processes will result in two important outputs: (1) the statistical methods section of the grant proposal; and (2) the biostatistics personnel assessment (e.g., specific methodological expertise and analytic skills; estimated percent effort needed to support the proposed scope of work). Recent years have seen an explosion of the availability of 'big data', along with methods and software tools to analyze such data. DMSC members have led recent efforts to develop and apply new methods in these

areas and innovative statistical methods include the application of multilevel models, high-dimensional feature selection, integrated omic analyses, interaction and mediation analysis, and the implementation of causal modeling. In addition to the Data Management support mentioned above, during the post-award phase, the design and statistics team will contribute to the data base development and help the funded PIs to identify/recruit the personnel with the specific data skills needed for the project performance.

C.3.3 Interaction and Data Access Across ADRC. The DMSC works closely with staff and investigators in other ADRC Cores, in the provision of multiple data support functions. These include constructing/maintaining core databases in REDCap with trackable data entry, establishing quality checking rules in REDCap to guide and supervise data entry, developing statistical programs for data quality checking and data reconciliation, enabling timely data evaluation and prioritizing schedule of study participants for Clinical Core, assisting qualified investigators to gain access to project data stored in LONI IDA, as well as compiling appropriate data sets and data dictionaries in response to requests from ADRC or ADRC-affiliated investigators. Importantly both data management and statistical support includes considerable expertise in analyzing multimodal data (Ning et al., 2018; Duncan et al., 2018; Weiner et al., 2017; Neu et al., 2017; Yew and Nation, 2017; Nation et al., 2019).

The DMSC meets monthly with key personnel of Administrative and Clinical Cores to go over issues identified in participant tracking, data entry, acquisition, transferring and access, and any other concerns. Working closely with staff and investigators in Administrative, Clinical, Neuropathology, Imaging and Biomarker Cores, the DMSC helps examine and address data issues identified including missing data, erroneous data, and inconsistencies between longitudinal visits or across multiple data sources in a timely manner. These regular interactions greatly help ensure the quality of the data stored and timeliness in sharing the NACC-related UDS and LONI IDA.

In 2023, the DMSC further established a centralized electronic recruitment website as part of the ORE core. In the next 5 years, DMSC will work with ORE core to maintain this system and further improve its function to accommodate the increasing needs and versatile requests in cooperation with other ADRC Cores.

C.4 Aim 4. Education & Training.
The DMSC will: (1) provide education support for USC investigators, staff, and trainees (e.g., minority researchers referred by the REC) new to ADRC; and (2) contribute to the planning and coordination of research seminars and workshops on epidemiologic and statistical methods relevant to AD research. For these educational and training activities, we leverage the resources available from the newly-funded T32 (T32AG058507; PI: Toga) as well as the infrastructure of education/training activities already developed by the Dept. of Preventive Medicine (see the intellectual environment section), AD-RCMAR (2P30AG043073; PI: Goldman, Zissimopoulos), T32 Gerontology (5T32AG000037; PI: Crimmins), plus the CTSI educational resources (5UL1TR001855; PI: Buchanan). Also, by leveraging other NIA-funded projects that include the development/applications of statistical modules, in the next funding cycle the DMSC plans to develop several statistical modules to implement quantitative methods for longitudinal data analyses (e.g., linear mixed modeling; Bayesian linear growth modeling; Cox proportional hazards) relevant to ADRD research. Initial applications of these sophisticated methods to study the impact of vascular factors on the trajectories of AD pathophysiologic change, neurodegeneration and cognitive decline have recently been published (Sible et al.,2022,2023,2024), and we plan to expand their applications to clinical studies using UDS data.

C.4.1 ADRC Website. Access to the database, notification of events, and research progress are all described within the site. The website is a dynamic and useful tool to help link the cores and projects. The website helps communicate ongoing clinical trials, activities, data, people, tools, and progress. It is a resource for trainees, staff and faculty.



Figure 6. Timeline of activities for each Aim. Although the emphasis in development of Aim 1 decreases somewhat over time, the activities continue for all years. Similarly, the effort in Aims 3 and 4 increases somewhat over time as projects mature.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	○No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	○No
Is the Project Exempt from Federal regulations?	○Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447176.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Neuropathology Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: Health Science Campus		Street2: 2011 Zonal Ave. Bldg. HMR
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

Project/Performance Site Location 1

Organization Name: Children's Hospital Los Angeles

* Street1: 4650 Sunset Boulevard		Street2:
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90027-0000
UEI: DVL1CMRMWRN9	* Project/Performance Site Congressional District: CA-028	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract Neuropathology_Abstract1044447043.pdf Mime Type: application/pdf

8. Project Narrative

9. Bibliography & References Cited Neuropathology_References1044447064.pdf Mime Type: application/pdf

10. Facilities & Other Resources

11. Equipment

ABSTRACT: Core D Neuropathology

The Neuropathology (NP) Core provides gold-standard neuropathologic diagnoses for the USC ADRC participants and banks fresh and frozen nervous system tissue for distribution to support ADRD researchers inside and outside of USC. The NP Core houses tissue from >1,100 cases. Tissue provided by the Core is unique among ADRC repositories in both the exceptional variety of tissue (e.g., eyes, spinal cord) and in its diversity with 30% of the sample deriving from underrepresented minorities including Latinx individuals, who are a focus population of the USC ADRC. Each case is reviewed independently by the two Core leaders (Drs. Annie Hiniker and Debra Hawes, both board-certified neuropathologists) who apply NACC 11 standards for diagnosis and then re-review together to reach consensus if needed. Each case will be stained for H&E, phospho-tau (AT8), beta-Amyloid (4G8), α-synuclein, and TDP-43. Diagnostic stains available and utilized as needed are AT8, RD3, and RD4 for tau species, 4G8 for Abeta, anti-alpha-synuclein for Lewy bodies, Iba-1 for macrophages, GFAP for gliosis, and neuronal markers including NeuN and NSE. Drs. Hiniker and Hawes are expert in developing and implementing new immunostains and new markers will be added as appropriate for refined diagnoses and research applications. Under the leadership of Dr. Hiniker and Dr. Michael Bienkowski, an expert in digital pathology, the Core will collaborate with the DMSC Core to develop an accessible digital pathology library of our autopsy cohort of >1,100 cases that will be ultimately integrated with genetic and biomarker data from the Biomarker Core. The NP Core contributes to our Center's studies of the effects of *APOE* genotype on ADRD, particularly focusing on the relationship of *APOE* genotype to vascular pathology and perivascular inflammation. The Core will also measure the detailed contributions of *APOE* genotype to pathology and cognitive impairment in collaboration with the Clinical Core. Finally, NP Core faculty support mentoring and training in ADRD and collaborate with the REC to educate students, residents and fellows in the pathologic diagnosis of ADRD.

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2. Coughlin DG, Dryden I, Goodwill VS, Pizzo DP, Wright B, Lessig S, et al. Long-standing multiple system atrophy-Parkinsonism with limbic and FTLD-type alpha-synuclein pathology. Neuropathol Appl Neurobiol. 2021. doi: 10.1111/nan.12783. PubMed PMID: 34847258.
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RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Anne Middle Name Elizabeth Last Name*: Hiniker Suffix:

Position/Title*:	Associate Professor of Clinical Pathology
Organization Name*:	University of Southern California
Department:	KSOM Pathology
Division:	KSOM Pathology
Street1*:	Health Sciences Campus
Street2:	2011 Zonal Ave. Bldg. HMR
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764-3341 Fax Number: E-Mail*: anne.hiniker@med.usc.edu

Credential, e.g., agency login: hiniker

Project Role*: Other (Specify) Other Project Role Category: Neuropathology Core Lead

Degree Type: MD, PhD Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Hiniker_0529241044446845.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Michael Middle Name Last Name*: Bienkowski Suffix:

Position/Title*:	Assistant Professor of Physiology & Neuroscie
Organization Name*:	University of Southern California
Department:	KSOM Neuroimaging and Informatics Institute
Division:	
Street1*:	ZNI, 1501 San Pablo St.
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-7246 Fax Number: (323) 442-2230 E-Mail*: bienkows@usc.edu

Credential, e.g., agency login: BIENKOWSKI2

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year:

	File Name
Attach Biographical Sketch*:	Bios-ketch_Bienkowski_ADRC1044446846.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Debra Middle Name Last Name*: Hawes Suffix:

Position/Title*:	Associate Professor of Clinical Pathology
Organization Name*:	University of Southern California
Department:	KSOM Pediatrics CHLA
Division:	KSOM Pediatrics
Street1*:	2011 Zonal Ave.
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-1156 Fax Number: (323) 442-1154 E-Mail*: hawes@usc.edu

Credential, e.g., agency login: DHAWES

Project Role*: Other (Specify) Other Project Role Category: Neuropathology Co-Core Lead

Degree Type: MD Degree Year:

	File Name
Attach Biographical Sketch*:	Hawes_Biosketch1044446857.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Annie Hiniker

eRA COMMONS USER NAME (credential, e.g., agency login): hiniker

POSITION TITLE: Associate Professor of Pathology at USC

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Harvard University, Cambridge, MA	A.B.	06/1999	Honors Biology
University of Michigan Medical School, Ann Arbor, MI	M.D./Ph.D.	06/2008	Medicine, Cellular and Molecular Biology
Princeton University, Princeton, NJ	Postdoctoral	05/2009	Molecular Biology
UCSF, San Francisco, CA	Residency	06/2011	Pathology
UCSF, San Francisco, CA	Fellowship	06/2013	Neuropathology
UCSF, San Francisco, CA	Postdoctoral	08/2016	Molecular Biology

A. Personal Statement

I am a board-certified neuropathologist focused on molecular mechanisms and neuropathology of neurodegenerative diseases. I am an Associate Professor of Pathology at USC and Director of the USC ADRC Neuropathology Core. I also run a basic research lab focused on the molecular underpinnings and cell biology of neurodegeneration, particularly as related to the degradation of neurotoxic proteins, such as tau and LRRK2. My research strives to utilize my expertise in neurodegenerative neuropathology and basic molecular biology to delineate the mechanisms of neuronal death in Alzheimer's Disease (AD) and Parkinson's Disease (PD), with the long-term goal of developing novel rational therapeutics. I have a strong interest in defining how the molecular interaction and neuroanatomic colocalization of diverse neuropathologic proteins (tau, α-synuclein,TDP-43) influences clinical manifestations and progression of AD. Much of my recent work combines quantitative analysis of patient postmortem tissue, including digital neuropathology and biochemical assays, with precise neuropsychological cognitive data obtained during life. These studies allowed us to define neuropathologic underpinnings of clinical phenotypes including the role of tau neurofibrillary tangle distribution in driving the disproportionate executive dysfunction of early onset sporadic AD and the role of TDP-43 pathology in driving clinical dementia in early-stage AD neuropathologic change. I have ongoing collaborations with key members of the USC ADRC including Drs. Mike Bienkowski, Debra Hawes, and John Ringman to study the neuropathology of AD, including a unique Latino cohort with the Presenilin 1 A431E (Jalisco) mutation. I have successfully competed for research grants including a recent R01 from NINDS, the American Parkinson Disease Association, the Department of Defense Parkinson's Research Program, and the Alzheimer's Association, among others.

The following recent peer-reviewed publications highlight my experience and qualifications to lead the USC Neuropathology Core:

a. Coughlin DG, Goodwill VS#, Standke HG#, Kim Y, Coley N, Pizzo DP, Galasko D, Kraus A*, **Hiniker A**.* Selective tau seeding assays and isoform-specific antibodies define neuroanatomic distribution of progressive supranuclear palsy pathology arising in Alzheimer's disease. Acta Neuropathologica. 2022 Aug 18. doi: 10.1007/s00401-022-02480-x. PMCID: PMC9467964

b. Stormo AEDS, FitzGibbon M, Shavarebi F, Earley EM, Lum LS, Verschueren E, Swaney DL, Skibinski G, Ravisankar A, van Haren J, Davis EJ, Johnson JR, Von Dollen J, Iaccarino C, Dauer WT, Nichols RJ, Wittmann T, Cox TC, Finkbeiner S, Krogan NJ, Oakes SA*, **Hiniker A***. The E3 ligase TRIM1 ubiquitinates LRRK2 and controls its localization, degradation, and toxicity. Journal of Cell Biology. 2022 Apr 4;221(4):e202010065. PMID: 35266954 PMCID: PMC8919618

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d. Smirnov DS, Salmon DP, Galasko D, Edland SD, Pizzo DP, Goodwill V, **Hiniker A**. TDP-43 Pathology Exacerbates Cognitive Decline in Primary Age-Related Tauopathy. Annals of Neurology. 2022 Jun 13. doi: 10.1002/ana.26438. Online ahead of print. PMID: 35696592 PMCID: PMC9391297

B. Positions, Scientific Appointments, and Honors

Positions and Employment

01/2024-	Associate Professor of Pathology, University of Southern California
2018-2023	Assistant Professor of Neuropathology, University of California, San Diego
2016-2018	Adjunct Assistant Professor of Neuropathology, University of California, San Francisco
2013-2016	Research Fellow, Division of Neuropathology, University of California, San Francisco
2011-2013	Fellow, Division of Neuropathology, University of California, San Francisco
2009-2011	Resident, Department of Anatomic Pathology, University of California, San Francisco
2008-2009	Research Fellow, Tom Silhavy lab, Princeton University, Princeton, NJ.
1999-2008	Medical Scientist Training Program Fellow, James Bardwell lab, University of Michigan
1995-1999	Undergraduate Researcher; Vernon Ingram lab, M.I.T., Cambridge, MA.

Other Experience and Professional Memberships

2023-	Neuropathologist, UCSD California NeuroAIDS Tissue Network
2022-	Co-Director, UCSD Shiley-Marcos ADRC Neuropathology Core
2022-	Neuropathologist (sole), UCSD ALLFTD Study
2021-	VA SHIELD COVID-19 Biorepository National Programmatic and Scientific Review Board
2020-	Neuropathologist (sole), UCSD Target ALS Brain Bank
2019-	Director, American Association of Neuropathologists Neurodegenerative Scholars (R13) Mentoring Program and Co-Organizer Annual Workshop
2019-2021	ACGME Neuropathology Milestones 2.0 Working Group Member
2018-	Neuropathologist (sole), UCSD Shiley-Marcos Alzheimer's Disease Research Center
2018-	Neuropathologist (sole), San Diego VA Medical Center
2014-	Grant reviewer, ASEE National Defense Science and Engineering Grad Fellowship (2014-2018), NSF Grad fellowship (2017-2019), Alzheimer's Association (2021-), National Alzheimer's Coordinating Center Scientific Review Board (2021-)
2014-	Ad-Hoc reviewer, JCI Insight, Acta Neuropatholgica, JAMA Neurology, Modern Pathology, Acta Neuropathologica Communications, European Journal of Neurology, among others
2014-2018	Assistant State Director of California Patient Advocacy, Parkinson's Action Network (now MJFF)
2012-2013	UCSF Pathology Dept Research Interest Group (RIG), Co-Chair; Women in RIG, Co-Founder
2009-	American Association of Neuropathologists, Member (2009-present), Program Committee (2016-19), Awards Committee (2020-present), Education Committee (2020-present)
2001-	Alpha Omega Alpha Medical Honor Society

Honors

2023	Faculty mentor of American Association of Neuropathologists (AANP) 2023 Women and Diversity Awardee, Melanie Estrella (Senior author on abstract).
2021	Flurida Alzheimer's Disease Emerging Scholars Award
2021	Junior Faculty Award, 15th International Conference on Alzheimer's and Parkinson's Diseases (Senior author on abstract)
2021	Hirano Award for the Best Paper on Neurodegenerative Diseases, American Association of

Neuropathologists Annual Meeting (Senior author on abstract)

2019 Invited speaker, Junior Investigator Session, 2019 National Alzheimer's Disease Center Meeting
2015 Invited speaker, Research Spotlight, Parkinson's Action Network Forum, Washington DC
2015 Nominated, White House Champions of Change for Parkinson's Disease
2015 2014-2015 UCSF Medical School Essential Core Teaching Award (nominated by students)
2015 2014-2015 Kaiser Award for Excellence in Teaching (nominated by students)
2014 Paul B. Beeson Career Development Award in Aging Research (K08)
2013 NINDS Research Education Grant (R25)
2012 American Association of Neuropathologists Richard C Davis Trainee Travel Award (one of three awards for best trainee clinical research papers)
2008 George R. DeMuth Medical Scientist Award for Excellence (to the graduating MSTP fellow who best exemplifies all of the attributes and characteristics desired in a physician scientist)
2007 MCAS M3 Scholarship Service Award (to one student in M3 class), University of Michigan
2007 Award for Clinical Excellence, University of Michigan Medical School
2007 Rackham Distinguished Dissertation Award (1 of 7 best PhD dissertations in any field), University of Michigan
2005 University of Michigan Medical School Excellence in Research and Service Award
2005 Diane H. Russell Award, American Women in Science Educational Foundation
2005 Rackham Predoctoral Fellowship, University of Michigan
2002 National Defense Science and Engineering Graduate Fellowship
2001 Alpha Omega Alpha Medical Honor Society, 1 of top 6 students, U. Michigan Medical School
1999 *Magna cum Laude* graduate with Highest Honors in Biology, Harvard University
1999 *Summa cum Laude* undergraduate thesis, Harvard University

C. Contributions to Science

The unifying theme of my research has been to define pathways of protein misfolding and clearance as they pertain to neurodegenerative diseases, including Parkinson's Disease and Alzheimer's Disease, among others.

1. Defining neuropathologic features underlying clinical variation in Alzheimer's disease. We have quantified neuroanatomic distribution of neuropathology using immunohistochemistry/immunofluorescence and RT-QuIC with a focus on how tau neuropathology drives clinical and neuropsychiatric symptoms in AD. New and unfunded work, in collaboration with Allison Kraus at Case Western University Medical Center, strives to test the hypothesis that RT-QuIC can precisely and quantitatively anatomically map disease-driving tau oligomers. Additional work characterizes the detailed tau neuropathology underlying AD clinical subtypes (collaboration with David Salmon and Doug Galasko). I have ongoing collaborations with John Ringman on PSEN1 AD and Jorg Hanrieder on mapping Abeta and tau species in human brain. This work has won multiple awards at meetings including in 2021 (Hirano Award for Best Paper on Neurodegenerative Disease and Junior Faculty Award at AD/PD) and 2023 (AANP Women and Diversity Presentation Award).

   a. Coughlin DG, Goodwill VS[#], Standke HG[#], Kim Y, Coley N, Pizzo DP, Galasko D, Kraus A*, **Hiniker A**.* Selective tau seeding assays and isoform-specific antibodies define neuroanatomic distribution of progressive supranuclear palsy pathology arising in Alzheimer's disease. Acta Neuropathologica 2022 Aug 18. doi: 10.1007/s00401-022-02480-x. PMCID: PMC9467964

   b. Smirnov DS, Salmon DP, Galasko D, Goodwill VS, Hansen LA, Zhao Y, Edland SD, Léger GC, Peavy GM, Jacobs DM, Rissman RA, Pizzo DP, **Hiniker A**. Association of Neurofibrillary Tangle Distribution with Age at Onset-Related Clinical Heterogeneity in Alzheimer's Disease: An Autopsy Study. Neurology. 2022 Feb 1. PMCID: PMC8826459

   c. Chen X, Calandrelli R, Girardini J, Yan Z, Tan Z, Xu X, **Hiniker A**, Zhong S. PHGDH expression increases with progression of Alzheimer's disease pathology and symptoms. Cell Metabolism. 2022 May 3;34(5):651-653. PMID: 35508105 PMCID: PMC9531314

   d. Smirnov DS, Ashton N, Blennow K, Zetterberg H, Simrén J, Rodriguez-Lantero J, Karikari T, **Hiniker A**, Rissman R, Salmon D, Galasko DR. Plasma biomarkers for Alzheimer's Disease in relation to

neuropathology and cognitive change. Acta Neuropathologica 2022 Apr;143(4):487-503. PMCID: PMC8960664

2. Identifying therapeutic targets in Parkinson's Disease (PD). Work in my lab has identified pathways important in degradation of LRRK2 and possible transmission routes of alpha-synuclein from the gut to the brain. I performed large-scale global interactome studies of LRRK2 and identified TRIM1 as a microtubule-associated E3 ubiquitin ligase that binds LRRK2, causes its microtubule localization, and ultimately ubiquitinates LRRK2 leading to proteasomal degradation. We have identified Rab38 as a factor that activate LRRK2's phosphorylation of substrates. Ongoing work studies novel pathways that drive LRRK2 degradation after kinase inhibitor treatment, which is critical to evaluating clinical utility and mechanism of LRRK2 kinase inhibitors as therapeutics for Parkinson's Disease.

a. Unapanta A, Shavarebi F, Porath J, Shen Y, Balen C, Nguyen A, Tseng J, Leong WS, Liu M, Lis P, Di Pietro SM, **Hiniker A**. Endogenous Rab38 regulates LRRK2's membrane recruitment and substrate Rab phosphorylation in melanocytes. J Biol Chem. 2023 Aug 23:105192. PMID: 37625589

b. Stormo AEDS, FitzGibbon M, Shavarebi F, Earley EM, Lum LS, Verschueren E, Swaney DL, Skibinski G, Ravisankar A, van Haren J, Davis EJ, Johnson JR, Von Dollen J, Iaccarino C, Dauer WT, Nichols RJ, Wittmann T, Cox TC, Finkbeiner S, Krogan NJ, Oakes SA*, **Hiniker A***. The E3 ligase TRIM1 ubiquitinates LRRK2 and controls its localization, degradation, and toxicity. Journal of Cell Biology. 2022 Apr 4;221(4):e202010065. PMCID: PMC8919618

c. Usmani A, Shavarebi F, and **Hiniker A.** The cell biology of LRRK2 in Parkinson's Disease. Molecular and Cellular Biology. 2021 Apr 22;41(5):e00660-20 PMCID: PMC8088268

d. Chandra R, **Hiniker A,** Kuo, YM, Nussbaum RL, Liddle R. Identification of alpha-synuclein in enteroendocrine cells suggests a direct route of pathologic spread in Parkinson's disease. JCI Insight. 2017 June 15 2(12). PMCID: PMC5470880

3. Neuropathology of human neurodegenerative diseases. My recent research has produced insights into the neuropathology of diverse human neurodegenerative disorders—including Limbic-predominant Age-related TDP-43 encephalopathy (LATE), Primary Age-related Tauopathy (PART), Multiple System Atrophy and Frontotemporal Lobar Degeneration. I am a member of the International FTLD-FUS consortium.

a. Smirnov DS, Salmon DP, Galasko D, Edland SD, Pizzo DP, Goodwill V, **Hiniker A**. TDP-43 Pathology Exacerbates Cognitive Decline in Primary Age-Related Tauopathy. Annals of Neurology 2022 Jun 13. doi: 10.1002/ana.26438. Online ahead of print. PMCID: PMC9391297

b. Goodwill VS, Coughlin DG, Pizzo DP, Galasko D, Hansen LA, Yuan SH, **Hiniker A**. A Case of Frontotemporal Lobar Degeneration With FUS-Positive Pathology (FTLD-FET) With Corticobasal Features and Language Deficits. Journal of Neuropathology and Experimental Neurology 2021 May 9. PMCID: PMC8476078

c. Coughlin DG, Dryden I, Goodwill VS, Pizzo DP, Wright B, Lessig S, Galasko D, Mac Kenzie IR, **Hiniker A**. Longstanding Multiple System Atrophy-Parkinsonism with Limbic and FTLD-type α-Synuclein Pathology. Neuropathology and Applied Neurobiology 2021 Nov 30. PMCID: PMC9097824

d. Arnold MR, Coughlin DG, Brumbach BH, Smirnov DS, Concha-Marambio L, Farris CM, Ma Y, Kim Y, Wilson EN, Kaye JA, **Hiniker A**, Woltjer RL, Galasko DR, Quinn JF. α-synuclein seed amplification in CSF and brain from patients with different brain distributions of pathological α-synuclein in the context of co-pathology and non-LBD diagnoses. Annals of Neurology. 2022 Jul 9. PMCID: PMC9489647

4. Markers of autophagy and aggregation distinguish the protein misfolding disorder Inclusion Body Myositis from Polymyositis. During my clinical training at UCSF, I became interested in the degenerative disorder Inclusion Body Myositis (IBM) which also shows striking features of pathologic protein aggregation and disruption of cellular protein folding machinery. With Dr. Marta Margeta, I developed new methods to differentiate IBM from the related disorder polymyositis. We found that markers of autophagy (LC3, p62) and aggregation (TDP-43) can distinguish IBM from PM with strong sensitivity and specificity.

a. **Hiniker A**, Daniels BH, Lee HS, Margeta M. Comparative utility of LC3, p62, and TDP-43 immunohistochemistry in differentiation of inclusion body myositis from polymyositis and related inflammatory myopathies. Acta Neuropathologica Communications.1:29, 2013. PMCID: PMC3893502

b. **Hiniker A**, Wong LJ, Berven S, Truong C, Adesina AM, Margeta M. Axial Mitochondrial Myopathy in a Patient with Rapidly Progressive Adult-Onset Scoliosis. Acta Neuropathologica Communications. 2(1):137, 2014. PMCID: PMC4180433

c. **Hiniker A**, Daniels BH, Margeta M. Polymyositis and inclusion body myositis in HIV-positive individuals: A study of 19 cases. Journal of Neuropathology and Experimental Neurology. 2016 Feb 3. PMCID: PMC5009472

5. The disulfide isomerase DsbC is essential under conditions of copper stress and folds proteins with non-consecutive disulfide bonds. My PhD thesis was performed in the lab of Dr. Jim Bardwell, where I studied the disulfide isomerase DsbC, which is an important enzyme contributing to correct protein folding in the *E. coli* periplasm. When I began my thesis, little was known about DsbC: its in vivo substrates, role in cell stress, and the structural elements differentiating it from the functionally distinct but similar proteins DsbA and DsbG were all unknown. I answered these questions in the six first author publications and four additional publications that composed my PhD thesis. For this work, I received the University of Michigan's Rackham Distinguished Dissertation Award for best dissertation of the year, one of 7 awards out of over 700 theses.

a. **Hiniker A,** Ren G, Heras B, Zheng Y, Laurinec S, Jobson RW, Stuckey J, Martin JL, Bardwell JC. (2007) Laboratory evolution of one disulfide isomerase to resemble another. Proc Natl Acad Sci USA. 104:11670-5. PMCID: PMC1906722

b. **Hiniker A**, Bardwell JC. (2004) In vivo substrate specificity of periplasmic disulfide oxidoreductases. Journal of Biological Chemistry. 279:12967-73. PMID 14726535.

c. **Hiniker A**, Collet JF, Bardwell JC. (2005) Copper stress causes an in vivo requirement for the Escherichia coli disulfide isomerase DsbC. Journal of Biological Chemistry. 280:33785-33791. PMID 16087673.

d. Bader MW, **Hiniker A**, Regeimbal J, Goldstone D, Haebel PW, Riemer J, Metcalf P, Bardwell JC.(2001) Turning a disulfide isomerase into an oxidase: DsbC mutants that imitate DsbA. EMBO Journal. 20:1555-62. PMID 11285220.

Complete List of Published Work (46 total): https://www.ncbi.nlm.nih.gov/pubmed?term=Hiniker%20A (of the 51 here, the heart association works (3) and the children and mobile devise works (3) are not me, the rest are my publications.) ORCID ID: 0000-0002-0055-6610

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Bienkowski, Michael S

eRA COMMONS USER NAME (credential, e.g., agency login): BIENKOWSKI2

POSITION TITLE: Assistant Professor of Physiology and Neuroscience

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of Pittsburgh, Pittsburgh, PA	B.S.	04/2007	Neuroscience / Chemistry minor
Center for Neuroscience at University of Pittsburgh, Pittsburgh, PA	Ph.D.	12/2012	Neuroscience
Center for Neural Basis of Cognition Graduate Training Program, Carnegie Mellon University, Pittsburgh, PA			
Laboratory of Neuro Imaging, University of California, Los Angeles (UCLA), Los Angeles, CA	Postdoctoral	08/2013	
USC Steven Neuroimaging and Informatics Institute, University of Southern California (USC), Los Angeles, CA	Postdoctoral	01/2019	

A. Personal Statement

Over a century of neuroscience research has led to the understanding that individual brain areas are not solely responsible for animal behavior. Rather, it is the connections between brain areas and the organized activity of distributed neural networks that determine how we act. I believe understanding the specific disconnection and disruption of brain organization is critical to treatment of degenerative diseases like Alzheimer's disease. Over my research career, I have studied brain architecture from brainstem to cortex, beginning with my graduate studies on rat viscerosensory circuits and continuing through my postdoctoral work on cortical and subcortical brain regions as part of the Mouse Connectome Project (MCP) and NIH Brain Initiative Cell Census Network (BICCN). My lab's multidisciplinary translational research employs neuroanatomical, computational, and spatial transcriptomics techniques to define cell type-specific degeneration in rodent disease models and human tissue using advanced high-throughput, high-resolution microscopy. I believe that a fundamental disconnect between mouse and human research underlies the failure to translate scientific research to cures for human disease. Therefore, it is critical to develop translational brain atlases and other tools which can bridge the gap between rodent models and human research at progressive disease stages and provide a guide to drug development and effective treatments for retinal degenerative diseases and Alzheimer's disease. Toward this goal, our lab has worked with the USC Alzheimer's Disease Research Center (ADRC) to create a high-throughput digital pathology workflow to better quantify, characterize, and map the distribution of pathology within post-mortem human brain tissue sections. By collecting a large pathology image database, we will better characterize the disease heterogeneity among the broad and diverse patient populations suffering from ADRD. In collaboration with Drs. John Ringman, Debra Hawes, and Annie Hiniker, we are analyzing the neuropathology of the A431E PSEN1 'Jalisco' mutation. As part of this ADRC renewal, we will leverage the USC ADRC's 40-year pathology slide collection to create an online, open-access digital pathology website for the ADRD research community and investigate the of the APOE4 allele on CAA, AD neuropathology, and perivascular inflammation.

Ongoing and recently completed projects that I would like to highlight include:

NIA K01 AG066847
Bienkowski (PI), Role: PI
08/01/21-07/31/26
A Translational 3D Map of Hippocampal Cell Types To Drive Investigations of Alzheimer's Disease
Alzheimer's disease (AD) drug development relies on both mouse and clinical research, but translational resources to bridge the gap between mouse and human AD are limited. Building upon my previous research creating the mouse Hippocampus Gene Expression Atlas (HGEA), this proposal will investigate translational gene expression patterns across hippocampal neuronal cell types in mouse AD models and humans (with and without AD). The completion of this research will produce a comprehensive human HGEA that, together with the mouse HGEA, can help guide AD drug development and reveal the hippocampal neuronal cell types that are primarily affected by AD.

NIA P01AG052350
Zlokovic/Toga (MPI), Role: Co-Investigator (Project 3)
05/01/22-03/31/27
Vascular contributions to dementia and genetic risk factors for Alzheimer's disease
Program project to study imaging and molecular biomarkers of neurovascular dysfunction in individuals at genetic risk for AD both familial and sporadic. Dysfunction in the blood-brain barrier (BBB) and each of the cellular components of the neurovascular unit (NVU) – vascular cells, glial cells, and neurons – has been linked to Alzheimer's disease (AD) evolution in experimental, imaging, pathological, and epidemiological studies. These findings have led to an emerging 'neurovascular hypothesis' of AD which holds that cerebrovascular dysfunction contributes to cognitive decline and dementia in AD. Project 3 supports the overall theme of P01 focused on apolipoprotein E (APOE) gene, the major genetic risk factor for AD, by interrogating the BBB/neurovascular hypothesis experimentally at the cellular, molecular and systems levels

NIDA RF1DA055665
Song (PI), Role: Co-Investigator
09//01/2021-08/31/2024
Combined Mechanistic and Input-Output Modeling of the Hippocampus During Spatial Navigation
The goal of this NIH BRAINI-funded project is to develop a novel modeling paradigm inspired by the generative adversarial network (GAN) that synergistically combines both mechanistic and input-output (machine learning) modeling techniques to build a large-scale realistic model of the hippocampus that are functionally indistinguishable from the real hippocampus. It seeks to provide a general computational framework for integrating biological knowledge, hypotheses, and input-output data to gain deeper and more quantitative understanding of cognitive functions.

Epstein Foundation Gene Therapy Program
Ringman (PI), Role: Co-Investigator
11/01/22-10/31/24
The Neuropathology of the A431E PSEN1 Mutation in Mouse and Man
The goal of this project is to characterize the human neuropathology caused by the A431E 'Jalisco' mutation affecting patients of Mexican heritage. In collaboration with the USC ADRC, our lab has developed a digital pathology pipeline to rapidly digitize and quantify neuropathology in whole slide images of human brain samples using machine learning classification and segmentation approaches.

NSF 2121164
Lazzi (PI/PD), Role: Co-PI
10/01/21-09/30/26
GCR: Reprogramming Biological Neural Networks with Field-Based Engineered Systems
Using Royal College of Surgeons (RCS) rats (a strain of rats affected by genetic blindness) and 5xFAD mice, we propose to slow and control the progression of Alzheimer's disease and retinal degeneration using bioengineered devices. The questions that we intend to ask in the proposed effort are: What are the morphological changes experienced by the neural network in neurodegenerative conditions? What are the underlying mechanisms for these changes? Can external or implantable bioengineered devices be exploited to

alter these changes? If so, what are the mechanisms for which this change is effected and do the morphologic changes induced by electrical stimulation lead to positive functional outcomes?

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2020 - Present	Director, USC Center for Integrative Connectomics
2020 - Present	Assistant Professor of Physiology and Neuroscience, USC Mark and Mary Stevens Neuroimaging and Informatics Institute and Zilkha Neurogenetic Institute, Keck School of Medicine of USC, University of Southern California, Los Angeles, CA
2021- Present	Member, iSTAART
2020 - Present	Ad hoc reviewer, NSF
2019	Assistant Professor of Research Neurology, USC Mark and Mary Stevens Neuroimaging and Informatics Institute, Keck School of Medicine of USC, University of Southern California, Los Angeles, CA
2017 - 2020	Member, BRAIN Initiative Cell Census Network (BICCN) Anatomy and Morphology working group
2017 - Present	Ad hoc reviewer, Brain Structure and Function
2013	Postdoctoral Scholar, Laboratory of Dr. Hong-Wei Dong, Laboratory of Neuro Imaging, University of California, Los Angeles (UCLA), Los Angeles, CA
2013 - 2019	Postdoctoral Fellow, Laboratory of Dr. Hong-Wei Dong, USC Mark and Mary Stevens Neuroimaging and Informatics Institute, Keck School of Medicine of USC, University of Southern California (USC), Los Angeles, CA
2009 - Present	Member, Society for Neuroscience

Honors

2023	NIH Brain Initiative “Show us Your Brains!” Photo and Video Contest Finalist
2018	NSF “Vizzies” Visualization Challenge Finalist, National Science Foundation
2017	Best Images of 2017, Spectrum News
2015	USC Provost’s Top-Off Program Research Award, University of Southern California
2011 - 2012	Andrew Mellon Predoctoral Fellowship, University of Pittsburgh
2009 - 2010	NIH Institutional Training Grant T32 DK063922, Digestive Diseases Training Program, University of Pittsburgh
2007	James E. Bradler Award for Excellence in Undergraduate Research, Department of Neuroscience, University of Pittsburgh
2007	Departmental honors, Department of Neuroscience, University of Pittsburgh

C. Contributions to Science

1. The major direction of my research program is to understand the cell-type organization of the hippocampus and it’s disruption in Alzheimer’s Disease. Decades of behavioral studies distinguished a “cold cognitive” dorsal hippocampus from the more “hot affective” ventral hippocampus. However, rat anatomical studies interpreted the hippocampus as a tripartite topographic gradient with dorsal, intermediate, and ventral hippocampal parts. More recent studies had identified multiple discrete genetic hippocampal domains, making it unclear how to relate these different behavioral, anatomical, and genetic data across the whole hippocampus. To understand the relationship between mouse hippocampal genetics and connectivity, I created the Hippocampus Gene Expression Atlas (HGEA) by manually annotating and analyzing over 255 gene expression patterns from the Allen Institute’s online *in situ* hybridization database. The HGEA outlines over 20 genetic domains of the dentate gyrus (DG), CA3, CA2, CA1, and subiculum (SUB) across the entire hippocampal axis and revealed many novel insights into hippocampal organization. Using the HGEA as a guide, I mapped each region’s input/output connectivity and created the most detailed, comprehensive wiring diagram of the mouse hippocampus yet. In contrast to previous interpretations based purely on anatomical data, our new analysis reveals that both hippocampal connectivity and gene expression are localized within discrete subregions. Using computational tools, I analyzed intra-hippocampal network connectivity to discover that the hippocampus is organized as a multiscale hierarchical network. Further analysis revealed that the hippocampal

subnetworks are uniquely connected to larger brain-wide networks implicated in spatial navigation, social behaviors, and homeostatic behaviors. Overall, my work has generated novel insights into the relationship between hippocampal genetics and connectivity and provides a foundation for the genetic dissection of hippocampal cell types and their contributions to behavior. More recently, I have begun to investigate how the organization of the hippocampus outlined by the mouse HGEA is present in the human hippocampus. By analyzing Allen Institute gene expression data in human tissue, I found evidence that the laminar organization of the SUB is conserved in mice and humans, despite a 3D rotation of the hippocampal axis between the two species. Using high resolution *ex vivo* MRI imaging of thick human tissue samples, we found differences in DTI fiber density strongly aligned with the distribution of gene expression patterns, suggesting a relationship between cell type gene expression and connectivity in the human SUB. Based on this data, we proposed a new 3D model of the human SUB and demonstrate its structural relationship to the mouse. Ultimately, my goal is to create translational mammalian cell type atlases and determine which cell types are disrupted in Alzheimer's disease.

a. **Bienkowski MS**, Bowman I, Song MY, Gou L, Ard T, Cotter K, Zhu M, Benavidez NL, Yamashita S, Abu-Jaber J, Azam S, Lo D, Foster NN, Hintiryan H, Dong H-W, Integration of gene expression and brain-wide connectivity reveals the multiscale organization of mouse hippocampal networks. *Nature Neuroscience*. 21: 1628-1643. 2018. PMID: 30297807. PMCID: PMC6398347.
b. Hahn, JD, Swanson, LW, Bowman, I, Foster, NN, Zingg, B, **Bienkowski, MS**, Hintiryan, H, Dong, HW, An open access mouse brain flatmap and upgraded rat and human brain flatmaps based on current reference atlases. *J Comp Neurol.* 2020. PMID: 32511750. PMCID: PMC7721992
c. **Bienkowski MS,** Sepeherband, F, Kurniawan, ND, Stanis, J, Korobkova, L, Khanjani, N, Hintiryan, H, Miller, CA, Dong, HW, Homologous laminar organization of the mouse and human subiculum. *Scientific Reports* 11(1), 1-14. 2021. PMID: 33580088. PMCID: PMC7881248
d. **Bienkowski, MS**, Further refining the boundaries of the hippocampus CA2 with gene expression and connectivity: Potential subregions and heterogenous cell types. *Hippocampus* 33(3), 150-160. 2023. PMID: 36786207; PMCID: PMC9987718

2. As part of the Brain Initative Cell Census Network (BICCN) and it's goal to develop the first cell-type census of the motor cortex across species, I had the opportunity to work alongside a number of prestigious laboratories. The BICCN recently published its initial research effort in 2021 as a collection of 27 research articles in Nature journals. As part of this massive collaborative effort, I worked with neuroscientists at Salk Institute and UCSD in the Center for Epigenomics of the Mouse Brain Atlas (CEMBA) with the goal of integrating gene expression and connectivity to identify distinct classes of cell types in the mouse motor cortex. In addition, to the motor cortex, our BICCN research projects extended to defining multiscale cell types within other areas of the brain including the amygdala, superior colliculus, and basal ganglia. Within the amydala, we defined three new subdivisions of the basolateral amygdala with distinct projections and delineated its wiring diagram (Hintiryan et al, 2021). In the superior colliculus, we defined four mediolateral zones based on cortico-tectal projections and described their input-output wiring diagram as it relates to visual and non-visual behaviors (Benavidez et al, 2021). In the basal ganglia, we established the closed cortco-basal ganglia loop circuit architecture (Foster et al, 2021). This research led the way toward understanding how to integrate connectivity, gene expression, morphology, physiology, and other characteristics to form a consensus on how to define neuronal cell types within the brain.

a. Hintiryan H, Bowman I, Johnson DL, Korobkova L, Zhu M, Khanjani N, Gou L, Gao L, Yamashita S, **Bienkowski MS**, Garcia L, Foster NN, Benavidez NL, Song MY, Lo D, Cotter KR, Becerra M, Aquino S, Cao C, Cabeen RP, Stanis, J, Fayzullina M, Ustrell SA, Boesen T, Tugangui AJ, Zhang Z-G, Peng B, Fanselow MS, Golshani P, Hahn JD, Wickersham IR, Ascoli GA, Zhang LI, Dong H-W. Connectivity characterization of the mouse basolateral amygdalar complex. *Nature Communications* 12 (1): 1-25. 2021. PMID: 34001873, PMCID: PMC8129205
b. Benavidez NL, **Bienkowski MS**, Zhu M, Garcia LH, Fayzullina M, Gao L, Bowman I, Gou L, Khanjani N, Cotter KR, Korobkova L, Becerra M, Cao C, Song MY, Zhang B, Yamashita S, Tugangui AJ, Zingg B, Rose K, Lo D, Foster NN, Boesen T, Mun HS, Aquino S, Wickersham IR, Ascoli GA, Hintiryan H, Dong HW. Organization of the inputs and outputs of the mouse superior colliculus. *Nature Communications* 12 (1), 1-20. 2021. PMID: 34183678, PMCID: PMC8239028

c. Muñoz-Castañeda R, Zingg B, Matho KS, Chen X, Wang Q, Foster NN, Li A, Narasimhan A, Hirokawa KE, Huo B, Bannerjee S, Korobkova L, Park CS, Parl Y-G, **Bienkowski MS**, Chon U, Wheeler DW, Li X, Wang Y, Naeemi M, Xie P, Liu L, Kelly K, An X, Attili SM, Bowman I, Bludova A, Cetin A, Ding L, Drewes R, D'Orazi F, Elowski C, Fischer S, Galbavy W, Gao L, Gillis J, Groblewski PA, Gou L, Hahn JD, Hatfield JT, Hintiryan H, Huang JJ, Kondo H, Kuang X, Lesnar P, Li X, Li Y, Lin M, Lo D, Mizrachi J, Mok S, Nicovich PR, Palaniswamy R, Palmer J, Qi X, Shen E, Sun Y-C, Tao HW, Wakeman W, Wang Y, Yao S, Yuan J, Zhan H, Zhu M, Ng L, Zhang LI, Lim BK Hawrylycz M, Gong H, Gee JC, Kim Y, Chung K, Yang XW, Peng H, Luo Q, Mitra PP, Zador AM, Zeng H, Ascoli GA, Huang ZJ, Osten P, Harris JA, Dong HW. Cellular anatomy of the mouse primary motor cortex. *Nature* 598**,** 159–166 2021. PMID: 34616071, PMCID: PMC8494646.

d. Foster NN, Barry J, Korobkova L, Garcia L, Gao L, Becerra M, Sherafat Y, Peng B, Li X, Choi J-H, Gou L, Zingg B, Azam S, Lo D, Khanjani N, Zhang B, Stanis J, Bowman I, Cotter K, Cao C, Yamashita S, Tugangui A, Li A, Jiang T, Jia X, Feng Z, Aquino S, Mun H-S, Zhu M, Santarelli A, Benavidez NL, Song M, Dan G, Fayzullina M, Ustrell S, Boesen T, Johnson DL, Xu H, **Bienkowski MS**, Yang XW, Gong H, Levine MS, Wickersham I, Luo Q, Hahn JD, Lim BK, Zhang LI, Cepeda C, Hintiryan H, Dong HW. The mouse cortico–basal ganglia–thalamic network. *Nature* 598**,** 188–194 2021. PMID: 34616074; PMCID: PMC8494639.

3. As a neuroanatomist working on the interdisciplinary Mouse Connectome Project, I have made major contributions toward understanding the systems-level network organization of the mouse connectome. The goal of the Mouse Connectome Project is to generate a comprehensive map of all neural connections in the mammalian brain. In 2014, we published the first wiring diagram of mouse cortico-cortical networks and our computational network analysis determined that the mouse cortex contains 5 functional cortico-cortical networks related to animal behavior (Zingg et al, *Cell*, 2014). This landmark study received much attention from many media sources (Fox News, Yahoo News, Scientific American, The Scientists, Medical Press, etc.) and featured by the NIH Director. Finally, this paper was selected as one of the top 10 research articles in Cell: Best of 2014, and was recognized in historic "40 years of Cell" timeline as one of only two landmark papers of 2014. Subsequently, in 2016, we showed how these cortico-cortical networks provide top-down input to striatum (Hintiryan, Foster, et. al, *Nature Neuroscience*, 2016). Our computational analysis of the distribution of cortico-striatal inputs revealed that the striatum could be subdivided into 29 subdomains. More recently, the Mouse Connectome Project has begun to analyze cortico-thalamic connectivity and our recent publication shows that the dorsolateral geniculate nucleus of the thalamus (LGd) has bidirectional connectivity with areas of the extrastriate visual cortex, not just primary visual cortex (VISp; Bienkowski et al, *J Comp Neurol.,* 2019). In this paper, we show that there is a long history of conflicting evidence for and against the dogmatic LGd-projects-to-VISp-only perspective and we provide new evidence in the mouse against this model. In addition to our scientific insights, the Mouse Connectome Project continues to develop advanced computational tools for dissecting neuroanatomical data. The studies above relied on computational software called Connection Lens we developed in the lab to register microscopy images to a standard mouse brain atlas and accurately quantify neuroanatomical data density and distribution.

   a. Zingg B, Hintiryan H, Gou L, Song MY, Bay M, **Bienkowski MS**, Foster NN, Yamashita S, Bowman I, Toga AW, Dong H-D, Neural networks of the mouse neocortex. *Cell*. 156: 1096-1111. 2014. PMID: 24581503, PMCID: PMC4169118
   b. Hintiryan H, Foster NN, Bowman I, Bay M, Song MY, Gou L, Yamashita S, **Bienkowski MS**, Zingg B, Zhu M, Yang WX, Shih JC, Toga AW, Dong H-W, The mouse cortico-striatal projectome. *Nature Neuroscience*. 19: 1100-1114. 2016. PMID: 27322419, PMCID: PMC5564682.
   c. **Bienkowski MS**, Benavidez NL, Wu K, Gou L, Becerra M, Dong H-W. Extrastriate connectivity of the mouse dorsal lateral geniculate thalamic nucleus. *J Comp Neurol*. 527: 1419-1442. 2019. PMID: 30620046, PMCID: PMC6992387
   d. Li, R, Zhu, M, Li, J, **Bienkowski, MS**, Foster, NN, Xu, H, Ard, T, Bowman, I, Hintiryan, H, Zhou, C, Veldman, M, Yang, X, Zhang, J, and Dong, HW. G-Cut: Precise segmentation of densely connected neuron clusters. *Nature Communications* 10: 1549. 2019. PMID: 30948706, PMCID: PMC6449501

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Hawes, Debra

eRA COMMONS USER NAME (credential, e.g., agency login): DHAWES

POSITION TITLE: Associate Professor of Clinical Pathology; Faculty Director of the Immunohistochemistry Laboratory

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
California State University, Fullerton	B.A.	01/1986	French
Ross University School of Medicine, Dominica	M.D.	06/1991	Medicine
LAC+USC Medical Center, Los Angeles		06/1996	Pathology Residency
LAC+USC Medical Center, Los Angeles		06/1997	Surgical Pathology Fellowship
LAC+USC Medical Center, Los Angeles		06/1998	GYN and Immunohistochemistry Fellowships
LAC+USC Medical Center, Los Angeles		06/2015	Neuropathology Fellowship

A. Personal Statement

I have over 15 years of experience in cancer research with expertise in immunopathology working directly under Clive R. Taylor, MD, PhD an internationally recognized expert on immunohistochemistry at the University of Southern California (USC) Health Science Campus. These studies included extensive experience in novel antibody development and optimization, assessment of immunohistochemical markers on normal human and animal tissue as well as in numerous types of tumors. In 2013 I refocused my activities to concentrate on neuropathology focusing on pediatric neuropathology but with a long-term special interest in neurodegenerative diseases. In June of 2015 I completed a 2 year ACGME accredited neuropathology fellowship at the LAC+USC Medical Center which included extensive work with Dr. Carol Miller, the head of the fellowship program and the then director of the Neuropathology Core for the USC ADRC. During that time I received extensive instruction and experience in assessing brains from patients with clinically diagnoses dementia on the gross and microscopic levels and to diagnose Alzheimer's and other related neurodegenerative diseases. I have maintained an interest and increased my expertise in the field of neurodegenerative diseases and in July 2021 I became director of the USC ADRC Neuropathology Core and then in January 2024 with the recruitment of Dr. Anne Hiniker from the UCSD ADRC I became co-Director of the NP Core. In July of 2015 I transferred from the USC Health Science Campus to Children's Hospital Los Angeles to pursue my other interest in pediatric brain tumors working with Alexander R. Judkins, MD, a recognized expert in the area of pediatric neuropathology. I am currently the Faculty Director of the CHLA Immunohistochemistry laboratory which provides service for both clinical and research applications of immunohistochemistry. This background supplies the expertise needed to successfully complete the neuropathological histologic and immunohistochemical aspects of this project.

Ongoing and recently completed projects that I would like to highlight include:

P30 AG066530, (PI) H Chui/A. Toga/ B. Zlokovic Dates of Award:6/2020 to 6/2025
Agency: NIH
Title: USC Alzheimer's Disease Research Center (ADRC)
Description: NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores. This Alzheimer's center will focus on vascular and metabolic risk factors (VMRF), the neurovascular unit, brain structure and function, and the treatment and prevention of Alzheimer disease.

Epstein Breakthrough Alzheimer's Research Fund, (PI) John Ringman, MD Dates of Award: 07/01/2022
Agency: Epstein Family Foundation
Title: Neuropathology of the A431E PSEN1
Description: Histologic and immunohistochemical analysis of brain tissue from patients with PSEN1 mutated Alzheimer's disease and compare to other forms of ADAD and sporadic dementia

Citations:

1. Guzman S, Koronyo-Hamaoui M, Koronyo Y, Shi H, Hiniker A, **Hawes D**, Kashani AH, Ringman J. In-vivo and post-mortem retinal and brain pathological findings in a homozygote for the A431E *PSEN1* mutation. Alzheimer's Association International Conference (AAIC) Amsterdam, Netherlands, July 16-20, 2023
2. Shih N-C, Kurian ND, Cabeen RP, Korobkova L, Wong E, Chui HC, **Hawes D**, Clark KA, Miller CA, Jones KT, Sephrband F. Microstructural mapping of dentate gyrus pathology in Alzheimer's disease: a 16.4 Tesla magnetic resonance imaging study. Jan 5, 2023 *Neuroimage: Clinical.* 37*(*2023) 103318https://doi.org/1016/j.nicl.2023.103318. PMID: 36630864.

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2023-present	Member, Los Angeles County Medical Association
2016-Present	Member, American Association of Neuropathologists
2015-Present	Associate Professor of Clinical Pathology, Children's Hospital Los Angeles, Department of Pathology and Laboratory Medicine, Children's Hospital Los Angeles, Keck School of Medicine of USC, Los Angeles, CA
2015-Present	Faculty Director, Immunohistochemistry Neuropathologist, Children's Hospital Los Angeles, Department of Pathology and Laboratory Medicine, Los Angeles, CA
2015-Present	Member, American Society for Clinical Pathology
2015-Present	Member, College of American Pathologists
2013-2015	Neuropathology Fellowship, LAC+USC Medical Center
2010-2015	Associate Professor of Clinical Pathology, Department of Pathology, University of Southern California, Los Angeles, CA
2009-2010	Member, AACR Tumor Microenvironment Working Group
2006-2013	Member, American Association for Cancer Research
2006-2008	Member, Association of Clinical Research Professionals
1998-2010	Assistant Professor of Clinical Pathology, Department of Pathology, University of Southern California, Los Angeles, CA
1997-1998	Fellow, GYN and Immunopathology, Department of Pathology, University of Southern California
1996-1997	Fellow, Department of Surgical Pathology, Los Angeles County + University of Southern California Medical Center, Los Angeles, CA
1992-1996	Resident, Department of Pathology, Los Angeles County-University of Southern California Medical Center, Los Angeles, CA
1991-1992	Research Technician, University of Southern California, Department of Hematopathology, Los Angeles, CA

C. Contributions to Science

1. My early work centered around the detection of early metastatic breast, lung cancer, and prostate cancer identifying novel markers of disease progression in bladder, prostate, ovary, lung and breast cancers using immunohistochemical techniques. The earlier works were based on simple identification and quantitation. Later work focused on the identification of a putative cancer stem cell profile in the circulating cells to predict the biologic potential of these cells to proliferate into disseminated disease.
   a. **Hawes D**, Neville AM, Cote RJ. Occult metastases in patients with breast cancer. Ann Surg Oncol 8: 60-63, 2001.
   b. Pagliarulo V, **Hawes D**, Brands FH, Groshen S, Cai J, Stein JP, Lieskovsky G, Skinner DG, Cote RJ. Detection of occult lymph node metastases in patients with locally advanced (pT3) node – negative prostate cancer. J Clin Oncol 24(18): 2735-42, 2006. PMID: 16782914.
   c. Giuliano AE, **Hawes D**, Ballman KV, Whitworth PW, Blumencranz PW, Reintgen DS, Morrow M, Leitch AM, Hunt KK, McCall LM, Abati A, Cote R. Association of occult metastases in sentinel lymph nodes and bone marrow with survival among women with early-stage invasive breast cancer. JAMA 306(4): 385-95, 2011. PMCID: PMC5389856.
   d. Rusch VW, **Hawes D**, Decker P, Martin SE, Abati A, Landreneau RJ, Patterson GA, Inculet RI, Jones DR, Malthaner RA, Cohen R, Ballman K, Putnam JB, Cote RJ. Occult metastases in lymph nodes predict survival in resectable non-small cell lung cancer Report of the ACOSOG Z0040 trial J Clin Oncol 29(32), 2011. Doi: 10.1200/jco.2011.35.2500

2. In addition to work described above I was also involved in collaborative studies with the Departments of Medicine, Radiology, Pathology, and Preventive Medicine at the University of Southern California. I was part of a team of researchers looking for ways to predict which women were likely to progress and what dietary factors such as soy and green tea have the potential to ameliorate disease progression in patients with DCID. My role involved determining biologic markers of proliferation and apoptosis by immunohistochemistry.
   a. Taylor D, Pearce CL, Hovanessian-Larsen, Downey S, Spicer DV, Bartow S, Pike MC, Wu AH, **Hawes D.** Progesterone and estrogen receptors in pregnant and premenopausal non-pregnant normal human breast. Breast Cancer Res Treat 118(1): 161-8, 2009. PMID: 19205874.
   b. Chung K,* Hovanessian-Larsen LJ*, **Hawes D*,** Taylor D, Downey S, Spicer DV, Stanczyk FZ, Patel S, Anderson AR, Pike MC, Wu AH, Pearce CL. Breast epithelial cell proliferation is markedly increased with short-term high levels of endogenous estrogen secondary to controlled ovarian hyperstimulation. Breast Cancer Treat 132(2): 653-60, 2012. (* co-first authors) UI 22083231. PMCID: PMC3303045.
   c. Hovanessian-Larsen L, Taylor D, **Hawes D**, Spicer DV, Press MF, Wu AH, Pike MC, Pearce CL. Lowering oral contraceptive norethindrone dose increases estrogen and progesterone receptor levels with no reduction in proliferation of breast epithelium: a randomized trial. Contraception 86(3): 238-43, 2012 UI 22325110. PMCID: PMC3355208.
   d. Wu AH, Spicer D, Garcia A, Tseng C-C, Hovanessian-Larsen L, Sheth P, Martin SE, **Hawes D**, Russell C, MacDonald H, Tripathy D, Su M-Y, Ursin G, Pike MC. Double-blind randomized 12-month soy intervention had no effect on breast MRI fibroglandular tissue density or mammographic density. Cancer Pre Res 8(10): 942-51, 2015. doi 10.1158/1940-6207. PMCID: PMC4596769

3. Now at Children's Hospital Los Angeles I have been involved in clinical service work in the area of pediatric brain tumors; however, I have continued my research efforts which include working with Drs. Venneti and Judkins in the area of H3K27me3 in childhood ependymomas and the role of SWI/SNF ATPase in H3.3K27M diffuse midline gliomas. I also have an ongoing collaboration with Dr. A. Erdreich-Epstein working to optimize immunohistochemical markers in the BarTeL mouse medulloblastoma model as well as performing marker studies for other projects and case reports. In addition to the work in pediatric tumors the immunohistochemistry laboratory I lead continues to provide the USC ADRC Neuropathology Core with the special stains for the diagnosis of Alzheimer disease and related dementias. As Co-director and Director of the USC ADRC NP Core I have provided detailed autopsy reports of the ADRC cases and participated in ongoing related research projects using the ADRC cohort including collaboration with Dr. Maya Koronyo-Hamaoui at Cedars-

Sinai looking at the retina in patients with autopsy confirmed AD and with John Ringman at USC who studies rare autosomal dominant forms of Alzheimer disease.

a. Panwalker P, Clark J, Ramaswamy V, **Hawes D**, Yang F, Dunham C, Yip S, Hukin J, Sun Y, Schipper MJ, Chavez L, Margol A, Pekmexci Y, Chung C, Banda A, Bayliss JM, Curry SJ, Santi M, Rodriguez FJ, Snuderl M, Karajannis MA, Saratsis AM, Horbinski CM, Carret A-S, Wilson B, Johnston D, Lafay-Cousin L, Zelcer S, Eisenstat D, Silva M, Scheinemann K, Jabado S, McNeely PD, Kool M, Pfister SM, Taylor MD, Hawkins C, Korshunov A, Judkins AR, Venneit S. Immunohistochemical analysis of H3K27me3 demonstrates global reduction in group is a powerful Immunohistochemical analysis of H3K27me3 demonstrates global reduction in group-A childhood posterior fossa ependymoma and is a powerful predictor of outcome. Acta Neuropathol, 134, 2017 doi: 10.1007/s00401-017-1752-4 . doi 10.1007/s00401-017-1752-4.
b. Ji J, Kaneva K, Hiemenz MC, Dhall G, Davidson TB, Erdreich-Epstein A, **Hawes D**, Hurth K, Margol AS, Mathew A, Robison NJ, Schmidt RJ, Tran HN, Judkins AR, Cotter JA, Biegel JA. Clinical utility of comprehensive genomic profiling in central nervous system tumors of children and young adults. Clinical utility of comprehensive genomic profiling in central nervous system tumors of children and young adults. Neuro-Oncology Advances, 2021; 3 (1). https://doi.org/10.1093/noajnl/vdab037
c. Mota M, ; Sweha SR, Stefan R. ; Pun M, Natarajan SK, Ding Y, Chung D, **Hawes D**, Yang F, Judkins AR, Samajdar S, Cao X, Xiao L, Parolia A, Chinnaiyan AM, Venneti S. Targeting SWI/SNF ATPase in H3.3K27M diffuse intrinsic pontine gliomas. PNAS, 2023, Vol.120 (118), p.e2221175120-e2221175120, Article e2221175120
d. Shi H, Koronyo Y, Fuchs D-T, Sheyn J, Jallow O, Mandalia K, Graham SL, Gupta VK, Mirzaei M, Kramerov AA, Ljubimov AV, **Hawes D**, Miller CA, Black KL, Carare RO, Koronyo-Hamaoui M. Retinal arterial Aβ40 deposition is linked with tight junction loss and cerebral amyloid angiopathy in MCI and AD patients. *Alzheimers Dement.* 2023; 1- 13. https://doi.org/10.1002/alz.

Complete List of Published Work in MyBibliography:
https://www.ncbi.nlm.nih.gov/myncbi/collections/bibliography

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Annie Hiniker

eRA COMMONS USER NAME (credential, e.g., agency login): hiniker

POSITION TITLE: Associate Professor of Pathology at USC

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Harvard University, Cambridge, MA	A.B.	06/1999	Honors Biology
University of Michigan Medical School, Ann Arbor, MI	M.D./Ph.D.	06/2008	Medicine, Cellular and Molecular Biology
Princeton University, Princeton, NJ	Postdoctoral	05/2009	Molecular Biology
UCSF, San Francisco, CA	Residency	06/2011	Pathology
UCSF, San Francisco, CA	Fellowship	06/2013	Neuropathology
UCSF, San Francisco, CA	Postdoctoral	08/2016	Molecular Biology

A. Personal Statement

I am a board-certified neuropathologist focused on molecular mechanisms and neuropathology of neurodegenerative diseases. I am an Associate Professor of Pathology at USC and Director of the USC ADRC Neuropathology Core. I also run a basic research lab focused on the molecular underpinnings and cell biology of neurodegeneration, particularly as related to the degradation of neurotoxic proteins, such as tau and LRRK2. My research strives to utilize my expertise in neurodegenerative neuropathology and basic molecular biology to delineate the mechanisms of neuronal death in Alzheimer's Disease (AD) and Parkinson's Disease (PD), with the long-term goal of developing novel rational therapeutics. I have a strong interest in defining how the molecular interaction and neuroanatomic colocalization of diverse neuropathologic proteins (tau, α-synuclein,TDP-43) influences clinical manifestations and progression of AD. Much of my recent work combines quantitative analysis of patient postmortem tissue, including digital neuropathology and biochemical assays, with precise neuropsychological cognitive data obtained during life. These studies allowed us to define neuropathologic underpinnings of clinical phenotypes including the role of tau neurofibrillary tangle distribution in driving the disproportionate executive dysfunction of early onset sporadic AD and the role of TDP-43 pathology in driving clinical dementia in early-stage AD neuropathologic change. I have ongoing collaborations with key members of the USC ADRC including Drs. Mike Bienkowski, Debra Hawes, and John Ringman to study the neuropathology of AD, including a unique Latino cohort with the Presenilin 1 A431E (Jalisco) mutation. I have successfully competed for research grants including a recent R01 from NINDS, the American Parkinson Disease Association, the Department of Defense Parkinson's Research Program, and the Alzheimer's Association, among others.

The following recent peer-reviewed publications highlight my experience and qualifications to lead the USC Neuropathology Core:

a. Coughlin DG, Goodwill VS#, Standke HG#, Kim Y, Coley N, Pizzo DP, Galasko D, Kraus A*, **Hiniker A**.* Selective tau seeding assays and isoform-specific antibodies define neuroanatomic distribution of progressive supranuclear palsy pathology arising in Alzheimer's disease. Acta Neuropathologica. 2022 Aug 18. doi: 10.1007/s00401-022-02480-x. PMCID: PMC9467964

PHS 398 Cover Page Supplement

1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

PHS 398 Cover Page Supplement

5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

Introduction	
1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	Neuropathology_Aims1044447062.pdf
3. Research Strategy*	Neuropathology_Research_Plan1044447068.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
9. Letters of Support	
10. Resource Sharing Plan(s)	
11. Other Plan(s)	
12. Authentication of Key Biological and/or Chemical Resources	
13. Appendix	
12. Appendix	

NEUROPATHOLOGY CORE (CORE D) - SPECIFIC AIMS

The Neuropathology (NP) Core (**Core D**) plays an essential role in the gold-standard diagnosis of AD and AD-related dementias (ADRD) and distributes tissue internally, nationally, and internationally to support ADRD research. The NP Core's biorepository houses >1,100 autopsy cases dating back to the 1980s, which includes nearly 15,000 tissue blocks, >50,000 slides, >150 eyes, and frozen aliquots of >30,000 unique neuroanatomic samples and >5,000 unique systemic tissue samples. Since our last renewal, the NP Core performed 90 autopsies, distributed 3,835 tissue allocations, and supported 71 projects and 34 publications. In the upcoming cycle, the NP Core will modernize its robust support of affiliated projects and investigators worldwide with tissue distribution enhanced by a newly implemented online tissue request system and a multi-Core tissue request review committee. Additionally, in collaboration with the Clinical Core, we will increase autopsies from diverse populations, with a particular focus on our large Latinx cohort and *APOE* ε4 positive cases. In collaboration with the DMS Core, we will create a freely accessible online digital neuropathology database of our slides; in collaboration with the REC Core, we will support the training and education of residents, fellows, and students in ADRD research, neuroanatomy, and neuropathologic diagnosis.

The NP Core is now directed by Dr. Annie Hiniker, a board-certified R01-funded neuropathologist who was recruited to USC in January 2024. She is joined in neuropathologic diagnostic skills and leadership by co-Director Dr. Debra Hawes as well as by Dr. Michael Bienkowski, who provides expertise in digital pathology analysis and spatial transcriptomics. With Dr. John Ringman (ORE Core), Drs. Hiniker, Hawes, and Bienkowski form an established research team with funding to study the neuropathology of AD. Since Dr. Hiniker's arrival, the NP Core was entirely updated including a move to new space in the Zilkha Neurogenetic Institute, the same building as key Clinical and Biomarker Core resources. The NP Core was renovated with new freezers and monitoring; new storage for slides, blocks, and fixed tissue; and institutional commitment to support full staging of the entire tissue bank to NACC 11 standards. The NP Core has begun creating a digital pathology library of scanned slides, reliant on the combined expertise of Drs. Bienkowski and Hiniker. Also aligned with the Center's overall goals, the team is utilizing their combined expertise in classical and digital neuropathology and spatial transcriptomics to investigate the relationship of *APOE* ε4 to amyloid angiopathy, perivascular microgliosis, and fibrinogen deposition. The NP Core will contribute to the Center through the following aims:

Aim 1: ***Rapid Autopsies from Diverse Populations with a focus on APOE ε4.*** With the Clinical Core, we will increase autopsies from minorities, especially our Latinx cohort, and prioritize collecting brain tissue from cases with the *APOE* ε4 allele, in keeping with Center mission. During the prior cycle, 30% autopsies came from underrepresented minority enrollees, and >32% of the cohort consented for autopsy are minorities.

Aim 2: ***Comprehensive Neuropathological Diagnosis.*** In addition to the standard stains for diagnosis of AD and full staging for ADNC, we utilize comprehensive evaluation for co-pathologies including α-synuclein and pTDP-43 for LATE, and tau for ARTAG at NACC11 standards. There will be independent review of all cases by both Dr. Hiniker and Hawes with re-review to reach consensus diagnosis. New markers will be developed and implemented as needed for diagnostic accuracy and research purposes.

Aim 3: ***Distribute Tissue to Catalyze ADRD Research.*** In addition to fixed hemibrains, eyes, and spinal cord the biorepository will continue to collect blocks, slides, and frozen aliquots and distribute our unique collection of neuroanatomic and systemic regions to researchers to further ADRD diagnosis and therapeutics.

Aim 4: ***Develop a freely accessible de-identified digital library.*** Dr. Hiniker and Dr. Bienkowski, an expert in digital neuropathology and spatial transcriptomics, will lead the creation of an online, open-access digital pathology library of all >1,100 cases in our Center at NACC 11 standards, in collaboration with the DMS Core. This digital library will expand the NP Core's ability to provide high-resolution AD pathology data to the wider AD community as both a research and education resource.

Aim 5: ***Interrogate the effect of the ApoE4 allele on CAA, AD neuropathology and perivascular inflammation.*** Using our digital pathology machine learning approach, we will quantify and map the distribution of fibrinogen leakage, CAA, and microgliosis in the hippocampus and cortex of 80 ADRC patients (40 APOE *ε4+* and 40 APOE *ε4-*). In a smaller cohort of 20 patients, we will perform Multiplexed Error Robust Fluorescent *In Situ* Hybridization (MERFISH) spatial transcriptomics on hippocampal tissue sections (n=10 APOE *ε4+* and n=10 APOE *ε4-)* with a 1000 gene panel to determine APOE differential cell-type specific changes to brain cell types in low and high stage AD (n=5 per group). This study will provide new insights into the effect of APOE genotype on the microglial response to fibrinogen and amyloid deposition in AD and ADRD.

Aim 6: ***Mentor the Next Generation of ADRD Researchers and Neuropathologists.*** The NP Core will increase engagement with the REC Core and expand the formal lectures on neurodegenerative neuropathology available to neuropathology and neurology fellows and resident, and medical and graduate students.

NEUROPATHOLOGY CORE- RESEARCH STRATEGY

A. SIGNIFICANCE

The Neuropathology (NP) Core (**Core D**) provides gold-standard neuropathologic diagnosis for USC ADRC patients; banks nervous system tissue (>1,100 brain autopsies, including fixed and frozen brain tissue, eyes, and spinal cord); and distributes these tissues to support ADRD research nationally and internationally. In the last cycle, we provided tissue to researchers within USC, to outside Centers (Washington University, Boston University, UCSD), and to outside researchers both nationally (UCSF, Harvard, UCLA, Cedars-Sinai, U. Colorado, and others) and internationally (U. Gothenburg). We contributed to national initiatives through GWAS, NIAGADS, NCRAD, ADNI, and NACC. We have a unique focus on neuropathologic manifestations of *APOE* ε4 genotypes, digital resources, and a uniquely diverse cohort, both in the tissue bank and consented for autopsy.

Leadership: new to this renewal, the NP Core is led by Dr. Annie Hiniker, a board-certified neuropathologist who was recruited to USC in January 2024. In Dr. Hiniker's recruitment, the NP Core was entirely updated, with consolidation of the NP Core space to a suite of labs on the 5th floor of Zilkha Neurogenetic Institute (ZNI), which also houses the Biomarker Core's Biorepository, the Yassine lab, and the USC Center for Integrative Connectomics, led by Dr. Michael Bienkowski. The NP Core received new -80°C freezers and Wi-Fi monitoring for the entire tissue bank; new storage for slides, cassettes, and fixed hemibrains; and designated institutional funding to support Dr. Hiniker's efforts to update pathologic staging of all cases to NACC 11 standards and to create a freely available digital resource of the NP Core slide collection in collaboration with Dr. Bienkowski.

Dr. Hiniker is an R01-funded Associate Professor of Pathology, NACC Scientific Review Committee member, and co-Director of the AANP Neurodegenerative Scholars program (R13), a national program to increase the number and diversity of neurodegenerative neuropathologists. She oversees daily operations of the NP Core and provides diagnostic and scientific expertise. Previously, Dr. Hiniker was the sole neuropathologist (2018-2023) and co-Director (2022-2023) of the UCSD NP Core with Dr. Robert Rissman. Importantly, Dr. Rissman also moved to USC to lead the USC Biomarker Core, maintaining Dr. Hiniker and Rissman's productive, collaborative relationship and fostering integration of the USC Cores. The NP core is co-Directed by Dr. Debra Hawes, a board-certified neuropathologist, who independently reviews all Center autopsies for consensus diagnosis with Dr. Hiniker. Dr. Hawes provides essential continuity to the NP Core as prior co-Director (2020-21) and Director (2021-23). The team is completed by Dr. Bienkowski, an expert in digital neuropathology and spatial transcriptomics, who will lead the creation of an online, open-access digital pathology library. With Dr. John Ringman (ORE Core), Drs. Hiniker, Hawes, and Bienkowski form an energetic team with established funding to study the neuropathology of AD. In this cycle, they will utilize their combined expertise in classical and digital neuropathology, transcriptomics, and data analysis to investigate the relationship of *APOE* ε4 to amyloid angiopathy, perivascular microgliosis, and fibrinogen deposition (**Aim 5**).

Biorepository: The NP core performs all aspects of autopsy tissue procurement, diagnosis, and distribution. Our autopsy cohort is exceptionally diverse: 30% of the prior cycle's autopsies came from minority enrollees and >32% of the cohort consented for autopsy are minorities (**Fig. 1**). The USC NP Core is also unique in that it harvests and banks additional CNS and non-CNS tissues not routinely collected by other sites, and houses >150 fixed eyes, >500 frozen and ~1,000 fixed spinal cords, and >500 cases from which frozen tissue from 8 non-nervous system organs is available (**Table 3**). A major initiative of Dr. Hiniker's recruitment (thru institutional funds) is to update the >1,100 cases in the NP Core to NACC 11 staging, digitize the collection, and ensure our resources are more widely available to ADRD researchers, as described in **Aims 2 and 4** below.

Integration of the NP Core within the Center: In this renewal, the NP Core will: i) catalyze ADRD research by providing definitive diagnosis of ADRD at NACC 11 standards on the entire existing cohort (>1,100 patients since 1984) complementing the standardized biospecimens, clinical, and imaging data from the Biomarker, Clinical and Imaging Cores; ii) prioritize collection of detailed neuropathologic data on the diverse populations accessible to our ADRC, particularly Latinx, in collaboration with the Clinical and ORE Cores; iii) work closely with the DMS Core to develop an accessible digital pathology library of our autopsy cohort that will later integrate with genetic and biomarker data from the Biomarker Core; iv) measure the detailed contributions of A*POE* genotype to pathology and cognitive impairment in collaboration with the Clinical Core; and v) provide training in neuropathology and diagnosis of ADRD, supporting the training efforts of the Research and Education Core.

B. INNOVATION

- Serve as a national resource for not only brain but other CNS and systemic tissues from patients with ADRD.
- Prioritize neuropathologic evaluation of diverse populations in collaboration with Clinical and ORE Core.
- Create an integrated and accessible digital pathology database at NACC11 standards with the DMS Core.
- Define effects of A*POE* ε4 on AD neuropathology using digital pathology, machine learning, transcriptomics.

APPROACH

Aim 1: *Rapid Autopsies from Diverse Populations with a focus on APOE ε4.*

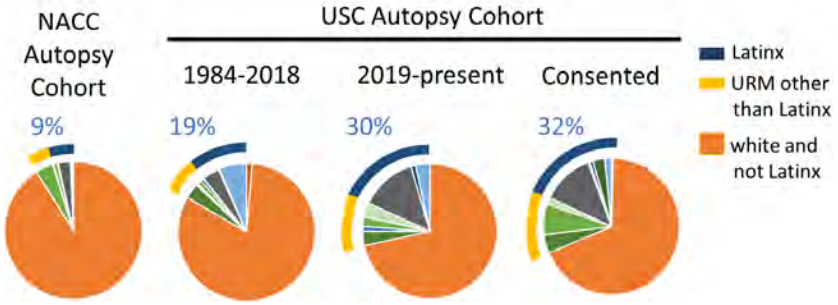


Figure 1. USC Autopsy Cohort captures diverse participants. Cohort demographics from entire NACC autopsy cohort (left, 9% URM, 4% Latinx) vs. USC cohort from 1984-2018 (19% URM, 12% Latinx), last cycle (30% URM, 20% Latinx), and consented living subjects (32% URM, 18% Latinx).

Progress Report on Case Collection: Two notable strengths of Center are (1) the robust and increasing representation of minority populations and (2) the prevalence of *APOE* ε4 genotype in our autopsy cohort. 30% of the prior cycle's autopsies came from minority enrollees and >32% of the cohort consented for autopsy are minorities compared to 9% of NACC autopsies (**Fig. 1**). With our Center's focus on *APOE* ε4, we also prioritize collecting brain tissue from cases with at least one *APOE* ε4 allele: 52% of our banked and genotyped cases have ≥1 *APOE* ε4 allele (compared to 44% of autopsies with *APOE* ε4 genotype in the NACC dataset and 29% of UK Biobank autopsies; **Fig. 2**). Our Center also has longstanding expertise in ophthalmic manifestations of AD, as evidenced by our ongoing collaboration with Dr. Maya Koronyo-Hamaoui (2R01AG055865), which Dr. Hiniker is continuing; this cycle we collected and banked 138 eyes (74% of cases).

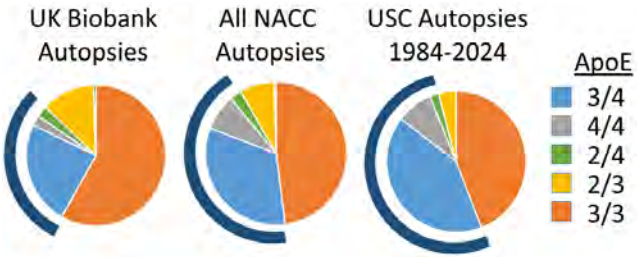


Figure 2. Autopsied cases with ≥1 A*POE* ε4 allele (blue line) in UK Biobank (29%) vs NACC (44%) vs our cohort.

Despite the COVID-19 pandemic, since the counts submitted for the prior renewal (Jan 2019-present) the NP Core performed 90 diagnostic brain autopsies (Table 2); 44 cases were male (48%); 46 female (52%) and with an age range of 39 to 102 years (average 83.5 years). All cases were enrolled in the Brain Research Study (BRS), the designation for the autopsy arm of our Center's studies (enrollment criteria in **Tissue Collection Workflow** section). The first year of the COVID-19 pandemic was challenging for case acquisition and autopsy. Our NP Core implemented safety changes to our workflow including: (1) purchase of a vacuum saw and shroud to keep the autopsy suite free from pathogens and (2) routine COVID-19 testing at autopsy and PCR results in 24-48 hours (we switched to a 15 minute COVID-19 antigen rapid test in 2023). Autopsies were resumed in August 2020 with these precautions; 7 BRS participants who died 3/2020-7/2020 could not be autopsied.

Tissue Collection Workflow: ADRC patients who consent to an autopsy become part of the BRS, our Center's name for our autopsy cohort. Current BRS enrollment is 256 patients; we aim to acquire 18-20 autopsies/year with 20-25% cognitively non-impaired, 20-25% mild cognitive impairment, and the remainder with dementia. To enroll in BRS, patients must meet one of the following criteria: (a) initial diagnosis of MCI or mild AD; (b) > 60 years with subjective cognitive decline; (c) enrolled as cognitively normal control; (d) participant in ADRC-affiliated study; or (e) diagnosis of autosomal dominant AD and have undergone a UDS and been enrolled in the Clinical Core. All patients in the Clinical Core are approached for consent to autopsy, prioritizing Latinx and others underrepresented in research and going forward, those in CLARiTI. The family/care facility receives a single number to alert the team of patient's demise. The diener and transport service (24-hour coverage) transport the body to USC Keck Hospital, where the brain is removed. Autopsy consent forms are stored in a locked file in the NP Core with access limited to Drs. Hiniker, Hawes, and on-call staff. Drs. Hiniker and Hawes are on call to oversee the dissection of the fresh specimens as needed, immediately after removal. Average death-to-autopsy interval of our cohort of <9 hours, largely dependent on communication from families.

At autopsy, the brain is divided sagittally, with the left hemibrain fixed in 10% buffered formalin and the right hemibrain sampled for the 8 most-requested anatomic regions, which are immediately frozen at -80°C. Requests for specific tissue sites are accommodated if feasible. CSF is routinely collected postmortem. Fresh tissues are obtained rapidly at autopsy for ongoing projects, including from CNS (Hussein, RF1AG076124) and eyes (Koronyo-Hamaoui, 2R01AG055865). Following ≥3 weeks fixation, the left hemibrain is serially sectioned in 1 cm slices and assessed by Drs. Hiniker and Hawes in a scheduled brain cutting conference attended by Neuropathology fellows, residents, medical students and other physicians. Tissue blocks from 18 standard regions are processed for histopathological evaluation (see below) and remaining fixed tissue stored in formalin in our fixed brain bank. Upon evaluating the prior tissue requests and noting a trend of increased researcher interest in frozen tissue, including diverse regions and larger quantities, Dr. Hiniker revised the S.O.P. for frozen tissue such that after sampling the most-requested regions, the remaining right hemibrain is frozen in six ~2 cm coronal slabs (2 frontal, 3 temporo-parietal, 1 occipital) along with cerebellum/brainstem. These are frozen overnight on a flat surface to retain morphology and individually stored in 7 labelled bags at -80°C (1-6, anterior to posterior, 7 cerebellum). Tissue distribution to researchers who request this material is detailed in **Aim 3**.

Aim 2: ***Comprehensive Neuropathological Diagnosis.***

Progress Report on Diagnosis: On her arrival in Jan 2024, Dr. Hiniker, assisted by Dr. Qiulan Ma, reviewed all 90 cases from the previous cycle and performed comprehensive evaluation for co-pathologies including pTDP-43 staining for Limbic Age-Related TDP-43 Encephalopathy (LATE), full α-synuclein staining, evaluation for Aging-Related Tau Astrogliopathy, and full review for ADNC stage (**Table 1**). Diagnostic improvements implemented since Dr. Hiniker's arrival include: addition of routine pTDP-43 LATE staging on all cases; routine sampling of olfactory bulb (in cassette with anterior cingulate) for α-synuclein staining/H&E, full α-synuclein staging on all cases regardless of clinical history; and dual, independent review of all cases by both Dr. Hiniker and Hawes with re-review to reach consensus diagnosis as needed. Neuroanatomic regions added during this cycle include motor cortex, amygdala, cingulate, and thalamus.

Primary Neuropathology Diagnosis	N	Additional NP diagnosis (N)		
		TDP	CAA	Vasc
ADNC High	30	15	27	14
ADNC Intermediate	17	13	12	9
ADNC Low	10	5	2	5
Normal Control	1	1	0	1
Lewy Body Disease (LBD)	3	1	0	2
ADNC (High/Intermed) + LBD	15	9	14	9
FTLD (Tau, TDP-43, U)	4	1	0	2
Primary Age Related Tauopathy	3	3	0	3
Misc. (Fahr's, ALS, AGD, LATE)	4	4	2	2
Pending to Diagnose	3	0	0	0
Total	90	49	57	47

Table 1. Autopsy cases analyzed by NP Core January 2019-present. TDP = TDP-43 LATE pathology; CAA = cerebral amyloid angiopathy; Vasc = vascular pathology.

Drs. Hiniker and Hawes are expert in developing and implementing new immunostains. Dr. Hawes is Director of Immunohistochemistry at the Center for Pathology Research Services (CPRS) at Children's Hospital of Los Angeles (CHLA), where immunostaining of our cases is performed. Diagnostic stains available at CHLA CPRS and utilized as needed include AT8, RD3, RD4 tau species, 4G8 Aβ, α-synuclein, Iba-1 for microglia, GFAP for gliosis, and neuronal markers including NeuN and NSE. At UCSD, Dr. Hiniker worked closely with Dr. Don Pizzo at the UCSD Tissue Technology Center (TTC) to implement new AD antibodies including AD-specific tau GT38 (as well as RD3, RD4) and Aβ antibodies including Aβ1-40, Aβ1-42, and Aβ1-43 which have been used both diagnostically and in her research of co-occurring AD and other tauopathies (e.g. *Acta Neuropathologica*, 2022).(1) With Dr. Pizzo, she developed FUS and TAF-15 antibodies allowing diagnosis and reporting of a novel FTLD-FUS case (2) and diagnosis of a second (clinically unidentified) FTLD-FUS case. Dr. Hiniker is part of the International aFTLD-U Consortium. Dr. Hiniker also developed TDP-43 and phospho-TDP-43 with Dr. Pizzo at TTC; the TTC is performing pTDP-43 staining on USC ADRC cases while the stain is validated at USC.

Tissue Processing, Neuropathological Assessment, and Reporting to NACC. Drs. Hiniker and Hawes jointly examine gross specimens and separately evaluate the microscopic pathology for each case before discussing at consensus conference attended by trainees. Each brain receives a primary diagnosis and additional contributing diagnoses according to NACC 11 NP guidelines. 18 brain regions are sampled, including 9 cortical and subcortical regions (midfrontal, superior temporal, inferior parietal, primary visual/occipital, motor, anterior cingulate, and 3 regions of subcortical white matter); 3 brainstem (medulla, pons with locus coeruleus, midbrain with substantia nigra), as well as cerebellar cortex with dentate, basal ganglia with nucleus basalis of Meynert, hippocampus/entorhinal cortex, amygdala, thalamus, and olfactory bulb. Tissue is embedded in paraffin and sectioned at 6 μm for our standard H&E and immunohistochemistry (IHC) workup using antibodies to phosphorylated tau (AT8), Aβ (4G8), α-synuclein, and pTDP-43. We immunostain for tau isoforms (RD3, RD4, GT-38), FUS pathology (FUS, TAF-15), ubiquitin, and other diagnostic markers as needed. For any stains getting set up at USC, Dr. Hiniker has arranged send-out to UCSD's TTC, and this is already in use.

We follow NACC 11 guidelines, using NIA-AA "ABC" reporting for ADNC (3) and categorizing Lewy pathology according to NIA-AA recommendations (none, brainstem predominant, limbic, neocortical, or amygdala predominant) (4, 5). We know and utilize the diagnostic requirements and consensus statements for PART (6) and LATE (7, 8). All FTLD cases undergo tau, TDP-43, and if necessary, ubiquitin, FUS, and TAF-15 staining. Cerebral vascular disease (gross and microscopic) and amyloid angiopathy are enumerated and graded.(9) NACC 11 introduced assessment of ARTAG using tau IHC on amygdala and middle frontal sections which we have incorporated into our standard workup. After neuropathological consensus, Drs. Hawes and Hiniker's report which is sent to the Clinical Core and is synthesized with the clinical history to create a detailed report sent to the next of kin with a letter of appreciation. Neuropathologic data are sent to NACC and entered into the NP Core and Center databases (data management described in DMS Core).

Aim 3: ***Distribute Tissue to Catalyze ADRD Research.***

Progress Report on Biorepository and Tissue Distribution: During the last cycle, the NP core distributed nearly 4,000 tissue allocations (**Table 2**), supported 71 projects to generate 34 publications and contributed to neuropathologic Working Groups and consortiums, including aFTLD-U (international, Hiniker), Autosomal Dominant AD ("ADAD") (international, Hiniker), and Digital Pathology (national, Hiniker and Bienkowski). Tissue was provided to researchers within the Center and to outside Centers (Washington University, Boston University, UCSD), as well as to researchers at USC, nationally (UCSF, Harvard, UCLA, Cedars-Sinai, University of Colorado, and others) and internationally (University of Gothenburg). Our resources contribute to national

initiatives (including GWAS, NIAGADS, NCRAD, DVR, ADNI, and NACC). We continued an ongoing collaboration with retinal expert Dr. Koronyo-Hamaoui (2R01AG055865) and began to collaborate with Dr. Diego Sepulvada-Falla on ADAD. With Dr. Hiniker's recruitment, the NP Core also began collaborations with Drs. Allison Kraus (Case Western), Sheng Zhong (UCSD), and John Ravits (UCSD).

Tissue Resource	Distributed Last cycle
CNS, slides	2,843
CNS, frozen	550
CNS, blocks	148
CNS, fresh	52
Eyes, fixed	104
Eyes, fresh	18
Spinal cord, slides	120

Table 2: Tissue distributed from 2019-present.

Since January 2024, near-total upgrades were made to the NP Core, much of it through institutional support, including: consolidation of the NP Core (with frozen and fixed biorepositories) to newly renovated lab space on the 5th floor of Zilkha Neurogenetic Institute (ZNI); organization and consolidation of all NP Core frozen tissues into six new PHCbi freezers with Wi-Fi alarm systems in an NP Core freezer farm; organization of all existing slides and blocks from our >1,100 cases (collected since 1984) into two new Tissue-Tek Lab Aid Ultra II storage cabinets; purchase of Spacesaver rolling shelves to organize the ~700 existing fixed hemibrains and ~150 fixed eyes; and a designated NP Core brain dissection laboratory and NP staff office. In addition to the fixed hemibrains and eyes, the current Biorepository houses nearly 15,000 blocks, >50,000 slides, and frozen aliquots of >30,000 unique neuroanatomic and >5,000 unique systemic regions (**Table 3**). A unique attribute our biorepository is that it also contains a collection of brain and spinal cord from 11 *PSEN1* ADAD cases, including 6 cases of the neuropathologically uncharacterized *PSEN1* A431E mutation, which originated as a founder effect from Mexico and causes a severe young onset AD with spastic paraparesis. Pathological and clinical characterization of these cases, as well as those of two others from a Mexican-American family with PSP, are being performed in conjunction with Dr. Ringman (ORE Core).

Institutional support is designated for ongoing immunostaining of all cases to NACC 11 standards and creation of a digital pathology library (**Aim 4**). Upon her arrival, Drs. Hiniker and Yassine implemented an online tissue request form in Redcap to allow broad and equitable access to tissue. A Tissue Request Committee was formed and reviews all tissue requests for scientific merit and feasibility, and includes Drs. Hiniker, Hawes (co-Core lead), Rissman (Biomarker), and Ringman (ORE). A new scientific technician, Josh Tseng, MS, who trained in Dr, Hiniker's lab, aids in tissue requests and organization, and Dr. Qiulan Ma, an MD, PhD neurologist, and Dr. Hawes assist Dr. Hiniker in reviewing older cases to allow complete NACC11 staging on all cases.

Biorepository organization for the upcoming cycle. Six new PHCbi -86°C Freezers on emergency power and with IntelliSense Wi-Fi monitoring system house the entire NP Core frozen tissue collection in a rigorously organized system with an empty backup freezer for emergencies. Four freezers are in ZNI 508, in a dedicated freezer farm with double air conditioning system maintaining air temperature $\leq$ 70°C. A second freezer farm in the basement (ZNI B102) houses two PHCbi freezers containing the oldest cases as well as two 2019 Thermo Fisher ULT freezers with samples from ADRC-affiliated research (including R01AG062687, PI: Schneider, Clinical Core; R01AG051346, MPI: Schneider). The frozen repository houses CNS tissue (~700 cases), spinal cord (>500 cases), and lung, liver, lymph nodes, spleen, kidney, pancreas, adrenal, deltoid, intercostal and psoas muscle (>500 cases). Formalin-fixed tissue, including ~700 fixed hemibrains and 154 eyes, is being transferred to a designated fixed brain bank (ZNI 510) where we have ongoing installation of high-density shelving, allowing storage of ~1,200 cases. All slides and paraffin blocks are organized in ascending order by case number and housed in two new locked Tissue-Tek Lab Aid Ultra II storage cabinets, with space for additional slides and blocks as the NP team completes immunohistochemistry of the >1,100 ADRC brain autopsies to restage older cases at NACC 11 standards.

Tissue Resource	Unique Regions	Unique Cases
CNS blocks	13,248	1104
CNS frozen	28,697	678
CNS fixed hemibrain	~700	~700
Spinal cord blocks	960	960
Spinal cord fixed	519	519
Spinal cord frozen	1,557	519
Systemic tissue frozen	5,190	519
Eyes fixed	154	154
CSF at autopsy	1104	1104

Table 3: Tissue banked and available for distribution from our biorepository.

The NP Core will utilize its online tissue request form to receive requests reviewed by the Tissue Request Committee. We have frozen tissue for genomics, transcriptomics (evaluation this month of 5 cases demonstrated average RIN of 9.2), CHIP-seq, and seed amplification assays; fixed tissue for immunostaining and transcriptomics; and provide consultation regarding neuroanatomy and experimental design. Confidentiality is maintained via distinct neuropathology core accession ID and Center patient IDs. NACC Neuropathology Forms as well as USC-specific neuropathology forms are maintained on DMS Core databases. DMS Core data storage, confidentiality measures, server size, and functionalities are described in detail in the DMS Core section.

Aim 4: ***Develop a freely accessible deidentified digital pathology library***.

This is a new aim building on the synergistic strengths of the NP and DMS Cores. Our Center foresees that a shared digital pathology library of our collection could be invaluable to researchers and synergize with digitization efforts at other ADRC NP cores. Drs. Bienkowski and Hiniker's research programs each involve a significant component of digital pathology (1, 10, 11) and both are members of the NACC Digital Pathology Working Group. This aim also leverages the strength of our computational resources in the DMS Core, which will also host digital pathology images from ADNI4 (Toga).

Over its nearly 40-year history, the NP Core has processed and expertly evaluated brain tissue from >1,100 cases. There is new institutional commitment to support complete NP staging at NACC 11 standards of the full USC autopsy brain bank. We have therefore initiated creation of a digital library of our collection to encompass all >1,100 cases; this project will be a major focus of our Center. Drs. Hiniker and Bienkowski have already commenced this work with full neuropathologic staging of the 90 cases from the last cycle at NACC 11 (**Table 1**), which Dr. Hiniker accomplished in the setting of moving the NP Core to ZNI and in <5 months since her start. As part of an ADRC pilot grant project, Dr. Bienkowski has already digitized images from >150 ADRC patients (1500+ images) and developed a machine learning-based classification and quantification approach for high-throughput pathology mapping (**Fig. 3**).(10)

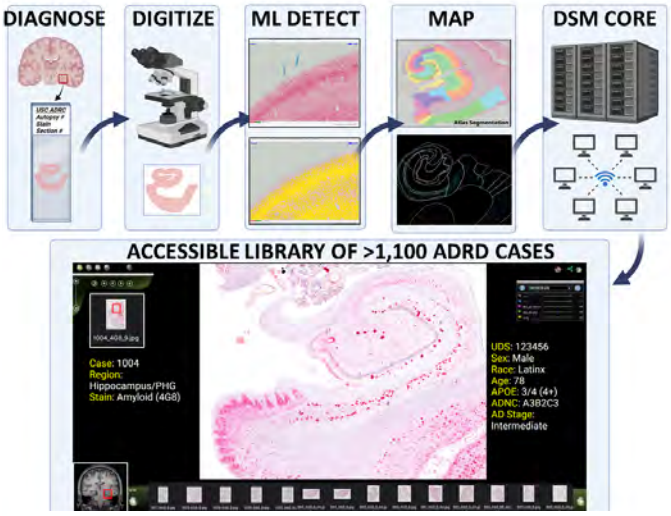


Figure 3: Ongoing workflow to create a freely accessible digital pathology database of all cases.

This cycle, Drs. Hiniker, Bienkowski, and Hawes will digitize the USC ADRC pathology slide archive and create an online repository for visualization and access to image data, with all data hosted by the DMS Core (see DMS Core for details of storage). As a previous member of the USC Mouse Connectome Project (MCP) and the Brain Initiative Cell Census Network (BICCN), Dr. Bienkowski has developed extensive experience in high-throughput microscopy, neuroinformatics, and online data visualization for open access science. Whole slide images (WSIs) from each brain area will be collected from all BRS cases. Coronal tissue sections will be imaged in brightfield using a 20X objective with an Olympus VS120 slide scanning microscope (preliminary imaging indicates ~20 min per slide and ~30 slides per patient case = ~10 hrs/case). We estimate we will have >500 cases stained and scanned at NACC 11 standards by 2027. Image tiles of the whole tissue section are automatically stitched, and the resultant VSI image files are stored on ADRC DMS Core servers (2-4GB per image). After imaging, all associated ADRC patient metadata is logged such that all ADRC-documented patient demographics and manually scored pathology from the ADRC NP report for a given patient is associated with the appropriate images. The resultant image database will serve as a valuable platform for advanced deep learning and artificial intelligence based digital pathology. Together with images from ongoing DMS Core projects including ADNI4, this website will be a centralized nexus allowing scientists in the AD community to visualize and analyze neuropathology data from diverse patient populations.

Aim 5. ***Interrogate the effect of the APOE ε4 allele on AD neuropathology and perivascular inflammation.***

This is a new aim that utilizes our expertise in digital pathology to investigate the Center's overall interests in *APOE ε4*. In a preliminary cohort, we noted that diffuse pericapillary fibrinogen appears to be present more strongly in *APOE ε4+* brain tissue compared to *APOE ε4-* brain tissue and further that the capillary-associated microglia (CAM) response around these vessels appears divergent, with *APOE ε4-* pericapillary microglia much more prominent (**Fig. 4**). Within the neurovascular unit, CAMs are positioned between astrocytic endfeet to regulate blood brain barrier (BBB) gaps and leakage of fibrinogen into brain parenchyma due to vascular damage signals microglia recruitment and phagocytic activities associated with cognitive impairment.(12-14) Fibrinogen also interacts with Aβ in the vasculature (CAA) as well as in parenchymal plaque deposits.(15-17) We hypothesize that the lack of CAMs around vasculature in *APOE ε4+* carriers may reflect a dysfunctional immune response to AD pathology and BBB leakage and explain regional vulnerability of the hippocampus to BBB breakdown. A dysfunctional microglial response to fibrinogen in *APOE ε4+* carriers may explain why this population is at greater risk of ARIA with treatment of Aβ monoclonal antibodies; co-treatment with fibrinogen-targeted antibodies is currently being developed to limit this risk.(18)

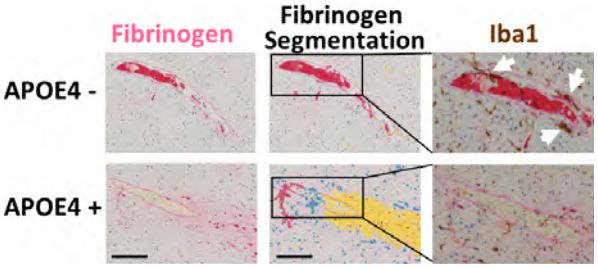


Figure 4: APOE *ε4* may drive fibrinogen leakage and abnormal microglial function. For fibrinogen segmentation panel (center), yellow indicates diffuse (leaking) fibrinogen while red indicates fibrinogen in intact vasculature. White arrows in right panel highlight CAM in APOE *ε4-* but not APOE *ε4+* cortex.

Using our digital pathology machine learning approach, we will quantify and map the distribution of fibrinogen leakage, CAA (4G8), and IBA-1+ microglia density in the hippocampus and cortex of 80 ADRC patients (40 *APOE ε4+* and 40 *APOE ε4-*; cohort partially examined in Stephen et al 2023).(10) Fibrinogen leakage will be quantified by pixel density within each brain region and comparisons of IBA1 and amyloid between fibrinogen+ zones and fibrinogen- zones will be made to determine differential differences in amyloid and microglia distribution. This study will provide new insights into the effect

of *APOE* genotype on the microglial response to fibrinogen and amyloid deposition and identify potential hippocampal/temporal lobe subregion susceptibility to vascular damage in ADRD.

Following up on the digital pathology analysis, we will perform Multiplexed Error Robust Fluorescent *In Situ* Hybridization (MERFISH) spatial transcriptomics (a technique in place in the Bienkowski lab, see **Fig. 5**) using a 1,000-gene panel on a cohort of 20 hippocampal tissue sections (n=10 *APOE ε4+* and n=10 *APOE ε4-)* to investigate *APOE* differential cell-type specific changes in low and high stage AD (n=5 per group). Genes in the 1,000-gene panel will be selected based on Dr. Bienkowski's expertise in hippocampal neuronal cell type markers utilized previously to create the Hippocampus Gene Expression Atlas. We will also focus on endothelial, vascular, and microglial cell types associated with fibrinogen leakage. Ultimately, these data will also be made available to the AD research community through the ADRC digital pathology website. Our efforts into assessing the effects of *APOE* on CAA and vascular inflammation will be facilitated through consultation with long-time ADRC collaborator Harry Vinters, a world-renowned expert in CAA at UCLA (see letter of support).

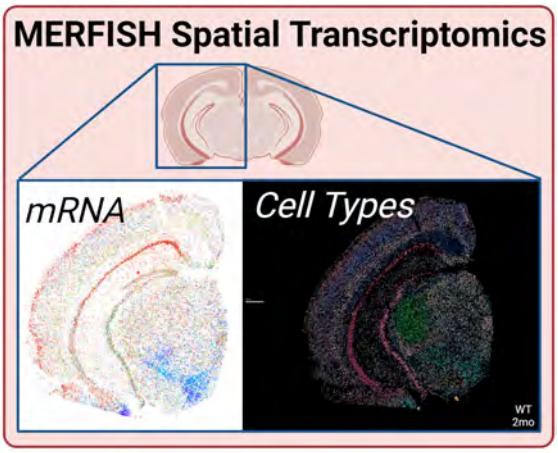


Figure 5: MERFISH spatial transcriptomics on mouse coronal section (performed in the Bienkowski lab) identifies thousands of individual mRNA transcripts to map cell types.

Aim 6. *Mentor the Next Generation of ADRD Researchers and Neuropathologists*.

Progress Report on Mentoring: Drs. Hiniker, Bienkowski and Hawes recognize that it is essential to develop the next generation of researchers and neuropathologists with an emphasis on diversity, equity, and inclusion—and all genuinely enjoy mentoring and teaching. Nationally, Dr. Hiniker is Co-Director of the American Association of Neuropathologists (AANP) Neurodegenerative Scholars (R13) program (renewed in 2023 for 5 years, 4th percentile), whose goal is to increase number and diversity of neurodegenerative neuropathologists. Dr. Hiniker closely supervised the UCSD NP clinical fellows such that each of the 4 fellows who graduated during her time at UCSD co-authored at least one, often multiple, publications or posters with her. At UCSD, Dr. Hiniker also mentored and published extensively with MD/PhD student Denis Smirnov, who chose neuropathology as his career and entered AP/NP residency/fellowship at Harvard MGH/Brigham in July 2023.

The USC Keck ACGME-accredited clinical programs in Neuropathology (in which Drs. Hiniker and Hawes are teaching faculty) and Neurology include training in gross/microscopic findings of neurodegenerative diseases and criteria for diagnosis. These sessions expose trainees to current advances in the field, both diagnostically and in translational research. Neuropathology conferences provide opportunities to review the clinical-pathologic correlations of AD and other dementias. During the last cycle, Dr. Samuel Guzman, a USC NP fellow, received a Diversity Supplemental Grant at USC to begin his research endeavors in ADAD and is now faculty at U. Colorado. Dr. Guzman continues to collaborate with Drs. Hiniker, Hawes, and Ringman. Dr. Bienkowski's PhD student, Maricarmen Pachicano, was awarded a NIA Diversity Supplement on the ADRC parent grant in 2021 and received an IF:24 on a pending NIA R36 Aging Research Dissertation Award to Promote Diversity.

Training and Mentoring Plan. We strive to make the NP Core a hub for neurodegeneration teaching and to provide outstanding research opportunities to young scientists and physicians. As such, we have both formal and informal didactics. NP Core formal didactics include brain cutting and slide sign-out sessions in which the NP fellows, pathology residents, and USC/visiting medical students participate. The first-year Clinical NP fellow, Sarah Edminster, was selected as a 2023 AANP Neurodegenerative Scholar and is part of the current cohort for this two-year program. Dr. Bienkowski is a teaching faculty of the Master's in Neuroimaging and Informatics program and Neuroscience PhD program and lectures frequently on neuroanatomy. He also mentors undergraduates through the Neurocognitive Aging and Analytics Research Education program which provides diverse undergraduates a comprehensive didactic and hands-on research experience, in order to support skilled new scientists entering the field of neurocognitive aging and ADRD research. Less formally, Dr. Edminster is beginning collaborative ADRD research projects utilizing Core tissue and Dr. Hiniker anticipates continuing her trajectory of strong mentoring of NP Clinical fellows here at USC. Dr. Hiniker is currently training Dr. Qiulan Ma in neurodegenerative neuropathology so Dr. Ma can assist in staging our cohort as in **Aims 2 and 4**. Over this cycle, Drs. Hiniker and Bienkowski will also increase involvement of the Core in formal lectures such as Clinicopathologic Correlation (CPC) Conferences, will increase engagement of the NP Core with the REC Core, and will expand the formal lectures on neurodegenerative neuropathology available to neuropathology and neurology fellows and residents, and medical and graduate students. Finally, we anticipate that the online, open-access digital pathology library in **Aim 4** will serve as a broad-reaching educational resource to allow users world-wide to explore AD pathology and learn neuroanatomy, especially in disadvantage areas with limited access to medical education.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	○No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	○No
Is the Project Exempt from Federal regulations?	○Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447177.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Outreach, Recruitment, and Engagement (ORE) Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: University Park Campus		Street2: MRF 321
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90089-0411
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	ORE_Abstract1044447030.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	ORE_References_0613241044447096.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Outreach, Recruitment, and Engagement Core

The overall objective of the Outreach, Recruitment and Engagement Core (ORE Core) is to serve as the primary bilingual and bicultural resource and liaison between the USC ADRC, our study participants, their caregivers/partners, and the larger community, i.e., the general public, people living with dementia and caregivers, professional care providers, and the lay community in Los Angeles County. The ORE Core facilitates bidirectional communication through our signature outreach and engagement activities, *USC GeneScreen* registry, and recruitment from the LA County Department of Health Services (DHS). We will: ***Aim 1:*** **Engage in Participant Recruitment & Develop a Research Registry:** The ORE Core will maintain and further develop *USC GeneScreen*, a registry of persons in the LA County area interested in participating in studies and enriched for *APOE* $\varepsilon 4$ carriers. We will augment Latinx participation (English, Spanish, low socioeconomic status) in the registry given the higher prevalence and disease burden in this population and our catchment area composition. During the next 5 -years, our goal is to genotype and register a diverse pool of 1500 participants who are "study-ready" to enroll in AD/ADRD research. ***Aim 2:*** **Enhance Recruitment from the LA County Department of Health Services (DHS) at the Rancho Los Amigos National Rehabilitation Center (RLANRC).** Our ADRC is affiliated with the California Alzheimer's Disease Center (CADC) based at RLANRC, an integral part of the DHS safety-net community health system that cares for a diverse, underserved, and largely Latinx population. As part of our *Plan to Enhance Diverse Perspectives* (PEDP-DHS), we will identify and invite RLANRC patients with MCI or mild dementia to participate in ADRC research studies. ***Aim 3:*** **Continue to Engage in Bidirectional Education, Information and Resource Sharing:** In collaboration with our extensive network of community partners, we will continue to serve as a community resource in targeted engagement and educational outreach to the public-at-large, healthcare and human service providers, potential study participants, and current participants. We will communicate the latest ADRC- and AD/ADRD research findings and timely information on brain health, aging and wellness.

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RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Maria Middle Name P Last Name*: Aranda Suffix:

Position/Title*:	Professor of Social Work
Organization Name*:	University of Southern California
Department:	Dworak-Peck Business Office
Division:	USC Suzanne Dworak-Peck School of Social Work
Street1*:	669 West 34Th St.
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (213) 740-0286 Fax Number: (213) 740-0789 E-Mail*: aranda@usc.edu

Credential, e.g., agency login: ARANDA

Project Role*: Other (Specify) Other Project Role Category: ORE Core Lead

Degree Type: PhD, MPA, MSW Degree Year:

	File Name
Attach Biographical Sketch*:	Aranda_Biosketch_5_17_241044446890.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: John Middle Name M. Last Name*: Ringman Suffix:

Position/Title*:	Professor of Clinical Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-6845 Fax Number: (323) 442-5595 E-Mail*: john.ringman@med.usc.edu

Credential, e.g., agency login: RINGMAN2

Project Role*: Other (Specify) Other Project Role Category: Co-Core Lead

Degree Type: Degree Year:

	File Name
Attach Biographical Sketch*:	
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Amytis Middle Name Last Name*: Towfighi Suffix:

Position/Title*:	Professor of Neurology (Clinical Scholar)
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	OCH Rncho Los Amigos
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (323) 442-5710 Fax Number: (323) 442-7691 E-Mail*: towfighi@usc.edu

Credential, e.g., agency login: atowfighi

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: MD Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Towfighi_ADRC1044446909.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Aranda, Maria P.

eRA COMMONS USER NAME (credential, e.g., agency login): ARANDA

POSITION TITLE: Professor of Social Work

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
California State University, Los Angeles, CA	B.S.W.	06/1979	Social Work
University of Southern California, Los Angeles, CA	M.S.W.	05/1985	Social Work
University of Southern California, Los Angeles, CA	M.P.A.	05/1991	Public Administration
University of Southern California, Los Angeles, CA	Ph.D.	05/1995	Social Work

A. Personal Statement

Enhancing the quality of life of persons living with declines in cognitive functioning and their family care partners continues to be the focus of my academic and professional career. Understanding the mechanisms of cognitive decline in health disparities populations is a key public health and scientific focus of our R21 application: Ecology of Support and Dementia Caregiver Health. **I am honored to serve as Co-Lead of the USC ADRC Outreach, Recruitment and Engagement Core alongside Dr. John Ringman.** As a BSS Researcher, I hold USC faculty appointments in Social Work, Gerontology and Neurology, and am an expert in the areas of Alzheimer's disease, psychosocial interventions for individuals and family care partners, recruitment and retention of underrepresented groups in observational and clinical trials, and health equity-driven research. I have extensive experience in randomized trials, ethnographic methods, large-scale epidemiological research, stakeholder engagement, and innovative recruitment and retention protocols. My research and teaching expertise address life course issues with regards to cognition, depression, and family caregiving with a specific focus on Latinx populations in the U.S. I have served as PI/Co-PI on several key studies/projects funded by the National Institute on Aging, National Institute of Mental Health, National Cancer Institute, Patient-Centered Outcomes Research Institute (PCORI), the Southern California – CTSI, The John A. Hartford Foundation/ The Gerontological Society of America, National Institute of Rehabilitation and Research, the Alzheimer's Association/ health Resources and Services Administration, Los Angeles County Department of Mental Health, the Partners in Care Foundation, and the California Community Foundation. I served on the Steering Committee of the NIA 2020 National Research Summit on Care, Services, and Supports for Persons with Dementia and their Caregivers. Currently, I am a member of the NIH Director's NExTRAC Working Group on Engaging the Public as Partners in Clinical Research. Given the shortage of culturally-responsive social-behavioral scientists dedicated to cognitive aging and family relations, I am an active mentor to early career investigators from underrepresented groups, and have received awards for my ongoing commitment to scientific mentorship.

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b. Aranda, M.P. Marquez, D.X., Gallagher-Thompson, D., Pérez, A., Rojas, J.C., Hill, C.V., Reyes, Y., Dilworth-Anderson, P., & Portacolone, E. (2023). A call to address structural barriers to the representation of Hispanics/Latinos in clinical trials on AD/ADRD: A micro-meso-macro perspective.

Alzheimer's & Dementia: Translational Research & Clinical Interventions, 9(2), e12389. doi.org/10.1002/trc2.12389

c. Aranda, M.P., Quiñones, A.R., Hinton, L., Peak, K.D., Jackson, J.D., Dilworth-Andersen, P., Gitlin, L.N., Jutkowitz, E., & Baier, R. (2023). Preparing for pragmatic trials in dementia care: Health equity considerations. Journal of the American Geriatrics Society. doi.org/10.1111/jgs.18568

d. Castellon-Lopez, Y., Carson, S.L., Ward, K.T., Ramirez, K.D., Phan Vo, L., Kuo, T., Seeman, T., Vassar, S.D., *Trejo, L., Eidem, E., Aranda, M.P., & Brown, A.F. (accepted). Understanding the implementation and sustainability needs of evidence-based programs for racial and ethnic minoritzed older adults in under-resourced communities with limited aging services. BMC Health Services Research, 24(1), 466. https://doi.org/10.1186/s12913-024-10925-0

Ongoing and recently completed projects that I would like to highlight include:

24-10004
Aranda (PI)
07/01/2024 - 06/30/2028
State of California Department of Public Health, Chronic Disease Control Branch, Alzheimer's Disease Program
Leveraging New and Existing Data to Evaluate the Efficacy and Effectiveness of Caregiver Interventions

2 P30AG043073
MPI: Zissimopoulos, Ailshire, Aguila, Role: Co-Investigator, Research and Education Core (REC)
07/01/2023 - 06/30/2028
USC Alzheimer's Disease and Alzheimer's Disease Related Dementias Resource Centers for Minority Aging Research (USC AD/ADRD RCMAR)

5 P30AG066530-04
Aranda (PI), Lead, Outreach, Recruitment and Education Core (ORE)
07/01/2020 - 03/31/2025
USC Alzheimer's Disease Research Center (USC ADRC; MPI: Chui, Toga, Zlokovic)

1 U54AG063546
Mor, Mitchell (MPI), Role: Co-Investigator, Health Equity Team
07/01/2019 – 06/30/2024
NIA AD/ADD Health Care Systems Research Collaboratory

1R24AG071456-01
Portacolone (PI), Role: Co-Investigator
04/01/2021 – 03/31/2024
National Institute on Aging

22-10079
Aranda (PI)
11/01/2022 - 06/30/2023
State of California Department of Public Health, Chronic Disease Control Branch, Alzheimer's Disease Program
Leveraging New and Existing Data to Evaluate the Efficacy and Effectiveness of Caregiver Interventions

B. Positions, Scientific Appointments, and Honors
Positions and Scientific Appointments

2023 – present	Margaret W. Driscoll/Louise M. Clevenger Professorship in Social Policy and Administration
2022 – present	Professor, USC Suzanne Dworak-Peck School of Social Work, CA
2018 – present	Executive Director, USC Edward R. Roybal Institute on Aging
2022	Issue Expert, AARP Global Brain Health Initiative

2021 – 2025	Member, External Advisory Board, Diverse VCID, NINDS
2012 – 2021	Committee Member, National Academies of Science, Engineering and Medicine
2021 – 2023	Member, External Advisory Board, Center for Aging in Diverse Communities, UCSF
2020 – 2023	Member, External Advisory Board, Michigan Center for Contextual Factors in AD
2020 – 2023	Member, External Advisory Board, San Diego ADRC for Minority Aging Research.
2002 – 2021	Associate Professor, USC School of Social Work, Los Angeles, CA
1995 – 2002	Assistant Professor, School of Social Work, USC, Los Angeles, CA
1993 – 1995	Lecturer, School of Social Welfare, University of California, Los Angeles, CA
2012 – 2021	Fellow, Gerontological Society of America, Social Research, Policy and Practice
2019 – 2020	Member, California Governor's Alzheimer's Prevention and Preparedness Task Force
2008 – 2016	Editor-in-Chief, *Home Health Care Services Quarterly*
2017	Reviewer, Implementation Science for the Prevention and Treatment of Mental and/or Substance Use Disorders in Low- and Middle-Income Countries (U01), NIMH/NIDA
2017	Reviewer, Aging with Mental Illness: How older Californians are faring in the Public Mental Health Delivery System. UCLA Center for Health Policy Research
2015	Reviewer, Global Mental Health - Scaling Up Mental Health Interventions in Low- and Middle-Income Countries (U19), NIMH
2015	Reviewer, Special Emphasis Panel, Scientific Review Group, NIMHD
2015	Reviewer, Health Disparities and Equity Promotion Study Section, NIH
2013	Reviewer, Standing Committee on Mental Health Services, NIMH 2012 Reviewer, Grand Challenges, NIMH

Honors

2022	James Jackson Outstanding Mentorship Award, GSA
2018	Pioneer Award, Latino Dementia Network, Alzheimer's Los Angeles, CA
2014	Civic Engagement Team Award, University of Southern California, CA
2013	Excellence in Teaching, Rainbow Alliance Student Caucus, USC School of Social Work
2012	Older Adult Advocacy Award, California Elder Mental Health and Aging Coalition
2011	Mentoring Award, CSWE, Council on the Role of and Status of Women, Arlington, VA
2009	Mellon Mentoring Award for Graduate Student Mentoring, USC, CA

C. Contributions to Science

1. My research addresses two overlapping foci: 1) identifying the determinants of health and psychiatric conditions among middle-age and older adult populations with focus on racial and ethnic minority populations; and 2) targeting and dissemination of interventions for historically underserved racial and ethnic adults with co- occurring medical and psychiatric conditions.

One focus of study is how we address the cultural, linguistic, and health equity issues that are increasingly salient in overall medical, cognitive and psychological wellbeing. Thus, we discussed the role of heath equity in dementia research, the role of neighborhood contexts in caregiving, caregiving in Latino families and other racial/ethnic minorities, and the impact of dementia on diverse populations. Of significance is the finding that living in densely populated areas with more Latinos/as was a protective factor for physical frailty using the H-EPESE sample of older US Mexicans.

a. Aranda, M.P., Kremer, I.N., Hinton, L., Zissimopoulos, J., Whitmer, R., Huling Hummel, C., Trejo, L., & Fabius, C. (2021). Impact of dementia: Health disparities, population trends, care interventions, and economic costs. Journal of the American Geriatrics Society, doi: doi.org/10.1111/jgs.17345
b. Quiñones, A., Mitchell, S., Jackson, J., Aranda, M.P. et al. (2020). Achieving health equity in embedded pragmatic trials for people living with dementia and their family caregivers. Journal of the American Geriatrics Society, 68, S8-S13.
c. Dilworth-Anderson, P., Moon, H., & Aranda, M.P. (2020). Dementia caregiving research: Expanding and reframing the lens of diversity, inclusivity, and intersectionality. The Gerontologist, 60(3), 797-805.
d. Aranda, M.P., Ray, L.A., Soham, A.S., Ottenbacher, K.J., & Markides, K.S. (2011). The protective effect of neighborhood composition on increasing frailty among older Mexican Americans. Journal of Aging and Health, 23(7), 1189–1217. doi: doi.org/10.1177/0898264311421961

2. Our work identified the implications of racial and ethnic factors in the health and psychological wellbeing of older minorities with a specific focus on communities of color. We documented sociocultural factors related to research and practice strategies to promote acceptability and accessibility in both depression and dementia care.
    a. Larson, E.B., & Stroud, C. (Eds.), and the Committee on Care Interventions for Individuals with Dementia and Their Caregivers: Albert, M. Aranda, M.P., Callahan, C.M., et al. (2021) Meeting the Challenge of Caring for Persons Living with Dementia and Their Care Partners and Caregivers: A Way Forward. Washington, DC: The National Academies Press. doi: doi.org/10.17226/26026.
    b. Camacho, D., Estrada, E., Lagomasino, I.T., Aranda, M.P. & Green, J. (2018). Descriptions of depression and depression treatment in older Hispanic immigrants in a geriatric collaborative care program. *Aging and Mental Health, 22*(8), 1056-1062.
    c. Hansen, M.C., Fuentes, D., & Aranda, M.P. (2017). Re-engagement into Care: The Role of social support on service use for recurrent episodes of mental health distress among primary care patients. *The Journal of Behavioral Health Services & Research,* 1-15.
    d. Fuentes, D. & Aranda, M.P. (2012). Interventions for depression among racial and ethnic minority older adults: A systematic review of the literature across 20 years. American Journal of Geriatric Psychiatry

3. We completed comparative effectiveness studies using Problem Solving Treatment to mitigate depression among low-income, primarily Spanish-speaking adults with chronic medical conditions (diabetes, CVD, cancer, etc.). A Helping Hand (n=348) showed that our promotora-mediated depression treatments mitigated depression in both treatment groups (PST/EUC) comprised of Latina/o patients with significant depression and diabetes and/or cardiovascular disease. We completed a second trial, Programa Esperanza, using socioculturally-modified PST. Our preliminary results indicate effectiveness for our Spanish-speaking sample (n=267).
    a. Aranda, M.P., Mack, W.J., Vega, W.A., Lee, P-J., & Gamulin, L. (2018). Comparing Two Ways to Help Latinx Patients with Depression Who Are Age 50 or Older: Programa Esperanza. Patient-Centered Outcome Research Institute. Washington, DC. March 2019. pcori.org/sites/default/files/Aranda238-Final-Research-Report.pdf
    b. Ell, K., Aranda, M.P., b. b. Wu, S., *Oh, H., Lee, P.J., & Guterman, J. (2017). Promotora assisted depression and self-care management among predominantly Latinos with concurrent chronic illness: Safety net care system clinical trial results. Contemporary Clinical Trials. doi: doi.org/10.1016/j.cct.2015.11.012
    c. Ell, K. M.P., Wu, S., Oh, H. Lee, P.J., & Guterman, J. (2016). Promotora Assisted depression care among predominately Hispanic patients with concurrent chronic illness: Public care system clinical trial design. Contemporary Clinical Trials, 46, 39-47. doi: doi.org/10.1016/j.cct.2015.11.012

4. We have published extensively on the Stress and Coping framework including stress and coping among Latinos/as and Chinese populations. For example, we addressed the relationship among financial strain, social support financial strain, negative interaction, coping styles, and the outcome of depression; we determined that financial strain was directly associated with depression and mediated the effects of sociocultural determinants (nativity, years of US residence) and social status factors (age, gender) on depression. Our study was the first to show that older Latinos/as who use avoidant styles of coping we more likely depressed while respondents who were “problem-solvers” were less like to endorse depression—key findings which later informed our choice of PST as an evidenced-based intervention for racial and ethnic minorities. We found in our subsequent work using longitudinal data that institutionalization rates for Latinas/os were relatively low, and that death was a more likely outcome than institutionalization. Living with family and high social support decreased the probability of mortality in a nursing home. Increased rates of institutionalization were seen mostly in persons living alone. The take-home message was that although living with family may mediate institutionalization, the personal and financial costs of living with a family member with high functional limitations may burden the family’s coping abilities, and therefore, may augment risk for depression.
    a. Liang, J., Aranda, M.P., & Lloyd, D. (2020). Association between role overload and sleep disturbance among dementia caregivers: The impact of social support and social engagement. Journal of Aging and Health. doi: doi.org/10.1177/0898264320926062

b. Aranda, M.P., & Lincoln, K. (2011). Financial strain, negative interaction, coping styles and mental health among low-income Latinos. Journal of Race and Social Problems, 3, 280-297.
c. Angel, R., Angel, J., Aranda, M.P., Miles, T.P. (2004). “Can the family still cope? Social support and health as determinants of nursing home use in the older Mexican origin population Journal of Aging and Health, 16(3), 338-354.
d. Guo, M., Aranda, M.P., & Silverstein, M. (2009). The impact of the out-migration on the intergenerational support and psychological well-being of older people in China. Aging and Society, 29, 1-20. doi: doi.org/10.1017/S0144686X0900871X

5. In our first RCT with older adults in home health care (Project HOPE-D) we showed that routine screening and identification of depression by existing nursing personnel was feasible in managed care, home health settings (Ell et al. 2005). This study was one of only several projects in the US that addressed home-bound elders using collaborative care models in “real-world settings (enhanced screening, training, guideline concordant pharmacological treatments, PST; Ell et al., 2007); although better depression outcomes were reported by the intervention group, the improvements were similar to those found in the IMPACT primary care study. The qualitative studies that I conducted indicated key barriers to depression care implementation included adjusting care to medical severity, cultural needs and preferences; anticipating consequences of stigma and shame over being labeled “crazy;” and potential treatment contamination due to training all staff on screening and psychoeducation. In our subsequent work we pooled data from Project HOPE-D and two NCI- supported collaborative care RCTs. Our major focus was to compare older and younger adults with physical illness in a pooled sample of 1,081 patients. We found no significant group differences in the reduction of depression symptoms, improving quality of life, or age (60 years cutoff). Both age groups demonstrated comparative decreases in depression symptom reduction and remission indicating that collaborative care in individuals with comorbid illness as effective in reducing depression in older as well younger low-income, minority patients.
    a. Ell, K., Unützer, R., Aranda, M.P., Gibbs, N.E., Lee, P.J., & Xie. B. (2007). Managing depression in home health care: A randomized clinical trial. Home Health Care Services Quarterly, 26(3), 81-104.
    b. Ell, K.R, Unutzer, R., Aranda, M.P., Sanchez, K., & Lee, P.J. (2005). Routine PHQ-9 depression screening in home health care: Depression prevalence, clinical and treatment characteristics. Home Health Care Services Quarterly, 24(4), 1-19.

Complete List of Published Work in MyBibliography:
https://www.ncbi.nlm.nih.gov/myncbi/maria.aranda.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: AMYTIS TOWFIGHI, MD

eRA COMMONS USER NAME (credential, e.g., agency login): atowfighi

POSITION TITLE: Associate Professor of Neurology and Preventive Medicine (Clinical Scholar)

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
Massachusetts Institute of Technology, Cambridge, MA	SB	06/1998	Biology; Minor: Political Science
Johns Hopkins School of Medicine, Baltimore, MD	MD	06/2002	Medicine
Harvard-Massachusetts General Hospital, Boston, MA		06/2003	Internal Medicine Internship
Harvard-Partners Neurology (Massachusetts General and Brigham & Women's Hospitals), Boston, MA		06/2006	Neurology Residency
University of California, Los Angeles, CA		06/2007	Vascular Neurology Fellowship

A. Personal Statement

I have extensive experience conducting research and designing and implementing coordinated systems of care within the Los Angeles County – Department of Health Services (LAC-DHS) public safety-net system, the second largest public healthcare system in the United States. I have the knowledge, experience, and leadership roles necessary to ensure that the USC ADRC builds on its existing strengths in the renewal. I will ensure that the PEDP is fully implemented in the coming cycle and the ADRC increases its emphasis on recruiting and retaining a cohort that represents the diversity of Los Angeles.

My previous research has focused on addressing inequities in cardiovascular risk factor control, particularly in LAC-DHS. In recent years, I have been engaged in 4 randomized controlled trials (RCTs) of secondary stroke prevention interventions LAC-DHS. As site PI of the principal enrolling site of SUSTAIN (Systematic Use of STroke Averting Interventions) trial, I oversaw enrollment, management, and evaluation of 2/3 of the trial participants. SUSTAIN tested a chronic-care model-based intervention for blood pressure control after stroke (American Heart Association [AHA] Spina Outcomes Center). I was the PI of the HEALS (Healthy Eating And Lifestyle after Stroke) trial, which pilot tested the impact of a 6-session occupational therapist-led intervention aimed at changing diet and physical activity after stroke (AHA 11SDG7590160). I was MPI of SUCCEED (Secondary stroke prevention by Uniting Community and Chronic care model teams Early to End Disparities), a multicenter trial testing the effectiveness of a multimodal, multilevel complex care manager/community health worker team intervention on improving risk factor control after stroke (NINDS 1U54NS081764-01). I am MPI of the BP REACH Study (Blood Pressure disparities Reduction, Equity, and Access among safety net patients with Cardiovascular Health risk) (NIMHD 5P50MD017366-02), testing a team-based intervention on improving blood pressure control after stroke and myocardial infarction, and I serve as MPI for the Race / Ethnicity, Hypertension and Prevention of VCID and Stroke after Intracerebral Hemorrhage (REACH ICH) study (NINDS 2R01NS093870-06A1).

I have worked in the LAC-DHS safety-net system since 2007, holding various administrative positions, including Associate Chief Medical Officer (CMO) at two LAC-DHS hospitals (Rancho Los Amigos and LAC+USC Medical Center), and have overseen neurological care in LAC-DHS since 2014, ensuring patients receive evidence-based, coordinated, efficient care. Currently, I serve as Associate CMO of Research at

LAC+USC Medical Center. I am Director of Healthcare Delivery Science at the SC CTSI, serving as a liaison and mentor for researchers developing, testing, and implementing novel care delivery interventions in LAC-DHS. I received the Clinical Systems Development Award from Kaiser Permanente, California Association of Public Hospitals and Health Systems, and California Health Care Safety Net Institute, which recognizes programs that improved care across complex systems. My research on inequities was recognized by the AHA Robert G. Siekert New Investigator Award in Stroke and American Academy of Neurology Michael S. Pessin Stroke Leadership Prize. I am the immediate past Chair of the AHA Stroke Council's Quality and Outcomes Committee, the Vice-Chair of AHA's Secondary Stroke Prevention Guidelines writing committee, Chair of the Writing Committee for AHA Scientific Statement on Strategies to Reduce Inequities in Stroke Preparedness, Care, and Risk Factor Control section editor for Health Equity for the journal *Stroke*, and serve on the Health Equity steering committee for the NINDS (leading the social determinants of health subgroup).

1. **Towfighi A,** Cheng EM, Ayala-Rivera M, et al. Effect of a Coordinated Community and Chronic Care Model Team Intervention vs. Usual Care on Systolic Blood Pressure in Patients with Stroke or Transient Ischemic Attack: the SUCCEED Randomized Clinical Trial. *JAMA Network Open*, 2021. 4(2):e2036227
2. **Towfighi A,** Cheng EM, Hill VA, et al. Results of a Pilot Trial of a Lifestyle Intervention for Stroke Survivors: Healthy Eating and Lifestyle after Stroke. *J Stroke Cerebrovasc Dis.* 2020 Sep 28;29(12):105323.
3. Lin AM, Vickrey BG, Barry F, Lee ML, Ayala-Rivera M, Cheng E, Montoya AV, Mojarro-Huang E, Gomez P, Castro M, Corrales M, Sivers-Teixeira T, Tran JL, Johnson R, Ediss C, Shaby B, Willis P, Sanossian N, Mehta B, Dutta T, Razmara A, Bryg R, Song S, **Towfighi A.** Factors Associated with Participation in the Chronic Disease Self-Management Program: Findings from the SUCCEED Trial. *Stroke*. 2020 Oct;51(10):2910-2917.
4. **Towfighi A,** Orechwa A, Aragón T, et al. Bridging the gap between research, policy, and practice: Lessons learned from academic–public partnerships in the CTSA network. *J Clin Transl Sci.* 2020 Mar 10;4(3):201-208. doi: 10.1017/cts.2020.23.
5. Cheng EM, Cunningham WE, **Towfighi A**, et al. Efficacy of a Chronic Care-Based Intervention on Secondary Stroke Prevention Among Vulnerable Stroke Survivors: A Randomized Controlled Trial. *Circ Cardiovasc Qual Outcomes.* 11(1):e003228, 2018.

B. Positions and Honors

Positions and Employment

2021 –	Director, Southern California Healthcare Delivery Science Center, Keck School of Medicine of USC
2021 –	Professor of Neurology Neurology (Clinical Scholar), University of Southern California
2020 –	Director, Healthcare Delivery Science, SC CTSI
2020 –	Associate Medical Director of Research, LAC+USC Medical Center
2018 –	Associate Medical Director and Chief of Neurological Services, LAC+USC Medical Center
2016 – 2020	Co-Director, Research Development and Director, Implementation Science, Southern California Clinical and Translational Science Institute (SC CTSI)
2016 – 2018	Director of Chronic Care Management, Rancho Los Amigos
2015 – 2021	Associate Professor of Neurology (Clinical Scholar), University of Southern California
2014 –	Director, Neurological Services, Los Angeles County Department of Health Services
2011 – 2016	Director, Neurology System of Care, Rancho Los Amigos
2011 – 2016	Director, Patient-Centered Medical Home for Stroke, Rancho Los Amigos
2011 – 2016	Associate Chief Medical Officer, Rancho Los Amigos
2007 – 2016	Director, Acute Neurology Unit, Rancho Los Amigos
2007 - 2018	Chair, Department of Neurology, Rancho Los Amigos National Rehabilitation Center
2007 – 2015	Assistant Professor of Clinical Neurology, University of Southern California

Other Experience and Professional Membership

National Institutes of Health

2020 –	NINDS Health Equity Steering Committee
2014 –	Primary and Secondary Prevention Working Group, NIH StrokeNet

2011 Co-Chair, Health Services Implementation, Stroke Progress Review Group, National Institute of Neurological Disorders and Stroke (NINDS), 2011

Professional Organizations

2020 – Executive Planning Committee, International Conference on Neurology and Epidemiology
2020 – Stroke and Neuroscience Chair, AHA Scientific Sessions Planning Committee
2018 – 2020 AHA Scientific Sessions Planning Committee
2018 – Vice-Chair, Writing Committee for AHA Guidelines for Secondary Stroke Prevention
2018 – 2020 Chair, Quality and Outcomes Committee, Council on Stroke, AHA
2017 – Member, Award Committee, David G. Sherman Lecture, International Stroke Conference
2017 – Member, Award Committee, William M. Feinberg Award, AHA
2017 – Member, Award Committee, Thomas Willis Award, AHA
2016 – Member, Award Committee, Robert G. Siekert New Investigator Award in Stroke, AHA
2016 – Member, Michael S. Pessin Stroke Leadership Prize Workgroup, AAN
2015 – Advisory Committee, California Department of Public Health - the California Stroke Registry/California Coverdell Program
2015 – Member, Stroke Council's International Stroke Conference Program Committee; Category Chair: Health Services, Quality Improvement, and Patient-Centered Outcomes, AHA
2016 – 2018 Vice-Chair, Quality and Outcomes Committee, Council on Stroke, AHA
2014 – 2016 Member, Quality and Outcomes Committee, Council on Stroke, AHA
2015 – 2016 Chair, AHA Writing Group for Post-Stroke Depression
2013 – Member, Council on Stroke Scientific Statements Oversight Committee, AHA
2013 – Member, AHA “Stroke Statistics” Committee/Writing Group
2012 – 2013 Member, AHA Writing Group for “Guidelines for the Prevention of Stroke in Women”
2012 AHA Executive Steering Committee, 2012 Women’s Triennial Health Tracking Study in Cardiovascular Disease & Stroke
2012 Member, AHA Writing Group for “Factors Influencing the Decline in Stroke Mortality”
2011 Member, AHA Writing Group for Scientific Statement “Assessing Adiposity”
2009 Stroke Advisory Panel, California Office of Statewide Health Planning & Development
2007 – 2011 Member, Task Force, LAC Acute Ischemic Stroke Clinical Resource Management Pathway
2006 – Premium Professional Member, AHA
2005 – Member, American Academy of Neurology

Honors

2019 James and Dorothy Williams Stroke Scholar, University of Southern California
2018 Fellow of American Heart Association
2013 Michael S. Pessin Stroke Leadership Prize, American Academy of Neurology
2012 Robert G. Siekert New Investigator Award in Stroke, American Heart Association
2011 Clinical Systems Development Award, Kaiser Permanente/California Association of Public Hospitals and Health Systems/California Health Care Safety Net Institute
2011 Junior Academic Neurologist, American Neurological Association
2009 American Board of Psychiatry and Neurology Certification, Vascular Neurology
2008 American Board of Psychiatry and Neurology Certification, Adult Neurology
2007 Will Rogers Fellowship, University of Southern California
2006 Massachusetts General Hospital Clinical Research Day Neurology Award
2002 American Academy of Neurology Prize for Excellence in Neurology
2002 Harold Lamport Research Prize, Johns Hopkins School of Medicine
1997 Burchard Scholar, MIT
1995 Peter J. Eloranta Undergraduate Research Fellowship, MIT

C. Contributions to Science

1. A primary barrier to translation of clinical research discoveries into care delivery and population health is the lack of sustainable infrastructure **bringing researchers, policymakers, practitioners, and communities together to reduce silos in knowledge and action**. As National Institutes of Health’s mechanism to advance translational research, Clinical and Translational Science Award (CTSA) awardees

are uniquely positioned to bridge this gap. Delivering on this promise requires sustained collaboration and alignment between research institutions and public health and healthcare programs and services. I led a writing group composed of seven CTSA hubs in four cities collaborating with city, county, and state healthcare and public health organizations striving to realize this goal. We identified key components, common and unique themes, and barriers in academic–public collaborations.

a. Towfighi A, Orechwa AZ, Aragon TJ, et al. 2020 Bridging the gap between research, policy, and practice: Lessons learned from academic–public partnerships in the CTSA network. *Journal of Clinical Translational Science*.

2. Seven ideal health characteristics – dubbed Life's Simple 7s – **controlled blood pressure, cholesterol and diabetes, and healthy lifestyle habits (not smoking, eating a healthy diet, normal weight, and engaging in physical activity)** are associated with lower risk of stroke. We found that achieving a greater number of ideal cardiovascular health metrics was associated with lower mortality after stroke. There are **disparities in Life's Simple 7s** among stroke survivors. One third of stroke survivors have undiagnosed diabetes or prediabetes, with highest prevalence among non-Hispanic blacks. Non-Hispanic blacks were twice as likely as whites to have poorly controlled blood pressure. Among US stroke survivors, prevalence of poor blood pressure and cholesterol control decreased whereas prevalence of obesity, poor diet and physical inactivity increased. After adjustment, black race, poverty and low education were associated with low likelihood of ideal health.
   a. Lin MP, Ovbiagele B, Markovic D, **Towfighi A.** 2015. "Life's Simple 7" and Long-Term Mortality After Stroke. *JAHA.* 4(11)
   b. Lekoubou A, Ovbiagele B, Markovic D, Sanossian N, **Towfighi A**. 2017 Age, sex, and race/ethnic temporal trends in metabolic syndrome prevalence among individuals with myocardial infarction or stroke in the United States. *J Neurol Sci.* 376:24-28.
   c. Kim O, Ovbiagele B, Valle N, Markovic D, **Towfighi A** 2017 Race-Ethnic Disparities in Cardiometabolic Risk Profiles among Stroke Survivors with Undiagnosed Diabetes and Prediabetes in the United States. *J Stroke Cerebrovasc Dis* S1052- 3057(17)30331-2.
   d. Lin AM, Lin MP, Markovic D, Ovbiagele B, Sanossian N, **Towfighi A** 2019 Less than Ideal: Trends in Cardiovascular Health Among US Stroke Survivors. *Stroke*. 50:5-12.

3. Blood pressure is the greatest risk factor for stroke; in fact, by controlling blood pressure, more than half of all strokes could be averted. Yet, in a national sample of US stroke survivors, we found that **less than half of stroke survivors achieved blood pressure control**, with significant inequities between groups. **The most vulnerable groups included women, non-Mexican Hispanics, blacks, diabetics, and older individuals.** Although the importance of blood pressure control after stroke is well known, the optimal goal blood pressure after stroke is unknown. In an analysis of a nationally representative sample of the US population, we found that intensive control was associated with higher all-cause and vascular mortality after stroke, whereas gentle control was associated with lower mortality.
   a. **Towfighi A,** Markovic D, Ovbiagele B. 2014. Consistency of Blood Pressure Control After Ischemic Stroke: Prevalence and Prognosis. *Stroke.* 45(5):1313-1317.
   b. Lin MP, Ovbiagele B, Markovic D, **Towfighi A.** 2015. Systolic Blood Pressure and Mortality After Stroke: Too Low, No Go? *Stroke.* 46(5):1307-1313. PMID: 25765723
   c. Razmara A, Ovbiagele B, Markovic D, **Towfighi A**. 2016 Patterns and Predictors of Blood Pressure Treatment, Control, and Outcomes among Stroke Survivors in the United States. *J Stroke Cerebrovasc Dis*. 2;25(4):857-865.

4. When assessing **temporal trends in stroke hospitalizations in the United States**, we found that stroke hospitalizations declined from 2000 to 2010, but there were **sex, age, and race/ethnic inequities.** The reduction was more pronounced among older individuals, women, Hispanics, and whites, whereas middle-aged individuals and blacks had increases in their hospitalization rates. Analysis of trends in transient ischemic attack admissions revealed a decline, but the reduction was more pronounced among older individuals, men, whites, and Hispanics. Lack of insurance was associated with higher mortality after stroke.
   a. McManus M, Ovbiagele B, Markovic D, **Towfighi A.** 2015. Association of Insurance Status with Stroke-Related Mortality and Long-term Survival after Stroke. *J Stroke Cerebrovasc Dis*. 24(8):1924-1930.
   b. Ramirez L, Kim-Tenser MA, Sanossian N, Cen S, Wen G, He S, Mack WJ, **Towfighi A.** 2016 Trends in Acute Ischemic Stroke Hospitalizations in the United States. *JAHA*. 5(5).

c. Ramirez L, Kim-Tenser MA, Sanossian N, Cen S, Wen G, He S, Mack WJ, **Towfighi A.** 2016 Trends in Transient Ischemic Attack Hospitalizations in the United States. *JAHA.* 5(9).

5. Using the National Center for Health Statistics mortality data, we assessed leading organ- and disease-specific causes of death from 1996 to 2005 and found that age-adjusted stroke death rates declined by 25%. The improvements were not uniform across subgroups of the population; while stroke had declined to the 5th leading cause of death in men and 4th leading cause of death in whites, it **remained the 2nd leading cause of death in women and blacks**. In 2011, as we had forecasted, stroke declined from the 3rd to the 4th leading cause of death in the United States. I served on the writing panel for the AHA Scientific Statement on stroke mortality.
   a. **Towfighi A** & Saver JL 2011 Stroke declines from third to fourth leading cause of death in the United States: historical perspective and challenges ahead. *Stroke,* 42:2351-5.
   b. **Towfighi A**, Tai W, Markovic D, Ovbiagele B. 2011 Sex-specific temporal trends in in-hospital mortality after stroke among middle-aged individuals in the United States. *Stroke*,42:2740-5.
   c. Lackland DT, Roccella EJ, Deutsch AF, Fornage M, George MG, Howard G, Kissela BM, Kittner SJ, Lichtman JH, Lisabeth LD, Schwamm LH, Smith EE, **Towfighi A**. 2014 Factors influencing the decline in stroke mortality: A statement from the American Heart Association/American Stroke Association. *Stroke.* 45(1):315-53.

R&R SUBAWARD BUDGET ATTACHMENT(S) FORM

Subaward IDC Ceiling: 25,000

Period	Subaward Direct Costs	Subaward Indirect Costs	Subaward Costs	Allocated To IDC Base
All Subawards				
1				
2				
3				
4				
5				
All				

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

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5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

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1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	ORE_Aims1044447032.pdf
3. Research Strategy*	ORE_Research_Plan1044447033.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
9. Letters of Support	
10. Resource Sharing Plan(s)	
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12. Authentication of Key Biological and/or Chemical Resources	
13. Appendix	
12. Appendix	

SPECIFIC AIMS

The Outreach, Recruitment, and Engagement Core (ORE Core) serves as the primary bilingual and bi-cultural resource and liaison between the USC ADRC, our study participants, their caregivers/partners, and the larger community, i.e., the general public, people living with dementia and caregivers, professional care providers, and the lay community in Los Angeles County. The ORE Core facilitates bi-directional communication through our signature outreach and engagement activities, USC GeneScreen registry, and recruitment from the LA County Department of Health Services (DHS). Based on prior needs assessment of our local population, we will continue to focus on Latinx communities given that: a) the center's catchment area is comprised overwhelmingly of low-income and working class Latinx people, and b) Latinx are at higher risk for AD/ADRD, diagnosed at later stages, often have low AD/ADRD literacy, and are underrepresented in AD/ADRD research. As a result of this underrepresentation, we have an incomplete understanding of the role of *APOE* genotype in risk for Alzheimer disease (AD) and cerebral amyloid angiopathy (CAA), and of the behavior of biomarkers in the Latinx population (1). Our aims are as follows:

Aim 1: Engage in participant recruitment & develop a research registry. It is well documented that the *APOE* ε4 genotype is the strongest genetic risk factor for late-onset AD and that homozygotes for this allele may be at a twelve times higher risk for AD (2), though the risk conferred among persons of mixed Amerindian or African descent is less clear, with conflicting results having been reported (3, 4). The ORE Core will maintain, characterize, and further develop the USC Alzheimer's Prevention Registry, "USC GeneScreen," a registry of persons in the LA County area interested in participating in studies and enriched for *APOE* ε4 carriers. We will augment Latinx participation (English, Spanish, low SES) in the registry given the higher prevalence and disease burden in this population and our catchment area composition. Through online enrollment/consent available in English and Spanish and *APOE* genotyping, USC GeneScreen will genotype and register a diverse pool of 1500 participants who are "study-ready" to enroll in AD/ADRD research.

Aim 2: Enhance recruitment from the LA County Department of Health Services (DHS) at the Rancho Los Amigos National Rehabilitation Center (RLANRC). The USC ADRC hosts two California Alzheimer's Disease Centers (CADCs), one of which is at RLANRC, part of the DHS safety-net community health system that provides health services to a diverse, underserved, and largely Latinx population. The CADC at RLANRC will expand recruitment of this population into the Clinical Core and associated studies at the USC ADRC. In addition, the USC ADRC will partner with the newly established Dementia Care Aware (DCA) program within the DHS, the goal of which is earlier identification of older patients with cognitive impairment in the primary care setting and refer to specialists as appropriate.

Aim 3: Continue to engage in bi-directional education, information and resource sharing: In collaboration with our extensive network of community partners, we will continue to serve as a community resource in targeted engagement and educational outreach to the public-at-large, healthcare and human service providers, potential study participants, and current participants. We will communicate the latest ADRC- and AD/ADRD research findings and timely information on brain health, aging and wellness, ADRC study participation opportunities, and community resources for diverse audiences in English and Spanish.

Impact: Our focus on *APOE* ε4 and dementia primarily among Latinx individuals provides the unique opportunity to advance an understanding of these focal areas in historically underrepresented groups in AD/ADRD research: differential effects of *APOE* ε4 genotype on risk of AD/ADRD, cerebral amyloid angiopathy (CAA), biomarkers, disease progression, the effects of anti-amyloid treatment among diverse groups, and recruitment/retention. All specific aims of the ORE core are responsive to the required Plan to Enhance Diversity Perspectives (PEDP-DHS).

SIGNIFICANCE

The ORE Core's role is to serve the ADRC by 1) providing participant referrals to the ADRC and affiliated studies, 2) conducting retention activities for current participants and study partners, and 3) offering information, education, and networking to the general public, people with AD/ADRD and their families, and professional and lay providers on topics related to AD/ADRD, brain health, aging and wellness, community resources, as well as elucidating the phenomenon of participation in research. Considered one of the most racially and culturally diverse communities in the world (5), the ADRC is located in the City of Los Angeles with 3.9 million residents. 48.1% of its population is Hispanic/Latino (a, e, x) descent (herein, Latinx) of which 80% report Mexican heritage. The neighborhood surrounding the USC Health Sciences Center is 90.4% Latinx, while neighboring East Los Angeles is 95.5% (5). Los Angeles County has the highest number of people with AD (6), wherein 30.8% are Latinx. Our collaborating clinics and hospitals are well dispersed in the ADRC target area (**Fig. 1**). The Latinx population is at a higher risk for developing dementia, though it is not clear to what degree this risk, the risk of CAA, or the risk of ARIA due to anti-amyloid therapies is related to *APOE* ε4 genotype (4, 7-10). Nonetheless, a national consensus report reinforced that lowering *APOE* ε4 expression should be of therapeutic benefit and prioritized (11). The ADRC is well-positioned — both geographically and scientifically — to continue to service a robust source of participant referrals.

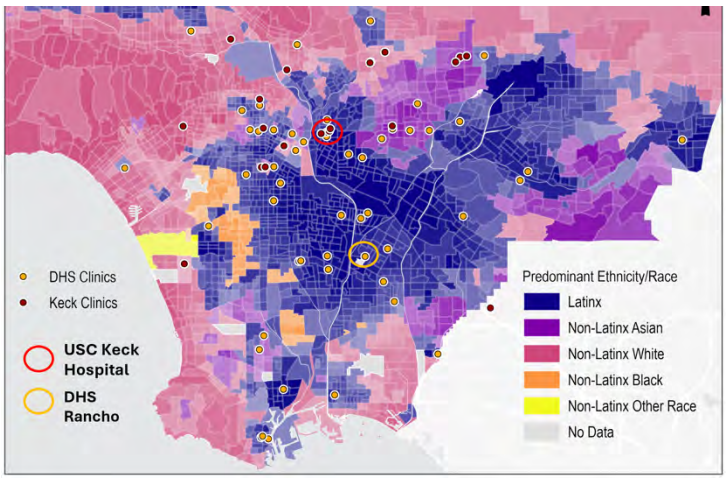


Figure 1: Los Angeles County percent ethnicity/race and clinic locations.

The Latinx population has been shown to have the highest population attributable fraction for dementia than any of the groups studied (33% vs. 22% in non-Latinx Whites), likely reflecting a greater exposure to vascular, metabolic, or educational risk factors (12). There is therefore tremendous opportunity for dementia prevention in the Latinx population. However, there are substantial barriers to optimal dementia prevention, identification, diagnosis and treatment, and overall care in this group (13). Cultural beliefs regarding cognitive decline as a normal part of aging, stigma, help-seeking behaviors [14],[15-17], financial barriers and immigration can impact AD/ADRD diagnosis and treatment. A lack of Spanish-speaking personnel in dementia care contributes to under- or late diagnosis of dementia,(18, 19) and the assessment of cognition across the spectrum of cultural, linguistic, educational, and acculturation levels in the Latinx population presents challenges (13). Several needs assessments co-authored by ORE Core leaders and associates (Drs. Aranda, Ringman, Segal-Gidan/RLANRC) highlight key takeaways: 1) A RLANRC study highlighted the substantial stress of caregiving and lack of information and support services among Latinx caregivers of persons with young onset AD (20); 2) In collaboration with LA County, and AARP, two needs assessments regarding older adults and caregivers' needs, (21, 22) indicated the need for dementia awareness, a countywide system of care to provide services to Latinx families facing dementia and other chronic conditions (diabetes, cardiovascular disease), and providing psychosocial caregiver interventions. The higher prevalence of dementia, modifiable risk factors among this historically disenfranchised group, and the ORE Core's deep understanding of the population, provide crucial opportunities for enhanced AD education, diagnosis, caregiving, and medical care.

INNOVATION

To better understand and innovate dementia prevention, diagnosis, treatment, and recruitment among the Latinx population, it is critical to meaningfully engage them in AD/ADRD research. The challenges of translating dementia care in the Latinx population are compounded by their historical underrepresentation in observational clinical (23) and biomarker studies (24, 25), genetic studies (26), and clinical trials (27). The ORE Core has decades-long expertise in bi-directional community engagement with high impact, and will expand these efforts in the new project cycle. The ORE Core contributes to the centers' theme of elucidating the effect of *APOE* ε4 on cognitive decline and ATN(v) biomarkers through enriched engagement of our local communities, with a

special focus on people of Latinx descent. The core focuses on participant recruitment enriched for *APOE* ε4, contributes to the participant registry, enhances recruitment from public safety-net hospitals, and engages in bi-directional education, information and resource sharing in English and Spanish. Based on existing frameworks described below, our overarching aim is to continue our long and deep history of serving as the primary resource and liaison between the ADRC, our study participants, their caregivers/partners, and the larger community, i.e., general public, people living with dementia who are not enrolled in our center, professional care providers, the lay community, and students/trainees.

In this renewal, we propose to extend these traditional strengths through the innovative use of a study-ready registry, USC GeneScreen (herein, GeneScreen), which has been successfully implemented in English and Spanish, and with diverse SES groups. We will continue the GeneScreen registry which is the first of its kind in the nation, and in the Los Angeles County region that has high representation of Latinx populations and tailored for Spanish-speakers and those with low SES. Our work with DHS and EHR data based on the Dementia Aware provider screening/identification and referral program is the first of its kind in the nation. The Brain Health for Me brain education study will provide registry referrals and is the first of its kind to provide online, accessible, feasible, scalable brain health information in English and Spanish led by bilingual/bicultural facilitators. Under the leadership of Drs. Aranda and Ringman, the core will facilitate ongoing evaluations and needs assessments including direct engagement with our Community Advisory Board members with lived experience. Persons referred to the Clinical Core will be contacted and relevant information including referral source and disposition will be documented in the Clinical Core's web-based research electronic data capture application: REDcap. Our core leaders bring significant expertise to the ADRC: Dr. Aranda is a leading expert in health equity and inclusion of under-represented minoritized populations in clinical and observational studies and trial research. She collaborates with the NACC ORE Core team and conferences, and other NIH-funded national initiatives such as the Latinos & AD Symposium steering committee, and the Knight ADRC conference proceedings on minoritized groups and AD/ADRD research. As a clinician-scholar, she is a member of the NIH Director's NExTRAC Working Group on Engaging the Public in Clinical Research which will develop a national framework for NIH-funded research. Dr Ringman, a clinician and researcher with a focus on the Latinx population, will be a co-leader and liaison with the Clinical Core. He is an expert in ADAD, genetics and phenotyping Mexican-origin people in the US and Mexico. Guiding our GeneScreen work, Dr Yassine (Associate Dir. ADRC) is the Director of the Center of Personalized Health, and an expert in the role of lipid metabolism and nutrition affecting cognition and AD risk.

APPROACH

(A) Approach overview. Considering our center's goals to enroll persons across the dementia spectrum, enriched for *APOE* ε4, and Latinx enrollment, we pursue multiple and diverse complementary strategies for participant engagement, recruitment, and retention. These include innovations such as the GeneScreen registry (Aim 1), recruitment from DHS primary care safety-net clinics and enriched Dementia Aware early identification (Aim 2), community engagement opportunities (Brain Health for Me, and other widely popular community information and education events in English and Spanish) (Aim 3). ORE Core leadership attends monthly Executive Committee meetings in which enrollment numbers are reviewed, study needs and recruitment plans are discussed, and feedback on action steps are reviewed.

(B) Guiding frameworks. Integrating principles from the *Trust in Community-Academic Research Partnerships (TICARP)* (28) and Micro-Meso-Macro Conceptual Frameworks (3Ms) (29), we exercise innovative core-specific approaches addressing facilitators/barriers to research engagement. The major principle that consistently emerges in participatory research among minoritized racial/ethnic groups is *the establishment of trust with the target population* — given the historical (and continued) harms committed in the name of science. Specific guidelines on how establishing and maintaining trust across highly complex, multi-component studies such as those represented in the ADRC are overlooked, yet our group has established expertise in this area as elucidated in national publications and successful target accruals. There is consensus that community-academic research partnerships must amplify collaborations in planning and implementing studies with mutually beneficial participation processes and outcomes. TICARP and 3Ms serve as frameworks to address this gap and form the

basis of guiding our multiple and diverse approaches with communities in LA County. **Table 1** lists the TICARP dimensions which inform process activities to establish trust in community-academic research partnerships including making decisions with community partners on issues related to communication, relationship-building and accountability, and committed partnerships through sustained involvement. To elucidate how we align our approach with this framework: 1) For the last three decades, we have built deep connections with community residents, providers, and agency representatives from a variety of non-profit, faith-based, educational, human service, healthcare, and community-convened groups, among others;

Table 1. TICARP Dimensions to Establish Trust in Community Academic Research Partnerships

		CONTEXT	SAMPLE ACTIVITIES
1	***Authentic, effective, and transparent communication***	Broad importance of ongoing, effective communication between community and academic research partnerships	Maintaining feedback loop between participants and researchers through bidirectional communication throughout all phases of research; honesty, full disclosure, no hidden agenda; Humility, modesty Clear expectations
2	***mutually respectful and reciprocal relationships***	on-going exchange of information and resources that take place between the individuals involved in community-academic research partnerships	Assuring a mutual "win" for all at the table; equal participation in and prioritization of projects by academic and community partners; minimize or address perceived power differentials and imbalances; community partners welcoming and taking time with academic researchers to show them their community
3	***sustainability***	capacity and resources of community-academic research partnerships to continue for long periods of time	Equitable funding arrangements and compensation funding including buying out a portion of partners' time commitments; training programs for researchers, which includes how to use community-engaged approaches and the many aspects to consider when working with communities; capacity building and empowerment of community partners
4	***committed partnerships***	the level of engagement of the partners involved in community-academic research partnerships and their investment in ensuring its maintenance.	Inclusion of community representation on research team in addition to community advisory board; academic researcher is present in the community including showing up at community events not directly related to research; having community partners participate in all phases of research - hypothesis development, study design, analysis, presentation of results, and as coauthors on publications
5	***communication, credibility, and problem-solving methodology***	the infrastructure in place to manage on-the-ground and day-to-day realities of research within community-academic partnership	Academic researcher experience and track-record with community and engaged research; ease of access to services; dissemination of information/data/ results/outcomes to community in methods other than publications; Realistic and clearly defined expectations about the research project for all parties

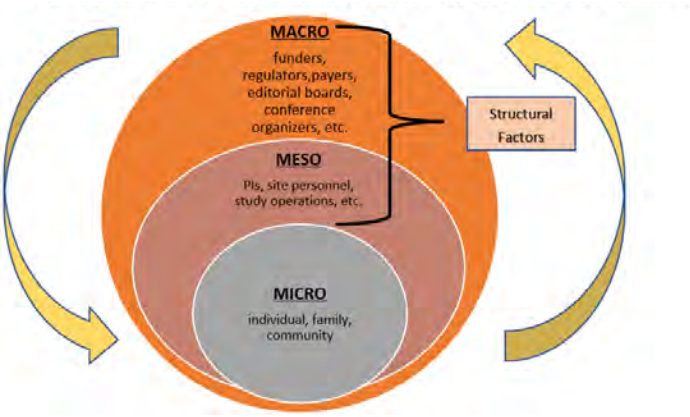


Figure 2: Micro-Meso-Macro framework for diversifying AD/ADRD trial recruitment.

2) The ability to sustain these respected connections is due to the transparent interactions with clear expectations of our respective roles during event planning and sponsorships, research study solicitations, needs assessments, including clear remuneration and disclosure policies around study participation. Complementary to TICARP is the 3Ms framework developed by Dr. Aranda and associates (**Fig. 2**), which addresses *individual+family+community (micro), research site+organizational (meso), and structural+institutional (macro) level facilitators/barriers* to the inclusion of Latinx in AD/ADRD trials. Efforts to understand recruitment barriers remain largely at the micro-level (cultural beliefs, help-seeking), resulting in diminished attention to upstream institutional- and policy-level barriers where decisions around scientific policies and funding allocations are ultimately made. To illustrate, our core: 1) Balances protocol refinements/adaptations to address all levels of outreach/engagement; 2) Supports robust ORE core outreach funding commitments; 3) Hires bilingual/bicultural personnel who are intensively trained to competency; 4) Establishes interagency collaborations to promote community resources addressing social determinants of health and resource navigation). Bilingual/bicultural outreach specialists will be trained in TICARP+3Ms frameworks using protocols developed by Dr. Aranda in community studies (30, 31), including staff from RLANRC who will enroll DHS patients.

(1) Aim 1 Approach: Participant Recruitment & Registry Development: GeneScreen. The purpose of the USC Alzheimer's Prevention Registry, herein GeneScreen, is to: 1) Identify and genetically characterize a large cohort of people at greater dementia risk; 2) Match individuals with genetic, vascular and metabolic risk factors into USC studies. The registry serves as a resource of "study-ready" minoritized people who are well-informed and engaged with our core. We recognize that *APOE* ε4 prevalence is lower in Latinx individuals (32), yet by facilitating recruitment of Latinx in English and Spanish we can elucidate the relationship of *APOE* genotype to AD risk in this population.

Table 2. USC GeneScreen (#participants)

Online Surveys	1470		
Leads	911		30 ε2/ε3
Completed Surveys	597		6 ε2/ε4
Registry Members	404		248 ε3/ε3
			99 ε3/ε4
			21 ε4/ε4

(1.1) Procedures. The registry, led by Drs. Yassine (Associate Dir.) and Aranda (core co-leader), matches cognitively-healthy adults to AD/ADRD studies based on their *APOE* genotype. 1,500 non-demented adults LA residents aged 40-90 will undergo screening for *APOE* genotype. Regardless of *APOE* allele variant, registry participants are offered participation in matched studies. Though short-term studies do not reveal a significant risk of adverse outcomes (anxiety, depression) to *APOE* genotype disclosure in asymptomatic persons (33), this varies on an individual basis (34), thus we will not disclose *APOE* test results. However, considering the growing role of *APOE* genotype in treatment implications (ARIA risk, potential for preventative treatment in preclinical populations), we will establish disclosure protocols and hire a bi-cultural, Spanish-speaking genetic counselor early in the next funding period (35) (Clinical Core budget).

(1.2) Identification, enrollment. Recruitment strategies will be used in concert with ORE Core personnel to enroll participants, including emails, mail, brochures, flyers, social media advertisements, community events/activities, a website, and phone calls (**Fig. 3**). Participants will screen themselves via an online application (English, Spanish; accessible by smartphone or computer) based on eligibility requirements listed on the registry website (5): aged 40-90; ability to provide basic demographics, medical history and contact information, live in the LA area, and consent to *APOE* genotyping without receiving results, and sharing de-identified data with other USC studies. Materials are reviewed by our community advisory board to assure readability and acceptability. Study coordinators call interested participants and perform a screening questionnaire. Eligible participants will be invited to be screened and enrolled through the website (**Fig. 3**): A screening questionnaire based on eligibility criteria, providing contact information, study consenting and providing demographic information, and confirming the contact/demographic information for enrollment. Genetic results may match them to certain studies that may have the discretion to disclose the *APOE* results as part of their respective protocols. Consent, HIPAA, and brief initial questionnaires are available online. Participants receive a $25 gift card for completing online enrollments which includes returning their buccal swab kit. ***APOE* Genotyping:** Study personnel mail or hand a cheek swab kit that includes easy-to-read instructions, including mailing the kit back in a pre-addressed and postage-paid envelope to a lab accredited by the College of American Pathologists and certified through Clinical Laboratory Improvement Amendments. To assure quality, the genetic sex identified from each sample is verified with the information provided on the requisition form provided during enrollment. Once a sample is processed, results are forwarded to staff by uploading a secure file. **Data source & recruitment yield:** Registrants will be referred to the ADRC active cohort and affiliated studies. The database can be queried by Clinical Core personnel or others approved by the IRB to identify persons appropriate for specific studies. When they are contacted by Clinical Core personnel, their information and disposition is entered into the Clinical Core REDCap Participant Tracking database. We registered 404 individuals from 1,470 online surveys with genotype results noted in **Table 2**. 31% are of Latinx descent. We will continue targeting advertising in Spanish-language social media outlets (Facebook, Instagram) that focus on the LA area. To date, we have 158 Spanish-language completed surveys of which 96 are registered [64 females; 61 years: +/-10.6 SD, and 25 are ε4 carriers (5 ε4/ε4)].

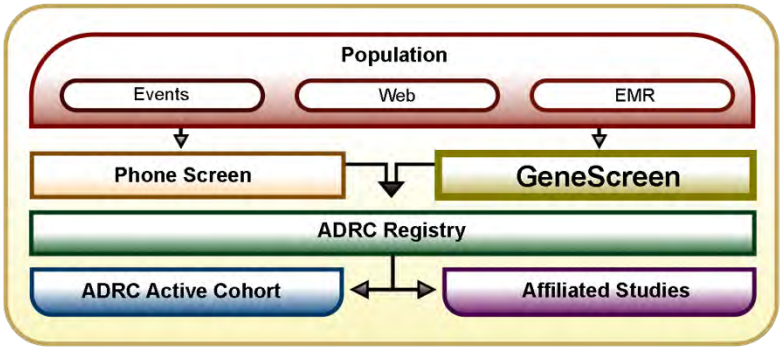


Figure 3: USC GeneScreen recruitment to referral flow.

(2) Aim 2 Approach: Recruitment from the Los Angeles Department of Health Services (DHS) at the Rancho Los Amigos National Rehabilitation Center (RLANRC). We will enroll persons with MCI or mild dementia into the ADRC (including SCAN/CLARiTI) through the Geriatric Neurology Clinic at RLANRC. The Rancho Geriatric Neurology Clinic is run by the CADC-Rancho. It is the largest dementia clinic in the LAC DHS which sees ~100 new consultations; follows 300 annually, of which 70% are Latinx. This increased focus recruitment from RLANRC will help further diversify our increasingly diverse ADRC cohort.

(2.1) Identification/Referral Procedures. After undergoing a work-up, patients are reviewed at bi-monthly consensus conferences, attended by Drs. Segal-Gidan, Chui, Wong, Ringman, and Joe as well as social workers and other CADC staff. Studies for which each patient is eligible are considered and noted on a list of potential referrals in their chart with a form the provider can share with the patient/family at the next clinic visit. With the

patient or surrogate permission, their contact information is shared with ADRC research coordinators and contact for study participation is initiated by the Clinical Core. Barriers to the participation of RLANRC patients in the ADRC have been attributed to the lack of ADRC research personnel on site and the 14-mile distance between the sites. We will therefore assign an ADRC research coordinator to present in-person to patients/families twice monthly at RLANRC to facilitate communication regarding study opportunities and initiate the consent process. We will initiate research activities (neuropsychological testing, UDS) at RLANRC and provide transportation support for imaging and other procedures at the USC ADRC.

To identify mildly affected persons, we will partner with the new Dementia Care Aware (DCA) program in DHS primary care clinics. Recognizing the under-diagnosis of cognitive impairment in older adults within the state MediCal system, the DCA program was established in 2022 to systemize and document a Cognitive Health Assessment (CHA) screen in the Electronic Health Record (EHR) for all MediCal participants at or above the age of 65 years in primary care. Led by Dr. Schickedanz, Chair of Family Medicine, Harbor-UCLA Medical Center, with the involvement of Dr. Segal-Gidan, four clinics (including RLANRC), have yielded 260 persons with cognitive impairment, triggering further assessment. The USC ADRC will partner with Dr. Schickedanz to train primary care physicians to identify and treat cognitive impairment and facilitate referrals to the RLANRC. We will partner with Dr. Towfighi, Director of Neurological Services, LAC DHS, to explore ways to use the EHR to identify persons with cognitive impairment. For all referrals, we will document if they are appropriate for research participation, as well as the ultimate disposition for each patient with regards to research involvement. Those contacted by Clinical Core personnel will be entered into the REDCap Participant Tracking registry, along with the referral source, to enable tracking of their disposition. Drs. Ringman and Joe will supervise the warm hand-off for patient and care partner transition from the RLANRC to the ADRC Clinical Core.

(3) Aim 3 Approach: Continue to engage in Bi-directional education, information and resource sharing.

(3.1) Advisory Board. Drawing from principles of the TICARP/3Ms frameworks, we assembled an impressive Community Advisory Board, in collaboration with the USC Roybal Institute on Aging (Dr. Aranda, Executive Director), to optimize engagement activities targeted specifically at increasing participation among Latinx and minoritized racial/ethnic groups. Our nine members (**Fig. 4**) will: 1) Gain foundational knowledge about AD/ADRD, brain health, aging, caregiving, center findings, recruitment/retention challenges; 2) Advise on matters related to ongoing needs assessment and strategic planning on outreach, recruitment and engagement; 3) Co-design community events and materials. All live in the local area, and most have lived experience with AD/ADRD, and have submitted letters of support. Members will meet three times per year with core leaders and receive remuneration for their participation ($100/meeting). The Chair will be remunerated for planning activities. Interpretation/translation services will be provided for all meetings.

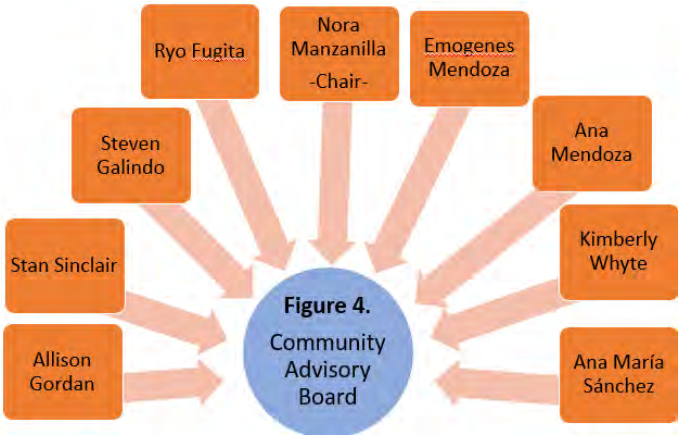


Figure 4: Community Advisory Board.

(3.2) Outreach, Recruitment, and Retention Activities.

(3.2.1) Outreach and recruitment: Communicating the latest research findings to the public. The ADRC has an extensive partner network which we cultivate through numerous community talks, workshops, conferences, symposia, health fairs, flyer distribution, social media, and radio announcements, among others. Our menu of education and information topics includes Alzheimer's and Dementia 101, Brain Health Tips, Research Participation Opportunities, Donating Your Brain, Caring for the Caregiver, Savvy Caregiver Program, *Unidos en el Cuidado* (United in Caring), Brain Health, and Older Adults Month Celebration, among others. We will continue these engagement and outreach activities in English and Spanish to further our enrollments, and trusted community engagement goals. **Figure 5** summarizes our expansive network of collaborators,

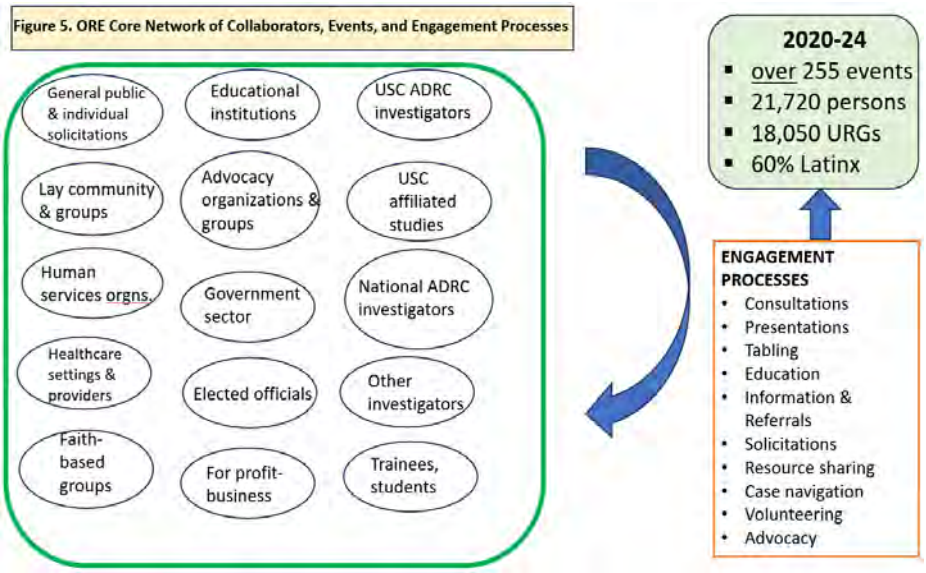


Figure 5: ORE Core network of collaborators, events, and engagement processes.

engagement processes (consultations, presentations, tabling, etc.), and data on our 2020-24 events: Over the past four years, we conducted over 255 events, reaching over 21,720 individuals. At least 18,050 individuals were from underrepresented groups, about 60% Latino, and 85% woman. The contacts yielded upwards of 100 referrals to the ADRC. One of our innovative strategies is the *Brain Health for Me (BH4Me): A Healthy Lifestyles Program,* hosted by the USC Roybal Institute on Aging, and co-designed with community representatives. BH4Me is a free online webinar covering a variety of topics related to brain health including dementia knowledge, physical activity, social engagement, sleep, and stress management. Participants will be queried about their interest in study participation and referred to the ADRC. The program will be translated to Spanish. Participants will be remunerated $20 gift cards for their participation and asked to complete brief follow-up questionnaires to ascertain lifestyle behavior changes.

(3.2.2) Signature retention event. Our signature retention, the annual ADRC Batey Appreciation Day, is funded by a former study participant and draws over 100 ADRC participants who hear about the center's scientific advances and future study participation. All Batey events are recorded and available for viewing upon request. We reached over 500 participated in recent years, and last year's evaluation data indicate that 98.5% were "very/somewhat satisfied" with the event. Over 33 participants requested to be contacted for study participation. We will continue to host the event and will add a similar event in Spanish.

(3.2.3) Provider events. We will continue our highly successful provider education events: 1) Annual hybrid Brain Health Forum with USC Verdugo Hills Hospital which highlights the latest in brain research and clinical practice from our physicians and scientists; 2) Biennial, hybrid Many Faces of Dementia Conference hosted by RLARC, targeting health professionals working with people living with dementia and their families which typically supports over 400 attendees.

(3.2.4) Continuous assessment of recruitment efforts and return on investment. Success of the various outreach activities are assessed from data curated in the REDCap data base. The registry documents the referral source (e.g. GeneScreen, DHS or other clinician referral, ORE Core outreach activities) and disposition, including the reasons for disqualification or participation refusal. Total enrollment and progress in the past month are reviewed by ORE Core staff and discussed at monthly ADRC Executive Committee meetings. This allows for the identification of recruitment needs and barriers, and for the modification of outreach efforts and study protocols as necessary. The current status of studies, eligibility requirements, and recruitment materials are also discussed with CAB members for their input and suggested modifications.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	❍No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	❍No
Is the Project Exempt from Federal regulations?	❍Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447178.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Biomarker Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: 1501 San Pablo St.		Street2:
* City: Los Angeles	County: Los Angeles	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	Biomarker_Abstract1044447034.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	Biomarker_References1044447035.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Biomarker Core

The Biomarker Core will unite ADRC efforts to standardize the collection and analysis of biofluid samples and brain imaging, along with the sharing of biomarker resources and derived data. In addition to serving as a hub to support these efforts for ADRC participants, we also support collaboration with academic and commercial collaborators. Our ADRC has previously focused on vascular factors related to AD and in this renewal we expand that focus to characterize factors that modify the effect of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers, emphasizing vascular, lipid, inflammatory, genetic, and ethnic influences. The Biomarker Core is led by Robert Rissman PhD, and supported by resources at the USC Zilkha Neurogenetics Institute (ZNI) in Los Angeles, and the USC Alzheimer Therapeutics Research Institute (ATRI) in San Diego. The Biomarker Core has 3 specific aims: **Aim 1**: **Biofluid collection, processing, storage and sharing**. 1) Coordinate collection, processing and storage of CSF, blood, plasma and DNA from ADRC Clinical Core participants using national ADRC Best Practices SOPs; sending plasma, PBMC, and DNA to NCRAD, **2)** Perform analyses of CSF, plasma and DNA relevant to ADRD using validated methods to facilitate characterization of the ADRC cohort. **3**) Provide ADRC samples and core resources (bioassay, DNA analyses, extractions, processing pipelines) to a broad array of investigators in support of new research projects and foster career development of diverse scientists in biomarker research. Provide expertise for training in biomarker methods. **Aim 2**: **Facilitate biomarker research through an integrated informatics toolkit**: Maintain a state-of-the-art biosample inventory linked with the ADRC DMSC to provide internal and external investigators with CSF and plasma samples, DNA, and derived data in support of ADRD research and novel biomarker discovery. **Aim 3**: **Accelerate local, national and international biomarker discovery efforts**: Leverage ADRC biospecimens to develop new methods that inform on neuropathology, diagnosis, progression, prognosis and other research applicable to ADRD. **1)** Leverage ADRC specimens for discovery projects at USC and nationally. **2)** Work with Industry partners to develop and validate novel assays and platforms. We will also continue to support extensive collaborations with national and international investigators and train fellows, residents, graduate and undergraduate students in biomarker discovery.

REFERENCES

1. Desikan RS, Schork AJ, Wang Y, Thompson WK, Dehghan A, Ridker PM, et al. Polygenic Overlap Between C-Reactive Protein, Plasma Lipids, and Alzheimer Disease. Circulation. 2015;131(23):2061-9. Epub 2015/04/12. doi: 10.1161/circulationaha.115.015489. PubMed PMID: 25862742; PubMed Central PMCID: PMCPMC4677995.

2. Desikan RS, Schork AJ, Wang Y, Witoelar A, Sharma M, McEvoy LK, et al. Genetic overlap between Alzheimer's disease and Parkinson's disease at the MAPT locus. Molecular psychiatry. 2015;20(12):1588-95. Epub 2015/02/18. doi: 10.1038/mp.2015.6. PubMed PMID: 25687773; PubMed Central PMCID: PMCPMC4539304.

3. Yokoyama JS, Wang Y, Schork AJ, Thompson WK, Karch CM, Cruchaga C, et al. Association Between Genetic Traits for Immune-Mediated Diseases and Alzheimer Disease. JAMA Neurol. 2016;73(6):691-7. doi: 10.1001/jamaneurol.2016.0150. PubMed PMID: 27088644; PubMed Central PMCID: PMCPMC4905783.

4. Ferrari R, Wang Y, Vandrovcova J, Guelfi S, Witeolar A, Karch CM, et al. Genetic architecture of sporadic frontotemporal dementia and overlap with Alzheimer's and Parkinson's diseases. Journal of neurology, neurosurgery, and psychiatry. 2017;88(2):152-64. Epub 2016/12/03. doi: 10.1136/jnnp-2016-314411. PubMed PMID: 27899424; PubMed Central PMCID: PMCPMC5237405.

5. Yokoyama JS, Karch CM, Fan CC, Bonham LW, Kouri N, Ross OA, et al. Shared genetic risk between corticobasal degeneration, progressive supranuclear palsy, and frontotemporal dementia. Acta neuropathologica. 2017;133(5):825-37. doi: 10.1007/s00401-017-1693-y. PubMed PMID: 28271184; PubMed Central PMCID: PMCPMC5429027.

6. Witoelar A, Jansen IE, Wang Y, Desikan RS, Gibbs JR, Blauwendraat C, et al. Genome-wide Pleiotropy Between Parkinson Disease and Autoimmune Diseases. JAMA Neurol. 2017;74(7):780-92. doi: 10.1001/jamaneurol.2017.0469. PubMed PMID: 28586827; PubMed Central PMCID: PMCPMC5710535.

7. Karch CM, Wen N, Fan CC, Yokoyama JS, Kouri N, Ross OA, et al. Selective Genetic Overlap Between Amyotrophic Lateral Sclerosis and Diseases of the Frontotemporal Dementia Spectrum. JAMA Neurol. 2018. Epub 2018/04/10. doi: 10.1001/jamaneurol.2018.0372. PubMed PMID: 29630712.

8. Winston CN, Langford O, Levin N, Raman R, Yarasheski K, West T, et al. Evaluation of Blood-Based Plasma Biomarkers as Potential Markers of Amyloid Burden in Preclinical Alzheimer's Disease. J Alzheimers Dis. 2023;92(1):95-107. Epub 2023/01/31. doi: 10.3233/JAD-221118. PubMed PMID: 36710683.

9. Alvarez XA, Winston CN, Barlow JW, Sarsoza FM, Alvarez I, Aleixandre M, et al. Modulation of Amyloid-β and Tau in Alzheimer's Disease Plasma Neuronal-Derived Extracellular Vesicles by Cerebrolysin® and Donepezil. J Alzheimers Dis. 2022;90(2):705-17. Epub 2022/09/27. doi: 10.3233/jad-220575. PubMed PMID: 36155516; PubMed Central PMCID: PMCPMC9697063.

10. Winston CN, Goetzl EJ, Akers JC, Carter BS, Rockenstein EM, Galasko D, et al. Prediction of conversion from mild cognitive impairment to dementia with neuronally derived blood exosome protein profile. Alzheimers Dement (Amst). 2016;3:63-72. Epub 2016/07/14. doi: 10.1016/j.dadm.2016.04.001. PubMed PMID: 27408937; PubMed Central PMCID: PMCPMC4925777.

11. Winston CN, Goetzl EJ, Baker LD, Vitiello MV, Rissman RA. Growth Hormone-Releasing Hormone Modulation of Neuronal Exosome Biomarkers in Mild Cognitive Impairment. J Alzheimers Dis. 2018;66(3):971-81. Epub 2018/10/30. doi: 10.3233/jad-180302. PubMed PMID: 30372675; PubMed Central PMCID: PMCPMC6487872.

12. Rissman RA, Langford O, Raman R, Donohue MC, Abdel-Latif S, Meyer MR, et al. Plasma Aβ42/Aβ40 and phospho-tau217 concentration ratios increase the accuracy of amyloid PET classification in preclinical Alzheimer's disease. Alzheimers Dement. 2024;20(2):1214-24. Epub 2023/11/07. doi: 10.1002/alz.13542. PubMed PMID: 37932961; PubMed Central PMCID: PMCPMC10916957.

13. Nelson PT, Lee EB, Cykowski MD, Alafuzoff I, Arfanakis K, Attems J, et al. LATE-NC staging in routine neuropathologic diagnosis: an update. Acta Neuropathol. 2023;145(2):159-73. Epub 2022/12/14. doi: 10.1007/s00401-022-02524-2. PubMed PMID: 36512061; PubMed Central PMCID: PMCPMC9849315.

14. Nelson PT, Dickson DW, Trojanowski JQ, Jack CR, Boyle PA, Arfanakis K, et al. Limbic-predominant age-related TDP-43 encephalopathy (LATE): consensus working group report. Brain. 2019;142(6):1503-27. Epub 2019/05/01. doi: 10.1093/brain/awz099. PubMed PMID: 31039256; PubMed Central PMCID: PMCPMC6536849.

15. Winston CN, Sukreet S, Lynch H, Lee VM, Wilcock DM, Nelson PT, et al. Evaluation of blood-based, extracellular vesicles as biomarkers for aging-related TDP-43 pathology. Alzheimers Dement (Amst). 2022;14(1):e12365. Epub 2022/12/22. doi: 10.1002/dad2.12365. PubMed PMID: 36540894; PubMed Central PMCID: PMCPMC9753157.

16. O'Bryant SE, Zhang F, Johnson LA, Hall J, Petersen M, Oh ES, et al. Precision Medicine for Preventing Alzheimer's Disease: Analysis of the ADAPT Study. J Alzheimers Dis. 2023;95(4):1609-22. Epub 2023/09/18. doi: 10.3233/JAD-230317. PubMed PMID: 37718801.

17. O'Bryant SE, Zhang F, Petersen M, Johnson L, Hall J, Rissman RA. A Precision Medicine Approach to Treating Alzheimer's Disease Using Rosiglitazone Therapy: A Biomarker Analysis of the REFLECT Trials. J Alzheimers Dis. 2021;81(2):557-68. Epub 2021/04/06. doi: 10.3233/jad-201610. PubMed PMID: 33814447; PubMed Central PMCID: PMCPMC8203239.

18. O'Bryant SE, Zhang F, Johnson LA, Hall J, Edwards M, Grammas P, et al. A Precision Medicine Model for Targeted NSAID Therapy in Alzheimer's Disease. J Alzheimers Dis. 2018;66(1):97-104. Epub 2018/09/11. doi: 10.3233/jad-180619. PubMed PMID: 30198872; PubMed Central PMCID: PMCPMC6428063.

19. Huang LK, Kuan YC, Lin HW, Hu CJ. Clinical trials of new drugs for Alzheimer disease: a 2020-2023 update. J Biomed Sci. 2023;30(1):83. Epub 2023/10/03. doi: 10.1186/s12929-023-00976-6. PubMed PMID: 37784171; PubMed Central PMCID: PMCPMC10544555.

20. Mattsson N, Andreasson U, Persson S, Arai H, Batish SD, Bernardini S, et al. The Alzheimer's Association external quality control program for cerebrospinal fluid biomarkers. Alzheimers Dement. 2011;7(4):386-95 e6. doi: 10.1016/j.jalz.2011.05.2243. PubMed PMID: 21784349; PubMed Central PMCID: PMCPMC3710290.

21. Mattsson N, Andreasson U, Persson S, Carrillo MC, Collins S, Chalbot S, et al. CSF biomarker variability in the Alzheimer's Association quality control program. Alzheimers Dement. 2013;9(3):251-

OMB Number: 4040-0001

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Robert Middle Name Last Name*: Rissman Suffix:

Position/Title*:	Professor of Physiology and Neuroscience
Organization Name*:	University of Southern California
Department:	KSOM Alzheimer's Therapeutic Research Institute
Division:	Keck School of Medicine of USC
Street1*:	Health Sciences Campus
Street2:	1501 San Pablo St.
City*:	Los Angeles
County:	California
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764-3341 Fax Number: E-Mail*: r.rissman@usc.edu

Credential, e.g., agency login: rrissman

Project Role*: Other (Specify) Other Project Role Category: Biomarker Core Lead

Degree Type: PhD Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Rissman1044446932.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Rissman, Robert A.

eRA COMMONS USER NAME (credential, e.g., agency login): rrissman

POSITION TITLE: Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of California, San Diego (UCSD)	B.S.	06/1996	Physiological Psych
Drexel University, College of Medicine	Ph.D.	12/2001	Neuroscience
University of California, Irvine	Postdoctoral	01/2005	Neurodegeneration
The Salk Institute for Biological Studies	Postdoctoral	09/2008	Neurodegeneration

A. Personal Statement

The goal of my basic science research is on novel biomarker discovery and experimental neuropathology in Alzheimer's disease and related dementias (ADRD), Parkinson's disease (PD), and preclinical animal models. Other research includes understanding how central corticotropin-releasing factor (CRF) neuromodulatory pathways interplays with peripheral stress signaling and contributes to neuronal vulnerability and AD neuropathology using in vivo pharmacology in transgenic animal models and human biospecimens.
In addition to my NIA funded research, I am the Founding Director of the Neuroscience Translational Research Division at USC Alzheimer's Therapeutic Research Institute (ATRI) as well as Founding Director of Alzheimer's Clinical Trials Consortium (ACTC) Biomarker Unit.

Ongoing and recently completed projects that I would like to highlight include:

R01AG079303 (PD/PI: Rissman, RA) 04/01/2023 – 03/31/2028
Title: Impact of TBI and Cognitive Decline on Alzheimer's Disease Brain-Derived Exosome Cargo
Major Goals: The objective of this project is to determine if circulating proteins carried in neuronal exosomes have utility as biomarkers for neurodegeneration and how this is impacted by TBI, PTSD in twins from the VETSA Study.

R01AG073979 (PD/PI: Rissman, RA) 09/30/2021 – 08/31/2024
Title: Precision Medicine for Inflammatory Treatment for Alzheimer's Disease in Down Syndrome
Major Goals: The major goal of this project is to explore the biological makeup of DS-AD is as heterogeneous as the sporadic AD population and that identifying specific subsets of DS-AD for particular treatments will yield meaningful therapeutic responses.

U24AG057437 (Aisen P, Petersen R, Sperling R) 07/01/2023 – 06/30/2028
Title: Alzheimer's Clinical Trials Consortium (ACTC)
Major Goals: This proposal aims aim to create an optimized infrastructure to optimally design, efficiently launch, rapidly recruit with improved diversity, and successfully complete NIH supported clinical trials in AD and related dementias.

Citations:

1. **Rissman RA**, Langford O, Raman R, Donohue MC, Abdel-Latif S, Meyer MR, Wente-Roth T, Kirmess KM, Ngolab J, Winston CN, Jimenez-Maggiora G, Rafii MS, Sachdev P, West T, Yarasheski KE, Braunstein JB, Irizarry M, Johnson KA, Aisen PS, Sperling RA; AHEAD 3-45 Study team. Plasma Aβ42/Aβ40 and phospho-tau217 concentration ratios increase the accuracy of amyloid PET classification in preclinical Alzheimer's disease. Alzheimers Dement. 2023 Nov 6. doi: 10.1002/alz.13542. Epub ahead of print. PMID: 37932961.

2. Winston CN, Langford O, Levin N, Raman R, Yarasheski K, West T, Abdel-Latif S, Donohue M, Nakamura A, Toba K, Masters CL, Doecke J, Sperling RA, Aisen PS, **Rissman RA**. Evaluation of Blood-Based Plasma Biomarkers as Potential Markers of Amyloid Burden in Preclinical Alzheimer's Disease. J Alzheimers Dis. 2023;92(1):95-107. doi: 10.3233/JAD-221118. PMID: 36710683.

3. Winston CN, Romero HK, Ellisman M, Nauss S, Julovich DA, Conger T, Hall JR, Campana W, O'Bryant SE, Nievergelt CM, Baker DG, Risbrough VB, **Rissman RA** (2019). Assessing Neuronal and Astrocyte Derived Exosomes From Individuals With Mild Traumatic Brain Injury for Markers of Neurodegeneration and Cytotoxic Activity. Front Neurosci. 2019 Oct 2;13:1005. doi: 10.3389/fnins.2019.01005. PMID: 31680797; PMCID: PMC6797846.

4. O'Bryant SE, Zhang F, Petersen M, Johnson L, Hall J, **Rissman RA** (2021). A Precision Medicine Approach to Treating Alzheimer's Disease Using Rosiglitazone Therapy: A Biomarker Analysis of the REFLECT Trials. J Alzheimers Dis. 2021;81(2):557-568. doi: 10.3233/JAD-201610. PMID: 33814447; PMCID: PMC8203239.

B. Positions, Scientific Appointments, and Honors

2022 – Present	Professor of Physiology and Neuroscience, WM Keck Endowed Chair in Medicine, University of Southern California, San Diego, CA
2016 – Present	Principal Investigator/Research Biologist, VA San Diego Healthcare System, San Diego, CA
2008 – 2024	Professor of Neurosciences, University of California, San Diego, CA (from Associate Professor, 2021; Assistant Professor, 2015)
2008 – Present	Clinical Trials Alzheimer’s Disease (CTAD), Scientific Board
2003 – Present	American Neuroendocrine Society
2001 – Present	American Association for the Advancement of Science
1999 – Present	Society for Neuroscience
2012	Outstanding Young Investigator, Charleston Conference on AD
2001 – 2003	National Research Service Award, NIA
1992 – 1996	Psi Chi Honor Society

C. Contributions to Science

1. AD NOVEL BIOMARKER DISCOVERY RESEARCH
   My lab is focused on research to identify and characterize novel biomarkers of AD. The primary focus of this research is on blood-based biomarkers, with particular interest in changes in neuronally-derived exosomes, inflammatory pathways, neuroimaging and retinal imaging. The samples we use are typically from the biospecimen repositories of the ADCS and ADRC.

   a. Allué JA, Pascual-Lucas M, Sarasa L, Castillo S, Sarasa M, Sáez ME, Abdel-Latif S, Rissman RA, Terencio J (2023). Clinical utility of an antibody-free LC-MS method to detect brain amyloid deposition in cognitively unimpaired individuals from the screening visit of the A4 Study. Alzheimers Dement (Amst). 2023 Jun 1;15(2):e12451. doi: 10.1002/dad2.12451. PMID: 37274930; PMCID: PMC10236000.

b. Winston CN, Sukreet S, Lynch H, Lee VM, Wilcock DM, Nelson PT, Rissman RA. Evaluation of blood-based, extracellular vesicles as biomarkers for aging-related TDP-43 pathology. Alzheimers Dement (Amst). 2022 Dec 15;14(1):e12365. doi: 10.1002/dad2.12365. PMID: 36540894; PMCID: PMC9753157.

c. Alvarez XA, Winston CN, Barlow JW, Sarsoza FM, Alvarez I, Aleixandre M, Linares C, García-Fantini M, Kastberger B, Winter S, **Rissman RA** (2022). Modulation of Amyloid-β and Tau in Alzheimer's Disease Plasma Neuronal-Derived Extracellular Vesicles by Cerebrolysin® and Donepezil. J Alzheimers Dis. 2022;90(2):705-717. doi: 10.3233/JAD-220575. PMID: 36155516; PMCID: PMC9697063.

d. O'Bryant SE, Zhang F, Petersen M, Hall JR, Johnson LA, Yaffe K, Braskie M, Vig R, Toga AW, **Rissman RA**; HABS-HD Study Team (2022). Proteomic Profiles of Neurodegeneration Among Mexican Americans and Non-Hispanic Whites in the HABS-HD Study. J Alzheimers Dis. 2022;86 (3):1243-1254. doi: 10.3233/JAD-210543. PMID: 35180110; PMCID: PMC9376967.

2. EXPERIMENTAL NEUROPATHOLOGY PRECLINICAL AND POSTMORTEM TISSUE WORK
The goal of my basic science research is on novel biomarker discovery and experimental neuropathology in AD, synucleinopathies and preclinical animal models. A major focus of my lab in AD and PD biomarkers is on the utility and mechanistic underpinnings of neuronal exosomes and our group has published manuscripts demonstrating the ability of pathologic proteins from neuronal exosomes to predict progression of disease and induce pathology in animal models.

a. Leitão ADG, Ahammad RU, Spencer B, Wu C, Masliah E, **Rissman RA**. Novel systemic delivery of peptide-conjugated antisense oligonucleotide to reduce α-synuclein in a mouse model of Alzheimer's disease. Neurobiol Dis. 2023 Sep 9;186:106285. doi: 10.1016/j.nbd.2023.106285. Epub ahead of print. PMID: 37690676

b. Leitão ADG, Spencer B, Sarsoza F, Ngolab J, Amalraj J, Masliah E, Wu C, **Rissman RA** (2023). Hippocampal Reduction of α-Synuclein via RNA Interference Improves Neuropathology in Alzheimer's Disease Mice. J Alzheimers Dis. 2023;95(1):349-361. doi: 10.3233/JAD-230232. PMID: 37522208.

c. Dutta AK, Armstrong C, Luo D, Das B, Spencer B, **Rissman RA** (2023). D-685 Reverses Motor Deficits and Reduces Accumulation of Human α-Synuclein Protein in Two Different Parkinson's Disease Animal Models. ACS Chem Neurosci. 2023 Mar 1;14(5):885-896. doi: 10.1021/acschemneuro.2c00655. Epub 2023 Feb 7. PMID: 36749600.

d. Spencer B, **Rissman RA**, Overk C, Masliah E (2022). Novel Brain-Penetrating Single Chain Antibodies Directed Against 3RTau for the Treatment of Alzheimer's Disease and Related Dementias. Methods Mol Biol. 2022;2383:447-457. doi: 10.1007/978-1-0716-1752-6_28. PMID: 34766306.

3. ALZHEIMER'S DISEASE CLINICAL TRIALS
In my role as ADCS and ATRI Biomarker Core leader, I have participated in many clinical trials. My lab received, processed, banked and analyzed the biospecimens that we received. The following publications resulted from this phase 1 and 2 human clinical trials work.

a. Rafii MS, Sperling RA, Donohue MC, Zhou J, Roberts C, Irizarry MC, Dhadda S, Sethuraman G, Kramer LD, Swanson CJ, Li D, Krause S, **Rissman RA**, Walter S, Raman R, Johnson KA, Aisen PS (2023). The AHEAD 3-45 Study: Design of a prevention trial for Alzheimer's disease. Alzheimers Dement. 2023 Apr;19(4):1227-1233. doi: 10.1002/alz.12748. Epub 2022 Aug 15. PMID: 35971310; PMCID: PMC9929028.

b. Kellar D, Register T, Lockhart SN, Aisen P, Raman R, **Rissman RA**, Brewer J, Craft S (2022). Intranasal insulin modulates cerebrospinal fluid markers of neuroinflammation in mild cognitive impairment and Alzheimer's disease: a randomized trial. Sci Rep. 2022 Jan 25;12(1):1346. doi: 10.1038/s41598-022-05165-3. PMID: 35079029; PMCID: PMC8789895.

c. Rafii MS, Sol O, Mobley WC, Delpretti S, Skotko BG, Burke AD, Sabbagh MN, Yuan SH, **Rissman RA**, Pulsifer M, Evans C, Evans AC, Beth G, Fournier N, Gray JA, Dos Santos AM, Hliva V, Vukicevic M, Kosco-Vilbois M, Streffer J, Pfeifer A, Feldman HH (2022). Safety, Tolerability, and Immunogenicity of the ACI-24 Vaccine in Adults With Down Syndrome: A Phase 1b Randomized Clinical Trial. JAMA Neurol. 2022 Jun 1;79(6):565-574. doi: 10.1001/jamaneurol.2022.0983. PMID: 35532913; PMCID: PMC9086937.

d. van Dyck CH, Nygaard HB, Chen K, Donohue MC, Raman R, Rissman RA, Brewer JB, Koeppe RA, Chow TW, Rafii MS, Gessert D, Choi J, Turner RS, Kaye JA, Gale SA, Reiman EM, Aisen PS, Strittmatter SM. Effect of AZD0530 on Cerebral Metabolic Decline in Alzheimer Disease: A Randomized Clinical Trial. JAMA Neurol. 2019 Oct 1;76(10):1219-1229. doi: 10.1001/jamaneurol.2019.2050. PMID: 31329216; PMCID: PMC6646979.

Complete List of Published Work in MyBibliography:
https://www.ncbi.nlm.nih.gov/myncbi/robert.rissman.1/bibliography/public/

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

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5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

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SPECIFIC AIMS

Biofluid biomarkers are critical for diagnosing and monitoring progression of Alzheimer's Disease (AD) and related disorders (ADRD). In this renewal, the Biomarker Core will use cutting edge, validated methods to characterize the ADRC cohort and support novel research, locally, nationally and internationally. In the past funding cycle, the Biomarker Core enabled research that was key for dramatically advancing the field of plasma biomarkers, which now have excellent diagnostic accuracy and can predict neuropathological changes that were previously only detectable using invasive methods. Through the leadership of Dr. Robert Rissman, the Biomarker Core has expanded biomarker assay and biorepository infrastructure through integration with NIA's Alzheimer's Clinical Trials Consortium (ACTC) Biomarker Unit. In this renewal, inventorying and sharing of ADRC biosamples will benefit from a single committee and centralized process leveraging the established working relationship of Drs. Rissman, Hiniker and Toga to facilitate wide sharing of biosamples and derived data to researchers, with linkage to clinical, pathological or other phenotypic data. The Core will continue to provide expertise and support affiliated projects and investigators and will harmonize procedures to avoid unnecessary duplication of procedural tests. The Biomarker Core will provide standardized methods for biofluid sample acquisition and storage and biomarker analysis for use by USC ADRC investigators.

The Biomarker core's primary goal is to serve as a central resource for the ADRC to collect, store, track and share plasma, serum, DNA and cerebrospinal fluid (CSF) biosamples. In addition, the Biomarker Core contributes to the goals of the ADRC by performing genetic and biochemical phenotyping of participants followed by the Clinical Core through analyses of *APOE* genotype and Amyloid, Tau and Neurodegeneration (A/T/N) biomarkers. Data from these gold standard analyses will facilitate characterization of longitudinal staging and progression of Clinical Core participants and provide a solid foundation to enable and support national ADRC research efforts and the rich environment of investigator-initiated research at USC. Biomarker phenotyping at this level of detail can allow the ADRC clinical core to personalize the analysis of risk, mechanisms and contributing pathologies, which has major implications for development of treatments and advancement of National Alzheimer's Project Act (NAPA) goals. Our large biobank and foundational biomarker data has historically been and continues to serve as a critical vehicle to attract and establish the careers of young investigators as well as support ongoing, multi-year collaborations with established USC ADRC-affiliated investigators. For example, we were fortunate in the past grant cycle to recruit MOSAIC K99/R00 awardee, Dr. C. Winston to USC, whose R00 research is focused on plasma exosome biomarkers in diverse populations using ADRC biofluids and biomarker data and the NIA Health and Aging Brain Study-Health Disparities (HABS-HD) program. Overall, through the recent recruitments of Drs. Rissman and Winston, both the expertise/infrastructure of ADRC Biomarker Core and the scope of biomarker research leveraging ADRC biofluids and biomarker data at USC has been greatly expanded. The work of Drs. Rissman and Winston have integrated well with the existing legacy biomarker research of established ADRC investigators, focusing on *APOE4*, the role of inflammation, cerebrovascular and BBB injury and dysfunction in ADRD.

Aim 1: **Biofluid collection, processing, storage and sharing**. **1)** Coordinate collection, processing and storage of CSF, PBMCs, blood, plasma and DNA from ADRC Clinical Core participants using national ADRC Best Practices SOPs. **2)** Perform analyses of CSF, plasma and DNA relevant to ADRD using validated methods to facilitate characterization of the ADRC cohort. **3**) Provide ADRC samples and core resources (bioassay, DNA analyses, extractions, processing pipelines) to a broad array of investigators in support of new research projects and foster career development of diverse scientists in biomarker research. Provide expertise for training in biomarker methods. Share plasma and PBMCs with the National Centralized Repository for Alzheimer's Disease and Related Dementias (NCRAD).

Aim 2: **Facilitate biomarker research through an integrated informatics toolkit**: Maintain a state-of-the-art biosample inventory linked with the ADRC DMSC to provide internal and external investigators with CSF and plasma samples, DNA, and derived data in support of ADRD research and novel biomarker discovery.

Aim 3: **Accelerate local, national and international biomarker discovery efforts**: Leverage ADRC biospecimens to develop new methods that inform on neuropathology, diagnosis, progression, prognosis and other research applicable to ADRD. **1)** Leverage ADRC specimens for discovery projects at USC and nationally. **2)** Work with Industry partners to develop and validate novel assays and platforms

A. SIGNIFICANCE

The Biomarker Core (**Core F**) will unite ADRC efforts to standardize the collection and analysis of biofluid samples and brain imaging, along with the sharing of biomarker resources and derived data. In addition to serving as a hub to support these efforts for ADRC participants, we also support collaboration with academic and commercial collaborators. Our ADRC has previously focused on vascular factors related to AD and in this renewal, we expand that focus to characterize factors that modify the effect of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers, emphasizing vascular, lipid, inflammatory, genetic, and ethnic influences. Over the 40-year history of continuous funding of the USC ADRC, the Biomarker Core activities have been well developed and expanded to support the mission and overall theme of the ADRC.

A.1 Leadership. This core will be directed by Dr. Robert Rissman. Dr. Rissman was recruited to USC from UC San Diego in 2023 to initiate a basic science research program at USC's Alzheimer's Therapeutic Research Institute (ATRI). Dr. Rissman took over leadership of the Biomarker Core in early 2024 from Dr. Zlokovic to expand the scope of the core and foster additional synergies with national efforts. Dr. Rissman is a member of the NACC Biomarker Steering Committee and an internationally recognized expert in biomarker discovery and assay standardization, who has been a critical contributor in several cross-ADC studies advancing research in biomarkers. Dr. Rissman has an active, federally funded research program on blood-based biomarker discovery and the application of biomarkers as secondary outcome measures in clinical trials. As founder and leader of the USC ATRI and ACTC Biomarker Cores (and formerly the UCSD ADRC, UCSD ADCS Biomarker Cores), he has managed the collection, analysis, sharing and management of NIH fluid biospecimens from multiple academic and industry-sponsored AD clinical trials and longitudinal cohorts. Dr. Rissman will leverage this expertise and manage the daily operations of the core.

A.2 Biorepository. The biomarker core in Los Angeles will receive, process, and bank blood (DNA and plasma), and CSF from ADRC participants (**Fig. 1**). These biosamples will be used for analyses and characterization of the cohort (e.g. *APOE*, CSF Aβ/tau, etc.) and to support discovery projects, large-scale genetic research coordinated by the National Cell Repository for AD (NCRAD), collaboration with industry/biotechnology companies for assay and methods development, and integrative research within the ADRC and ADC-network. For this renewal, the core infrastructure has been greatly expanded and will now include long term banking and storage at USC ATRI in San Diego. Because ATRI is an off-site facility of USC, on campus space constraints are alleviated. Dr. Rissman's biorepository at ATRI currently has 148 -80 freezers and has the capacity to over 250. Blood and CSF samples collected through the ADRC will be received and processed by Dr. Rissman's staff at an on-site laboratory in Los Angeles supervised by Dr. Abhay Sagare. The on-site laboratory contains all needed equipment, staff, and freezer infrastructure to manage ADRC samples. On a monthly basis, Dr. Sagare will ship ADRC biofluids to Dr. Rissman's ATRI facility for long-term storage, sharing, and analyses.

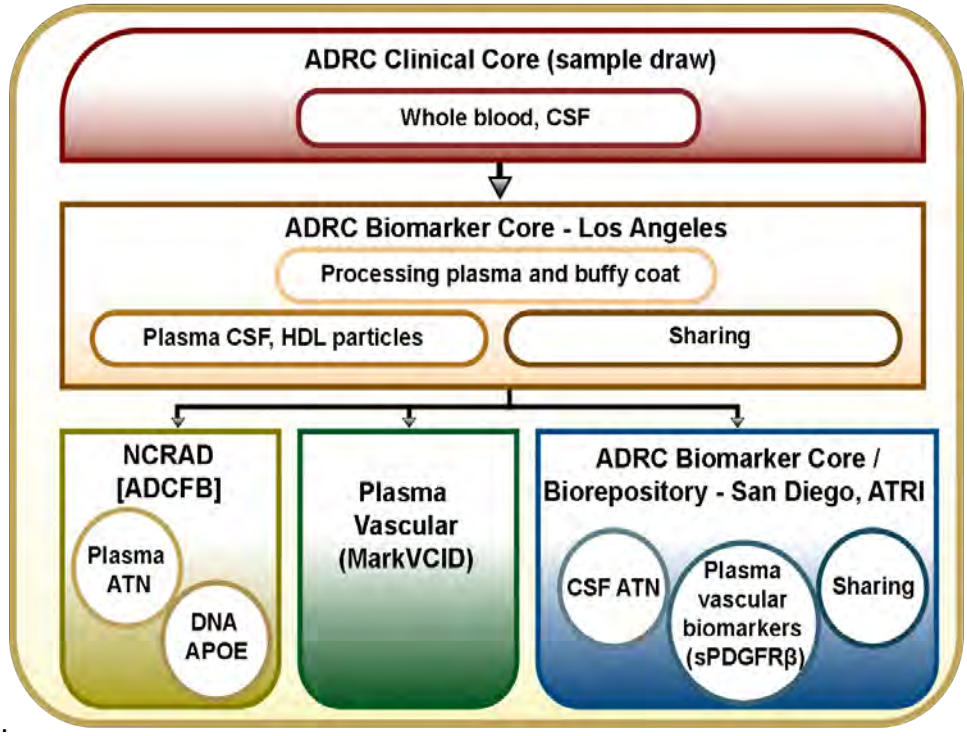


Figure 1. ADRC Biomarker Core workflow.

A.3 Integration of the core in the ADRC. The Biomarker Core will greatly enhance the overall mission of the ADRC by providing its associated researchers with extensive access to expertise and infrastructure,

standardized biosamples, and associated datasets that characterize the ADRC clinical cohort. The biomarker core will also provide novel capabilities for linking biomarker and genetic data to the quantitative pathology obtained in our Neuropathology Core and clinical data obtained in the Clinical Core. This will enable and catalyze research on a variety of topics in neurodegeneration, such as early and accurate diagnosis of AD and ADRD, staging across the AD spectrum, prognosis, heterogeneity of AD, and moving towards personalized profiles of disease risk and mechanisms. The core will ensure standardization and implementation of best practices for sample and imaging acquisition and analysis, and share this expertise with investigators. Rational prioritization and mechanisms regarding consent and collection for biomarkers is provided in the Clinical Core. Finally, the **core will provide resources for training in biomarker measurement and analysis methods, to support many cross-disciplinary training efforts including those of the RL5/REC**.

A.4 MOST IMPORTANT CONTRIBUTIONS TO RESEARCH

A.4.1 Development and validation of novel methods for integration of genetic, clinical, imaging, and pathology data. Support the development and validation of novel methods for boosting power for discovery in GWAS by leveraging pleiotropy among traits and conditions. This includes high-profile publications demonstrating genetic overlap between AD, c-reactive protein, plasma lipids (1), Parkinson's and AD (2), AD and immune-mediated diseases (3), AD and FTD (4), CBD, PSP, and FTD (5), Parkinson's and autoimmune disorders (6) ALS, FTD, and PSP(7). The ADRC also supported the development of vascular dementia and disorders through collaboration with MarkVCID (ref), Diverse VCID.

A.4.2 Biofluid biomarkers:
Neuronal exosome cargo as biomarkers of AD. Dr. Rissman has an active, federally funded program to study the role of plasma exosome biomarkers (8-11) and the use of plasma to identify ADRD neuropathology and predict amyloid-PET positivity (12). Recent data from this work demonstrates that plasma-based methods can identify amyloid-PET positive participants who are asymptomatic from the AHEAD trial (12). Investigations are also ongoing to develop new assays that target TDP43 in plasma to differentiate participants who may have Limbic Predominant TDP43 encephalopathy (LATE), a disease which was discovered/described in a work group Dr. Rissman participates in (13, 14). Dr. Rissman's lab is in the early stages of developing a screening assay to identify participants with TDP43 pathology (15).

Inflammatory biomarkers in plasma. In addition to traditional protein markers of neurodegeneration, Dr. Rissman is also involved in working to understand how subsets of patients with MCI and AD have altered levels of particular cytokines, which can be used to design algorithms to predict response to treatments targeting inflammatory pathways (16-18). Some of this work has led to the development of novel bioassays that, in collaboration with commercial partners, can be used as screening tools in clinical trials (19).

International CSF biomarkers QC program. His multicenter program coordinated by Drs. Blennow and Mattson (University of Gothenburg), with funding from the Alzheimer's Association, is part of an International CSF Quality Control Initiative. The program monitors the degree of variability within and between over 30 participating sites, across several assay platforms, by providing common CSF samples. Four times a year, since 2009, Dr. Rissman's lab has measured core CSF biomarkers on standard CSF samples and sent the resultant data to Dr. Blennow. We are co-authors on two manuscripts generated by this program (20, 21).

A.4.3 Linkage and Incorporation of Lipid and Vascular Biomarkers of relevance to *APOE* e4. We have developed several independent biomarkers to better understand the effects of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers: These include early stage biomarkers (1) metabolism (HDL particles) in both CSF and plasma(22) (2) *APOE* glycosylation profiles in CSF that strongly correlates with *APOE* ε4 neuropathology (22), (3) measures of BBB integrity [sPDGFRβ(23)] independently validated as very early biomarkers in AD (24), and later stage biomarkers (4), lipid inflammation biomarkers (eicosanoids) in CSF, plasma and brain tissues (25), and MarkVCID vascular biomarkers such as the Placental growth factor (PlGF) studied (24).

A.5 PROGRESS (2020-2024): USE OF BIOMARKER CORE RESOURCES

The center has banked DNA and longitudinal plasma samples from over 1000 ADRC participants and CSF from over 500, including samples from subjects who have come to autopsy. Sample collection, storage and distribution to other researchers for this funding cycle are summarized in **Table 3.**

During the past funding period, the ADRC analyzed over 450 CSF samples for core biomarkers of Aβ42, tau and pTau, and performed *APOE* genotyping on 537 DNA samples. The core provided samples for national and international collaborations. Highlights are presented in **Table 4** below.

Table 3. Biomarker Laboratory Sample Inventory and Allocation: 1/1/2020 to 4/30/2024								
	Collected and stored				**Samples distributed last cycle**			
Biofluid	**Buffy**	**Plasma**	**CSF**	**Serum**	**Buffy/blood/PBMC**	**Plasma**	**CSF**	**Serum**
Total aliquots	811	1675	2325	3542	59/61/9	1312	1311	115
Total participants	414		113	284	59/61/9	1091	382	115

Table 4. Area of Research, Investigator, Publication(s)	**Sample**	**N of subjects**	**Funding**
NCRAD Sample Distribution Report (1/1/2020 to 4/20/24) - During the current funding period we sent to NCRAD: 282 plasma, 148 buffy coat, 154 PBMC, 153 DNA, and 53 RNA. - Aliquots from USC ADRC were distributed to 12 investigators who were supported by 14 NIH Grants.	DNA	153	U24AG021886
	Plasma	282	
Yassine/Krauss: HDL particle size in CSF and plasma as an early AD biomarker.	CSF and Plasma	181	R01AG067063
Yassine/Nedelkov: *APOE* and clusterin glycosylation in CSF and plasma as an early AD biomarker.	CSF and Plasma	120	R01AG067063, R43AG079709, R43AG069552
Zlokovic/Toga: Vascular contributions to dementia and genetic risk factors for Alzheimer disease.	CSF	486	P01AG052350
Zandi, Ebrahim: Cost effective Luminex based plasma AD biomarker assay development.	Plasma and CSF	120	Grant in progress
Pinchas (Hassy) Cohen: Mitochondrial micropeptide SHMOOSE.	CSF	100	Epstein Grant
Marmarelis/Chui: Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression.	*APOE* Phenotype (Serum)	20	R01AG058162
DeCarli/Fornage: The clinical significance of incidental white matter lesions on MRI amongst diverse population with cognitive complaints (diverse VCID).	Serum and Plasma	30	1U19NS120384
University of Miami/VICO, Inc/Leuven University/NovoGene/ Amprion: Genetic and fluid biomarkers in genetically determined ADRD.	Plasma, CSF, DNA	100	R01AG069013 R01AG062007 CurePSP
MarkVCID: Validation of imaging and blood-based small vessel VCID biomarkers in a multiethnic population.	Plasma, DNA	300	UH2/UH3/U01 NS100614

B. INNOVATION

B.1 Fluid biomarkers research:

B.1.1 Novel blood-based biomarkers related to neurodegeneration, especially in exosomes. Dr. Rissman applies Nanosight tracking and FACs sorting of exosomes to quantify numbers and characterize cells of origin.

B.1.2 Analysis of inflammatory markers in plasma as predictors of progression of AD.

B.1.3 Discovery of novel CSF biomarkers such as synaptic markers (neuronal pentraxin 2 [NPTX2]); microRNA; proteomic studies in conjunction with other academic investigators.

B.1.4 Novel biomarkers for pathogenic forms of α-Synuclein and tau (collaboration with Amprion), and analysis of exosomes as mediators of spread of pathogenic α-Synuclein.

B.1.5 Analysis of the human microbiome in stool samples collected from Clinical Core participants.

B.1.6 HDL particles in both CSF and plasma using calibrated Ion Mobility platforms at the Center of Personalized Brain Health (CPBH) Biomarker Core, correlate with cognition and amyloid pathology.

B.1.7 *APOE* glycosylation biomarkers in CSF that correlate with AD neuropathology.

B.2 Standardization of core AD CSF biomarkers using new platforms including automation:

B.2.1 FujireBio Lumipulse 2000 instrument. Dr. Rissman will analyze ADRC CSF samples for Aβ42; tau and pTau. Dr. Rissman has advised on an FDA pre-submission based on preliminary data and helped validate this CSF assay for use clinically.

B.2.2 Inter-ADRC Biomarker Harmonization. Dr. Rissman is a steering committee member for the national ADRC Biomarker Core and a member of the ADRCs Best Practices working group.

B.2.3 Validation of mass spectrometry assays for CSF Aβ and tau (collaboration with C2N and Quest Diagnostics). Dr. Rissman has a long-standing collaboration with Quest and C2N Diagnostics in methods development towards new clinical assays for ADRD.

C. APPROACH

C.1 Collection and storage of biosamples. Samples are collected from all ADRC participants as described in the Clinical Core. CSF is obtained on as many subjects as possible at entry, and at follow-up after two years. Blood is drawn for plasma/serum and DNA at entry and every two years. This supports cross-sectional phenotyping of ADRC participants, and longitudinal analysis of changes in biomarkers over time. Collection standardization follows best practices and is managed by well-trained, full-time staff.

C.1.1 Blood/plasma collection. Venous blood samples (30 mL) are drawn by certified phlebotomy staff. DNA is prepared from buffy coat after whole blood samples are spun down to collect plasma. EDTA plasma and buffy coat is prepared and aliquoted into 0.5 mL aliquots in low bind polypropylene cryotubes (0.5 mL, Sarstedt) with barcoded labels; aliquots are then flash-frozen on dry ice and stored at -80°. Dr. Rissman is involved in national and international efforts to standardize plasma and serum biomarkers, focusing on pre-analytical factors, the review of protocols for available bioassays, and storage conditions. Current methodological work aims to foster cross-validation across cohorts and laboratories (26-28).

C.1.2 CSF. SOPs at the Clinical Core include LPs performed with Sprotte atraumatic needles. CSF is collected and initially processed by trained study coordinators at the ADRC, centrifuged (1500g x 10 minutes) using polypropylene tubes, aliquoted into low bind polypropylene cryotubes (0.5 mL, Sarstedt) with barcoded labels, flash-frozen on dry ice and stored at -80°. 2 mL of CSF is sent to the Keck clinical lab for total protein, glucose and cell counts.

C.1.3 Biosample storage. Cryotubes are coded with labels containing an anonymized subject ID, date of draw, and a barcode. The core has dedicated staff at both Los Angeles and San Diego campuses who catalog, and track received samples and share aliquots for analysis and collaboration, managed through an inventory database. Reports on the repository inventory are generated monthly for Biosample and Administrative Core leaders. The Core makes pools and aliquots of standard samples of CSF and plasma to be used for discovery and as common controls. Re-aliquoting of CSF and plasma samples for assays or collaborations that require < 500 uL is managed by core personnel.

All freezers are maintained with a preventative service contract (Sharpe Refrigeration) which replaces filters and display batteries and assesses functionality of compressors. The core only uses freezers with dual compressors (Panasonic) as an additional assurance against freezer failure. All freezer locations have one empty -80 freezer should a backup be needed. Freezers are monitored through the use of emergency alert system (Rees) in the event of failure. In the event of power failure, both Los Angeles and San Diego campuses have backup generators to maintain power to the freezers.

C.1.4 Biosample sharing. We have designed a weblink request form and internal committee review procedures to allow outside investigators to access ADRC samples for research. The website also describes the types of samples that are available and all associated data. Typically, investigators contact Dr. Rissman before official requests for guidance on preliminary data required, sample selection and phenotypic Clinical Core data; statistical consultation is available. For sample requests to be filled, we require background about the proposed assay(s) or studies to judge scientific merit and study design, a commitment to share results with us and to publish them where relevant (for a feasibility study, publication may not be mandated). Prior to moving to USC, Dr. Rissman led the Biospecimen Use Committee for the UCSD ADRC, ADCS and ACTC, and has extensive experience in the sharing of CSF, plasma and DNA samples nationally and internationally. Samples are shipped by trained personnel who follow established SOPs.

C.2 Biofluid biomarker analysis

We summarize the priority biomarkers in **Table 5**.

Table 5. Proposed Biomarkers		
NCRAD	**USC ADRC Biomarkers**	**MarkVCID panel**
Plasma ATN	CSF ATN (ATRI)	PlGF (MarkVCID biomarker lab)
APOE genotype	HDL particles in CSF and plasma (CPBH)	
	sPDGFRβ (MSD, ATRI)	

Plasma ATN and APOE DNA at NCRAD. Samples are processed in the clinic lab or Biomarker Core lab, plasma is aliquoted in 1.5 mL samples in low bind polypropylene cryovial tubes and frozen within two hours of draw. Split samples are shipped to NCRAD (Indiana University), usually in batch based on clinic flow, including WBCs/DNA for the ADGC, with remaining samples stored at the Biomarker Core. NCRAD assays are used to support ATN-informed diagnoses, phenotyping, quality control, validations, cross-validations, laboratory harmonizations, and novel assay development. White cells and iPSCs are stored at the Biomarker Core for future use. The NCRAD plasma sample is for Aβ40, Aβ42, pTau181, pTau217, NfL, GFAP (Quanterix Simoa HD-X) and Aβ42, Aβ40, p-Tau181 (Fujirebio Lumipulse) assays. *APOE* genotypes will be obtained from NCRAD.

C.2.1 CSF ATN at USC ATRI. Analysis of core biomarkers in CSF (Aβ42, tau and pTau181) supports the implementation of Amyloid/Tau/Neurodegeneration (A/T/N) classification as a framework for further research. *APOE* genotyping supports research using ADRC and affiliated cohorts. Dr. Rissman oversees technicians trained in conducting single and multiplex immunoassays with appropriate quality control. For measures of CSF Aβ, tau and other core markers, we have generated data on ADRC CSF using several assays. We now use the Fujirebio Lumipulse assay for Aβ40, Aβ42 and ptau181. Dr. Rissman's lab has a full complement of ELISA and immunoassay equipment also used in ADCS clinical trials, including the MesoScale, Luminex and standard ELISA platforms. With interest in fully automated assays, we are using the Fujirebio Lumipulse reader, and through Dr. Rissman's ACTC Biomarker Core infrastructure, we are exploring use of the Roche Cobas e601 automated immunoassay platform and mass spectrometry for prediction of amyloid-PET positivity (12).

C.2.2 Metabolic and Vascular biomarkers at USC Biomarker Core. Participants will receive additional biomarkers that focus on the *APOE* ε4 theme and lipid biomarkers (HDL particles measured in plasma and CSF using Ion Mobility in the CPBH, Biomarker Core), HgbA1c, standard lipid profiles (Keck Lab), and vascular risk markers (PIGF, MarkVCID) and sPDGFRβ performed at USC. Participants for in-depth biomarker characterization are selected to ensure enrichment for *APOE* ε4 carriers and to support pre-clinical and early AD diagnoses.

C.2.3 Core service to outside investigators. For biomarkers other than Aβ and tau, and for DNA analysis, the core will provide analysis as a recharge service, using research grade assays (e.g., ELISAs or MSD assays) to support affiliated investigators. *Developmental research for novel assays* will be done through independently funded grants, collaborations, and using philanthropic funding.

C.2.4 Standardization and Quality Control:

DNA and Genetic Analyses. In addition to genotype data from NCRAD, *APOE* genotyping will be performed on site by Dr. Rissman's lab. *APOE* genotyping uses a QPCR assay: genomic DNA from blood is extracted using QIAamp DNA blood maxi kit (Qiagen, Venlo, Netherlands) and *APOE* genotyping performed using Applied Biosystems (Foster City, CA) TaqMan SNP Genotyping Assay (29). Determination of genotypes is performed in Dr. Rissman's lab by real-time PCR (BioRad CFX96 C1000 ThermoCycler). In particular, Taqman® reagents are used for real-time analyses because we have found them to yield fast, highly reliable results, allowing for true automated allelic discrimination by 2D plot. Standards for real-time analyses were verified by sequencing to ensure accurate identification. On a bi-weekly basis, *APOE* results are interpreted independently by Dr. Rissman and Biomarker Core technicians. If agreement exists between the two methods and all three observers, the data are transmitted to the ADRC Data Core. If methods do not agree and/or there is disagreement among observers, samples in question are re-analyzed or may be sent out for sequencing.

Biofluids. Dr. Rissman has been part of the Alzheimer's Association International QC program since founding the UCSD ADCS Biomarker Core in 2008. Dr. Rissman led UCSD contributions to this QC program during his tenure at UCSD and this continues at USC, in which we compare our CSF data with that of other laboratories. We have reduced variability by automated pipetting and liquid handling and use four internal CSF standards on every plate.

C.2.5 Relationship with NCRAD. the Clinical Core provides blood samples directly to NCRAD for DNA analysis. The biorepository houses buffy coats, which have been sent to NCRAD on request to augment the amount of DNA; frozen brain samples have been sent from autopsy cases, from our brain bank as needed.

C.3 Training resources. Dr. Rissman will organize and provide didactic lectures and mentorship for center affiliated researchers and REC trainees. The core will train new investigators regarding implementation of best practices for biofluid sample collection and analysis, as well as study design and the biology underlying analytes that are measured in CSF and plasma.

C.3.1 Interaction with other Cores. Working with Dr. Toga, the Data Core's Dashboard provides real-time checking of imaging data quality within the ADRC workflow. The Data Core will provide secure access, authentication, and authorization for local and remote users, storage redundancy, and regular backup for the image repository along with other data objects in the study. The Data Core conversely benefits from expertise within the Biomarker Core to assure that uploaded objects are in compliance with SOPs and are free of personally identifiable information, including embedded metadata (DICOM headers) of uploaded images. To make data and analytics tools available to the local research community and researchers at large, explorative data analysis of integrated ADRC data (demographic, clinical, cognitive, genetic, imaging, etc.) will be provided by the Data Exploration and Analysis Portal (DEAP) web-based software (see Core C: Data Core), with ready linkage to available ACTC or other clinical trial datasets.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	○No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	○No
Is the Project Exempt from Federal regulations?	○Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447179.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Research Education Component (REC)

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: 1540 Alcazar Street		Street2:
* City: Los Angeles	County: Los Angeles	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	REC_Abstract1044447038.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	REC_References_0613241044447055.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Research Education Component

The mission of the University of Southern California (USC) Alzheimer's Disease Research Center (ADRC) Research Education Component (REC) is to direct the scientific careers of scientists from diverse disciplines into a career in Alzheimer's disease and related dementias (ADRD) research. Co-directed by Hussein Yassine and Duke Han and leveraging unique training programs and research resources in the USC academic environment, REC scholars will be trained with an emphasis on age-associated vascular/metabolic processes that affect brain health and ADRD risk in *APOE* ε4 carriers and the Latinx population in Los Angeles (LA) County. Through the USC ADRC REC, trainees will be exposed to an annual program where they are introduced to mentors, oriented to existing ADRC resources (imaging data, clinical trial and longitudinal data, and tissue samples), and provided with an opportunity for pilot funding support. We will integrate current post-doctoral and junior faculty trainees at USC from relevant training and academic programs. ADRC scientists will be mentored in how to advance their career in ADRD research within the context of multidisciplinary collaborations with experts in biostatistics, neuropsychology, medicine, gerontology, sociology, demography, economics, computer science, and social work. This program will support culturally informed research on minority older adult populations and prepare scientists to become independent researchers through three Specific Aims: (1) Support ADRC scientists through a multidisciplinary training program integrated with the USC RCMAR led by Julie Zissomopoulos, pilot funding, and an annual symposium. (2) Support the career development of one post-doctoral investigator from an under-represented background who has shown professional promise for a 2-year post-doctoral fellowship sponsored by the USC Keck School of Medicine Dean's office. The fellowship will be focused on AD risk in diverse populations. (3) Monitor the progress of ADRC scientists during the period of the award as well as through their early career milestones (e.g., receipt of an independent grant). ADRC scientist alumni will be supported through continued involvement in activities that support and improve both their career opportunities and the experience of the next cohort of pilot grantees. To date, the USC ADRC REC has successfully contributed to the development of multiple early career and junior scholars devoted to ADRD research. The continued success of this program will be monitored by the degree to which the scientific career trajectories of the investigators are successful, specifically with respect to an ADRD research focus and extramural funding.

Literature Cited

1. Duron, E. & Hanon, O. Vascular risk factors, cognitive decline, and dementia. *Vascular health and risk management* **4**, 363-381 (2008).
2. Mayer-Davis, E. J. *et al.* Incidence Trends of Type 1 and Type 2 Diabetes among Youths, 2002–2012. *New England Journal of Medicine* **376**, 1419-1429, doi:10.1056/NEJMoa1610187 (2017).
3. Wang, L. *et al.* Ethnic Differences in Risk Factors for Obesity among Adults in California, the United States. *Journal of obesity* **2017**, 2427483-2427483, doi:10.1155/2017/2427483 (2017).
4. Cuellar, I., Arnold, B. & Maldonado, R. Acculturation Rating Scale for Mexican Americans-II: A Revision of the Original ARSMA Scale. *Hispanic Journal of Behavioral Sciences* **17**, 275-304, doi:10.1177/07399863950173001 (1995).
5. Ivanova, I., Salmon, D. P. & Gollan, T. H. The multilingual naming test in Alzheimer's disease: clues to the origin of naming impairments. *Journal of the International Neuropsychological Society : JINS* **19**, 272-283, doi:10.1017/S1355617712001282 (2013).
6. Mungas, D., Reed, B. R., Crane, P. K., Haan, M. N. & Gonzalez, H. Spanish and English Neuropsychological Assessment Scales (SENAS): further development and psychometric characteristics. *Psychological assessment* **16**, 347-359, doi:10.1037/1040-3590.16.4.347 (2004).
7. Teng, E. L. *et al.* The Cognitive Abilities Screening Instrument (CASI): a practical test for cross-cultural epidemiological studies of dementia. *International psychogeriatrics* **6**, 45-58; discussion 62 (1994).

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Hussein Middle Name Last Name*: Yassine Suffix:

Position/Title*:	Associate Professor of Medicine
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	KSOM Clinical
Street1*:	1540 Alcazar Street
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764-3341 Fax Number: E-Mail*: hyassine@usc.edu

Credential, e.g., agency login: hyassine

Project Role*: Other (Specify) Other Project Role Category: Lead

Degree Type: MD Degree Year:

File Name

Attach Biographical Sketch*:

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Duke Middle Name Last Name*: Han Suffix:

Position/Title*:	Professor of Psychology and Family Medicine
Organization Name*:	University of Southern California
Department:	Dornsife Psychology
Division:	Dornsife Life Sciences
Street1*:	University Park Campus
Street2:	HSA Unit22 A-7 7423 1000 S. Fremont Avenue
City*:	Alhambra
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	91803-0000

Phone Number*: (626) 457#5858 Fax Number: E-Mail*: dukehan@usc.edu

Credential, e.g., agency login: DHAN06

Project Role*: Other (Specify) Other Project Role Category: REC Co-Lead

Degree Type: PhD, MA Degree Year:

File Name

Attach Biographical Sketch*: Duke_Han_Biosketch1044446971.pdf

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Julie Middle Name Last Name*: Zissimopoulos Suffix:

Position/Title*:	Professor
Organization Name*:	University of Southern California
Department:	Sol Price Schaeffer Center
Division:	Sol Price Research Centers
Street1*:	University Park Campus
Street2:	VPD 412H
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (213) 821#7947 | Fax Number: | E-Mail*: zissimop@usc.edu

Credential, e.g., agency login: ZISS01

Project Role*: Co-Investigator | Other Project Role Category:

Degree Type: PhD, MA | Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Zissimopoulos1044446973.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Han, Duke

eRA COMMONS USER NAME (credential, e.g., agency login): DHAN06

POSITION TITLE: Professor of Family Medicine, Neurology, Psychology, and Gerontology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Duke University, Durham, NC	BS	05/1998	Psychology/Neurosci
University of Massachusetts Boston, Boston, MA	MA	06/2001	Clinical Psychology
University of Massachusetts Boston, Boston, MA	PhD	06/2004	Clinical Psychology
UCSD/VA San Diego Healthcare System, San Diego, CA	Intern	06/2004	Neuropsychology
VA San Diego Healthcare System, San Diego, CA	Postdoc	06/2005	Neuroimaging/Npsych
UCSD School of Medicine, San Diego, CA	Postdoc	06/2006	Neuroimaging/Npsych

A. Personal Statement

My research experiences and professional training have provided me the necessary expertise to help contribute to this project. I am a Diplomate of the American Board of Professional Psychology in Clinical Neuropsychology, a Fellow of the American Psychological Association and the National Academy of Neuropsychology, Director of the Neuropsychology Division in Family Medicine, and a tenured Professor of Family Medicine, Neurology, Psychology, and Gerontology at the Keck School of Medicine of the University of Southern California. Prior to completing a clinical neuropsychology internship and an NIH T32 postdoctoral fellowship in neuropsychology and neuroimaging at the University of California San Diego (UCSD), I trained with the Brigham Behavioral Neurology Group and the Surgical Planning Laboratory of Harvard Medical School's Brigham and Women's Hospital in advanced neuropsychological assessment and functional neuroimaging approaches respectively. I was awarded a Paul Beeson Career Development Award in Aging Research devoted to the study of financial and healthcare decision making in older age using multi-modal neuroimaging, behavioral economic, neuropsychological, and epidemiologic approaches. I am a PI of multiple grants, including an R01 grant (R01AG055430) investigating factors that mediate racial differences in decision making in older age, an RF1 grant (RF1AG068166) focused on examining the association of financial exploitation in older age with markers of Alzheimer's Disease, and a K24 grant (K24AG081325) focused on the association of financial vulnerability with Alzheimer's Disease. I am also a co-investigator or consultant on a number of federally-funded awards on a wide range of research topics including cholesterol mechanisms and cerebrovascular markers in preclinical Alzheimer's Disease, MRI markers of brain aging in older Black adults, driving decision making in older age, and mentoring in decision making neuroscience.

Ongoing projects I would like to highlight include:

K24AG081325
Han (PI)
4/1/23-1/31/28
Mid-Career Development and Mentoring in Financial Vulnerability and Alzheimer's Disease

RF1AG068166
Han (PI)
5/1/21-4/30/26
Finance, Cognition, and Default Network in Aging (FCDNA)

R01AG055430
Han (PI)
7/15/17-4/30/24
Racial Differences in Decision Making Among Older Adults

R01AG056405
Barnes (PI), Role: co-investigator
7/1/21–6/30/24
MRI Markers of Brain Aging and Risk Factors for Cognitive Decline in Older African Americans

R01AG059613
Betz (PI), Role: co-investigator
5/1/19-4/30/24
Decision Making Among Older Adults; the AUTO study

R01AG064228
Nation (PI), Role: co-investigator
6/1/19-5/31/24
Endothelial Progenitor Cells and Neurovascular Injury in the Aging Brain

R01AG060049
Nation (PI), Role: co-investigator
6/1/19-5/31/24
Cerebrovascular Resistance in Cognitive Aging and Alzheimer's Disease Risk

R24AG076847
Seaman (PI), Role: co-investigator
6/1/22-5/31/27
Research Network on Decision Neuroscience and Aging

Citations:

1. **Han, S.D.,** Gruhl, J.C., Beckett, L., Dodge, H.H., Stricker, N.H., Farias, S., & Mungas, D. (2012). Beta amyloid, tau, neuroimaging, and cognition: Sequence modeling of biomarkers for Alzheimer's disease. *Brain Imaging and Behavior, 6,* 610-620. PMCID: PMC3582336
2. **Han, S.D.,** Suzuki, H., Jak, A.J., & Bondi, M.W., Chang, Y.-L., Salmon, D.P., & Bondi, M.W. (2010). Hierarchical cognitive and psychosocial predictors of amnestic mild cognitive impairment. *Journal of the International Neuropsychological Society, 16(4),* 721-729. PMCID: PMC3005198
3. **Han, S.D.,** Boyle, P.A., Yu, L., Arfanakis, K., James, B.D., Fleischman, D.A., & Bennett, D.A. (2016). Grey matter correlates of susceptibility to scam in community-dwelling older adults*. Brain Imaging and Behavior, 10(2),* 524-532. PMCID: PMC4689662
4. **Han, S.D.,** Boyle, P.A., James, B.D., Yu, L., & Bennett, D.A. (2015). Mild cognitive impairment is associated with poorer decision-making in community-based older persons. *Journal of the American Geriatrics Society, 63,* 676-683. PMCID: PMC440679

B. Positions, Scientific Appointments, and Honors

Positions and Employment:

2020- Professor, Dept. of Family Medicine (primary appointment), University of Southern California, Alhambra, CA

2020- Professor, Dept. of Neurology (secondary appointment), University of Southern California, Los Angeles, CA
2020- Professor, Dept. of Psychology (courtesy appointment), University of Southern California, Los Angeles, CA
2020- Professor, Dept. of Gerontology (courtesy appointment), University of Southern California, Los Angeles, CA
2016-2020 Associate Professor, Dept. of Family Medicine (primary appointment), University of Southern California, Alhambra, CA
2016-2020 Associate Professor, Dept. of Psychology (courtesy appointment), University of Southern California, Los Angeles, CA
2016-2020 Associate Professor, Dept. of Neurology (secondary appointment), University of Southern California, Los Angeles, CA
2016-2020 Associate Professor, Dept. of Gerontology (courtesy appointment), University of Southern California, Los Angeles, CA
2016-2017 Visiting Professor, Dept. of Neurological Sciences, Rush University Medical Center, Chicago, IL
2016-2017 Visiting Professor, Dept. of Behavioral Sciences, Rush University Medical Center, Chicago, IL
2016- Director, Neuropsychology Division, Dept. of Family Medicine, University of Southern California, Alhambra, CA
2014-2015 Conjoint Associate Professor, Dept. of Neurological Sciences, Rush University Medical Center, Chicago, IL
2014-2015 Associate Professor, Dept. of Behavioral Sciences, Rush University Medical Center, Chicago, IL
2012-2015 Rush Alzheimer's Disease Center, Rush University Medical Center, Chicago, IL
2012-2015 Director of Neuropsychology, Mental Health Care Group, VA Long Beach Healthcare System
2012-2014 Conjoint Assistant Professor, Dept. of Neurological Sciences, Rush University Medical Center, Chicago, IL
2009-2014 Assistant Professor, Dept. of Behavioral Sciences, Rush University Medical Center, Chicago, IL
2007-2009 Clinical Assistant Professor, Dept. of Neurology, Loyola University Medical Center, Maywood, IL
2006-2009 Assistant Professor, Dept. of Psychology, Loyola University Chicago, Chicago, IL

Professional Memberships:

2016-present Member, Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment (ISTAART)
2012-present Member, American Board of Professional Psychology
2012-present Member, American Academy of Clinical Neuropsychology
2006-present Member, National Academy of Neuropsychology
2003-present Member, International Neuropsychological Society
2000-present Member, American Psychological Association

Honors:

2023 Elected Representative to American Psychological Association (APA) Council, Society for Clinical Neuropsychology
2022 Elected Co-Chair, Dementia Special Interest Group (SIG) of the International Neuropsychological Society (INS)
2022 Elected Fellow, National Academy of Neuropsychology
2022 Scientific Advisory Board, Alzheimer's Disease Neuroimaging Initiative
2021 Elected Fellow, American Psychological Association
2021- Executive Board Member, Aging Research in Criminal Justice Health (ARCH) Network
2020-2021 Chair of Review, NIH/NIA Clinical and Translational Research of Aging (NIA-T) Review Committee
2019-2020 Keck School of Medicine of USC Healthcare Leadership Academy
2019-2020 Chair of Review, NIH/NIA Neuroscience of Aging (NIA-N) Review Committee
2017 Clinical Scholar designation by the President of the University of Southern California
2017 Board of Directors nomination, American Academy of Clinical Neuropsychology
2016-2021 Oral Examiner for Board Certification, American Board of Clinical Neuropsychology
2016 Graduate Student Organization Honoree, Wayne State University Institute of Gerontology
2015 Best Poster Finalist, Alzheimer's Imaging Consortium

2015 Walking Tour Selection, Alzheimer's Imaging Consortium
2014 Alzheimer's Association International Conference Abstract Reviewer
2014 Paul B. Beeson Fellowship Planning Committee for Annual Meeting
2015- Inaugural Governance Committee Member, AARP Global Council on Brain Health
2015-2016 International Neuropsychological Society Annual Meeting Program Committee
2014-2017 National Academy of Neuropsychology Publications Committee
2013 Nominated for APA Division 40: Clinical Neuropsychology Member-At-Large
2012-2017 Paul B. Beeson Career Development Award for Aging Research
2012, 2019 American Psychological Association Division 40 Blue Ribbon Award
2012- International Neuropsychological Society Continuing Education Committee
2011 International Neuropsychological Society Annual Meeting Program Committee
2011 Advanced Psychometrics Methods Workshop: ADNI Participant
2011-2014 American Psychological Association Division 40 Program Committee
2008 NIA Summer Research Institute Participant
2006 International Neuropsychological Society Nelson Butters Award
2006 Alzheimer's Association New Investigator Neuroimaging Award Honorable Mention
2003 Elected Chief Intern of UCSD Internship Class 2003-2004

Reviewership: Ad-hoc journal reviewer for 35 peer-review scientific journals, editorial board member for 2 scientific journals (*Archives of Clinical Neuropsychology*, *Frontiers in Aging Neuroscience*)

C. Contributions to Science

1. My efforts to elucidate the cognitive, behavioral, and neural correlates of decision making in old age have yielded significant insights into possible risk and protective factors and potential neuroimaging biomarkers of financial abuse, fraud and exploitation of older adults.
    a. **Han, S.D.,** Boyle, P.A., Arfanakis, K., Fleischman, D.A., Yu, L., James, B.D., & Bennett, D.A. (2016). Financial literacy is associated with white matter integrity in old age. *NeuroImage, 130,* 223-229. PMCID: PMC4808430
    b. **Han, S.D.,** Boyle, P.A., Yu, L., Arfanakis, K., James, B.D., Fleischman, D.A., & Bennett, D.A. Grey matter correlates of susceptibility to scam in community-dwelling older adults. (In press). *Brain Imaging and Behavior.* PMCID: PMC4689662
    c. **Han, S.D.,** Boyle, P.A., Yu, L., Fleischman, D.A., Arfanakis, K., & Bennett, D.A. (2013). Ventromedial PFC, parahippocampal, and cerebellar connectivity are associated with temporal discounting in old age. *Experimental Gerontology, 48,* 1489-1498. PMCID: PMC3855854
    d. **Han, S.D.,** Boyle, P.A., James, B.D., Yu, L., & Bennett, D.A. (2015). Mild cognitive impairment is associated with poorer decision-making in community-based older persons. *Journal of the American Geriatrics Society, 63,* 676-683. PMCID: PMC440679

2. My application of advanced statistical and neuroimaging approaches to better understand risk for Alzheimer's Disease, preclinical Alzheimer's Disease, or recovery from brain injury has yielded new insights into these conditions.
    a. **Han, S.D.,** Suzuki, H., Jak, A.J., & Bondi, M.W., Chang, Y.-L., Salmon, D.P., & Bondi, M.W. (2010). Hierarchical cognitive and psychosocial predictors of amnestic mild cognitive impairment. *Journal of the International Neuropsychological Society, 16(4),* 721-729. PMCID: PMC3005198
    b. **Han, S.D.,** Suzuki, H., Drake, A.I., Cessante, L.M., Jak, A.J., Houston, W.S., & Bondi, M.W. (2009). Clinical, cognitive, and genetic predictors of change in job status following traumatic brain injury in a military population. *Journal of Head Trauma and Rehabilitation, 24(1),* 57-64. PMCID: PMC3319716
    c. **Han, S.D.,** Gruhl, J.C., Beckett, L., Dodge, H.H., Stricker, N.H., Farias, S., & Mungas, D. (2012). Beta amyloid, tau, neuroimaging, and cognition: Sequence modeling of biomarkers for Alzheimer's disease. *Brain Imaging and Behavior, 6,* 610-620. PMCID: PMC3582336
    d. **Han, S.D.,** Arfanakis, K., Fleischman, D., Leurgans, S., Tuminello, E.R., Edmonds, E.C., & Bennett, D.A. (2012). Functional connectivity variations in mild cognitive impairment: Associations with cognitive function. *Journal of the International Neuropsychological Society, 18,* 39-48. PMCID: PMC3368801

3. My use of neuroimaging and neurocognitive assessment approaches to better understand the association of apolipoprotein E genotype with cognition across the lifespan has informed theoretical models linking genetics, cognition, and brain compensatory mechanisms.
    a. **Han, S.D.**, Houston, W.S., Jak, A.J., Eyler, L.T., Nagel, B.J., Fleisher, A.S., Brown, G.G., Corey-Bloom, J., Salmon, D.P., Thal, L.J., & Bondi, M.W. (2007). Verbal paired-associate learning by APOE genotype in non-demented adults: fMRI evidence of a right hemispheric compensatory response. *Neurobiology of Aging, 28(2),* 238-247. PMCID: PMC1705815
    b. **Han, S.D.,** Drake, A.I., Cessante, L.M., Jak, A.J., Houston, W.S., Delis, D.C., Huynh, D.V., Filoteo, J.V., & Bondi, M.W. (2007). Apolipoprotein E and traumatic brain injury in a military population: evidence of a neuropsychological compensatory mechanism? *Journal of Neurology, Neurosurgery, and Psychiatry, 78,* 1103-1108. PMCID: PMC2117544
    c. **Han, S.D.,** & Bondi, M.W. (2008). Revision of the Apolipoprotein E Compensatory Mechanism Recruitment Hypothesis. *Alzheimer's and Dementia, 4(4),* 251-254.
    d. **Han, S.D.,** Bangen, K., & Bondi, M.W. (2009). Functional Magnetic Resonance Imaging of Compensatory Neural Recruitment in Aging and Risk for Alzheimer's Disease: Review and Recommendations. *Dementia and Geriatric Cognitive Disorders, 27(1),* 1-10. PMCID: PMC2820574

4. My early work focused on the long-term effects of chronic schizophrenia in old age. By integrating computer modeling, cognitive neuroscience, and functional neuroimaging approaches to investigate the semantic language disturbance of schizophrenia in old age, I was able to associate auditory hallucinations with aberrant functional activity in brain language regions among older schizophrenia patients.
    a. **Han, S.D.**, Nestor, P.G., Shenton, M.E., Niznikiewicz, M., Hannah, G., & McCarley, R. (2003). Associative memory in chronic schizophrenia: A computational model. *Schizophrenia Research, 61(2-3),* 255-263. PMCID: PMC2849103
    b. **Han, S.D.**, Nestor, P.G., Hale-Spencer, M., Cohen, A., Niznikiewicz, M., McCarley, R.W., & Wible, C.G. (2007). Functional neuroimaging of word priming in males with chronic schizophrenia. *NeuroImage, 35,* 273-282. PMCID: PMC1852450
    c. **Han, S.D.,** & Wible, C.G. (2010). Neuroimaging of Semantic Processing in Schizophrenia: A Parametric Priming Approach. *International Journal of Psychophysiology, 75(2),* 100-106. PMCID: PMC2827622
    d. Wible, C.G., **Han, S.D.**, Spencer, M.H., Kubicki, M., Niznikiewicz, M.H., Jolesz, F.A., McCarley, R.W., & Nestor, G. (2006). Connectivity among semantic associates: An fMRI study of semantic priming. *Brain and Language, 97,* 294-305.

5. My efforts in applying clinical neuropsychological assessment methods to real-world outcomes have produced a number of studies that inform community health and clinical practice.
    a. Luboyeski, E.J., **Han, S.D.,** Lansing, A.E., Holdnack, J.A., & Delis, D.C. (2009). Development of Set-Shifting Ability From Late Childhood through Early Adulthood. *Archives of Clinical Neuropsychology, 24(6),* 565-574.
    b. Schiehser, D.M., **Han, S.D.,** Zizak, V., Song, D.D., Lessig, S., & Filoteo, J.V. (2009). Predictors of health status in nondepressed and nondemented Individuals with Parkinson's Disease. *Archives of Clinical Neuropsychology, 24,* 699-709. PMCID: PMC2770862
    c. Ruppert, P., Edmonds, E.C., Brook, M., Musil, S., & **Han, S.D.** (2012). Neuropsychological assessment in a case of adult-onset hemophagocytic lymphohistiocytosis (HLH). *The Clinical Neuropsychologist, 26(12),* 1038-1052.
    d. Wingo, J.M., Kalkut, E.J., Tuminello, E.R., Asconape, J., & **Han, S.D.** (2013). Executive functions, depressive symptoms, and college adjustment in women. *Applied Neuropsychology: Adult, 20(2),* 136-144.

Complete List of Published Work in My Bibliography:

https://www.ncbi.nlm.nih.gov/myncbi/s.%20duke.han.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Zissimopoulos, Julie

eRA COMMONS USER NAME: ZISS01

POSITION TITLE: Professor

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date	FIELD OF STUDY
Boston College	B.A.	06/1991	Political Science, Spanish
Columbia University	M.A.	06/1993	International and Public Affairs
University of California, Los Angeles	M.A., Ph.D.	06/1998, 2000	Economics

A. Personal Statement

I have the leadership and mentorship skills, expertise and motivation to support USC's ADRC. My own research focuses on identifying opportunities to reduce racial/ethnic disparities in risk of Alzheimer's disease and other dementias and improve care for diverse persons living with dementia. In particular, my research addresses pharmaceutical innovation in Alzheimer's disease, caregiving and health care costs of dementia, and medical care of persons living with dementia. It also includes research on medication adherence, insurance design, medical expenditures, and work and savings. I approach these topics from the perspective of an empirical economist with expertise in health economics, labor economics and econometrics and with a multidisciplinary approach that comes from collaborations with colleagues from myriad disciplines including gerontology, social work, medicine, pharmacology, psychology, and demography. I currently lead a research program at USC's Schaeffer Center on Aging and Cognition. With funding from the National Institute on Aging, in collaboration with a multidisciplinary group of scholars, I direct a research team that includes predoctoral and postdoctoral fellows, and a programming team with expertise in Medicare claims data and large panel studies. I am an experienced administrator, director and mentor and have heavily invested in training the next generation of social scientist scholars working in dementia research. I have over 20 years of experience directing post- and pre-doctoral training programs. I am Principal Investigator of the NIH-funded Resource Center for Minority Aging Research (RCMAR) focused on Alzheimer's disease and other dementias, Principal Investigator of the NIA-funded Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease and Related Dementias (CeASES-ADRD). I have a track-record of mentoring diverse scholars in aging research. I have served as mentor to 17 postdoctoral fellows and 17 predoctoral fellows, and 11 assistant professor. All of my mentees who have completed training have secured positions at accredited universities or research positions at research organizations.

Ongoing and completed projects that I would like to highlight include:

United States Cost of Dementia Model: Quantifying Current and Future Impacts of Alzheimer's Disease and Related Dementias
6/1/2024 – 05/31/2029
U01AG086827 (Zissimopoulos, Goldman)

The goal is to provide annual, national, and comprehensive estimates of the cost of dementia in the U.S, and to build research infrastructure to support dementia research.

Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease and Related Dementias (CeASES-ADRD)
07/01/20 – 06/30/25
P30 AG066589-01 (Zissimopoulos/Goldman/Hayward/Bhattacharya)
The goal is to advance innovative social science ADRD research, to increase knowledge and technical capability of social science researchers, to broaden and diversify the set of researchers working in the field and, to disseminate resources and communicate findings for impact.
Role: Principal Investigator

USC Resource Centers for Minority Aging Research (RCMAR)
09/01/12 – 06/30/23
P30 AG043073 (Zissimopoulos/ Goldman)
The goal is to increase the number, diversity and success of researchers focusing on the health and economic wellbeing of the elderly. Special topic area is Alzheimer's disease and other dementias.
Role: Principal Investigator

USC Roybal Center for Behavioral Interventions in Aging
08/01/17 – 05/01/21
P30 AG024968 (Doctor/Goldman)
The Roybal Center for Behavioral Interventions in Aging aims to encourage safe and effective medical decisions through behavioral economic intervention. The Center will support a Pilot Core for research and a Management and Administrative Core that oversees all Center activities. The Center will sustain and grow the community of researchers conducting behavioral interventions to encourage appropriate use of medical treatments that affect people aging through the lifespan.
Role: Co-Investigator

The Importance of the Neighborhood Environment in Determining Health and Well-being among Persons with Dementia and their Caregivers
07/01/19 – 06/30/24
NIA R01AG064491 (Ailshire)
The proposed project examines the importance of neighborhood-based supports and stressors in the health and well-being of people with dementia and their caregivers.
Role: Co-Investigator

Citations:

1. Zissimopoulos JM, Barthold D, Brinton RD, Joyce G. Sex and Race Differences in the Association Between Statin Use and the Incidence of Alzheimer Disease. JAMA Neurol. 2017; 74(2):225–232.
2. Barthold D, Joyce G, Wharton W, Kehoe P, Zissimopoulos J (2018) The association of multiple anti-hypertensive medication classes with Alzheimer's disease incidence across sex, race, and ethnicity. PLoS ONE 13(11): e0206705. https://doi. org/10.1371/journal.pone.0206705.
3. Drabo E., D. Barthold, G. Joyce, P. Ferido H. Chui, J. Zissimopoulos (2019), "Longitudinal Analysis of Dementia Diagnosis and Specialty Care Among Racially Diverse Medicare Beneficiaries," *Alzheimer's and Dementia.* Vol. 15(11), 1402-1411.
4. Jacobson, M., J. Thunell, J. Zissimopoulos (2021), ""Cognitive Screening at Medicare's Annual Wellness Visit Among Beneficiaries in Fee-for-Service and Medicare Advantage Plans," Health Affairs. 39(11): 1935-1942.

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2020 – present	Health Policy Section Editor, Alzheimer's and Dementia, Jo. of the Alzheimer's Association
2020 - present	Director, Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease and Related Dementias (NIA), USC Schaeffer Center

2019 – present	Associate Professor by Courtesy, USC, Davis School of Gerontology
2019 – present	Deputy Editor, Demography
2018 – present	Director, Aging and Cognition Research Program, USC Schaeffer Center
2017 – present	Director, Research Training, USC Schaeffer Center for Health Policy and Economics
2017 – present	Associate Professor, University of Southern California, Sol Price School of Public Policy
2016 – 2019	Vice Dean Academic Affairs, Price School of Public Policy, University of Southern California
2014 – 2017	Assistant Professor, University of Southern California, Sol Price School of Public Policy
2012 – present	Co-Director, NIA Resource Center for Minority Aging Research, USC
2011 – 2017	Associate Director, Leonard D. Schaeffer Center for Health Policy and Economics
2011 – 2014	Research Associate Professor, University of Southern California, School of Pharmacy
2010 – 2011	Senior Economist, RAND Corporation, Santa Monica, CA
2007 – 2011	Professor of Economics, Pardee RAND Graduate School
2006 – 2011	Director, RAND Postdoctoral Training Program in the Study of Aging
2004 – 2010	Economist, RAND Corporation, Santa Monica, CA
2000 – 2004	Associate Economist, RAND Corporation, Santa Monica, CA
1999 – 2000	Teaching Assistant Consultant, Department of Economics, UCLA
1997 – 2000	Teaching Associate, Department of Economics, UCLA
1996 – 2000	Consultant, RAND

Honors

2024	High Impact Research Award
2021	Journal of Gerontology, Social Sciences, Editor's Choice Paper
2019-2024	USC Price School of Public Policy Gold Level Research Award
2018	USC Price School of Public Policy Silver Level Research Award
2017	USC Price School of Public Policy Bronze Level Research Award
2016	USC Price School of Public Policy Bronze Level Research Award
1991	Summa Cum Laude, Boston College
1991	Phi Beta Kappa, Boston College

C. Contributions to Science

1. Alzheimer's disease and related dementias (ADRD) is costly for individuals and society and the burden is higher for non-Whites than Whites. We estimated the prevalence and incidence of ADRD in diverse populations. We quantified the medical care and caregiving costs associated with it and projected future cost under different treatment scenarios. We examined how therapeutics for chronic conditions provide opportunities to reduce disparities in risk. We quantified disparities in dementia diagnosis and medical care of persons living with dementia. The research informs clinical and policy opportunities to reduce racial and ethnic disparities across the dementia continuum.
    a. Chen, Y., E. Crimmins, P. Ferido, J. Zissimopoulos (2022), "Racial/Ethnic Disparities in Length of Life after Dementia Diagnosis: an 18-Year Follow-up Study of Medicare Beneficiaries," *The Lancent Regional Health - Americas*
    b. Zhu, Yingying, Y Chen, E. Crimmins, J. Zissimopoulos, (2020), "Sex, Race and Age Differences in Prevalence of Dementias in Medicare Claims and Survey Data," *Journal of Gerontology, Social Sciences.*
    c. Zissimopoulos, Julie, B. Tysinger, P. StClair, E. Crimmins (2018), "The Impact of Changes in Population Health and Mortality on Future Prevalence of Alzheimer's Disease and Other Dementias in the United States." *The Journals of Gerontology: Series B*, Vol.73(1), 16 April 2018, S38–S47.
    d. Zissimopoulos, Julie, E. Crimmins and P. St.Clair (2014). "The Value of Delaying Alzheimer's Disease Onset," *Forum for Health Economics and Policy. Nov; 18(1):25-39.*

2. Threats to leading secure lives at older ages come from multiple sources including longevity, poor health and financial risk. My research shows how government and employers can help families mediate risk by providing, for example, financial safety nets and savings vehicles (e.g. Social Security and pensions), and health insurance (e.g. Medicaid, Medicare and employer provided plans). My research shows how incentives in social insurance programs and changes to them (Social Security and Medicare) impact work and health and older ages.

a. Goldman, D., C. Chen, J. Zissimopoulos, J. Rowe (2018), “Measuring how Nations Adapt to Societal Aging,” *Proceedings of the National Academies of Science*, 115(3): 435-437. DOI:10.1073/pnas.1720899115.
b. Joyce G., L. Henkhaus, L. Gascue, J. Zissimopoulos (2018), “Generic Drug Price Hikes and Patient Out-of-Pocket Costs for Medicare Beneficiaries.” Health Affairs 37(10): 1578-1586.
c. Zissimopoulos, Julie, G. Joyce, L. Scarpati*, D. Goldman (2015), “Did Medicare Part D Reduce Disparities?” *American Journal of Managed Care*, 21(2): 119-128.
d. Maestas, Nicole and Julie Zissimopoulos (2010). “How Longer Work Lives Ease the Crunch of Population Aging,” *Journal of Economic Perspectives* 24(1): 139-60.

3. In a series of articles, my colleagues and I shed new light on how individuals and families support health and financial well-being over their lifecycle and across generations.
    a. Chen C, Thunell J, Zissimopoulos J. Changes in physical and mental health of Black, Hispanic, and White caregivers and non-caregivers associated with onset of spousal dementia. Alzheimers Dement (N Y). 2020 Nov 2;6(1):e12082. doi: 10.1002/trc2.12082. PMID: 33163612; PMCID: PMC7606182.
    b. Zissimopoulos, J., J. Thunell, S. Mudrazija (2019) “Parental Income and Wealth Loss and Transfers to their Young Adult Children,” *Journal of Family and Economic Issues.* https://doi.org/10.1007/s10834-019-09645-z
    c. Lee, H., Myers, D., Painter, G., Thunell, J., & Zissimopoulos, J. (2018). The role of parental financial assistance in the transition to homeownership by young adults. Journal of Housing Economics. doi:https://doi.org/10.1016/j.jhe.2018.08.002
    d. Asch, E., Steven Haider, and Julie Zissimopoulos (2005). “Financial Incentives and Retirement: Evidence from Federal Civil Service Workers” *Journal of Public Economics:* 89(3) 427-440.

Complete List of Published Work in MyBibliography

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

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5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

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2. Specific Aims	REC_Aims_0613241044447056.pdf
3. Research Strategy*	REC_Research_Plan_0613241044447057.pdf
4. Progress Report Publication List	REC_pubs_FINAL_6_14_241044447226.pdf
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
9. Letters of Support	
10. Resource Sharing Plan(s)	
11. Other Plan(s)	
12. Authentication of Key Biological and/or Chemical Resources	
13. Appendix	
12. Appendix	

SPECIFIC AIMS

The mission of the University of Southern California (USC) Alzheimer's Disease Research Center (ADRC) Research Education Component (REC) is to direct the scientific careers of scientists from diverse disciplines into a career in Alzheimer's disease and related dementias (ADRD) research in general, and with an emphasis on age-associated vascular/metabolic processes that affect brain health and ADRD risk in *APOE* ε4 carriers and the Latinx population in Los Angeles (LA) County.

Obesity, diabetes, dyslipidemia and hypertension have been found at disproportionally high rates in Latinx communities. These metabolic and vascular risk factors have also been associated with an increased AD incidence and age-related cognitive decline[1]. However, very few interventions have been designed to target the risk factors specific to these populations. We propose to target research areas that are focused on Latinx older brain health through training programs that include (1) lifestyle risk factors and interventions to counteract the effects obesity and diabetes by interventions such as weight loss and exercise, (2) family-based and economic/health care interventions, (3) psychometric testing applicable to the Latinx population, and (4) population and genetic research focusing on AD risk in ethnic minority groups.

The transition from junior investigator to independent researcher after completion of a doctoral program is challenging. Launching a new research program involves forming new collaborations, obtaining grant funding, and publishing. These tasks require significant mentorship and a tailored training program. The success of this program will be monitored by the degree to which the scientific career trajectories of the investigators are successful, specifically with respect to an ADRD research focus.

Through the REC, trainees are exposed to an annual program where they are introduced to mentors, oriented to existing ADRC resources (imaging data, clinical trial and longitudinal data, and tissue samples), and provided with an opportunity for pilot funding support. We will integrate current post-doctoral and junior faculty trainees at USC from relevant training and academic programs. Specifically, junior investigators, known as ADRC scientists, will be recruited from several training programs at USC that focus on aging and ADRD. These include AD-RCMAR, the Schaeffer Center and the Roybal Center for Health Policy Simulation, Roybal Center for Health Decision Making and Financial Independence in Old Age, and Roybal Institute on Aging, The Center for Economic and Social Research, Brookings Institute, the Andrus Gerontology Center, the Clinical Psychology PhD program, Multi-Modal imaging T32 program and USC Alzheimer's Therapeutic Research Institute (ATRI) Biostatistics Training Core. This provides an extraordinarily rich environment of early career investigators

ADRC scientists will be mentored in how to advance their career in ADRD research within the context of multidisciplinary collaborations with experts in biostatistics, neuropsychology, medicine, gerontology, sociology, demography, economics, computer science, and social work. This program will support culturally informed research on minority older adult populations and prepare scientists to become independent researchers. To this end, we have three Specific Aims:

1. Support ADRC scientists through a multidisciplinary training program integrated with the USC AD-RCMAR, pilot funding, development projects, and an annual symposium. Our training emphasis will be focused in four areas: (1) lifestyle risk factors and interventions, (2) family and healthcare decisions, (3) cross-cultural neuropsychology, and (4) population and genetic research. Our goal is to make them competitive in securing an award for an independent, investigator-initiated research project. Integration with RCMAR is part of our Plan to Enhance Diverse Perspectives (**PEDP-RCMAR**)

2. Support the career development of one post-doctoral investigators from an under-represented background who has shown professional promise for a 2-year post-doctoral fellowship sponsored by the Keck School of Medicine Dean's office. The fellowship will be focused on ADRD studies in underserved populations, as part of our Plan to Enhance Diverse Perspectives (**PEDP-RCMAR**).

3. Monitor the progress of ADRC scientists during the period of the award as well as through their early career milestones (e.g., receipt of an independent grant). ADRC scientist alumni will be supported through continued involvement in activities that support and improve both their career opportunities and the experience of the next cohort of pilot grantees.

REC Research Strategy

The risk of dementia among *APOE* ε4 carriers varies considerably particularly among different ethnicities and vascular risk factors. Type 2 diabetes prevalence has dramatically increased over the last decade; especially among ethnic minority groups in which prevalence is typically 1.5- to 2-fold higher than non-Hispanic whites[2]. It was estimated that 25.8 million Americans (~8.3% of the population) had diabetes in 2011, adding an increased fiscal burden for Americans (www.cdc.gov/diabetes/pubs/factsheet11.htm). This trend is reflected in ethnically diverse metropolitan areas such as Los Angeles (**Fig. 1**), where the prevalence of diabetes increased by 38% from 1997 to 2007 [3], despite diabetes/obesity prevention efforts. However, our understanding of the contributions of obesity, diabetes, vascular, and genetic risk factors to Alzheimer's disease and related dementias (ADRD) risk in the Latinx population remains unclear and highlights a major gap in the scientific literature. One way to address this gap is to support early career scientists interested in conducting research on AD risk in underrepresented populations.

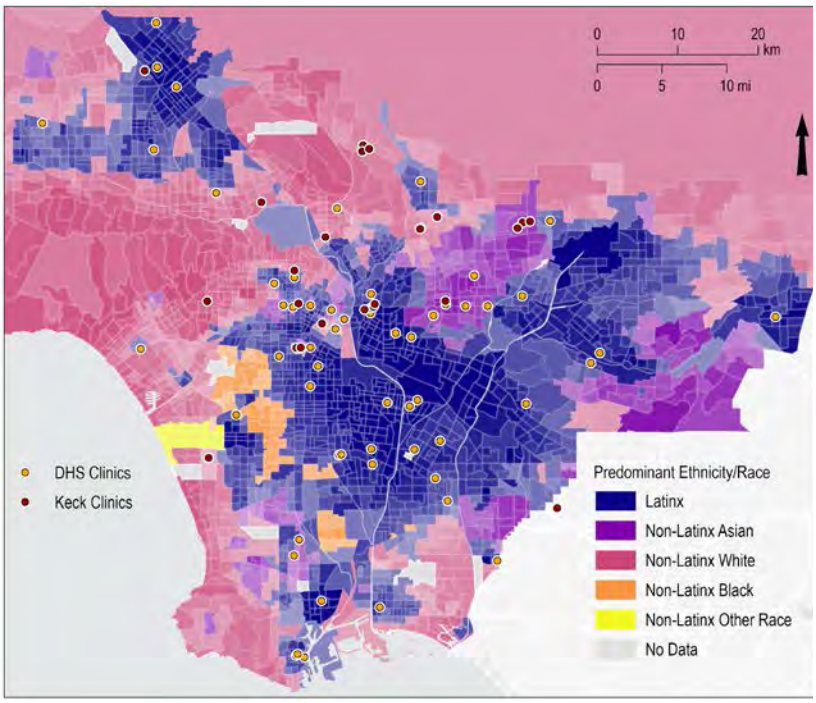


Figure 1 shows the ethnic diversity in LA area surrounding USC research and clinical sites.

The transition from junior investigator to independent researcher requires a significant investment of time, resources, and effort by senior faculty. The REC training program will be customized to complement existing ADRD training programs at USC and fill a key gap. We will leverage the outstanding resources of USC's investigators to provide top-rate training and mentorship in this important area.

USC has outstanding research programs in population studies of obesity, diabetes, aging and ADRD-related studies. The Roybal Clinic Alzheimer's Disease Research Program is located at the Roybal Comprehensive Healthcare Clinic and is led by Dr. Yassine (REC Co-Leader). This program is conducting studies focused on addressing how lifestyle, metabolic and vascular risk factors affect the risk of cognitive decline and AD in LA's Latinx population using both longitudinal cohorts and clinical trials. This is in collaboration with USC Alzheimer's Disease Research Center (ADRC) investigators (Aranda, Ringman, Chui D'Orazio, and others), with the overarching goal of studying multiple aspects of AD risk factors in Latinx populations as described below.

The USC ADRC REC will advance the ADRC center goals by offering ADRC scientists multidisciplinary training and mentorship necessary to: (1) design clinical studies (trials, longitudinal studies), (2) assess cross-cultural neuropsychology outcomes, (3) utilize family-based approaches, and (4) use the rich, longitudinal population-level data sources including administrative and survey data on chronic disease challenges of Latinx older adults. The REC will prepare mentees with the essential skills needed to become successful independent researchers. It will also establish mechanisms for cultivating improved representation by supporting scientists from diverse backgrounds who bring an added level of insight to AD risk research. The activities of the REC will have a significant and long-lasting impact on the quality of research conducted to improve the well-being of underrepresented aging populations.

The faculty involved in the REC program are currently involved in a wide array of academic activities as part of various existing training programs at USC. USC has an extremely rich training ground for the next generation of research leaders in the area of vascular/metabolic risk and AD among underrepresented older adults. Specifically, the REC program activities will:

- Provide information to facilitate access to tissues (blood, CSF or DNA), and data from (1) longitudinal studies and clinical trials at the USC ADRCS and all ADRCs, (2) MRI, and PET imaging through LONI (3) population data through the RCMAR program.
- Provide an intensive training course to provide a multidisciplinary perspective that will enable mentees to recognize the impact of obesity, diabetes, and metabolic diseases on populations and make the best use of available data. ADRC scientists will have intensive training in methods and theories to ask questions and pursue the best possible answers during an intensive training course. This is a key benefit, given the interdisciplinary nature of the field of aging.
- Support ADRC scientists with pilot development project funding (see Admin Core). ADRC scientists will be given the opportunity to compete for development pilot funding to produce excellent preliminary data. Pilot funding applications will undergo competitive review to select the best candidates.
- Provide an interactive learning experience. Through our clinical research at the Roybal Comprehensive Clinic, and at the ADRC clinical core, ADRC scientists will be involved in hands-on training focused on clinical recruitment, neuropsychological testing, novel imaging modalities, and clinical trial interventions.
- Support grant application development and the transition to independence. ADRC scientists will be advised through the grant proposal and review processes, and they will see that their collaborative and independent research, and other professional activities, are continued in the years following their appointment as scientists. Depending on the background of the applicant, the scientists will be mentored early on to apply for KL2, K01, K08, K23 or R01 applications.
- Send ADRC scientists to Bootcamp for REC trainees. This is organized by the NIA ADC parent program (Contact: Luo Yuan PhD) and held in Bethesda, Maryland, around the time of the annual AD/ADRD summit. The selected nominees will have the opportunity to learn the NIH peer review process presented by NIH Program Officers and Scientific Review Officers and will serve as reviewers in a mock study section.

Training Environment for the USC ADRC REC

USC's Provost's Neuroscience Hiring Initiative in 2011 substantially accelerated the growth of the USC ADRC. The Hiring Initiative identified areas within the field that were crucial for raising USC Neuroscience to a higher level of academic excellence and the recruitment of transformative faculty. This is reflected by expansions in neuroimaging and informatics and the development of plasma and CSF biomarkers for ADRD. This rich environment will support ADRC scientists to pursue research training in the study of AD in diverse minority populations, where SEDoH, cardiovascular and metabolic risk factors frequently interact with genetic risk factors (e.g., *APOE ε4*) on risk of ADRD. The proposed program is interdisciplinary, involving multiple schools, departments, and centers at USC. The REC brings together the resources and researchers from:

A) P30AG043073: USC Alzheimer's Disease and Alzheimer's Disease Related Dementias Resource Center for Minority Aging Research (RCMAR) is housed under the USC Schaeffer Center for Health Policy & Economics. Since its establishment in 2012, this center has been dedicated to advancing the careers of early-stage investigators, particularly scholars from underrepresented racial and ethnic populations and women economists who are underrepresented in their discipline. Funding from the National Institutes of Health, through the National Institute on Aging (NIA), has totaled $9.7 million. The center receives generous financial support from the USC Price School of Public Policy and Leonard Davis School of Gerontology. Other participating centers, schools, and institutions include the Alfred E. Mann School of Pharmacy and Pharmaceutical Sciences, Howard University, Spelman College, Cal State Fullerton, University of Texas Medical Branch, and other NIA-funded centers at USC, including the Roybal Center for Behavioral Interventions of Aging, Alzheimer Disease Research Center, Center for Economic and Sociodemographic Study of AD/ADRD, and USC/UCLA Center on Biodemography and Population Health. **AD-RCMAR will partner with the ADRC in jointly sponsoring development projects and training a post-doctoral minority scholar (Plan to Enhance Diverse Perspectives [PEDP-RCMAR])**

B) P30AG066589: The Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease (CeASES ADRD) is an interdisciplinary research center established through a grant from the National Institute on Aging. The center is housed at the Schaeffer Center for Health Policy and Economics at the University of

Southern California (USC) and is led by scholars from USC, Stanford University, and the University of Texas, Austin. It is led by Drs. Julie Zissimopoulos, Dana Goldman and Mark Hayward.

C) CeASES ADRD supports innovative research on the health, economic and social impacts of dementia for identifying opportunities to reduce the risk and costs of dementia, improve access and care, and advance solutions for achieving health equity. A special interest is in development of global, dynamic microsimulation models of ADRD for informing policymakers, other stakeholders and persons affected by dementia. Goals are accomplished through (1) network building to broaden and diversify social scientist researchers of dementia; (2) funding support for pilot projects; (3) convening workshops and the annual Science of ADRD for Social Scientists Program to improve the knowledge and capability of researchers.

D) T32 Gerontology (5T32AG000037): Eileen Crimmins is the AARP Professor of Gerontology in the Davis School of Gerontology at USC. She is currently the director of the USC/UCLA Center on Biodemography and Population Health, one of the NIA Demography of Aging Centers which includes AD research. She is the Director of the Multidisciplinary Training in Gerontology Program (one of the oldest and longest-running T32 institutional training programs in the country) and the NIA-sponsored Network on Biological Risk. Crimmins is also a co-investigator of the Health and Retirement Study in the U.S. She has also been elected to the National Academy of Science and the Institute of Medicine.

E) T32 Gerontology (5T32AG052374): Kelvin Davies directs the USC-Buck Geroscience Training Program in the Biology of Aging, which supports both Ph.D. students and post-doctoral fellows. This innovative and unique geroscience training program is a fully joint venture of the University of Southern California's Leonard Davis School of Gerontology and the Buck Institute for Research on Aging. The bi-institutional nature of our USC-Buck Program in the Biology of Aging is unique and allows us to broadly foster a pipeline of investigators engaged in basic and translational research in aging biology throughout California. This collaborative effort has been in existence in various ways for several years and has proven to be an important force in advancing the aging research agenda both at the University of Southern California and at the Buck Institute for Research on Aging

F) T32 Multi-Modal Imaging (5T32AG058507): Arthur Toga is Provost Professor of Ophthalmology, Neurology, Psychiatry and the Behavioral Sciences, Radiology, and Engineering; Ghada Irani is Chair in Neuroscience, and Director, USC Mark and Mary Stevens Neuroimaging and Informatics Institute. Dr. Toga directs the T32 USC Training in the Multiscale and Multimodal Analysis of Biomarkers in AD program which includes four integrated themes: Molecular and Cellular Methods, Imaging Tools and Informatics, Quantitative Methods for Clinical Trials Research, and Large-Scale Population Analyses.

G) KL2-SC CTSI (5UL1TR001855): Cecilia M. Patino-Sutton MD, Med, PhD is an Associate Professor of Preventive Medicine and Multiple-PI and Director of the NIH NCATS sponsored KL2 Program in Clinical Translational Research and Director of Workforce Development within the Southern California Clinical Translational Research Institute (PI: Thomas Buchanan MD) at USC and Children's Hospital. She has developed and implemented key research career development opportunities to advance the careers of junior faculty who seek research independence across health specialties and the translational spectrum for almost 10 years. These include individual development plans, courses in clinical translational research, research in diverse populations, a career development seminar series, and a state-of-the-art recording studio to support online education.

H) The USC Alzheimer's Therapeutic Research Institute (ATRI) Biostatistics training core is led by Dr. Rema Raman. The Alzheimer's Clinical Trials Consortium (ACTC: 1U24AG057437) is a clinical trials infrastructure launched in December, 2017 and designed to accelerate and expand studies for therapies in Alzheimer's disease and related dementias. ACTC IDEA-CT committee (the mentoring and training committee for the ACTC) is led by Dr. Raman and aims to make researchers more representative of the community through online and in-person training workshops in biostatistics and data management. It includes a biostatistics seminar for trainees and features a statistics concept or a study design each month, followed by a discussion of one of ATRI's ongoing projects. Examples of seminar topics include the application of machine learning in cohort selection (creating an adaptive risk algorithm to efficiently screen participants), progression modeling (with a focus on clinical trials), and AD biomarker prediction. The seminars also introduce software applications for dynamic programming and modeling.

I) R13 Conference grant led by Maria Aranda, Director of our ADRC ORE core. This 2-year project will support a two-installment conference series that will build the next generation of scientists (n=10) specializing in health disparities research related to ADRD. This will be accomplished by developing an incubator of scientific knowledge and networks of interdisciplinary scholars and mentors specializing in rigorous team-based approaches to ADRD health disparities. Based on the NIA Health Disparities Framework, we will address the full range of biological, biomedical, behavioral, and health sciences supported by NIH to close the scientific gap affecting millions of minority older adults and their family caregivers. Dr. Aranda is the Executive Director of the USC Edward R. Roybal Institute on Aging, a multidisciplinary research hub with an applied community focus across diverse underrepresented groups.

J) K24 Mentoring Grant (K24AG081325): Duke Han, PhD, is REC Co-Leader and a Professor of Psychology, Family Medicine, Neurology, and Gerontology. He oversees the clinical neuropsychological assessment training program within the Clinical Psychology PhD program. He has been awarded a 5-year grant to mentor early career investigators in researching cognitive and decision-making outcomes among those with ADRD with a particular focus on financial and cross-cultural considerations.

Components of the Training Program

A) Training Course.

An intensive training program to be at the start of the academic year of every year and will include training to facilitate:

1) Access to ADRC resources (Yassine, Chui, Schneider and Toga)
2) Understanding AD risk factors in LA Latinx groups (Yassine, Chui)
3) Cross-cultural neuropsychological assessment in Latinx older adults (D'Orazio)
4) Introduction to brain imaging in AD (Toga, Braskie)
5) Designing clinical trials in AD with emphasis on racial and ethnic groups (Schneider, Raman, Yassine)
6) Sample size estimation for pilot clinical trials (Raman, Schneider)
7) Animal studies and vascular risk factors in AD (Pike)
8) Developing a successful grant application (Yassine, Chui and others)
9) Health care policies (Faculty of the RCMAR, Segal-Gidan, Crimmins)
10) Long-term services and supports – frontline care (Aranda)
11) Responsible Conduct of Research (Yassine)

The content of the training course will also be tailored to fit the needs of existing USC training programs.

Progress to date 2020-2024. Tables on REC scholars who successfully obtained pilot development funding, career development awards or first R01 fundings are included in the **admin section Tables 4 and 5**. Tables 3 and 4 below summarize our active USC seminar series and symposia with significant progress/success over the past 5 years. The large number of mentees over the last two years (728) is a testament to our success in attracting scholars not just from USC but from outside institutions and ADRCs.

Table 3: ADRC REC Seminar Progress

	Fall 2020 – Spring 2021	**Fall 2021 – Spring 2022**	**Fall 2022 – Spring 2023**	**Fall 2023 – Spring 2024**
Seminars per year	11	20	33	36
External speakers per year	0	2	9	15
Scholars attending seminars per year	318 total (Avg: 29 / seminar)	462 total (Avg: 23 / seminar)	1463 Total (Avg 44 / seminar)	1656 Total (Avg 47 / seminar)
Symposia per year (UCI and ZNI)	0	1	2	2

Table 4: ADRC REC Mentees				
	Fall 2020 – Spring 2021	**Fall 2021 – Spring 2022**	**Fall 2022 – Spring 2023**	**Fall 2023 – Spring 2024**
Mentees per year	140	203	643	728

B) Support for Development Projects.

General selection criteria

ADRC scientists will be encouraged to apply for pilot development project funding to study racial and ethnic health disparities and mechanisms of increased AD risk in the setting of observational or interventional trials focused on lifestyle (diet, exercise, weight loss), family-care, imaging biomarkers, and cognitive outcomes. The PI must be a promising junior scientist or a mid-level scientist transitioning into this research area and a member of an underrepresented sociodemographic group in the representative scientific field. Priority will be given to applications that (1) make the best use of the wealth of resources the ADRC has to offer in terms of research expertise of faculty, data resources, methodological expertise of participating mentors, and research centers; and (2) are most likely to lead to an independent investigator award. Additional criteria for consideration include whether the pilot proposes interdisciplinary research. The research proposed should be completed within one year (two years maximum) and with a $50K budget.

A plan is in place to Integrate one ADRC pilot award with P30AG043073 (RCMAR) and P30AG066589 (CeASES ADRD) for a common pool of applicants with joint interests (**PEDP-RCMAR**). ADRC REC faculty with work closely with Drs. Julie Zissimopoulos, Dana Goldman and Mark Hayward to select and mentor successful candidates.

C) Supporting a minority post-doctoral candidate (PEDP-RCMAR).

We propose to recruit and support one investigator from underrepresented backgrounds at USC for a two-year post-doctoral fellowship. The fellowship will be focused on leveraging ADRC resources (data analysis or existing tissue sample use) to study ADRD in underserved populations, as part of our Plan to Enhance Diverse Perspectives (PEDP-RCMAR).

We take advantage of resources from the Dean's Office for Faculty Diversity. This program expands upon our efforts to enhance the diversity of postdoctoral scholars at the university and build a pool of exceptional underrepresented minority candidates for the next generation of faculty across higher education. These scholars will link the expertise of USC faculty and doctoral students with the knowledge and unique experience gained through their own research and scholarship. These appointments are for up to two years.

D) Training Plan.

Through the training program, the mentor will assist ADRC scientists in obtaining access to existing databases and resources, develop a pilot project proposal, provide a platform for presenting the data, and write an effective grant application. Trainees will be encouraged to work with mentors on: (1) lifestyle risk factors and interventions to counteract the effects of obesity and diabetes through interventions such as weight loss and exercise, (2) family-based and economic/health care outcomes and interventions, (3) psychometric testing applicable to the Latinx population, and (4) population research focusing on AD risk in ethnic minority groups. Training resources of relevance to the Latinx population are summarized below:

1) USC GeneScreen and GeneScreen Spanish: In 2023, Dr. Yassine launched USC GeneScreen which screens and enriches USC ADRC-affiliated studies for *APOE* ε4 from LA country and over 200 *APOE* ε4 carriers have been identified and recruited to date. More recently, USC GeneScreen Spanish was started to increase representation from Latinx within LA.

2) Genetic Alzheimer's Disease Subtypes in Latinos (GADSIL, U01 AG051218; R01AG062007, PI: John Ringman, MD) is an ADRC-affiliated study to characterize pre- and symptomatic autosomal dominant AD

(ADAD) due to two specific mutations (A431E in *PSEN1* and V717I *APP*) in relation to late-onset AD (LOAD). All subjects are of Latinx descent and undergo the Human Connectome Protocol (HCP) MRI scanning, tau PET imaging with flortaucipir, and comprehensive clinical and cognitive assessments including the UDS 3, the goals being to define phenotype and brain connectivity in genetic subtypes of AD among the Latinx population.

3) The Cross-Cultural Neuropsychology Program (CCNP) was established in 2015 and led by Dr. Lina D'Orazio with the goals of: (1) providing bilingual and culturally informed neuropsychological assessment and intervention services in the neurology clinic, and (2) providing expert consultation and training services to investigators interested in working with Latinx populations across KSOM departments.

4) Psychological, social, and family-based intervention research program led by Maria Aranda. Postdoctoral scholars will have the opportunity to advance knowledge on the psychosocial issues related to dementia diagnosis (psychological response and adaptation, crisis intervention, diagnosis disclosure, psychological adaptation, and coping), and family processes related to decision-making, caregiver stress, and long-term services and support. Other opportunities include community-based interventions for families affected by dementia; ADRD health disparities among underrepresented racial and ethnic groups, including Latinx, African-American, and Asian American families.

5) LONI's Image and Data Archive (IDA) led by Dr. Toga is a secure data archiving system, currently housing data from more than 30,000 subjects. The IDA uses a robust infrastructure to provide researchers with a flexible and simple interface for de-identifying, searching, retrieving, converting, and disseminating their biomedical data. With thousands of investigators across the globe and more than 21 million data downloads to dat, the IDA guarantees reliability with a fault-tolerant network comprising multiple switches, routers, and Internet connections to prevent system failure.

6) USC RCMAR and CeASES-ADRD led by Dr. Julie Zissimopoulos provides access and training to utilize data from Health and Retirement Study, National Health and Aging Trends Study, and Medicare Claims data.

7) USC Clinical Psychology Doctoral Training Program led by Dr. Duke Han and accredited since 1948 by the American Psychological Association, the PhD program in Clinical Science is dedicated to the integration of science and practice. Dr. Duke Han (REC Co-Leader) oversees the clinical neuropsychological assessment training program for the PhD program, which incorporates an emphasis on cross-cultural and cognitive aging considerations. The program admits between 8-12 graduate students per year, many of whom have interests in ADRD, dementia caregiving, cross-cultural neuropsychological assessment, vascular and metabolic contributions to cognitive aging, and financial and health consequences of AD.

Trainees are not restricted to these resources or programs and will have the opportunity to work in other ADRD-related areas. We will monitor the progress of ADRC scientists during the period of the award as well as through their early career milestones (e.g., receipt of an independent grant). ADRC scientist alumni will be supported through continued involvement in activities that support and improve both their career options and the experience of the next cohort of pilot grantees.

Component Leader(s)

Hussein Yassine and Duke Han will co-direct the REC program. Members of the ADRC REC Executive Committee include Julie Zissimopoulos, Maria Aranda, Lina D'Orazio, Helena Chui, Arthur Toga and Rema Raman. Additional external members of the executive committee will be invited to assist with the recruitment of under-represented minority scientists. Key faculty members of the REC program are Christian Pike, Lon Schneider, John Ringman, Meredith Braskie, and Eileen Crimmins (**Table 1**).

A) *Relevant scientific background.*

1) Dr. Yassine is a Professor at Keck School of Medicine, Department of Neurology and has an outstanding record of training junior scholars. Over the course of 15 years as an independent investigator, Dr. Yassine has been fully devoted to the training and championing of a diverse population of graduate and undergraduate students and post-doctoral fellows. He has mentored more than 50 scholars, and all have gone on to pursue further training in graduate, medical school, residency or fellowship. Of the seven faculty, four are currently doing an academic postdoc, one is an Associate Professor on the teaching track and two are at the Assistant Professor level. Dr. Yassine was the 2022 chair of NIA's Research and Education Component's (REC) steering committee, past co-chair of the Nutrition, Metabolism and Dementia Professional Interest Area (PIA) of the Alzheimer's Association, and sits on the ISTAART Advisory council 2022-2024, championing ADRC REC scholars.

2) Dr. Han is a diplomate of the American Board of Professional Psychology, a fellow of the National Academy of Neuropsychology and the American Psychological Association, and a Professor of Psychology, Family Medicine, Neurology, and Gerontology in the Dornsife College of Letters, Arts, and Sciences of the University of Southern California. He has trained over 60 trainees of multiple levels (undergraduates, MA and PhD graduates, medical students, postdoctoral fellows, and junior and mid-career faculty), many of whom have obtained academic medical positions at institutions such as UCSD, Duke University, UC Irvine, and Banner Alzheimer's Institute. He is Co-Chair of the International Neuropsychological Society's Dementia Special Interest Group, Co-Director of the Scientific Research Network on Decision Neuroscience and Aging, Editor-In-Chief of The Journal of Gerontology: Psychological Sciences, and a member of the Scientific Advisory Board for the Alzheimer's Disease Neuroimaging Initiative. He has mentorship roles in multiple professional societies and was recently awarded a K24 mentoring grant (K24AG081325) by the National Institute on Aging because of his mentoring success.

3) Dr. Zissimopoulos is a Professor in the Sol Price School of Public Policy at the University of Southern California. In addition to her faculty appointment, she is a Senior Fellow and Director of Aging and Cognition Research Program and Research Training at the Schaeffer Center for Health Policy and Economics. She is a Principal Investigator of USC's Resource Center for Minority Aging Research (USC AD RCMAR), and Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease and Related Dementias (CeASES-ADRD), both focused on reducing the burden of Alzheimer's disease and funded by the National Institute on Aging. She has mentored 10 junior faculty, 16 postdoctoral fellows and 17 predoctoral fellows, over half of whom are from underrepresented backgrounds.

4) Dr. Aranda has trained a large number of professionals and graduate students in strengths-based, evidence-based practice for older adults and their family members. She has received multiple awards for her mentoring efforts. She received the USC Mellon Award for Excellence in Mentoring Students in 2009, and the Council on Social Work Education Mentoring Award in 2011. Under her guidance, students have discovered their passion for working with older populations from underserved communities. Aranda also teaches classes in clinical social work practice with older adults and is a mentor for the Latinx Social Work Caucus.

5) Dr. D'Orazio is the Director of Cross-Cultural Neuropsychology and Assistant Professor of Clinical Neurology in the Department of Neurology, Keck School of Medicine of USC. Dr. D'Orazio is a bilingual (English/Spanish), first-generation Latina American licensed clinical psychologist specializing in neuropsychology and health psychology, with a particular focus on culture and neurocognitive changes with age and those within the context of chronic illness. Dr. D'Orazio specializes in the neuropsychological assessment of Spanish speakers and Spanish/English bilinguals, as well as in health psychology assessment and interventions for Latinx populations. Dr. D'Orazio has over 15 years of experience in neuropsychological assessment with English and Spanish speakers from clinical and research positions that include neurology research by the USC Memory and Aging Center (a NACC Alzheimer's Disease Research Center), cognitive and psychophysiological studies in Columbia University's Department of Psychiatry Behavioral Medicine Program, psycho-oncology research in the USC Department of Psychology and LAC Hospital, applied behavioral medicine research in the West Los Angeles VA Health

Psychology Fellowship program, and cultural psychiatry/global mental health research in the Department of Psychiatry and Human Behavior of the Warren Alpert Medical School of Brown University.

B) ***REC Executive Committee.***

ADRC Executive committee will participate in REC/RCMAR and will be ultimately will be responsible for:

- Recruitment of ADRC scientists
- Building pool of potential ADRC scientists
- Day-to-day administration of the REC (assisted by a Program Coordinator)
- Data collection on scientist progress and evaluation of scientists, mentors, and ADRC program
- Ensuring continuing mentoring activities

C) ***REC Faculty Mentors***

A strong roster of distinguished faculty will be taking part in the REC. **Table 1** below lists all the participating faculty members, their appointments and research interests. Faculty members' joint appointments and their collaborations on research projects link the institutions, departments, and centers participating in the ADRC CEC. These links will ensure ADRC scientists' access to all available resources. Mentoring faculty in the REC regularly collaborate on research projects as is demonstrated in their publication lists. Mentors will be appreciated in the annual Finch AD symposium.

Some examples follow. Crimmins and Zissimopoulos have ongoing research on the value of delaying the onset of AD for individuals, families and society. Yassine, Peters and Beale collaborate on Obesity Research studies. D'Orazio is an expert in cross-cultural neuropsychology and cognitive assessment of Spanish speakers and is involved in cognitive research in under-represented populations. Schneider and Yassine work together on AD clinical trials. Aranda is working with Yassine to design family-based approaches for improving brain health and reducing cognitive decline as part of the Roybal Clinic AD Program.

As seen in **Table 1** below, the mentoring faculty is multidisciplinary and racially and ethnically diverse. Mentors come from the medical sciences, exercise physiology, nutrition, clinical trials, economics, gerontology, sociology, social work, engineering and computer science and their research expands across the ADRC areas of focus. The faculty are experts in obesity and diabetes (Yassine) and racial and ethnic differences in aging and brain health (Aranda, Chui, Ringman, Crimmins, Zissimopoulos). All the faculty have significant mentoring experience. Drs. Yassine, Han, Aranda, and Chui, have mentored large numbers of postdoctoral fellows and junior faculty.

Table 1. Participating REC faculty

Last Name	First Name	Title	Institution/School and department	Theme	Research Area
Yassine*	Hussein	Professor	USC/Neurology	*APOE*	Lipid metabolism
Han*	Duke	Professor	USC/Psychology, Neurology and Family Medicine	Cognitive aging and decision making	Financial decision making as an early neurocognitive marker of AD
Zissimopoulos*	Julie	Professor	USC/Policy School	Costs of dementia	Dementia risk, financial and health care implications of dementia
Aranda*	Maria	Associate Professor	USC/Social Work	Family based care	Family-based care in the elderly
Ringman*	John	Professor	USC/Neurology	Biomarkers, cognitive and behavioral outcomes	Genetics and Clinical manifestations of ADRD in Latinx
D'Orazio*	Lina	Assistant Professor	USC/Neurology	Cognitive outcomes	Cross-cultural neuropsychological assessment, specialty in assessment of Latinx populations and Spanish-speakers
Chui	Helena	Professor	USC/Neurology	Cognitive outcomes	Diabetes and AD
Toga	Arthur	Professor	USC/INI	Imaging biomarkers	Imaging biomarkers access in LONI databased

Raman	Rema	Professor	USC/ATRI	Biostatistics/clinical trials	Clinical trial design, database management
Schneider	Lon	Professor	USC/Psychiatry	Clinical trials	Clinical trial simulation in AD
Braskie	Meredith	Associate Professor	USC/INI	Imaging biomarkers	Imaging the diabetes brain
Pike	Christian	Professor	USC/Gerontology	Lifestyle interventions	Animal models of obesity and AD risk
Finch	Caleb	Professor	USC/Gerontology	Gene by environment AD interactions	Animal models of AD risk and environmental risk factors
*Members of the REC executive committee					

D) *Faculty Members*

Program faculty members will meet weekly with their assigned ADRC scientists to guide, train, and direct them. This will be their primary task, aimed at producing independent scientists. Additionally, they will review scientists' progress and their own mentoring. Faculty members will also participate in the training and research activities previously described.

E) *External Advisory Panel*

The REC will seek the input of an external advisory panel that monitors the overall ADRC program. The external advisory panel will meet in person once per year and review: (1) the training program, (2) strategies to obtain scientists from diverse backgrounds, (3) success of pilots of selected ADRC scientists, (4) the composition of primary faculty mentor and secondary mentor for scientists, (5) help with the review of scientists and mentors, and (6) continuously provide advice and options for enhancing the REC. The EC will also conduct reviews of the REC. Finally, EC members will also be part of the mentor faculty, working with ADRC scientists on research projects, reviewing papers, grant proposals, presentations, and providing professional guidance.

F) *Tracking and Evaluation of Success of ADRC scientists*

A Project Coordinator (50% dedicated time budgeted) collects data supplied by all ADRC scientists and faculty mentors to track the success of scientists. The PIs will use this information to aid the scientists, inform the program and make appropriate changes to the program. Evaluations focus on: (1) scientists, (2) mentors, and (3) the REC program. The EC will meet twice a year (initially quarterly) to evaluate progress. The EC will use the data collected on scientist recruitment, scientist and mentor evaluations, publications, presentations, and grant proposal outcomes of scientists to evaluate the REC and suggest improvement for both the short and long term. Mentors who do not produce successful scientists will be asked to step down from this role and will not be assigned scientists in the future. REC EC (along with scientist mentors) will follow the careers of former scientists closely for the next several years and interact with them at least once a year about their career progress. All feedback will be communicated to the EC in regularly scheduled meetings. All evaluations and feedback will be logged into a continuous evaluation table maintained by the Project Coordinator.

Program Participants: Sources of ADRC Scientists

The ADRC REC will reach out to integrate trainees of six main programs at USC summarized in **Table 2**. We expect 20 ADRC scientists to participate in the training course, and to support up one REC scientist through our Dean's Program for post-doctoral minorities.

Table 2: USC training programs that include ADRD research and minority scholars

Program	Number of post-doctoral trainees/year	% minority researchers	Focus	Preceptor
RCMAR	2 post-doctoral fellows /year	70%	Population research	Julie Zissimopoulos
Psychology (Han)	1-3 post-doctoral fellows, 1-5 pre-doctoral fellows	30%	Neuropsychology, neuroimaging, and decision making	Duke Han
T32 Gerontology	5-10 post-doctoral fellows	25%	Biodemography and population health exposure to ADRD	Eileen Crimmins

T32 Multi-Modal Imaging	2 post-doctoral fellows (program started in 2018).		Imaging research	Arthur Toga
SC CTSI-KL2 Institutional Scholars	KL2 Scholars – 2 post-doctoral (i.e., junior faculty)	15%	Clinical translational research	Cecilia Patino-Sutton, MD, Med, PhD
Translational genomics U54 REC	Program was initiated in 2019		Cancer research	John Carpten
ACTC IDEA-CT	Post-doctoral scholars		Biostatistics-clinical trials	Rema Raman
T32 Gerontology	Post-doctoral scholars	11%	Aging research	Kelvin Davies

Recruitment Plan of Underrepresented Minority Scientists to Enhance Diversity:

Given one of the focus areas of the REC training program is studying AD risk in LA's Latinx population, we believe this will help in recruiting scientists from under-represented ethnic groups. Our first approach to recruitment is to reach out to all training programs at USC. There are many research centers with an ADRD focus at USC: we will use these as well as our Key faculty to disseminate information to potential scientists and people who will know scientists. In addition to our partners (**Table 2**), we will reach out to external groups and institutions such as the National Medical Association representing African American physicians, the National Hispanic Medical Association, and two historically African American colleges. Additional resources to identify URM scientists include:

1) USC ADRC Outreach and Education Core (ORE): The USC ADRC ORE serves as the coordinating unit for all ADRD-related research training, outreach, and education. We will invite URM from the ORE core to participate in REC.

2) Diabetes and Obesity Research Institute Bridging-the-Gap (BTG) and SPIDOR Programs: The Keck School of Medicine (KSOM) of USC recruits locally and nationally to achieve a diverse student body. In 2011, a summer research program known as Bridging the Gaps (BTG) was instituted to increase URM studying at KSOM. The objectives were to increase awareness of the roles of physician-scientists and biomedical scientists, thereby serving as a catalyst to ignite interest in choosing these particular career paths, and to increase interest in those areas of research in which minorities are severely impacted, serving as a potential mechanism to reduce health disparities. The SPIDOR program led by Joyce Richey and Richard Watanabe complements and expands the stated objectives of BTG. The novelty of SPIDOR is our interdisciplinary model that emphasizes diabetes-related research in basic science, clinical, and engineering fields. The preceptors of this grant represent a balanced mix of full, associate, and assistant professors who are members of our USC Diabetes and Obesity Research Institute (DORI). These faculty members have proven records of student training and a history of collaborative research.

3) Integrate local Hispanic-serving institutions within the California State University and University of California systems into ADRC REC. Joyce Richey MD (Associate Dean for Diversity) will assist us with recruitment. We will foster strong relationships with California schools that train URM trainees. Our primary objectives are to recruit and expose URM students to an integrated approach to studying AD in the LA Latinx population to increase their competitiveness for independent funding.

4) Integrating ADRC REC/RCMAR Efforts (**PEDP-RCMAR**): We will cast as wide a net as possible, as well as target particular opportunities, to develop the pool of underrepresented minority applicants who would benefit the most from participation in the USC AD/ADRD RCMAR program. When appropriate individuals have been identified, we will seek them out and cultivate their interest in joining our RCMAR. Scientists will be recruited from the main participating USC schools, departments and centers, our collaborators from HBCU's and Hispanic Serving Institutions, and qualified applicants from other institutions. Previous experiences have shown that RCMAR scientists are willing to fly to Los Angeles monthly to be part of the RCMAR program and can be successfully integrated. For instance, past scientists have been from USC, UCLA, the LA County Health Department, Brown, Cornell, Duke, Indiana, Miami, U Penn, Morehouse, South Carolina, Stanford, Spelman, and Vanderbilt. Multiple RCMAR scientists received new USC appointments after their RCMAR experience. While we have a substantial pool of applicants within USC, because of the specialized nature of our training in economics, health, and health care, the pool of aging researchers and individual investigators who are not at USC have shown to benefit more from the experience; by leveraging USC resources. Thus, our aim is to increase the recruitment of scientists at

USC, from underrepresented minority-serving institutions, and other institutions nationally with underrepresented racial and ethnic scientists interested in minority health, AD/ADRD, health care, and economic forces.

A) *USC Programs for Recruitment and Retention of Underrepresented Minority Investigators*

Diversity is critical to USC's strategy for excellence and impact. Given our location at the center of one of the most diverse cities in the world and our commitment to enhancing diversity, it is essential that USC be a leader in research that helps and engages diverse populations. In 2018 the USC Office of Research published its *Strategic Plan 2018, Fulfilling the Promise for Diversity and Inclusion in Research and Scholarship*. The Plan highlights existing, longstanding USC programs that underrepresented minority individuals at various stages in their career development. By leveraging the programs' existing procedures in place, ADRC members will engage with and participate in these programs to provide information to postdoctoral students and junior faculty about ADRD research and recruit them to the REC as ADRC scientists. Through these existing USC programs, the ADRC will strive to enhance diversity in ADRD and aging research and clinical practice. Dr. Yassine and Dr. Han will oversee these efforts for the ADRC and will liaise with the program leaders to participate in their activities. Furthermore, we have garnered support from several minority-serving academic institutions in Southern California that have agreed to promote the ADRC REC to their postdoctoral fellows and junior faculty. These individuals will be recruited to participate in REC activities.

B) *USC Programs for Recruitment and Retention of Underrepresented Minority Postdoctoral Fellows*

With over 650 postdoctoral scholars, USC's Office of Postdoctoral Affairs works closely with academic units to help recruit a diverse cadre of the best junior scholars to the university and facilitates the training and preparation of postdoctoral scholars for careers as independent researchers, academics, and leaders of their chosen fields. USC programs at the postdoctoral level include:

1) The Provost's Postdoctoral Scholars Program for Faculty Diversity Program enhances the diversity of postdoctoral scholars and builds a pool of exceptional underrepresented minority candidates for the next generation of faculty. These scholars link the expertise of USC faculty and doctoral students with the knowledge and experience gained through their own research. Appointments are for up to two years (with a research and travel account of $5,000 per year).

2) Training for the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease T32 (Keck School, PI: Arthur Toga). This program aims to fill the critical need for a broad range of expertise in biomarker analysis at several levels of analysis. USC is home to leading experts who utilize biomarkers in their work investigating AD through a myriad of methods and analyses. Drawing from USC's eminent resources and expertise, the program focuses on preparing investigators for independent research careers in the multiscale and multimodal analysis of biomarkers in AD, by integrating molecular and cellular methods, imaging tools and informatics, quantitative methods for clinical trials research, and large-scale population analyses. The program draws preceptors and faculty from six schools, departments, institutes, and centers at USC: Mark and Mary Stevens Neuroimaging and Informatics Institute (INI) in the Keck School of Medicine; Department of Biomedical Engineering (BME) in the Viterbi School of Engineering; Zilkha Neurogenetic Institute (ZNI) in the Keck School of Medicine; Department of Psychology in the Dana and David Dornsife College of Letters, Arts and Sciences; and Leonard D. Schaeffer Center for Health Policy and Economics. The program includes initiatives to recruit individuals from underrepresented populations.

3) Multidisciplinary Research Training in Gerontology T32 (Davis School, PI Crimmins). This USC multidisciplinary research training program in Gerontology provides postdoctoral training within key disciplines in aging and familiarity with the theory and approaches of related disciplines. The intent is to inspire students to incorporate multidisciplinary models in their research. Postdoctoral students can be appointed through Economics, Sociology and Policy Planning and Development. The focus of the program is on physical and mental health over the lifecycle and factors influencing health. The program includes initiatives to recruit individuals from underrepresented populations.

C) ***USC Programs for Recruitment and Retention of Underrepresented Minority Junior Faculty***

Diversity in research can enhance the creativity of our faculty through a culture of transformative scholarship. USC has promoted increasing diversity among its outstanding faculty and expanded the pool of faculty who conduct research in diversity-related fields. This occurs through recruitment, retention, and mentoring. Each school and several central university units have instituted mentoring programs and activities to foster the success of faculty, as well as initiatives related to faculty diversity in recruitment and retention. Deans of Faculty Affairs also serve as points of contact within each school to assist faculty at all levels in providing support for development and mentoring as well as guidance for appointment, tenure, and promotion. They also advise department chairs and administrative offices on procedures regarding recruitment of new faculty, compensation, and leaves of absence. USC programs for junior faculty include:

1) Provost's Diversity and Inclusion Council, spearheaded by the Provost's Office to address foundational changes to improve the campus climate at USC, one of the charges of the Diversity and Inclusion Council (previously known as the Diversity Task Force) is to provide recommendations to the provost to improve recruitment, retention, support, and climate for a diverse and inclusive faculty, student body, and university staff.

2) Women in Science and Engineering (WISE) increases the representation and success of women faculty members in science and engineering at USC through a number of programs that enable women to thrive in their careers by developing university policies and building a supportive university environment.

3) UCAPT workshops: The Office of the Vice Provost for Academic and Faculty Affairs, in conjunction with the University Committee on Appointments, Promotion, and Tenure, hosts a number of workshops for faculty each academic year to provide transparency and guidance on the tenure process.

4) Center for Excellence in Research within the Office of Research provides training for faculty on obtaining research funding and on novel approaches to conducting research, including diversifying research programs.
5) USC Alzheimer's Disease and Alzheimer's Disease Related Dementias Resource Center for Minority Aging Research (USC RCMAR) (MPIs Julie Zissimopoulos, Jennifer Ailshire, Emma Aguila) in USC's Schaeffer Center for Health Policy and Economics, works to increase the number, diversity, and academic success of researchers focusing on the health and economic well-being of minority older adult populations. At USC AD/ADRD RCMAR, there is a strong commitment to rigorous research investigating the complex pathways through which social, behavioral, environmental and economic factors, along with policies and healthcare systems, influence disparities in the risk of dementia. Additionally, research explores how these factors impact the health, healthcare, and economic well-being of individuals living with dementia.

Plan for Instruction in the Responsible Conduct of Research. We will provide a face-to-face lecture on responsible conduct of research during the training course. This will include training on the conflict of interest, authorship, data management, human subjects and animal use, laboratory safety, research misconduct, and research ethics. This lecture will be given by Dr. Yassine and All REC faculty will be encouraged to attend. The lecture will last two hours and will be conducted every year.

Evaluation Plan. During the first quarter, we plan an intensive training course that will include a grant writing workshop and detailed instructions on how to access databases relevant to AD. The applicants will be encouraged to apply for pilot funding that emphasizes research in Latinx and AD risk factors. In the last quarter, applicants will be required to present their research progress during the annual Finch AD symposium. A diagram is provided to illustrate the plan. The short or long-term success of the research education program in achieving its objectives will be monitored through publications (abstracts, papers), awards, or independent research funding. ADRC scientists' future career achievements will be tracked and included in the progress report. We will obtain feedback from participants to help identify weaknesses and provide suggestions for improvements.

Table. REC Scholars Publications (n=40)			
Name	**Role of the ADRC in bringing new investigators to the field**	**Funding**	**Publication Title**
Michael Bienkowski PhD	Development Award (Mentor: Toga)	K01AG066847	• Bienkowski MS. Further refining the boundaries of the hippocampus CA2 with gene expression and connectivity: Potential subregions and heterogeneous cell types. Hippocampus. 2023;33(3):150-60. Epub 2023/02/15. doi: 10.1002/hipo.23508. PubMed PMID: 36786207; PubMed Central PMCID: PMCPMC9987718.
Lirong Yan PhD	K25 (Mentor: Chui)	K25AG056594 RF1AG072490	• Tang J, Heidari Pahlavian S, Joe E, Gamez MT, Zhao T, Ma SJ, et al. Assessment of arterial pulsatility of cerebral perforating arteries using 7T high-resolution dual-VENC phase-contrast MRI. Magn Reson Med. 2024;92(2):605-17. Epub 2024/03/05. doi: 10.1002/mrm.30073. PubMed PMID: 38440807. • Sun K, Chen Z, Dan G, Luo Q, Yan L, Liu F, et al. Three-dimensional echo-shifted EPI with simultaneous blip-up and blip-down acquisitions for correcting geometric distortion. Magn Reson Med. 2023;90(6):2375-87. Epub 2023/09/05. doi: 10.1002/mrm.29828. PubMed PMID: 37667533; PubMed Central PMCID: PMCPMC10903279. • Heidari Pahlavian S, Geri O, Russin J, Ma SJ, Amar A, Wang DJJ, et al. Semiautomatic cerebrovascular territory mapping based on dynamic ASL MR angiography without vessel-encoded labeling. Magn Reson Med. 2021;85(5):2735-46. Epub 2020/12/22. doi: 10.1002/mrm.28623. PubMed PMID: 33347641. • Shao X, Yan L, Ma SJ, Wang K, Wang DJJ. High-Resolution Neurovascular Imaging at 7T: Arterial Spin Labeling Perfusion, 4-Dimensional MR Angiography, and Black Blood MR Imaging. Magn Reson Imaging Clin N Am. 2021;29(1):53-65. Epub 2020/11/26. doi: 10.1016/j.mric.2020.09.003. PubMed PMID: 33237015; PubMed Central PMCID: PMCPMC7694883. • Heidari Pahlavian S, Cen SY, Bi X, Wang DJJ, Chui HC, Yan L. Assessment of carotid stiffness by measuring carotid pulse wave velocity using a single-slice oblique-sagittal phase-contrast MRI. Magn Reson Med. 2021;86(1):442-55. Epub 2021/02/06. doi: 10.1002/mrm.28677. PubMed PMID: 33543788.
Isaac Asante PhD	Development Award (Mentor: Yassine)	CTSI KL2 Scholar REC Scholar	• Asante I, Louie S, Yassine HN. Uncovering mechanisms of brain inflammation in Alzheimer's disease with APOE4: Application of single cell-type lipidomics. Ann N Y Acad Sci. 2022;1518(1):84-105. Epub 2022/10/07. doi: 10.1111/nyas.14907. PubMed PMID: 36200578; PubMed Central PMCID: PMCPMC10092192.
Matt Borzage PhD	Promoted to Associate Professor (Mentor: Yassine)	RF1AG076124	• Shah P, Doyle E, Wood JC, Borzage MT. Imputation models and error analysis for phase contrast MR cerebral blood flow measurements. Front Physiol. 2023;14:1096297. Epub 2023/03/10. doi: 10.3389/fphys.2023.1096297. PubMed PMID: 36891147; PubMed Central PMCID: PMCPMC9988286.

Name	Role of the ADRC in bringing new investigators to the field	Funding	Publication Title
Kristan Leech, PhD	Assistant Professor in Physiology (Mentor: Duke Han)	K01AG073467	• Bonilla Yanez M, Kettlety SA, Finley JM, Schweighofer N, Leech KA. Gait speed and individual characteristics are related to specific gait metrics in neurotypical adults. Sci Rep. 2023;13(1):8069. Epub 2023/05/19. doi: 10.1038/s41598-023-35317-y. PubMed PMID: 37202435; PubMed Central PMCID: PMCPMC10195830. • Kettlety SA, Finley JM, Reisman DS, Schweighofer N, Leech KA. Speed-dependent biomechanical changes vary across individual gait metrics post-stroke relative to neurotypical adults. J Neuroeng Rehabil. 2023;20(1):14. Epub 2023/01/27. doi: 10.1186/s12984-023-01139-2. PubMed PMID: 36703214; PubMed Central PMCID: PMCPMC9881336. • Leech KA, Roemmich RT, Gordon J, Reisman DS, Cherry-Allen KM. Author Response to Macpherson et al. Phys Ther. 2022;102(8). Epub 2022/06/18. doi: 10.1093/ptj/pzac084. PubMed PMID: 35713528; PubMed Central PMCID: PMCPMC10071573. • Rossi C, Roemmich RT, Schweighofer N, Bastian AJ, Leech KA. Younger and Late Middle-Aged Adults Exhibit Different Patterns of Cognitive-Motor Interference During Locomotor Adaptation, With No Disruption of Savings. Front Aging Neurosci. 2021;13:729284. Epub 2021/12/14. doi: 10.3389/fnagi.2021.729284. PubMed PMID: 34899267; PubMed Central PMCID: PMCPMC8664558.
Joseph Saenz, PhD	Promoted to Assistant Professor at Arizona State University (Mentor: Duke Han)	K99AG058799	• Doyle C, Andel R, Saenz J, Crowe M. Correlates of SuperAging in Two Population-Based Samples of Hispanic Older Adults. J Gerontol B Psychol Sci Soc Sci. 2024;79(6). Epub 2024/04/06. doi: 10.1093/geronb/gbae058. PubMed PMID: 38581241; PubMed Central PMCID: PMCPMC11075730. • Farina MP, Saenz J, Crimmins EM. Does adding MRI and CSF-based biomarkers improve cognitive status classification based on cognitive performance questionnaires? PLoS One. 2023;18(5):e0285220. Epub 2023/05/08. doi: 10.1371/journal.pone.0285220. PubMed PMID: 37155663; PubMed Central PMCID: PMCPMC10166486. • Saenz JL, Milani SA, Mejía-Arango S. Gender, Personality, and Cognitive Resilience Against Early-Life Disadvantage. J Gerontol B Psychol Sci Soc Sci. 2023;78(5):913-24. Epub 2023/01/31. doi: 10.1093/geronb/gbad017. PubMed PMID: 36715207; PubMed Central PMCID: PMCPMC10174201. • Saenz J, Beam CR, Kim AJ. Development of a latent dementia index in the aging, demographics, and memory study: Validation and measurement invariance by sex. Alzheimers Dement (Amst). 2023;15(2):e12433. Epub 2023/05/16. doi: 10.1002/dad2.12433. PubMed PMID: 37187808; PubMed Central PMCID: PMCPMC10175944. • Franco Y, Jang Y, Saenz JL, Ho JY. The Relationship Between Multimorbidity and Types of Chronic Diseases and Self-Rated Memory. Res Aging. 2023;45(2):149-60. Epub 2022/04/08. doi: 10.1177/01640275221087612. PubMed PMID: 35387519.

Name	Role of the ADRC in bringing new investigators to the field	Funding	Publication Title
Gali Weissberger, PhD	Promoted to Assistant Professor at Bar Ilan University (Mentor: Duke Han)	T32AG000037	• Weissberger GH, Goodman MC, Mosqueda L, Schoen J, Nguyen AL, Wilber KH, et al. Elder Abuse Characteristics Based on Calls to the National Center on Elder Abuse Resource Line. J Appl Gerontol. 2020;39(10):1078-87. Epub 2019/08/01. doi: 10.1177/0733464819865685. PubMed PMID: 31364442; PubMed Central PMCID: PMCPMC6992470. • Weissberger GH, Gollan TH, Bondi MW, Nation DA, Hansen LA, Galasko D, et al. Neuropsychological Deficit Profiles, Vascular Risk Factors, and Neuropathological Findings in Hispanic Older Adults with Autopsy-Confirmed Alzheimer's Disease. J Alzheimers Dis. 2019;67(1):291-302. Epub 2019/01/15. doi: 10.3233/jad-180351. PubMed PMID: 30636736; PubMed Central PMCID: PMCPMC6925532.
Kayla Tureson, MS	Clinical Core (Mentors: Ringman, Christopher Beam)	NIA Predoctoral Fellowship F31	• Tureson KN, Beam CR, Medina LD, Segal-Gidan F, D'Orazio LM, Chui H, et al. Use of the Spanish English Neuropsychological Assessment Scale in older adult Latines and those at risk for autosomal dominant Alzheimer's disease. J Clin Exp Neuropsychol. 2023;45(6):553-69. Epub 2023/11/22. doi: 10.1080/13803395.2023.2284971. PubMed PMID: 37990912; PubMed Central PMCID: PMCPMC10926998.
Charisse Winston PhD	Promoted to Assistant Professor of Physiology and Neuroscience (Mentor: Robert Rissman)	K99MD015777	• Rissman RA, Langford O, Raman R, Donohue MC, Abdel-Latif S, Meyer MR, et al. Plasma Aβ42/Aβ40 and phospho-tau217 concentration ratios increase the accuracy of amyloid PET classification in preclinical Alzheimer's disease. Alzheimers Dement. 2024;20(2):1214-24. Epub 2023/11/07. doi: 10.1002/alz.13542. PubMed PMID: 37932961; PubMed Central PMCID: PMCPMC10916957. • Sloley SS, Main BS, Winston CN, Harvey AC, Kaganovich A, Korthas HT, et al. High-frequency head impact causes chronic synaptic adaptation and long-term cognitive impairment in mice. Nat Commun. 2021;12(1):2613. Epub 2021/05/12. doi: 10.1038/s41467-021-22744-6. PubMed PMID: 33972519; PubMed Central PMCID: PMCPMC8110563.
Douglas Barthold, PhD	Assistant Professor, University of Washington (Mentor: Julie Zissimopoulos)	K01AG071843	• Kaggiah A, Maina CN, Kinuthia J, Barthold D, Hauber B, Tran J, et al. Key informant views on potential acceptability and feasibility of long-acting antiretroviral treatment for HIV in Kenya. BMC Infect Dis. 2024;24(1):415. Epub 2024/04/20. doi: 10.1186/s12879-024-09309-w. PubMed PMID: 38641565; PubMed Central PMCID: PMCPMC11027348. • Barthold D, Saldarriaga EM, Brah AT, Hauber B, Banerjee P, Fuller SM, et al. Preference for daily oral pills over long-acting antiretroviral therapy options among people with HIV. Aids. 2023;37(10):1545-53. Epub 2023/06/08. doi: 10.1097/qad.0000000000003620. PubMed PMID: 37289570; PubMed Central PMCID: PMCPMC10355802

Name	Role of the ADRC in bringing new investigators to the field	Funding	Publication Title
Kexin Yu PhD	Started new Post doctoral fellowship at Oregon Health & Science University (Mentor: Maria Aranda)	F99AG068492	• Yu K, Jiang H, Liu M, Wu S, Jordan-Marsh M, Chi I. Chinese Immigrant Caregivers: Understanding Their Unmet Needs and the Co-Design of an mHealth App. Can J Aging. 2024:1-8. Epub 2024/05/17. doi: 10.1017/s0714980824000187. PubMed PMID: 38757195. • Dodge HH, Yu K, Wu CY, Pruitt PJ, Asgari M, Kaye JA, et al. Internet-Based Conversational Engagement Randomized Controlled Clinical Trial (I-CONECT) Among Socially Isolated Adults 75+ Years Old With Normal Cognition or Mild Cognitive Impairment: Topline Results. Gerontologist. 2024;64(4). Epub 2023/11/08. doi: 10.1093/geront/gnad147. PubMed PMID: 37935416; PubMed Central PMCID: PMCPMC10943511. • Yu K, Liang J, Tung YH, Zhang M, Wu S, Chi I. Social network and mental health of Chinese immigrants in affordable senior housing during the COVID-19 pandemic: a mixed-methods study. Aging Ment Health. 2023;27(10):1956-64. Epub 2023/05/22. doi: 10.1080/13607863.2023.2213646. PubMed PMID: 37212623; PubMed Central PMCID: PMCPMC10529124. • Palinkas LA, De Leon J, Yu K, Salinas E, Fernandez C, Johnston J, et al. Adaptation Resources and Responses to Wildfire Smoke and Other Forms of Air Pollution in Low-Income Urban Settings: A Mixed-Methods Study. Int J Environ Res Public Health. 2023;20(7). Epub 2023/04/14. doi: 10.3390/ijerph20075393. PubMed PMID: 37048007; PubMed Central PMCID: PMCPMC10094253. • Yu K, Siang Ng TK. Investigating Biological Pathways Underpinning the Longitudinal Association Between Loneliness and Cognitive Impairment. J Gerontol A Biol Sci Med Sci. 2023;78(8):1417-26. Epub 2022/10/07. doi: 10.1093/gerona/glac213. PubMed PMID: 36200979; PubMed Central PMCID: PMCPMC11015400. •
Hoda Abdel Magid, PhD	Assistant Professor, USC, (Mentor: Julie Zissimopoulos)	R00HL161479	• Abdel Magid HS, Jaros S, Li Y, Steinman MA, Lee S, Jing B, et al. Effects of residential socioeconomic polarization on high blood pressure among nursing home residents. Health Place. 2024;87:103243. Epub 2024/04/26. doi: 10.1016/j.healthplace.2024.103243. PubMed PMID: 38663339; PubMed Central PMCID: PMCPMC11102837. • Milliren CE, Sajjad OM, Abdel Magid HS, Gooding HC, Richmond TK, Nagata JM. Adolescent individual, school, and neighborhood influences on young adult diabetes risk. Health Place. 2023;83:103047. Epub 2023/06/11. doi: 10.1016/j.healthplace.2023.103047. PubMed PMID: 37301169; PubMed Central PMCID: PMCPMC10798047 •

Azizi Seixas, PhD	Associate Professor, University of Miami (Mentor: Julie Zissimopoulos)	R01AG075007	• Rajabli F, Seixas AA, Akgun B, Adams LD, Inciute J, Hamilton KL, et al. African Ancestry Individuals with Higher Educational Attainment Are Resilient to Alzheimer's Disease Measured by pTau181. J Alzheimers Dis. 2024;98(1):221-9. Epub 2024/02/23. doi: 10.3233/jad-231116. PubMed PMID: 38393909; PubMed Central PMCID: PMCPMC11091636. • Jean-Louis G, Jin P, Moise R, Blanc J, Rogers A, Bubu OM, et al. Effectiveness of peer-delivered sleep health education and social support in increasing OSA evaluation among at-risk blacks. J Sleep Res. 2024:e14213. Epub 2024/05/22. doi: 10.1111/jsr.14213. PubMed PMID: 38773705. • Seixas A, Richards S, Moore JQ, Izeogu C, Hollimon LA, Jin P, et al. Precision Recruitment and Engagement of Individuals at Risk for Diabetes and Hypertension in Clinical Trials (PREDHICT): A Randomized Trial for an E-Persuasive Mobile Application to Inform Decision Making about Clinical Trials. Int J Environ Res Public Health. 2023;20(23). Epub 2023/12/08. doi: 10.3390/ijerph20237115. PubMed PMID: 38063545; PubMed Central PMCID: PMCPMC10706176. • Olaye IM, Seixas AA. The Gap Between AI and Bedside: Participatory Workshop on the Barriers to the Integration, Translation, and Adoption of Digital Health Care and AI Startup Technology Into Clinical Practice. J Med Internet Res. 2023;25:e32962. Epub 2023/05/02. doi: 10.2196/32962. PubMed PMID: 37129947; PubMed Central PMCID: PMCPMC10189623. • Rogers A, Chung A, Seixas A, Chung D, Zizi F, Jean-Louis G. Strategies to Engage Blacks in Sleep Medicine: Lessons Learned from Three Studies Applying Community-Based Participatory Research Principles. J Sleep Disord Ther. 2023;12(4). Epub 2023/07/10. PubMed PMID: 37425370; PubMed Central PMCID: PMCPMC10327646.
Sadaf Milani, PhD	Assistant Professor, University of Texas Medical Branch, (Mentor: Julie Zissimopoulos)	K01AG075254	• Hernández-Pérez JG, Taha S, Torres-Sánchez LE, Villasante-Tezanos A, Milani SA, Baillargeon J, et al. Association of sleep duration and quality with serum testosterone concentrations among men and women: NHANES 2011-2016. Andrology. 2024;12(3):518-26. Epub 2023/07/15. doi: 10.1111/andr.13496. PubMed PMID: 37452666; PubMed Central PMCID: PMCPMC10788378. • Milani SA, Sanchez C, Kuo YF, Downer B, Al Snih S, Markides KS, et al. Pain and incident cognitive impairment in very old Mexican American adults. J Am Geriatr Soc. 2024;72(1):226-35. Epub 2023/10/05. doi: 10.1111/jgs.18618. PubMed PMID: 37794825; PubMed Central PMCID: PMCPMC10842321. • Robertson MC, Downer B, Schulz PE, Samper-Ternent R, Lyons EJ, Milani SA. Social and Leisure Activities Predict Transitions in Cognitive Functioning in Older Mexican Adults: A Latent Transition Analysis of the Mexican Health and Aging Study. J Gerontol B Psychol Sci Soc Sci. 2023;78(10):1625-35. Epub 2023/05/25. doi: 10.1093/geronb/gbad082. PubMed PMID: 37227927; PubMed Central PMCID: PMCPMC10561883. • Momin KN, Sheehan C, Samper-Ternent R, López DS, Wong R, Milani SA. The impact of insomnia symptoms on obesity among Mexicans aged 50 and older. Salud Publica Mex. 2023;65(5, sept-oct):530-41. Epub 2023/12/07. doi: 10.21149/14759. PubMed PMID: 38060913; PubMed Central PMCID: PMCPMC10962277.

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	❍No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	❍No
Is the Project Exempt from Federal regulations?	❍Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447180.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R Multi-Project Cover

5. APPLICANT INFORMATION

Legal Name: University of Southern California
Department:
Division:
Street1: Department of Contracts and Grants
Street2: 3720 South Flower Street

City: Los Angeles	County/Parish:	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0701

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Dr.	Rey		Dominguez	

Position/Title: Contracts and Grants Officer
Street1: 3720 S. Flower Street
Street2: Fl 3

City: Los Angeles	County/Parish: Los Angeles	State: CA: California
Province:	Country: USA: UNITED STATES	ZIP / Postal Code: 90089-0000
Phone Number: 2137401182	Fax Number:	Email: reymundd@usc.edu

7. TYPE OF APPLICANT: O: Private Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Imaging Core

12. PROPOSED PROJECT:

Start Date	Ending Date
04/01/2025	03/31/2030

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: University of Southern California

* Street1: Health Sciences Campus		Street2: SHN 2025 Zonal Avenue
* City: Los Angeles	County:	* State: CA: California
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 90033-0000
UEI: G88KLJR3KYT5	* Project/Performance Site Congressional District: CA-034	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ● Yes ❍ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number 00005906

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	Imaging_Abstract1044447012.pdf	Mime Type: application/pdf
8. Project Narrative		
9. Bibliography & References Cited	Imaging_References1044447011.pdf	Mime Type: application/pdf
10. Facilities & Other Resources		
11. Equipment		

Abstract: Imaging Core

As a part of the USC Alzheimer Disease Research Center competitive renewal, the Imaging Core will continue to serve the overall mission to provide high-quality MR and PET imaging resources**. The main goal of the Imaging Core is to provide a standardized imaging protocol across ADRC following the Standardized Centralized Alzheimer's & Related Dementias Neuroimaging guidelines (SCAN).** The ADRC Imaging Core will also provide core resources, including MR physics and PET expertise, multi-scanner sequence and protocol development, image data quality control and management, research collaboration, and a global framework for conducting AD neuroimaging research. The Imaging Core is led by a well-integrated and collaborative team, including **Arthur Toga (Core Leader), Ioannis Pappas (Core Co-Leader), Nasim Sheikh-Bahaei, Yonggang Shi, Meredith Braskie, Priya Rajagopalan, and Ararat Chakhoyan** who together provide expertise in vascular, functional, and structural imaging and its analyses for 3T and 7T MRI, and PET. We propose 5 specific aims where the Imaging Core will provide a standardized imaging framework for all ADRC; perform high quality control of all imaging data; develop technical and scientific approaches for ultra-high field 7T MRI; assist in PET protocol development and quality control; and conduct education and training in modern imaging techniques. Taken together, these goals will serve the USC ADRC's overarching goals to **1) Recruit and follow a clinical cohort that reflects the ethnic diversity of Los Angeles and is enriched for *APOE* ε4,** focusing on factors that modify the effect of *APOE* ε4 on cognitive decline and AD biomarkers, **2) promote local ADRD research at USC,** and **3) further national collaborations.**

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RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Arthur Middle Name W. Last Name*: Toga Suffix:

Position/Title*:	Provost Professor
Organization Name*:	University of Southern California
Department:	KSOM Neuroimaging and Informatics Institute
Division:	Keck School of Medicine of USC
Street1*:	SSB 102
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (323) 442-7246 Fax Number: (323) 442-2230 E-Mail*: toga@usc.edu

Credential, e.g., agency login: TOGA22

Project Role*: Other (Specify) Other Project Role Category: Imaging Core Lead

Degree Type: PhD, MS, BS Degree Year:

File Name

Attach Biographical Sketch*:

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Ioannis Middle Name Last Name*: Pappas Suffix:

Position/Title*:	Assistant Professor of Research Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neuroimaging and Informatics Institute
Division:	
Street1*:	Health Sciences Campus
Street2:	SHN 2025 Zonal Avenue
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764-3341 Fax Number: E-Mail*: ipappas@usc.edu

Credential, e.g., agency login: IOANNIS.PAPPAS

Project Role*: Other (Specify) Other Project Role Category: Imaging Co-Core Lead

Degree Type: PhD, MSc Degree Year:

File Name

Attach Biographical Sketch*: Biosketch_Pappas1044446987.pdf

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Nasim Middle Name Last Name*: Sheikh-Bahaei Suffix:

Position/Title*:	Assistant Professor of Clinical Radiology
Organization Name*:	University of Southern California
Department:	KSOM Neurology
Division:	
Street1*:	1500 San Pablo Street
Street2:	Bldg. UNH
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (213) 764-3341 Fax Number: E-Mail*: nasim.sheikh-bahaei@med.usc.edu

Credential, e.g., agency login: nasim.sheikh-bahaei

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: MD, PhD Degree Year:

File Name

Attach Biographical Sketch*: Biosketch_Sheikh_Bahaei1044446992.pdf

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Yonggang Middle Name Last Name*: Shi Suffix:

Position/Title*:	Associate Professor of Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neuroimaging and Informatics Institute
Division:	
Street1*:	SSB 102
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (323) 442-7246 Fax Number: (323) 442-2230 E-Mail*: yonggans@usc.edu

Credential, e.g., agency login: YONGGANGSHI

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD, MS Degree Year:

File Name

Attach Biographical Sketch*: Biosketch_Shi_May20241044446997.pdf

Attach Current & Pending Support:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Meredith Middle Name N. Last Name*: Braskie Suffix:

Position/Title*:	Assistant Professor of Neurology
Organization Name*:	University of Southern California
Department:	KSOM Neuroimaging and Informatics Institute
Division:	

Street1*:	4676 Admiralty Way, ISI 430
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90089-0000

Phone Number*: (323) 442-7246 Fax Number: (323) 442-2230 E-Mail*: braskie@usc.edu

Credential, e.g., agency login: mbraskie

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Braskie1044447000.pdf
Attach Current & Pending Support:	

PROFILE - Senior/Key Person

Prefix: First Name*: Priya Middle Name Last Name*: Rajagopalan Suffix:

Position/Title*:	Assistant Professor of Clinical Radiology
Organization Name*:	University of Southern California
Department:	KSOM Radiology
Division:	
Street1*:	1500 San Pablo St.
Street2:	
City*:	Los Angeles
County:	Los Angeles
State*:	CA: California
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	90033-0000

Phone Number*: (323) 442-7600 Fax Number: E-Mail*: priyaraj@usc.edu

Credential, e.g., agency login: PRRAJAGO

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: MBBS, MPH Degree Year:

	File Name
Attach Biographical Sketch*:	Biosketch_Rajagopalan1044447009.pdf
Attach Current & Pending Support:	

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Pappas, Ioannis

eRA COMMONS USER NAME (credential, e.g., agency login): IOANNIS.PAPPAS

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
National Technical University of Athens, Greece	BS	05/2013	Engineering
University of Florida, Gainesville, FL	MSc	08/2015	Engineering
University of Cambridge, Cambridge, England	PhD	01/2019	Clinical Neurosciences
University of California, Berkeley, Berkeley, CA	Postdoctoral	06/2021	Clinical Neurosciences

A. Personal Statement

I am an Assistant Professor of Research Neurology, and my research is focused on using brain imaging markers to understand changes in brain organization in health and disease. I have a broad background in clinical neurosciences with technical expertise in magnetic resonance (MR) imaging data acquisition and analysis. I have developed novel MR spectroscopy, blood flow, and functional MR imaging sequences for the 3 Tesla and 7 Tesla scanners at USC LONI. I am the Neuroimaging Core Co-Lead for the USC Alzheimer's Disease Research Center (ADRC) where I am responsible for all ADRC imaging data acquisition and analysis. My focus is on functional and vascular imaging for the purpose of early detection of dementia. During these projects, I have successfully collaborated with leading researchers in the field and produced peer-reviewed publications. Thus, I strongly believe that I can provide the required expertise on this project.

Ongoing projects that I would like to highlight include:

A) P01AG052350-06
Toga (PI), Role: Co-Investigator (Project lead)
09/2016-03/2027
Vascular Contributions to Dementia and Genetic Risk Factors for Alzheimer's Disease

I am the Project 2 lead that examines how structural and functional connectivity markers change in individuals at risk of developing Alzheimer's disease and what is their relationship to vascular biomarkers. I am also the Co-lead of the Neuroimaging Core that develops neuroimaging techniques for obtaining imaging data at the 3T and 7T scanners, assess their reliability, and is responsible for the analysis of the acquired data sets.

B) P30AG066530-02
Chui, Toga (PIs), Role: Co-Investigator (Imaging Core Co-lead)
09/2020-02/2025
Alzheimer Disease Research Center-USC

The scientific theme of USC-ADRC focuses on vascular contributions to dementia and Alzheimer's disease, vascular and metabolic risk factors, the neurovascular unit, brain structure and function, and the treatment and prevention of Alzheimer's disease. I am the Co-Lead of the Neuroimaging Core that is responsible for neuroimaging data acquisition and processing across the ADRC with focus on vascular imaging and connecting

markers. I oversee the development and delivery of the imaging protocols at the 3T and 7T MRI scanners, as well as the storage and processing of the imaging data.

Research products:
1. Contreras, J. A., Fujisaki, K., Ortega, N., **Pappas, I.**, Barisano, G., Sagare, A., Zlokovic, B., Toga, A. W., & Pa, J. (2022, July 31 – August 4) Functional hyperconnectivity is associated with higher levels of sPDGFRβ, a marker of blood brain barrier breakdown, in older adults, *Alzheimer's Association International Conference, San Diego, USA.*

C) R01MH127176-03
Ho (PI), Role: Co-Investigator (Subcontract-PI)
06/2022-04/2027
Inflammatory and Glutamatergic Mechanisms of Sustained Threat in Adolescents with Depression: Toward Predictors of Treatment Response and Clinical Course

This proposal is a longitudinal, multimodal neuroimaging study that uses spectroscopic methodologies and validated behavioral metrics to identify predictors of no response to depression treatment. I am responsible for developing the spectroscopic sequences and analyzing the obtained data with focus on estimating glutamate.

Research products:
1. Coury, S. M., López, V., Bajwa, Z., Garcia, J. M., Teresi, G. I., Kuhlman, K. R., Li, Y., Cole, S., Miklowitz, D.J., **Pappas, I.**, & Ho, T. C. (2024). Protocol for teen inflammation glutamate emotion research (TIGER): Toward predictors of treatment response and clinical course. *Brain, Behavior, & Immunity-Health* (2024) 35:100718

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2021 - Present Assistant Professor of Research Neurology, University of Southern California, Los Angeles
2019 - 2021 Postdoctoral Research Fellow, University of California, Berkeley
2017 - 2018 Visiting Research Scholar, Harvard Medical School

Other Experience and Professional Memberships

2021 – Present Member, Cognitive Neuroscience Society
2021 – Present Member, International Society for Magnetic Resonance in Medicine
2019 – Present Member, Society for Neuroscience

Honors

2016 Downing College, Cambridge University Oon Khye Beng Ch'Hia Tsio studentship for research in preventive medicine.
2016 Keystone Symposia of Science Fund Scholarship, National Institute of Neurological Disorders and Stroke (NINDS).
2014 Gerondelis Foundation Graduate scholarship award.
2013 Graduate school award, University of Florida.

C. Contributions to Science

1. Brain imaging and big data in Alzheimer's disease
I provide expertise with respect to imaging acquisition and analysis imaging markers for Alzheimer's disease (e.g., amyloid and tau PET, structural MRI). I am also responsible for big data collection and analysis thought the Global Alzheimer's Association Interactive Network (GAAIN).

1a. Leuzy, A., Dore, V., Raket, L. R., Klein, G., Baker, S. L., Carrillo, M. C., Charil, A., Collins, E. C., Collins, J. A., Haeberlein, S. B., Higuchi, M., Hostetler, E. D., Hutchison, R. M., Iaccarino, L., Irizarry, M. C., Jagust, W. J.,

Johnson, K. A., Karten, Y., Kolb, H. C., Lopresti, B. J., Mathotaarachchi, S., Mintun, M. A., Ossenkoppele, R., **Pappas, I.**, Pascoal, T. A., Pontecorvo, M. J., Rabinovici, G. D., Rosa-Neto, P., Saad, Z. S., Bohorquez, S. S., Stephens, A. W., Sivakumaran, S., Tonietto, M., Smith, R., Rowe, C. C., Villemagne, V. L., & Hansson, O. Harmonization of tau-PET in Alzheimer's disease: comparison of methods to derive CenTauR units for [18F]RO948, [18F]Flortaucipir, and [18F]MK-6240. Accepted in Alzheimer's and Dementia (2024).

1b. Xiao, C., **Pappas, I.**, Aksman, L. M., O'Bryant S. E., & Toga, A.W., Health and Aging Brain Study (HABS-HD Study Team, Alzheimer's Disease Neuroimaging Initiative. Comparison of genetic and health risk factors for mild cognitive impairment and Alzheimer's disease between Hispanic and non-Hispanic white participants. Alzheimer's & Dementia (2023) PMID: 37104247

1c. Toga, A. W., Phatak, M., **Pappas, I.**, Thompson, S., McHugh, C. P., Clement, M. H. S, Bauermeister, S., Maruyama, T., & Gallacher, J. The pursuit of approaches to federate data to accelerate Alzheimer's disease and related dementia research: GAAIN, DPUK, and ADDI. Frontiers in Neuroinformatics (2023) PMID: 37304174 PMCID: PMC10248126

2. Functional imaging of human brain

I developed imaging acquisition and analysis methods of functional MRI that depend on blood oxygenated level dependent (BOLD) contrast. In a series of published studies, I used BOLD fMRI and functional connectivity to probe brain organization in the healthy and diseased brain.

2a. Lurie D. J., **Pappas, I.**, & D'Esposito, M. Cortical timescales and the modular organization of structural and functional brain networks. Human Brain Mapping (2024) 45 (2), e26587 PMID: 38339903 PMCID: PMC10823764

2b. Eichenbaum, A.*, **Pappas, I.***, Lurie, D., Cohen, J. R., & D'Esposito, M. (2020). Differential contributions of static and time-varying functional connectivity to human behavior. *Network Neuroscience*, 1–21. PMID: 33688610. PMCID: PMC7935045

2c. Lu, D., **Pappas, I.**, Menon, D. K., & Stamatakis, E. A. (2021). A precuneal causal loop mediates external and internal information integration in the human brain. *Journal of Neuroscience*, 41 (48), 9944-9956

2d. Toker, D., **Pappas, I.**, Lender, J., Frohlich, J., Mateos, D. M., Muthukumaraswamy, S., Carhart-Harris, R., Paff, M., Vespa, P. M., Monti, M. M., Sommer, F. T., Knight, R. T., & D'Esposito, M. Consciousness is supported by near-critical slow cortical electrodynamics. *PNAS*, 119, e2024455119 (2022) PMID: 35145021 PMCID: PMC8851554

3. Pulses sequence for for magnetic resonance

I provide expertise with respect to image acquisition methods for magnetic resonance spectroscopy. I developed broadband excitation pulses that accurately capture the localized metabolic content (such as glutamate) in vivo.

3a. Coury, S. M., López, V., Bajwa, Z., Garcia, J. M., Teresi, G. I., Kuhlman, K. R., Li, Y., Cole, S., Miklowitz, D.J., **Pappas, I.**, & Ho, T. C., Protocol for teen inflammation glutamate emotion research (TIGER): Toward predictors of treatment response and clinical course. Brain, Behavior, & Immunity-Health (2024) 35:100718 PMID: 38235411 PMCID: PMC10792689

3b. Kaiser, L. G., Veshtort, M., **Pappas, I.**, Deelchand, D. K., Auerbach, E. J., Marjańska, M., & Inglis, B.(2021) Broadband selective excitation radiofrequency pulses for optimized localization in vivo, Magnetic resonance in medicine 87(5), 2111–2119. PMID: 34866226 PMCID: PMC8847340.

Link to biography

https://www.ncbi.nlm.nih.gov/myncbi/ioannis.pappas.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: Sheikh-Bahaei, Nasim

eRA COMMONS USER NAME (credential, e.g., agency login): nasim.sheikh-bahaei

POSITION TITLE: Assistant Professor of Radiology and Neurology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
Tehran University of Medical Sciences , Tehran	MD	12/2003	Medicine
University of Cambridge , Cambridge	PHD	12/2018	Alzheimer’s disease imaging
Cambridge University Hospitals , Cambridge	Resident	02/2012	Radiology
Cambridge University Hospitals , Cambridge	Fellow	06/2016	Clinical lecturer and Neuroradiology fellow

A. Personal Statement

I am a double board-certified physician in internal medicine and radiology with subspecialty training in neuroradiology. During my clinical lecturer position in University of Cambridge, I started my PhD on Alzheimer’s disease (AD) imaging. My PhD was a multimodality imaging study focusing on positron emission tomography (PET), MR spectroscopy (MRS) and susceptibility weighted imaging (SWI) in AD, funded by National Institute for Health Research (NIHR). As the PI for my study and collaborator for many other studies on neurodegenerative diseases in the University of Cambridge, I read and analyzed all the images visually and also provided quantitative measurements for MRI, MRS and PET scans, and performed statistical analyses for several studies. After joining USC as a Neuroradiology faculty in 2018, I became a Co-I in Alzheimer’s Disease Research Center (ADRC)(P30 AG066530-02 (MPI: (Chui, Toga, Zlokovic)) at USC and started to work on different imaging cohorts in Laboratory of Neuroimaging (LONI) and Imaging Core of ADRC. As an expert in PET and MRI in neurodegenerative diseases, I am providing imaging insight as well as visual and quantitative PET and MRI reads for some of the studies in USC ADRC. In addition, over the last couple of years and because of my experience with PET and MRI imaging in AD, I was also invited to collaborate with ADRC and the 90plus Study in University of California, Irvine. I have reported more than 500 amyloid PET and more than thousands MRI for preclinical, MCI, AD and other dementia cases for various studies. Currently I am also conducting and collaborating with different research projects involving AD biomarkers such as amyloid and tau PET and I am collaborating with other projects on non-Alzheimer dementias such as primary progressive aphasia and hippocampal sclerosis. Recently I was awarded an R01 as mPI to harmonize large multimodal cohorts using artificial intelligence and machine learning to improve the accuracy of AD and ADRD diagnosis. I will work on defining and categorizing cases based in their biomarkers to harmonize this complex data set in order to develop algorithm and tools for better diagnosis of AD. I am also involved as a Co-I in another R21 looking into the radiological and pathological correlations in hippocampal sclerosis of aging utilizing antemortem and postmortem MRIs. I have received funding from Alzheimer's Association to investigate the risk factors of AD in MENA population. My career goal is to become a leader on imaging biomarkers in Alzheimer’s disease at USC and later nationally. The area of my research interest is focused on biomarkers which can explain the dissociation between pathological changes in AD (presence of amyloid and tau) and clinical AD syndrome. Synaptic dysfunction/loss and concurrent microvascular pathologies can both contribute to this picture. I am working on the role of microvascular changes in large cohorts of AD and normal cases to investigate how these microvascular changes (such as perivascular spaces, cerebral microbleeds and white matter lesions) contributes to conversion to clinical dementia. For this purpose, we developed an automated software to quantify the volume and severity of perivascular spaces and white matter changes. I am also working on different biomarkers for synaptic density and function including MR spectroscopy, DTI and PET. As a part of my clinical duty, I am also in charge of imaging and reporting the MRI of cases who are considered for anti-amyloid therapy at USC.

1. Woodworth DC, Sheikh-Bahaei N, Scambray KA, Phelan MJ, Perez-Rosendahl M, Corrada MM, Kawas CH, Sajjadi SA. Dementia is associated with medial temporal atrophy even after accounting for neuropathologies. Brain Commun. 2022;4(2):fcac052. PubMed Central PMCID: PMC8952251.
2. Sepehrband F, Barisano G, Sheikh-Bahaei N, Cabeen RP, Choupan J, Law M, Toga AW. Image processing approaches to enhance perivascular space visibility and quantification using MRI. Sci Rep. 2019 Aug 26;9(1):12351. PubMed Central PMCID: PMC6710285.
3. Sheikh-Bahaei N, Manavaki R, Sajjadi SA, Priest AN, O'Brien JT, Gillard JH. Correlation of Lobar Cerebral Microbleeds with Amyloid, Perfusion, and Metabolism in Alzheimer's Disease. J Alzheimers Dis. 2019;68(4):1489-1497. PubMed PMID: 30909210.
4. Sheikh-Bahaei N, Sajjadi SA, Manavaki R, McLean M, O'Brien JT, Gillard JH. Positron emission tomography-guided magnetic resonance spectroscopy in Alzheimer disease. Ann Neurol. 2018 Apr;83(4):771-778. PubMed PMID: 29518282.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2018 -	Assistant Professor of Radiology and Neurology, Keck school of Medicine, University of South California , Los Angeles, CA
2016 - 2018	PhD Candidate , University of Cambridge , Cambridge
2012 - 2016	Clinical lecturer and Neuroradiology Fellow, University of Cambridge, School of clinical medicine , Cambridge
2008 - 2012	Resident in Radiology, Cambridge University Hospitals , Cambridge
2006 - 2008	Resident in Internal Medicine , Blackpool and Birmingham NHS Foundation Trusts, Blackpool and Birmingham
2005 - 2006	Research associate , University College London, London
2004 - 2005	Research Associate , Endocrine Research Center , Tehran
2002 - 2003	Intern , Tehran University of Medical Science , Tehran

Honors

2021 - 2022	Stewart Dale Fordham Distinguished Faculty Award" for excellence in leadership, commitment to education, clinical excellence, and consistent and significant contribution to building the culture of inquiry in department of Radiology, Keck School of Medicine oF USC
2018	Gabriel H Wilson Award , Western Neuroradiological Society (WNRS)
2016	EG Fearnsides Scholarship for The Best Research Presentation in University of Cambridge , EG Fearnsides Fund, University of Cambridge
2013	Charles Freer Scholarship for Top Radiology Resident Research , Cambridge University Hospitals NHS Foundation Trust
2002	Best Medical Student Researcher , Tehran University of Medical Sciences

C. Contribution to Science

1. MR Spectroscopy and PET imaging in AD: As part of my PhD project, I performed the first PET-guided MRS in AD. I investigated the spatial and temporal associations between different brain metabolites and both amyloid and FDG PET. Our data showed that myoinositol could be used as state and total N-acetyl group as stage biomarkers in AD. MRS is a relatively available, inexpensive and non-invasive technique, which based on the associations that we demonstrated, could be used to improve the accuracy of diagnosis in early AD. We have also investigated different components that could be responsible for controversial MRS results in the literature, such as the effect of voxel position and tissue composition in AD.

    a. Sheikh-Bahaei N. MR Spectroscopy in Alzheimer's Disease. Biomedical Spectroscopy and Imaging. 2020 January; 9:13-21.

b. Sheikh-Bahaei N, Sajjadi S, Manavaki R, Mclean M, O'Brien jT, Gillard JH. The effect of voxel position and tissue composition on MR Spectroscopy in Alzheimer's disease. Western Neuroradiology meeting; 2018 October 11; Dana Point, CA, USA.
c. Sheikh-Bahaei N, Sajjadi SA, Manavaki R, McLean M, O'Brien JT, Gillard JH. Positron emission tomography-guided magnetic resonance spectroscopy in Alzheimer disease. Ann Neurol. 2018 Apr;83(4):771-778. PubMed PMID: 29518282.
d. Sheikh-Bahaei N, Sajjadi SA, Pierce AL. Current Role for Biomarkers in Clinical Diagnosis of Alzheimer Disease and Frontotemporal Dementia. Curr Treat Options Neurol. 2017 Nov 14;19(12):46. PubMed PMID: 29134465.

2. Imaging vascular changes in AD: Another aspect of my PhD was focused on assessment of vascular changes in AD. I have investigated the global and regional associations between lobar cerebral microbleeds with PET driven values of amyloid, perfusion and metabolism. Our study was the first to assess the regional association between microbleeds and PET imaging biomarkers in AD. I was also involved in a project on the association between cerebral small vessel disease and genetic predisposition in late onset AD. Currently, I am working on perivascular spaces and their association with microbleeds and other AD imaging biomarkers. In addition, I am analysing quantitative susceptibility mapping in AD cases and their relationship with PET.

   a. Sepehrband F, Barisano G, Sheikh-Bahaei N, Choupan J, Cabeen RP, Lynch KM, Crawford MS, Lan H, Mack WJ, Chui HC, Ringman JM, Toga AW. Volumetric distribution of perivascular space in relation to mild cognitive impairment. Neurobiol Aging. 2021 Mar;99:28-43. PubMed Central PMCID: PMC7902350.
   b. Barisano G, Sheikh-Bahaei N, Law M, Toga AW, Sepehrband F. Body mass index, time of day and genetics affect perivascular spaces in the white matter. J Cereb Blood Flow Metab. 2021 Jul;41(7):1563-1578. PubMed Central PMCID: PMC8221772.
   c. Sheikh-Bahaei N, Barisano G, Sepehrband F, Acharya J, Rajamohan A, Law M, Chui H, Toga A. Microvascular signature of preclinical Alzheimer's disease in a longitudinal cohort. American Society of Neuroradiology Meeting; 2019; Boston, USA.
   d. Stefaniak JD, Su L, Mak E, Sheikh-Bahaei N, Wells K, Ritchie K, Waldman A, Ritchie CW, O'Brien JT. Cerebral small vessel disease in middle age and genetic predisposition to late-onset Alzheimer's disease. Alzheimers Dement. 2018 Feb;14(2):253-258. PubMed PMID: 29156222.

3. Imaging of non-Alzheimer Dementia: I was also involved in research project to assess the accuracy of MRI visual classification to differentiate the subtypes of primary progressive aphasia. Our data showed apart from Semantic variant, visual read is not sensitive enough for other subtypes. I am actively involved in another project investigating the role of MRS in Frontotemporal dementia with the aim to improve the accuracy of diagnosis

   a. Donahue EK, Murdos A, Jakowec MW, Sheikh-Bahaei N, Toga AW, Petzinger GM, Sepehrband F. Global and Regional Changes in Perivascular Space in Idiopathic and Familial Parkinson's Disease. Mov Disord. 2021 May;36(5):1126-1136. PubMed Central PMCID: PMC8127386.
   b. Sheikh-Bahaei N, Sajjadi SA, Pierce AL. Current Role for Biomarkers in Clinical Diagnosis of Alzheimer Disease and Frontotemporal Dementia. Curr Treat Options Neurol. 2017 Nov 14;19(12):46. PubMed PMID: 29134465.
   c. Sajjadi SA, Sheikh-Bahaei N, Cross J, Gillard JH, Scoffings D, Nestor PJ. Can MRI Visual Assessment Differentiate the Variants of Primary-Progressive Aphasia?. AJNR Am J Neuroradiol. 2017 May;38(5):954-960. PubMed Central PMCID: PMC7960364.

4. Imaging vessel walls and atheroma plaques: I have been involved in various research projects on atheroma imaging. As a neuroradiologist with expertise in vessel wall imaging, I classified critical plaques based on their composition and also the presence of adjacent calcification. We have also conducted longitudinal studies to investigate if these features predict future cerebrovascular events. In another project, we investigated the effect of plaques deformation due to shear effect and pulse pressure and its association with future clinical symptoms.

a. Sheikh-Bahaei N, Matys T, Gillard J. Anatomy of the Neck Arteries. In: Saba L, Raz E, editors. Neurovascular Imaging [Internet] New York, NY: Springer New York; 2016. Chapter Chapter 287-94p. Available from: http://link.springer.com/10.1007/978-1-4614-9029-6_2 DOI: 10.1007/978-1-4614-9029-6_2
b. Teng Z, Sadat U, Wang W, Bahaei NS, Chen S, Young VE, Graves MJ, Gillard JH. Intraplaque stretch in carotid atherosclerotic plaque--an effective biomechanical predictor for subsequent cerebrovascular ischemic events. PLoS One. 2013;8(4):e61522. PubMed Central PMCID: PMC3634006.
c. Teng Z, He J, Degnan AJ, Chen S, Sadat U, Bahaei NS, Rudd JH, Gillard JH. Critical mechanical conditions around neovessels in carotid atherosclerotic plaque may promote intraplaque hemorrhage. Atherosclerosis. 2012 Aug;223(2):321-6. PubMed Central PMCID: PMC3437553.

Complete List of Published Work in My Bibliography:
https://www.ncbi.nlm.nih.gov/myncbi/nasim.sheikh-bahaei.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Shi, Yonggang

eRA COMMONS USER NAME (credential, e.g., agency login): YONGGANGSHI

POSITION TITLE: Associate Professor of Neurology, Electrical and Computer Engineering, and Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Southeast University, China	B.S.	07/1996	Electrical Engineering
Southeast University, China	M.S.	05/1999	Electrical Engineering
Boston University	Ph.D.	05/2005	Electrical Engineering
UCLA School of Medicine	PostDoc	06/2009	Neuroscience

A. Personal Statement

I direct the Neuro Image Computing Research (NICR) group at the Stevens Neuroimaging and Informatics Institute at USC. My research focuses on the development of general mathematical frameworks and computational algorithms in mapping brain structures and functions for studying neurological disorders. For 3D brain surface analysis, I have developed a unified approach based on the spectrum of the Laplace-Beltrami (LB) operator on manifolds and applied it to study brain development and atrophy in various projects. Under this framework, I have developed robust algorithms for intrinsic surface reconstruction, modeling, and mapping. These works have been published in a series of papers in the *IEEE Transactions on Medical Imaging* and resulted in software tools distributed publicly via our MOCA (Metric Optimization for Computational Anatomy) project on NITRC. More recently, we have extended our shape analysis tools for vessel segmentation and density mapping for various imaging modality. Another major area of my research is connectome modeling in human brain. With the advance of connectome imaging techniques that can provide diffusion MRI data with drastically increased spatial and angular resolutions, I have developed novel algorithms and software tools for distortion correction, harmonization, fiber orientation distributions (FODs) modeling, topography-preserving fiber tracking and filtering, and fiber bundle reconstruction. Using these novel computational methods, I have also invented algorithms for the construction and public distribution of probabilistic atlases of challenging fiber bundles in human brain such as the retinofugal visual pathway and brainstem pathways.

Ongoing and recently completed projects that I would like to highlight include:

R01EB022744
Shi (PI)
09/22/16-1/31/27
Surface-Based Fiber Tracking and Modeling Techniques for Mapping the Superficial White Matter Connectome with Diffusion MRI
In this project we will develop a series of surface-based modeling techniques to greatly advance superficial white matter fiber tracking and connectivity research.

RF1AG064584
Shi (PI)
09/01/20-08/31/24
Tau-induced connectome imaging markers of Alzheimer's disease
Our project will for the first time provide the comprehensive and in vivo characterization of the fiber pathways affected by tau pathology in Alzheimer's disease (AD). This systematic investigation is important for improving our understanding of the biological mechanism of AD and provide more targeted imaging makers for the early prediction of AD at the prodromal stage.

RF1AG077578
Shi (PI)
06/01/2023-05/31/2026
Shape-based personalized AT(N) imaging markers of Alzheimer's disease
We will develop novel imaging markers to allow more personalized characterization of the AT(N) status of individual subjects. Building upon our extensive experience in brain shape analysis, we will develop tools to quantify the topographic pattern of cortical tau and Aβ pathology and build a personalized analysis framework for the early detection of localized brain atrophy.

RF1AG056573
Shi (PI)
09/01/18-08/31/23
Brainstem connectomes related to Alzheimer's disease
The novel brainstem connectomes developed in this project will provide the enabling technology for the *in vivo* examination of the relation between brainstem connectivity, tau pathology, and behavior changes in AD. This will greatly improve our understanding of the anatomical basis underlying the development of neurodegeneration and behavior symptoms in AD.

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2019 - present	Associate Professor with Tenure, USC Mark and Mary Stevens Neuroimaging and Informatics Institute, Keck School of Medicine of USC
2013 - 2019	Tenure-track Assistant Professor, USC Mark and Mary Stevens Neuroimaging and Informatics Institute, Keck School of Medicine of USC
2009 - 2013	Assistant Professor, Department of Neurology, School of Medicine, UCLA
2005 - 2009	PostDoctoral fellow, Department of Neurology, School of Medicine, UCLA
2004	Intern, Siemens Corporate Research, Princeton, NJ
2003	Visiting scholar, I3S Laboratory, University of Nice-Sophia Antipolis, France
2000 - 2005	Research Assistant, Electrical and Computer Engineering Department, Boston University

Other Experience and Professional Memberships
- Member: IEEE, MICCAI, ISTAART
- Associate Editor, IEEE Transactions on Image Processing (2014-2019)
- Ad Hoc Member of NIH Study Sections: EITA (2019, 2020), EITN (2020, 2021, 2022), ZRG1 BBBP-D (2020), Brain Initiative Cell Census Network (BICCN) Review Panel (2021), Brain Initiative U24 Review Panel (2023).
- MICCAI Area Chair, 2020, 2021, 2022
- Guest Editor, Computerized Medical Imaging and Graphics. Special Issue on Ophthalmic Medical Image Analysis (2015-2016)

Honors

2016	MICCAI Young Scientist Award to my PostDoc Dogu Baran Aydogan for our paper "Probabilistic tractography for topographically organized connectomes".
2008	Best paper award, IEEE Workshop on Mathematical Methods in Biomedical Image Analysis (MMBIA).
2005	Winner of the student paper competition, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

C. Contributions to Science
My research focuses on the development of cutting-edge computational algorithms for the automated analysis of large-scale brain imaging data. For 3D medical imaging data, I developed various algorithms for segmentation and registration. Using the spectrum of the Laplace-Beltrami operator on manifolds, I have developed a series of work on automated surface reconstruction and mapping. I have also developed novel algorithms for studying brain connectivity using multi-shell diffusion MRI. More detailed contributions of my research are listed as follows.

1. **3D Surface-based Image Analysis**. We have developed novel algorithms for surface mapping that includes both cortical and sub-cortical structures. In particular, I have developed a general approach for mapping brain structures based on the Conformal Metric Optimization on Surfaces (CMOS) and Riemannian Metric Optimization on Surfaces (RMOS). This method achieves excellent performance for a wide range of brain structures. It is publicly available in our MOCA tool on NITRC. More recently, we extended our shape analysis method for studying vasculature imaging data.
   a. **Shi Y**, Lai R, Wang DJ, Pelletier D, Mohr D, Sicotte N & Toga AW 2014 Metric optimization for surface analysis in the Laplace-Beltrami embedding space. IEEE Trans Med Imaging, 33(7):1447-63. PMCID: PMC4079755
   b. Gahm JK & **Shi Y** 2018 Riemannian Metric Optimization on Surfaces (RMOS) for intrinsic brain mapping in the Laplace-Beltrami embedding space. Medical Image Analysis, 46:189-201. PMCID: PMC5910235
   c. Zhang J, Qiao Y, Sarabi MS, Khansari MM, Gahm JK, Kashani AH, **Shi Y 2020** 3D Shape Modeling and Analysis of Retinal Microvasculature in OCT-Angiography Images. IEEE Trans Med Imaging, 39(5):1335-1346. PMCID: PMC7174137.
   d. Zhang J, **Shi Y** 2023 Personalized Patch-based Normality Assessment of Brain Atrophy in Alzheimer's Disease. Med Image Comput Comput Assist Interv.14224:55-62. PMCID: PMC10948101.

2. **3D Image Segmentation and Registration**. I have developed various segmentation and registration algorithms for 3D medical images from multiple modalities. Using fast-level-set methods and Laplace-Betrami spectral analysis, we developed novel brain image segmentation methods for cortical and sub-cortical brain structures. We have developed novel image registration methods for challenging 3D images from optical coherence tomography (OCT) and fiber orientation distribution (FOD) images for distortion correction in diffusion MRI.
   a. **Shi Y**, Lai R, Toga AW & the Alzheimer's Disease Neuroimaging Initiative 2013 Cortical surface reconstruction via unified Reeb analysis of geometric and topological outliers in magnetic resonance images. IEEE Trans Med Imaging, 32(3):511-30. PMCID: PMC3785796
   b. Khansari MM, Zhang J, Qiao Y, Gahm JK, Sarabi MS, Kashani AH & **Shi Y** 2020 Automated deformation-based analysis of 3D optical coherence tomography in diabetic retinopathy. IEEE Trans Med Imaging, 39(1):236-245. PMCID:PMC6928449
   c. Qiao Y & **Shi Y** 2022. Unsupervised Deep Learning for FOD-Based Susceptibility Distortion Correction in Diffusion MRI. IEEE Trans Med Imaging, 41(5):1165-1175. PMCID: PMC9177803.
   d. Deng Z, Xu S, Zhang J, Zhang J, Wang DJ, Yan L, **Shi Y** 2023 Shape-Aware 3D Small Vessel Segmentation with Local Contrast Guided Attention. Med Image Comput Comput Assist Interv. 14223:354-363. PMCID: PMC10948105.

3. **Connectome Modeling with Multi-shell Diffusion Imaging**. For cutting-edge modeling of fiber orientations with multi-shell imaging, we have developed a general computational framework for the joint estimation of fiber orientation distribution (FOD) and compartmental parameters. We have also developed novel FOD-based tractography and groupwise filtering algorithms for the robust reconstruction of challenging brain pathways such as the retinofugal visual pathway with successful application in Argus® II Retinal Prosthesis Patients.
   a. Tran G & **Shi Y** 2015 Fiber orientation and compartment parameter estimation from multi-shell diffusion imaging. IEEE Trans Med Imaging, 34(11):2320-32. PMCID: PMC4657863.
   b. Kammen A, Law M, Tjan BS, Toga AW & **Shi Y** 2016 Automated retinofugal visual pathway reconstruction with multi-shell HARDI and FOD-based analysis. NeuroImage, 125:767-79. PMCID: PMC4691391.
   c. Xia Y, **Shi Y** 2024 Diffusion MRI harmonization via personalized template mapping. Hum Brain Mapp. 45(5):e26661. PMCID: PMC10960558.
   d. Nie X, Ruan J, Otaduy MCG, Grinberg LT, Ringman J, **Shi Y** 2024 Surface-Based Probabilistic Fiber Tracking in Superficial White Matter. IEEE Trans Med Imaging. 43(3):1113-1124. PMCID: PMC10917128.

4. **Structural and Functional Connectopy**. While the topographic organization of brain pathways, i.e., *connectopy*, as a general principle has been known for a long time, it has yet to be quantitatively studied with *in vivo* neuroimaging. To this end, we have developed three new algorithms to quantitively model the

connectopy in human brains using connectome imaging data. These algorithms can be used to reconstruct topography-preserving pathways in visual, auditory, somatosensory, cortical-striatal, and fiber pathways in other brain networks. They can also improve the robustness of functional independent component analysis with regularization from topography-preserving fiber tracts.

a. Aydogan D & **Shi Y** 2018 Tracking and validation techniques for topographically organized tractography. NeuroImage, 181:64-84. PMCID: PMC6139055.
b. Wang J, Aydogan D, Varma R, Toga AW & **Shi Y** 2018 Modeling topographic regularity in structural brain connectivity with application to tractogram filtering. NeuroImage, 183:87-98. PMCID:PMC6197890.
c. Aydogan DB & **Shi Y** 2021 Parallel Transport Tractography. IEEE Trans Med Imaging, 40(2):635-647. PMCID: PMC7931442.
d. Nie X, **Shi Y** 2023 Flow-based Geometric Interpolation of Fiber Orientation Distribution Functions. Med Image Comput Comput Assist Interv. 14227:46-55. PMCID: PMC10978007.

5. **Alzheimer's Disease (AD) Imaging Research**. We have collaborated extensively with AD researchers by contributing our brain imaging expertise. We have developed novel atlases for locus coeruleus and other brainstem pathways for AD imaging research. We also provided brain image analyses to multiple large-scale AD research projects including the Health & Aging Brain Study – Health Disparities (HABS-HD) project and the clinical trial of allopregnanolone for AD.
   a. Sun W, Tang Y, Qiao Y, Ge X, Mather M, Ringman JM & **Shi Y** 2020 A probabilistic atlas of locus coeruleus pathways to transentorhinal cortex for connectome imaging in Alzheimer's disease. Neuroimage. 223:117301. PMCID: PMC7797167.
   b. O'Bryant SE, Johnson LA, Barber RC, Braskie MN, Christian B, Hall JR, Hazra N, King K, Kothapalli D, Large S, Mason D, Matsiyevskiy E, McColl R, Nandy R, Palmer R, Petersen M, Philips N, Rissman RA, **Shi Y**, Toga AW, Vintimilla R, Vig R, Zhang F, Yaffe K; HABLE Study Team. 2021 The Health & Aging Brain among Latino Elders (HABLE) study methods and participant characteristics. Alzheimers Dement (Amst). 13(1):e12202. PMCID: PMC8215806.
   c. Dahl MJ, Mather M, Werkle-Bergner M, Kennedy BL, Guzman S, Hurth K, Miller CA, Qiao Y, **Shi Y**, Chui HC, Ringman JM 2022 Locus coeruleus integrity is related to tau burden and memory loss in autosomal-dominant Alzheimer's disease. Neurobiol Aging. 112:39-54. PMCID: PMC8976827.
   d. Yue J, **Shi Y** 2023 Uncovering Heterogeneity in Alzheimer's Disease from Graphical Modeling of the Tau Spatiotemporal Topography. Med Image Comput Comput Assist Interv. 14224:262-271. PMCID: PMC10951551.

Complete List of Published Work in MyBibliography:
https://www.ncbi.nlm.nih.gov/myncbi/yonggang.shi.1/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Braskie, Meredith Nicole

eRA COMMONS USER NAME (credential, e.g., agency login): mbraskie

POSITION TITLE: Assistant Professor of Neurology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
College of William and Mary, Williamsburg, VA	B.B.A.	05/1994	Accounting
University of California, Los Angeles	Ph.D.	12/2006	Neuroscience
University of California, Berkeley	Postdoc	01/2009	Neuroscience
University of California, Los Angeles	Postdoc	08/2011	Neuroscience

A. Personal Statement

As an Assistant Professor of Neurology at the University of Southern California, my lab uses multi-modal neuroimaging (structural MRI, fMRI, diffusion tensor imaging, white matter hyperintensity assessments on FLAIR images, and amyloid and tau PET) to study Alzheimer's disease and cognitive aging. My research mainly focuses on modifiable risk factors (with a particular interest in inflammation, metabolic, and vascular risk) for Alzheimer's disease and cognitive decline. Thus far, I have co-authored >60 papers that mainly use neuroimaging to study Alzheimer's disease risk and cognitive aging.

Ongoing projects (5) that I would like to highlight include:

RF1AG078 NIH/NIA (mPIs: Yassine, Braskie, Borzage) 9/30/22 – 8/31/25
Title: Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults
The specific aims of this study are to: (1) Relate type 2 diabetes diagnosis and biomarkers to neuroimaging measures (2) Evaluate how the brain responds to acute glucose ingestion in those with and without type 2 diabetes and (3) To assess which baseline neuroimaging biomarkers best predict brain changes over two years.
Role: Co-PI

U19AG078109 NIH/NIA (mPIs: O'Bryant, Johnson, Yaffe, Rissman, Toga) 9/30/22 – 8/31/27
Title: Health and Aging Study – Health Disparities (HABS-HD)
This study will examine both PET-based and blood-based biomarkers of amyloid (A), tau (T) and neurodegeneration (N) in Mexican Americans and non-Hispanic Whites to determine whether the 2018 AT(N) Framework requires ethnic adjustments, aid in screening Mexican Americans in a primary care setting, and support ethnically-appropriate tailored novel clinical trials in this underserved population. Project 1) Examine the timing, sequence, progression and clinical impact of AT(N) biomarkers across diverse populations; Project 2) Examine the impact of vascular, metabolic and inflammatory factors on the timing, sequence, progression and clinical impact on AT(N) biomarkers across diverse populations; and Project 3) Examine the impact of the exposome (via neighborhood disadvantage) and sociocultural factors on the timing, sequence, progression and clinical impact of AT(N) biomarkers across diverse populations.
Roles: Project 2 lead (Dr. Kristine Yaffe is co-lead); Neuroimaging core co-lead (Dr. Arthur Toga is lead).

R01AG058162 NIH (mPIs Marmarelis, Chui, Billinger, Zhang 9/1/2018 – 5/31/2023
Title: Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression. This project will evaluate cerebral hemodynamics, using transcranial Doppler

and Near Infrared Spectroscopy measures, can provide a biomarker of mild cognitive impairment (MCI) and Alzheimer's disease (AD) pathophysiology and progression.
Role: Co-Investigator in charge of MRI and amyloid PET scan analyses.

R01AG054073 NIH/NIA O'Bryant (PI) 09/01/2017 – 05/31/2023
Health Disparities in Alzheimer's Disease among Mexican Americans
We will examine neuroimaging biomarkers and blood-based biomarkers associated with MCI and AD among Mexican Americans and non-Hispanic whites.
Role: Co-Investigator – Helped design the imaging portion and aided in writing the grant. My team is performing structural MRI analyses

R01 AG058533 NIH/NIA O'Bryant, Johnson, Rissman, Toga (PIs) 08/01/20 – 07/31/2025
Title: Health and Aging. Brain Among Latino Elders (HABLE-AT(N)) Study
This study will examine both PET-based and blood-based biomarkers of amyloid (A), tau (T) and neurodegeneration (N) in Mexican Americans and non-Hispanic Whites to determine whether the 2018 AT(N) Framework requires ethnic adjustments, aid in screening Mexican Americans in a primary care setting, and support ethnically-appropriate tailored novel clinical trials in this underserved population.
Role: Co-Investigator in charge of amyloid and tau PET scan analyses.

B. Positions, Scientific Appointments, and Honors

Positions

2019 - present	Assistant Professor of Neurology, University of Southern California
2019 - present	Director of Education, Mark and Mary Stevens Neuroimaging and Informatics Institute, USC
2013 - 2019	Assistant Professor of Research, University of Southern California
2011 - 2013	Assistant Researcher, U California, Los Angeles
2009 - 2011	Postdoctoral Fellow, U. California, Los Angeles – Mentors: **Drs. Paul Thompson & John Ringman**
2006 - 2009	Postdoctoral Fellow, U. California, Berkeley – Mentor: **Dr. William Jagust**
2002 - 2006	Graduate Student Researcher, U. California, Los Angeles– Mentor: **Dr. Susan Bookheimer**
2000	Research Assistant, independent study, U. Massachusetts, Amherst
1999 - 2001	Research Assistant and volunteer, U. Massachusetts, Amherst
1998 - 2001	Certified Public Accountant, Joseph D. Kalicka & Co.
1994 - 1998	Certified Public Accountant, Arthur Anderson LLP

Honors

2023	Bosco S. Tjan Mentoring Award, Neuroscience Graduate Program
2011	Turken Award for Alzheimer's Disease Research
2009 - 2010	Ruth L Kirschstein National Research Service Award (NRSA) T32 NS048004, PI - Dr. Nelson Freimer, Neurobehavioral Genetics – UCLA
2004 - 2006	Achievement Rewards for College Scientists (ARCS) fellowship (Underwritten in part by John Douglas French Alzheimer's Foundation and the Erteszek Foundation through September 2006)
2003 - 2006	Ruth L Kirschstein National Research Service Award (NRSA), F31 NS45425, PI - Meredith Braskie

A. Contributions to Science

1. Genetic associations with brain measures:

   Understanding how genetic variants affect brain structure and connectivity is an important first step in designing treatments for the disease that are affected by the systems those genes influence. In my graduate work, I was involved in a study by Burggren et al., that was the first to show how the Alzheimer's disease (AD) genetic risk variant *APOE*4 related to the thickness of hippocampal area subregions. My role in that highly cited study was to help with the scanning, hippocampal unfolding, and data interpretation, and to edit the manuscript.

   In my postdoctoral work, I became a fellow supported by an institutional NRSA grant at UCLA in Neurobehavioral Genetics (PI Nelson Freimer) and continued to study genetic effects on the brain, this

time related to structural connectivity. We were the first to publish how the newly confirmed genetic risk allele C of the clusterin (*CLU*) gene variant rs11136000, related to brain measures in living humans. We found that each C allele of the *CLU* variant was associated with lower diffusion tensor imaging fractional anisotropy (DTI FA), a widely accepted measure of white matter integrity, broadly throughout the brain, including in regions having preferential degeneration in AD. Clusterin is implicated in β-amyloid transport. However, our 398 young adult subjects were likely too young to have amyloid deposits in their brains. Our results suggest another way by which the *CLU* C allele may make the brain vulnerable to AD-related cognitive decline, decades before an AD diagnosis would be expected. Our findings were reported by newspapers in at least 15 countries, widely on the internet, and by KCBS radio news in San Francisco. **The Journal of Neuroscience** paper that resulted has been cited >130 times.

In another paper published in **The Journal of Neuroscience**, we found that a genetic variant in the *NTRK1* (also known as *TRKA*) gene was associated with white matter integrity in a large sample of healthy young adults. NTRK1 is a high affinity receptor for nerve growth factor (NGF), which has been implicated in normal brain development, AD, schizophrenia, and other neurological disorders. My role in these two studies was to conceive the research idea, perform statistical analyses, interpret the results, write the initial draft of the paper, and to implement all subsequent revisions.

I was also lead author on two fMRI studies of people from families that carry rare genetic mutations that confer AD in middle age in an autosomal dominant manner. In one of these, published in **Neurobiology of Aging**, fMRI activity during a novelty encoding task increased the closer AD mutation carriers got to the expected age of disease onset, suggesting that the increase in activity related to disease processes rather than to developmental differences related to the gene mutation. In a follow-up paper, published in **Human Brain Mapping**, preclinical mutation carriers showed greater fMRI activation during memory retrieval, and this activation was greatest in those who performed most poorly on the task. This suggests that pathological processes rather than benign compensation are responsible for the increased activity that occurs prior to clinically relevant cognitive decline. In these studies, I planned and executed the analyses, interpreted the results, and wrote and revised the manuscript.

a. Burggren AC, Zeineh MM, Ekstrom AD, **Braskie MN**, Thompson PM, Small GW & Bookheimer SY 2008 Reduced cortical thickness in hippocampal sub-regions among cognitively normal apolipoprotein E e4 carriers. Neuroimage, 41(4):1177-83. PMCID: PMC2601686
b. **Braskie MN**, Jahanshad N, Stein JL, Barysheva M, McMahon KL, de Zubicaray GI, Martin NG, Wright MJ, Ringman JM, Toga AW & Thompson PM 2011 Common Alzheimer's disease risk variant within the CLU gene affects white matter microstructure in young adults.. The Journal of Neuroscience, 31(18):6764-70. PMCID: PMC3176803
c. **Braskie MN**, Medina LD, Rodriguez-Agudelo Y, Geschwind DH, Macias-Islas MA, Cummings JL, Bookheimer SY & Ringman JM 2010 Increased fMRI signal with age in familial Alzheimer's disease mutation carriers. Neurobiology of Aging, 33(2):424.e11-21. PMCID: PMC3097258
d. **Braskie MN**, Jahanshad N, Stein JL, Barysheva M, Johnson K, McMahon KL, de Zubicaray GI, Martin NG, Wright MJ, Ringman JM, Toga AW & Thompson PM 2012 Relationship of a variant in the NTRK1 gene to white matter microstructure in young adults. The Journal of Neuroscience, 32(17):5964-72. PMCID: PMC3393752

2. Amyloid PET in cognitively intact older adults:

I was the lead author on the first paper published online to demonstrate that amyloid PET signal was associated with cognition in cognitively intact older adults. This paper, published in **Neurobiology of Aging**, has been cited >100 times. We used the (18F)FDDNP-PET ligand, which labels amyloid plaques and, to a lesser extent, neurofibrillary tangles. Using MRI-derived cortical surface models, aligned across subjects with sulcal matching, we showed that in AD patients, those with MCI, and healthy older controls, AD pathology was significantly correlated with cognitive performance in a pattern that mirrored a classic pattern of pathological progression. When only cognitively intact older controls were considered, FDDNP signal in small regions of the frontal and parietal cortex was still significantly higher in those who had worse cognition. My role in this project was to conceive of the idea, perform MRI scanning and analysis, interpret the data, and write and revise the manuscript. A graduate individual NRSA grant on which I was the principal investigator funded this work. In Scott et al., 2016, we found in cognitively intact older adults, that greater amyloid burden at baseline was associated with greater WMH accrual over a two-year period of time. This association was especially pronounced in those with hypertension, but the relationship remained significant even after controlling for age and hypertension. Our results suggest that the contribution of possible amyloid deposition should be considered as an independent factor when evaluating white matter lesion etiology. In Scott et al., 2015, we found that amyloid deposition and hypertension were independently associated with white matter hyperintensity volume in older adults.

a. **Braskie MN**, Klunder AD, Hayashi KM, Protas H, Kepe V, Miller KJ, Huang S-C, Barrio JR, Ercoli LM, Siddarth P, Satyamurthy N, Liu J, Toga AW, Bookheimer SY, Small GW & Thompson PM 2010 Plaque and tangle imaging and cognition in normal aging and Alzheimer's disease. Neurobiology of Aging, 31(10):1669-78. PMCID: PMC2891885

b. Scott JA. **Braskie MN**, Tosun D, Maillard P, Thompson P, Weiner MW, DeCarli C, Carmichael OT. Cerebral Amyloid Is Associated With Greater White Matter Hyperintensity Accrual in Cognitively Normal Elderly 2016. Neurobiology of Aging. 48:48-52. PMCID in process.

c. Scott JA, **Braskie MN**, Tosun D, Thompson PM, Weiner M, DeCarli C, Carmichael OT Cerebral Amyloid and hypertension are independently associated with white matter lesions in the elderly. Frontiers in Aging Neuroscience. PMCID: PMC4664630

3. Somatic health and brain aging:
One of my major interests is how general health, particularly cardiovascular and metabolic health and the immune system, affects brain aging and AD risk. We previously discovered that even within the normal range of systolic blood pressure and body mass index, higher values were associated with greater fMRI activity in regions vulnerable to AD. Cardiovascular deficits and AD frequently co-occur. Our results suggest that part of the AD-related pattern frequently seen in those with increased risk may relate to cardiovascular risk. Conversely, the pattern of brain changes associated with cardiovascular risk may create a vulnerability to cognitive deficits when AD-related pathology is present. In this study, I conceived the idea for the study, performed all analyses, interpreted the results, and wrote and revised the manuscript. More recently, we examined the relationship of total brain volume and voxelwise brain volume (assessed using tensor-based morphology) with reported physical activity intensity and blood serum levels of TNFα, a pro-inflammatory cytokine. We found that lower physical activity intensity and higher levels of TNFα were both independently associated with smaller brains, and that higher TNFα was associated with lower voxelwise brain volume in the inferior parietal lobule. TNFα and physical activity intensity were not significantly correlated with one another. Our results suggest that the effect of physical activity on brain volume is largely independent of any effect it might have on inflammation. For this paper, I planned the study, conceived the idea for and wrote the grant that funded the study, performed all imaging analyses, interpreted the results, and wrote and revised the manuscript. In Corlier et al., 2018, we found that peripheral C-reactive protein levels helped mediate a relationship between higher metabolic risk and thinner cortex in AD-relevant brain regions 9 years later. For this paper, I planned the study, conceived the idea for and wrote the grant that funded the study, oversaw data analysis and project progress, helped interpret the results, and revised the manuscript. In Tubi et al., 2021, we found that abnormal CSF amyloid and tau levels strengthened a positive association between vascular endothelial growth factor (VEGF) and regional FDG-PET signal. My role was to help my PhD student to develop her proposal, supervise her and staff, and provide critical feedback throughout the analysis and manuscript writing process.

a. **Braskie MN**, Small GW & Bookheimer SY 2010 Vascular health risks and fMRI activation during a memory task in older adults. Neurobiology of Aging, 31(9):1532-42. PMCID: PMC2965069

b. **Braskie MN**, Boyle CP, Rajagopalan P, Gutman BA, Toga AW, Raji CA, Kuller LH, Becker JT, Lopez OL & Thompson PM 2014 Physical activity, TNFα, and volume of the aging brain. Neuroscience, 273:199-209. PMCID: PMC4076831
c. Corlier F, Hafzalla G, Faskowitz J, Kuller LH, Becker JT, Lopez OL, Thompson PM, **Braskie** MN. 2018. Systemic inflammation as a predictor of brain aging: Contributions of physical activity, metabolic risk, and genetic risk. NeuroImage. PMCID: PMC5954991
d. Tubi M, Feingold FW, Corlier F, Mostowfi M, Thompson PM, **Braskie MN, for ADNI.** Regional relationships between CSF VEGF levels and Alzheimer's disease brain markers and cognition (2021). ***Neurobiology of Aging.*** 105:241-251. PMCID in process

4. Dopamine synthesis capacity, early adult development, and cognitive aging:

In our **Journal of Neuroscience** paper, in which I was the lead author, we used the (PET) radiotracer 6-[(18)F]fluoro-l-m-tyrosine (FMT), a marker for activity of the dopamine-synthesizing enzyme, aromatic amino acid decarboxylase (AADC), to measure dopamine synthesis capacity in the caudate in healthy older and younger adults. We evaluated how FMT signal in the caudate related to frontal lobe function and age. Prior studies using FDOPA as a radiotracer to measure striatal dopamine synthesis had inconsistent results. However, FDOPA signal is subject to some aspects of post-release processing, which may themselves change with aging, while FMT signal is a purer measure of AADC function. Consistent with prior work in non-human primates, we found that FMT signal in the dorsal caudate increased with age, suggesting compensation for deficits elsewhere in the dopamine system. In contrast, in younger adults, FMT signal in the dorsal caudate was lower with age, possibly due to ongoing developmental in young adulthood. Younger adults who performed worse on tests of frontal lobe function showed greater FMT signal in the right dorsal caudate, independent of age effects. Together this suggests that higher striatal higher striatal FMT signal is associated with non-optimal dopamine processing.

In a follow-up study, we found older cognitively intact adults showed less fMRI deactivation in the posterior cingulate cortex and precuneus during a delayed recognition working memory task than younger adults. These regions are an integral part of the default mode network, which shows deficits in many disease and disorders, including AD. We also found that in young adult subjects, greater task-induced fMRI deactivations were associated with higher FMT signal in the caudate, and with worse memory performance. This suggests that dopamine synthesis may help modulate default mode network activity, and changes to this system may contribute to working memory deficits in older adults.

For both of these studies, I recruited and performed MRI scanning all the younger subjects in this study as well as coordinating their neuropsychological testing and PET scanning. I analyzed the PET and MRI scans included in the study and trained, supervised, and checked the analyses of the others who assisted me. I also wrote and revised the manuscripts.

a. **Braskie MN**, Wilcox CE, Landau SM, O'Neil JP, Baker S, Madison CM, Kluth JT & Jagust WJ 2008 Relationship of striatal dopamine synthesis capacity to age and cognition. Journal of Neuroscience, 28(52):14320-8. PMCID: PMC2880923

b. **Braskie MN**, Landau SM, Wilcox CE, Taylor SD, O'Neil JP, Baker S, Madison CM & Jagust WJ 2011 Correlations of striatal dopamine synthesis with default network deactivations during working memory in younger adults. Human Brain Mapping, 32(6):947-61. PMCID: PMC3176660

Complete List of Published Work in MyBibliography:
http://www.ncbi.nlm.nih.gov/sites/myncbi/1H5KN3R_gyckc/bibliography/44271641/public/?sort=date&direction=ascending

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Priya Rajagopalan

eRA COMMONS USER NAME (credential, e.g., agency login): PRRAJAGO

POSITION TITLE: Assistant Professor, Radiology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Kamineni Institute of Medical Sciences, India	MBBS	01/2007	Medical School
University of California, Berkeley	MPH	06/2010	Epidemiology
Indiana University School of Medicine	-	06/2015	Internal medicine
Indiana University School of Medicine	-	06/2019	Diagnostic Radiology
University of California, San Francisco	-	06/2021	Neuroradiology

A. Personal Statement

My area of interest is studying structural brain changes in relation to human genetics using magnetic resonance imaging (MRI), more particularly in aging and dementia. I focus on the effect of cardiovascular risk factors and their underlying genetic pathways, on brain structure. My work, thus far, has focused on analyzing peripheral blood-based risk factors including homocysteine, folate, cystatin C, stress-hormone cortisol on brain and cognition in healthy aging as well as in dementia. I have published in peer-reviewed journals, including the New England Journal of Medicine (as 1st author) and presented my work through talks and posters at national and international platforms, distinguishing myself as a budding researcher in the field of neuroimaging-genetics.

After earning a Bachelor's degree in Medicine and Surgery from India, I pursued a Master's degree in Epidemiology from the School of Public Health at University of California, Berkeley, where I trained in epidemiological and statistical methods. Subsequently, under the guidance of Dr. Paul Thompson at the Laboratory of Neuroimaging, University of California, Los Angeles, I lead authored several studies in the Alzheimer's Disease Neuroimaging (ADNI) cohort, studying brain structural and cognitive changes in relation to other peripheral blood based and genetic biomarkers.

After pursuing a Radiology residency at Indiana University, an ACGME Neuroradiology Fellowship followed by Clinical Instructor role at University of California, San Francisco, I joined the Neuroradiology Faculty at University of Southern California (USC). I am fortunate to belong to a department which values academic research and encourages me to pursue my research endeavors in addition to providing an enriching clinical environment. I continue to pursue neuroimaging genetics research and Big-Data analysis for identifying novel pathways for various diseases across the biological spectrum from Autism Spectrum Disorder to Alzheimer's disease under the mentorship of Dr. Paul Thompson, who is now the Associate Director of the Mary and Mark Stevens Neuroimaging and Informatics Institute, and Director of the Imaging Genetics Center at USC. I am also interested in studying the field of Sustainability as it pertains to Radiology and hopefully contribute towards Greener Radiology.

B. Positions, Scientific Appointments, and Honors

Positions

- 2021 to present**,** Assistant Professor of Clinical Radiology, Neuroradiology Division, University of Southern California
- 2021 to present, Associate Program Director, ACGME Neuroradiology Fellowship, University of Southern California
- 2021 to present, Medical Director, Center for Image Acquisition, Mark and Mary Stevens Neuroimaging and Informatics Institute, University of Southern California
- 2021 to present, Course Director, Visiting Medical Student Radiology Clerkship, University of Southern California
- 2019-21, Neuroradiology Fellowship (PGY-6 to 7), Department of Radiology, University of California, San Francisco
- 2015-19, Radiology Residency (PGY-2 to 5), Department of Radiology, Indiana University School of Medicine
- 2014-15, Preliminary Medicine Residency (PGY-1), Department of Internal Medicine, Indiana University School of Medicine
- 2013-14, Post-doctoral Researcher, Center for Neuroimaging, Indiana University
- 2010-13, Staff Research Associate, Laboratory of NeuroImaging, University of California, Los Angeles
- Fall 2009, Graduate Student Instructor, The Aging Brain, Department of Neuroscience, University of California, Berkeley
- Spring 2010, Graduate Student Instructor, Department of Nutritional Sciences and Toxicology, University of California, Berkeley
- 2009-10, Masters in Public Health, Epidemiology, University of California, Berkeley
- 2009-09, Certified HIV counselor, HIV Prevention Services, Berkeley Free Clinic, Berkeley, CA
- 2007-07, Visiting physician, San Jose Pacific Neurology Center, San Jose, CA
- 2006-07, Internship, Kamineni Institute of Medical Sciences, NTR UHS, India
- 2000-06, Medical School (MBBS), Kamineni Institute of Medical Sciences, NTR UHS, India

Honors

- Foundation of American Society of Neuroradiology Grant, Co-Investigator, 2024-2025
- Boerger Grant, Principal Investigator, 2023-2024
- American Society of Pediatric Neuroradiology, Guerbet Grant, Principal Investigator, 2023-2024
- Alzheimer's Association Grant, Principal Investigator, "Mapping differential genetic effects of APOE e2 on 40,083 healthy brains", 2023-2025.
- Zumberge Fund Individual Grant program, Principal Investigator, "Mapping of Differential Genomic effects of the Alzheimer's Protective Genotype (APOE2) in Aging: Big Data Analysis in 15,500 Subjects", 2022-2023
- Outstanding Teacher of the Year Award, Radiology Residency, USC, June, 2022
- RSNA Research Resident Grant, Principal Investigator, Cerebral perfusion alterations in cortisol gene carriers: A potential Alzheimer's disease pathway, 2018-2019
- RSNA Research Resident Grant, Principal Investigator, Increased White Matter Hyperintensity in Folate Gene Carriers: A Novel Amyloid Interaction Pathway, 2017-2018
- Robert H. Choplin & Marjorie A. Bowman Scholarship, Indiana University, 2017-2018
- Robert H. Choplin & Marjorie A. Bowman Scholarship, Indiana University, 2016-2017
- Outstanding Scientific and Translational Research Scholar Presentation, AUR, 2017
- ACR-AUR Research Scholar, Association of University Radiologists, 2017
- Introduction to Academic Radiology Scholarship, RSNA, Chicago, 2016
- Staff Assembly Professional Development Scholarship, UCLA, 2011-2012
- Kamineni Gold Medal All-round performance during medical school, 2005
- Sri. Gondi Venkataratnam, Gold Medal in ENT, 2004
- Oil Country Tubular Limited, Gold Medal in Forensic medicine, 2003

- Sri. Kolli Laxmaiah, Gold Medal in Physiology, 2002
- Dr. A.S.Rao Awards Council Science Talent Search Examination, 1998-1999, 1999-2000

C. Contributions to Science

TREM2 and neurodegenerative disease

Rajagopalan P, Hibar DP, Thompson PM. TREM2 and neurodegenerative disease. **N Engl J Med**. 2013 Oct 17;369(16):1565-7.

TREM2 gene roughly triples the lifetime risk of Alzheimer's disease (AD). Using tensor-based morphometry (1) on 478 brain MRI scans. We found carriers of the close proxy (r^2=0.492) of the *TREM2* risk gene variant annually lose brain tissue up to 1.4–3.3% faster than non-carriers, in a pattern that mirrors the profile of AD pathology in the brain. The *TREM2* risk variant may affect signaling by TREM2 receptors interfering with microglial anti-inflammatory functions and brain amyloid clearance, leading to AD-like neurodegeneration and accelerated cognitive decline. Selection of *TREM2* carriers in Alzheimer's disease intervention trials, should increase the power of such trials.

References:
1. Hua X, Hibar DP, Ching CR, Boyle CP, Rajagopalan P, et al. Unbiased tensor-based morphometry: improved robustness and sample size estimates for Alzheimer's disease clinical trials. Neuroimage. 2013;66:648-61.

Neural growth *SPON1* gene variant influencing connectivity and dementia severity

Jahanshad N, **Rajagopalan P**, Hua X, et al. Genome-wide scan of healthy human connectome discovers *SPON1* gene variant influencing dementia severity. **Proc Natl Acad Sci USA**. 2013 Mar 19;110(12):4768-73.

Aberrant connectivity is implicated in Alzheimer's disease. To identify the heritability of brain connectivity, we carried out a genome wide association study and identified SPON1 variant at rs2618516 on chromosome 11. *SPON1*, a member of the spondin family of genes, is highly expressed in the embryonic neural floor plate and is essential for neural growth and cell adhesion. Prior studies implicate this gene in brain structural development and neural connectivity, and its expression declines with aging (2). rs2618516 was shown to affect brain structure in an elderly population with varying degrees of dementia and older people who carried the connectivity variant had significantly milder clinical dementia scores and lower risk of Alzheimer's disease. The stronger white matter fiber connection associated with the *SPON1* variant in healthy young adults may dampen the severity of dementia in carriers of this protective variant, affecting brain connectivity and risk for brain diseases much later in life.

Accelerated brain atrophy in carriers of folate-pathway gene, *MTHFR*

Rajagopalan P, Jahanshad N, Stein JL, et al. Common folate gene variant, MTHFR C677T, is associated with brain structure in two independent cohorts of people with mild cognitive impairment. **Neuroimage Clin.** 2012 Oct 4;1(1):179-87.
A commonly carried *C677T* polymorphism in the *MTHFR* folate-related gene, is associated with higher plasma homocysteine (1), a well-known mediator of neuronal damage and brain atrophy. As homocysteine promotes brain atrophy, we set out to discover whether people carrying the *C677T* MTHFR polymorphism, might also show systematic differences in brain structure. Using tensor-based morphometry, we tested this association in elderly Caucasian subjects with mild cognitive impairment (MCI) in two independent cohorts. We noted detectable 2-8% white matter deficits and 0.5-1.5% accelerated annual brain atrophy per every risk allele, which weakened after controlling for homocysteine. We were thus able to spatially detail the underlying genetic pathway for homocysteine-mediated brain atrophy in elderly people with MCI.

References:
1. Rajagopalan P, Hua X, Toga AW, Jack Jr CR, Weiner MW, Thompson PM. Homocysteine effects on brain volumes mapped in 732 elderly individuals. Neuroreport. 2011;22(8):391.

Homocysteine mediated brain atrophy in mild cognitive impairment

Rajagopalan P, Hua X, Toga AW, Jack CR Jr, Weiner MW, Thompson PM. Homocysteine effects on brain volumes mapped in 732 elderly individuals. **Neuroreport**. 2011 Jun 11;22(8):391-5.

High homocysteine levels may facilitate a build-up of toxic β-amyloid and tau in the brain, and may compromise DNA repair and promote degenerative changes due to oxidative stress, which may in turn lead to excitotoxicity and apoptosis. Although homocysteine is associated with lower overall brain volumes, only region of interest analysis in subcortical structures had been carried out. We aimed to map a voxel-wise anatomical pattern of homocysteine effects. To do this, we created a three-dimensional map of the homocysteine related pattern of atrophy. Our findings have implications for drug trials that aim to lower homocysteine levels and thereby slow the onset of dementia or age-related neuro-degeneration through folate and other vitamin B supplements.

White matter brain deficits in impaired renal function

Rajagopalan P, Refsum H, Hua X, Toga AW, Jack CR Jr, Weiner MW, Thompson PM; Alzheimer's Disease Neuroimaging Initiative. Mapping creatinine- and cystatin C-related white matter brain deficits in the elderly. **Neurobiol Aging**. 2013 Apr;34(4):1221-30.

Since poor renal function is associated with increased risk of cognitive decline and generalized brain atrophy, we hypothesized impaired renal function would be associated with brain deficits in regions vulnerable to neurodegeneration. Using tensor-based morphometry, we related patterns of brain volumetric differences attributable to kidney biomarkers (creatinine, cystatin C levels, and glomerular filtration rate) in a large cohort of 738 elderly subjects. The volume deficits were predominantly in the white matter, specifically the anterior limb of internal capsule, bilaterally, helping us to explain the cognitive deficits. These findings also underscore the need for early diagnosis and treatment of subclinical kidney dysfunction.

Complete List of Published Work in PubMed:
https://www.ncbi.nlm.nih.gov/pubmed/?term=Rajagopalan%2C%20Priya%5BFull%20Author%20Name%5D&cmd=DetailsSearch

Name	Role of the ADRC in bringing new investigators to the field	Funding	Publication Title
Kexin Yu PhD	Started new Post doctoral fellowship at Oregon Health & Science University (Mentor: Maria Aranda)	F99AG068492	• Yu K, Jiang H, Liu M, Wu S, Jordan-Marsh M, Chi I. Chinese Immigrant Caregivers: Understanding Their Unmet Needs and the Co-Design of an mHealth App. Can J Aging. 2024:1-8. Epub 2024/05/17. doi: 10.1017/s0714980824000187. PubMed PMID: 38757195. • Dodge HH, Yu K, Wu CY, Pruitt PJ, Asgari M, Kaye JA, et al. Internet-Based Conversational Engagement Randomized Controlled Clinical Trial (I-CONECT) Among Socially Isolated Adults 75+ Years Old With Normal Cognition or Mild Cognitive Impairment: Topline Results. Gerontologist. 2024;64(4). Epub 2023/11/08. doi: 10.1093/geront/gnad147. PubMed PMID: 37935416; PubMed Central PMCID: PMCPMC10943511. • Yu K, Liang J, Tung YH, Zhang M, Wu S, Chi I. Social network and mental health of Chinese immigrants in affordable senior housing during the COVID-19 pandemic: a mixed-methods study. Aging Ment Health. 2023;27(10):1956-64. Epub 2023/05/22. doi: 10.1080/13607863.2023.2213646. PubMed PMID: 37212623; PubMed Central PMCID: PMCPMC10529124. • Palinkas LA, De Leon J, Yu K, Salinas E, Fernandez C, Johnston J, et al. Adaptation Resources and Responses to Wildfire Smoke and Other Forms of Air Pollution in Low-Income Urban Settings: A Mixed-Methods Study. Int J Environ Res Public Health. 2023;20(7). Epub 2023/04/14. doi: 10.3390/ijerph20075393. PubMed PMID: 37048007; PubMed Central PMCID: PMCPMC10094253. • Yu K, Siang Ng TK. Investigating Biological Pathways Underpinning the Longitudinal Association Between Loneliness and Cognitive Impairment. J Gerontol A Biol Sci Med Sci. 2023;78(8):1417-26. Epub 2022/10/07. doi: 10.1093/gerona/glac213. PubMed PMID: 36200979; PubMed Central PMCID: PMCPMC11015400. •
Hoda Abdel Magid, PhD	Assistant Professor, USC, (Mentor: Julie Zissimopoulos)	R00HL161479	• Abdel Magid HS, Jaros S, Li Y, Steinman MA, Lee S, Jing B, et al. Effects of residential socioeconomic polarization on high blood pressure among nursing home residents. Health Place. 2024;87:103243. Epub 2024/04/26. doi: 10.1016/j.healthplace.2024.103243. PubMed PMID: 38663339; PubMed Central PMCID: PMCPMC11102837. • Milliren CE, Sajjad OM, Abdel Magid HS, Gooding HC, Richmond TK, Nagata JM. Adolescent individual, school, and neighborhood influences on young adult diabetes risk. Health Place. 2023;83:103047. Epub 2023/06/11. doi: 10.1016/j.healthplace.2023.103047. PubMed PMID: 37301169; PubMed Central PMCID: PMCPMC10798047 •

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1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines? ☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period *Anticipated Amount ($) *Source(s)

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: https://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Cell Line(s): ☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "Yes" then provide the HFT Compliance Assurance: File Name: Mime Type:

If "Yes" then provide the HFT Sample IRB Consent Form File Name: Mime Type:

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5. Inventions and Patents Section (RENEWAL)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously reported: ☐ Yes ☐ No

6. Change of Investigator / Change of Institution Section

☐ Change of Project Director / Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

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1. Introduction to Application (or Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	Imaging_Aims_Final1044446984.pdf
3. Research Strategy*	Imaging_Research_Plan1044447010.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	
9. Letters of Support	
10. Resource Sharing Plan(s)	
11. Other Plan(s)	
12. Authentication of Key Biological and/or Chemical Resources	
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12. Appendix	

SPECIFIC AIMS

The overarching mission and function of the USC Alzheimer Disease Research Center (ADRC) - Imaging Core is to provide standardized, high-quality MR and PET imaging resources to the USC ADRC and the local and broader AD research community. During the last cycle, the Imaging Core: a) successfully provided imaging support for the ADRC including the development of novel acquisition methods and tools for robust vascular, functional, and structural imaging techniques including dynamic contrast-enhanced MRI (DCE-MRI), functional MRI (fMRI), diffusion tensor imaging (DTI) and blood-brain barrier water exchange/Arterial Spin Labeling (ASL) MRI. b) The Imaging Core initiated sequence and protocol development at 7T MRI for highly detailed structural and vascular mapping. c) The Imaging Core worked with the ADRC DMSC to provide a seamless workflow from imaging data acquisition to data storage in a secure repository (USC ADRC archive). d) Offered educational and training opportunities focused on neuroimaging and weekly imaging seminars and didactics for graduate students, postdoctoral fellows, residents, and early-stage investigators. For the competitive renewal, the ADRC Imaging Core will continue to provide core resources, including MR physics and PET expertise, multi-scanner sequence and protocol development, image data quality control and management, research collaboration, and a global framework for conducting AD neuroimaging research. Our main goal for this funding period is to ensure harmonized imaging data acquisition based on a unified MRI and PET protocol that will standardize data acquisition across ADRC and will adhere to the Standardized Centralized Alzheimer's & Related Dementias Neuroimaging (SCAN) guidelines.

The Imaging Core will be led by a well-integrated and collaborative team, including **Arthur Toga (Core Leader), Ioannis Pappas (Core Co-Leader), Nasim Sheikh-Bahaei, Yonggang Shi, Meredith Braskie, Priya Rajagopalan, and Ararat Chakhoyan** who together provide expertise in vascular, functional, and structural imaging and its analyses for 3T and 7T MRI, and PET. The Imaging Core currently serves an active cohort of 436 participants and supports over 20 active ADRC-affiliated research across USC. Participants that are a part of the Primary ADRC Cohort will be followed every two years and numbers will be adjusted based on recruitment wave. We propose the following five specific aims:

Aim 1: Provide an imaging framework for all ADRC. The Imaging Core will provide a standardized imaging data acquisition protocol in compliance with SCAN requirements. In addition, the Imaging Core will assist and tracking of all raw images in conjunction with the DMSC. Finally, the Imaging Core will assist with data dissemination of images to authorized users under the USC ADRC project.

Aim 2: Perform high quality control on all imaging data and derive MRI analytics. State-of-the-art workflow for the review and assessment of all imaging data. Quantitative MRI analytics including white matter hyperintensities load, perivascular spaces, volumetrics, vessel density, BBB integrity, and cerebral blood flow.

Aim 3: Develop technical and scientific approaches for ultra-high field 7T MRI. The Imaging Core will provide sequence development of novel 7T MRI sequences. The Imaging Core will assist with protocol development and study design for 7T MRI research that will allow structural, vascular, and functional mapping with unprecedented resolution.

Aim 4: Assist in PET protocol development and quality control and provide PET analytics. The Imaging Core will provide regulatory guidance and standardized protocol deployment for AD PET studies. In addition, the Imaging Core will provide qualitative review of PET scan data quality and PET data analytics including Standard Uptake Value Ratios (SUVRs).

Aim 5: Conduct education and training in modern imaging techniques. The Imaging Core will provide didactic and neuroimaging training for AD research across departments and campuses.

Impact: The Imaging Core serves an essential role on the theme of USC ADRC on *APOE* ε4 and dementia. It provides a comprehensive set of imaging markers for studying the differential effects of the *APOE* ε4 genotype on dementia and disease progression. The Imaging Core will provide guidance on compliance with national imaging initiatives as well conduct education and training across USC.

IMAGING CORE-RESEARCH STRATEGY

SIGNIFICANCE

The Imaging Core (IC) provides imaging support for the entire ADRC by acquiring, processing, and analyzing MRI and PET imaging data for all ADRC participants. One of the innovations of this proposal is the mission to elucidate how imaging biomarkers can further our understanding of the modifying effects of apolipoprotein E4 (*APOE* ε4) on cognitive decline. The IC provides a unique imaging protocol that is tailored to the precise acquisition of structural, vascular, and functional markers associated with *APOE* ε4. The IC has contributed to multiple publications including new sequences for mapping the BBB using water exchange[1], mapping small vessels using novel segmentation methods[2], perivascular space quantification[3], and tau PET analysis[4]. The IC protocol has been included in the ADRC and ADRC-affiliated projects including MarkVCID (multi-delay PCASL) and HABS-HD (structural MRI imaging). The IC has developed sophisticated imaging processing pipelines applied to ADRC data for ATN(vi) characterization of the ADRC cohort. The sophistication of the protocol enables the use of cutting-edge research techniques. The IC plays a critical role in connecting the USC ADRC to national AD efforts by facilitating data sharing with national initiatives such as SCAN, CLARiTI, and ADNI4.

INNOVATION

The IC contributes to the ADRC's central theme to elucidate factors that modify the effect of *APOE* ε4 on cognitive decline and ATN(vi) biomarkers. In a paradigm shift towards understanding early pathological changes, imaging markers are becoming increasingly important. We introduce novel imaging methods at high field 3T and ultra-high field 7T that can better elucidate the complex pathophysiology of *APOE* ε4. The mission of this ADRC is supported by the specific aims of the IC with new possibilities for fine-grained imaging at 7T. The IC also provides expertise and support to national initiatives such as SCAN, CLARiTI, and ADNI4. The IC connects to the USC ADRC archive (as described in the DMSC) which will be the key existing infrastructure used as support for imaging data collection and dissemination. The USC ADRC archive is used for safely de-identifying, storing, and indexing hundreds of ADRC imaging data.

(A) ASL/BBB Imaging. The IC has pioneered a new technique that allows simultaneous measurement of cerebral perfusion using pseudo-continuous spin labeling (ASL) perfusion MRI and BBB water permeability using diffusion-weighted imaging (DW-pCASL)[1]. Perfusion MRI permits noninvasive measurement of cerebral blood flow using magnetically labeled water as an endogenous tracer[5,6]. Perfusion MRI can provide measurements of Cerebral Blood Flow (CBF) and Arterial Transit Times (ATT) if the signal is measured at different delay times post water labeling. In addition, water exchange across the BBB can be quantified based on ASL signal fractions in the intra- and extravascular compartments[7,8]. Using appropriate gradients commonly used in diffusion imaging, this ASL technique can differentiate the fraction of labeled water in capillary and brain tissue based on their distinctive (pseudo) diffusion coefficients (high in capillary and low in tissue). Thus, this sequence can provide simultaneous measurement of the water exchange rate (k_W) across the BBB, ATT, and CBF derived from ASL MRI. To increase sensitivity and signal-to-noise ratio, the IC has developed a novel MR pulse sequence for DW-pCASL with improved test-retest repeatability by integrating a diffusion preparation module optimized for minimizing eddy current and spoiling of non-CPMG signals with 3D background suppressed 3D GRASE pCASL. Recent studies have shown age and sex-based variations in K_W, ATT, and CBF (**Fig. 1**). K_W is increased in subjects with type 2 diabetes and hypercholesterolemia, both of which have emerged as risk factors for small vessel disease and AD[9].

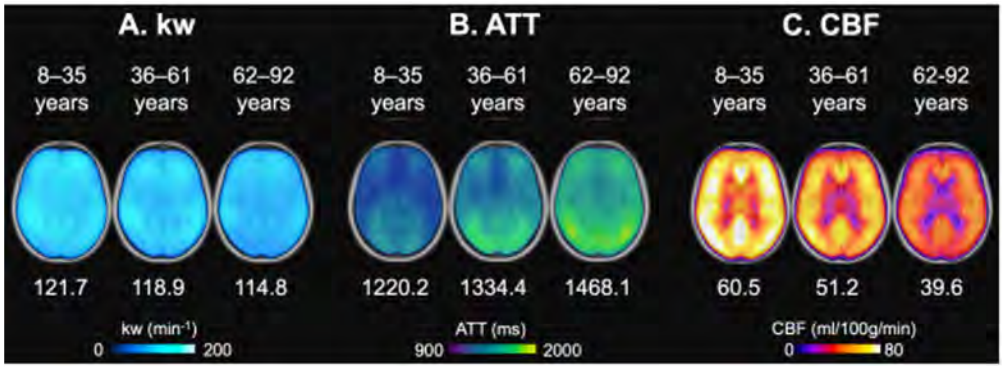


Figure 1: ASL/BBB imaging. The ADRC supports the PPG project, testing the effect of *APOE* ε4 on BBB. Example of individual differences in derived measurement of the water exchange rate (Kw), Arterial Transit Times (ATT), and Cerebral Blood Flow (CBF) using a diffusion weighted ASL sequence.

(B) Spectroscopy at 7T. Proton magnetic resonance spectroscopy (MRS) is a potential imaging marker that can quantify biochemical changes *in vivo*. Spectroscopy can be used to quantify different metabolites *in vivo* including myo-Inositol (mI), choline (Cho), N-acetyl aspartate (NAA), and creatine (Cr)[10–12]. MRS may prove to be useful for understanding neurochemical changes that can precede cognitive decline as well as help distinguish AD from other neurodegenerative diseases. For example, antemortem and postmortem studies have found that NAA, a marker of neuronal integrity, is decreased in AD and correlates with tangle pathology. In addition, mI associated with glia proliferation is elevated in AD and also changes in cognitively intact individuals with higher cortical amyloid load. Ultra-high fields at 7T allow better spectroscopic acquisition due to sharper linewidths and less overlap between metabolites. The IC has developed a spectroscopy sequence at the 7T that can be used to map spectroscopic data in the ADRC cohort. This sequence is a single-voxel (SV) magnetic resonance based on the sLASER sequence for metabolite detection *in vivo*[13]. Preliminary data from voxels show good quantification of metabolites *in vivo* (**Fig. 2**).

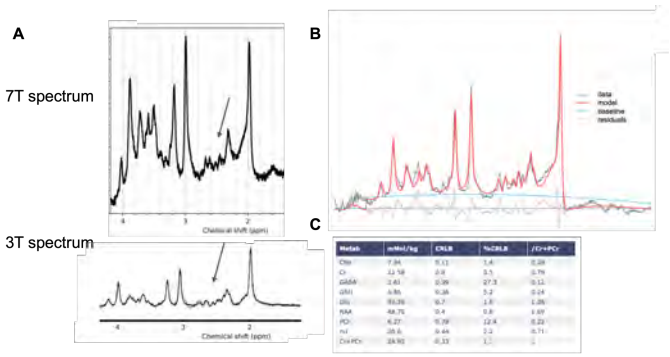


Figure 1: Spectroscopy at 7T. The ADRC is supporting the diabetes brain study to study the effect of diabetes on brain glucose MRS signatures A) Higher magnetic fields allows better linewidth and less overlap between metabolites compared to the 3T (e.g., in the glutamate/glutamine region). B) Data can be fitted using a custom basis and C) results can be obtained in a tabular format.

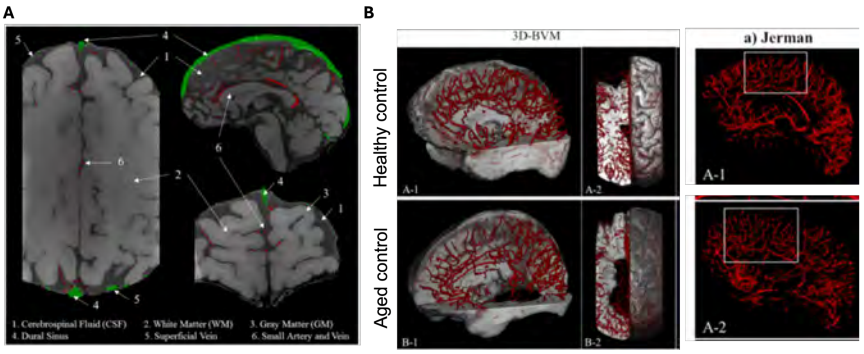


Figure 3: Vessel mapping. The ADRC supports the GADSIL project to study genetic forms of AD and vascular effects. A) Spatial distribution of arteries, veins, and dural sinuses in high-resolution black-blood MRI. B) Representative examples of vessel density image from two control subjects and performance of Hessian-based methods for small cerebral vessel segmentation.

(C) Vessel Mapping. Neuroimaging of cerebral small vessel disease and associated differences in the morphology of small arteries is critical in understanding potential risk factors for early-onset dementia. The IC has developed a specialized sequence, a T1-weighted 3D turbo spin-echo with a variable flip angle scheme (TSE-VFA), which is optimized for visualization of vessels and smaller arteries[14]. This sequence can be deployed at both the 3T and 7T. The TSE-VFA sequence employs a long echo train with variable flip angles to preserve the transverse magnetization. Flowing spins will experience overall dephasing during TSE-VFA scans due to the differences in flow-related phase accruals by the different spin and stimulated echo pathways, leading to black blood contrast. Using optimized vessel segmentation, large vessel pruning, and non-linear registration, a semiautomatic pipeline has been developed for the quantification of small vessel density across the brain and further for localized detection of small vessel changes across populations[2] (**Fig. 3**). The method relies heavily on a novel filtering method called the Jerman filter, which is based on the ratio of Hessian matrix eigenvalues, and has been proposed for enhancement of both 2D and 3D vasculatures[2,15]. Vessel density has been shown to differentiate between young controls and aged controls.

APPROACH

Table 1 describes the sequences included in the ADRC imaging protocol. This protocol reflects the core protocol but does not include all studies across ADRC that might require specific sequences (e.g., contrast-based DCE-MRI or phase contrast MRI).

(1) Aim 1: Provide an imaging framework for ADRC.

(1.A) ADRC imaging protocol. The IC has adopted a standardized imaging protocol across all ADRC studies. Two essential sequences are compliant with the SCAN protocol i) an Accelerated Sagittal 3D T1 Weighted sequence (ADNI4 sequence) with 1mm isotropic voxel resolution and ii) an Accelerated Sagittal 3D fluid-attenuated inversion recovery (FLAIR) sequence (ADNI4 sequences) with 1mm isotropic voxel resolution (**Fig 4. and Table 1**). For a standardized PET protocol please refer to Aim 4. The USC ADRC site has completed ADNI4 certification for MRI and PET acquisition as required by SCAN. The remaining sequences reflect a mix of structural, vascular, and functional imaging sequences that have been developed at the ADRC. The ADRC protocol harmonizes with CLARiTI, DVID, ADNI4, and SCAN. In **Table 1**, a star (*) indicates sequences that are SCAN/CLARiTI/ADNI4 compliant. We want to emphasize the use of novel sequences for measuring BBB integrity and ASL and the use of TSE-VFA for measuring vessel integrity. The IC is working on providing a robust ASL/BBB imaging sequence at the 7T which will hopefully be completed in the next year. The 7T protocol is also accompanied by a novel spectroscopic sequence. All sequences have been extensively validated and used across multiple projects. Parallel to MRI acquisition, the IC obtains heart rate variability (**HRV**) and blood pressure variability (**BPV**) via MR-compatible devices. The IC also uses capnography inside the scanner to measure etCO2 for cerebrovascular measurements[16,17]. The physiological data is collected in parallel to MRI and stored in the USC ADRC archive.

Table 1:	**3T**		**7T**		
Sequence	Voxel Size (mm)	Scan Time	Sequence	Voxel Size (mm)	Scan Time
MPRAGE*	1^3	5:12	MP2RAGE	0.75^3	8:50
FLAIR*	1^3	6:37	FLAIR	0.4x0.4x1.0	6:56
T2Hippo*	0.4x0.4x0.2	4:18	T2Hippo	0.2x0.2x2	5:59
ASL/BBB	3.4x3.4x3.4	10:00	Spectro	12x12x12	6:00
rsfMRI*	2.5^3	5:00	rsfMRI	2x2x1.5	10:54
DTI*	2^3	8:03	DTI	1.3^3	9:08
TSE-VFA	0.5x0.5x0.64	8:39	TSE-VFA	0.5X0.5	10:05
T2star/GRE*	0.5x0.5x1.8	8:44	T2 star	0.5^3	6:48

(1.B) Imaging data management system: The LONI NIX is used by the IC and aids in the direct interactions with the other ADRC Cores including the Clinical Core for subject tracking, the DMSC for data upload and data analytics, the Biomarker Core for cross-checking biological and imaging samples, and the NP Core for enabling *in vivo/ex vivo* studies. We anticipate collecting MR data on **120 subjects every other year** based on numbers estimated by the Clinical Core. Data is uploaded to a centralized USC ADRC archive in communication with the DMSC (**Fig 5**). The uploading system ensures: 1) data privacy; 2) strict access controls to ensure data is only accessible to authorized individuals; 3) tracking of all data accesses to provide an audit trail so that project managers may understand who and in what way data are being accessed; 4) ease of use through interactive, user-friendly interfaces. Overall, these features address the issue of trust in the archive by satisfying depositors that the data is being securely maintained and complies with privacy regulations. The IC will work closely with other cores to ensure stable, secure, and accessible access to ADRC research images. Importantly these systems enable the aggregation (with controlled access) of data and the provision of harmonized analytics. The IC will continue to monitor and adjust the imaging protocol as needed to be compliant with SCAN requirements.

(2) Aim 2: Quality control and MRI analytics. The LONI QC System is an imaging data review and assessment system specifically designed for neuroimaging research studies involving one or more centers[18]. This system allows users to upload their imaging data for automated pre-processing designed to generate a range of vector statistics and derived images directly relevant to gauging the relative quality of the data collected[19,20]. The LONI

QC system is designed for the review and assessment of DTI, fMRI, structural MRI imaging data, DCE-MRI (BBB permeability), and ASL perfusion (CBF). Metrics generated by the LONI QC pipeline provide a set of quality-sensitive metrics. If a value exceeds a normative range, the LONI QC system will flag these for closer inspection. The LONI QC is still expanding to accommodate more advanced features (for example carpetplots or ICA decomposition for resting stage fMRI). The LONI QC workflows interface with the IDA and are implemented on the Stevens Neuroimaging Institute computer grid.

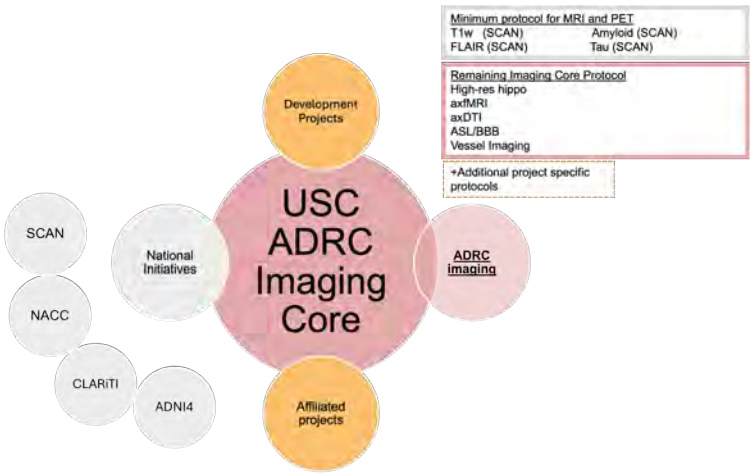


Figure 4: Interactions between the Imaging Core, the ADRC, projects, and national initiatives. The Imaging Core imposes a standardized protocol consisting of T1 and FLAIR, amyloid and tau PET acquisitions for providing data to National Initiatives (SCAN, CLARiTI, ADNI4 etc.) alongside a set of vascular, functional, and structural sequences. Additional sequences will be added on demand and in support for Development and Affiliated projects.

(2.A) MRI Analytics. The IC obtains a variety of analytics with a focus on imaging markers for the pathophysiology of *APOE* ε4 (see Overall: Aim 1). Volumetrics and cortical thickness per each ADRC participant is obtained using the Freesurfer pipeline that operates on each participant's T1-weighted image[21,22]. White matter hyperintensities (**WMH**) volume is obtained from an automated pipeline that segments FLAIR images[23]. The number of perivascular spaces (PVS) and perivascular space diameter is obtained using appropriate modules of the LONI pipeline[24] that operate on each participant's T1-weighted image[3,25]. Diffusion Tensor Image (DTI) Analysis of perivascular spaces (DTI-ALPS) is obtained using the DTI data[26]. Cerebral Amyloid Angiopathy (**CAA**) is quantified using the Boston Criteria white matter markers that rely on quantification of WMH-multispot patterns, and identification of severe centrum semiovale perivascular spaces[27]. Carotid pulse wave velocity (**PWV**) quantifications will be obtained from the phase contrast MRI when available[28]. Research-based neurodegeneration (N) characterization of ADRC participants is determined based on obtained cortical thickness values. N+ is determined based on a cut-off of 2.68 mm of the weighted average of the mean cortical thickness in individual ROIs of the entorhinal cortex, fusiform, inferior temporal gyri, and middle temporal gyri[29,30]. In addition, for (v) characterization of participants, vessel density, **CBF**, ATT, and Kw, are derived from the remaining sequences of the protocol. fMRI connectivity, white matter integrity, and resting-state cerebrovascular reactivity (**CVR**) are obtained using resting-state fMRI and etCO2. The IC will continue to disseminate these analytics as part of the USC ADRC archive. Pre-processing pipelines will also be disseminated via the LONI pipeline. During the past cycle, the IC has QC-ed and processed the following number of MRI images.

Table 2: MRI scan analytics	**T1w**	**DTI**	**fMRI**
Total scans (ADRC cohort)	278	278	278
QC-ed	183	96	89
Processed	117	96	89

(2.B) ALZ-NET and ARIA workflow. For participants that are on anti-amyloid treatment and for monitoring for amyloid-related imaging abnormalities (ARIA)-E/H, the IC: 1) will assist with clinical imaging data deposition to ALZ-NET and, when appropriate consent is in place; 2) will assist with storing de-identified clinical imaging data to the USC ADRC archive when UDS is available. For non-UDS participants, a different setup will be implemented where non-identification will be utilized via a different key stored in a dedicated clinical server. The IC will assist with the analysis of available FLAIR and SWI images for ARIA detection and connection of CAA risk to ARIA outcomes.

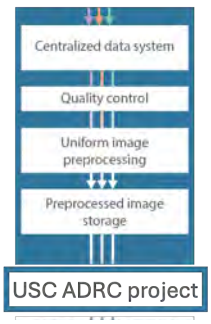


Figure 5: Schematic of imaging data flow.

(3) Aim 3: Ultra-high field 7T MRI. The Stevens Neuroimaging Institute possesses a Siemens 7T Terra ultra-high field MRI that is an FDA-approved instrument. The IC has developed novel pulse sequences that can be used for high-resolution structural, functional, vascular, and spectroscopic data acquisition. The system comes with an 8-channel transmit; 32-channel receive head coil by Nova Medical Inc. (Nova 8Tx/32Rx). Its capabilities will significantly improve the B1 field homogeneity and visualization of the temporal lobe and brain stem. By imaging subjects at both 3T and 7T, we will compare results and quantify advantages of higher field. We already obtained IRB approval to add this to existing AD imaging protocols for those subjects who agree to an additional scanning session. Our sequences are optimized to be more efficient in terms of time and subject comfort.

(4) Aim 4: PET acquisition and analytics. The IC investigators have developed a specific PET Protocol and respective analysis pipelines. For all ADCR PET data, the acquisition now follows the proposed SCAN protocol. The parameters proposed assure a standardized acquisition across all participants and follow national initiative and the RFA directives. For PET, we acquire amyloid and tau data using approved amyloid SCAN tracers (e.g., [18F]florbetaben (FBB) for amyloid and [18F]flortaucipir (AV-1451) for tau). Scans are obtained per SCAN protocol parameters at either 50-70 minutes (florbetapir) or 90-105 minutes (florbetaben or flortaucipir) post-injection. Reconstruction parameters adhere to SCAN requirements: Method Iterative: OSEM-3D), 4 iterations, 24 subsets, 400x400x81 grid, Zoom 2.0, “no smoothing” setup, and Match Slices “Off”. For PET data analytics, the IC has developed a pipeline that produces SUVRs from raw DICOM data (**Fig. 6**). The pipeline takes anatomical T1 weighted data and PET images within six months of each other. In turn, the Freesurfer segmentation is applied to obtain regions of interest and cerebellar segmentations for SUVR calculation. For amyloid, ADNI4 regions of interest are calculated for composite ROIs consisting of frontal, parietal, lateral temporal, and cingulate cortices. For tau, SUVRs are calculated for Braak stage regions[31,32]. Amyloid and tau PET data reference regions include the whole cerebellum and the inferior cerebellar gray matter respectively. The IC collects PET data using standardized amyloid and tau tracers across multiple ADRC studies including SCAN (PI: Chui), Genetic Alzheimer’s Disease Subtypes in Latinos (GADSIL) (PI: Ringman), and PPG (PI: Toga). The pipeline has been validated against the PET ADNI pipelines[33] (**Fig. 6**).

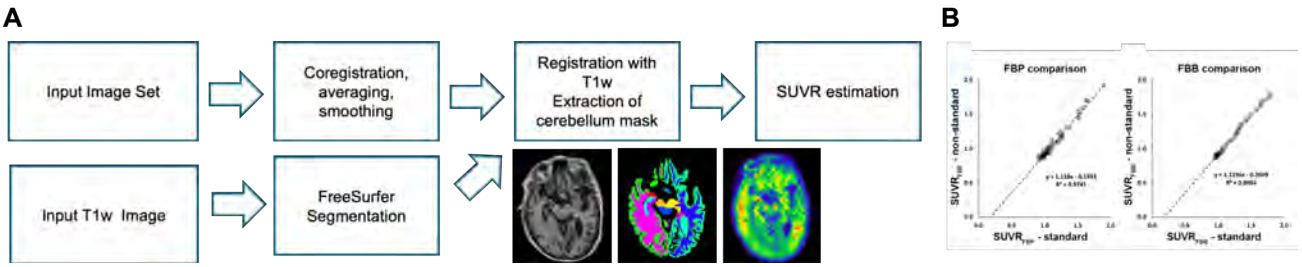


Figure 6: A) Schematic of the ADRC PET pipeline B) We found high correspondence between values obtained using the Core's pipeline and the PET ADNI4 pipeline on the same data.

Research-based A+/T+ characterization of ADRC participants will be done by the IC based on derived SUVRs and, in the case of amyloids, centiloids. For amyloid imaging, since the IC has processed both ADNI4 and ADRC data, a linear regression has been obtained that can convert ADRC SUVRs to centiloids. Specifically, the equations for each amyloid tracer presented (**Fig. 6**) are used to convert ADRC SUVR values to ADNI4-equivalent standard SUVR values, which are then converted to centiloids values using the equations determined for the ADNI data. An amyloid centiloid value of 19 is used as the cutoff for amyloid positivity, as the rate of change in amyloid PET has been shown to reliably increase at this level (34). For tau imaging, the IC is experimenting with a new “centiloid-like” scale called CenTauR that allows standardization of tau PET data in a universal standard scale[4,34]. So far efforts have included multiple tau tracers including those proposed by the ADRC, so we are hoping to provide CenTauR values for the ADRC participants. During the last cycle, the IC has QC-ed and analyzed the following amyloid and tau PET scans.

Table 3: PET scan analytics	**Amyloid-PET**	**Tau-PET**
Total scans (ADRC cohort)	111	60
Qc-ed	63	53
Analytics	33	31

(5) Aim 5: Conduct education and training in modern imaging techniques. The IC continues to engage students, postdocs, faculty, and community across departments and campuses. **USC INI provides a Master's program in Neuroimaging and Informatics** that began in 2015 and has successfully trained over 60 students. Graduates of this program have secured positions as graduate students, medical students, and working professionals at Stanford, Johns Hopkins, UC Irvine, Georgia Tech, USC, Wake Forest University, and within the private sector as data analysts and study consultants. **The USC INI also recently received a T32 postdoctoral program in Multiscalar, Multimodal, and Translational Research in AD**. Fellows in this program are exposed to multiple omics for conducting large studies involving imaging, genetics, and behavior. The IC, in concert with the Center for Image Acquisition (CIA), offers free pilot studies for applicants who are intent on developing a new study in AD and need preliminary data to test their methods and refine their design. Additionally, we encourage all new investigators to collaborate with a member of the IC team or the CIA for sequence optimization and deployment.

Table 4: Imaging Core supported studies	**PI**	**Imaging Core contribution**
ADRC-Pilots		
MRS pilot for AD pathology	Sheikh-Bahaei	Imaging protocol development
Training		
Neurocognitive Aging and Analytics Research Education (NAARE)	McEligot, Braskie, Toga	Imaging data harmonization
Training in the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease (AD)	Toga	Imaging data training
Clinical Trials		
ADNI4, DIAN TU	Schneider	Imaging protocol (MRI, PET)
Program projects		
Vascular Contributions to Dementia and Genetic Risk Factors for AD	Toga	Imaging protocol, QC, analysis
NIH Initiatives		
SCAN, CLARiTI	Chui, Schneider	Imaging protocol development (MRI, PET)
NIH Cooperative Grants		
Genetic Alzheimer's Disease Subtypes in Latinos (GADSIL)	Ringman	Imaging protocol, QC, analysis (MRI, PET analytics)
Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression (DVR)	Joe	Imaging analysis
MarkVCID: Validation of imaging and blood-based small vessel VCID biomarkers in multiethnic populations	Wang, Ringman	Imaging protocol, QC, analysis (DW-pCASL, DTI-ALPS)
Diverse VCID	Joe	Imaging protocol, QC, analysis

Table 4. Representative studies that are supported by the USC ADRC IC. This includes a combination of program project grants, parent research grants, clinical trials, training and fellowship awards, and cooperative grants. Collectively the USC ADRC IC supports over 20 active ADRC-affiliated research across USC. The LONI NIX is used by the IC and aids in the direct interactions with the other ADRC Cores including the Clinical Core for subject tracking, the DMSC for data upload to the USC ADRC archive and data analytics, the Biomarkers Core for cross-checking biological and imaging samples, and the Neuropathology Core for enabling in vivo-ex vivo studies. We anticipate collecting MRI and PET data from **120 participants to be scanned every other year as described in the Overview and Clinical Core timeline of subject recruitment.**

PHS Human Subjects and Clinical Trials Information

Use of Human Specimens and/or Data

* Does any of the proposed research involve human specimens and/or data?	●Yes	❍No
Provide an explanation of why the application does not involve human subjects research		
Are Human Subjects Involved	●Yes	❍No
Is the Project Exempt from Federal regulations?	❍Yes	●No
Exemption Number	❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8	
Other Requested Information	Human_Subjects_Research1044447181.pdf	

Human Subject Studies

Study#	Study Title	Clinical Trial?
The form does not have any study records		

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Human Subjects Research

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and other clinical studies. Clinical intervention trials are affiliated with, but are not a funded part of the ADRC.

Protection of human subjects is a primary concern for the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although the specific details may differ among the indivdiual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The specifics of the **Protection of Human Subjects** can be found in the **Overall** Section of this application.